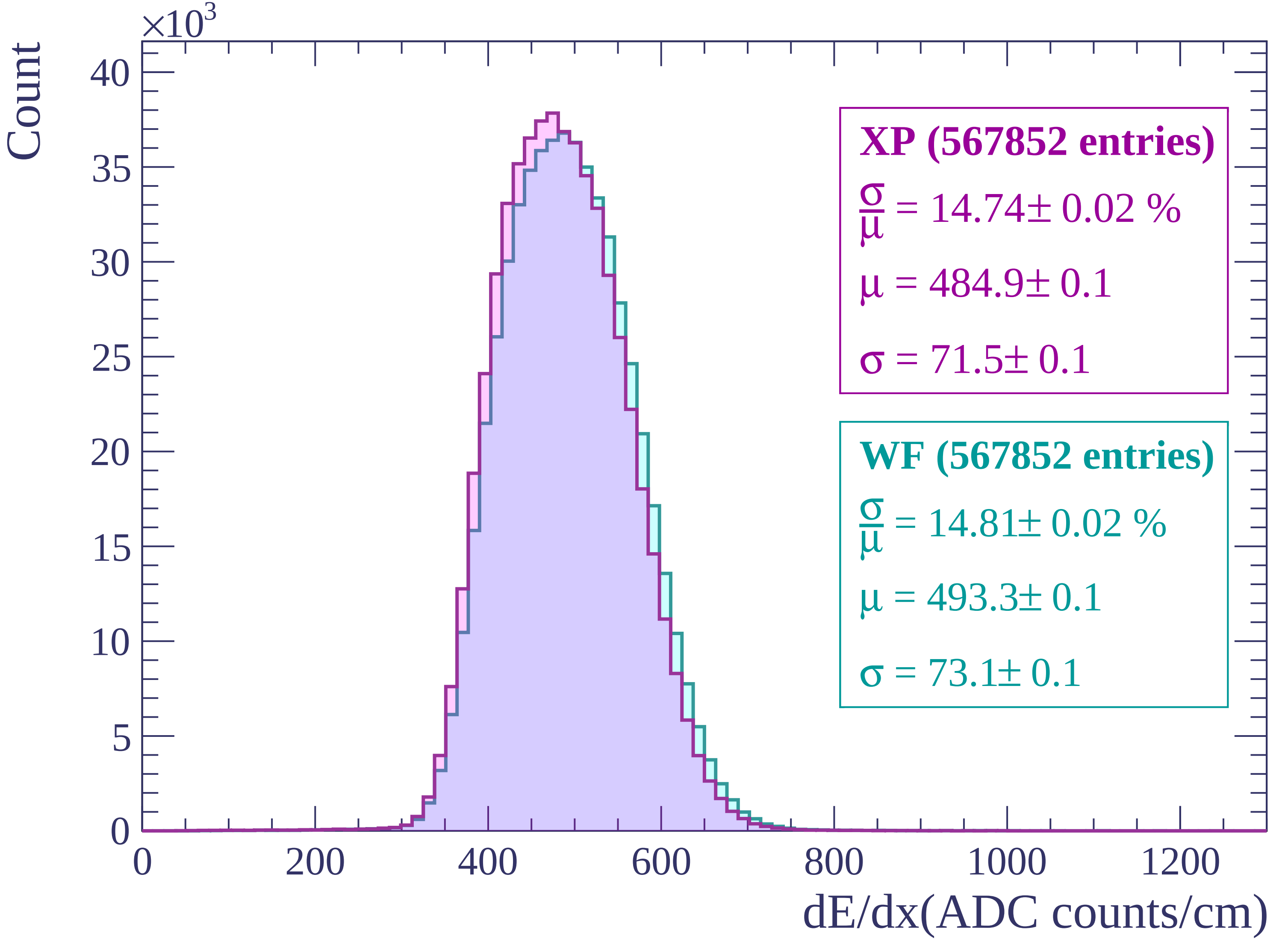
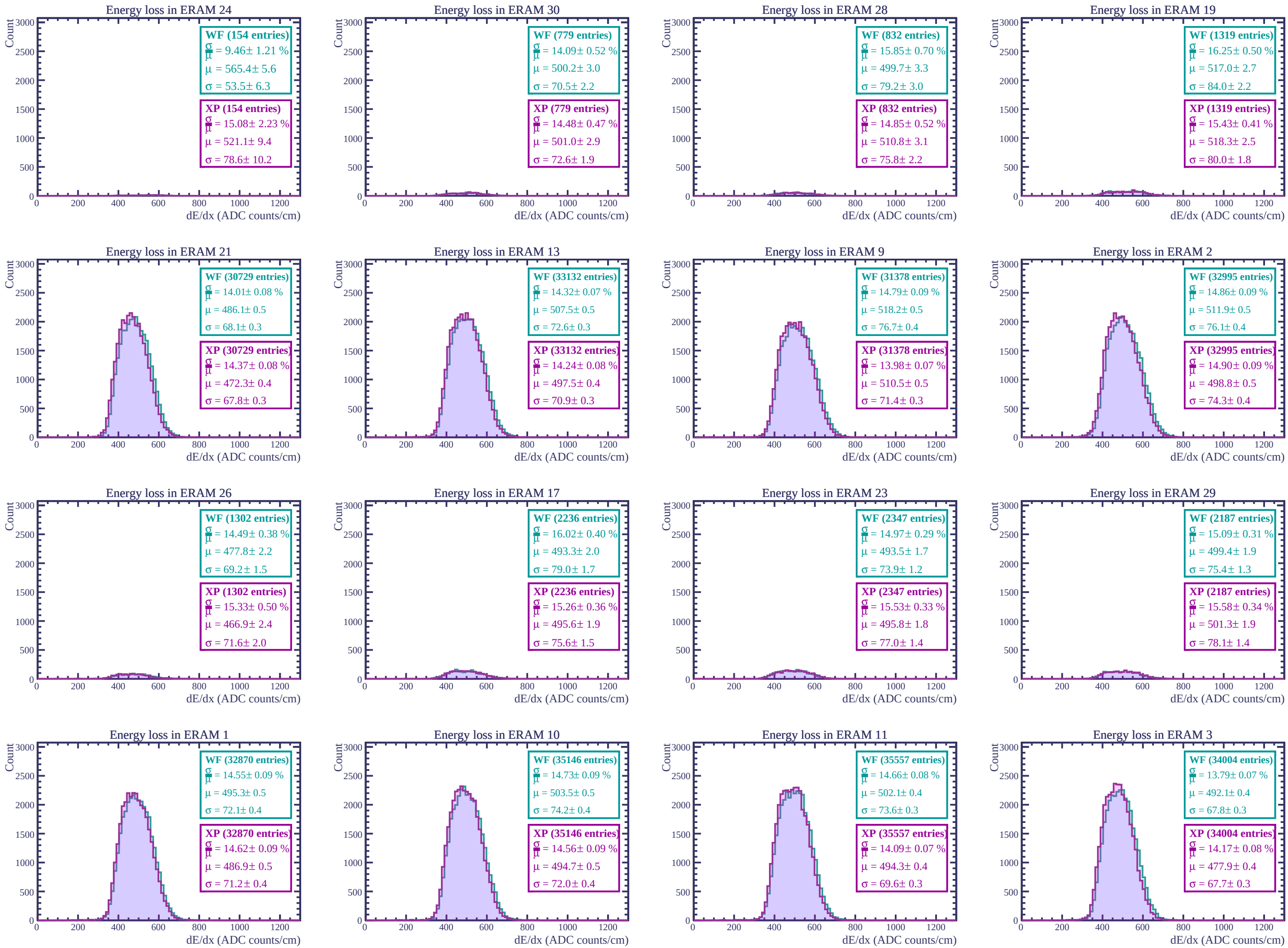
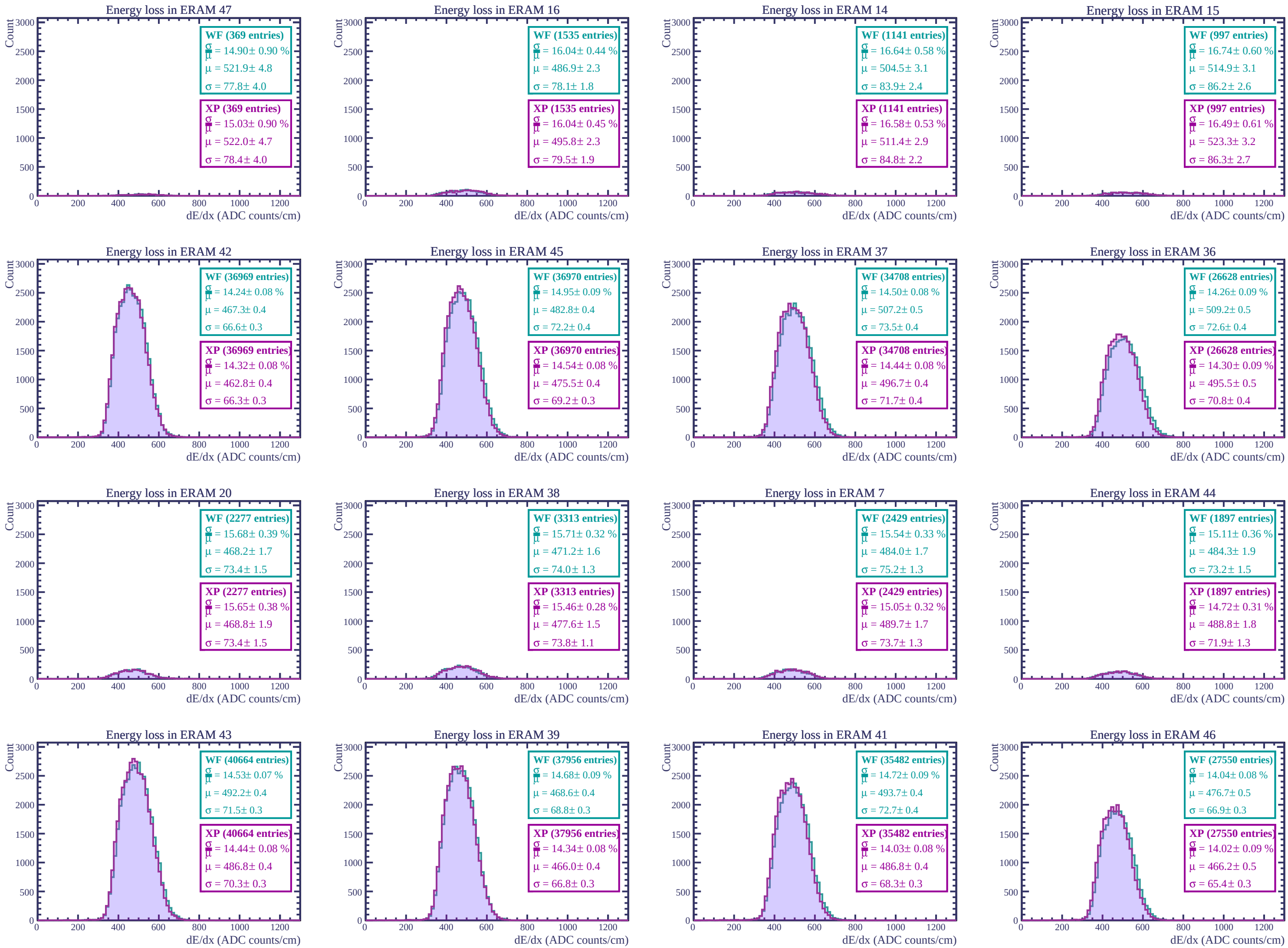
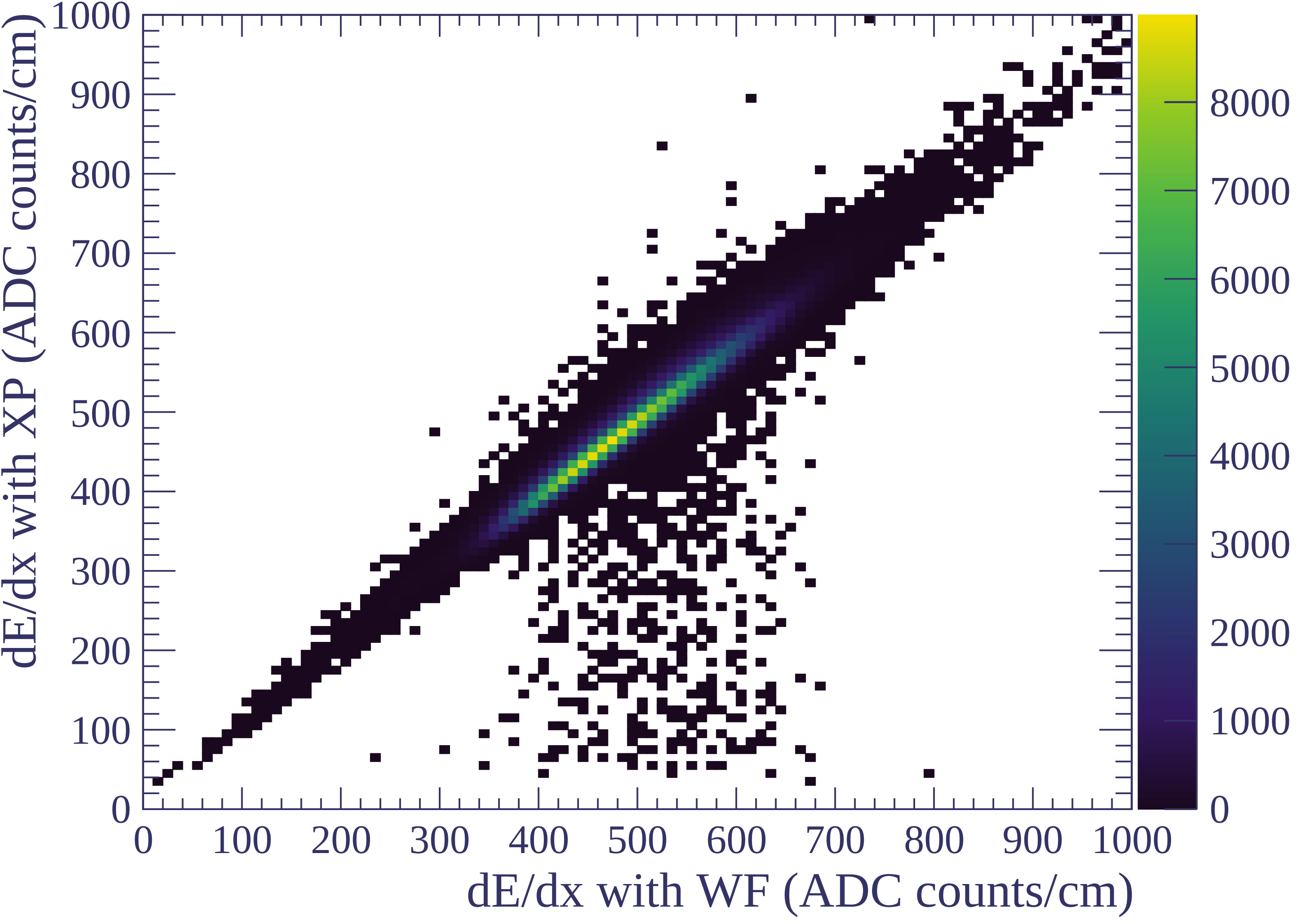


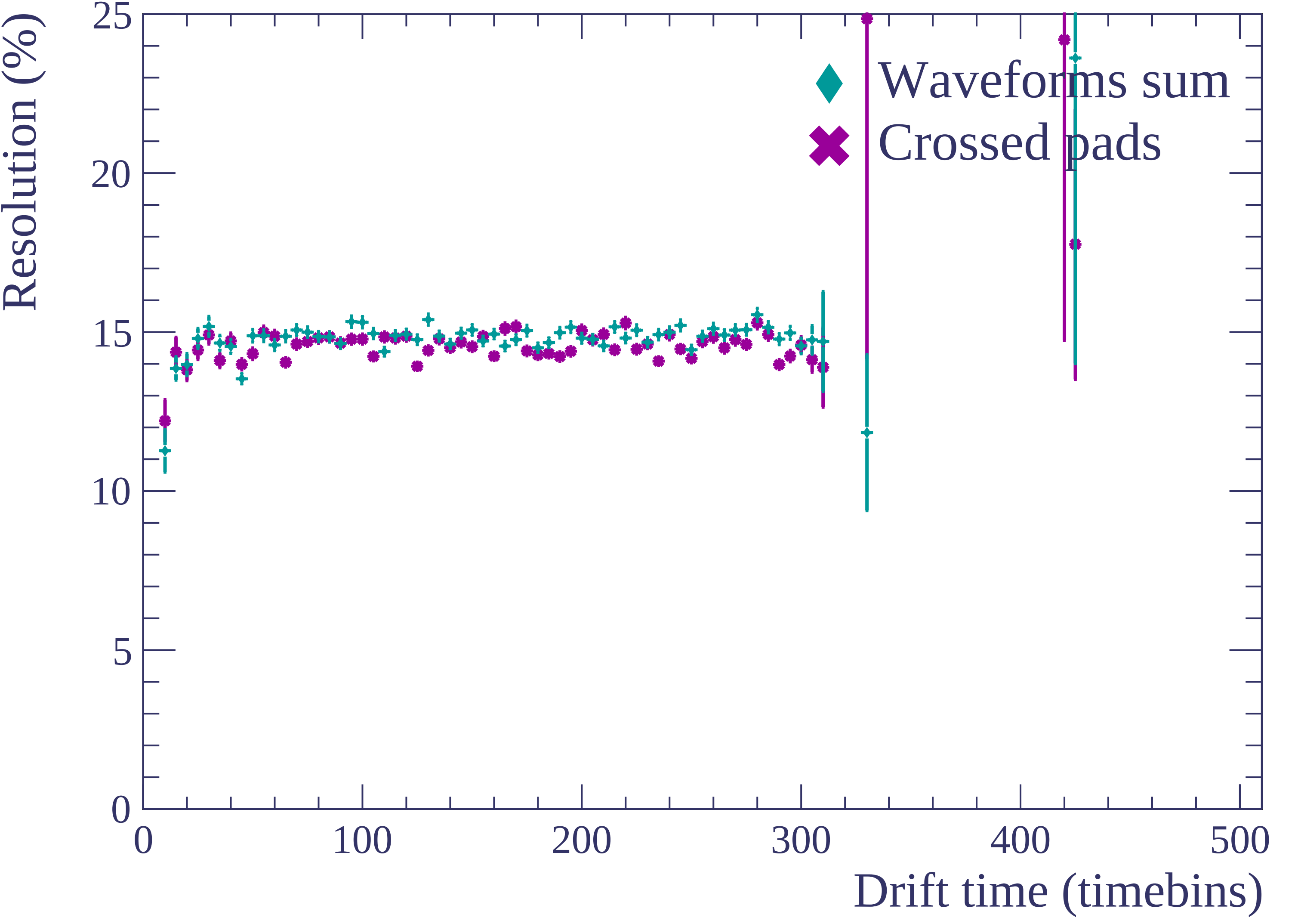
Count
$\times 10^3$
40
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (567852 entries)
$\frac{\sigma}{\mu}$ = 14.74± 0.02 %
μ = 484.9± 0.1
σ = 71.5± 0.1
WF (567852 entries)
$\frac{\sigma}{\mu}$ = 14.81± 0.02 %
μ = 493.3± 0.1
σ = 73.1± 0.1


Energy loss in ERAM 24
WF (154 entries)
σ/μ = 9.46± 1.21 %
μ = 565.4± 5.6
σ = 53.5± 6.3
XP (154 entries)
σ/μ = 15.08± 2.23 %
μ = 521.1± 9.4
σ = 78.6± 10.2
Energy loss in ERAM 30
WF (779 entries)
σ/μ = 14.09± 0.52 %
μ = 500.2± 3.0
σ = 70.5± 2.2
XP (779 entries)
σ/μ = 14.48± 0.47 %
μ = 501.0± 2.9
σ = 72.6± 1.9
Energy loss in ERAM 28
WF (832 entries)
σ/μ = 15.85± 0.70 %
μ = 499.7± 3.3
σ = 79.2± 3.0
XP (832 entries)
σ/μ = 14.85± 0.52 %
μ = 510.8± 3.1
σ = 75.8± 2.2
Energy loss in ERAM 19
WF (1319 entries)
σ/μ = 16.25± 0.50 %
μ = 517.0± 2.7
σ = 84.0± 2.2
XP (1319 entries)
σ/μ = 15.43± 0.41 %
μ = 518.3± 2.5
σ = 80.0± 1.8
Energy loss in ERAM 21
WF (30729 entries)
σ/μ = 14.01± 0.08 %
μ = 486.1± 0.5
σ = 68.1± 0.3
XP (30729 entries)
σ/μ = 14.37± 0.08 %
μ = 472.3± 0.4
σ = 67.8± 0.3
Energy loss in ERAM 13
WF (33132 entries)
σ/μ = 14.32± 0.07 %
μ = 507.5± 0.5
σ = 72.6± 0.3
XP (33132 entries)
σ/μ = 14.24± 0.08 %
μ = 497.5± 0.4
σ = 70.9± 0.3
Energy loss in ERAM 9
WF (31378 entries)
σ/μ = 14.79± 0.09 %
μ = 518.2± 0.5
σ = 76.7± 0.4
XP (31378 entries)
σ/μ = 13.98± 0.07 %
μ = 510.5± 0.5
σ = 71.4± 0.3
Energy loss in ERAM 2
WF (32995 entries)
σ/μ = 14.86± 0.09 %
μ = 511.9± 0.5
σ = 76.1± 0.4
XP (32995 entries)
σ/μ = 14.90± 0.09 %
μ = 498.8± 0.5
σ = 74.3± 0.4
Energy loss in ERAM 26
WF (1302 entries)
σ/μ = 14.49± 0.38 %
μ = 477.8± 2.2
σ = 69.2± 1.5
XP (1302 entries)
σ/μ = 15.33± 0.50 %
μ = 466.9± 2.4
σ = 71.6± 2.0
Energy loss in ERAM 17
WF (2236 entries)
σ/μ = 16.02± 0.40 %
μ = 493.3± 2.0
σ = 79.0± 1.7
XP (2236 entries)
σ/μ = 15.26± 0.36 %
μ = 495.6± 1.9
σ = 75.6± 1.5
Energy loss in ERAM 23
WF (2347 entries)
σ/μ = 14.97± 0.29 %
μ = 493.5± 1.7
σ = 73.9± 1.2
XP (2347 entries)
σ/μ = 15.53± 0.33 %
μ = 495.8± 1.8
σ = 77.0± 1.4
Energy loss in ERAM 29
WF (2187 entries)
σ/μ = 15.09± 0.31 %
μ = 499.4± 1.9
σ = 75.4± 1.3
XP (2187 entries)
σ/μ = 15.58± 0.34 %
μ = 501.3± 1.9
σ = 78.1± 1.4
Energy loss in ERAM 1
WF (32870 entries)
σ/μ = 14.55± 0.09 %
μ = 495.3± 0.5
σ = 72.1± 0.4
XP (32870 entries)
σ/μ = 14.62± 0.09 %
μ = 486.9± 0.5
σ = 71.2± 0.4
Energy loss in ERAM 10
WF (35146 entries)
σ/μ = 14.73± 0.09 %
μ = 503.5± 0.5
σ = 74.2± 0.4
XP (35146 entries)
σ/μ = 14.56± 0.09 %
μ = 494.7± 0.5
σ = 72.0± 0.4
Energy loss in ERAM 11
WF (35557 entries)
σ/μ = 14.66± 0.08 %
μ = 502.1± 0.4
σ = 73.6± 0.3
XP (35557 entries)
σ/μ = 14.09± 0.07 %
μ = 494.3± 0.4
σ = 69.6± 0.3
Energy loss in ERAM 3
WF (34004 entries)
σ/μ = 13.79± 0.07 %
μ = 492.1± 0.4
σ = 67.8± 0.3
XP (34004 entries)
σ/μ = 14.17± 0.08 %
μ = 477.9± 0.4
σ = 67.7± 0.3
Count
dE/dx (ADC counts/cm)

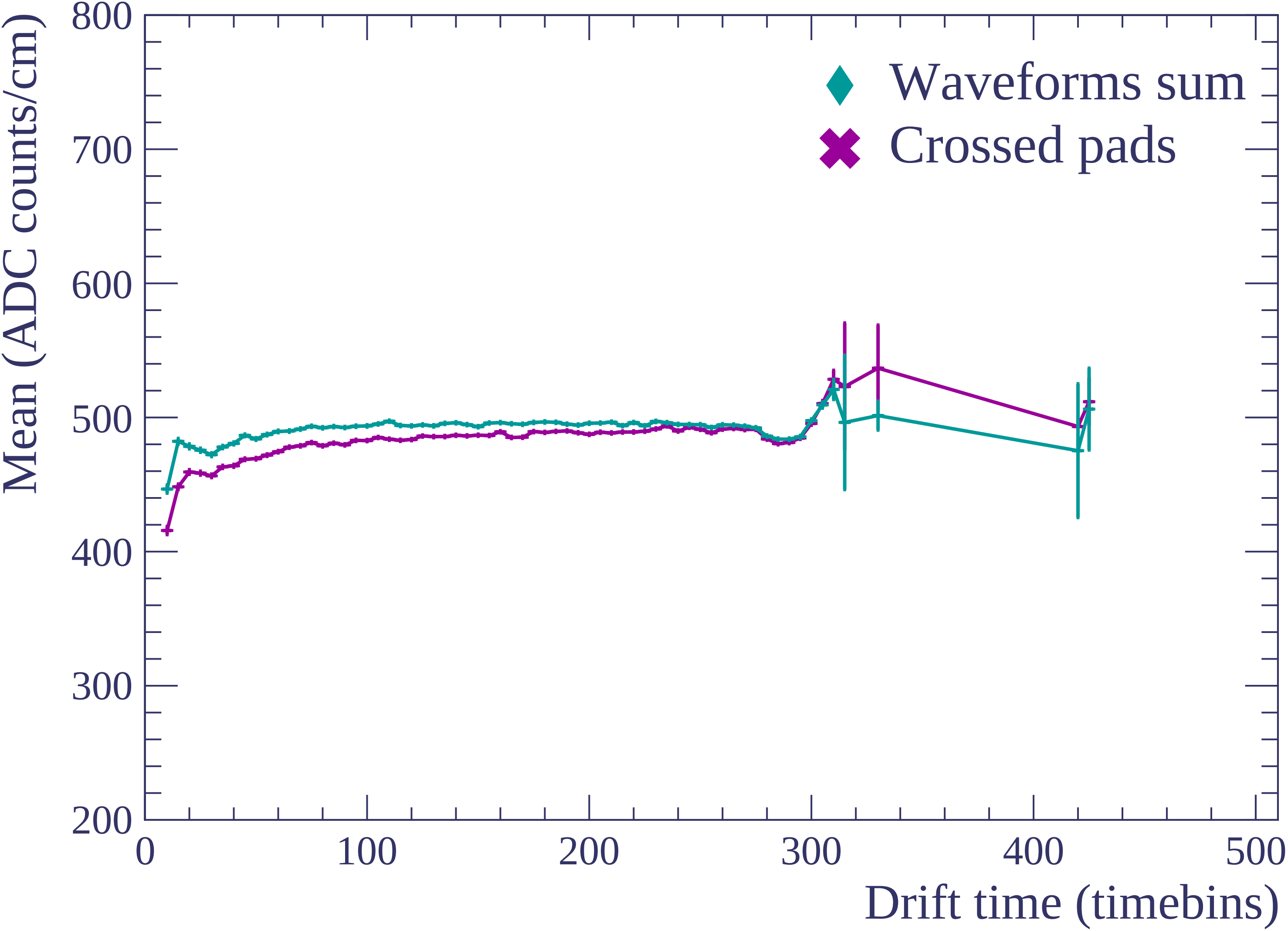

Energy loss in ERAM 47
WF (369 entries)
σ/μ = 14.90± 0.90 %
μ = 521.9± 4.8
σ = 77.8± 4.0
XP (369 entries)
σ/μ = 15.03± 0.90 %
μ = 522.0± 4.7
σ = 78.4± 4.0
Energy loss in ERAM 16
WF (1535 entries)
σ/μ = 16.04± 0.44 %
μ = 486.9± 2.3
σ = 78.1± 1.8
XP (1535 entries)
σ/μ = 16.04± 0.45 %
μ = 495.8± 2.3
σ = 79.5± 1.9
Energy loss in ERAM 14
WF (1141 entries)
σ/μ = 16.64± 0.58 %
μ = 504.5± 3.1
σ = 83.9± 2.4
XP (1141 entries)
σ/μ = 16.58± 0.53 %
μ = 511.4± 2.9
σ = 84.8± 2.2
Energy loss in ERAM 15
WF (997 entries)
σ/μ = 16.74± 0.60 %
μ = 514.9± 3.1
σ = 86.2± 2.6
XP (997 entries)
σ/μ = 16.49± 0.61 %
μ = 523.3± 3.2
σ = 86.3± 2.7
Energy loss in ERAM 42
WF (36969 entries)
σ/μ = 14.24± 0.08 %
μ = 467.3± 0.4
σ = 66.6± 0.3
XP (36969 entries)
σ/μ = 14.32± 0.08 %
μ = 462.8± 0.4
σ = 66.3± 0.3
Energy loss in ERAM 45
WF (36970 entries)
σ/μ = 14.95± 0.09 %
μ = 482.8± 0.4
σ = 72.2± 0.4
XP (36970 entries)
σ/μ = 14.54± 0.08 %
μ = 475.5± 0.4
σ = 69.2± 0.3
Energy loss in ERAM 37
WF (34708 entries)
σ/μ = 14.50± 0.08 %
μ = 507.2± 0.5
σ = 73.5± 0.4
XP (34708 entries)
σ/μ = 14.44± 0.08 %
μ = 496.7± 0.4
σ = 71.7± 0.4
Energy loss in ERAM 36
WF (26628 entries)
σ/μ = 14.26± 0.09 %
μ = 509.2± 0.5
σ = 72.6± 0.4
XP (26628 entries)
σ/μ = 14.30± 0.09 %
μ = 495.5± 0.5
σ = 70.8± 0.4
Energy loss in ERAM 20
WF (2277 entries)
σ/μ = 15.68± 0.39 %
μ = 468.2± 1.7
σ = 73.4± 1.5
XP (2277 entries)
σ/μ = 15.65± 0.38 %
μ = 468.8± 1.9
σ = 73.4± 1.5
Energy loss in ERAM 38
WF (3313 entries)
σ/μ = 15.71± 0.32 %
μ = 471.2± 1.6
σ = 74.0± 1.3
XP (3313 entries)
σ/μ = 15.46± 0.28 %
μ = 477.6± 1.5
σ = 73.8± 1.1
Energy loss in ERAM 7
WF (2429 entries)
σ/μ = 15.54± 0.33 %
μ = 484.0± 1.7
σ = 75.2± 1.3
XP (2429 entries)
σ/μ = 15.05± 0.32 %
μ = 489.7± 1.7
σ = 73.7± 1.3
Energy loss in ERAM 44
WF (1897 entries)
σ/μ = 15.11± 0.36 %
μ = 484.3± 1.9
σ = 73.2± 1.5
XP (1897 entries)
σ/μ = 14.72± 0.31 %
μ = 488.8± 1.8
σ = 71.9± 1.3
Energy loss in ERAM 43
WF (40664 entries)
σ/μ = 14.53± 0.07 %
μ = 492.2± 0.4
σ = 71.5± 0.3
XP (40664 entries)
σ/μ = 14.44± 0.08 %
μ = 486.8± 0.4
σ = 70.3± 0.3
Energy loss in ERAM 39
WF (37956 entries)
σ/μ = 14.68± 0.09 %
μ = 468.6± 0.4
σ = 68.8± 0.3
XP (37956 entries)
σ/μ = 14.34± 0.08 %
μ = 466.0± 0.4
σ = 66.8± 0.3
Energy loss in ERAM 41
WF (35482 entries)
σ/μ = 14.72± 0.09 %
μ = 493.7± 0.4
σ = 72.7± 0.4
XP (35482 entries)
σ/μ = 14.03± 0.08 %
μ = 486.8± 0.4
σ = 68.3± 0.3
Energy loss in ERAM 46
WF (27550 entries)
σ/μ = 14.04± 0.08 %
μ = 476.7± 0.5
σ = 66.9± 0.3
XP (27550 entries)
σ/μ = 14.02± 0.09 %
μ = 466.2± 0.5
σ = 65.4± 0.3
Count
dE/dx (ADC counts/cm)



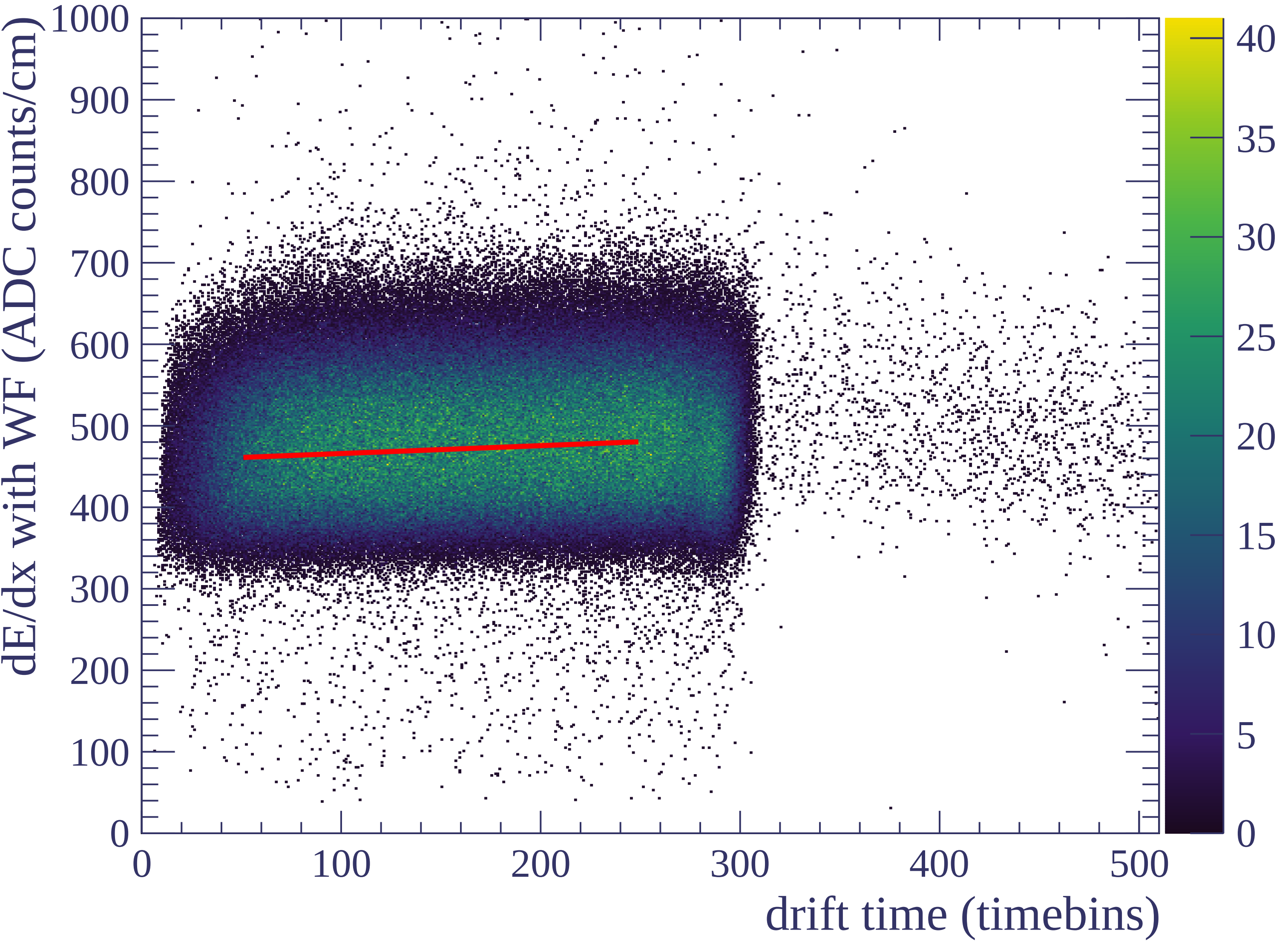
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
1000
2000
3000
4000
5000
6000
7000
8000



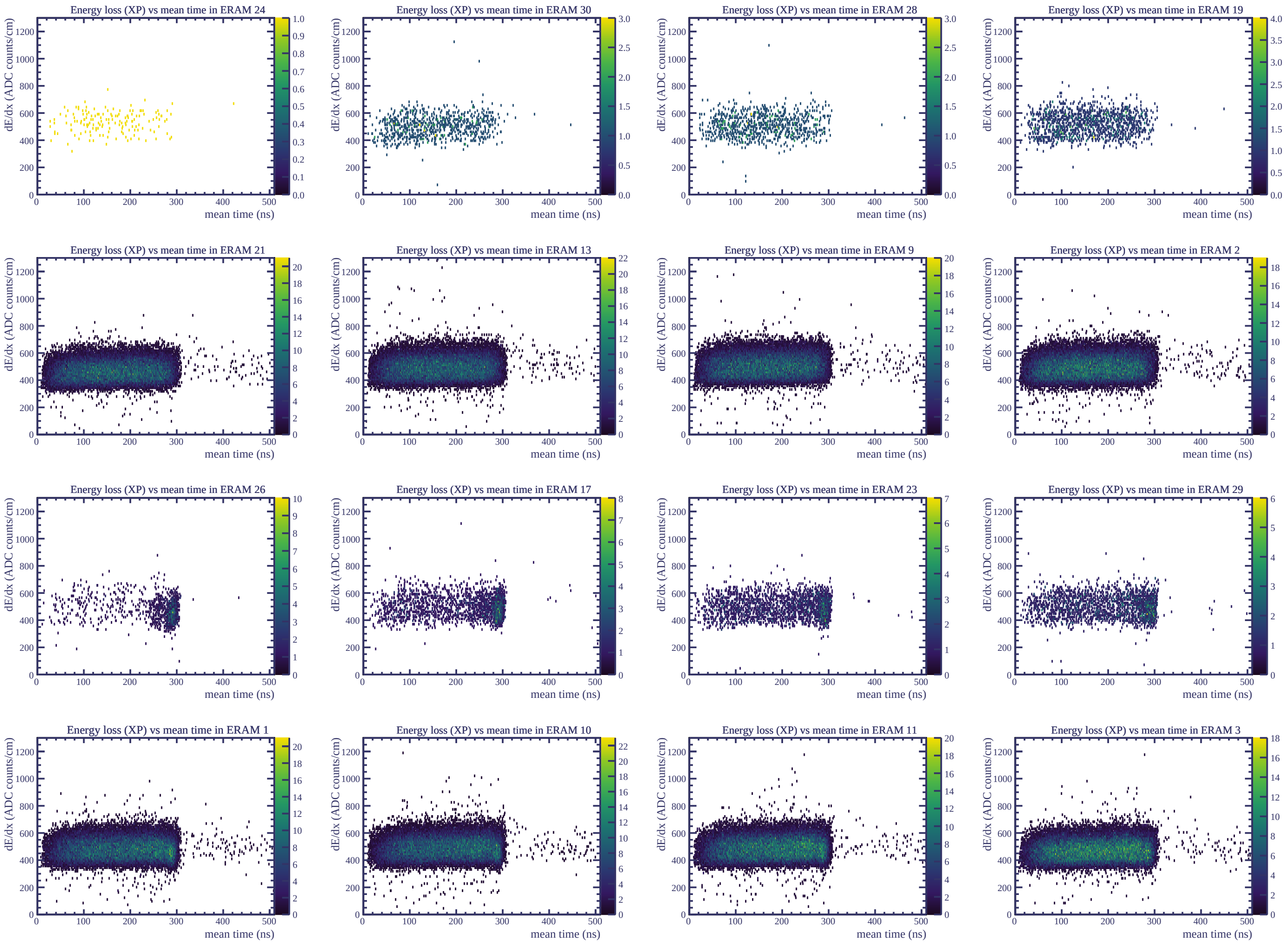
Resolution (%)
25
20
15
10
5
0
0
100
200
300
400
500
Drift time (timebins)
Waveforms sum
Crossed pads

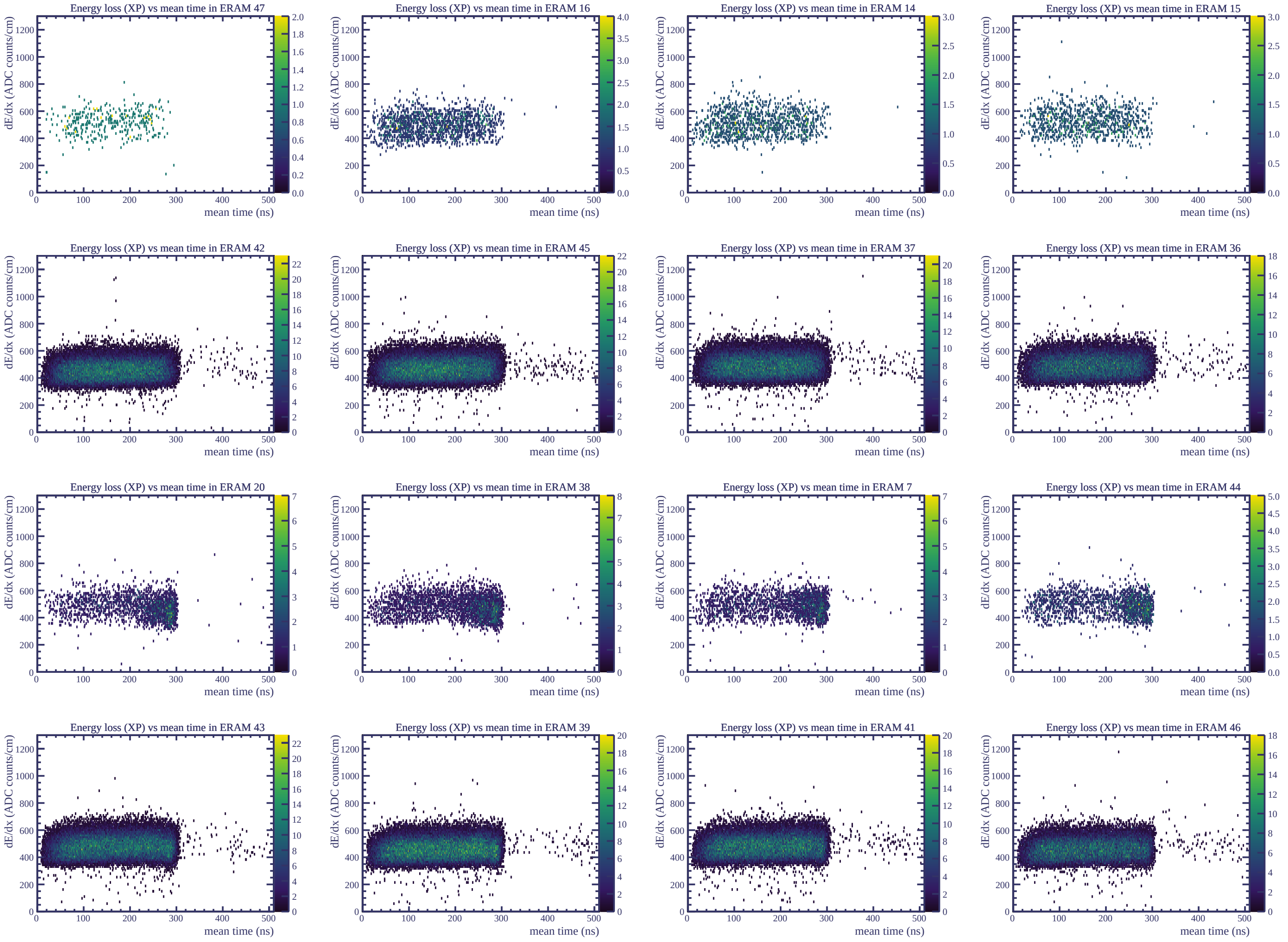


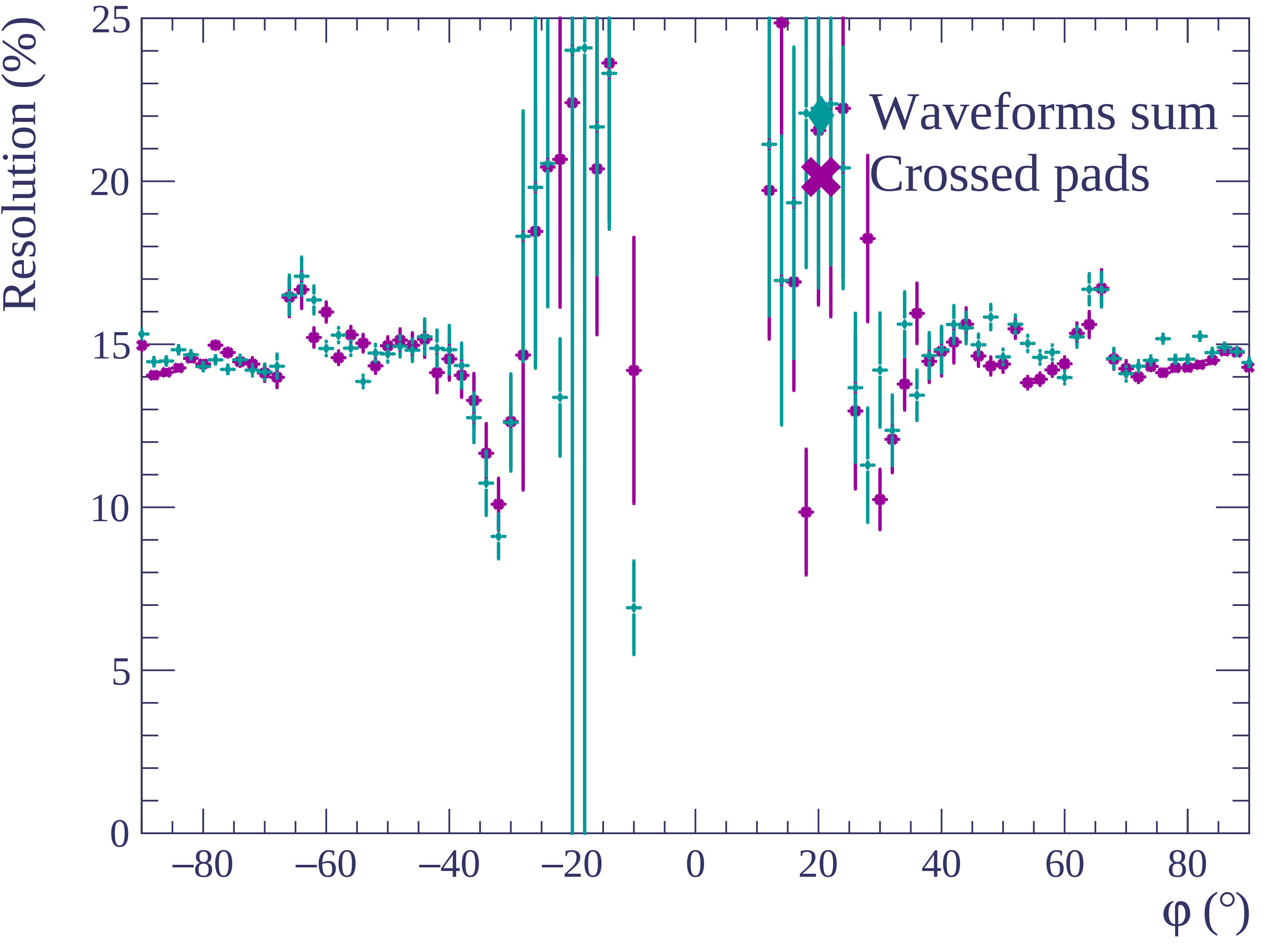
Mean (ADC counts/cm)
Drift time (timebins)
Waveforms sum
Crossed pads
200
300
400
500
600
700
800
0
100
200
300
400
500



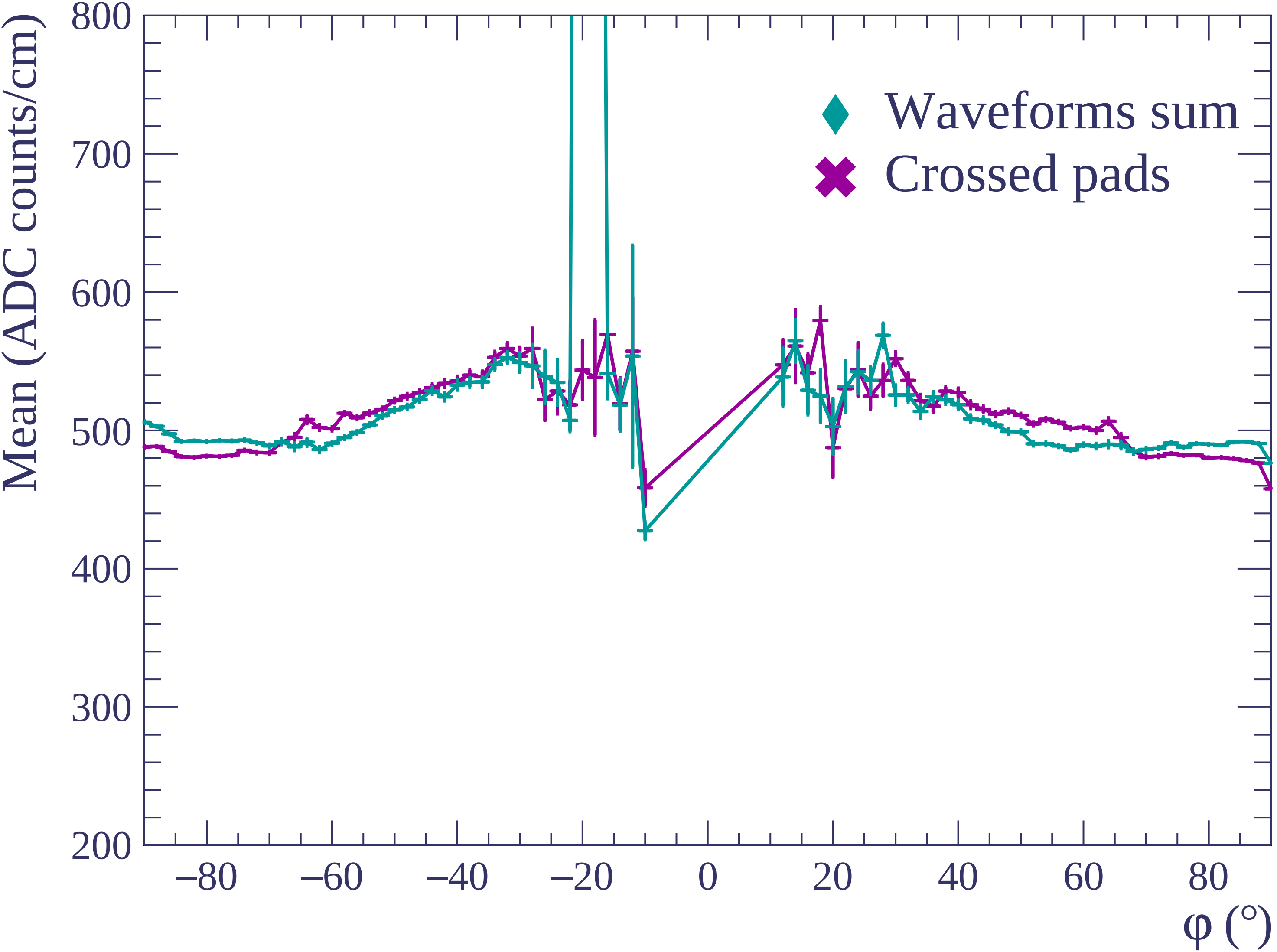
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
5
10
15
20
25
30
35
40


Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

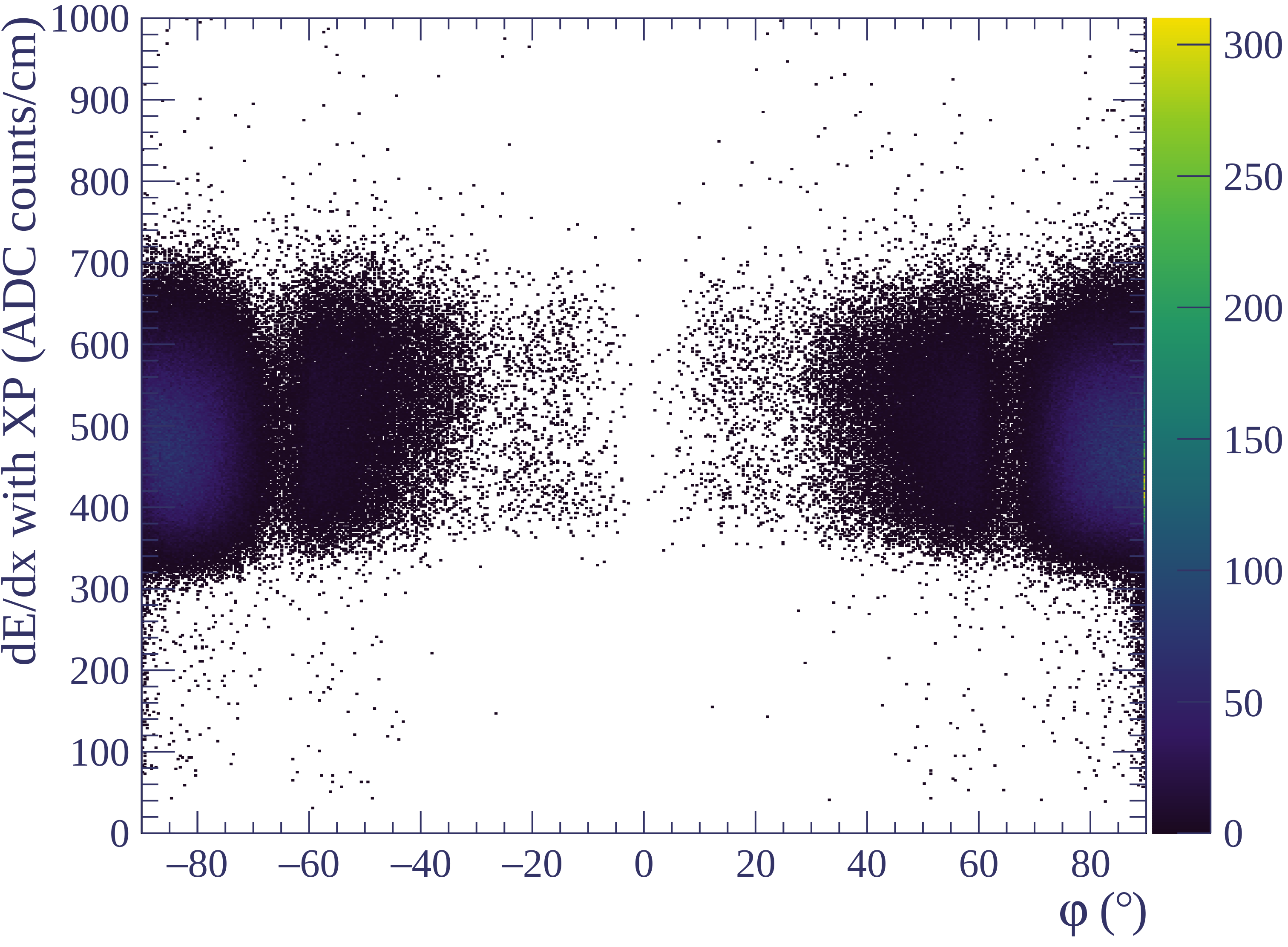

Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

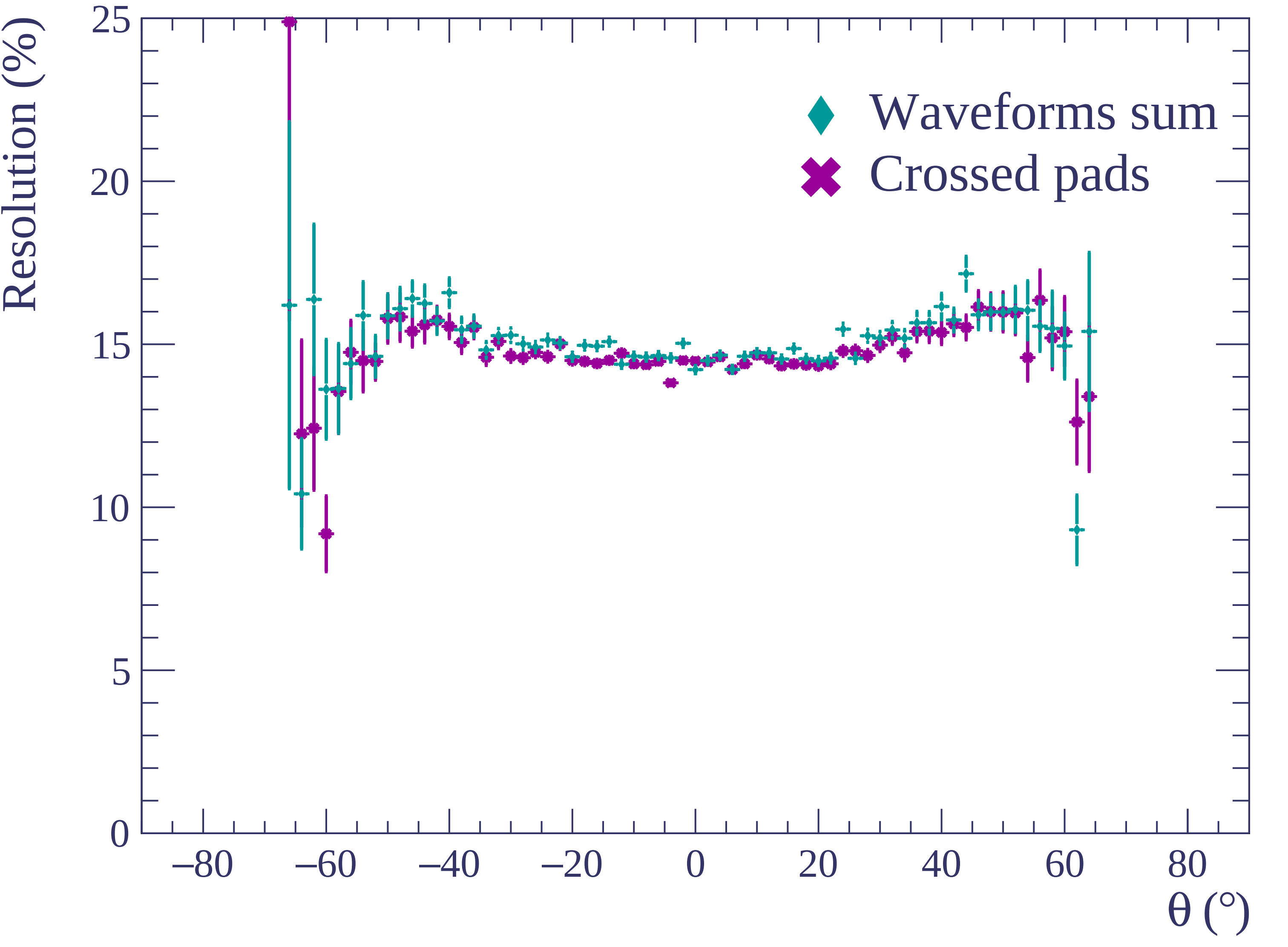


Resolution (%)
φ (°)
Waveforms sum
Crossed pads
0
5
10
15
20
25
−80
−60
−40
−20
0
20
40
60
80

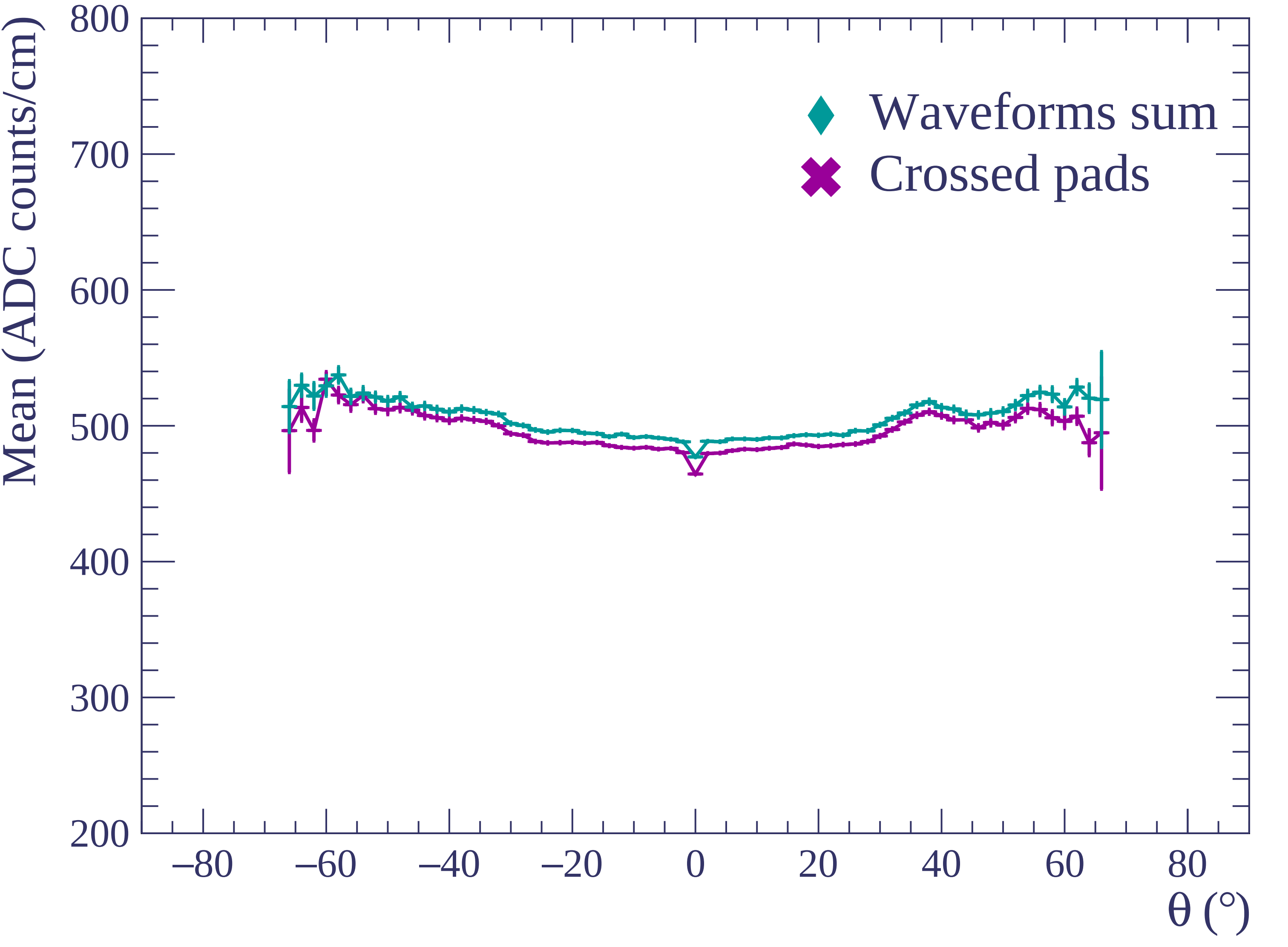


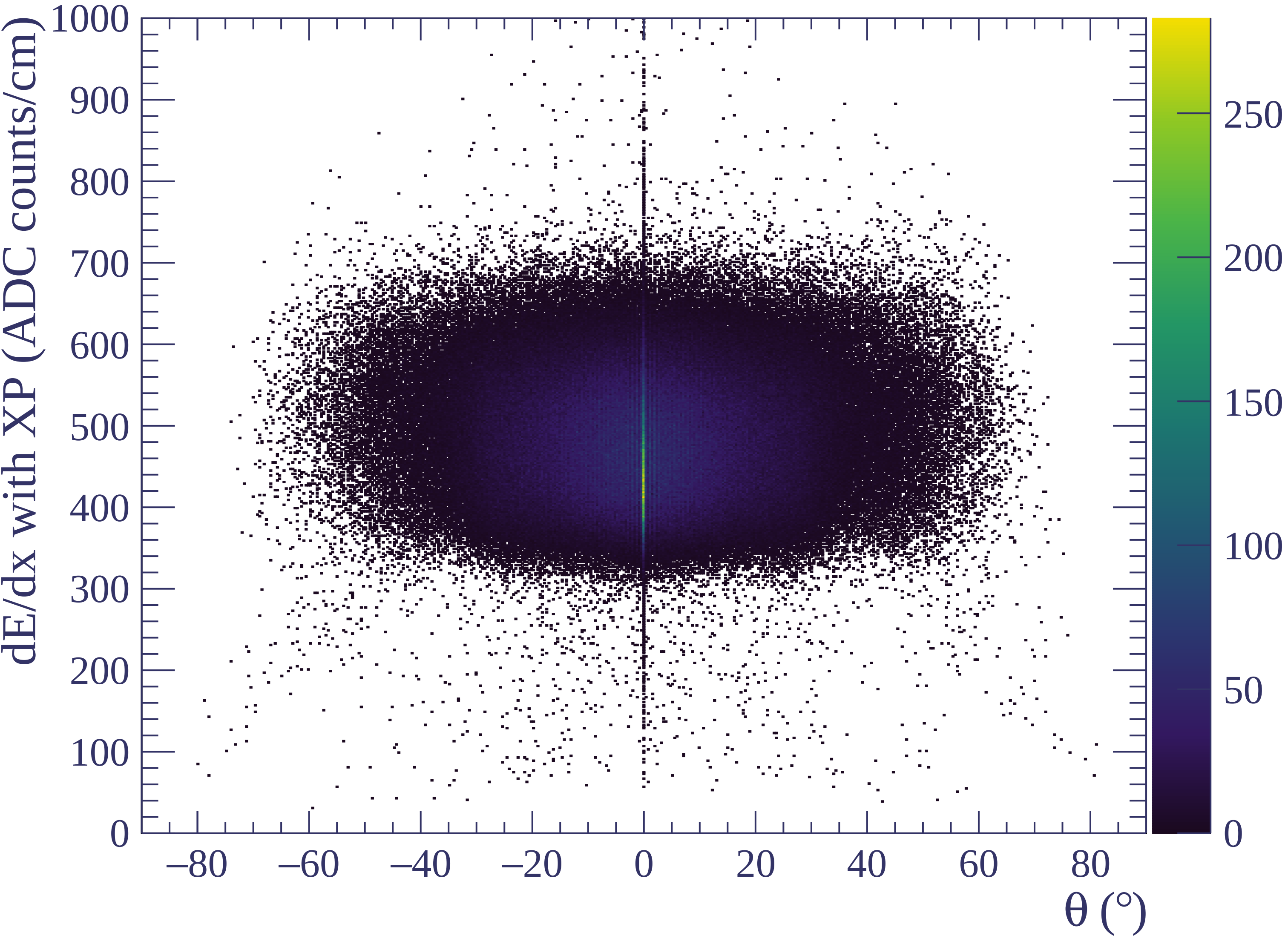
Waveforms sum
Crossed pads
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
φ (°)


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
300
250
200
150
100
50
0

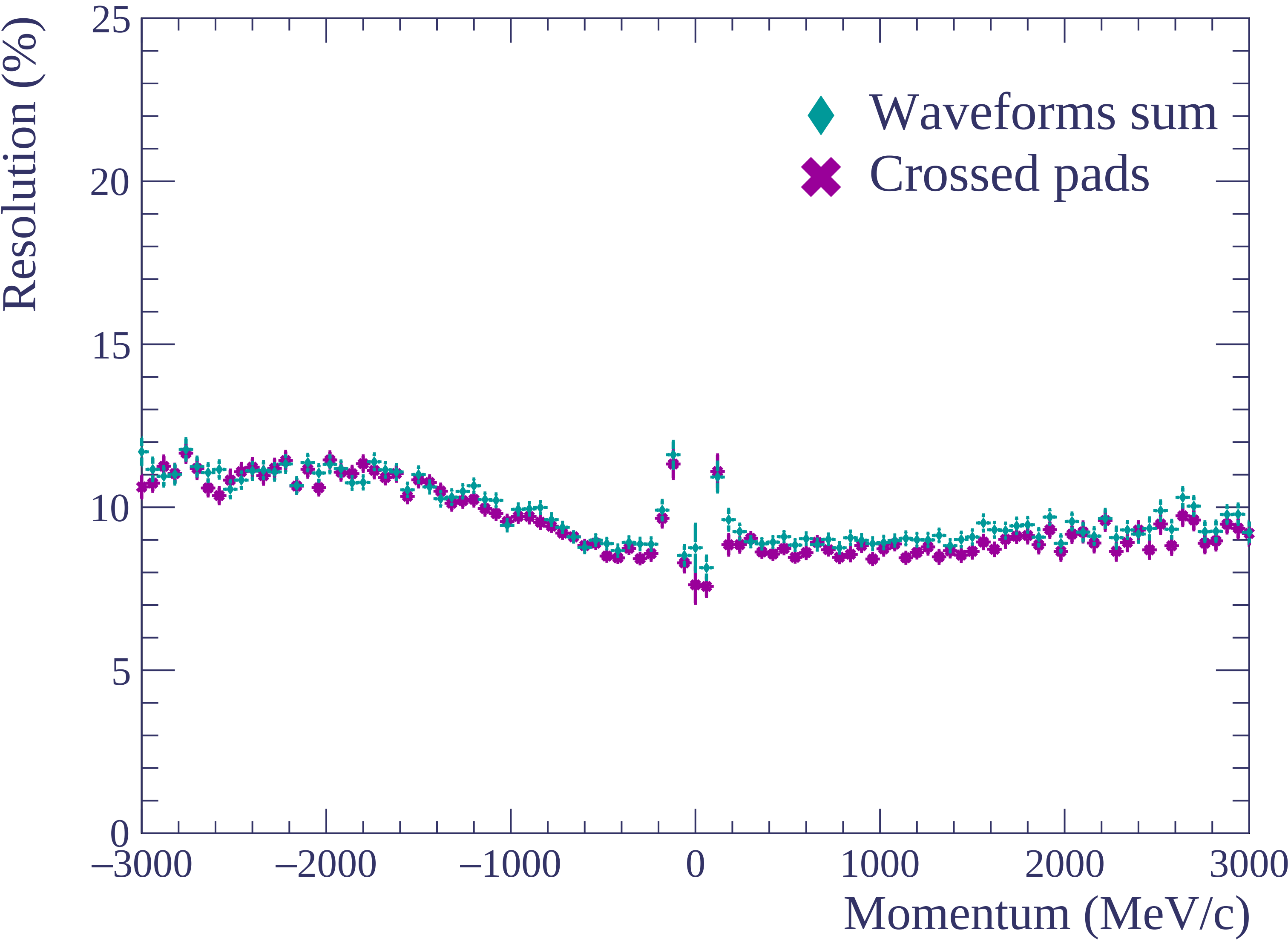


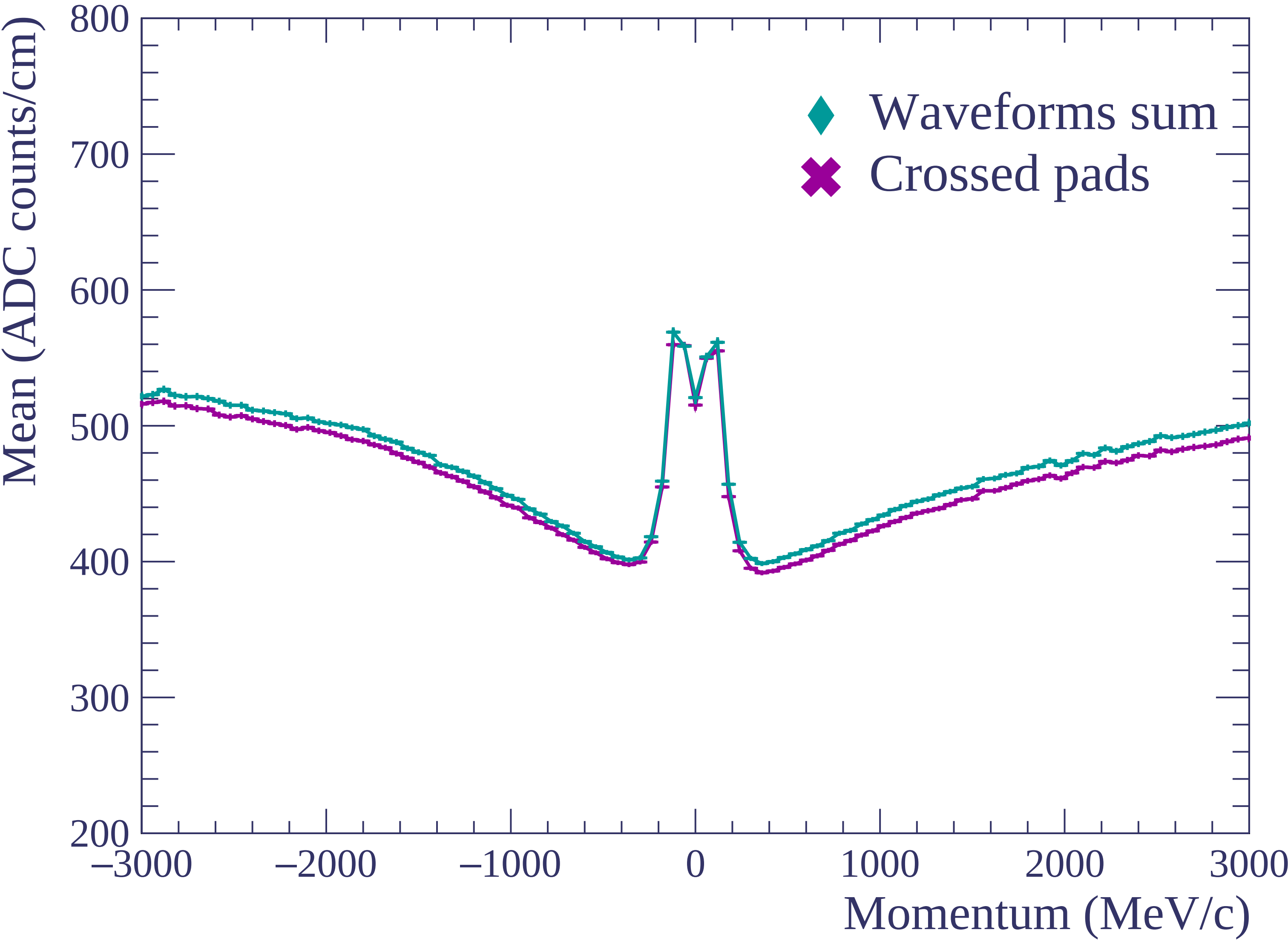
Resolution (%)
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
Waveforms sum
Crossed pads


Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
θ (°)
Waveforms sum
Crossed pads

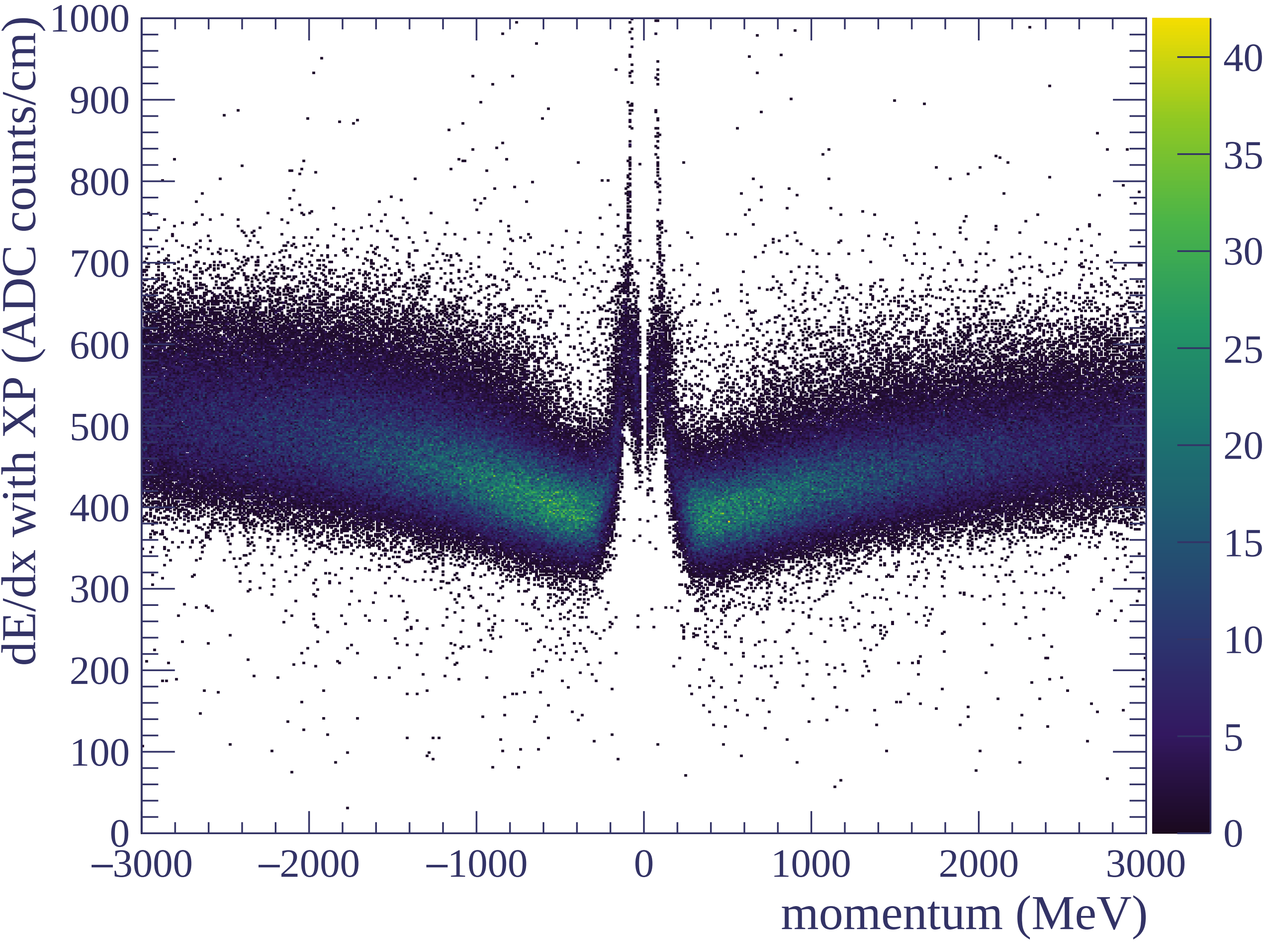


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
250
200
150
100
50
0

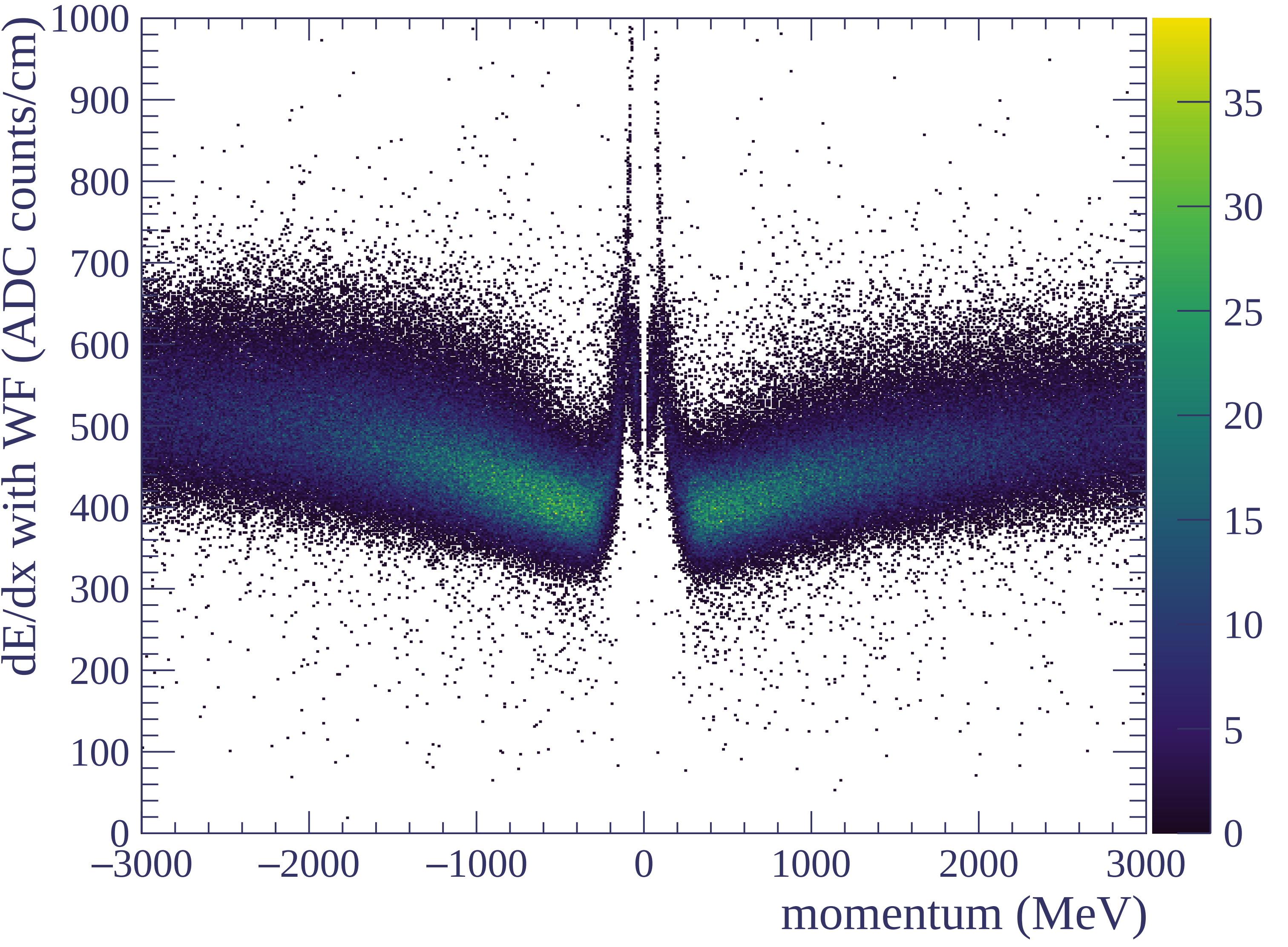

Resolution (%)
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads



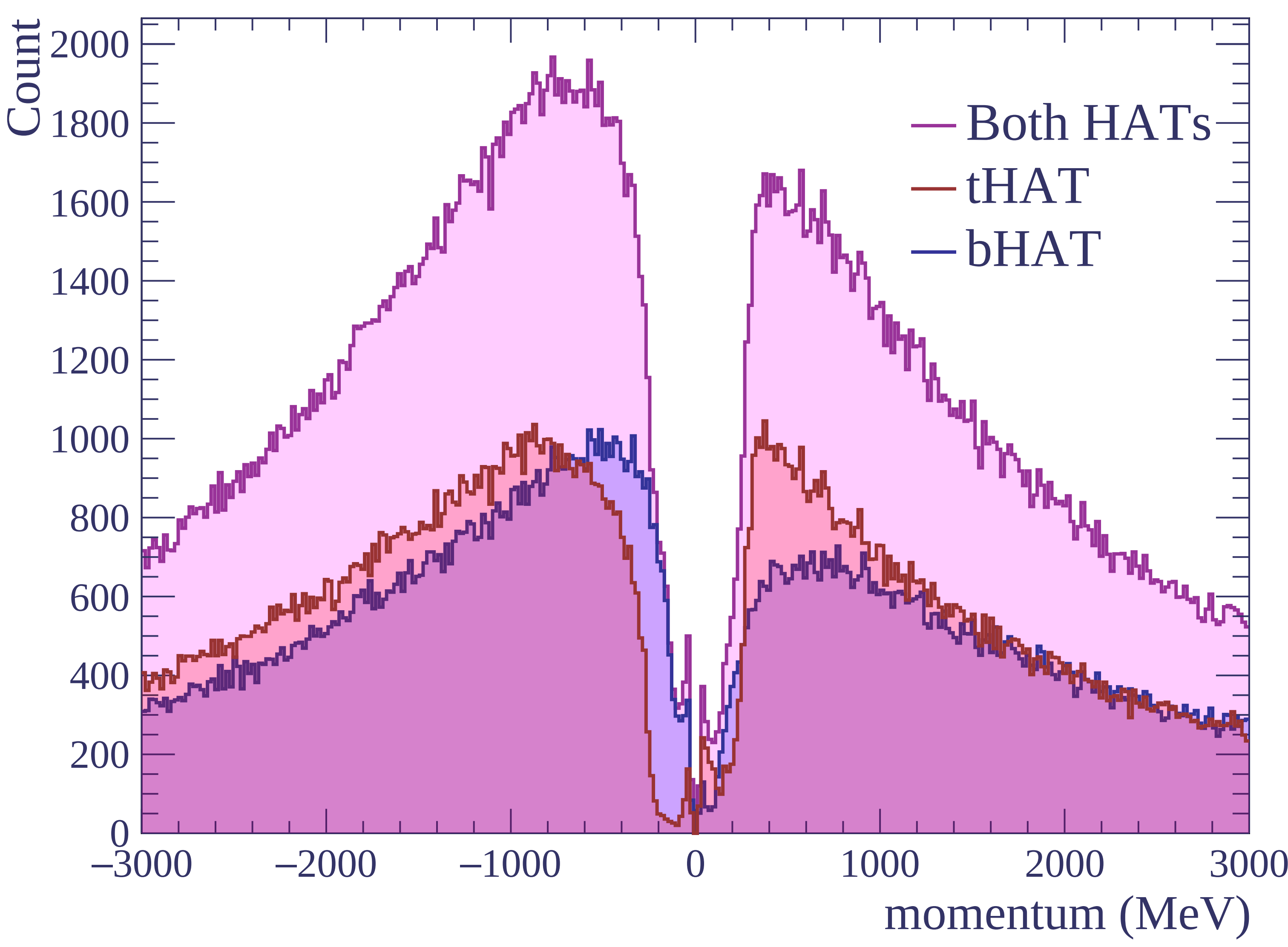
Waveforms sum
Crossed pads
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)



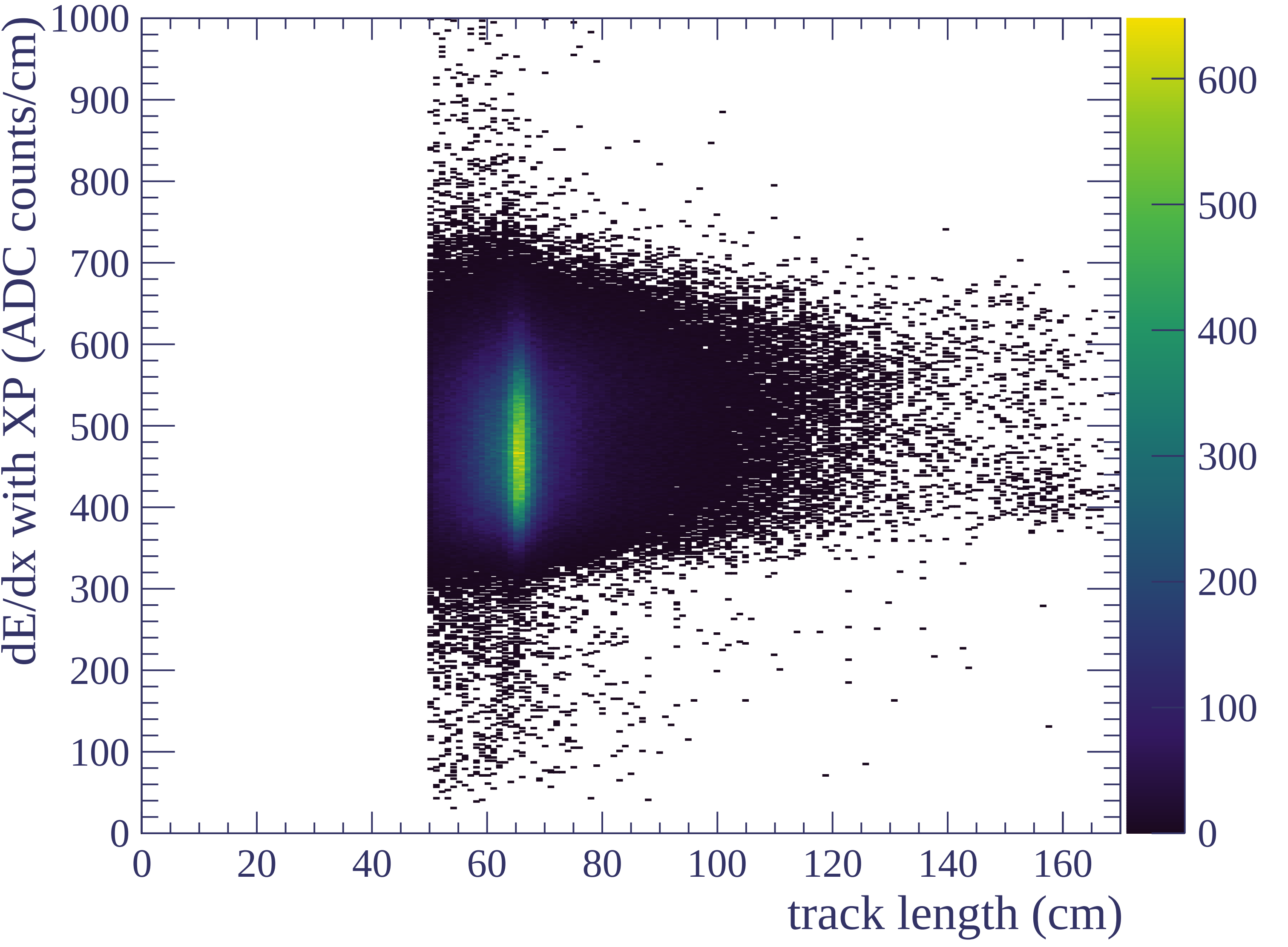
dE/dx with XP (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
0
5
10
15
20
25
30
35
40



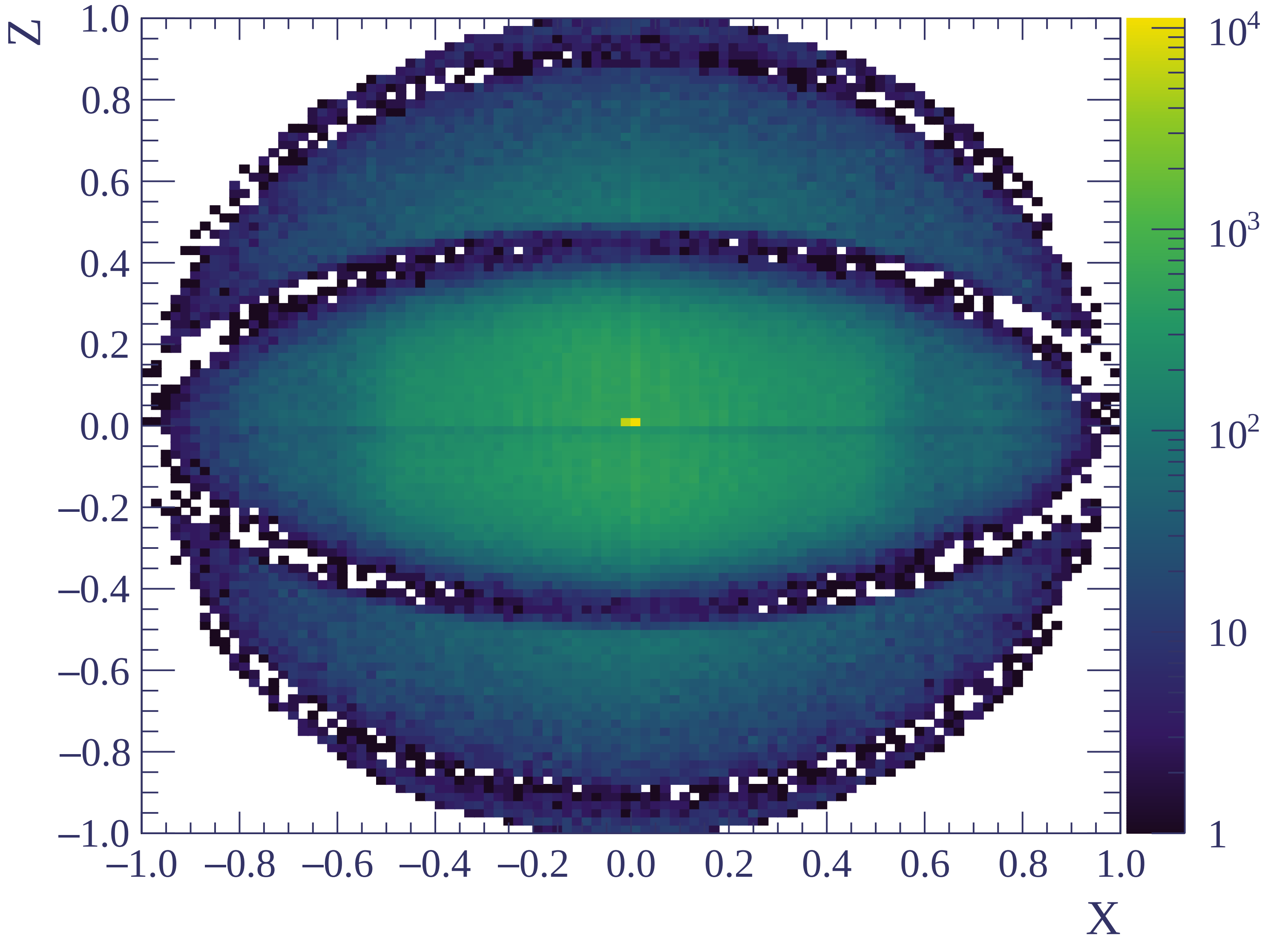
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
35
30
25
20
15
10
5
0



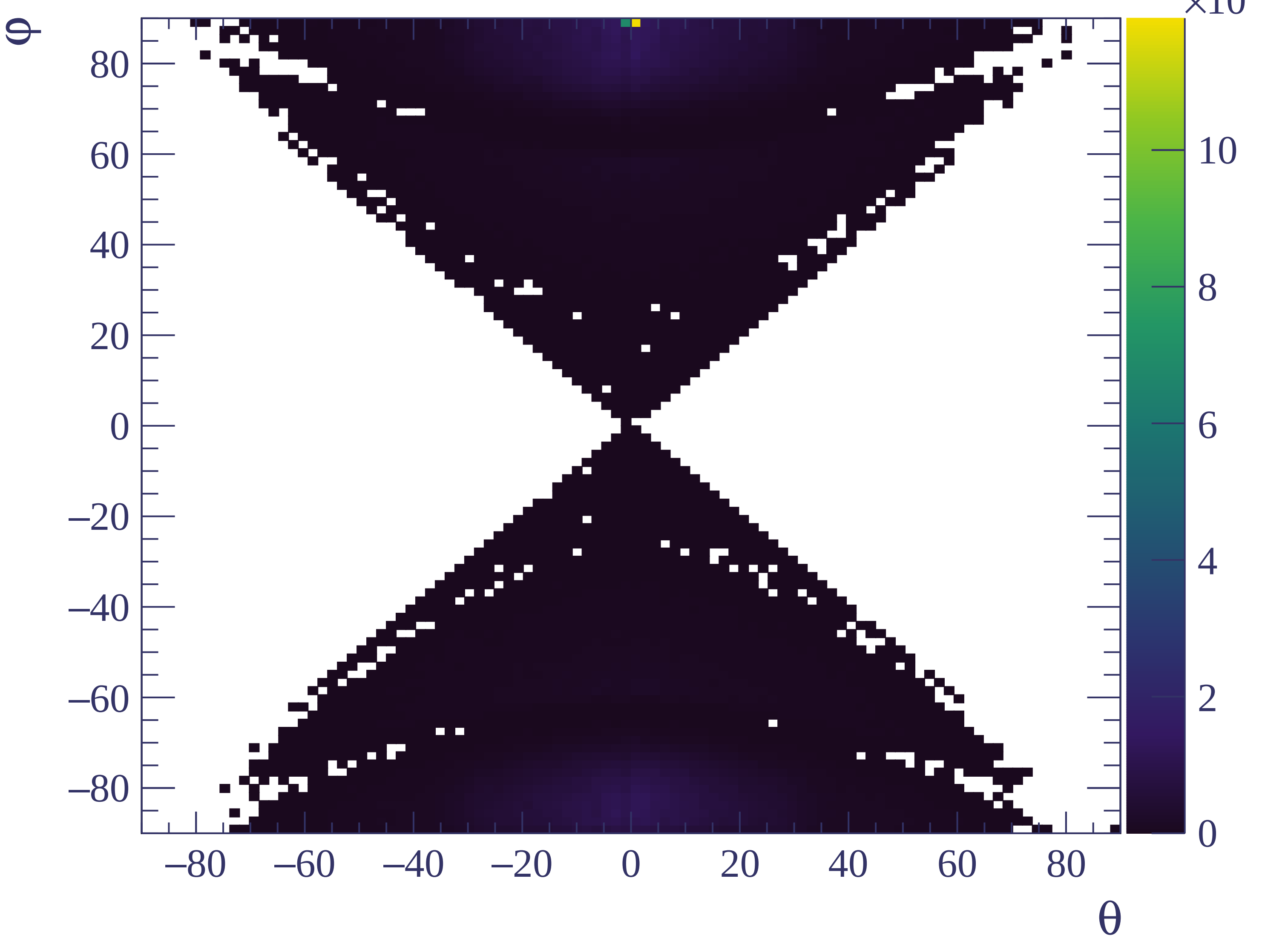
Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT



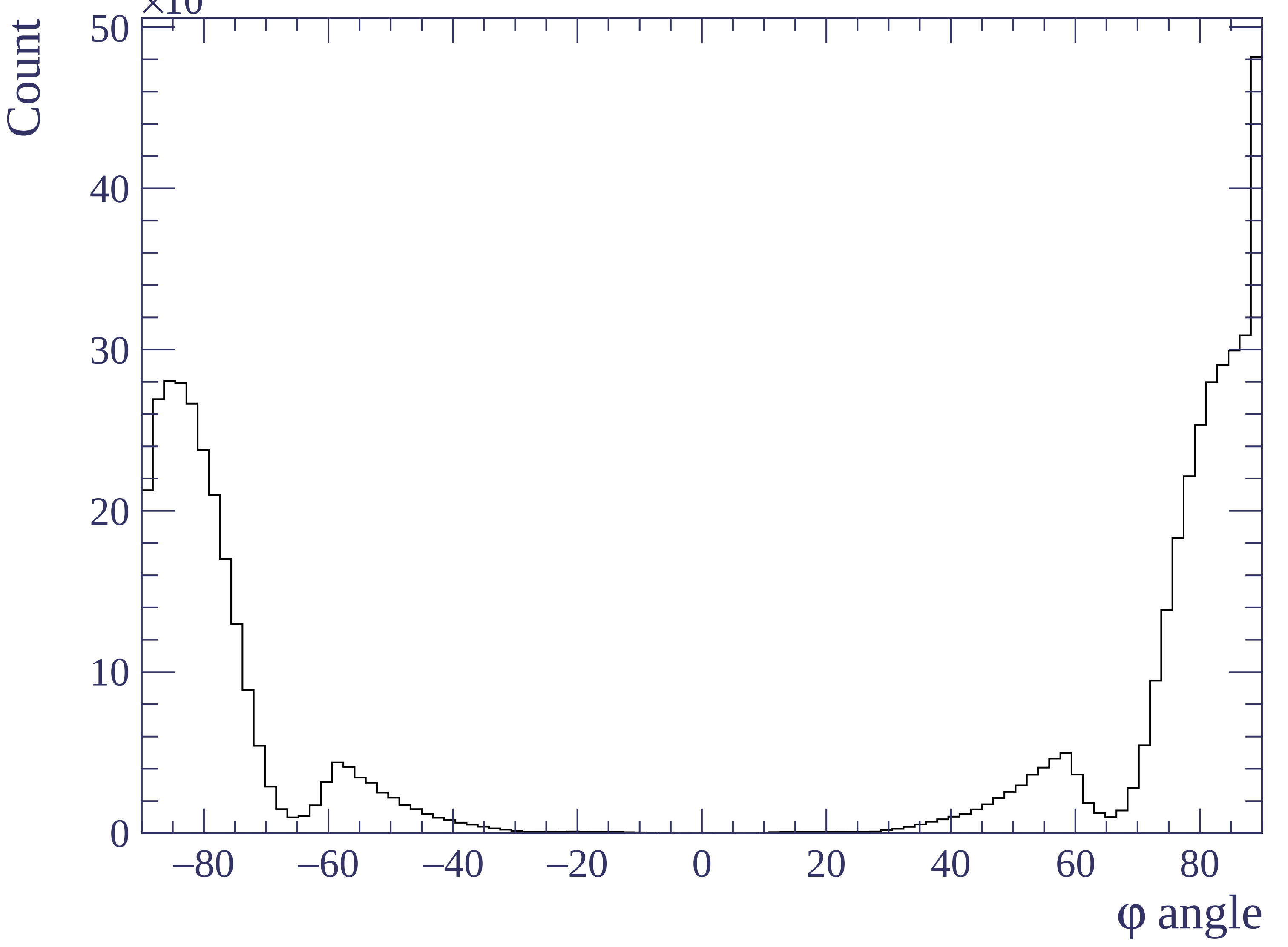
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
600



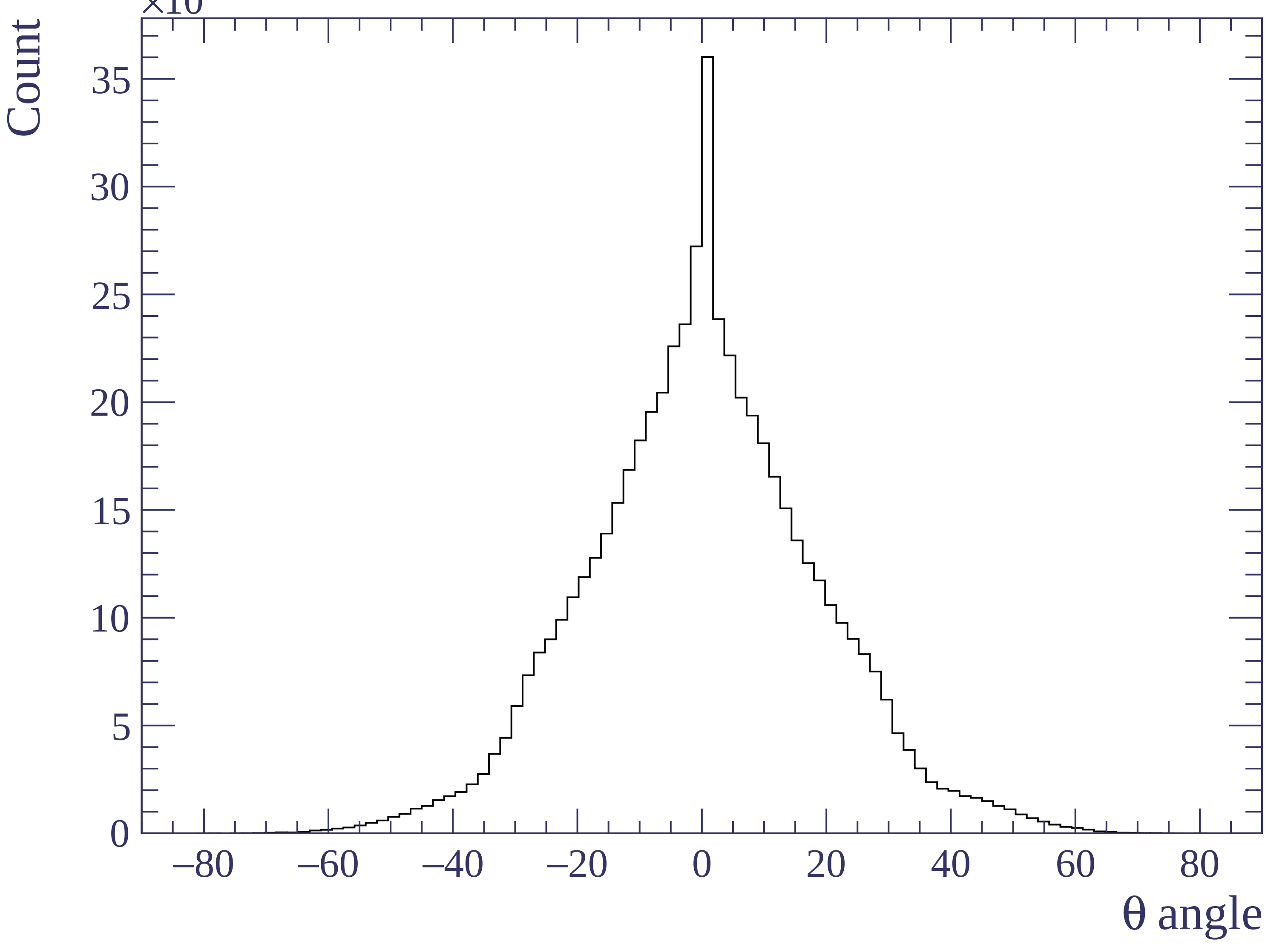
Z
X
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
10^4
10^3
10^2
10
1



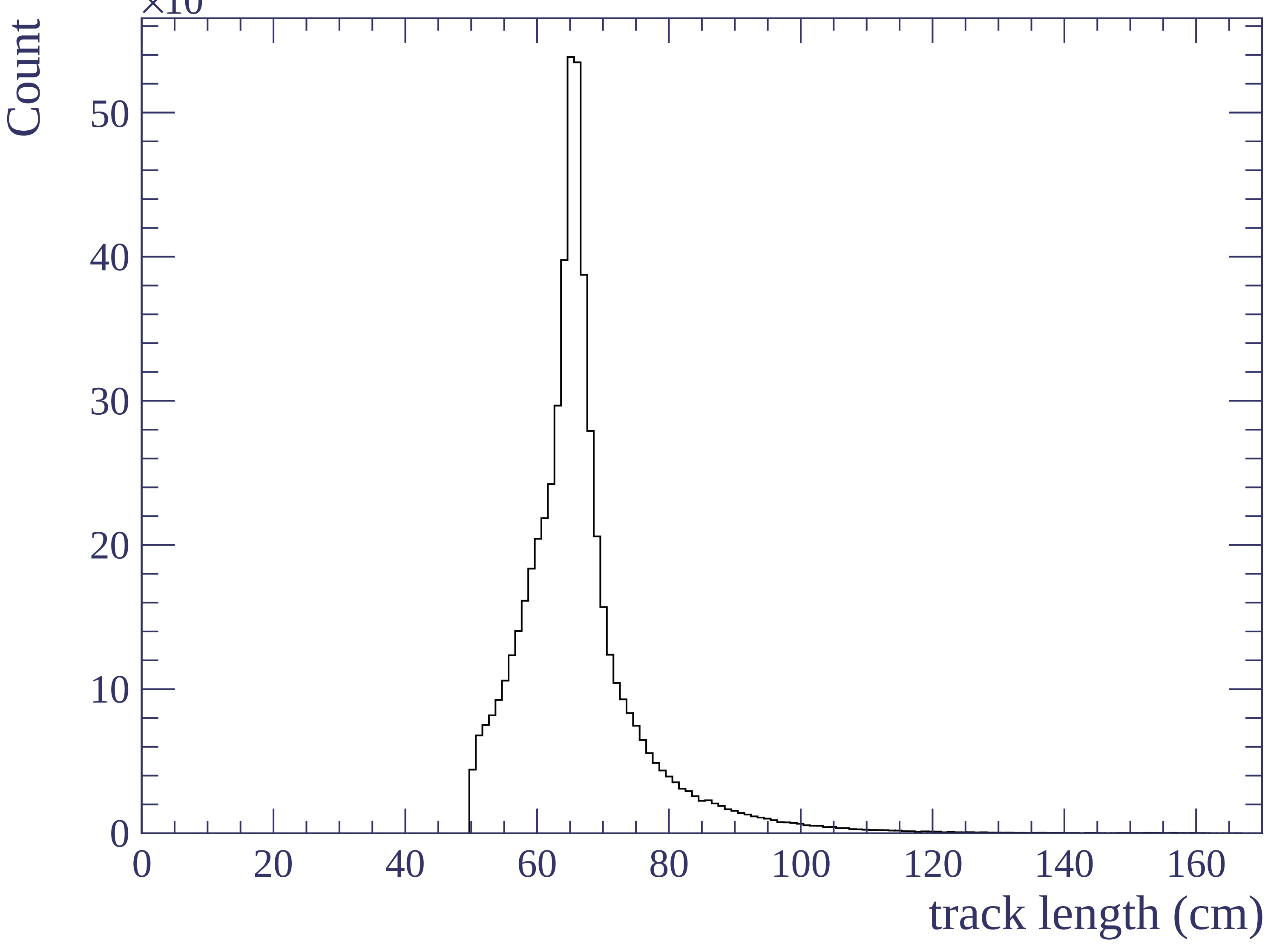
φ
θ
×10
80
60
40
20
0
−20
−40
−60
−80
10
8
6
4
2



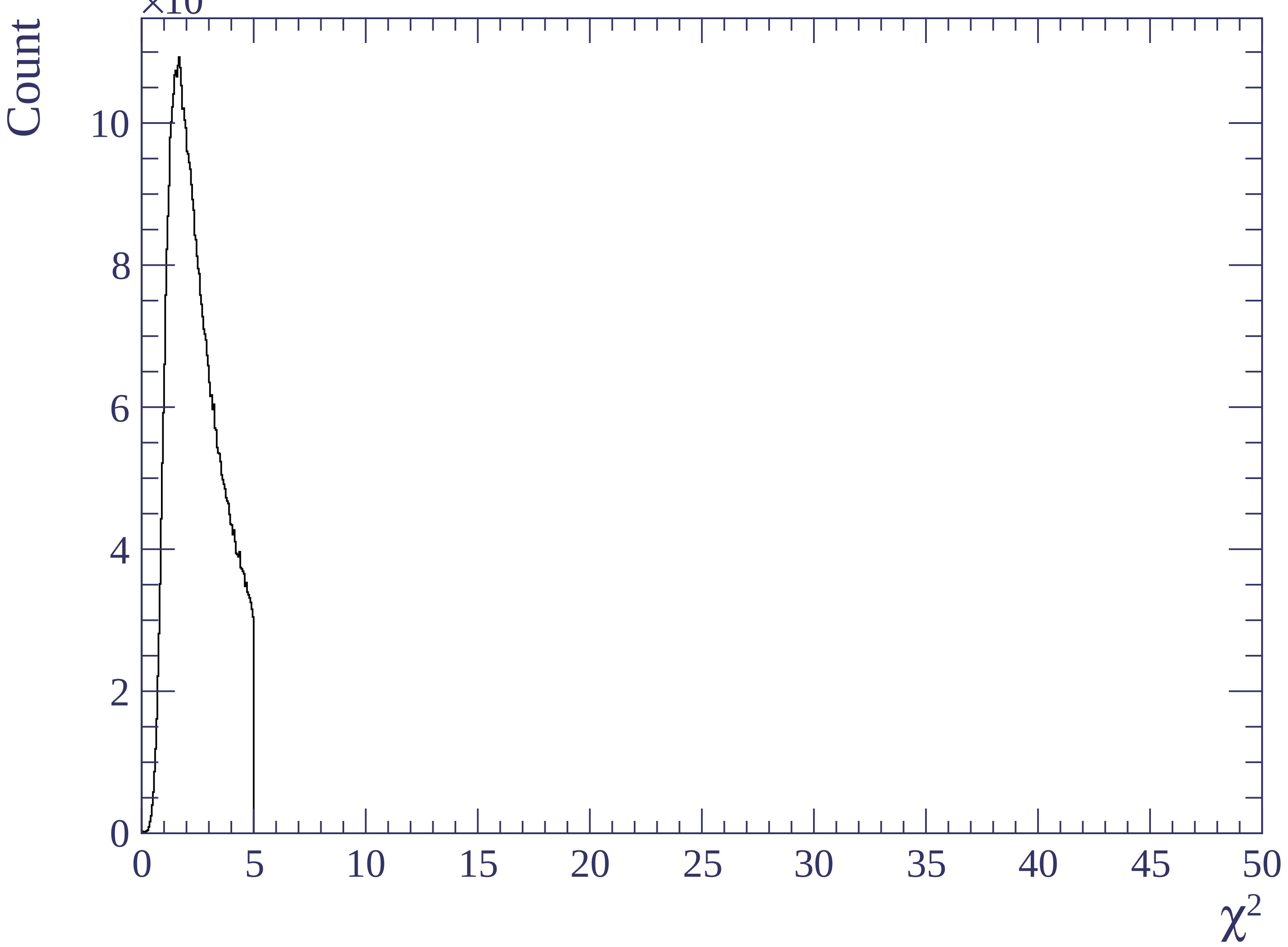
×10
Count
50
40
30
20
10
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



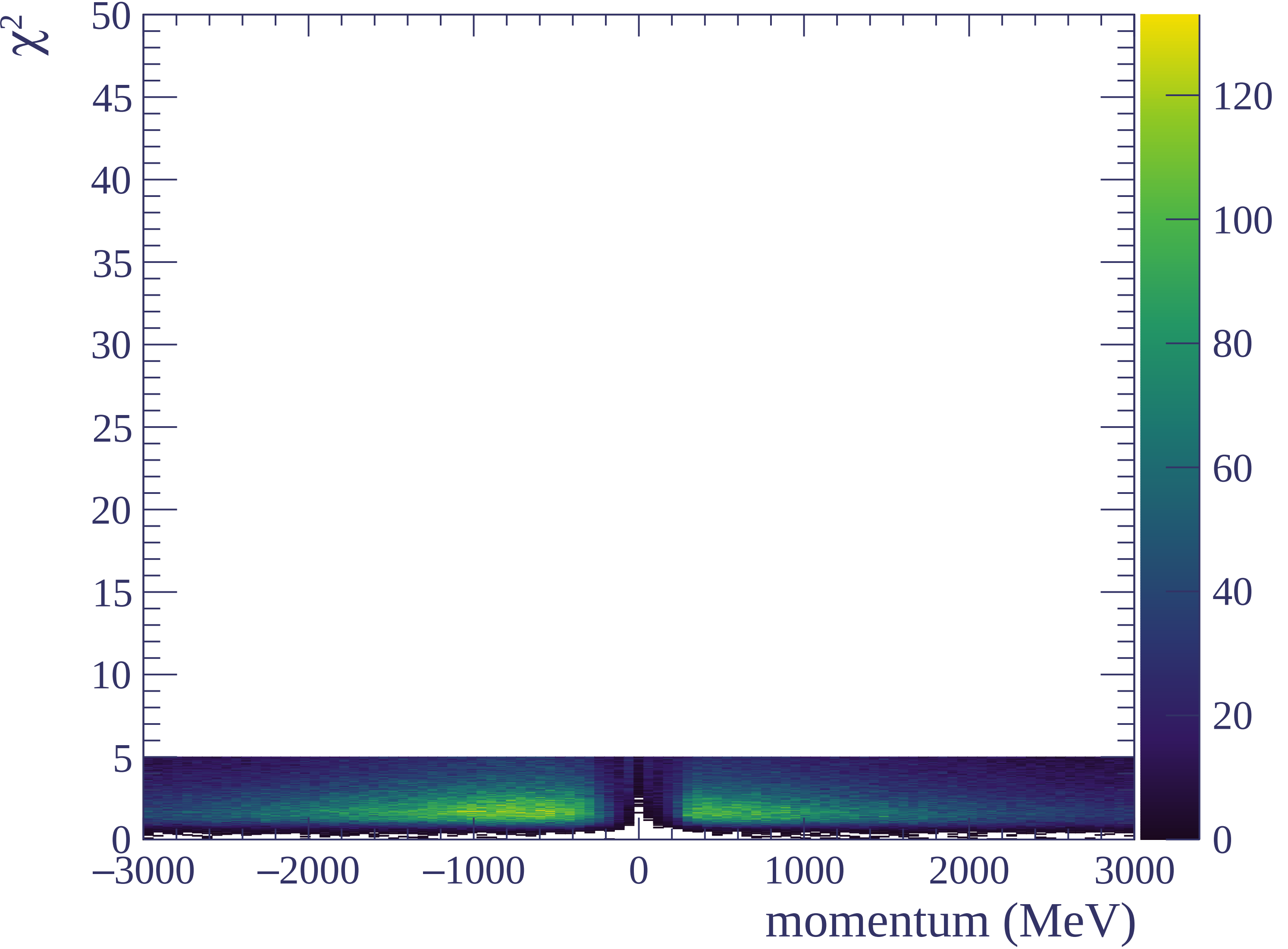
×10
Count
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
θ angle



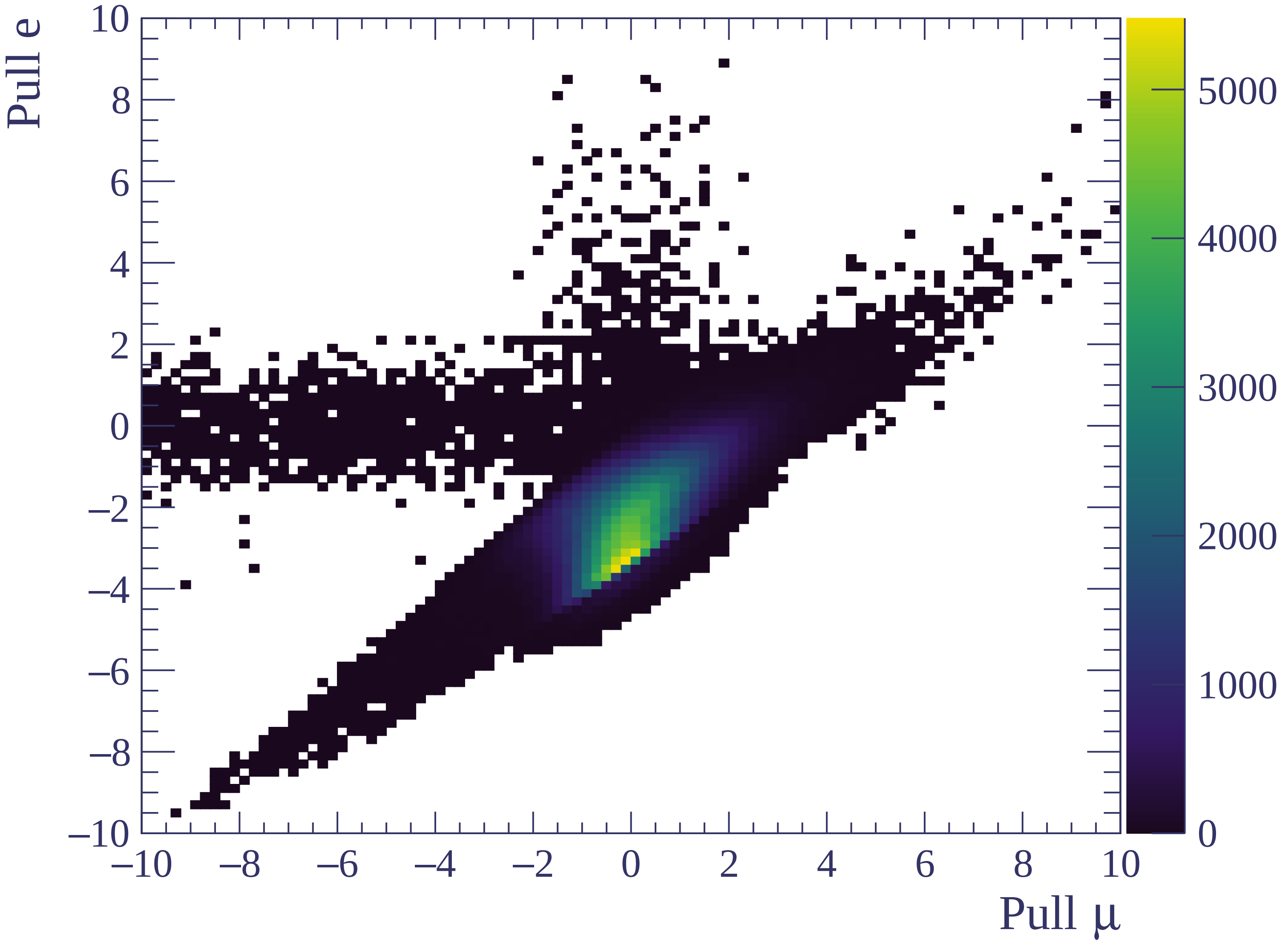
×10
Count
0
10
20
30
40
50
0
20
40
60
80
100
120
140
160
track length (cm)



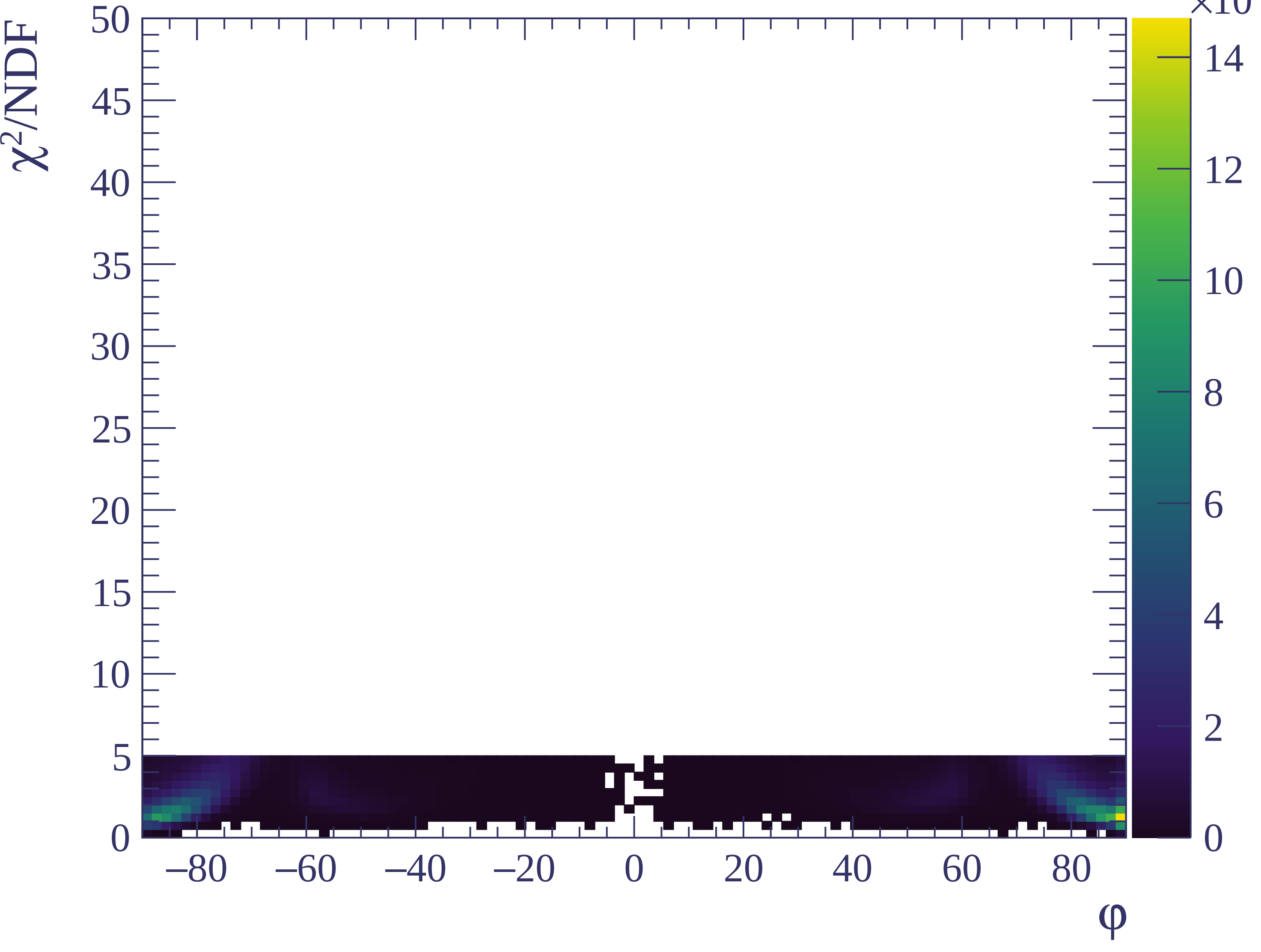
Count
×10
0
2
4
6
8
10
0
5
10
15
20
25
30
35
40
45
50
χ^2



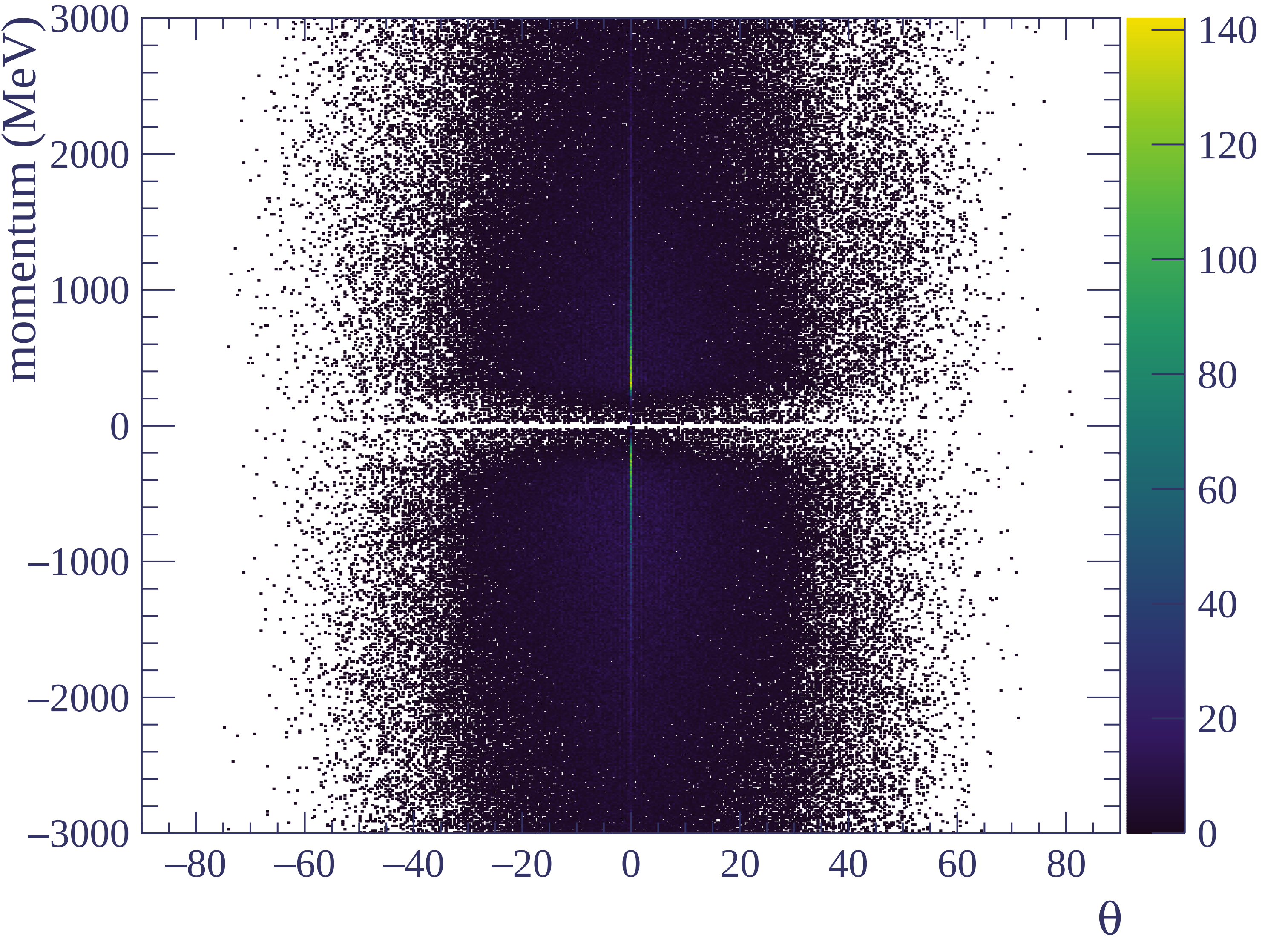
χ^2
50
45
40
35
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
120
100
80
60
40
20
0



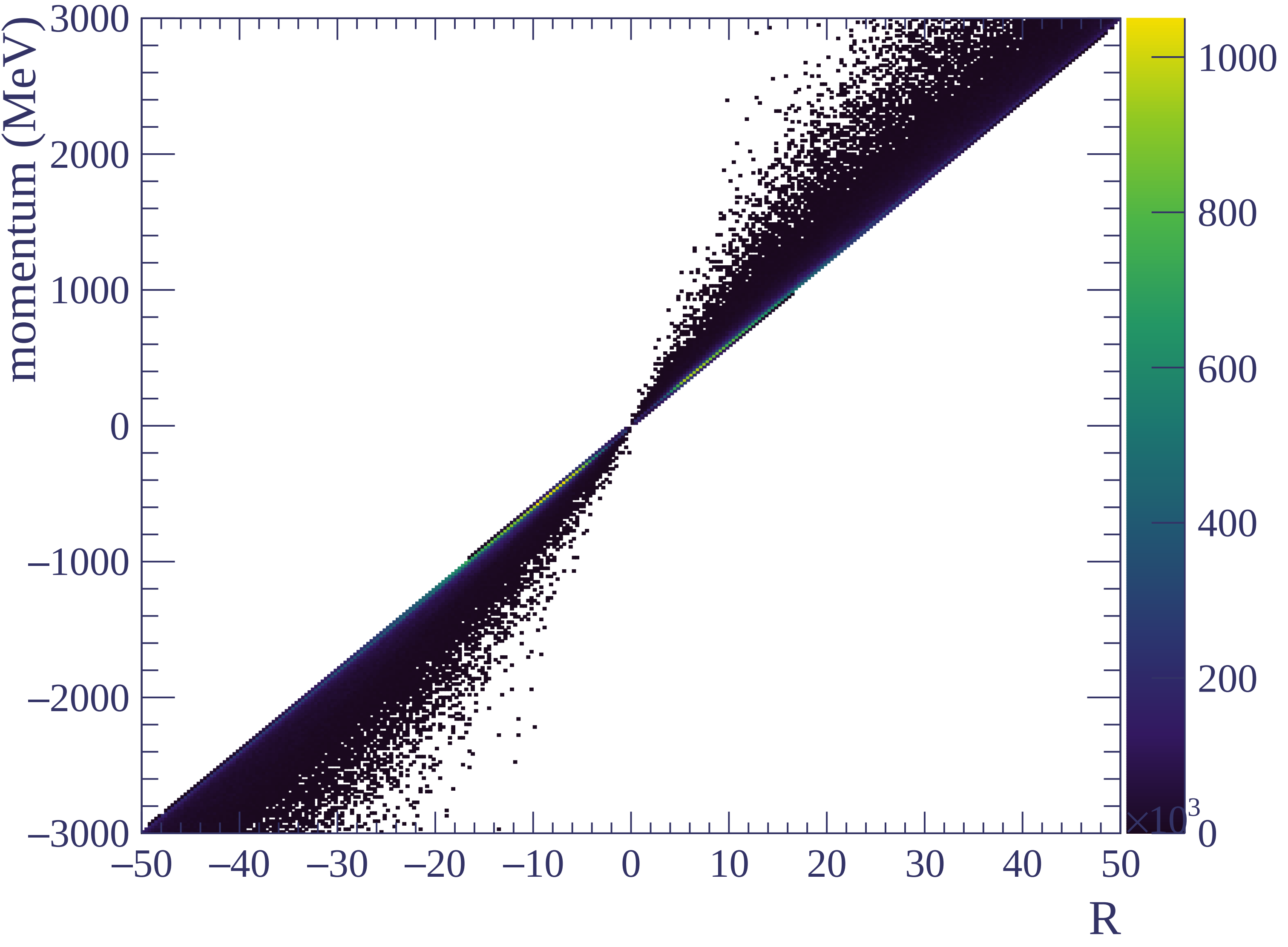
Pull e
Pull μ
10
8
6
4
2
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0
2
4
6
8
10
5000
4000
3000
2000
1000
0



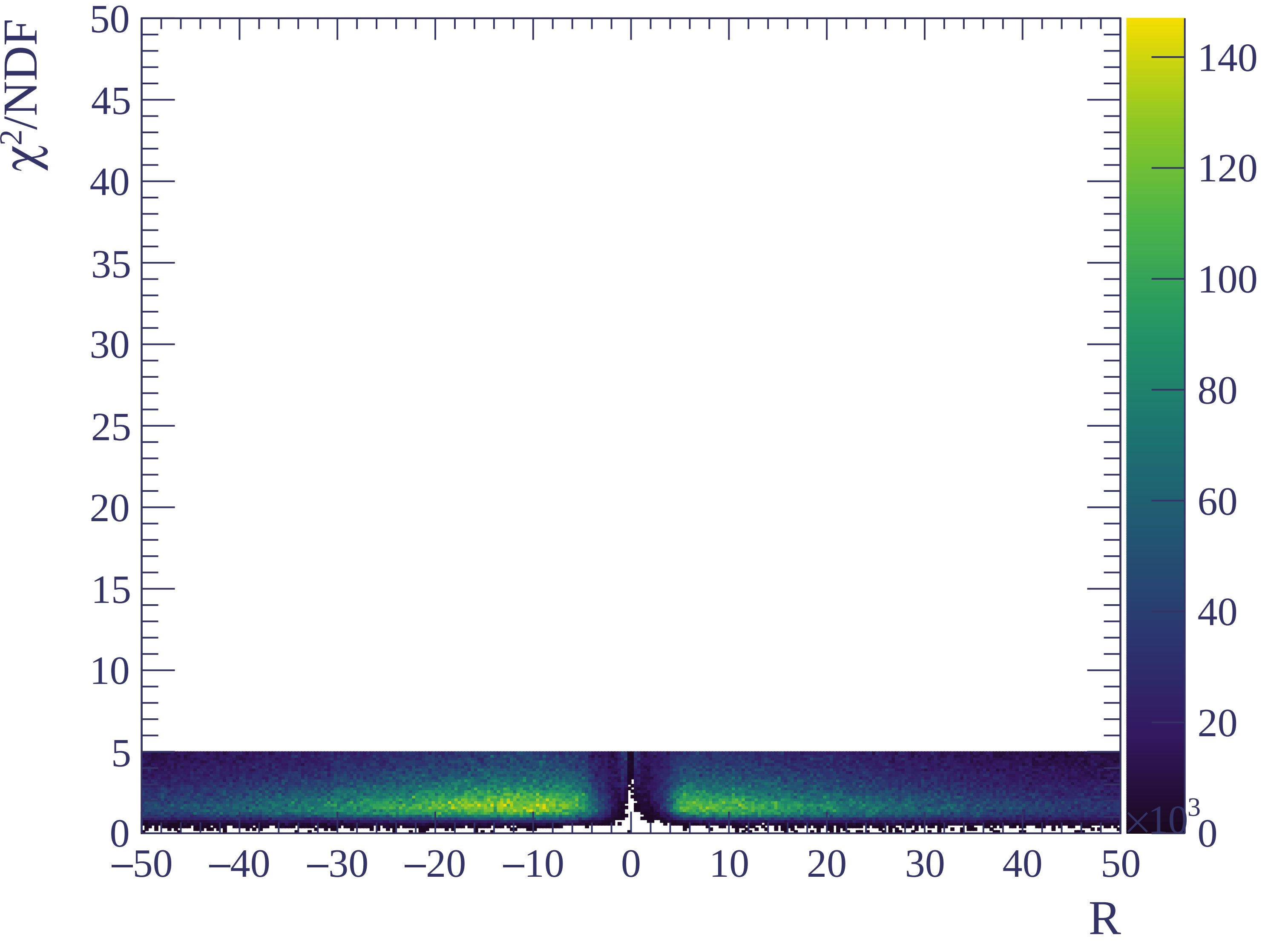
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
×10
14
12
10
8
6
4
2
0



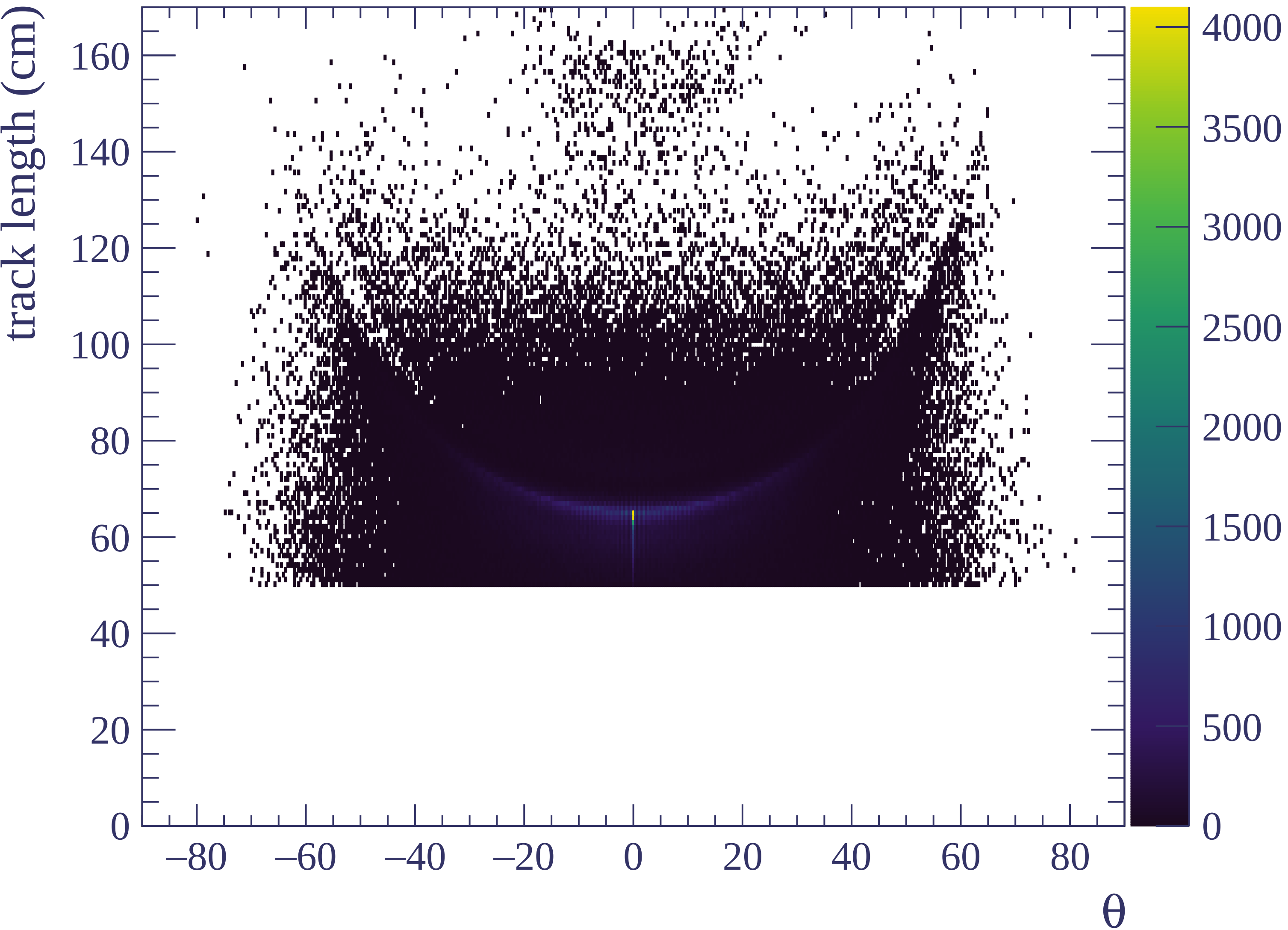
momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−80
−60
−40
−20
0
20
40
60
80
θ
140
120
100
80
60
40
20
0



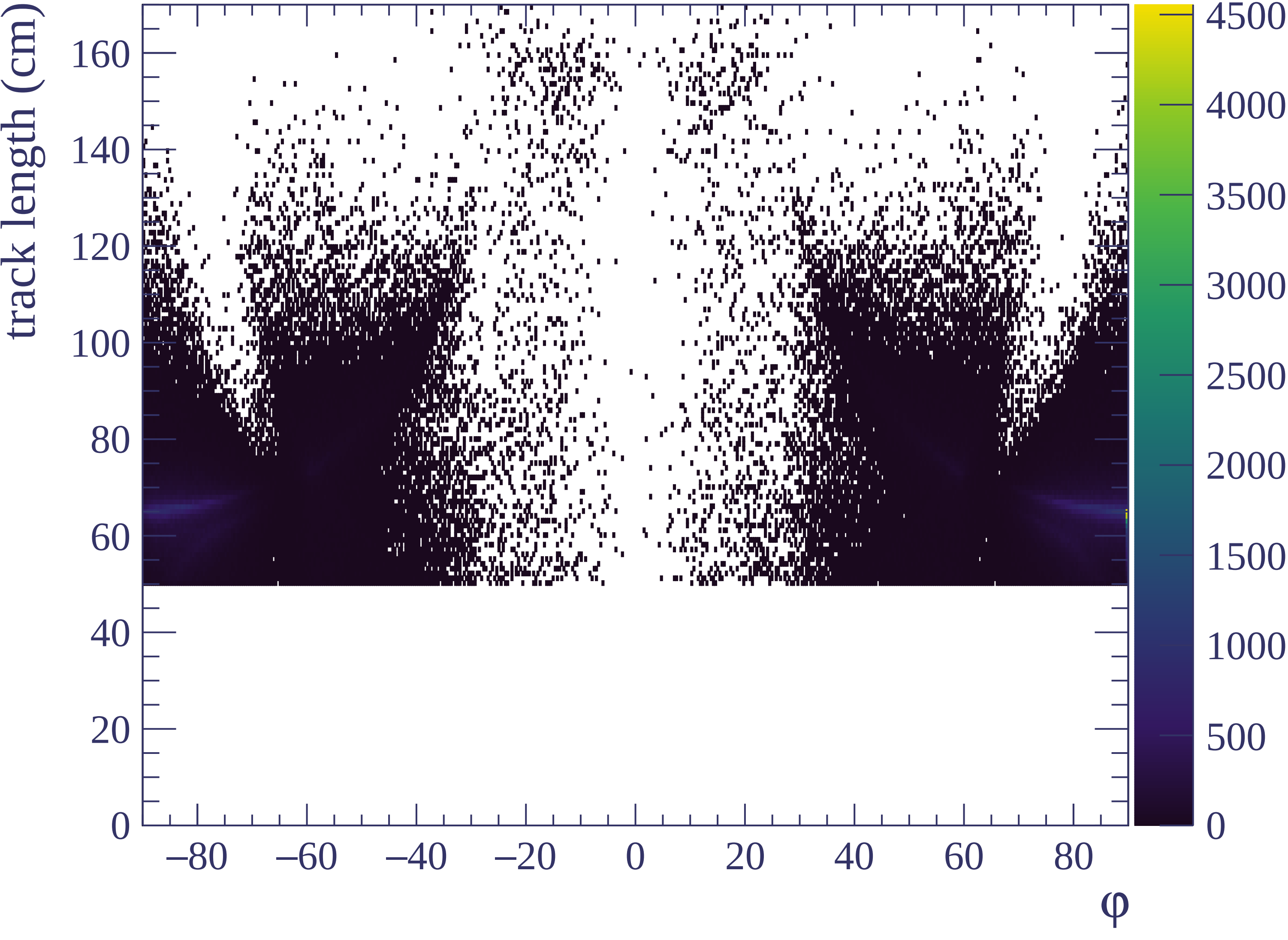
momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
1000
800
600
400
200
0
$\times 10^3$



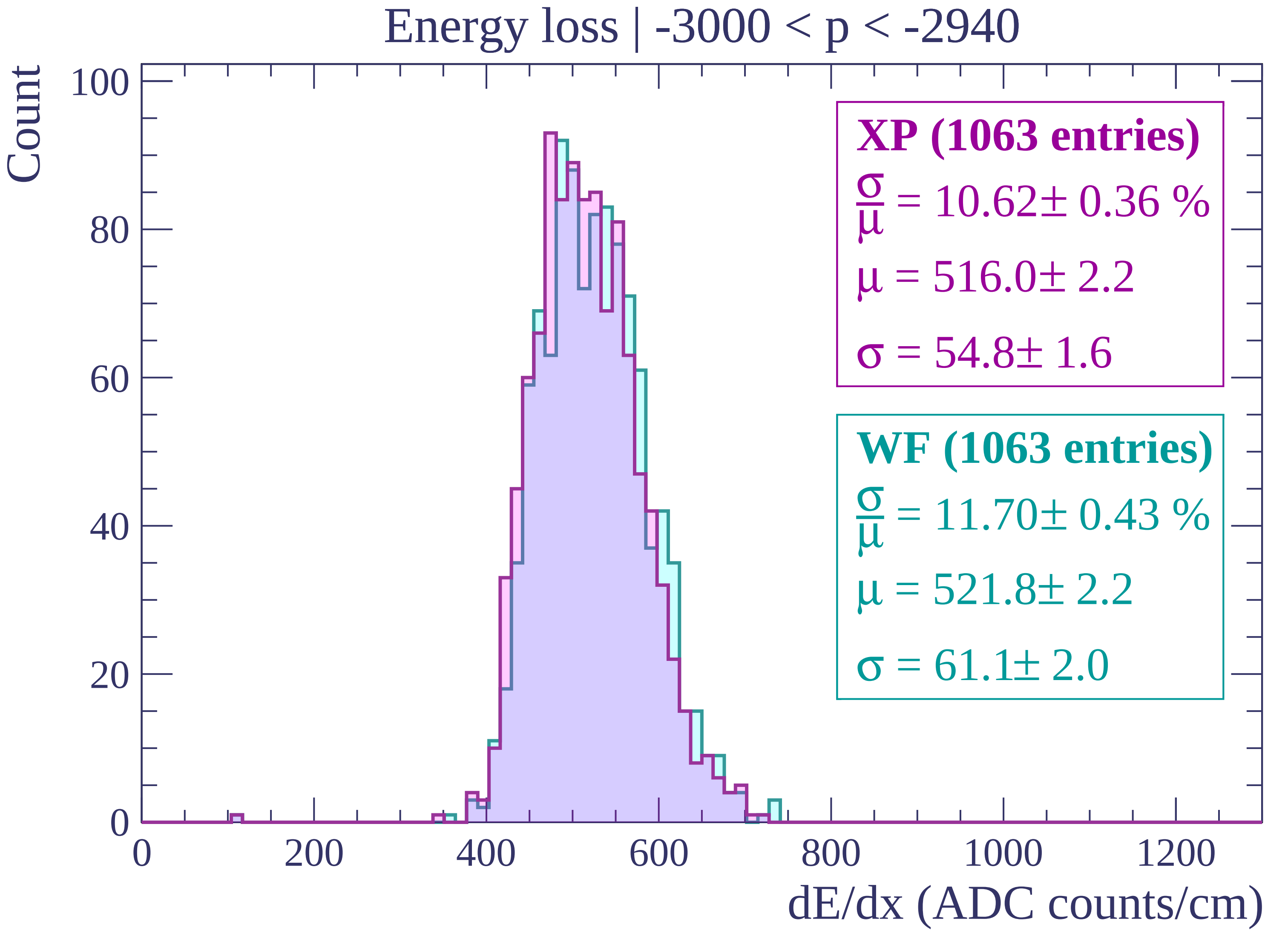
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
140
120
100
80
60
40
20
0
$\times 10^3$



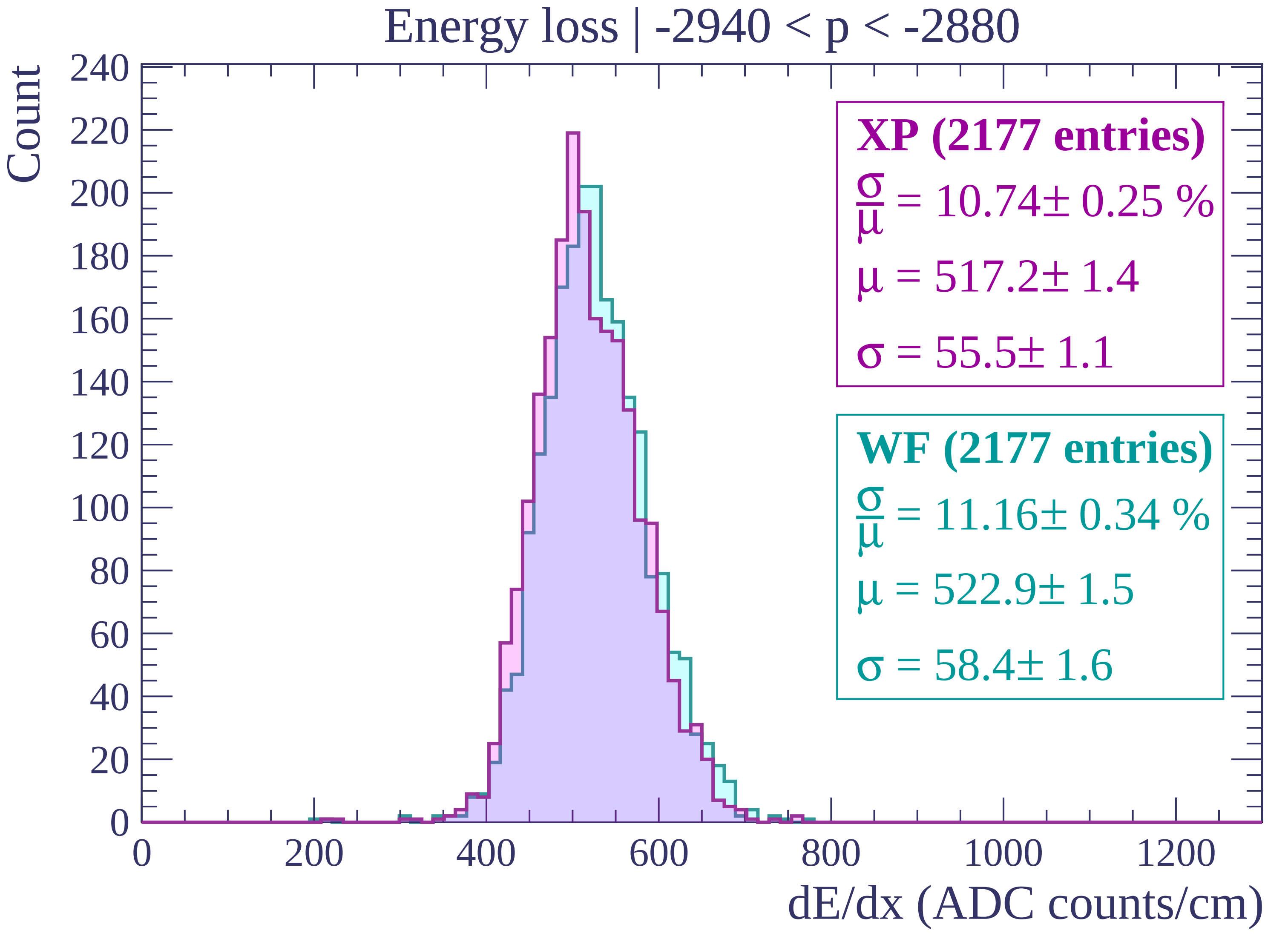
track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
θ
0
500
1000
1500
2000
2500
3000
3500
4000



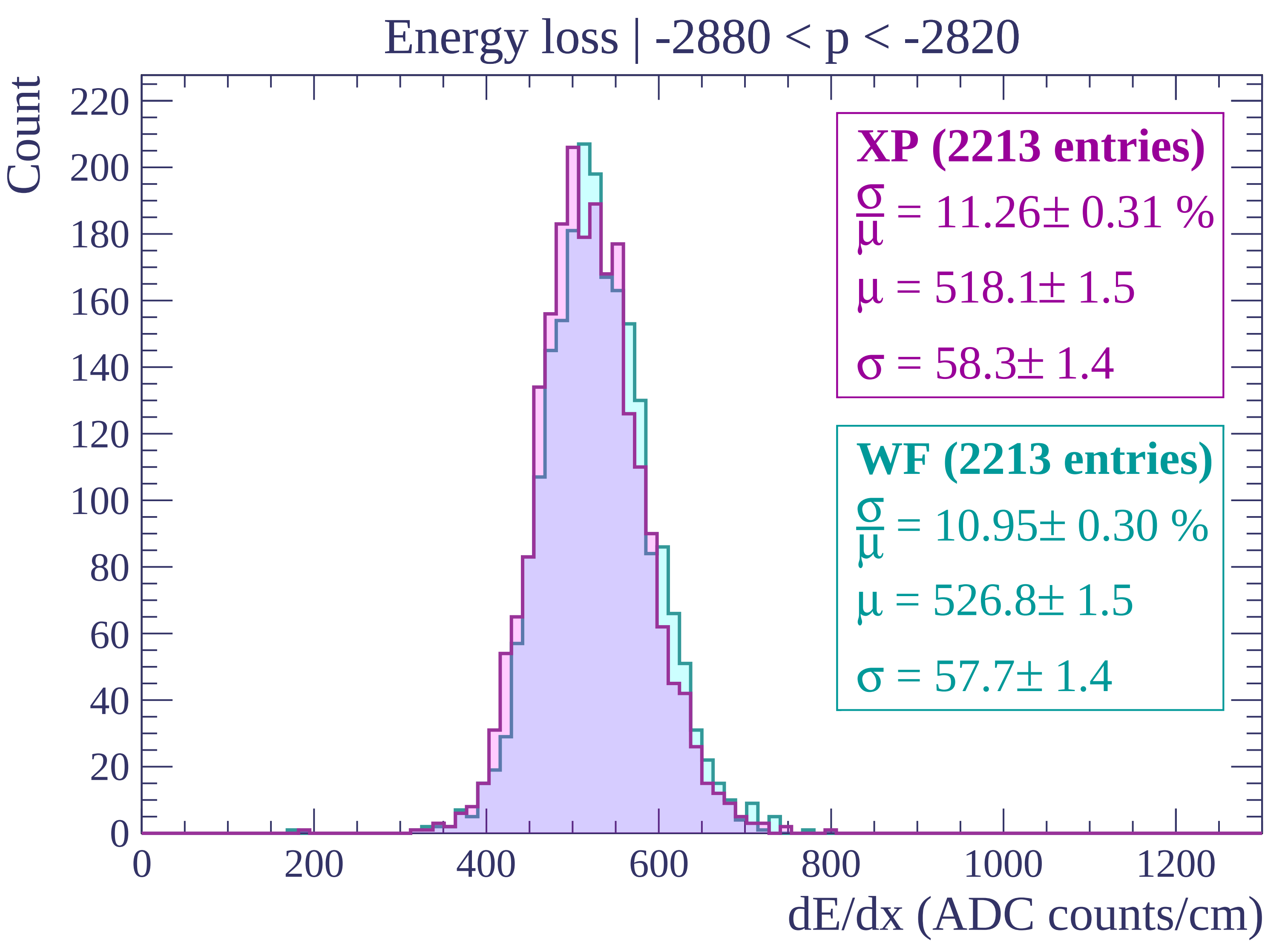
track length (cm)
φ
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
0
500
1000
1500
2000
2500
3000
3500
4000
4500



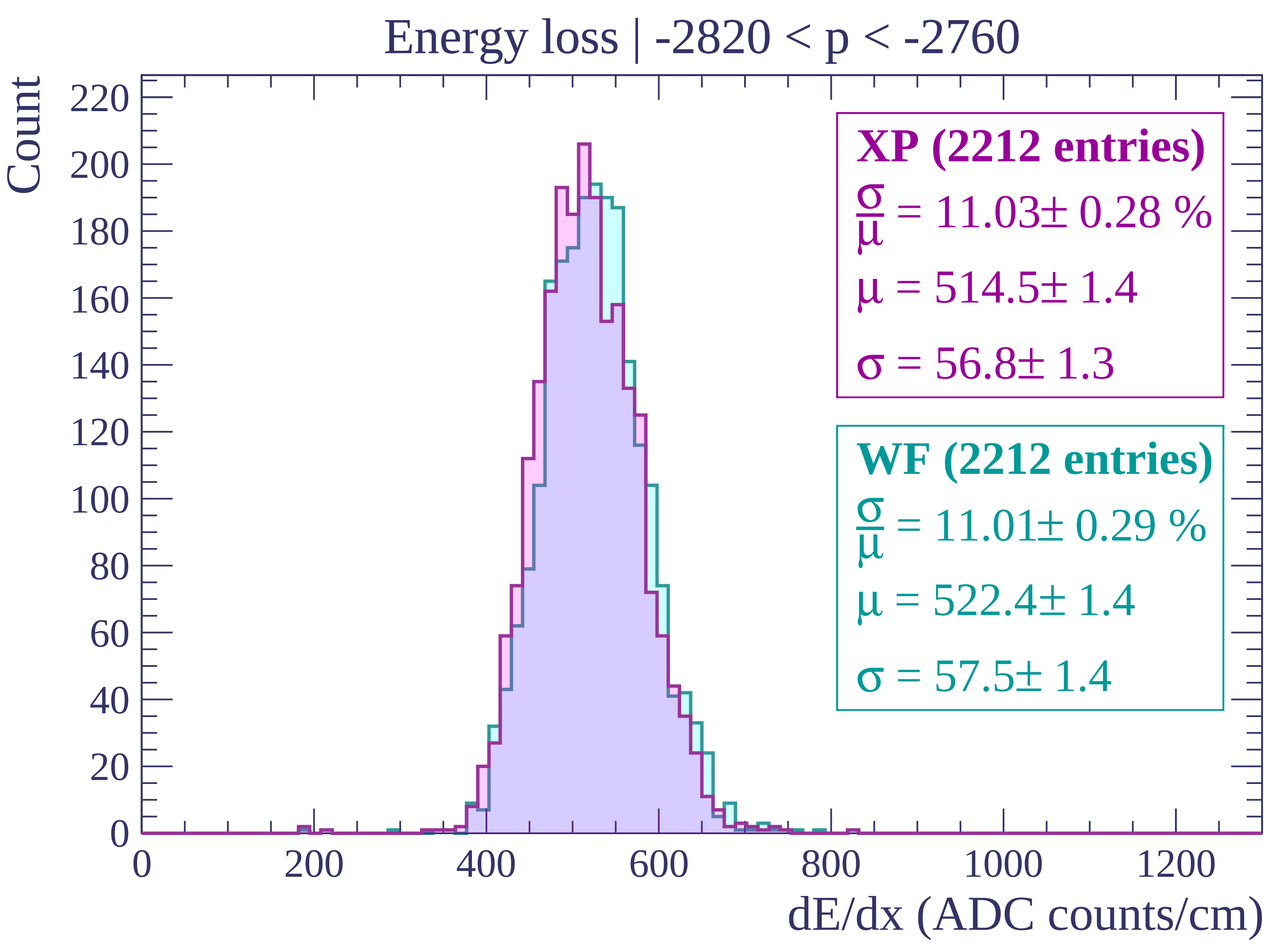
Energy loss | -3000 < p < -2940
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1063 entries)
σ/μ = 10.62± 0.36 %
μ = 516.0± 2.2
σ = 54.8± 1.6
WF (1063 entries)
σ/μ = 11.70± 0.43 %
μ = 521.8± 2.2
σ = 61.1± 2.0

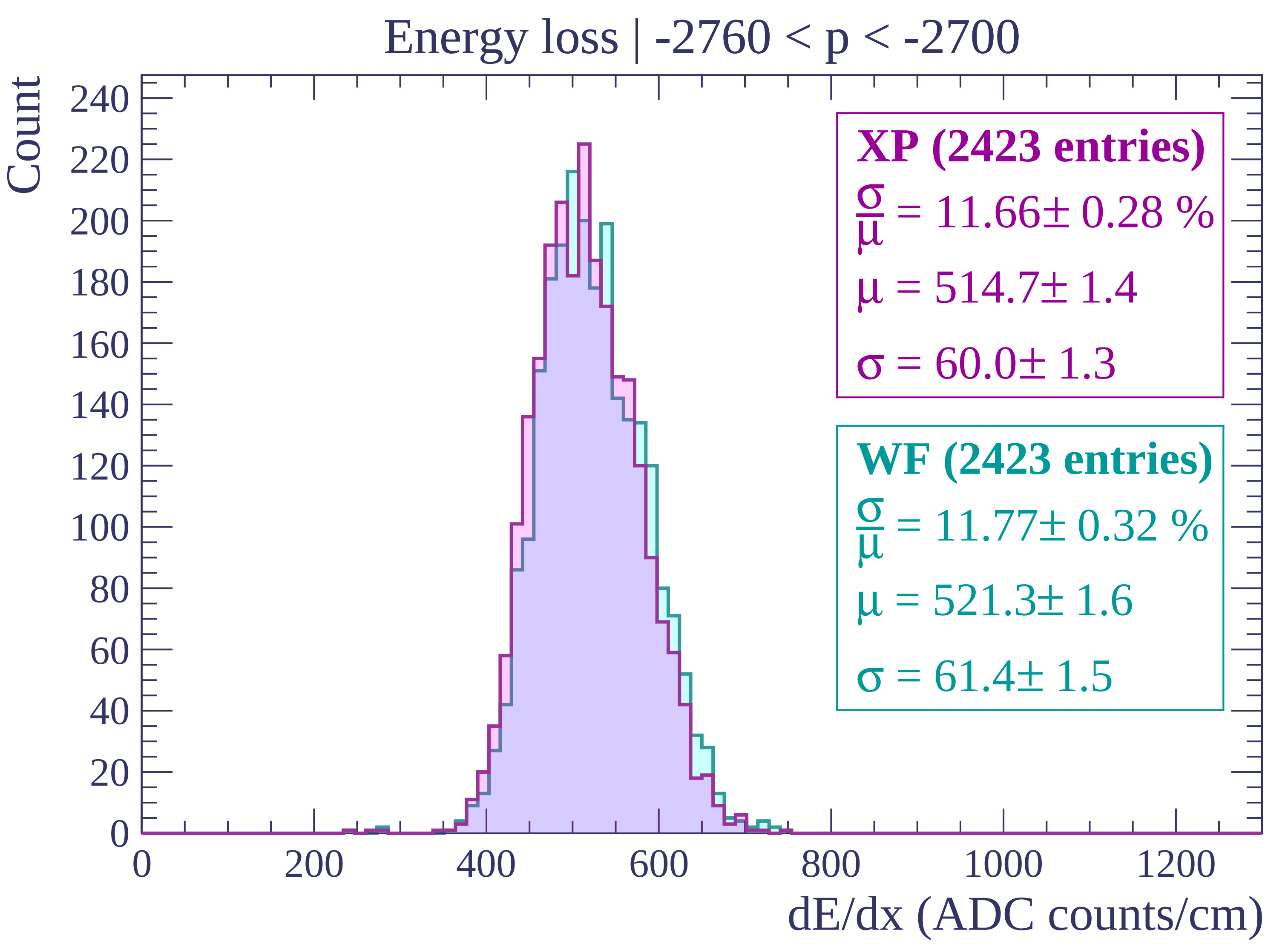


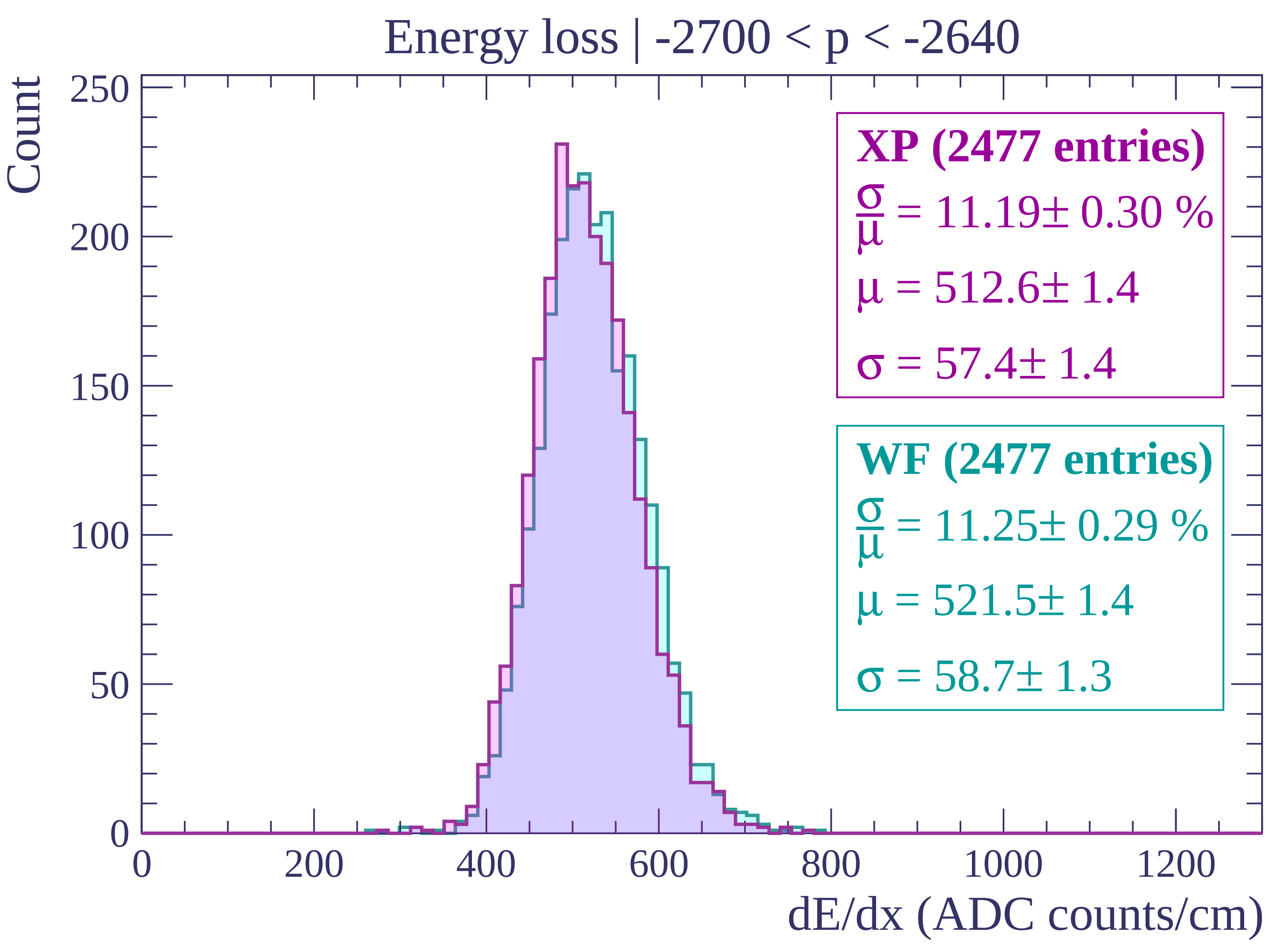
Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
240
XP (2177 entries)
σ/μ = 10.74± 0.25 %
μ = 517.2± 1.4
σ = 55.5± 1.1
WF (2177 entries)
σ/μ = 11.16± 0.34 %
μ = 522.9± 1.5
σ = 58.4± 1.6



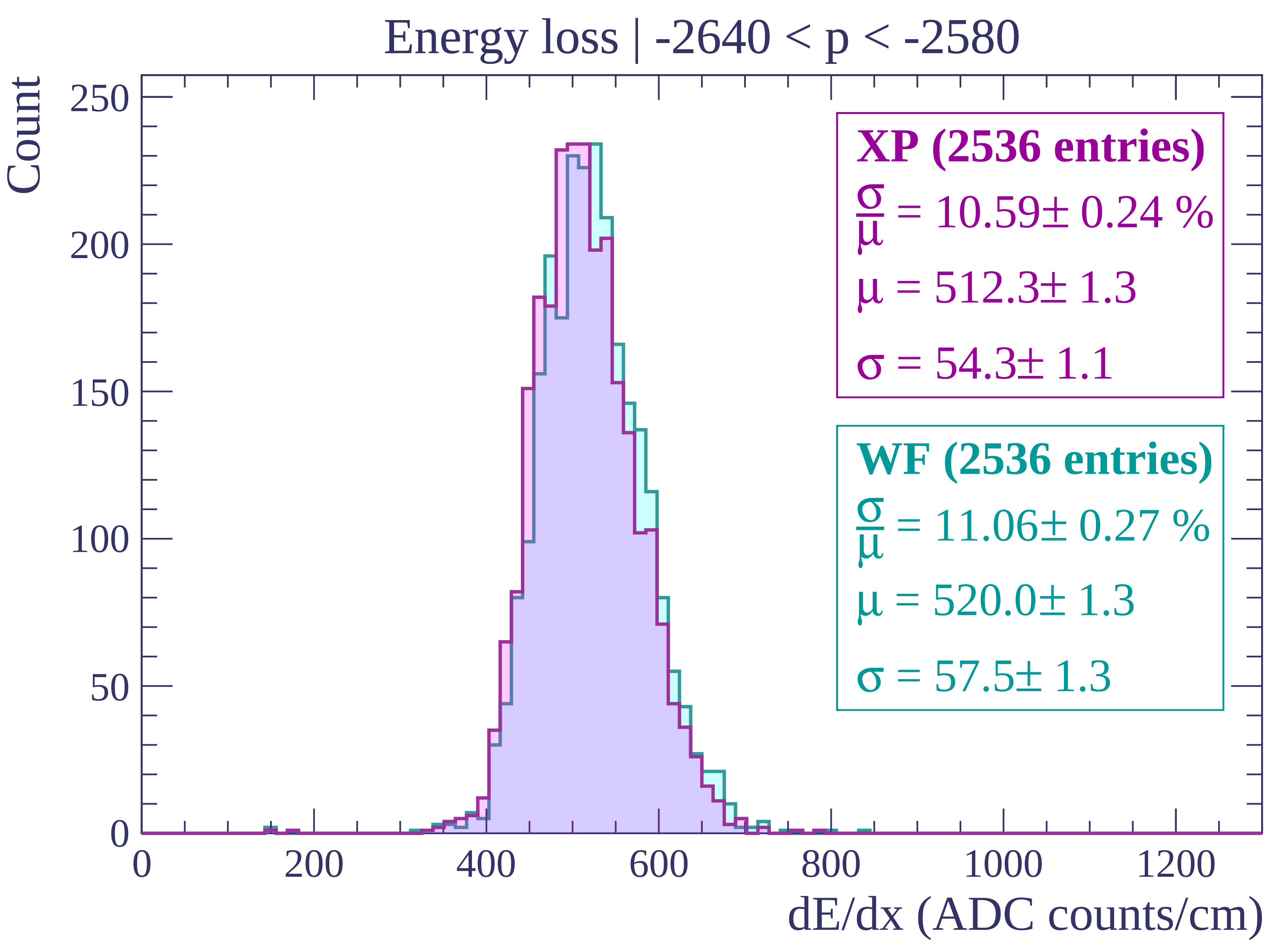
Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (2213 entries)
σ/μ = 11.26± 0.31 %
μ = 518.1± 1.5
σ = 58.3± 1.4
WF (2213 entries)
σ/μ = 10.95± 0.30 %
μ = 526.8± 1.5
σ = 57.7± 1.4


Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
XP (2212 entries)
σ/μ = 11.03± 0.28 %
μ = 514.5± 1.4
σ = 56.8± 1.3
WF (2212 entries)
σ/μ = 11.01± 0.29 %
μ = 522.4± 1.4
σ = 57.5± 1.4
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200

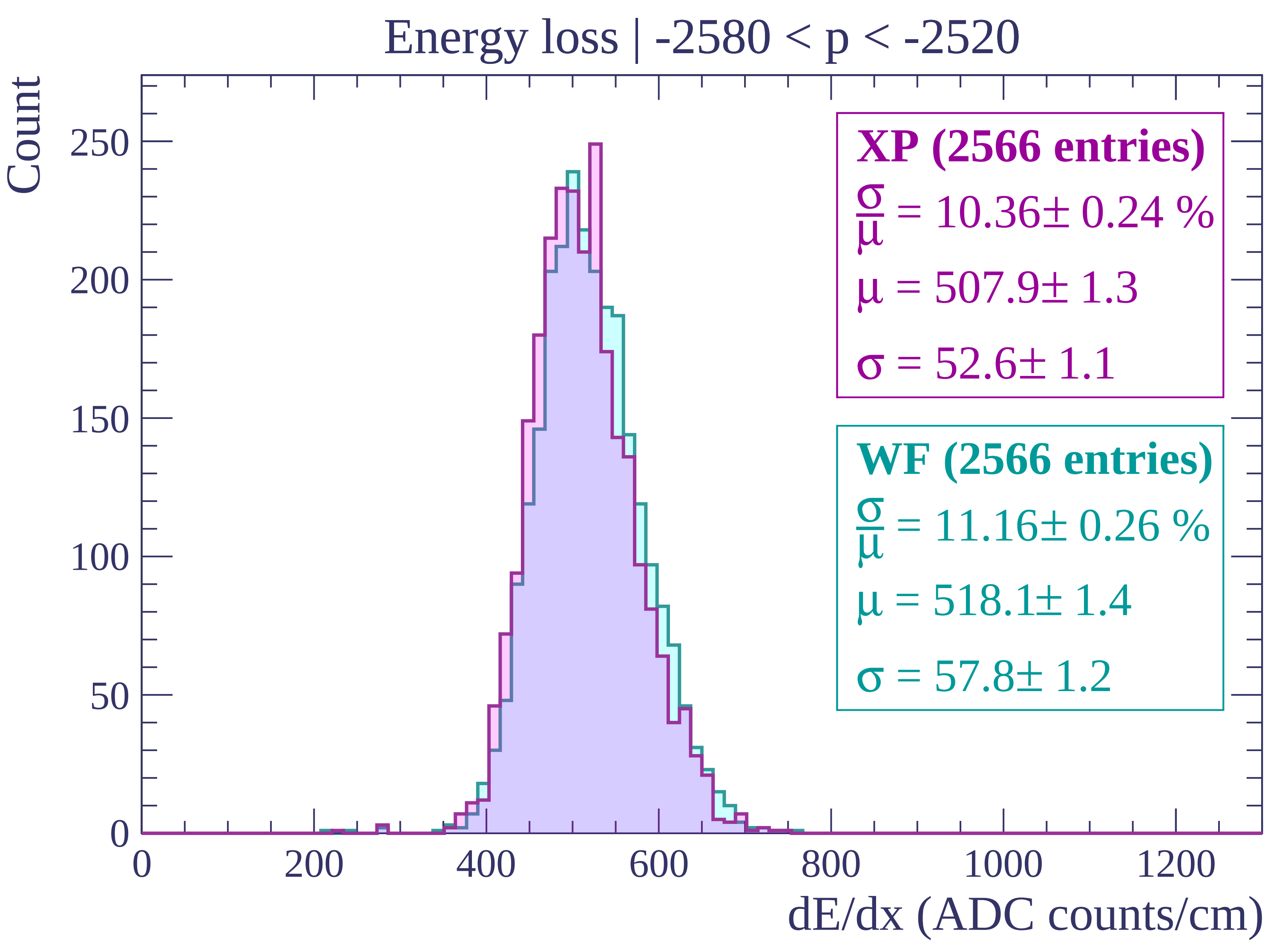

Energy loss | -2760 < p < -2700
Count
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2423 entries)
σ/μ = 11.66± 0.28 %
μ = 514.7± 1.4
σ = 60.0± 1.3
WF (2423 entries)
σ/μ = 11.77± 0.32 %
μ = 521.3± 1.6
σ = 61.4± 1.5

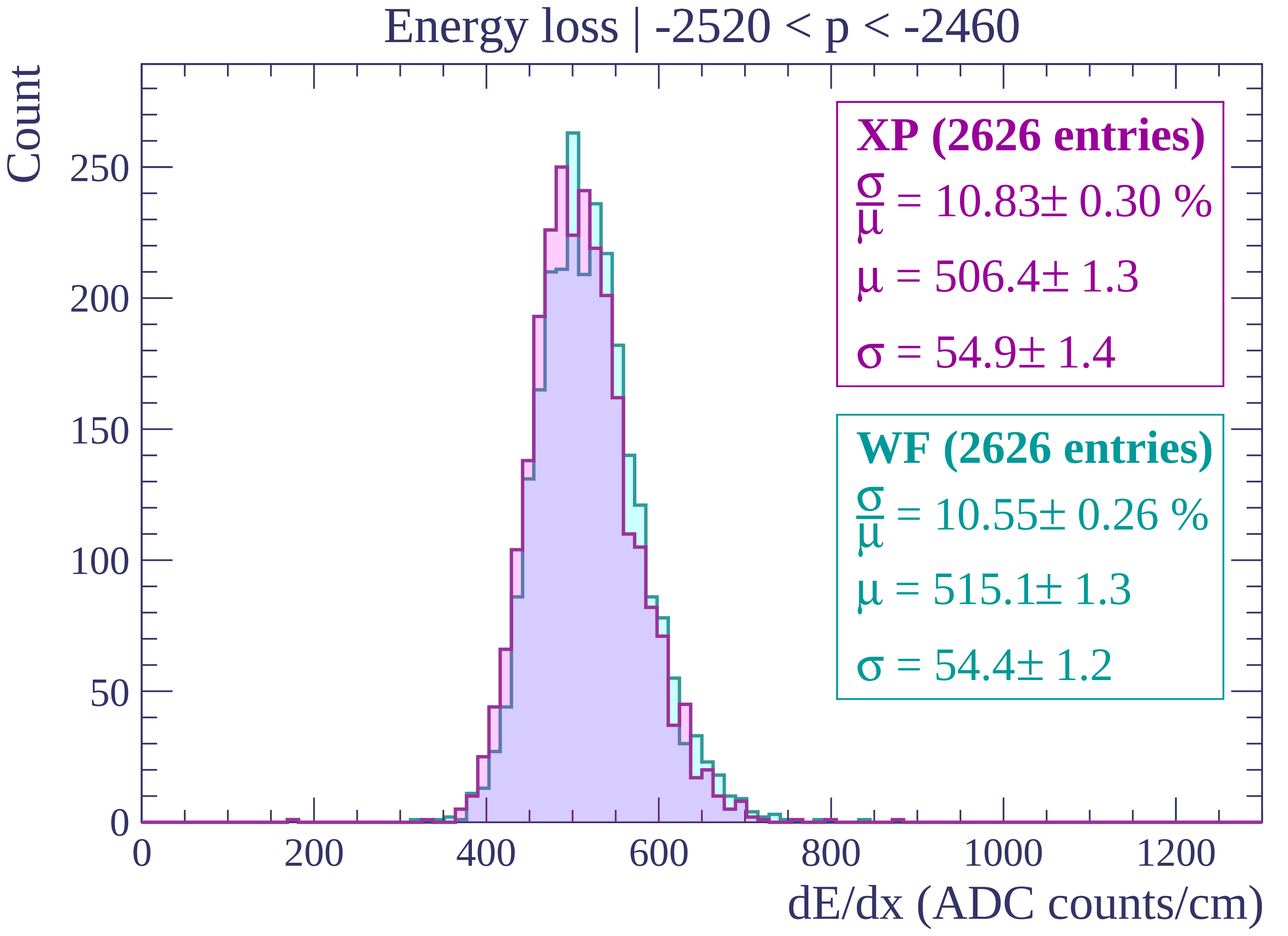


Energy loss | -2700 < p < -2640
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2477 entries)
σ/μ = 11.19± 0.30 %
μ = 512.6± 1.4
σ = 57.4± 1.4
WF (2477 entries)
σ/μ = 11.25± 0.29 %
μ = 521.5± 1.4
σ = 58.7± 1.3

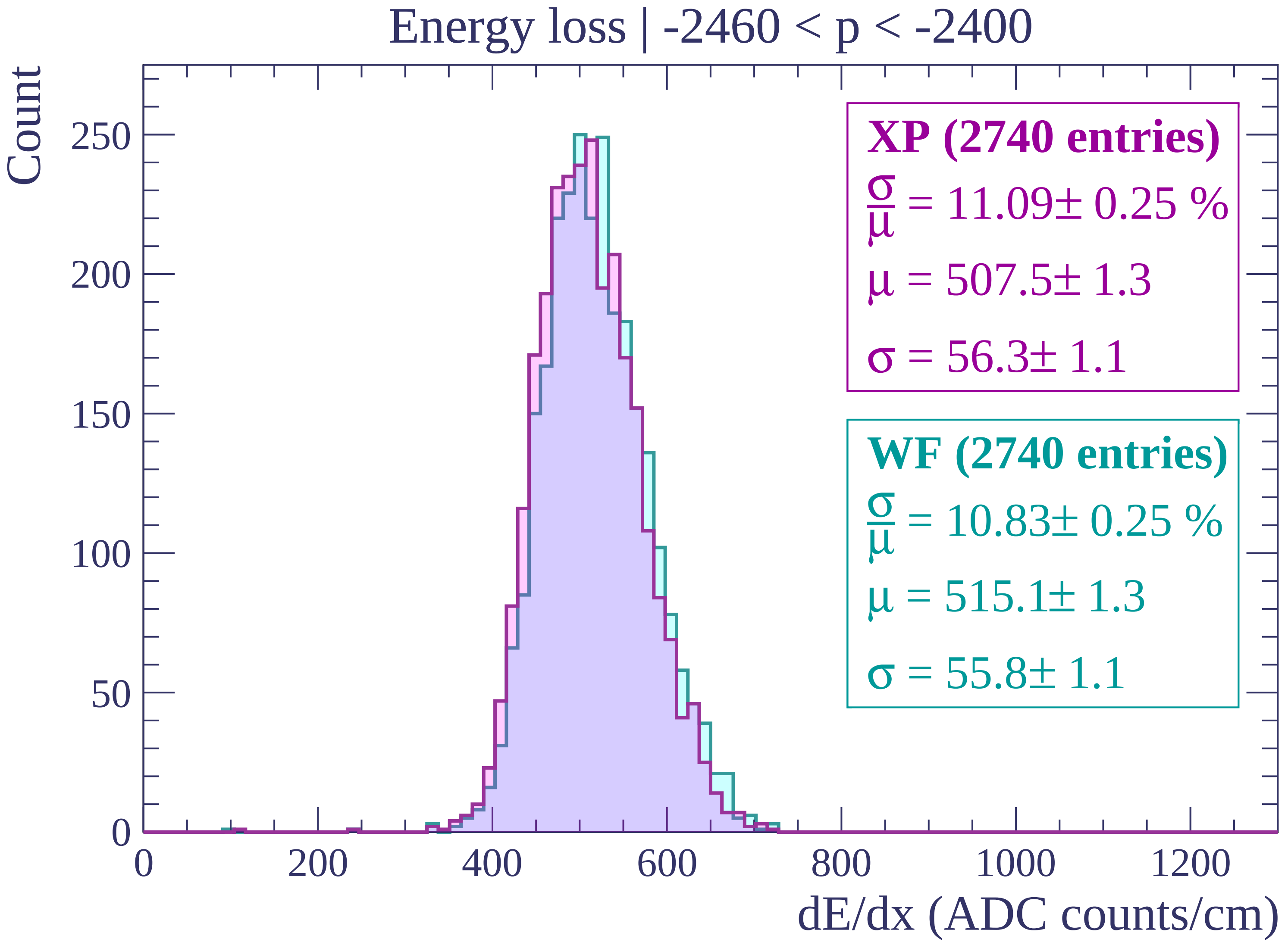


Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP (2536 entries)
$\frac{\sigma}{\mu}$ = 10.59± 0.24 %
μ = 512.3± 1.3
σ = 54.3± 1.1
WF (2536 entries)
$\frac{\sigma}{\mu}$ = 11.06± 0.27 %
μ = 520.0± 1.3
σ = 57.5± 1.3

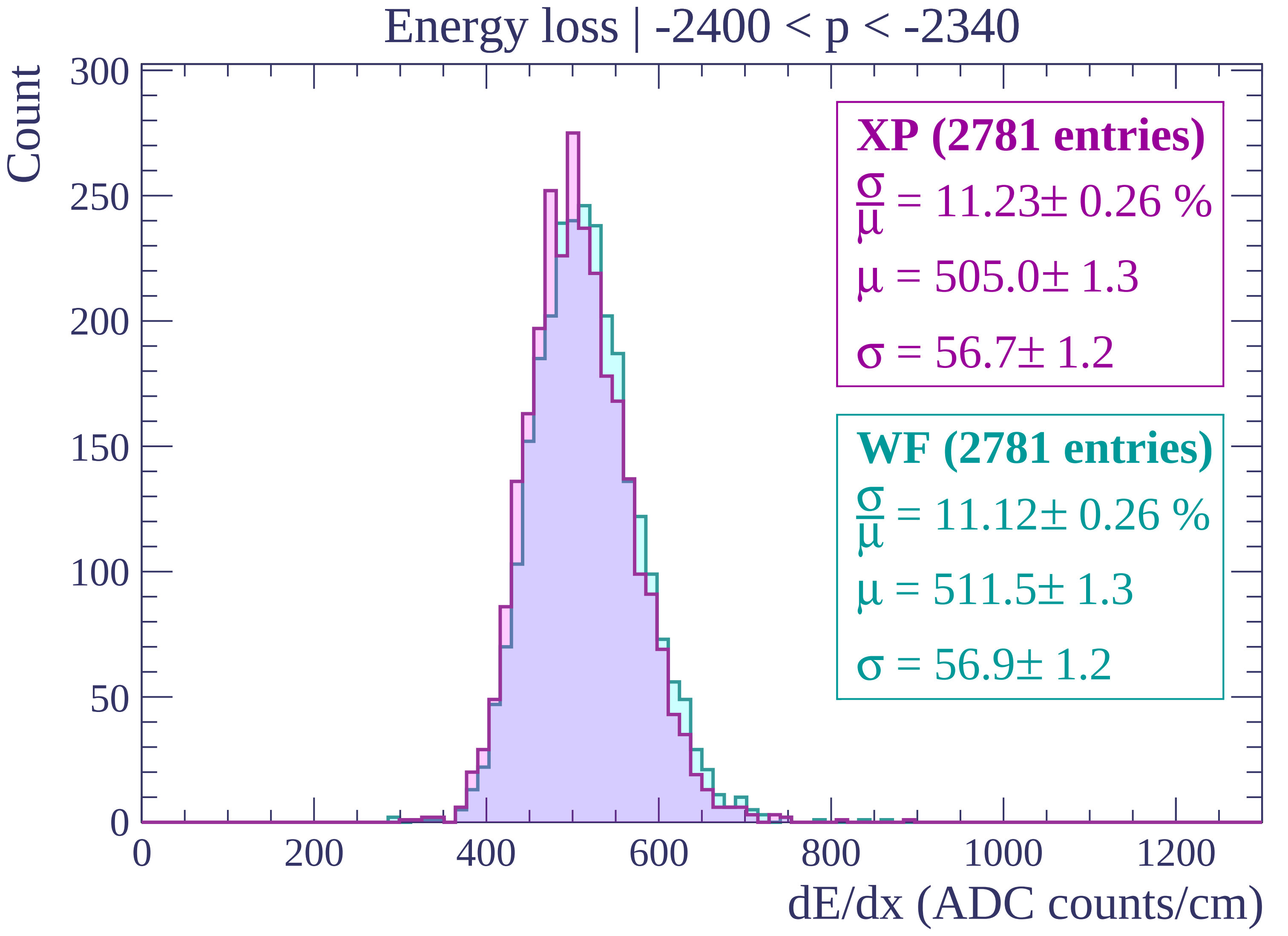

Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2566 entries)
σ/μ = 10.36± 0.24 %
μ = 507.9± 1.3
σ = 52.6± 1.1
WF (2566 entries)
σ/μ = 11.16± 0.26 %
μ = 518.1± 1.4
σ = 57.8± 1.2

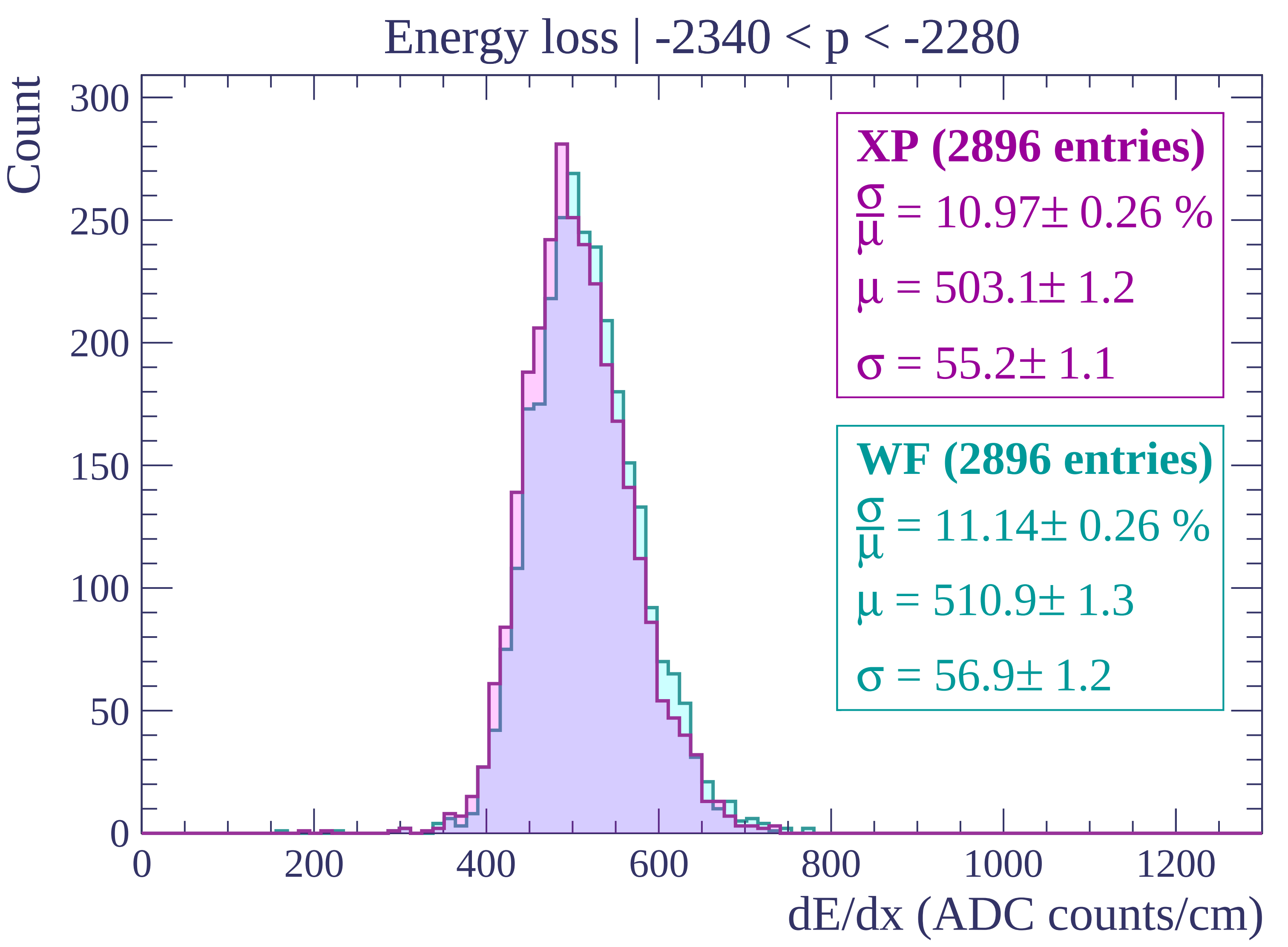


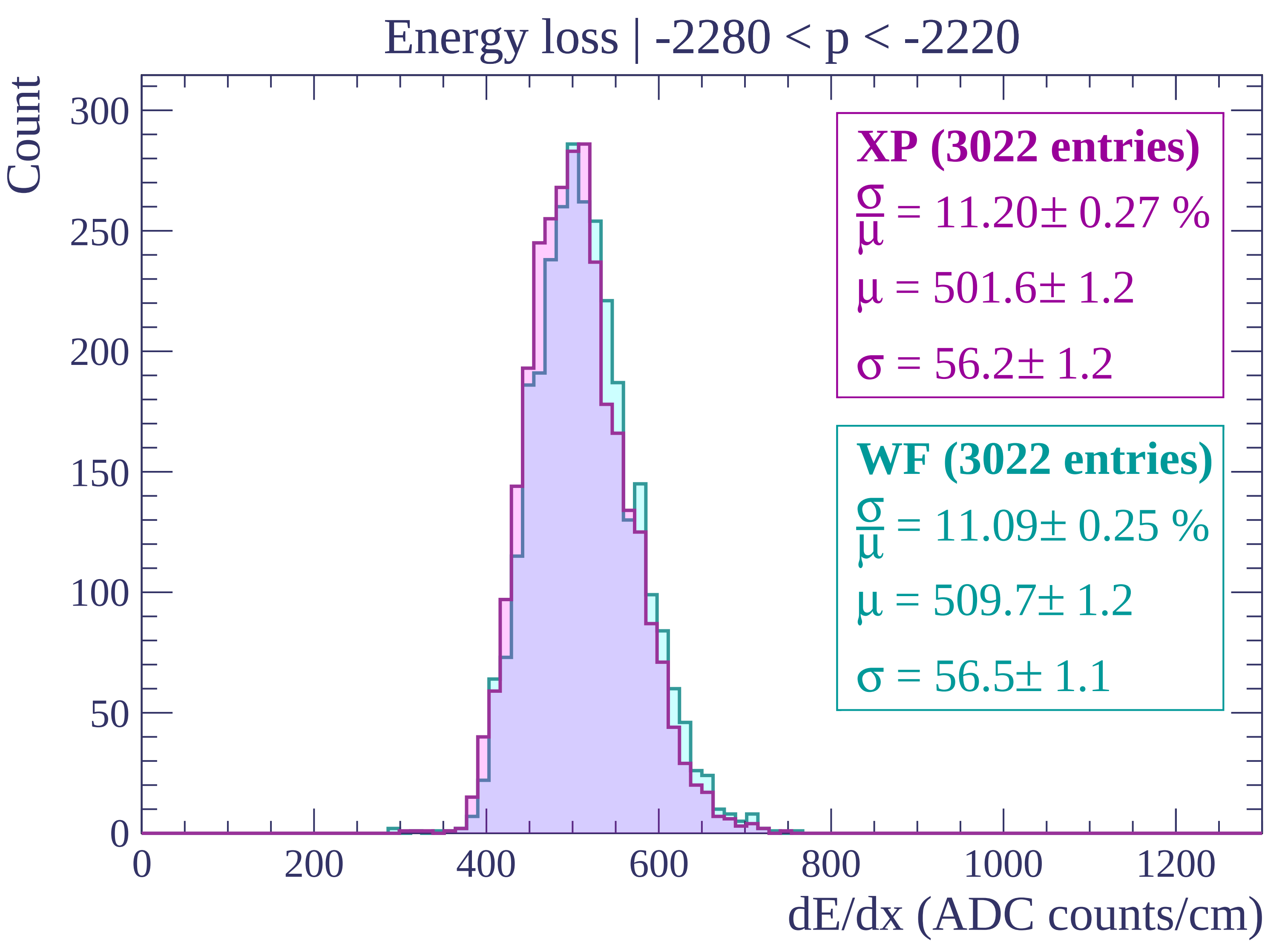
Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2626 entries)
σ/μ = 10.83± 0.30 %
μ = 506.4± 1.3
σ = 54.9± 1.4
WF (2626 entries)
σ/μ = 10.55± 0.26 %
μ = 515.1± 1.3
σ = 54.4± 1.2



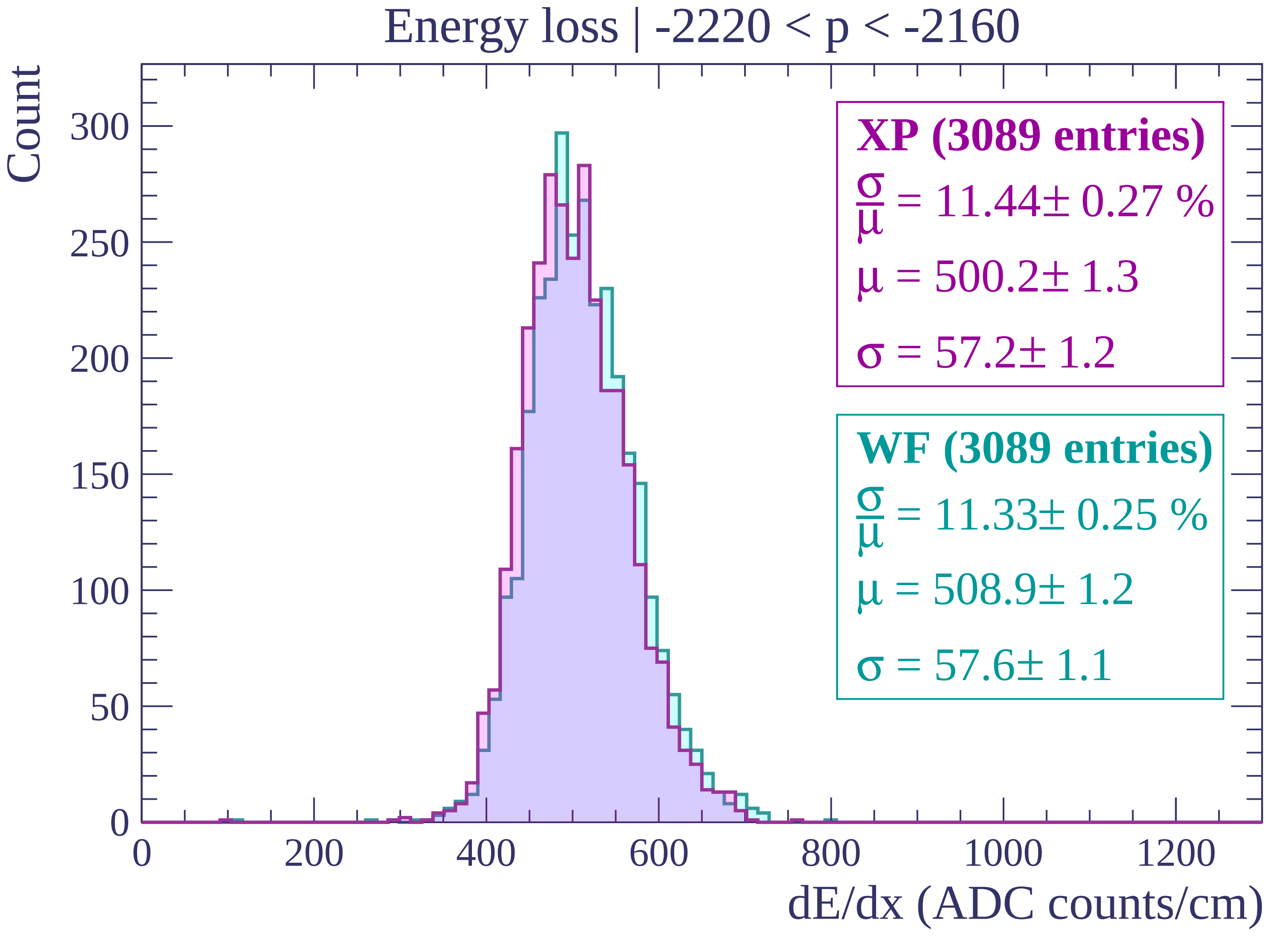
Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
XP (2740 entries)
σ/μ = 11.09± 0.25 %
μ = 507.5± 1.3
σ = 56.3± 1.1
WF (2740 entries)
σ/μ = 10.83± 0.25 %
μ = 515.1± 1.3
σ = 55.8± 1.1


Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2781 entries)
$\frac{\sigma}{\mu}$ = 11.23± 0.26 %
μ = 505.0± 1.3
σ = 56.7± 1.2
WF (2781 entries)
$\frac{\sigma}{\mu}$ = 11.12± 0.26 %
μ = 511.5± 1.3
σ = 56.9± 1.2

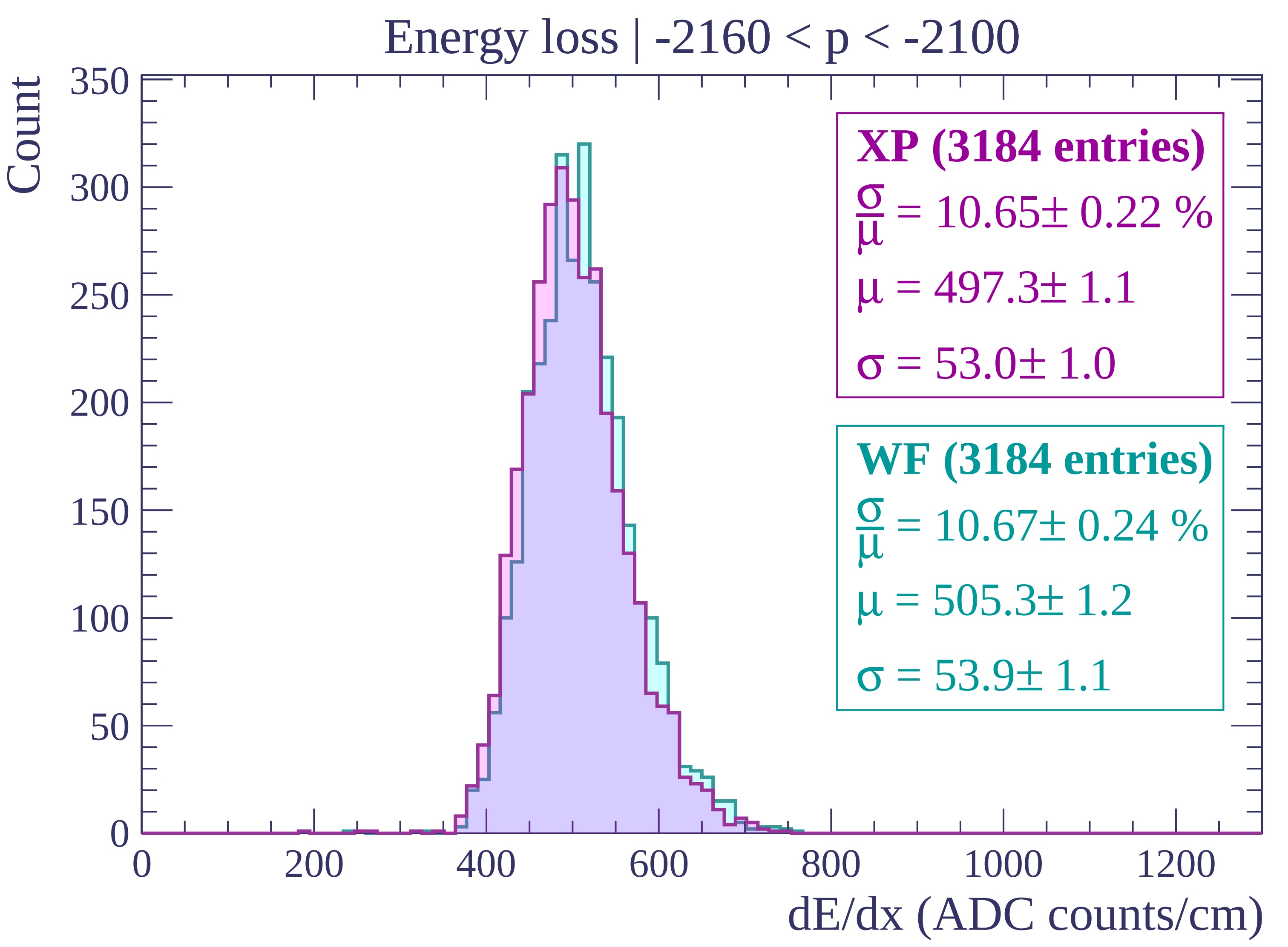

Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2896 entries)
σ/μ = 10.97± 0.26 %
μ = 503.1± 1.2
σ = 55.2± 1.1
WF (2896 entries)
σ/μ = 11.14± 0.26 %
μ = 510.9± 1.3
σ = 56.9± 1.2



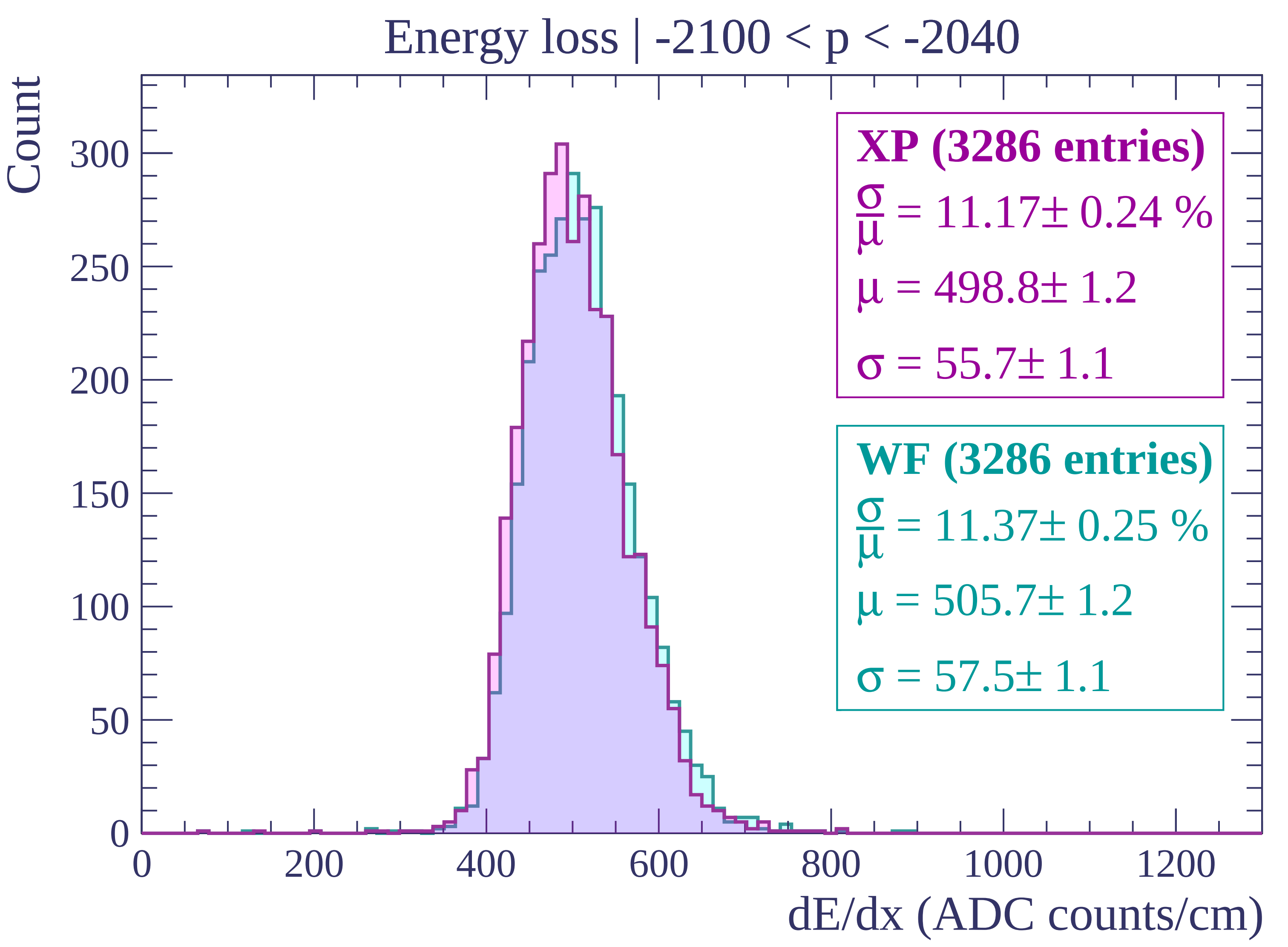
Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP (3022 entries)
σ/μ = 11.20± 0.27 %
μ = 501.6± 1.2
σ = 56.2± 1.2
WF (3022 entries)
σ/μ = 11.09± 0.25 %
μ = 509.7± 1.2
σ = 56.5± 1.1



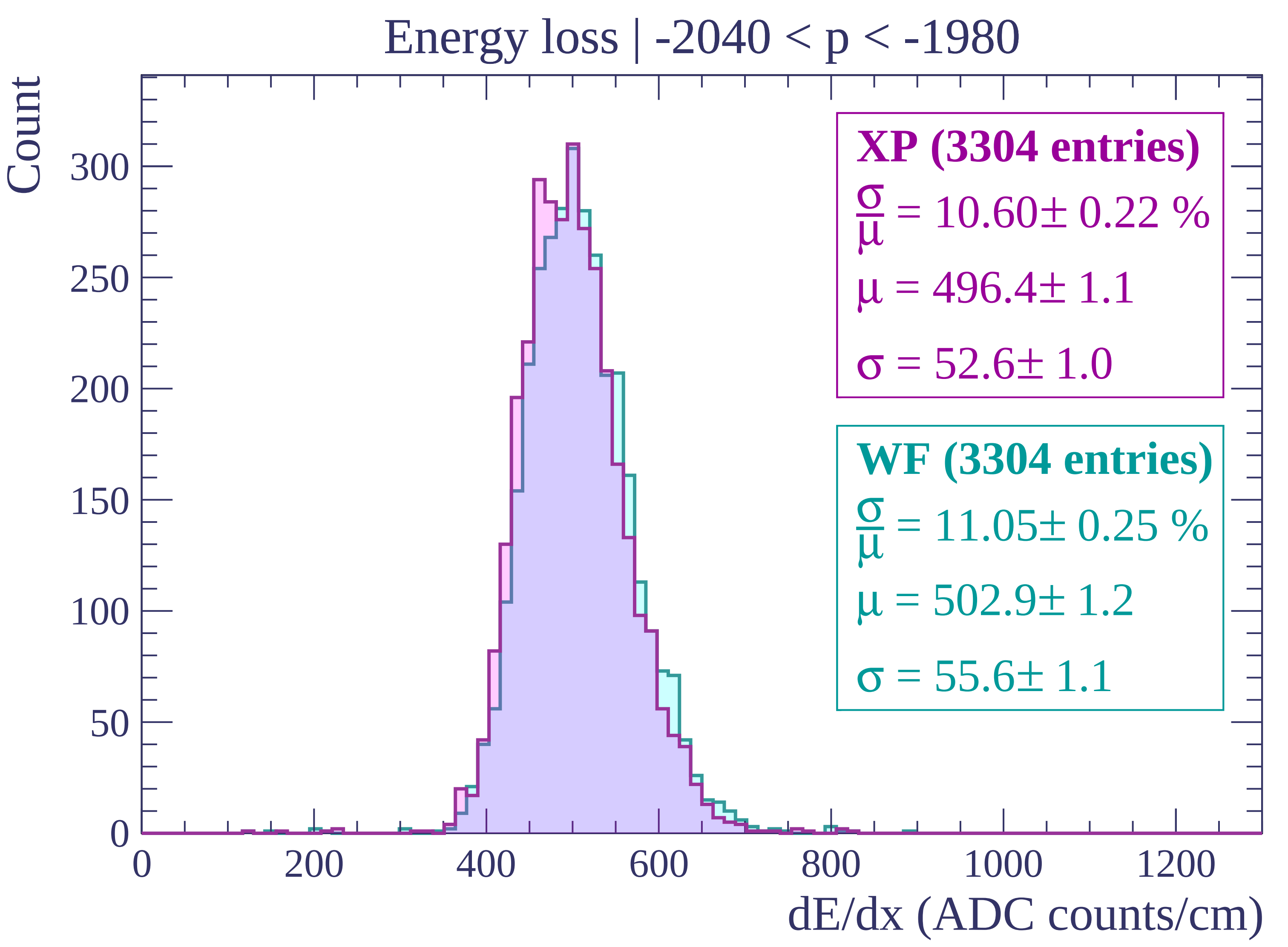
Energy loss | -2220 < p < -2160
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3089 entries)
σ/μ = 11.44± 0.27 %
μ = 500.2± 1.3
σ = 57.2± 1.2
WF (3089 entries)
σ/μ = 11.33± 0.25 %
μ = 508.9± 1.2
σ = 57.6± 1.1

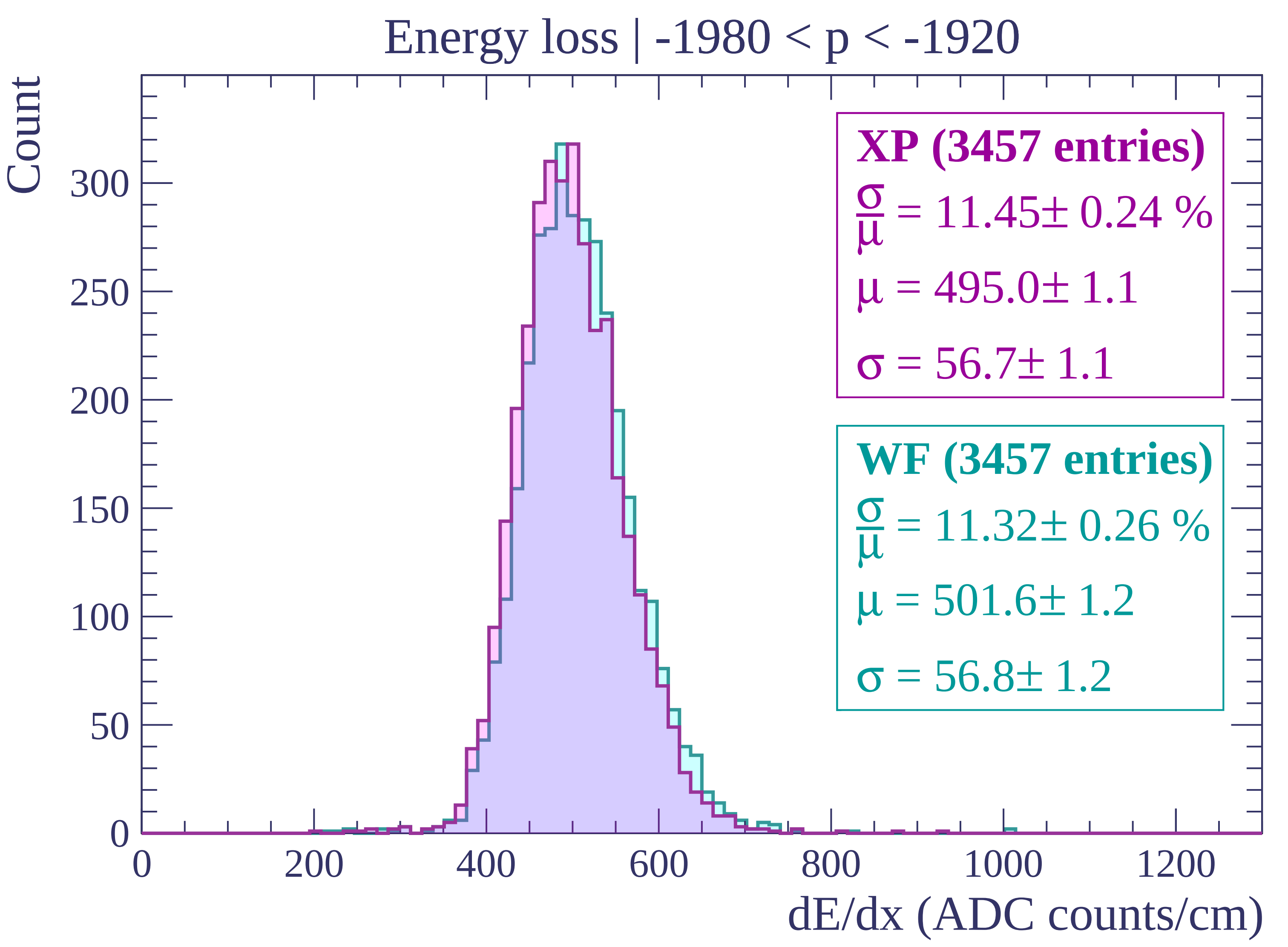


Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
XP (3184 entries)
σ/μ = 10.65± 0.22 %
μ = 497.3± 1.1
σ = 53.0± 1.0
WF (3184 entries)
σ/μ = 10.67± 0.24 %
μ = 505.3± 1.2
σ = 53.9± 1.1
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

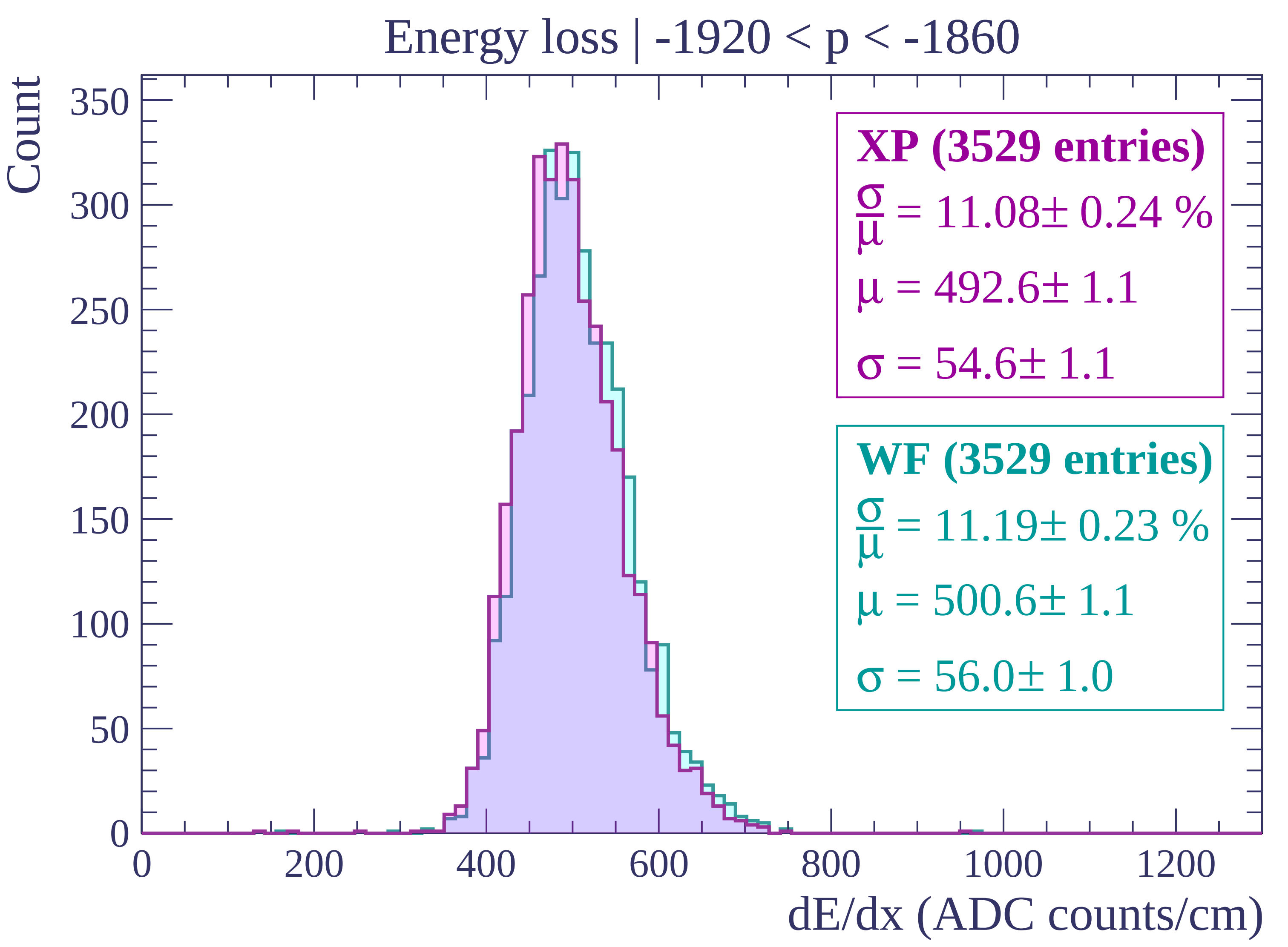


Energy loss | -2100 < p < -2040
Count
dE/dx (ADC counts/cm)
XP (3286 entries)
$\frac{\sigma}{\mu}$ = 11.17± 0.24 %
μ = 498.8± 1.2
σ = 55.7± 1.1
WF (3286 entries)
$\frac{\sigma}{\mu}$ = 11.37± 0.25 %
μ = 505.7± 1.2
σ = 57.5± 1.1
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

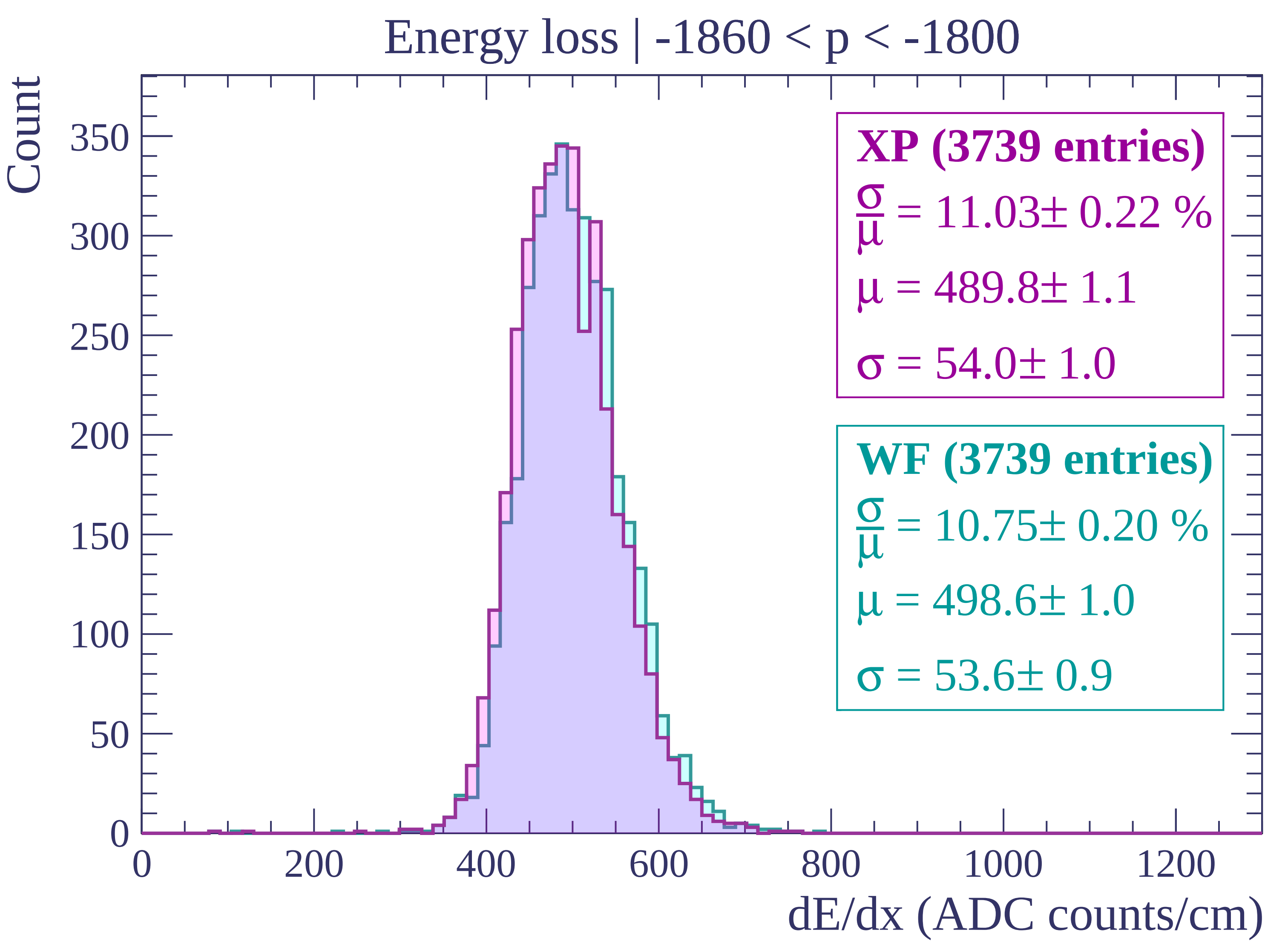

Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3304 entries)
σ/μ = 10.60± 0.22 %
μ = 496.4± 1.1
σ = 52.6± 1.0
WF (3304 entries)
σ/μ = 11.05± 0.25 %
μ = 502.9± 1.2
σ = 55.6± 1.1



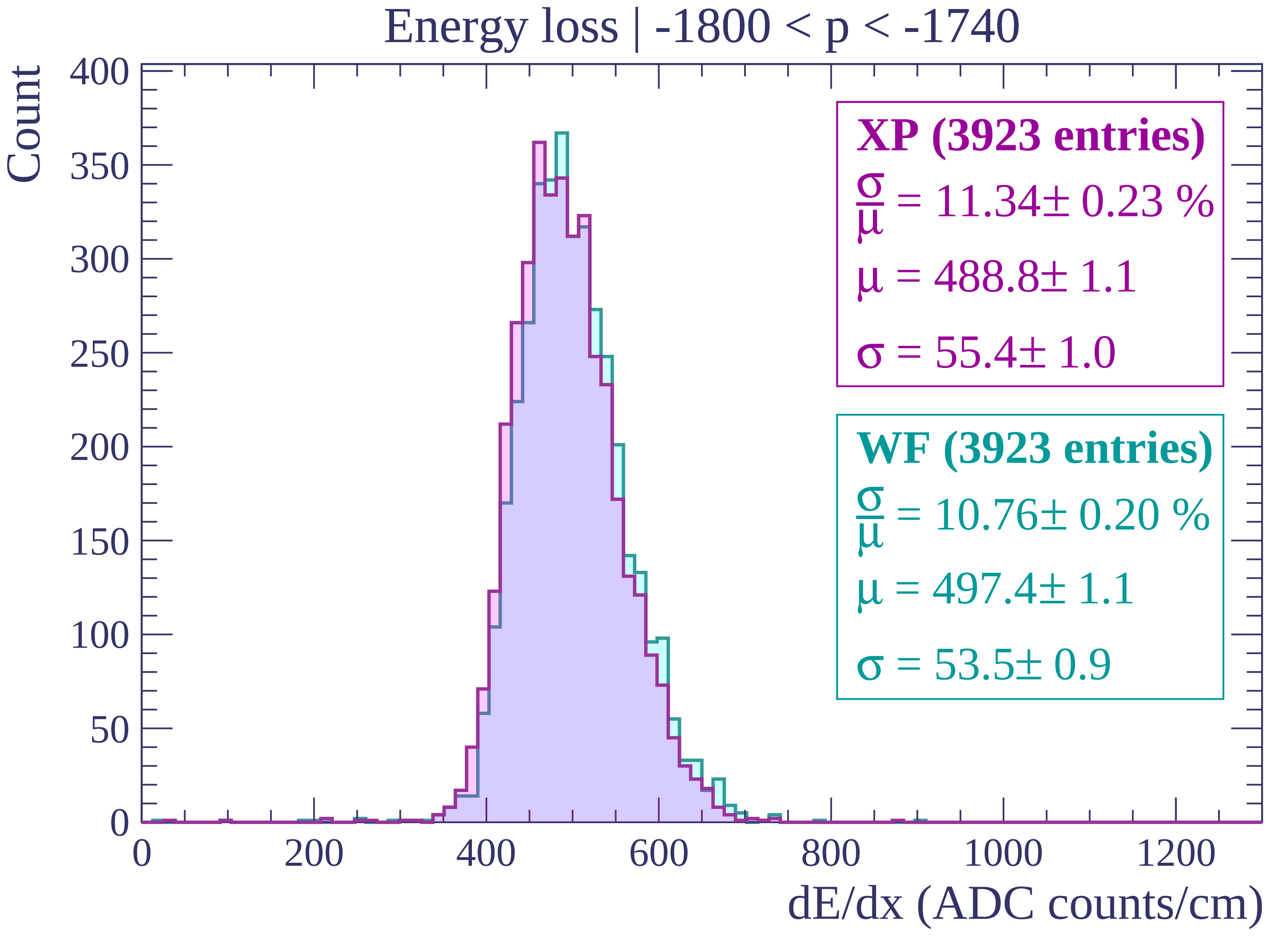
Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
XP (3457 entries)
σ/μ = 11.45± 0.24 %
μ = 495.0± 1.1
σ = 56.7± 1.1
WF (3457 entries)
σ/μ = 11.32± 0.26 %
μ = 501.6± 1.2
σ = 56.8± 1.2
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

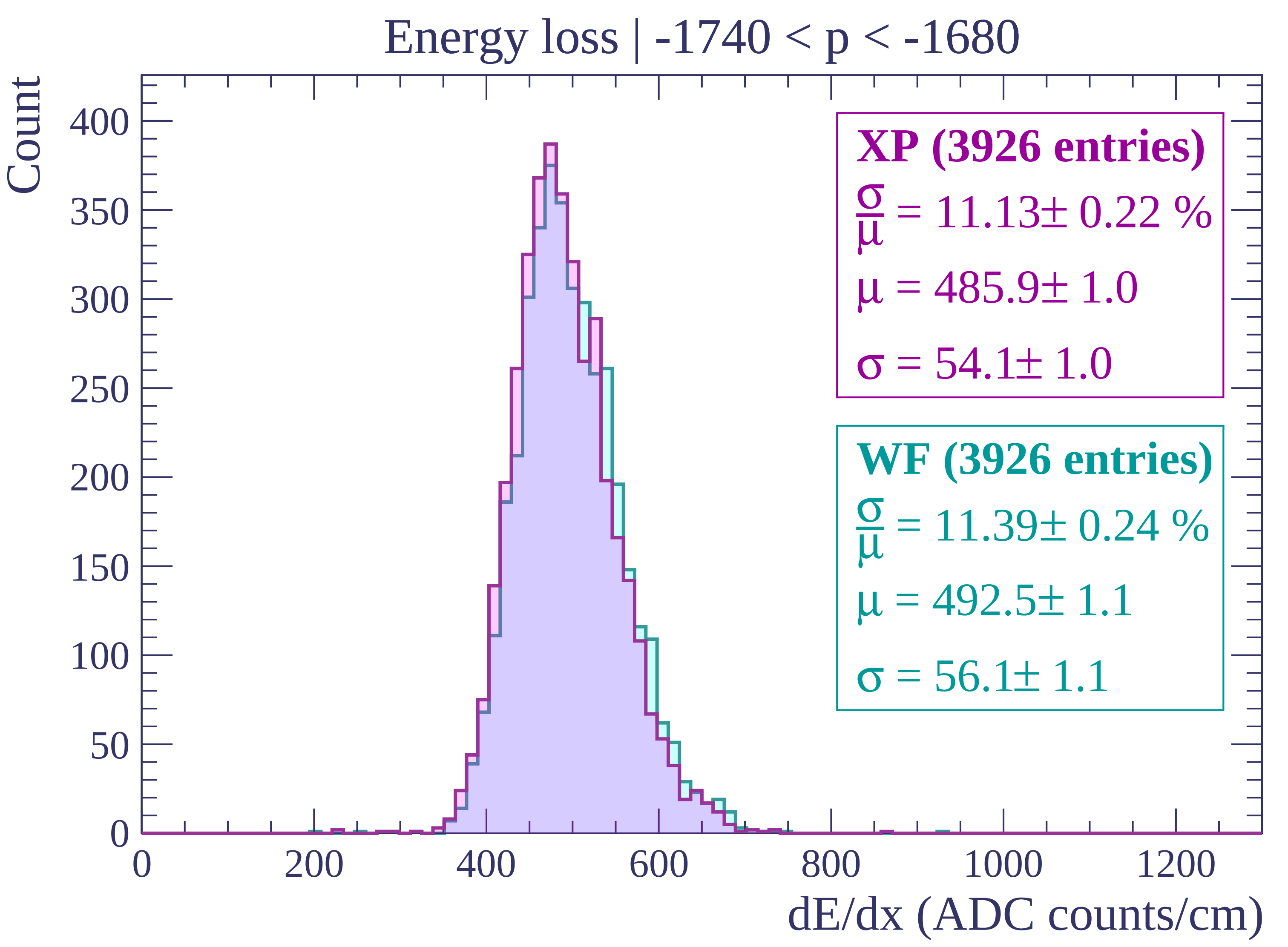


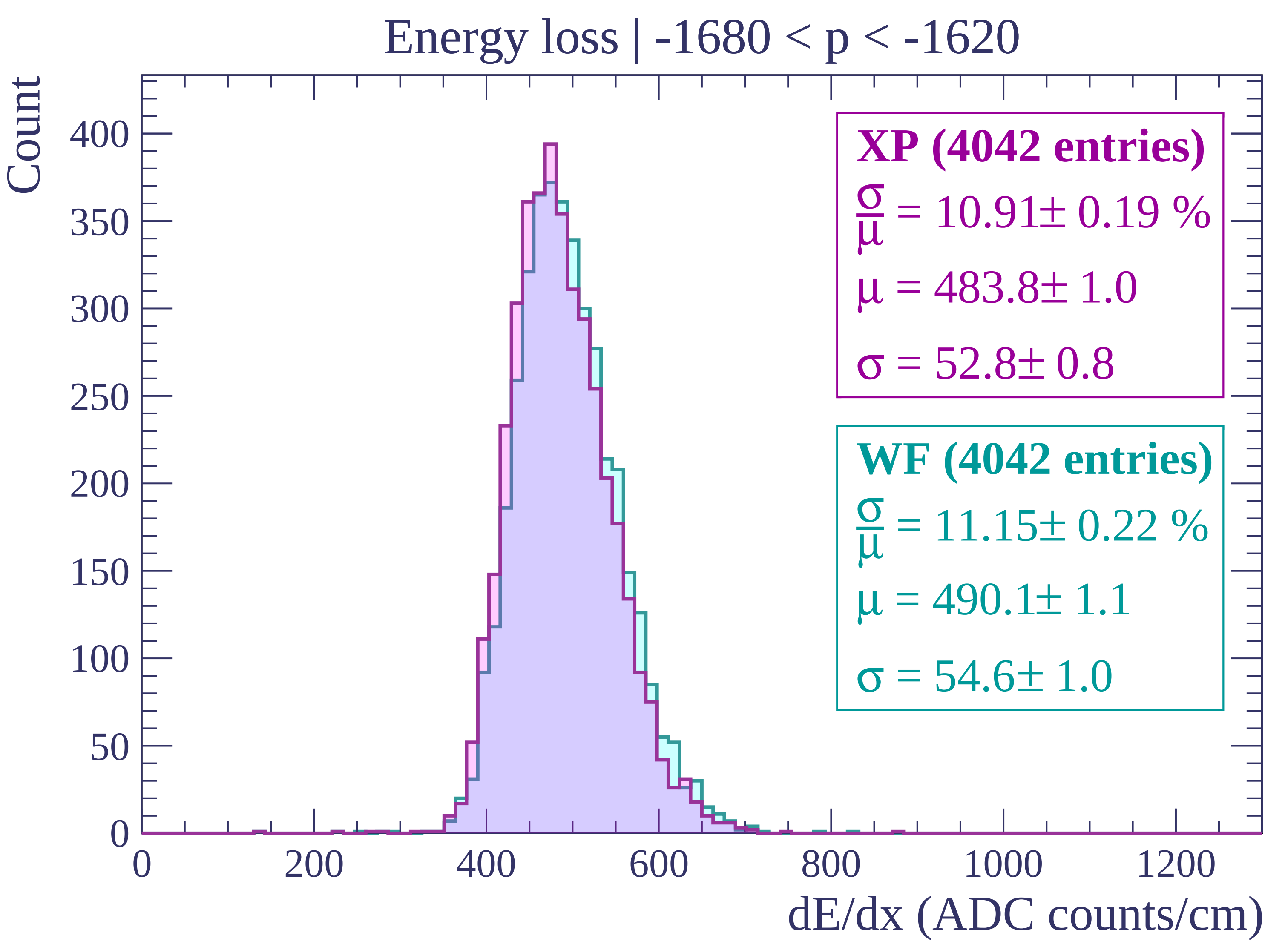
Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3529 entries)
σ/μ = 11.08± 0.24 %
μ = 492.6± 1.1
σ = 54.6± 1.1
WF (3529 entries)
σ/μ = 11.19± 0.23 %
μ = 500.6± 1.1
σ = 56.0± 1.0

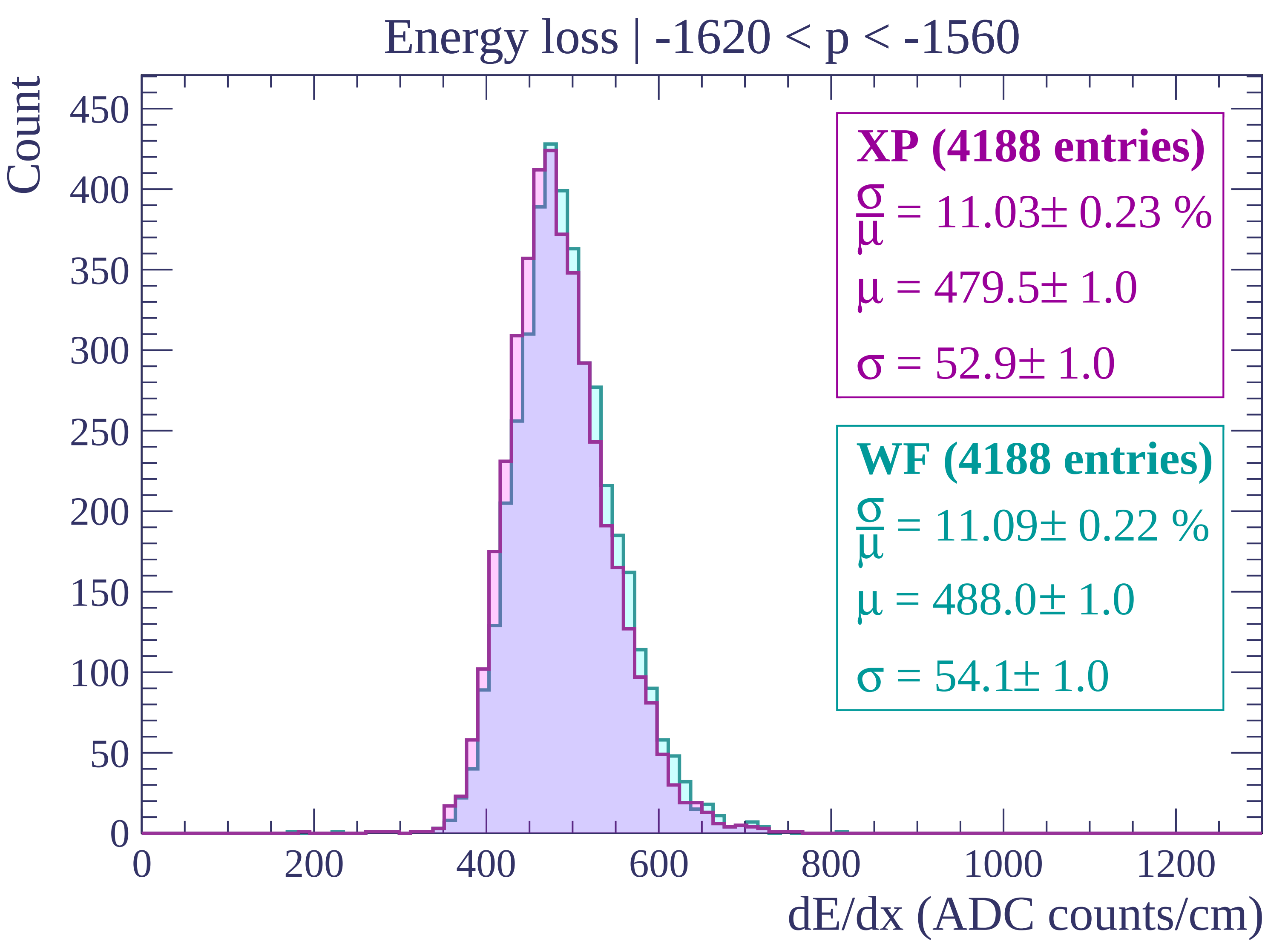


Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
XP (3739 entries)
σ/μ = 11.03± 0.22 %
μ = 489.8± 1.1
σ = 54.0± 1.0
WF (3739 entries)
σ/μ = 10.75± 0.20 %
μ = 498.6± 1.0
σ = 53.6± 0.9

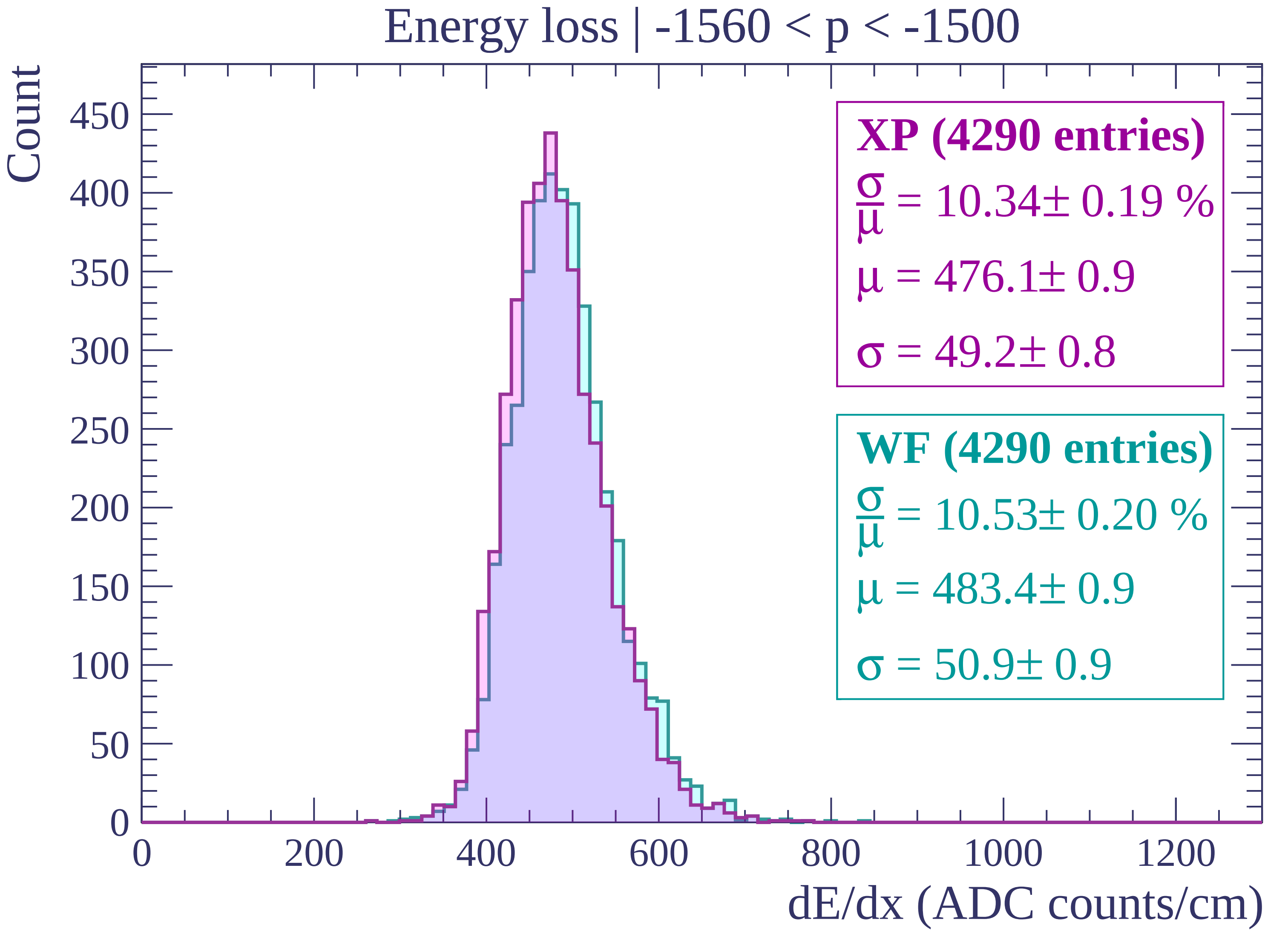

Energy loss | -1800 < p < -1740
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3923 entries)
σ/μ = 11.34± 0.23 %
μ = 488.8± 1.1
σ = 55.4± 1.0
WF (3923 entries)
σ/μ = 10.76± 0.20 %
μ = 497.4± 1.1
σ = 53.5± 0.9


Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
XP (3926 entries)
$\frac{\sigma}{\mu}$ = 11.13± 0.22 %
μ = 485.9± 1.0
σ = 54.1± 1.0
WF (3926 entries)
$\frac{\sigma}{\mu}$ = 11.39± 0.24 %
μ = 492.5± 1.1
σ = 56.1± 1.1
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

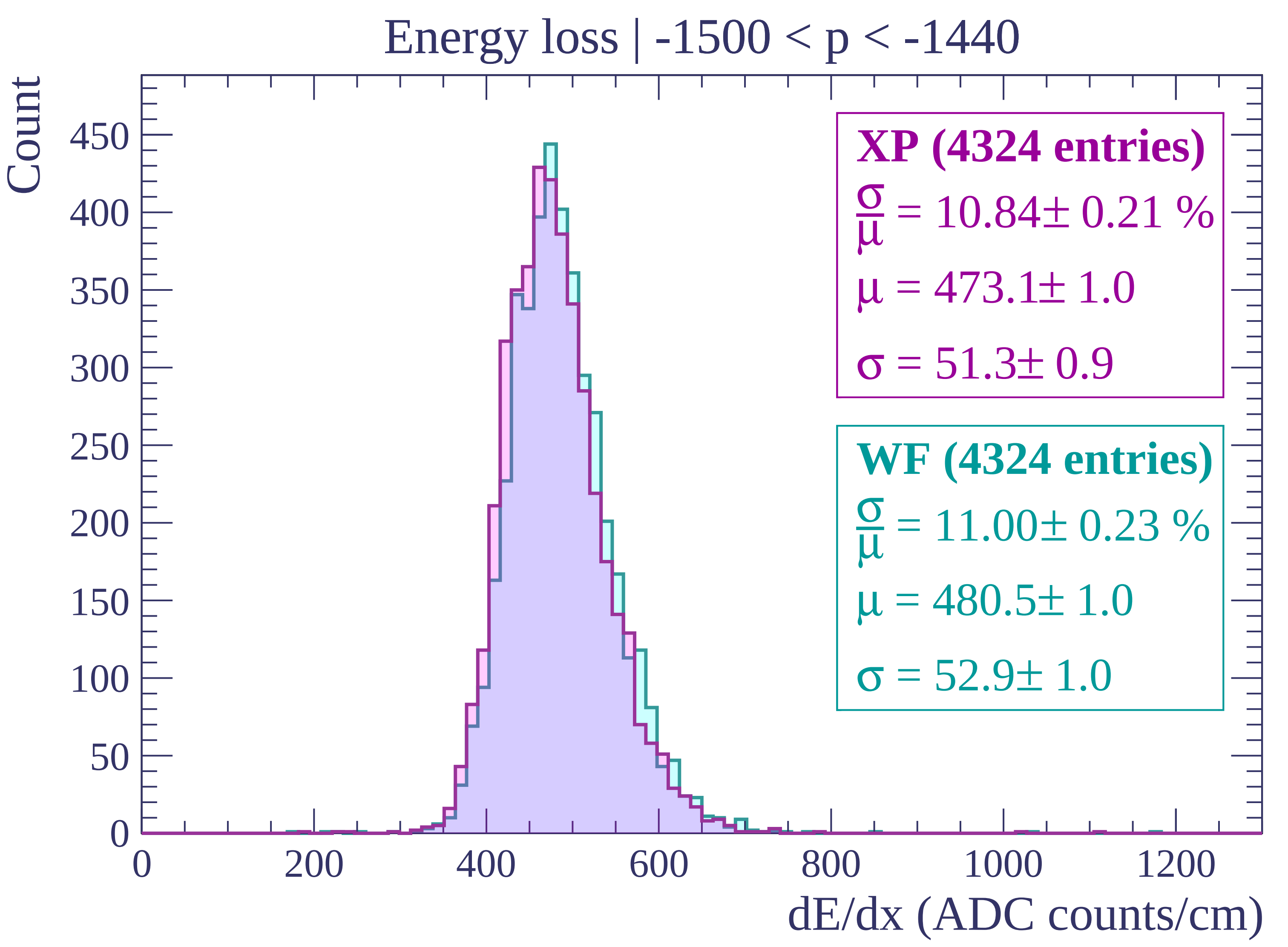

Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (4042 entries)
$\frac{\sigma}{\mu}$ = 10.91± 0.19 %
μ = 483.8± 1.0
σ = 52.8± 0.8
WF (4042 entries)
$\frac{\sigma}{\mu}$ = 11.15± 0.22 %
μ = 490.1± 1.1
σ = 54.6± 1.0



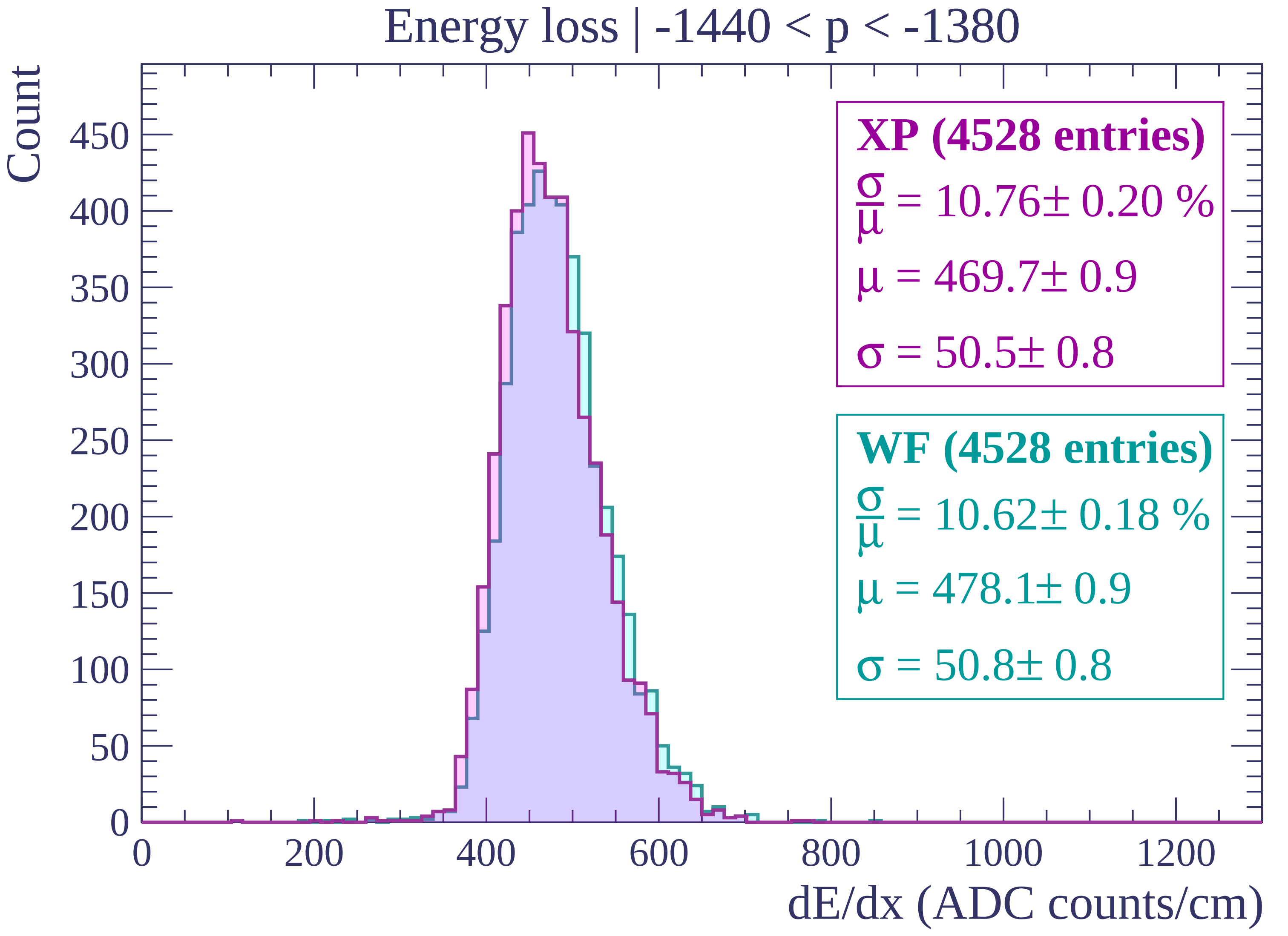
Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4188 entries)
σ/μ = 11.03± 0.23 %
μ = 479.5± 1.0
σ = 52.9± 1.0
WF (4188 entries)
σ/μ = 11.09± 0.22 %
μ = 488.0± 1.0
σ = 54.1± 1.0



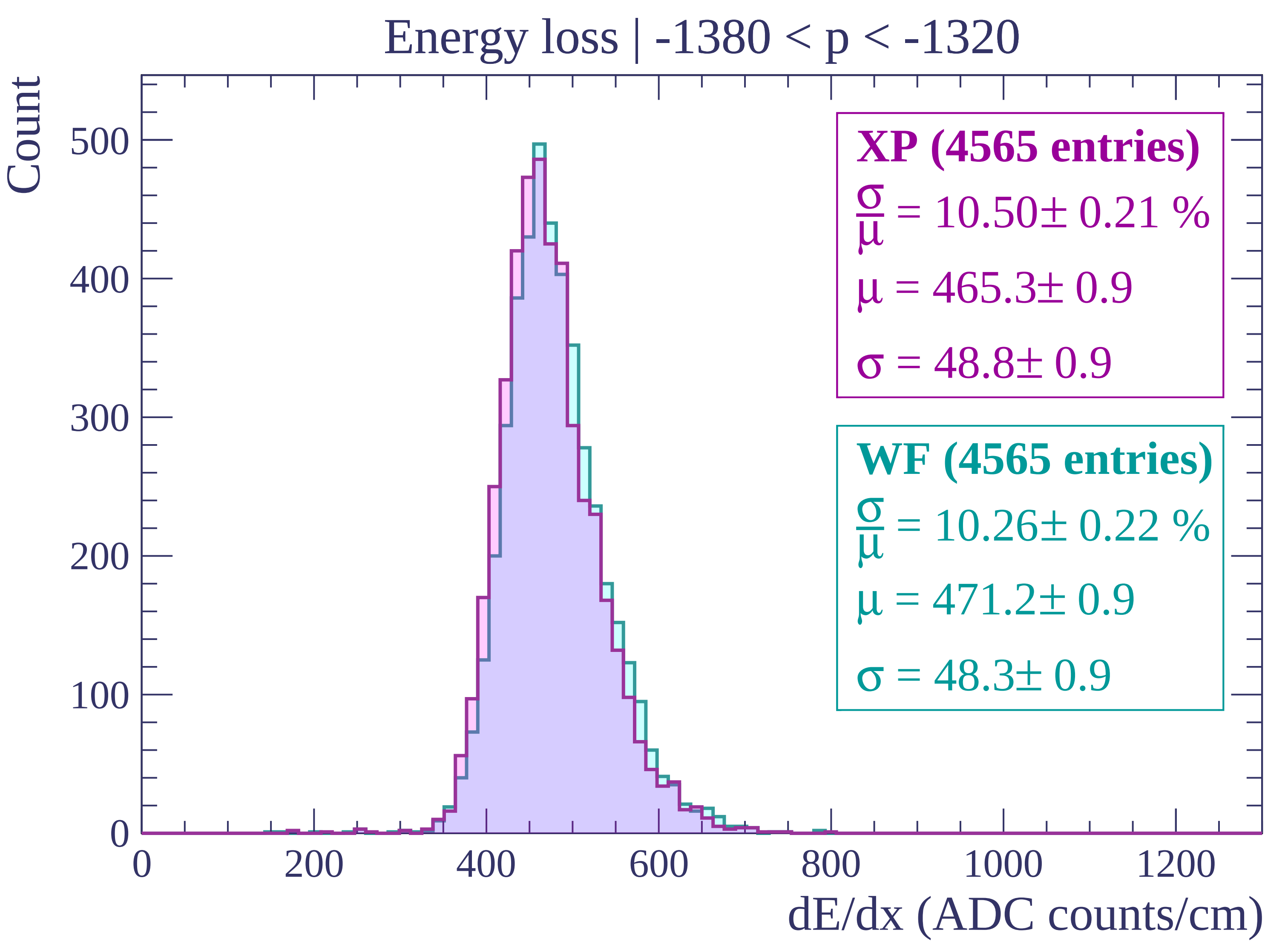
Energy loss | -1560 < p < -1500
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4290 entries)
σ/μ = 10.34± 0.19 %
μ = 476.1± 0.9
σ = 49.2± 0.8
WF (4290 entries)
σ/μ = 10.53± 0.20 %
μ = 483.4± 0.9
σ = 50.9± 0.9



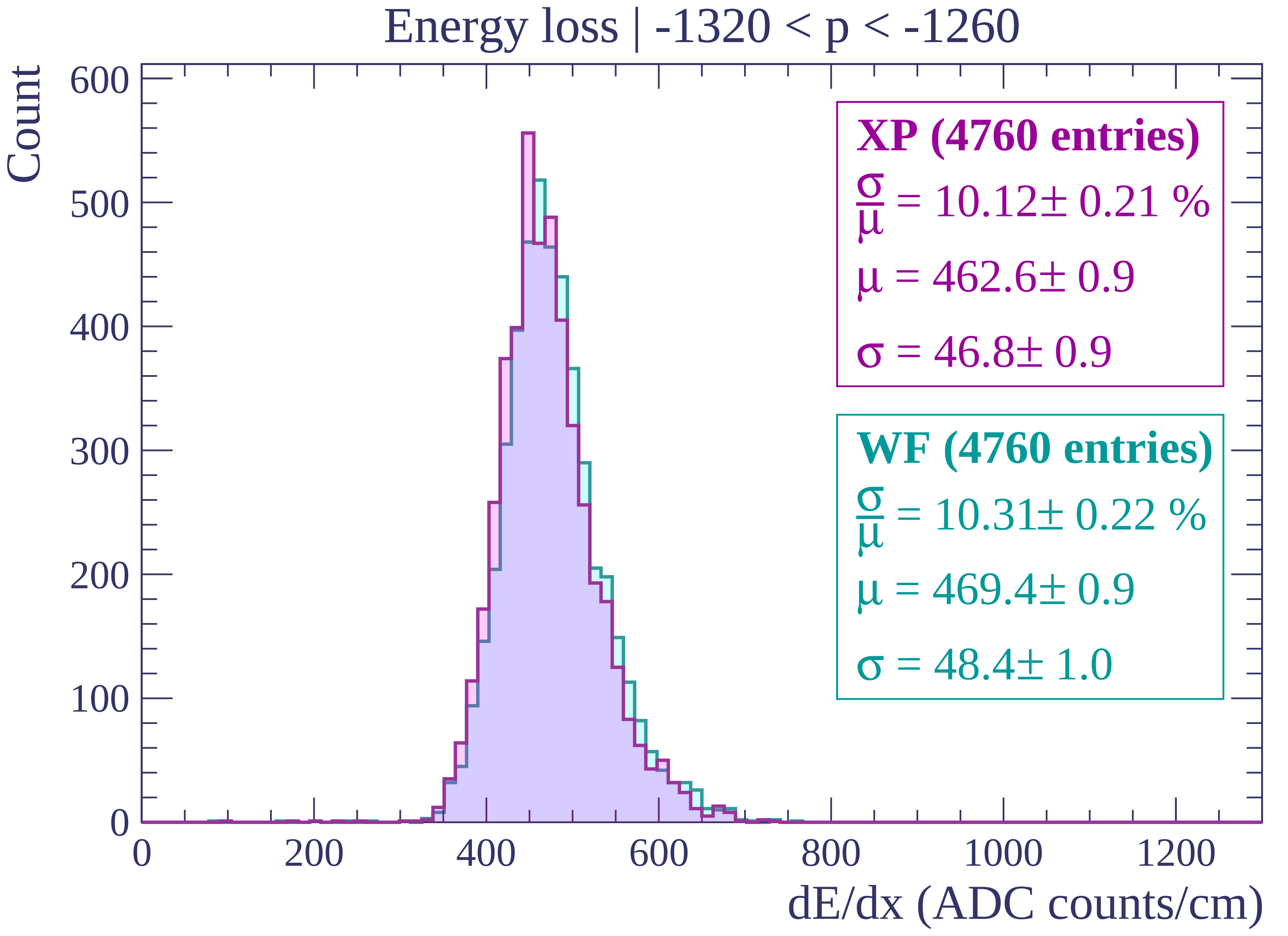
Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
450
XP (4324 entries)
$\frac{\sigma}{\mu}$ = 10.84± 0.21 %
μ = 473.1± 1.0
σ = 51.3± 0.9
WF (4324 entries)
$\frac{\sigma}{\mu}$ = 11.00± 0.23 %
μ = 480.5± 1.0
σ = 52.9± 1.0

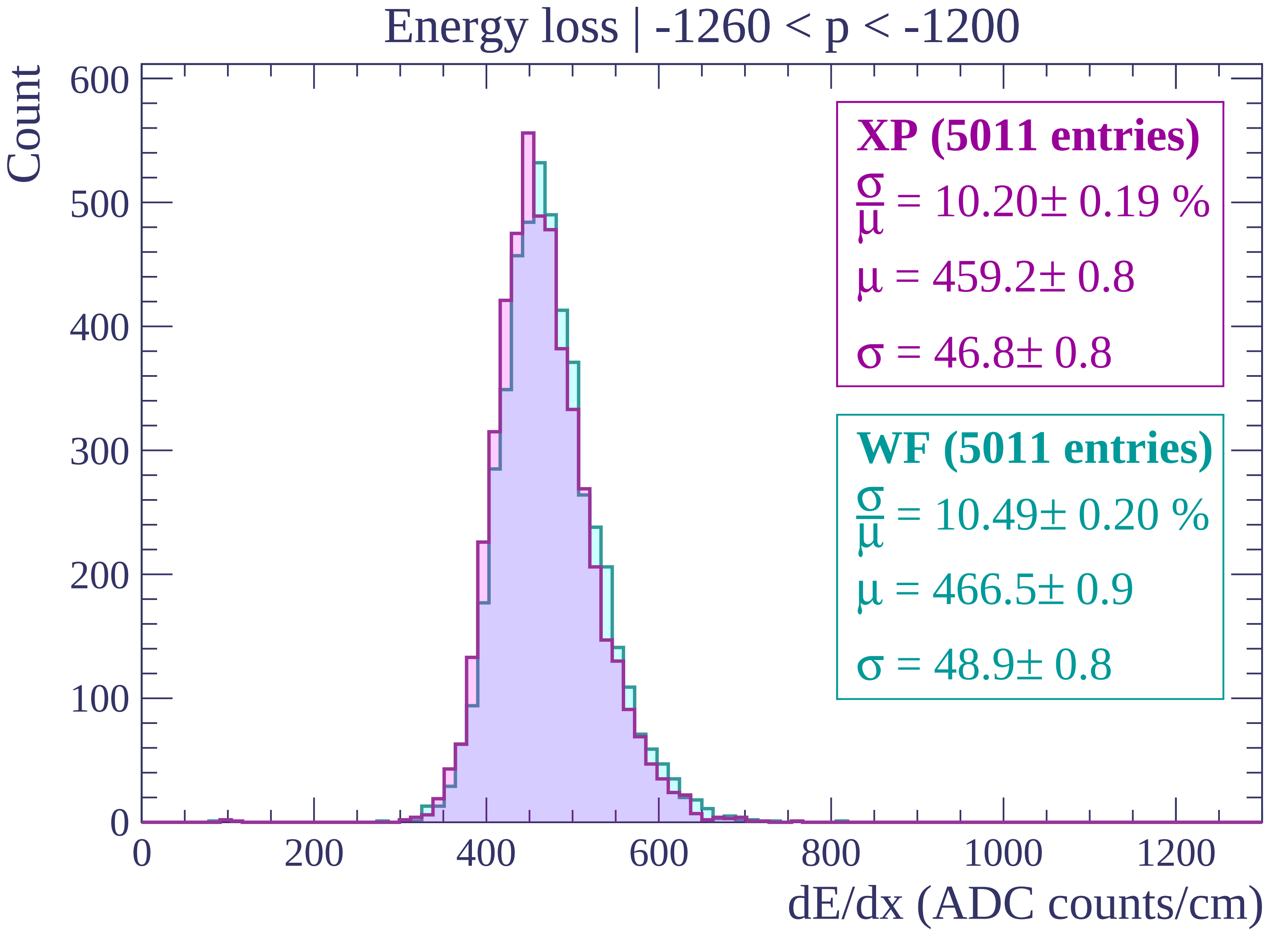


Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
XP (4528 entries)
σ/μ = 10.76± 0.20 %
μ = 469.7± 0.9
σ = 50.5± 0.8
WF (4528 entries)
σ/μ = 10.62± 0.18 %
μ = 478.1± 0.9
σ = 50.8± 0.8
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

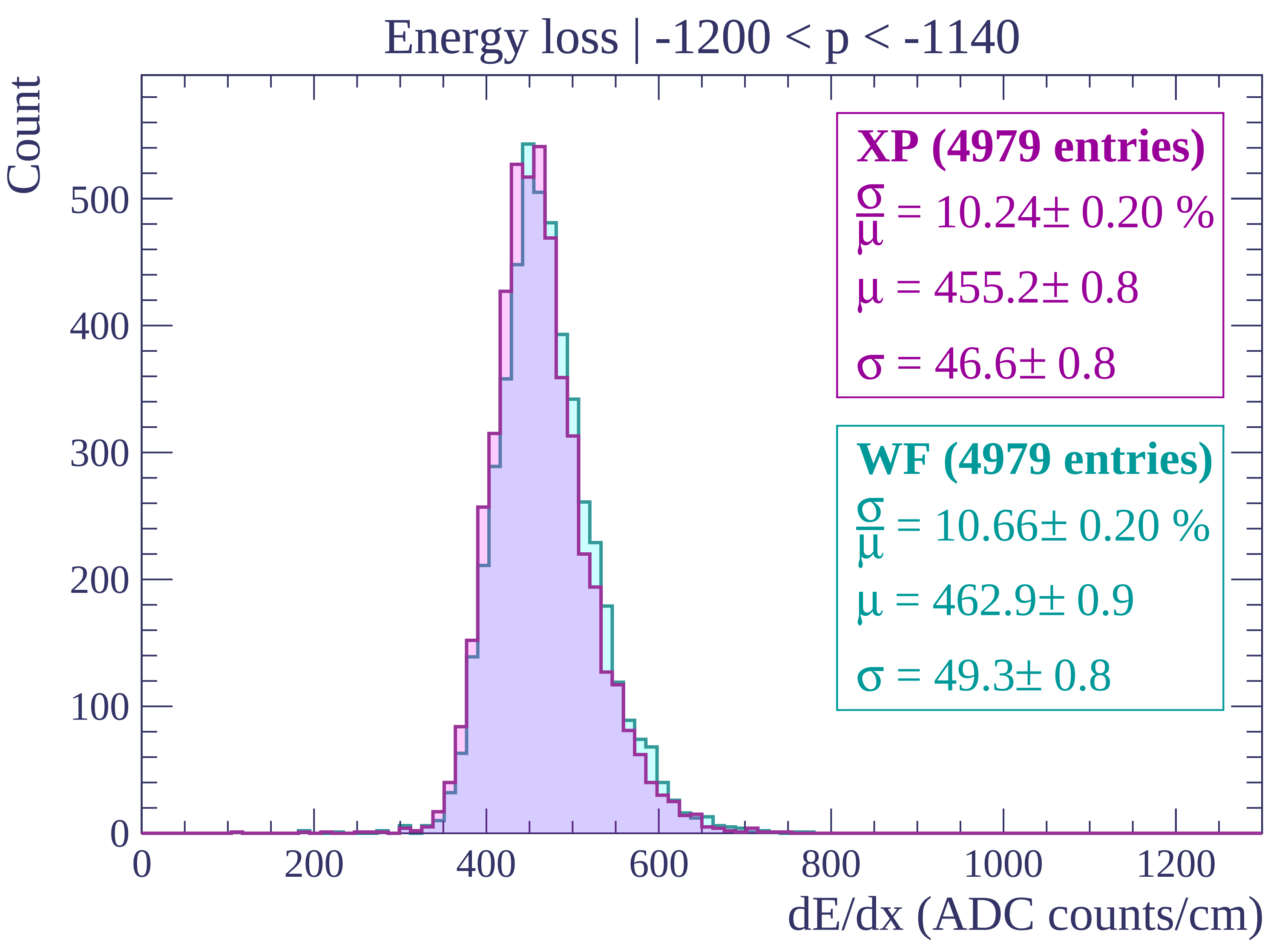


Energy loss | -1380 < p < -1320
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4565 entries)
σ/μ = 10.50± 0.21 %
μ = 465.3± 0.9
σ = 48.8± 0.9
WF (4565 entries)
σ/μ = 10.26± 0.22 %
μ = 471.2± 0.9
σ = 48.3± 0.9

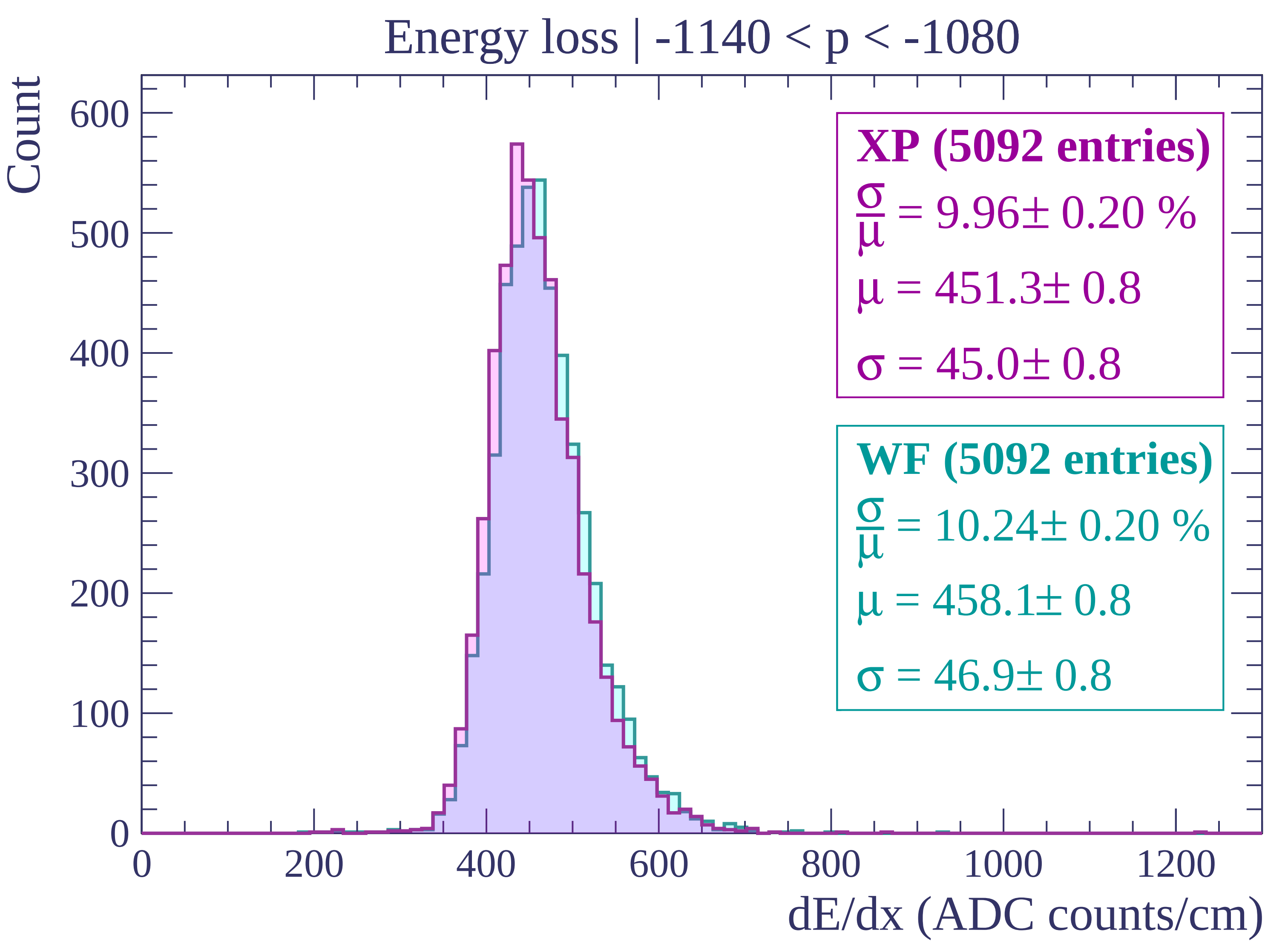

Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4760 entries)
σ/μ = 10.12± 0.21 %
μ = 462.6± 0.9
σ = 46.8± 0.9
WF (4760 entries)
σ/μ = 10.31± 0.22 %
μ = 469.4± 0.9
σ = 48.4± 1.0

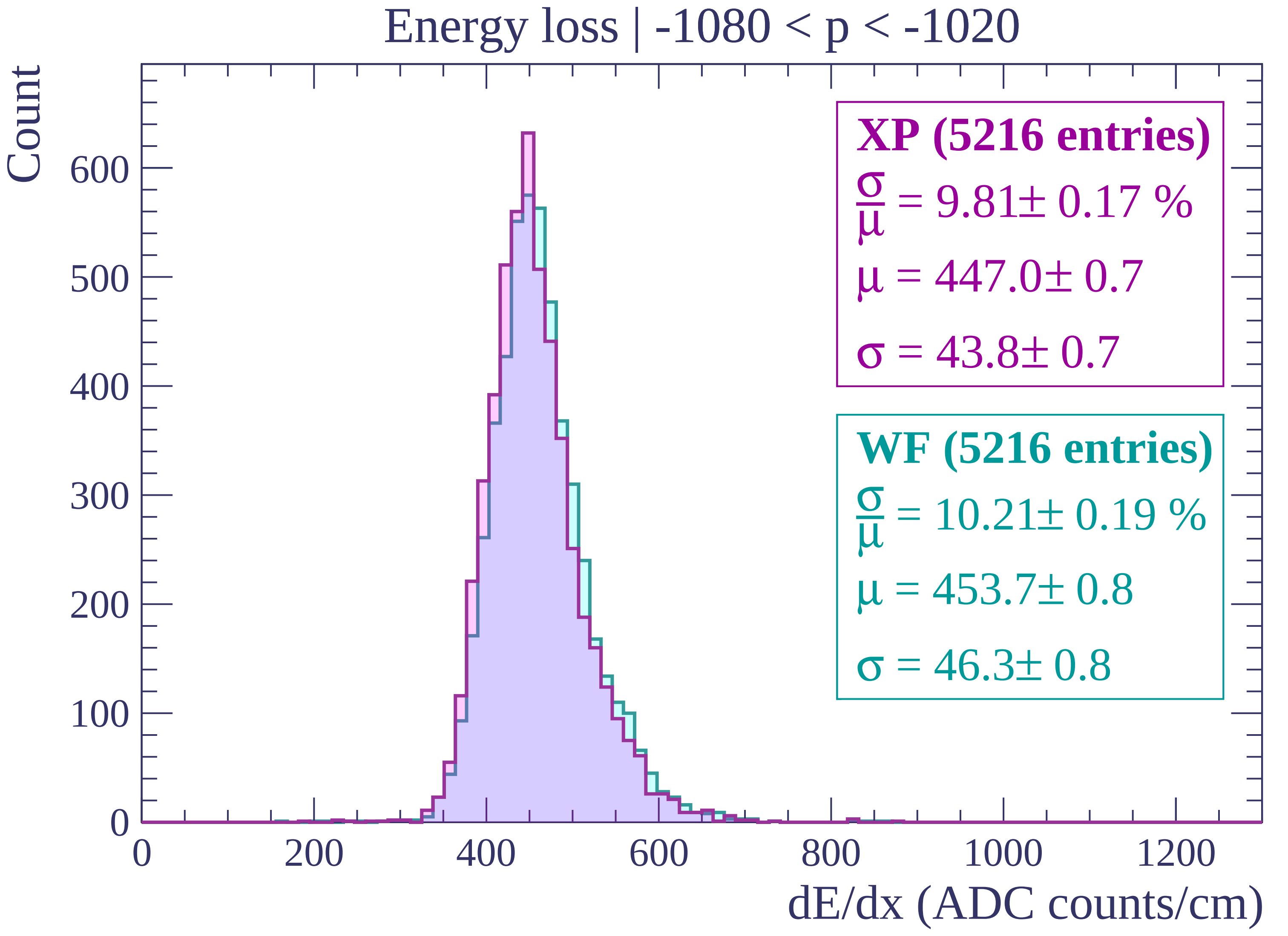


Energy loss | -1260 < p < -1200
Count
XP (5011 entries)
σ/μ = 10.20± 0.19 %
μ = 459.2± 0.8
σ = 46.8± 0.8
WF (5011 entries)
σ/μ = 10.49± 0.20 %
μ = 466.5± 0.9
σ = 48.9± 0.8
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

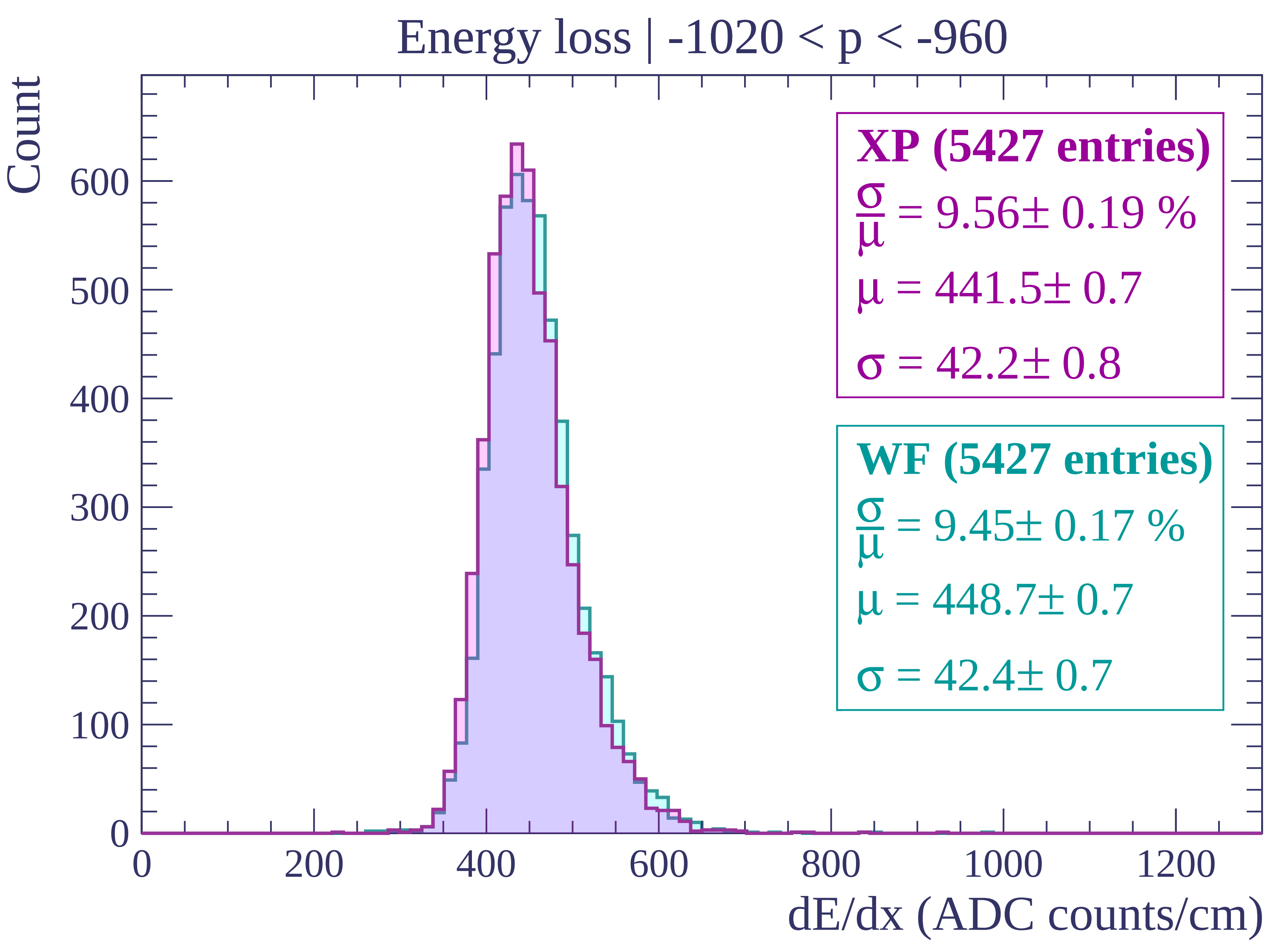


Energy loss | -1200 < p < -1140
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4979 entries)
$\frac{\sigma}{\mu}$ = 10.24± 0.20 %
μ = 455.2± 0.8
σ = 46.6± 0.8
WF (4979 entries)
$\frac{\sigma}{\mu}$ = 10.66± 0.20 %
μ = 462.9± 0.9
σ = 49.3± 0.8

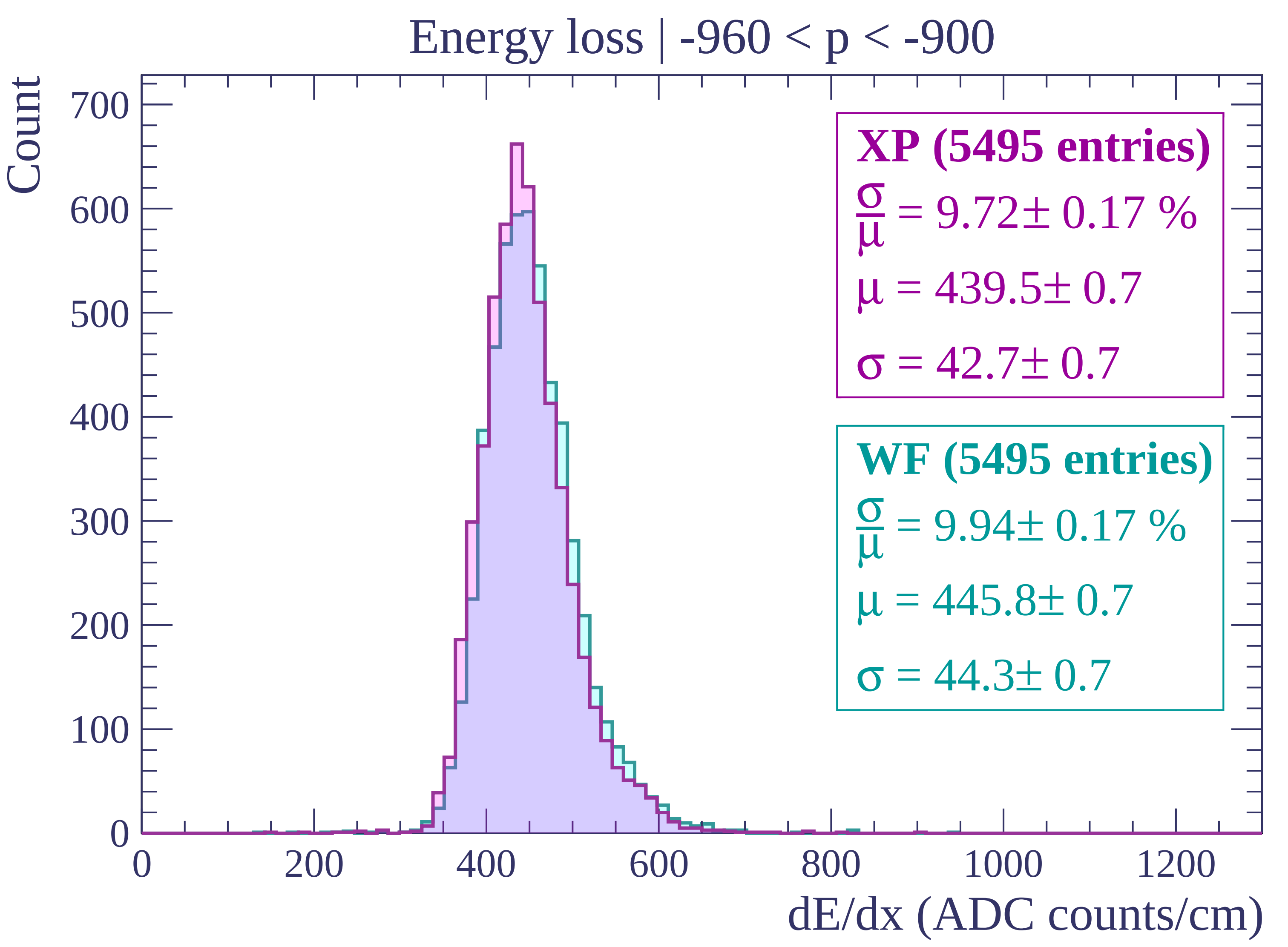

Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
XP (5092 entries)
$\frac{\sigma}{\mu}$ = 9.96± 0.20 %
μ = 451.3± 0.8
σ = 45.0± 0.8
WF (5092 entries)
$\frac{\sigma}{\mu}$ = 10.24± 0.20 %
μ = 458.1± 0.8
σ = 46.9± 0.8
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200



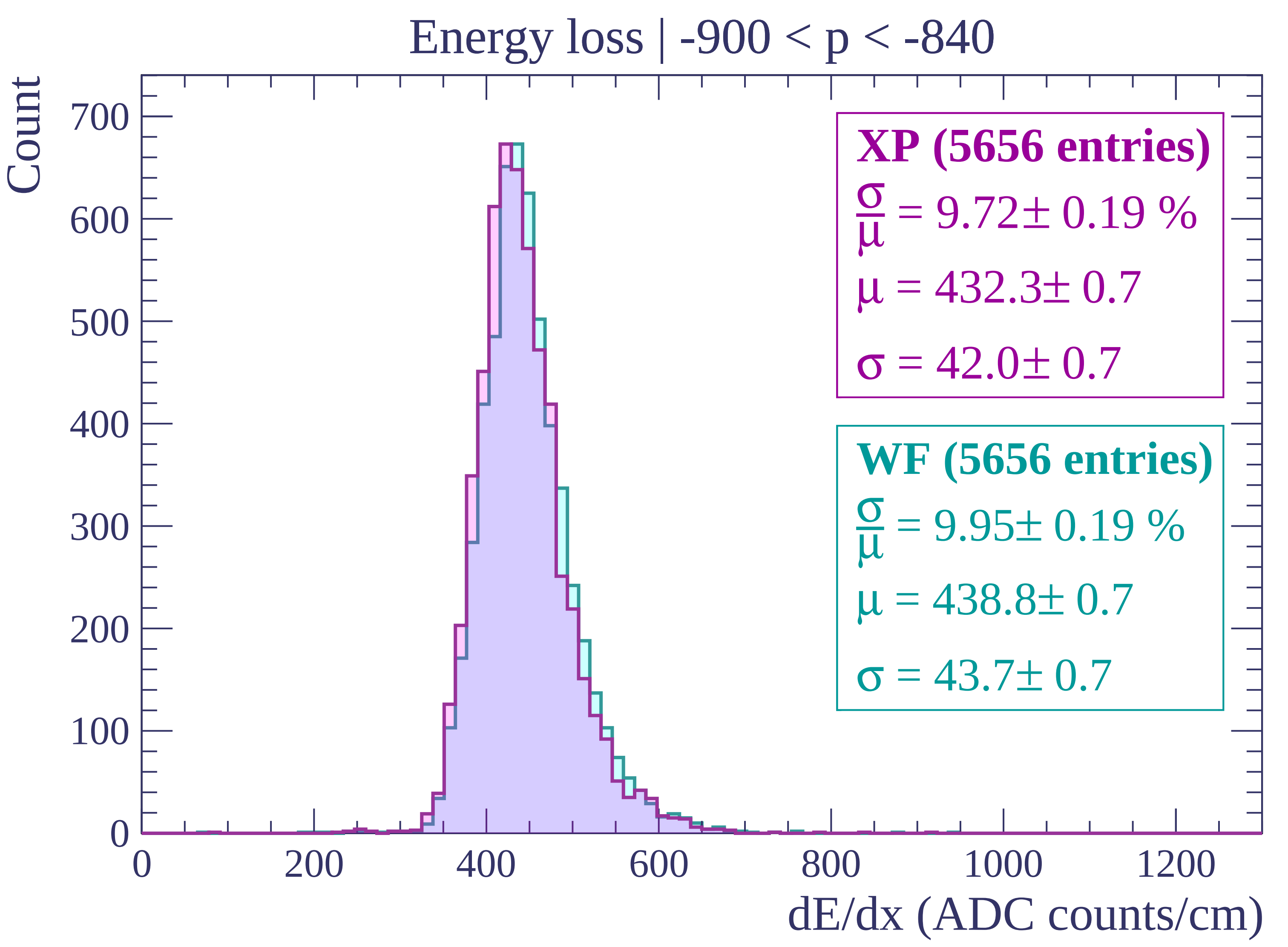
Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (5216 entries)
σ/μ = 9.81± 0.17 %
μ = 447.0± 0.7
σ = 43.8± 0.7
WF (5216 entries)
σ/μ = 10.21± 0.19 %
μ = 453.7± 0.8
σ = 46.3± 0.8



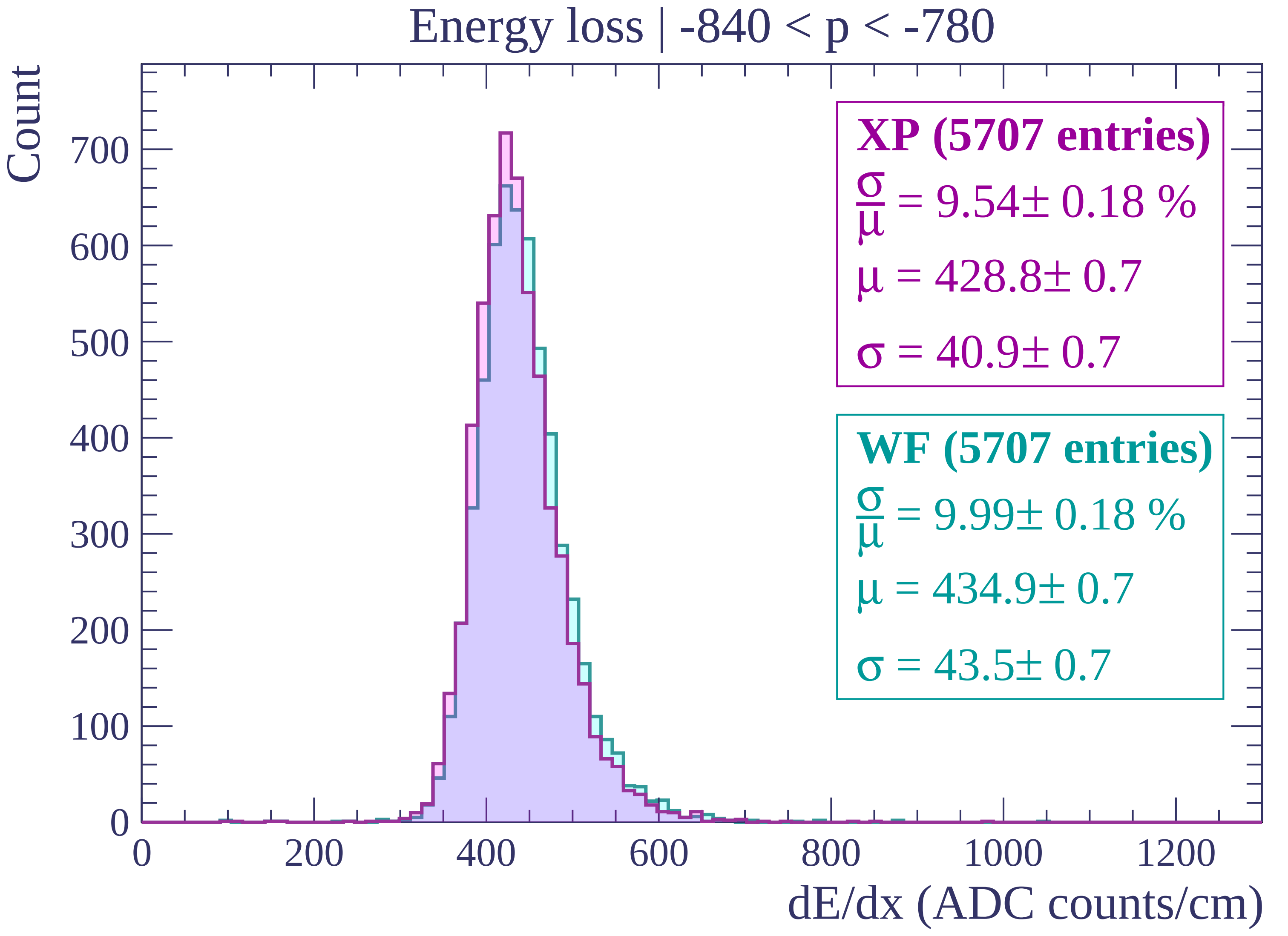
Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
XP (5427 entries)
σ/μ = 9.56± 0.19 %
μ = 441.5± 0.7
σ = 42.2± 0.8
WF (5427 entries)
σ/μ = 9.45± 0.17 %
μ = 448.7± 0.7
σ = 42.4± 0.7



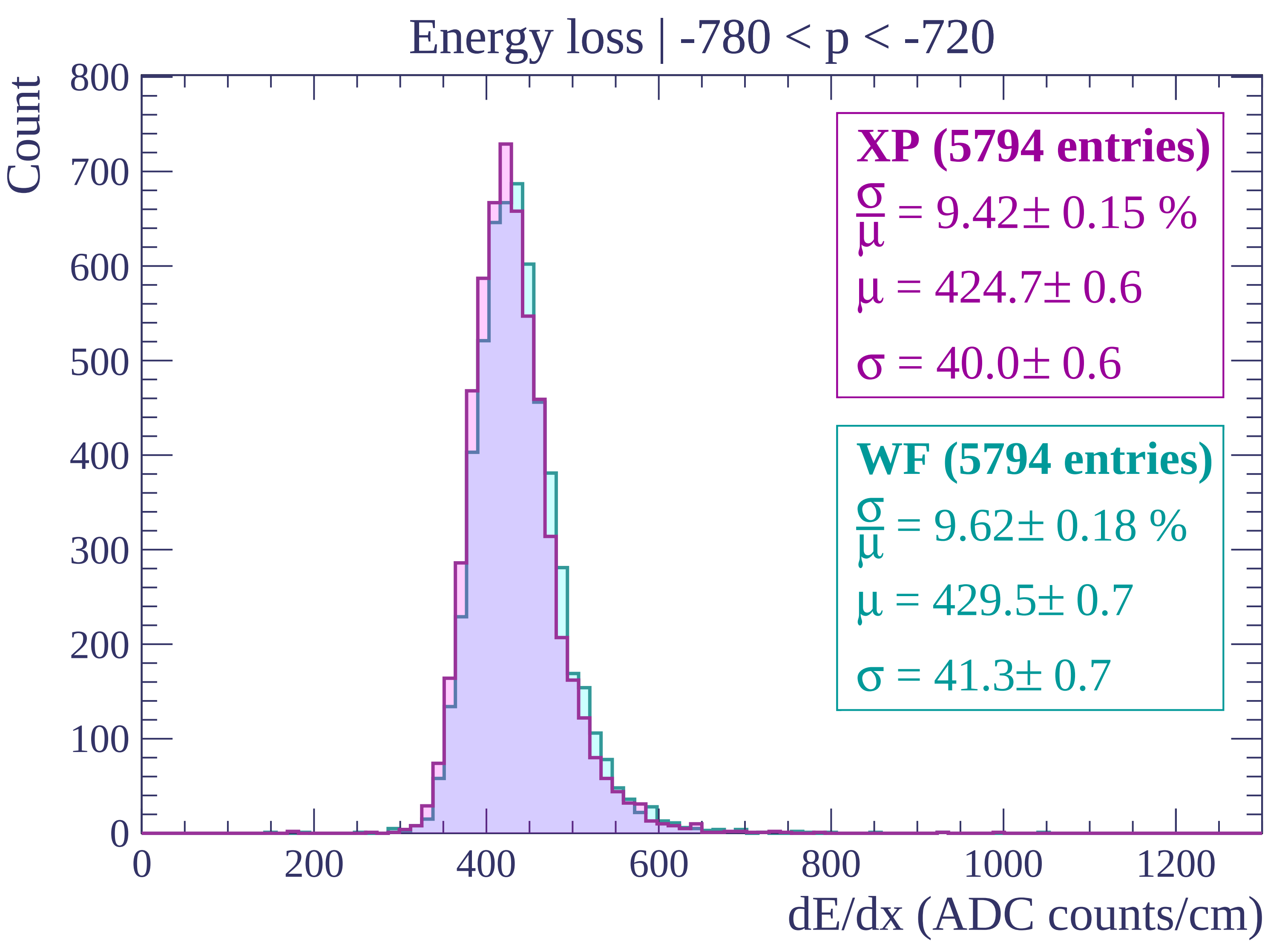
Energy loss | -960 < p < -900
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5495 entries)
σ/μ = 9.72± 0.17 %
μ = 439.5± 0.7
σ = 42.7± 0.7
WF (5495 entries)
σ/μ = 9.94± 0.17 %
μ = 445.8± 0.7
σ = 44.3± 0.7



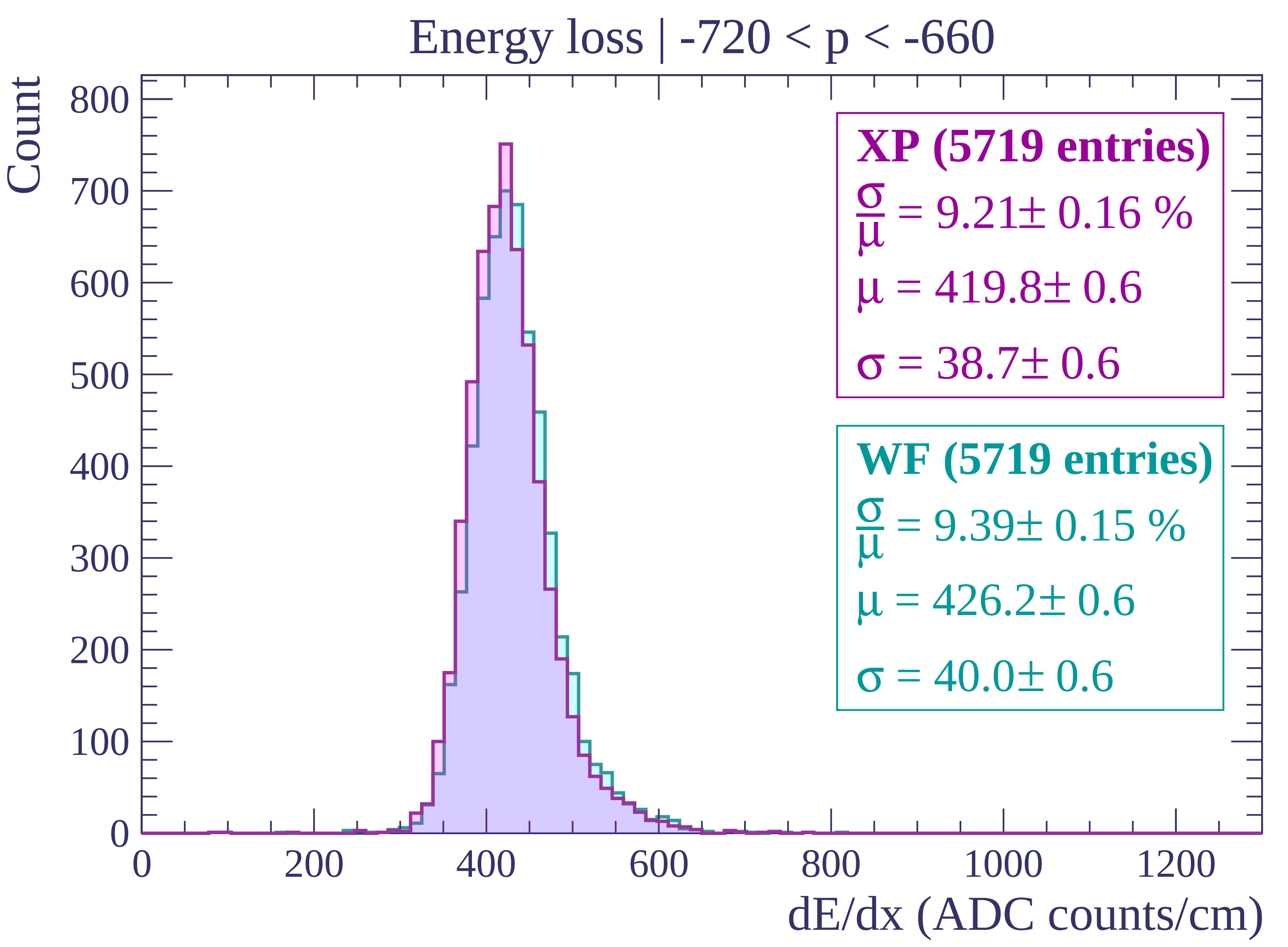
Energy loss | -900 < p < -840
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5656 entries)
$\frac{\sigma}{\mu}$ = 9.72± 0.19 %
μ = 432.3± 0.7
σ = 42.0± 0.7
WF (5656 entries)
$\frac{\sigma}{\mu}$ = 9.95± 0.19 %
μ = 438.8± 0.7
σ = 43.7± 0.7



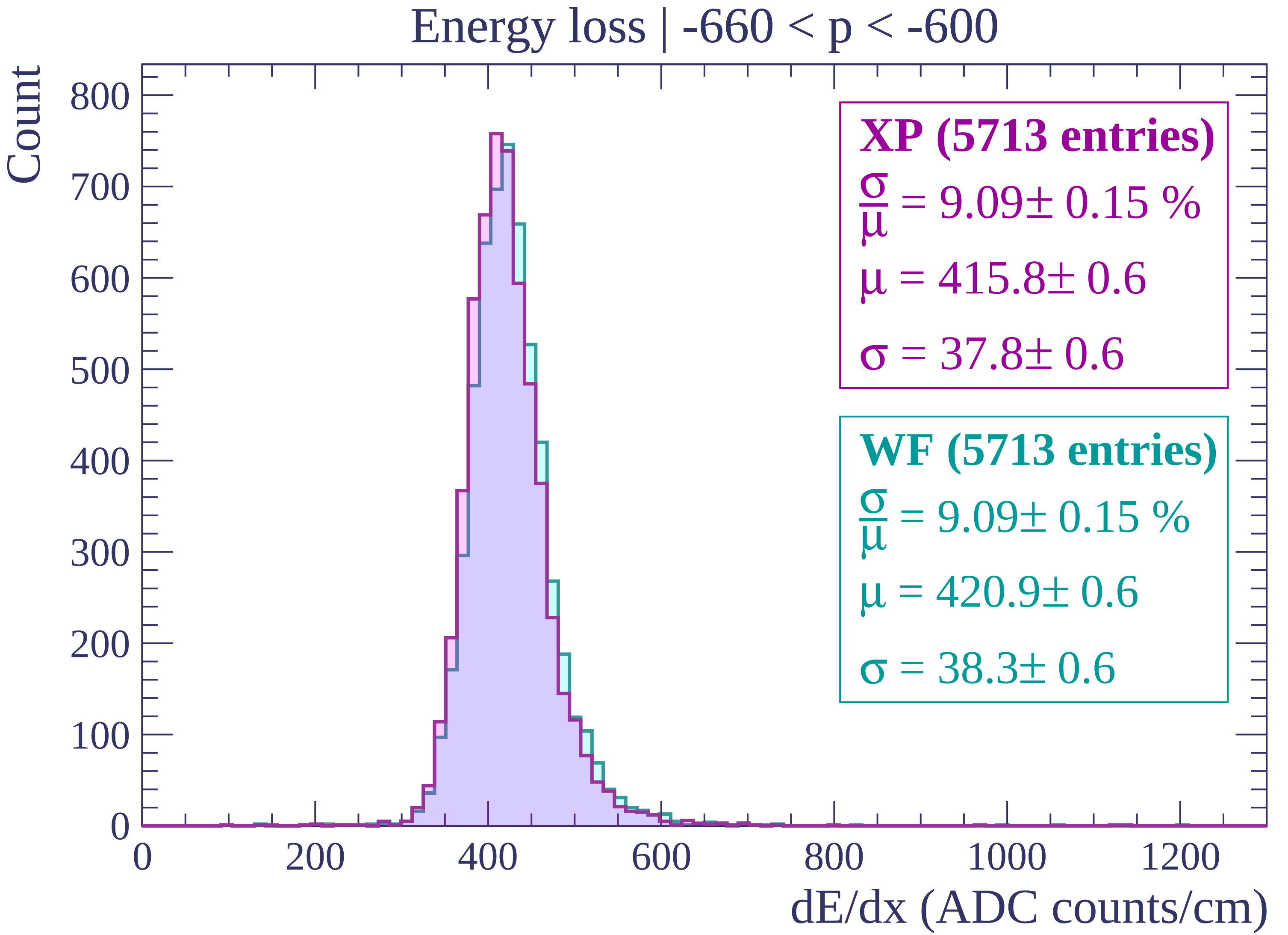
Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5707 entries)
$\frac{\sigma}{\mu}$ = 9.54± 0.18 %
μ = 428.8± 0.7
σ = 40.9± 0.7
WF (5707 entries)
$\frac{\sigma}{\mu}$ = 9.99± 0.18 %
μ = 434.9± 0.7
σ = 43.5± 0.7

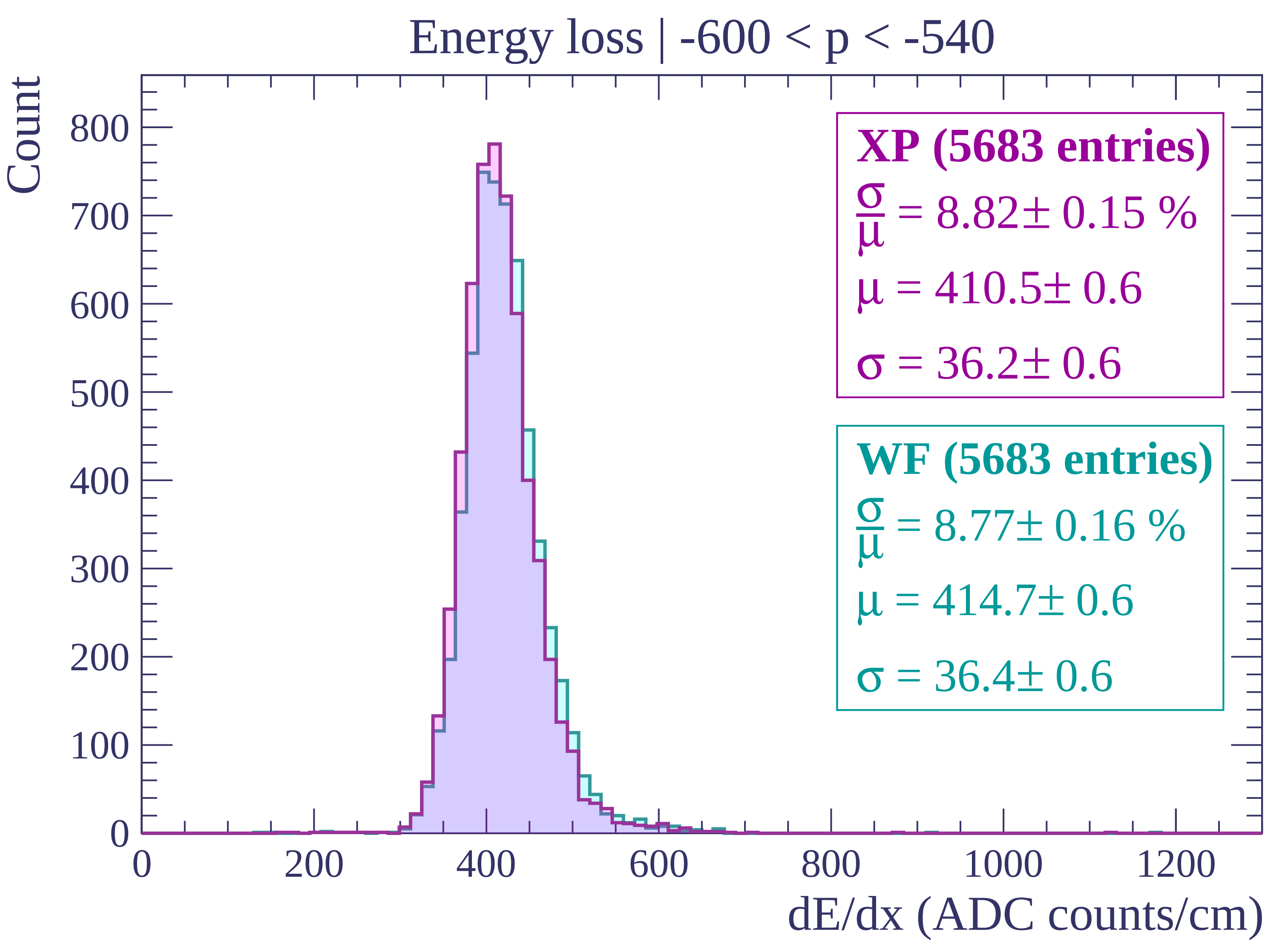


Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
XP (5794 entries)
$\frac{\sigma}{\mu}$ = 9.42± 0.15 %
μ = 424.7± 0.6
σ = 40.0± 0.6
WF (5794 entries)
$\frac{\sigma}{\mu}$ = 9.62± 0.18 %
μ = 429.5± 0.7
σ = 41.3± 0.7

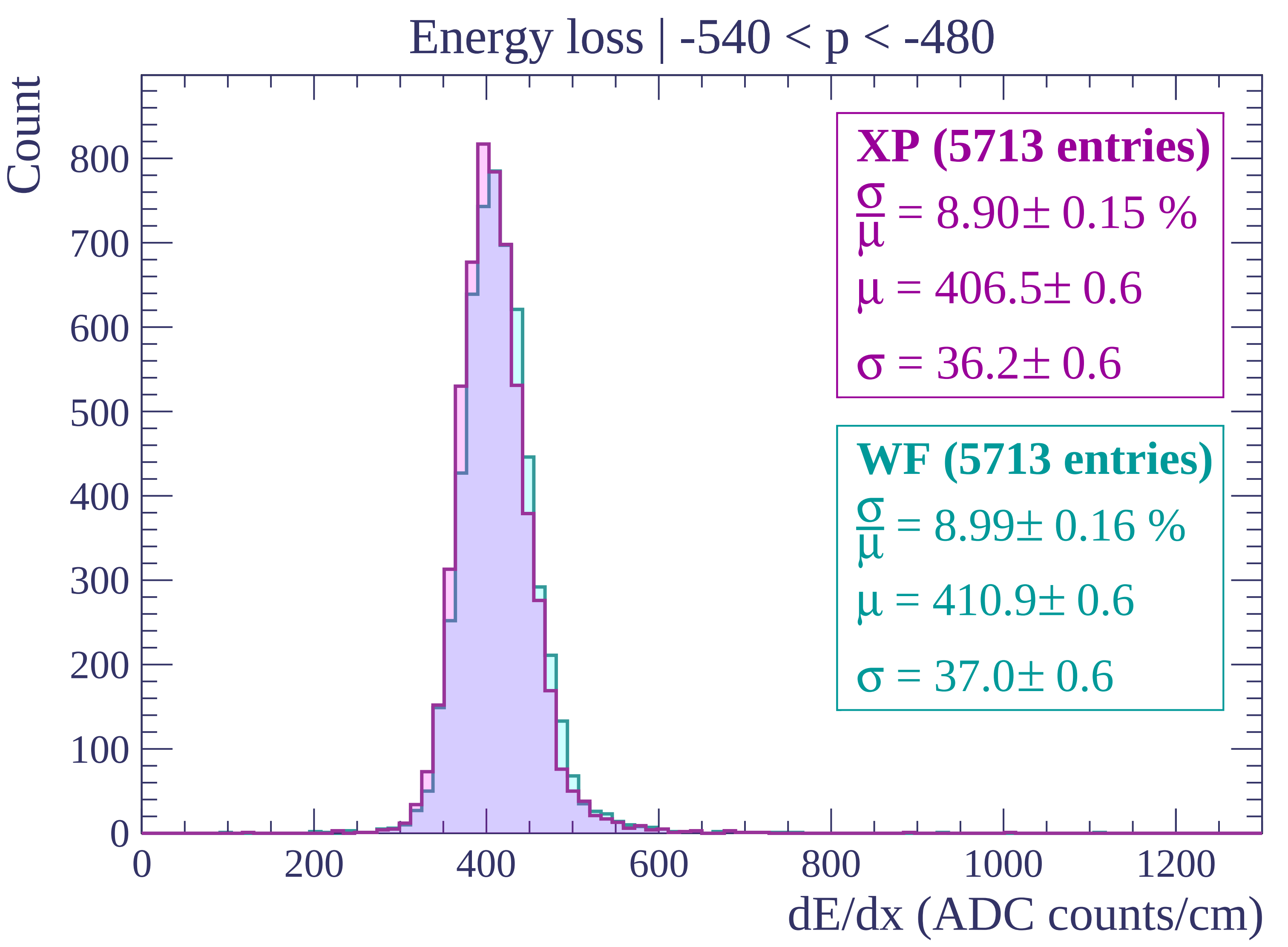


Energy loss | -720 < p < -660
Count
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5719 entries)
$\frac{\sigma}{\mu}$ = 9.21± 0.16 %
μ = 419.8± 0.6
σ = 38.7± 0.6
WF (5719 entries)
$\frac{\sigma}{\mu}$ = 9.39± 0.15 %
μ = 426.2± 0.6
σ = 40.0± 0.6

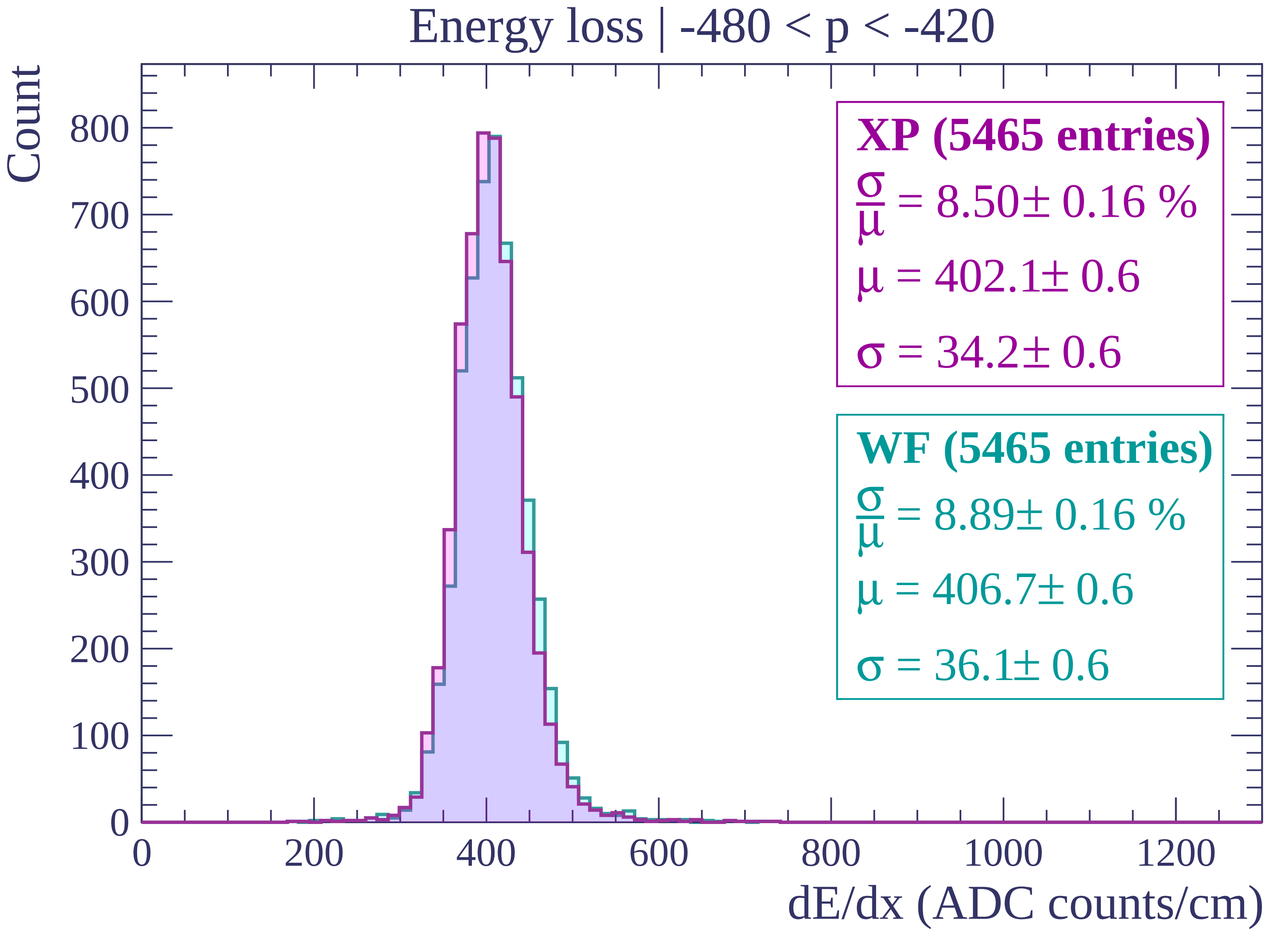

Energy loss | -660 < p < -600
Count
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5713 entries)
$\frac{\sigma}{\mu}$ = 9.09± 0.15 %
μ = 415.8± 0.6
σ = 37.8± 0.6
WF (5713 entries)
$\frac{\sigma}{\mu}$ = 9.09± 0.15 %
μ = 420.9± 0.6
σ = 38.3± 0.6



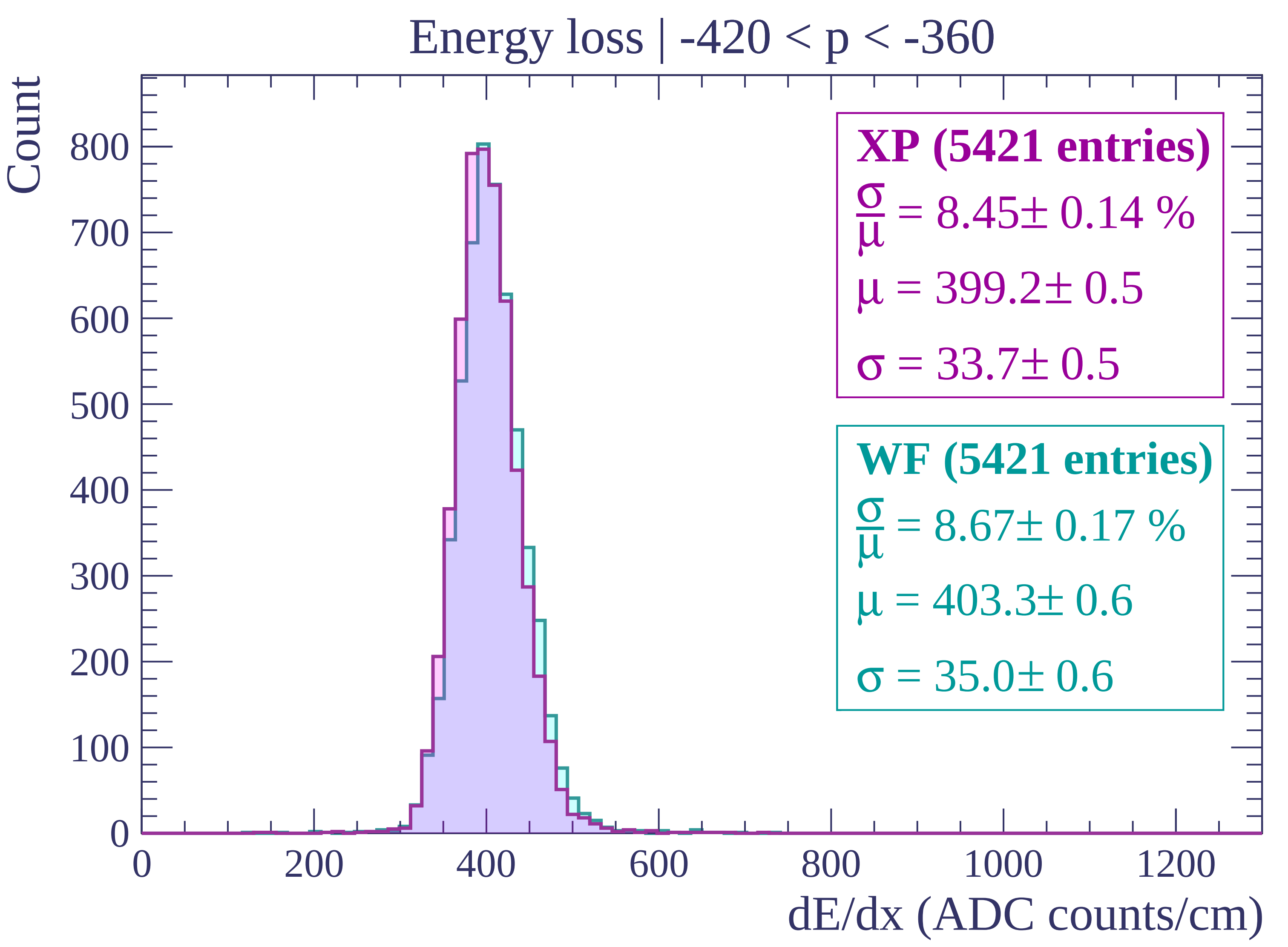
Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (5683 entries)
σ/μ = 8.82± 0.15 %
μ = 410.5± 0.6
σ = 36.2± 0.6
WF (5683 entries)
σ/μ = 8.77± 0.16 %
μ = 414.7± 0.6
σ = 36.4± 0.6



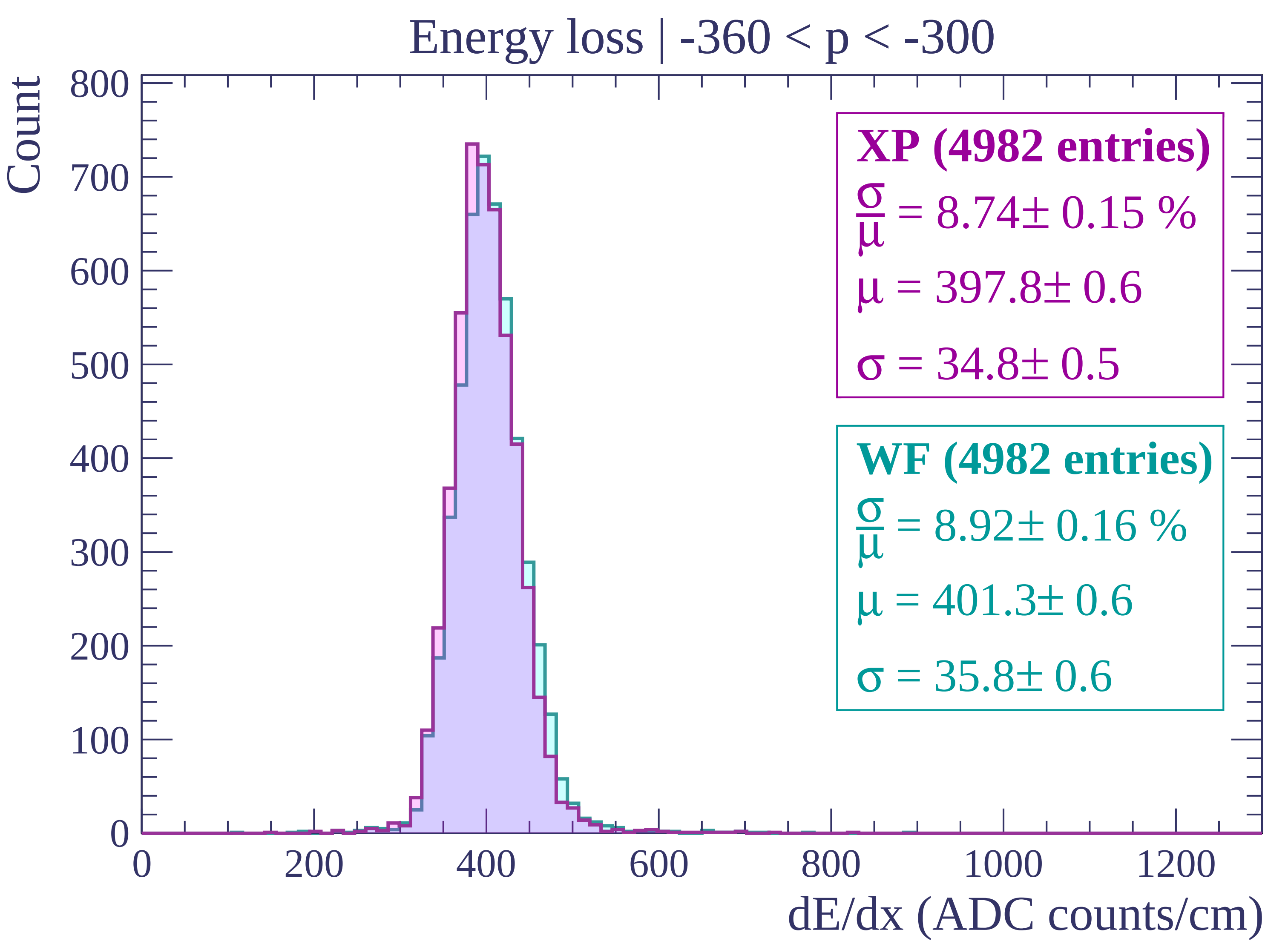
Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
XP (5713 entries)
σ/μ = 8.90± 0.15 %
μ = 406.5± 0.6
σ = 36.2± 0.6
WF (5713 entries)
σ/μ = 8.99± 0.16 %
μ = 410.9± 0.6
σ = 37.0± 0.6



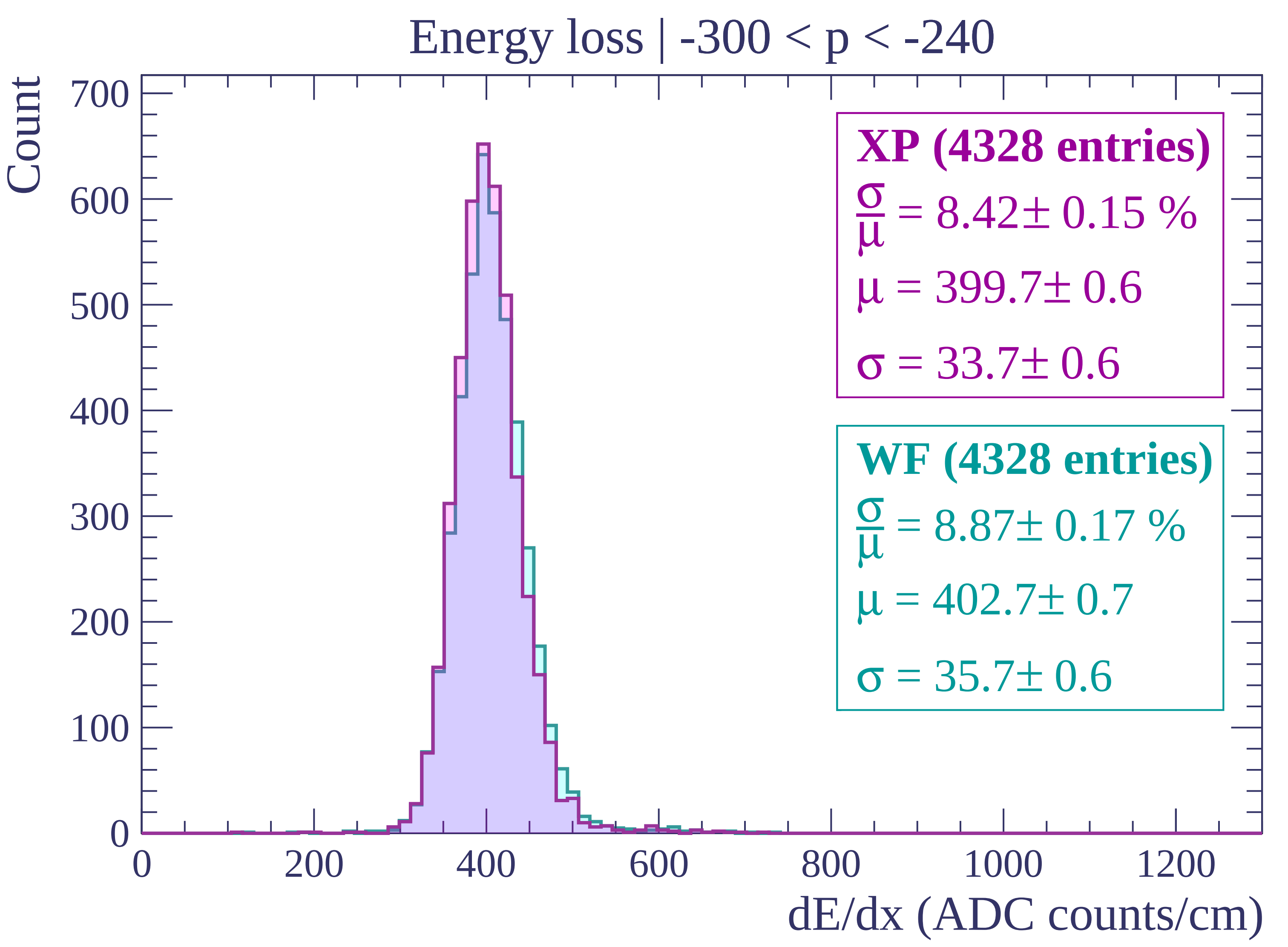
Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
XP (5465 entries)
$\frac{\sigma}{\mu}$ = 8.50± 0.16 %
μ = 402.1± 0.6
σ = 34.2± 0.6
WF (5465 entries)
$\frac{\sigma}{\mu}$ = 8.89± 0.16 %
μ = 406.7± 0.6
σ = 36.1± 0.6



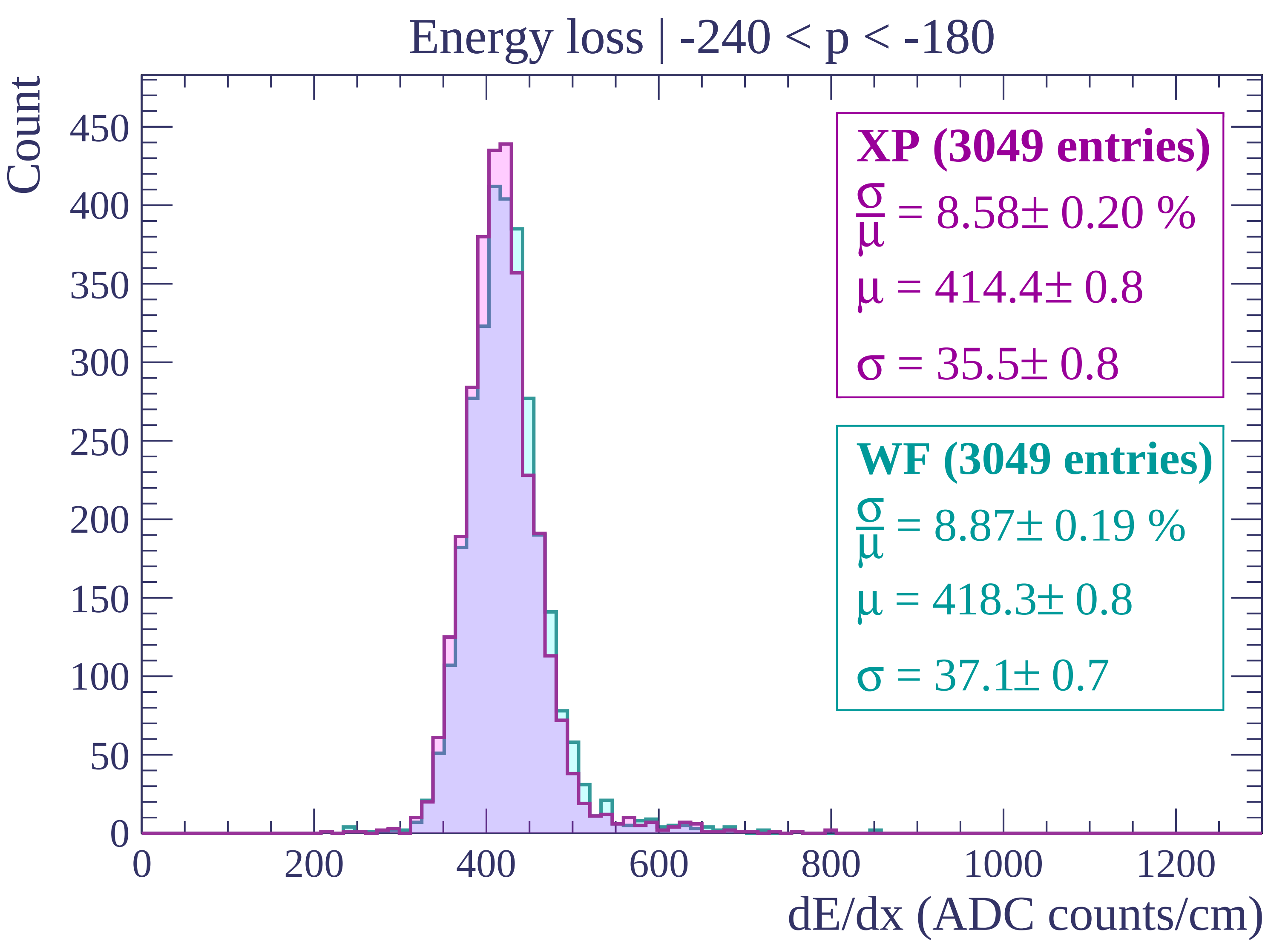
Energy loss | -420 < p < -360
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5421 entries)
σ/μ = 8.45± 0.14 %
μ = 399.2± 0.5
σ = 33.7± 0.5
WF (5421 entries)
σ/μ = 8.67± 0.17 %
μ = 403.3± 0.6
σ = 35.0± 0.6

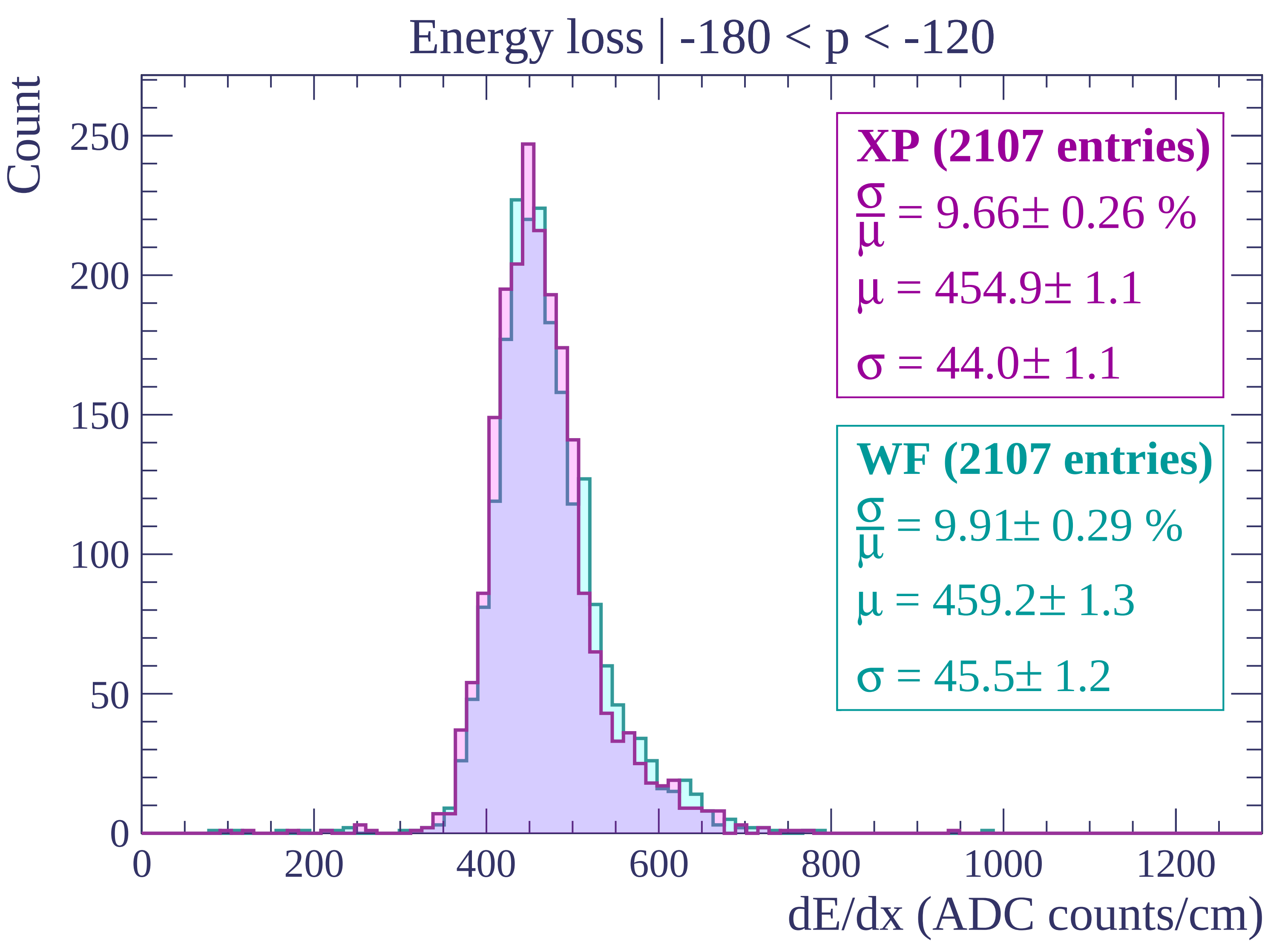


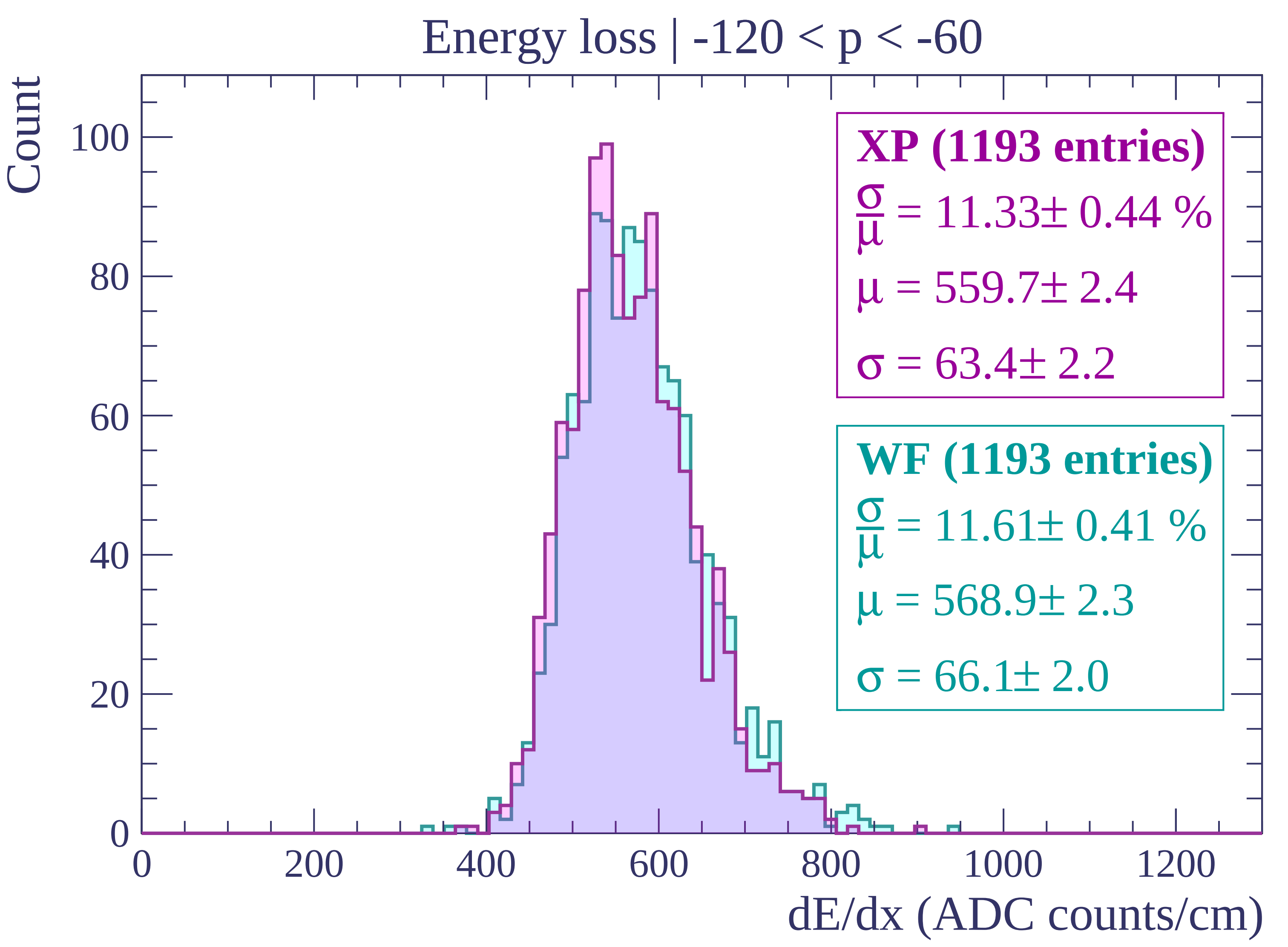
Energy loss | -360 < p < -300
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4982 entries)
$\frac{\sigma}{\mu}$ = 8.74± 0.15 %
μ = 397.8± 0.6
σ = 34.8± 0.5
WF (4982 entries)
$\frac{\sigma}{\mu}$ = 8.92± 0.16 %
μ = 401.3± 0.6
σ = 35.8± 0.6



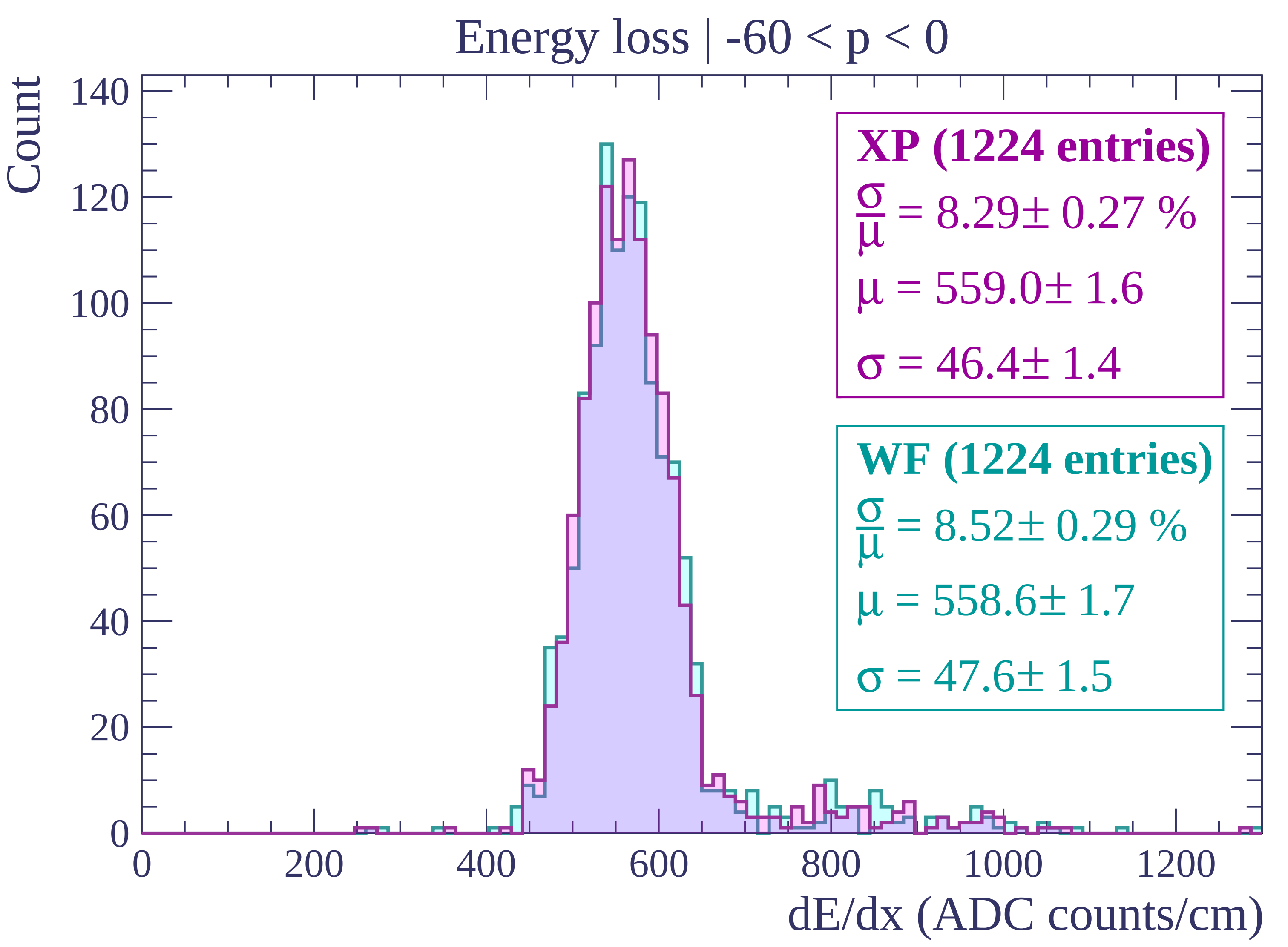
Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (4328 entries)
$\frac{\sigma}{\mu}$ = 8.42± 0.15 %
μ = 399.7± 0.6
σ = 33.7± 0.6
WF (4328 entries)
$\frac{\sigma}{\mu}$ = 8.87± 0.17 %
μ = 402.7± 0.7
σ = 35.7± 0.6


Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
XP (3049 entries)
σ/μ = 8.58± 0.20 %
μ = 414.4± 0.8
σ = 35.5± 0.8
WF (3049 entries)
σ/μ = 8.87± 0.19 %
μ = 418.3± 0.8
σ = 37.1± 0.7
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

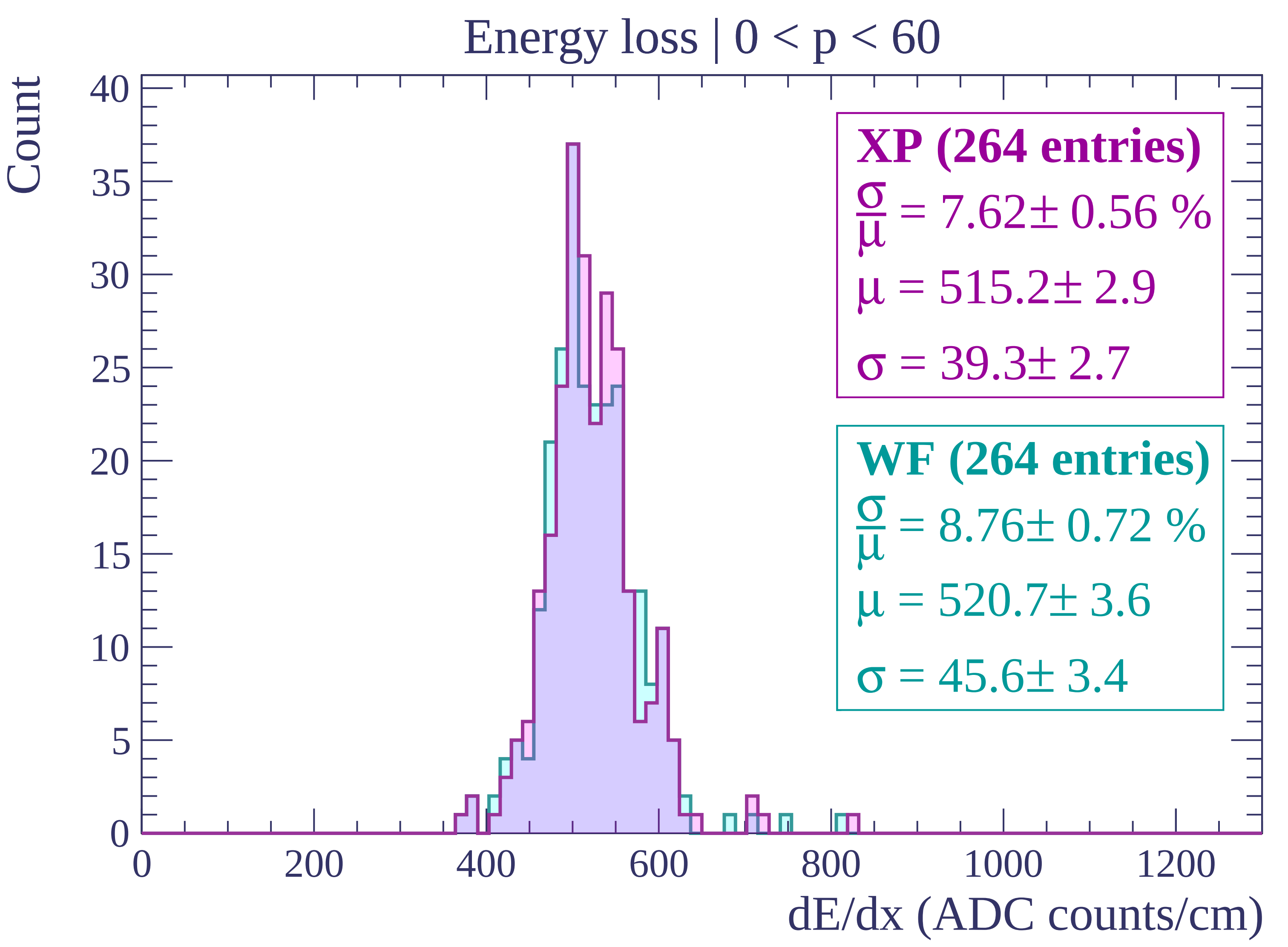

Energy loss | -180 < p < -120
Count
dE/dx (ADC counts/cm)
XP (2107 entries)
σ/μ = 9.66± 0.26 %
μ = 454.9± 1.1
σ = 44.0± 1.1
WF (2107 entries)
σ/μ = 9.91± 0.29 %
μ = 459.2± 1.3
σ = 45.5± 1.2
0
50
100
150
200
250
0
200
400
600
800
1000
1200



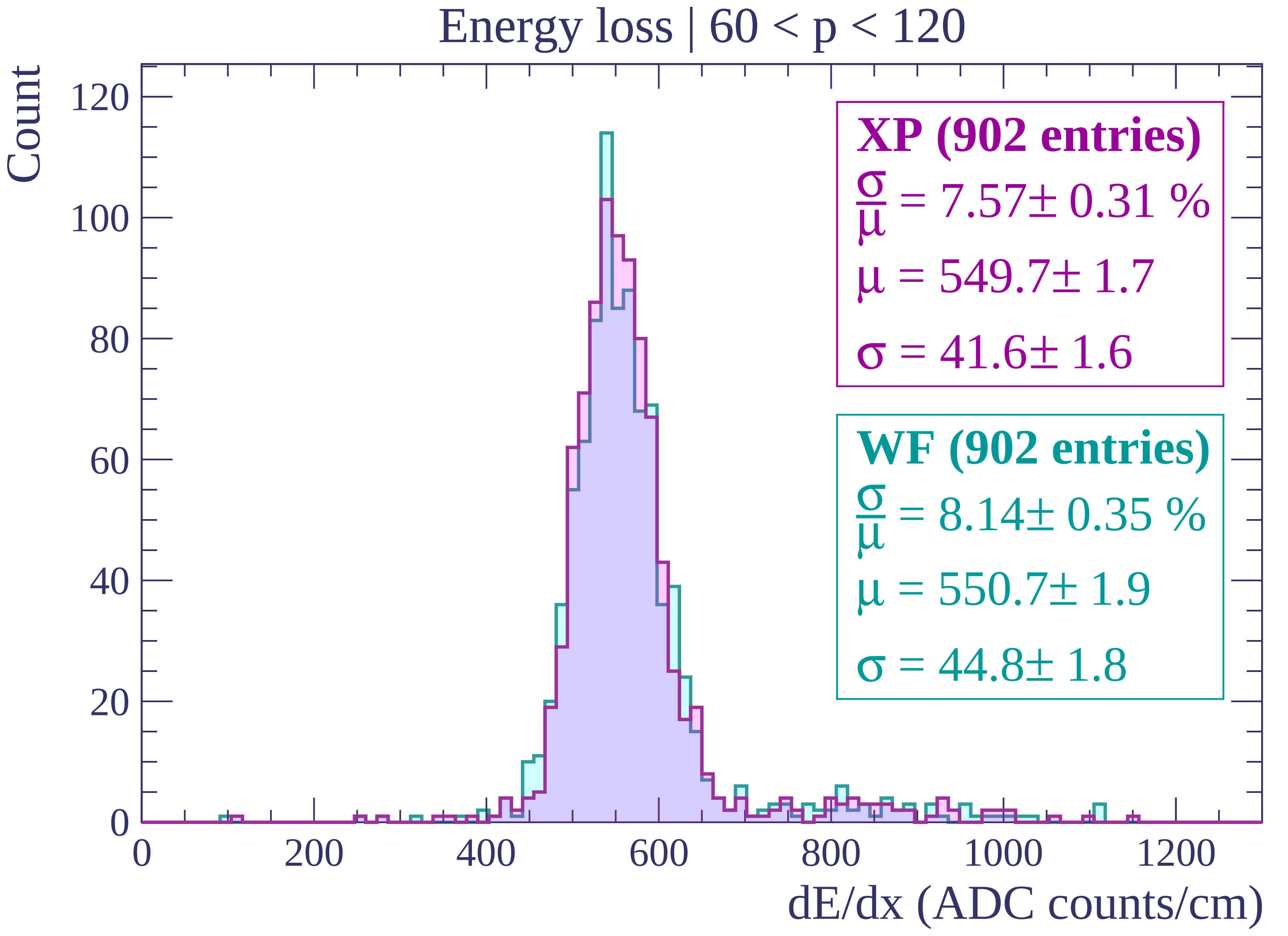
Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1193 entries)
σ/μ = 11.33± 0.44 %
μ = 559.7± 2.4
σ = 63.4± 2.2
WF (1193 entries)
σ/μ = 11.61± 0.41 %
μ = 568.9± 2.3
σ = 66.1± 2.0

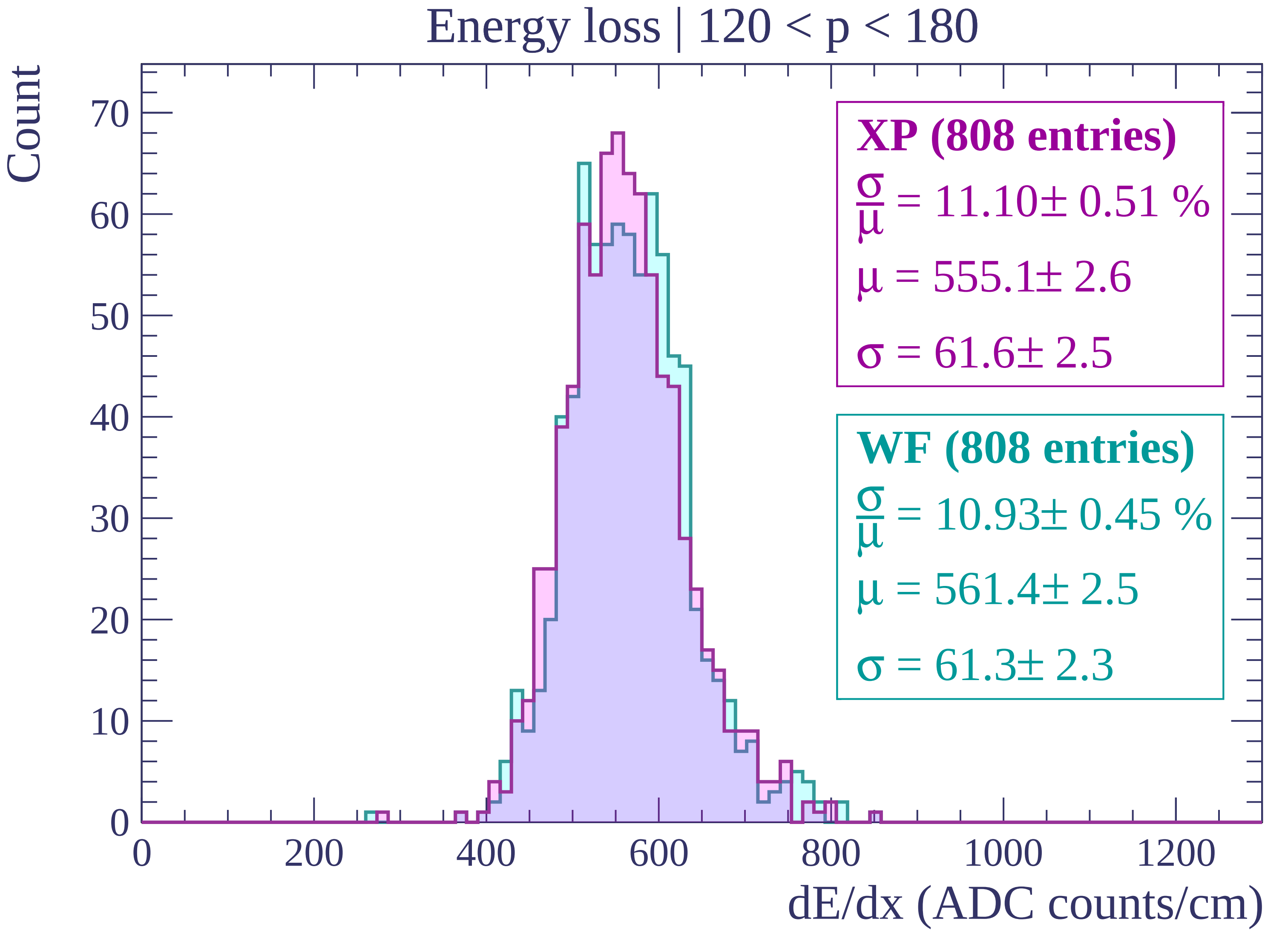


Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1224 entries)
$\frac{\sigma}{\mu}$ = 8.29± 0.27 %
μ = 559.0± 1.6
σ = 46.4± 1.4
WF (1224 entries)
$\frac{\sigma}{\mu}$ = 8.52± 0.29 %
μ = 558.6± 1.7
σ = 47.6± 1.5

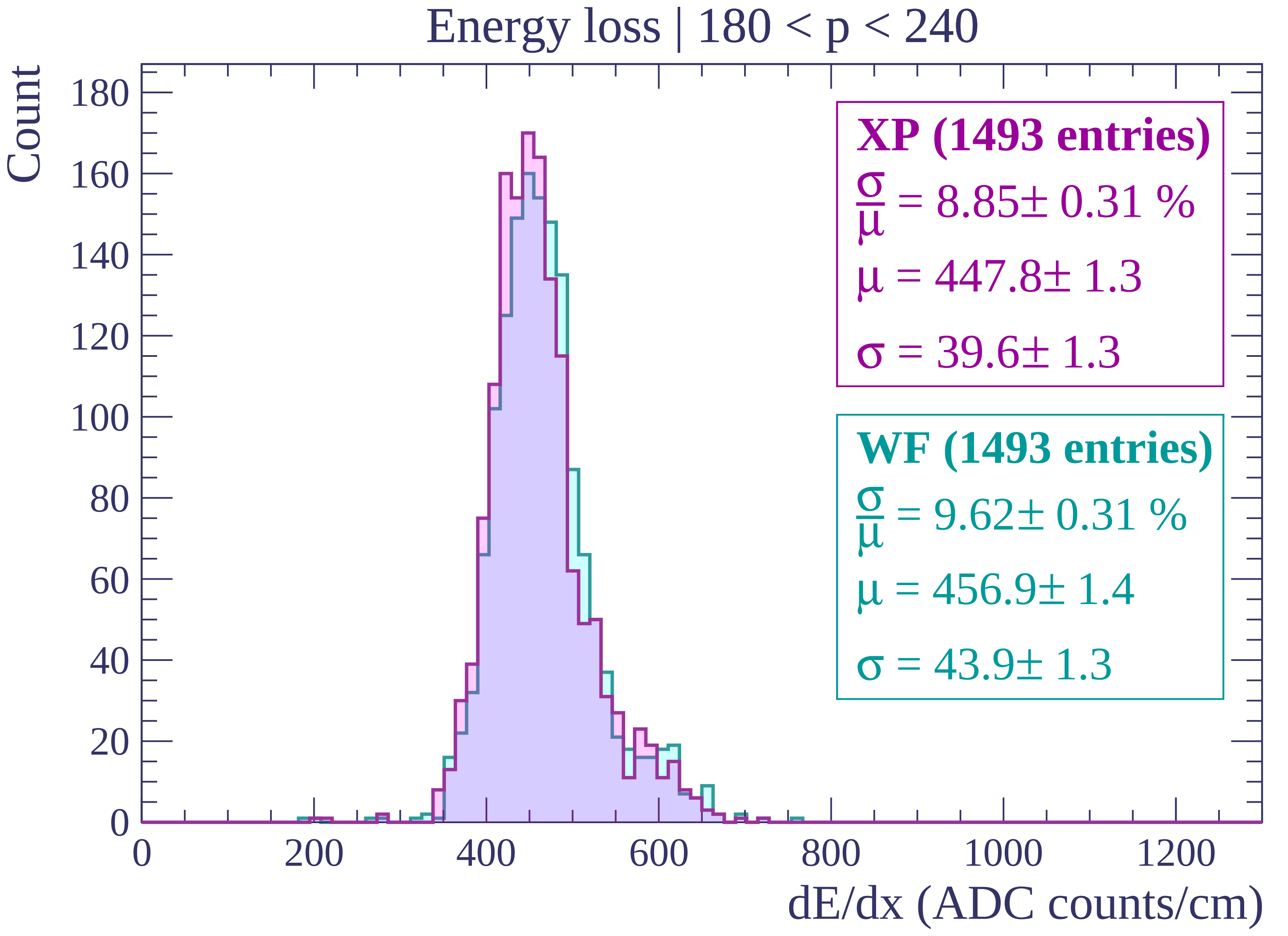


Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
XP (264 entries)
σ/μ = 7.62± 0.56 %
μ = 515.2± 2.9
σ = 39.3± 2.7
WF (264 entries)
σ/μ = 8.76± 0.72 %
μ = 520.7± 3.6
σ = 45.6± 3.4

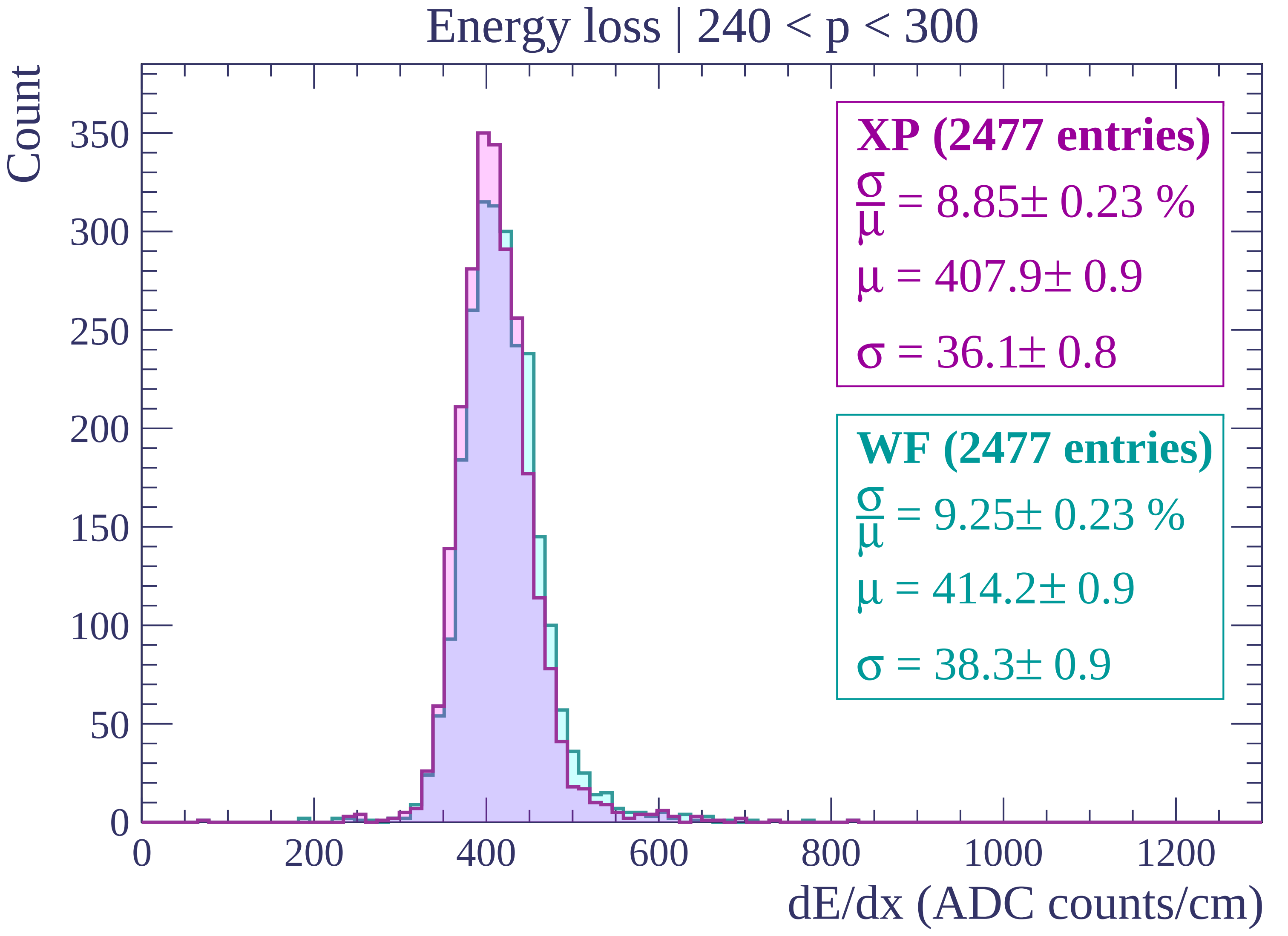

Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
XP (902 entries)
$\frac{\sigma}{\mu}$ = 7.57± 0.31 %
μ = 549.7± 1.7
σ = 41.6± 1.6
WF (902 entries)
$\frac{\sigma}{\mu}$ = 8.14± 0.35 %
μ = 550.7± 1.9
σ = 44.8± 1.8
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

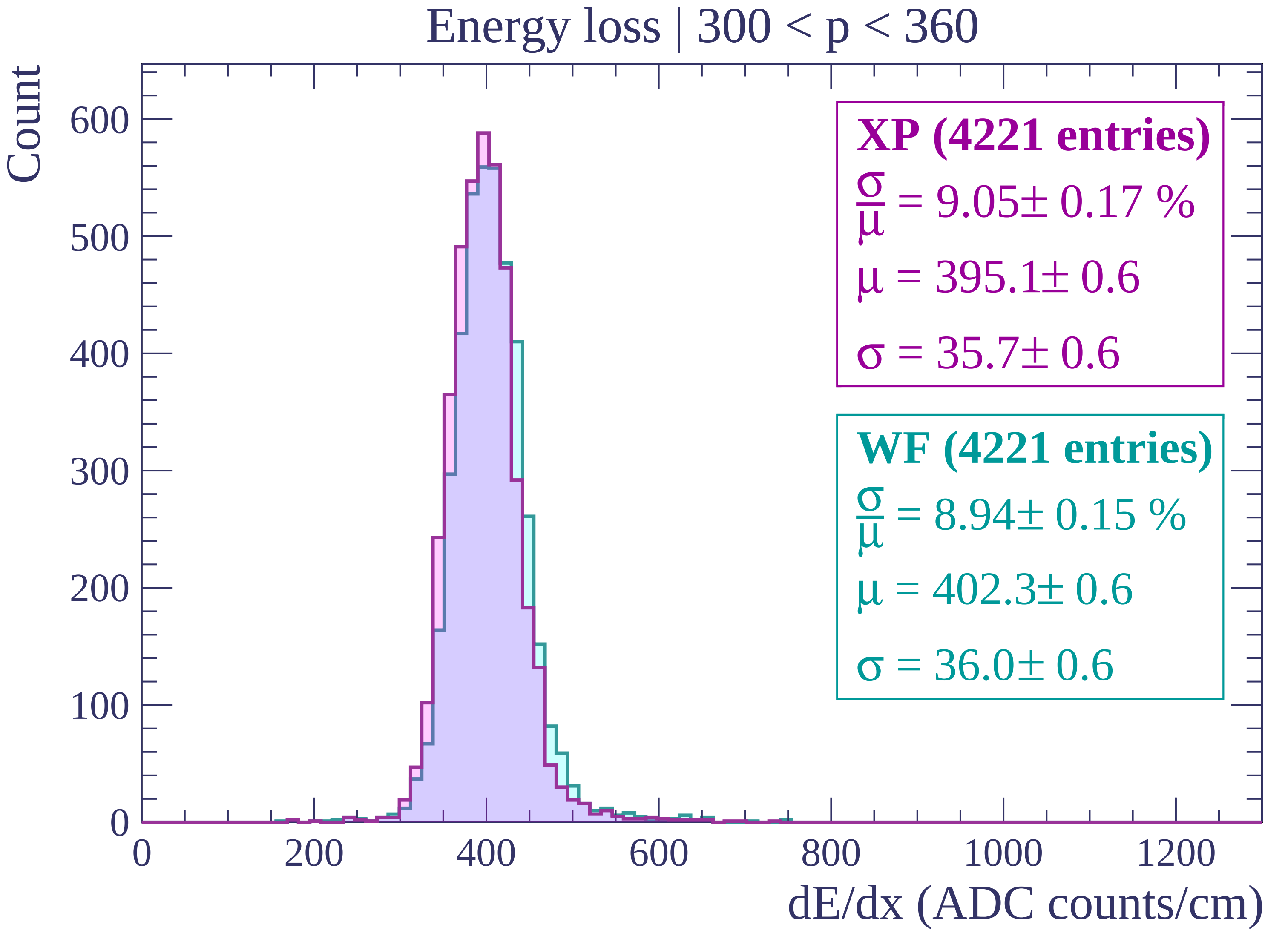


Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (808 entries)
$\frac{\sigma}{\mu}$ = 11.10± 0.51 %
μ = 555.1± 2.6
σ = 61.6± 2.5
WF (808 entries)
$\frac{\sigma}{\mu}$ = 10.93± 0.45 %
μ = 561.4± 2.5
σ = 61.3± 2.3

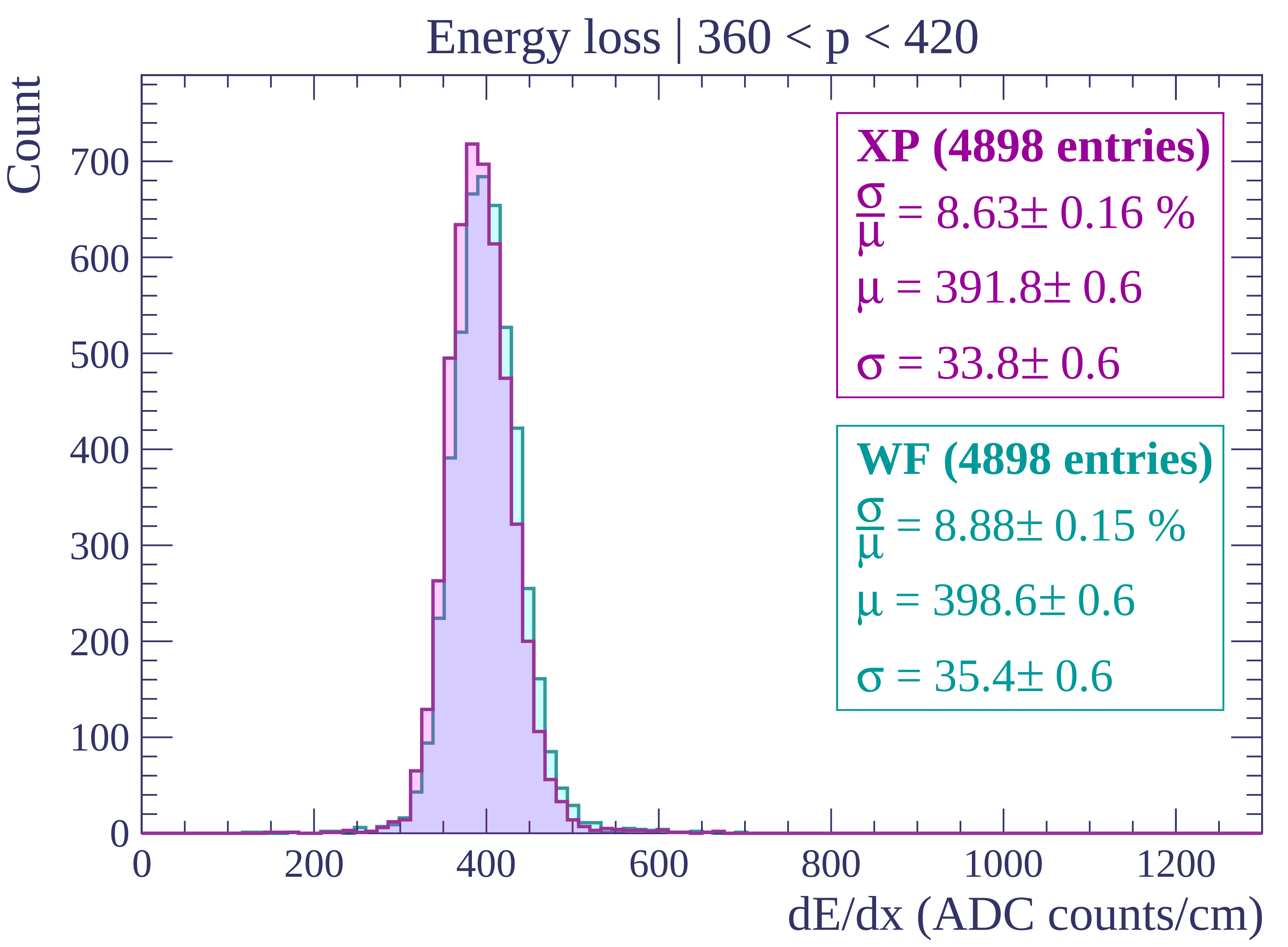


Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1493 entries)
σ/μ = 8.85± 0.31 %
μ = 447.8± 1.3
σ = 39.6± 1.3
WF (1493 entries)
σ/μ = 9.62± 0.31 %
μ = 456.9± 1.4
σ = 43.9± 1.3

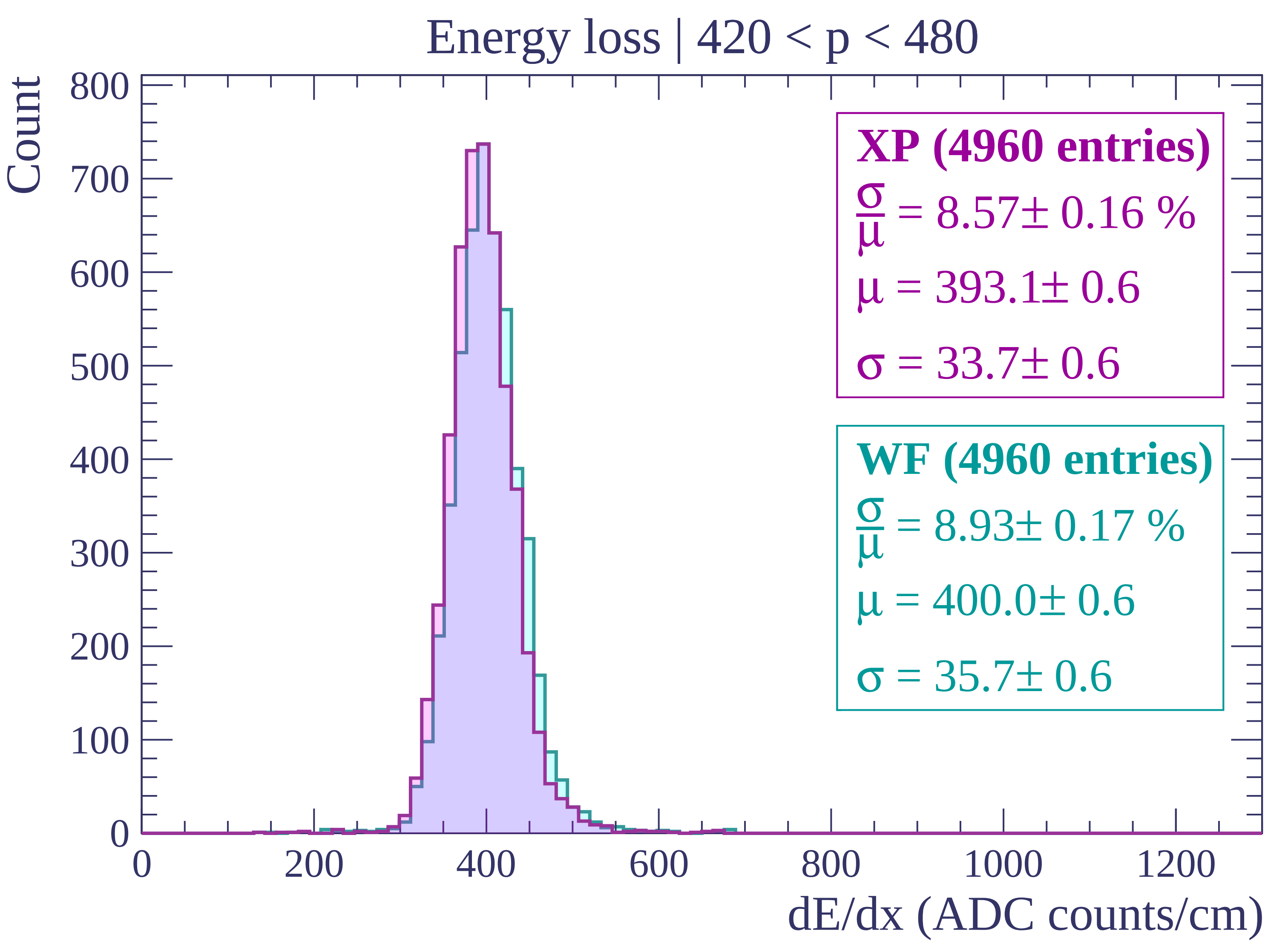

Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
XP (2477 entries)
σ/μ = 8.85± 0.23 %
μ = 407.9± 0.9
σ = 36.1± 0.8
WF (2477 entries)
σ/μ = 9.25± 0.23 %
μ = 414.2± 0.9
σ = 38.3± 0.9
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

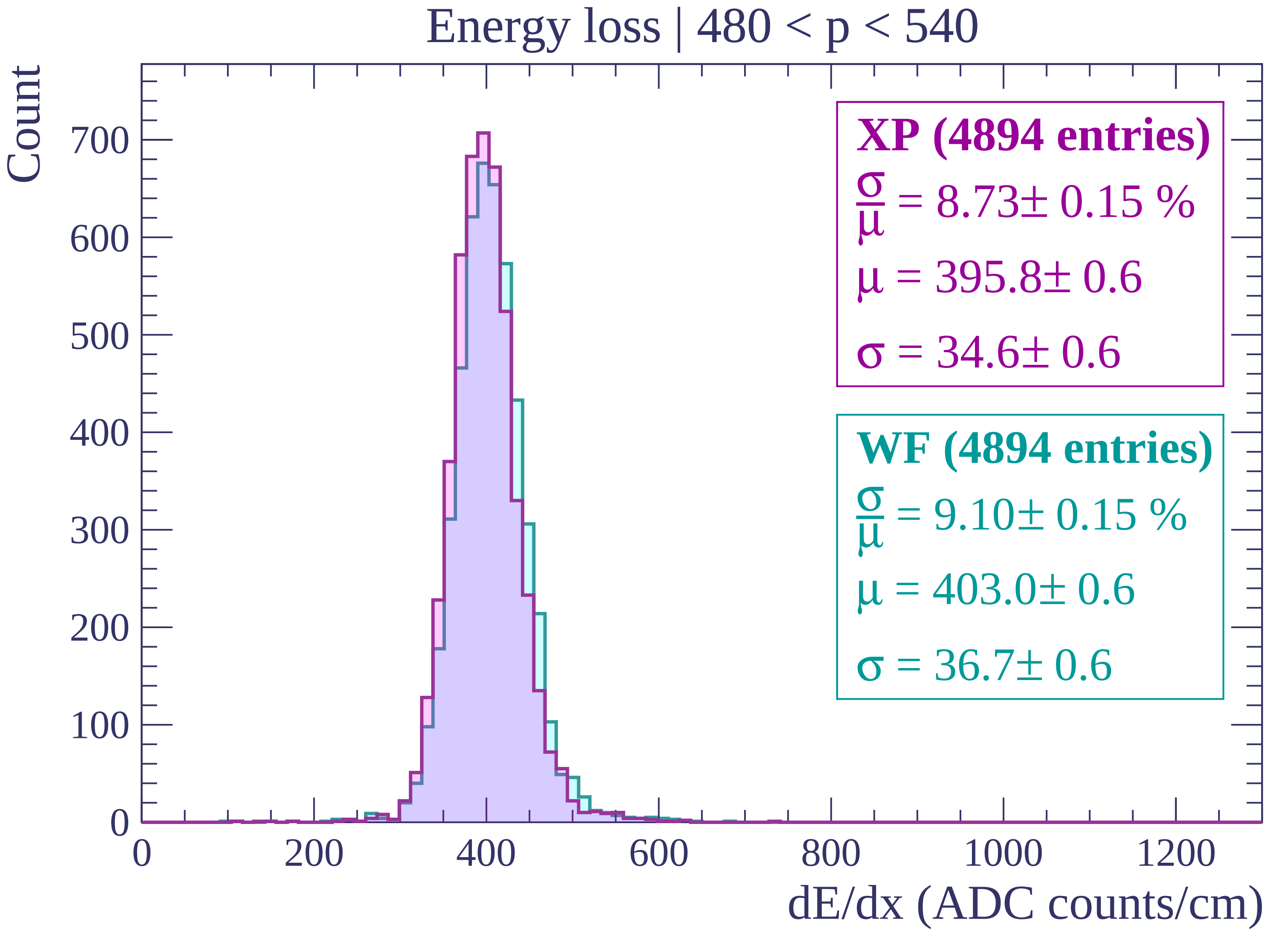


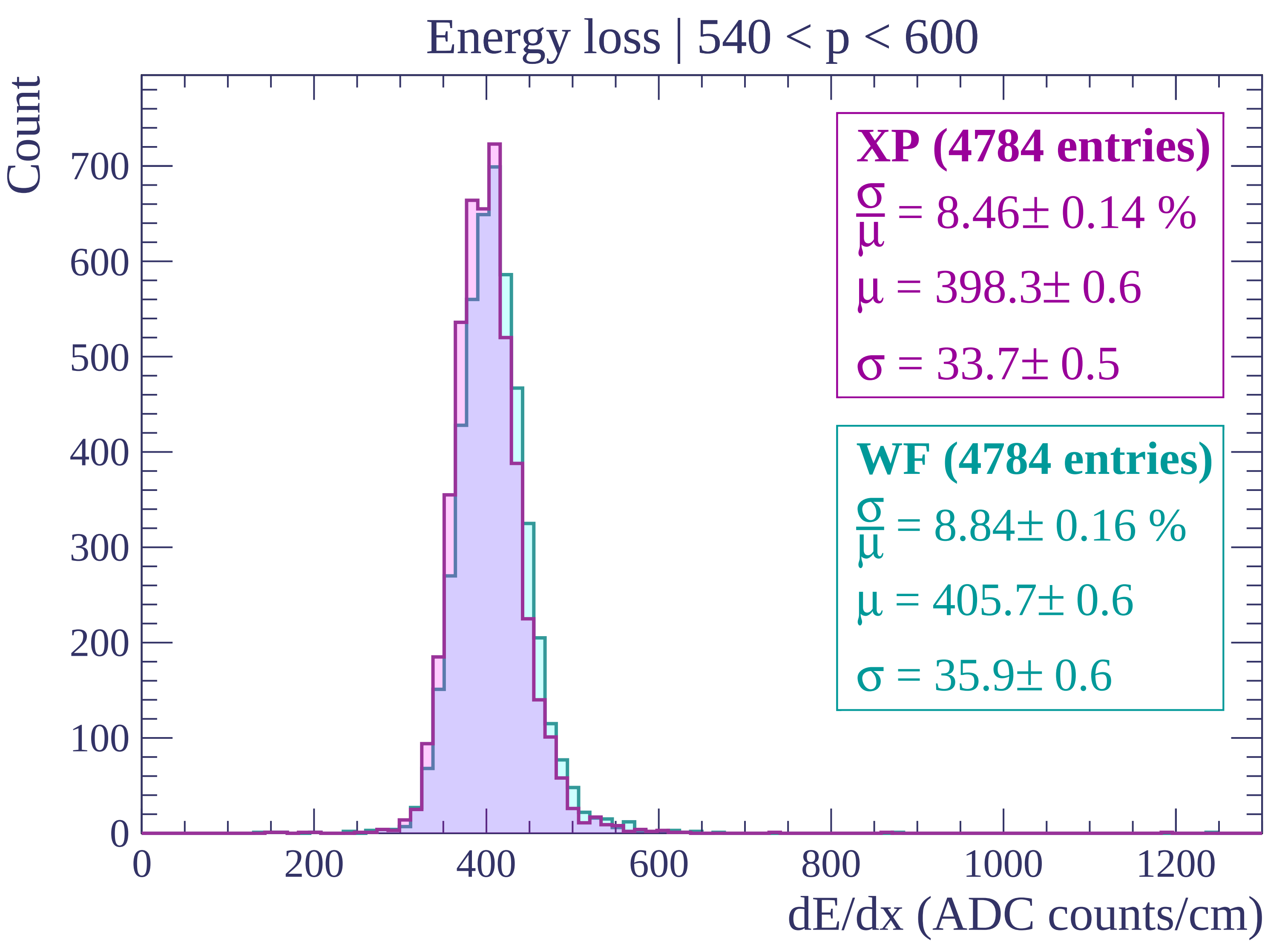
Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4221 entries)
σ/μ = 9.05± 0.17 %
μ = 395.1± 0.6
σ = 35.7± 0.6
WF (4221 entries)
σ/μ = 8.94± 0.15 %
μ = 402.3± 0.6
σ = 36.0± 0.6

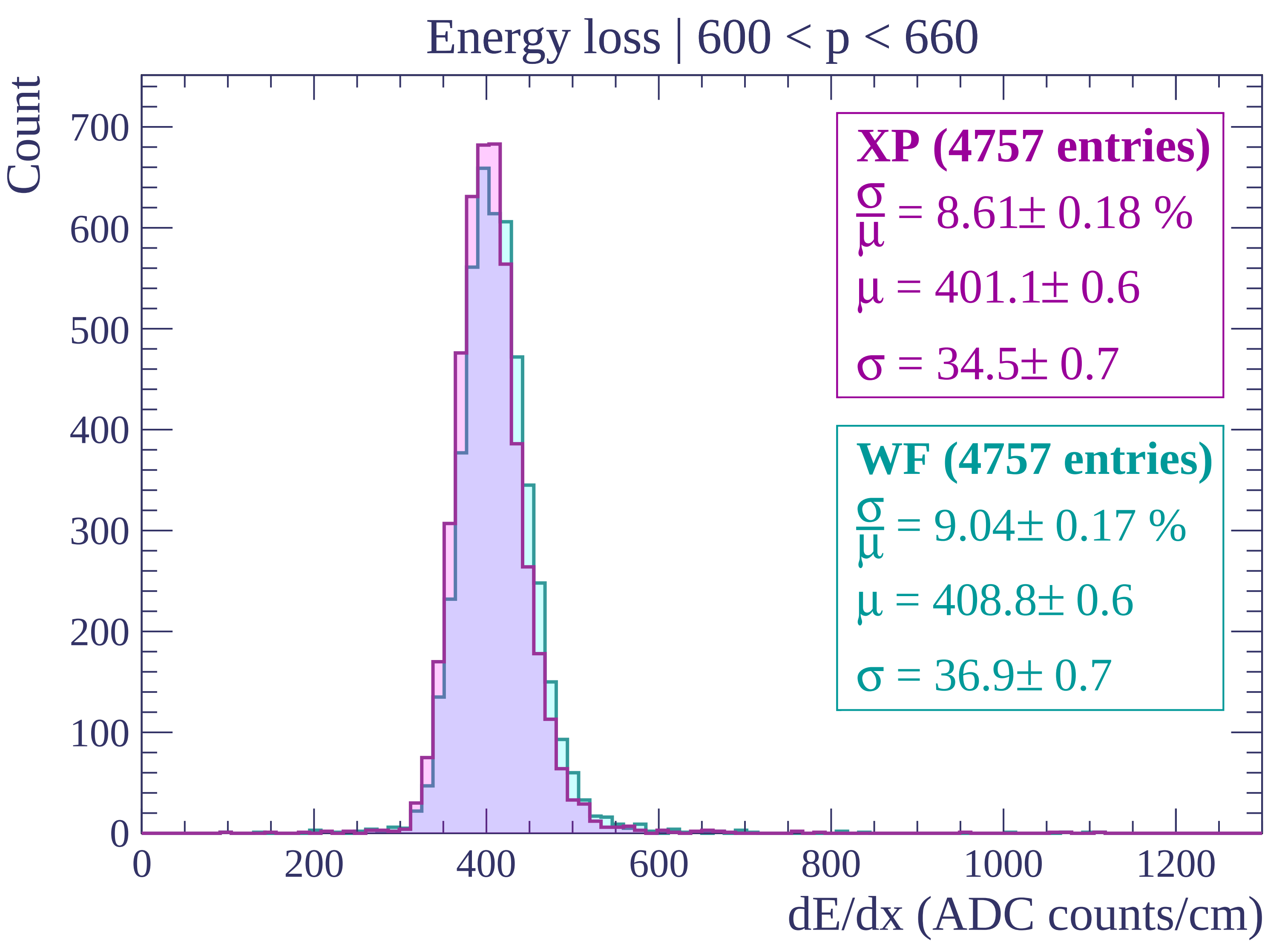


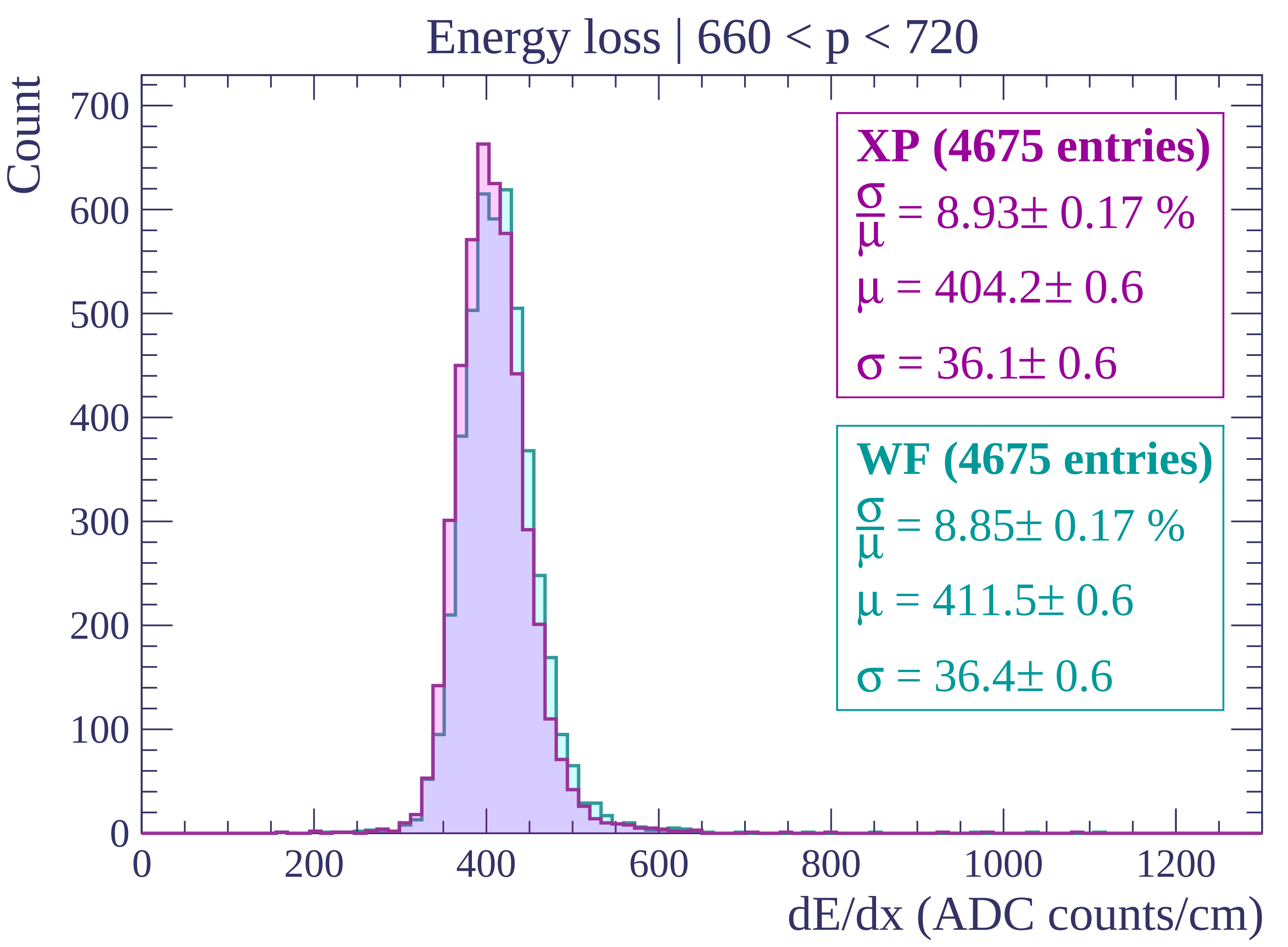
Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
XP (4898 entries)
$\frac{\sigma}{\mu}$ = 8.63± 0.16 %
μ = 391.8± 0.6
σ = 33.8± 0.6
WF (4898 entries)
$\frac{\sigma}{\mu}$ = 8.88± 0.15 %
μ = 398.6± 0.6
σ = 35.4± 0.6


Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
XP (4960 entries)
$\frac{\sigma}{\mu}$ = 8.57± 0.16 %
μ = 393.1± 0.6
σ = 33.7± 0.6
WF (4960 entries)
$\frac{\sigma}{\mu}$ = 8.93± 0.17 %
μ = 400.0± 0.6
σ = 35.7± 0.6
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

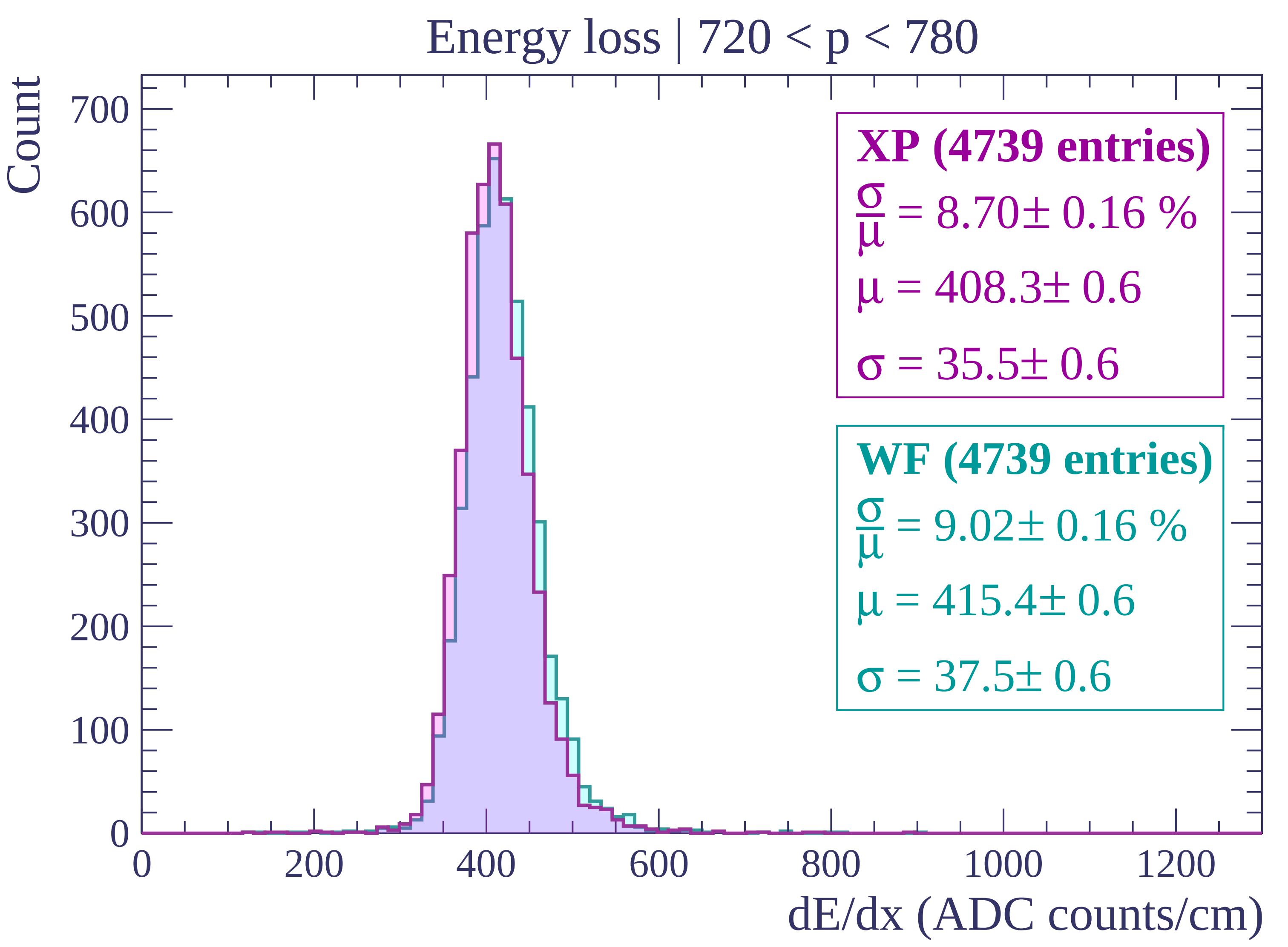

Energy loss | 480 < p < 540
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4894 entries)
σ/μ = 8.73± 0.15 %
μ = 395.8± 0.6
σ = 34.6± 0.6
WF (4894 entries)
σ/μ = 9.10± 0.15 %
μ = 403.0± 0.6
σ = 36.7± 0.6


Energy loss | 540 < p < 600
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (4784 entries)
$\frac{\sigma}{\mu}$ = 8.46± 0.14 %
μ = 398.3± 0.6
σ = 33.7± 0.5
WF (4784 entries)
$\frac{\sigma}{\mu}$ = 8.84± 0.16 %
μ = 405.7± 0.6
σ = 35.9± 0.6

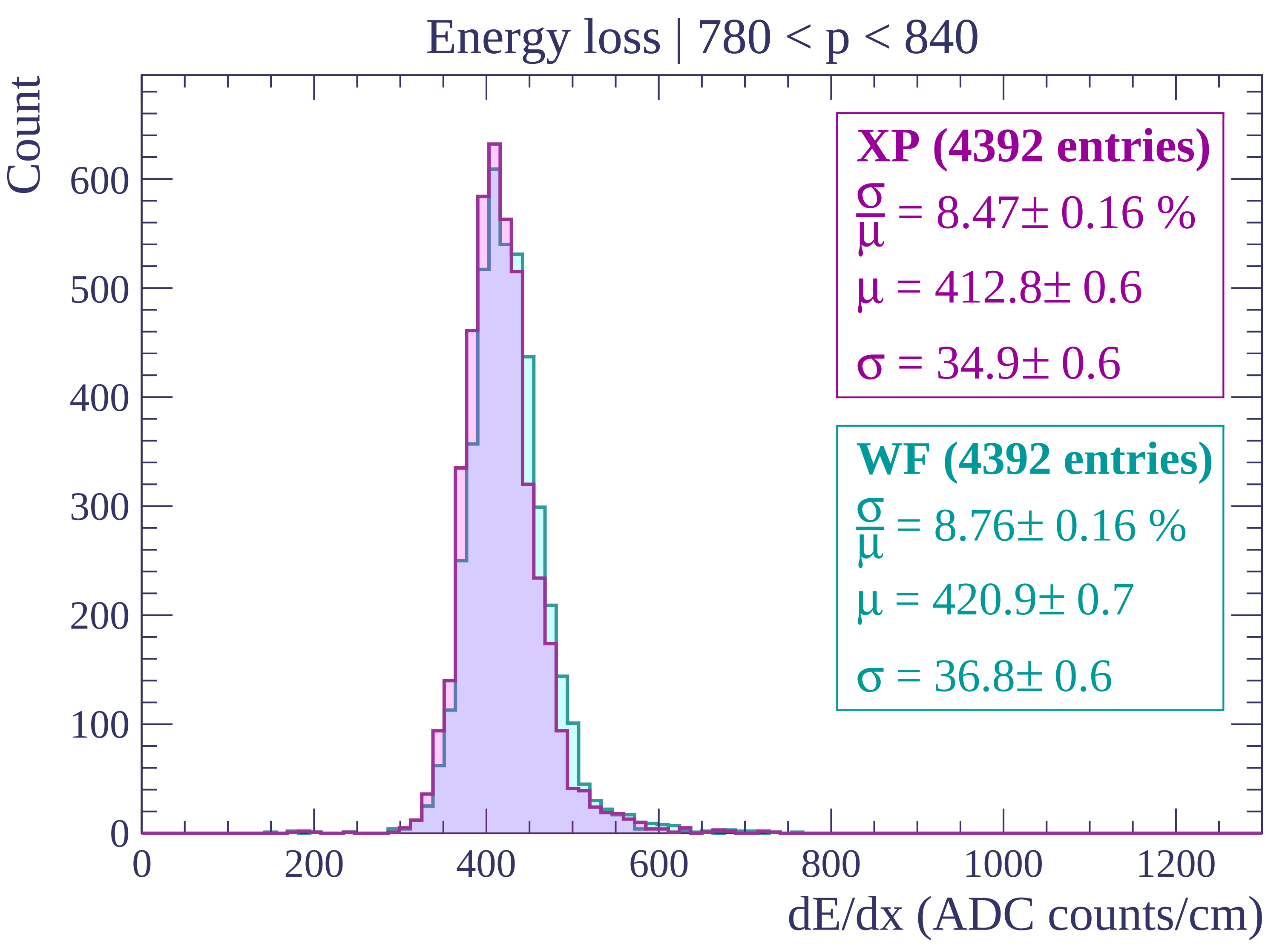

Energy loss | 600 < p < 660
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4757 entries)
$\frac{\sigma}{\mu}$ = 8.61± 0.18 %
μ = 401.1± 0.6
σ = 34.5± 0.7
WF (4757 entries)
$\frac{\sigma}{\mu}$ = 9.04± 0.17 %
μ = 408.8± 0.6
σ = 36.9± 0.7



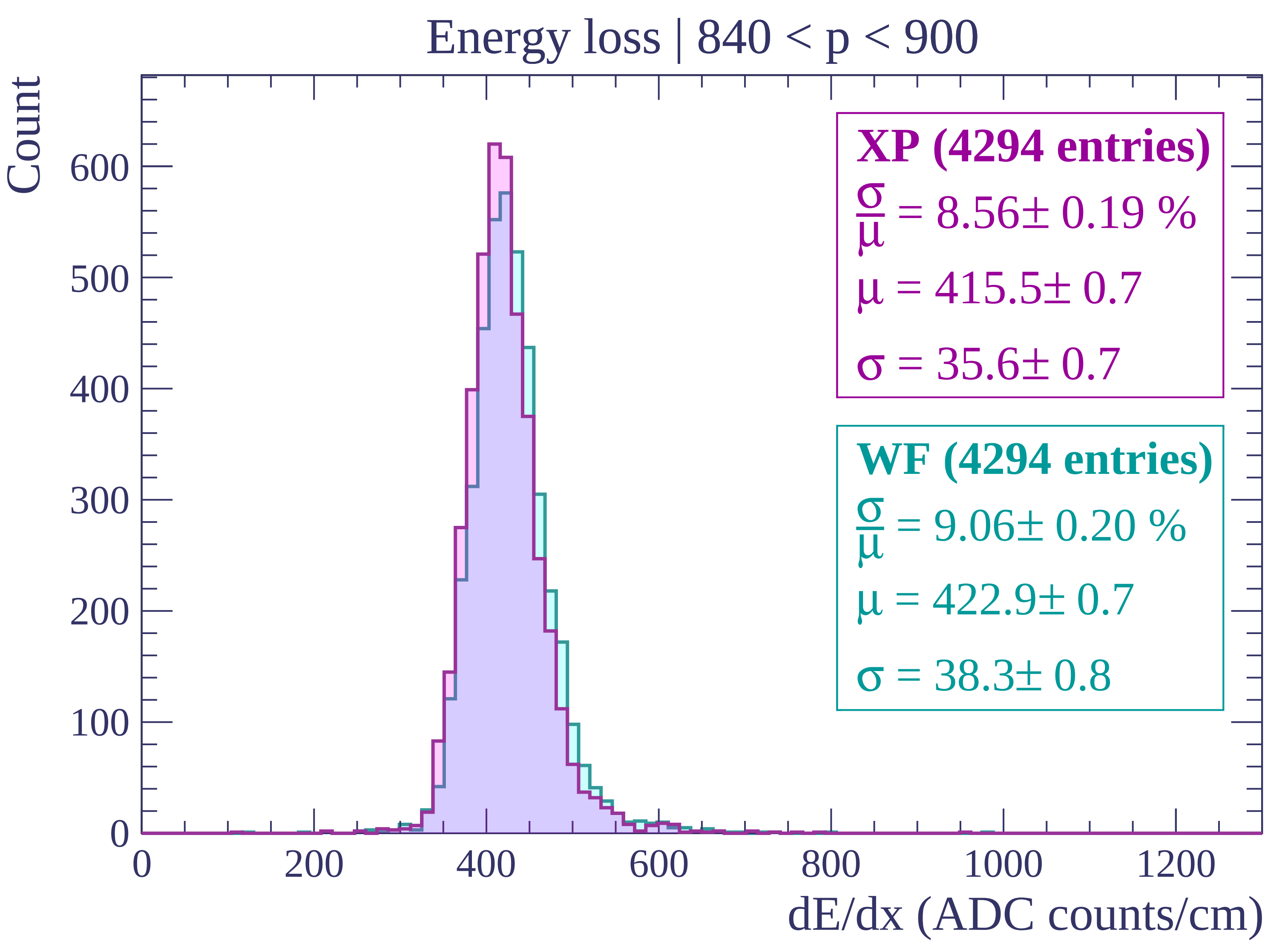
Energy loss | 660 < p < 720
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (4675 entries)
σ/μ = 8.93± 0.17 %
μ = 404.2± 0.6
σ = 36.1± 0.6
WF (4675 entries)
σ/μ = 8.85± 0.17 %
μ = 411.5± 0.6
σ = 36.4± 0.6



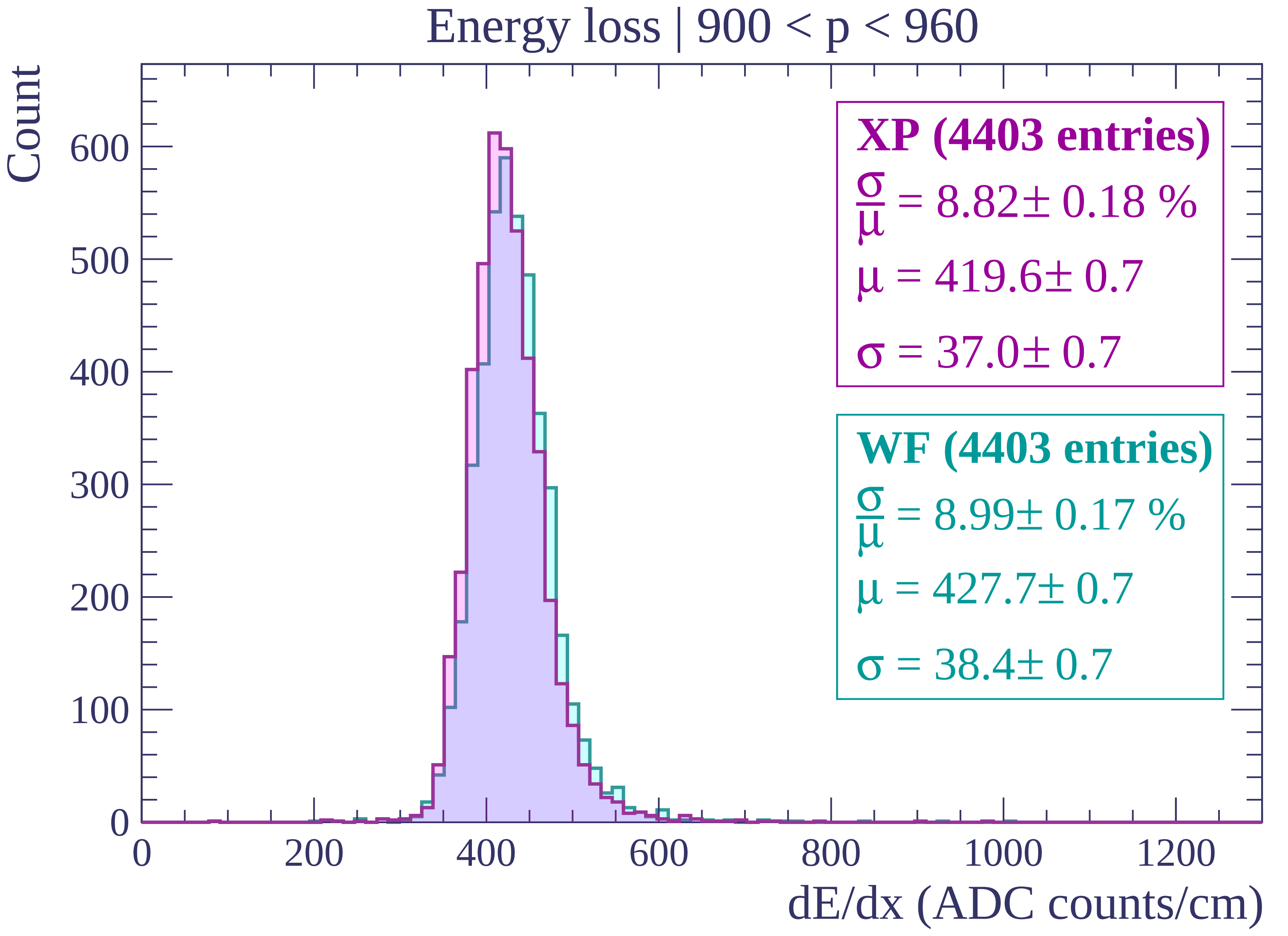
Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (4739 entries)
σ/μ = 8.70± 0.16 %
μ = 408.3± 0.6
σ = 35.5± 0.6
WF (4739 entries)
σ/μ = 9.02± 0.16 %
μ = 415.4± 0.6
σ = 37.5± 0.6



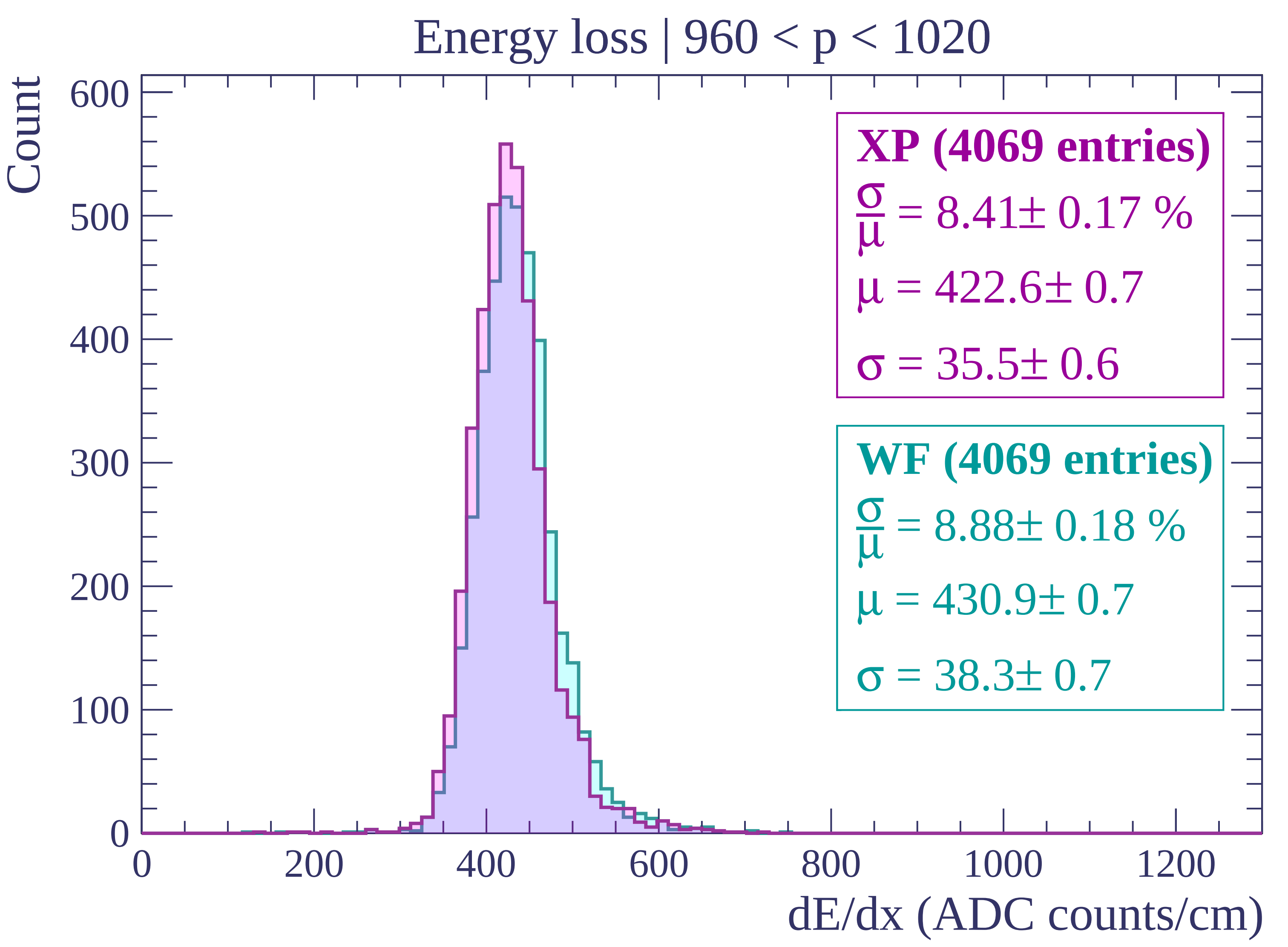
Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4392 entries)
$\frac{\sigma}{\mu}$ = 8.47± 0.16 %
μ = 412.8± 0.6
σ = 34.9± 0.6
WF (4392 entries)
$\frac{\sigma}{\mu}$ = 8.76± 0.16 %
μ = 420.9± 0.7
σ = 36.8± 0.6

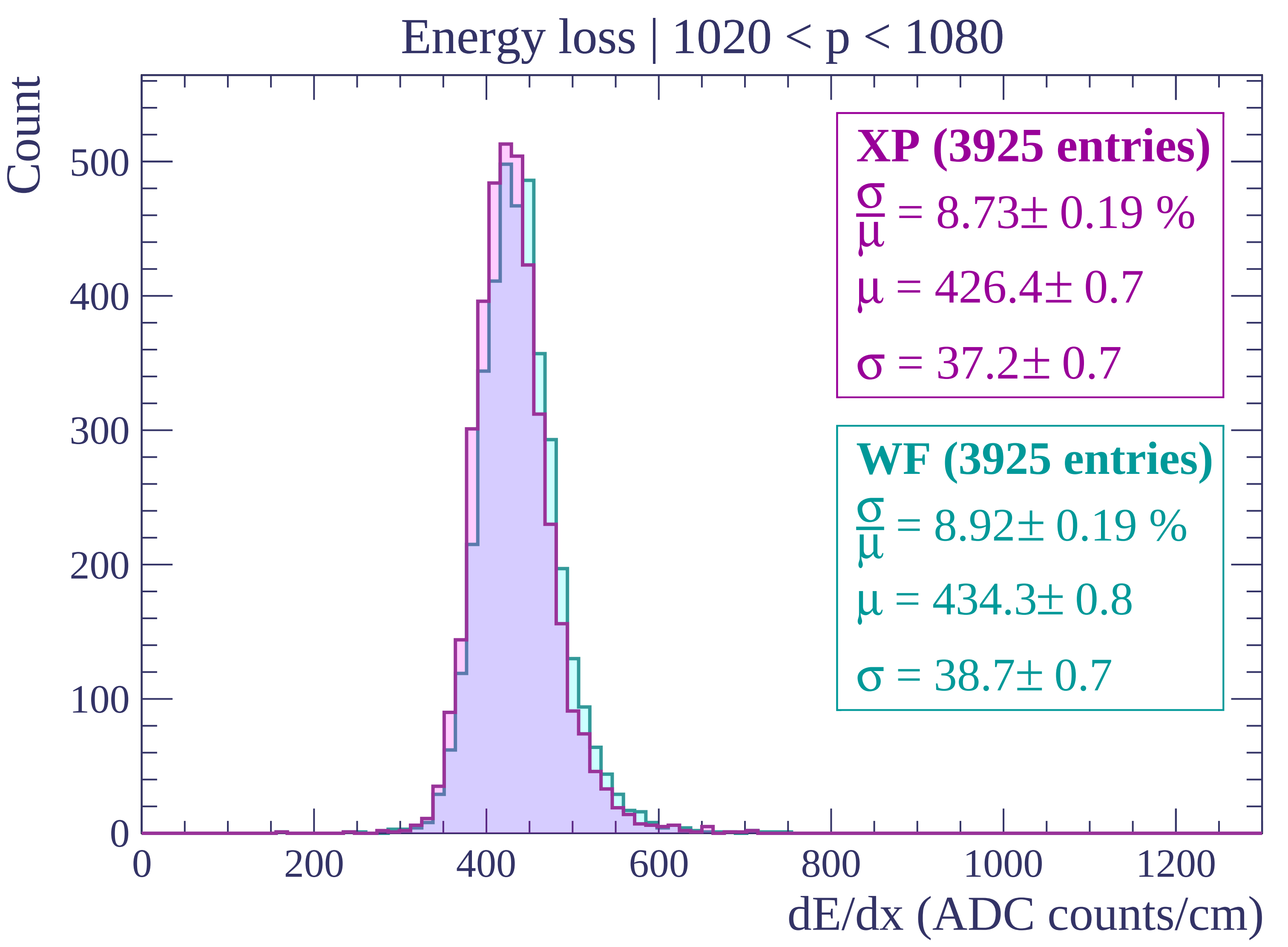


Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4294 entries)
$\frac{\sigma}{\mu}$ = 8.56± 0.19 %
μ = 415.5± 0.7
σ = 35.6± 0.7
WF (4294 entries)
$\frac{\sigma}{\mu}$ = 9.06± 0.20 %
μ = 422.9± 0.7
σ = 38.3± 0.8

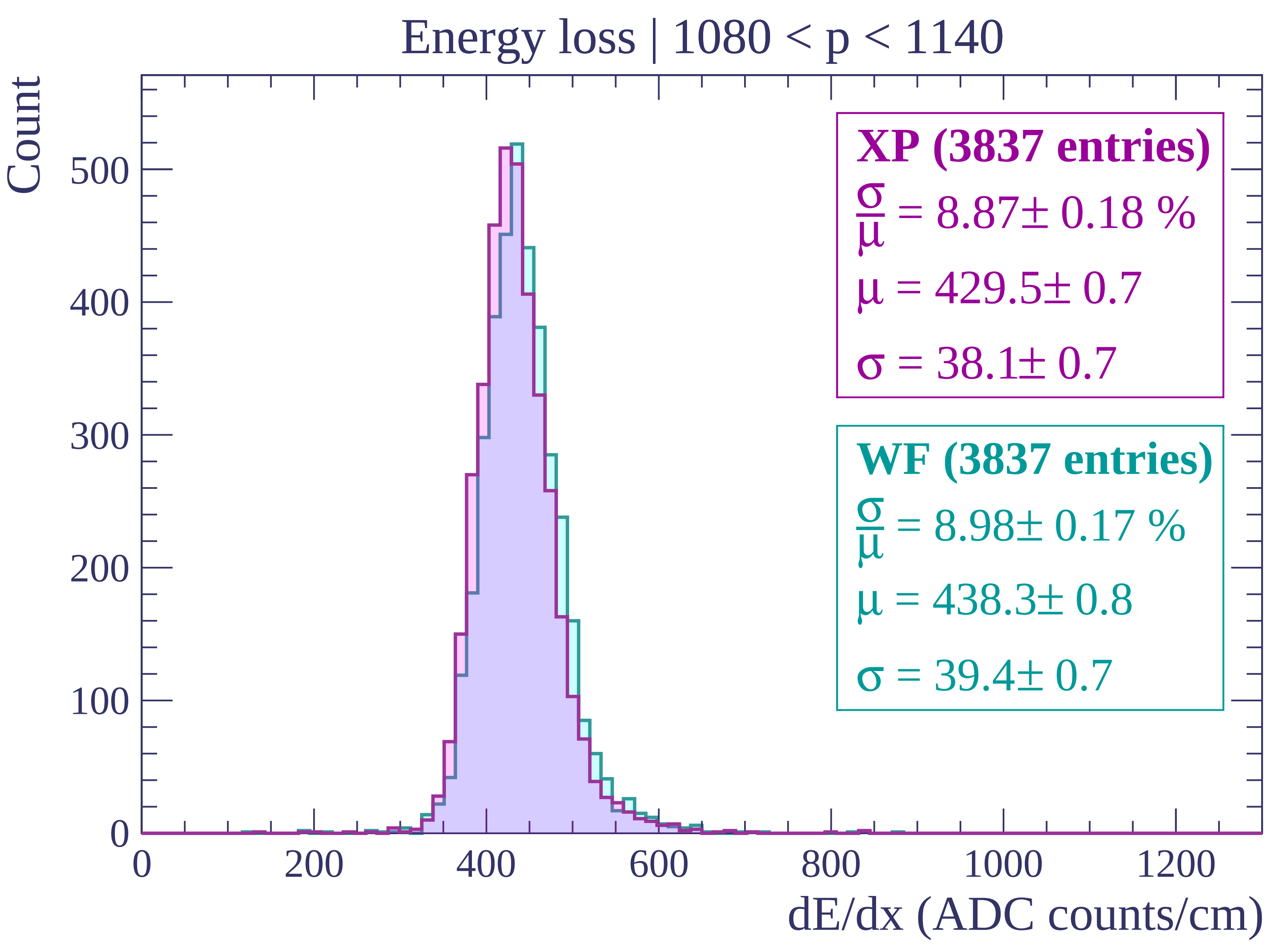


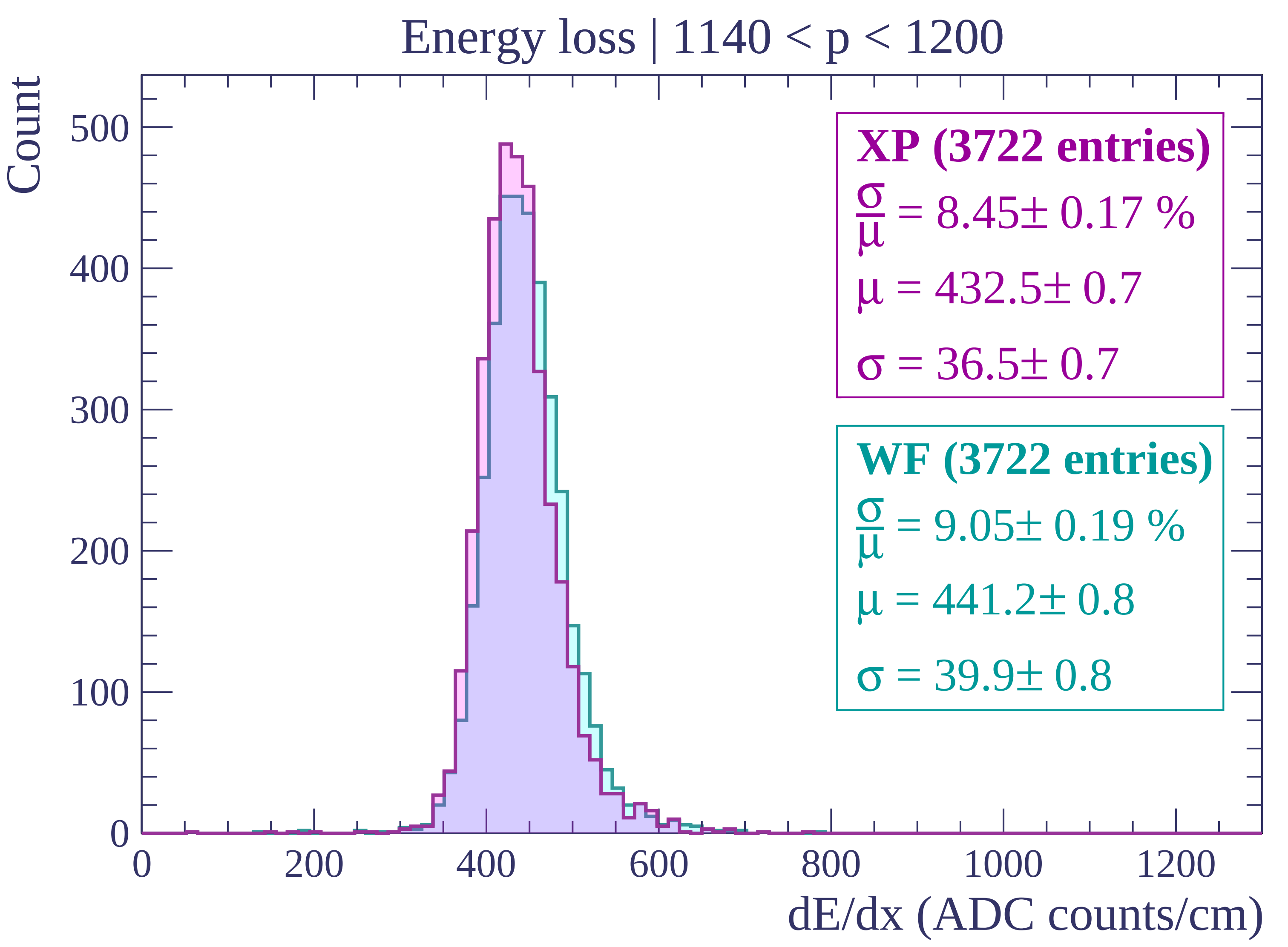
Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4403 entries)
$\frac{\sigma}{\mu}$ = 8.82± 0.18 %
μ = 419.6± 0.7
σ = 37.0± 0.7
WF (4403 entries)
$\frac{\sigma}{\mu}$ = 8.99± 0.17 %
μ = 427.7± 0.7
σ = 38.4± 0.7

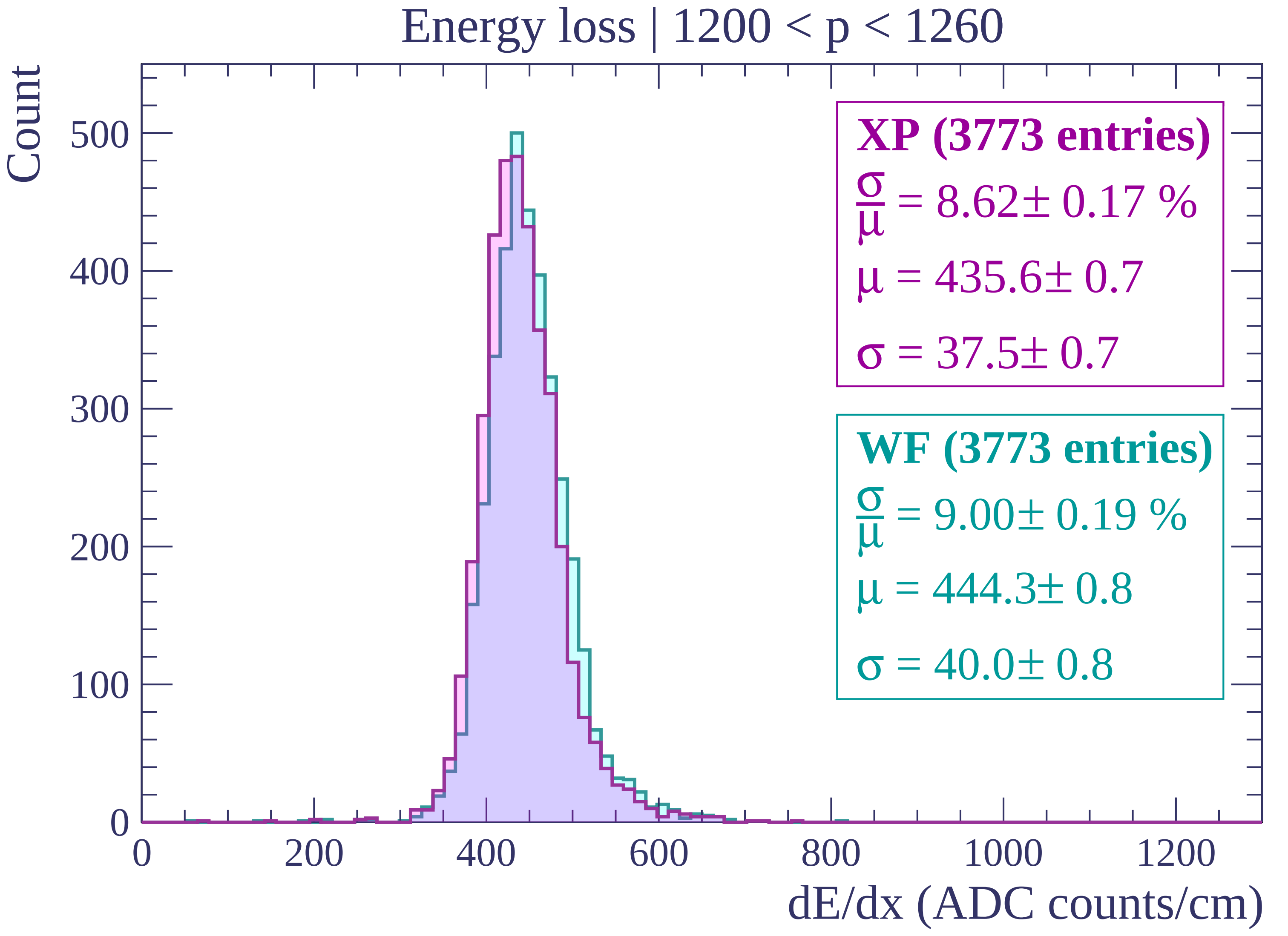

Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4069 entries)
$\frac{\sigma}{\mu}$ = 8.41± 0.17 %
μ = 422.6± 0.7
σ = 35.5± 0.6
WF (4069 entries)
$\frac{\sigma}{\mu}$ = 8.88± 0.18 %
μ = 430.9± 0.7
σ = 38.3± 0.7

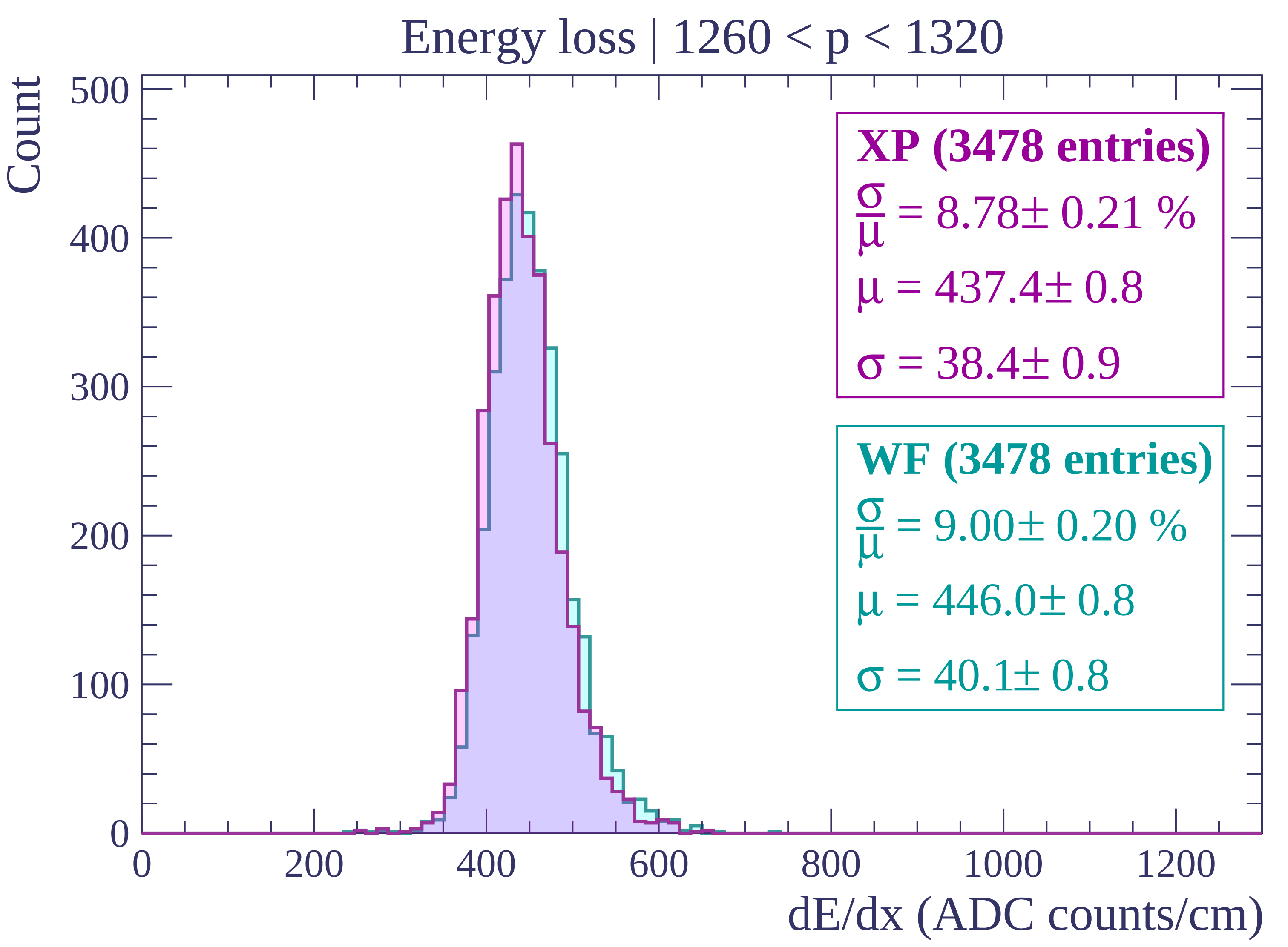


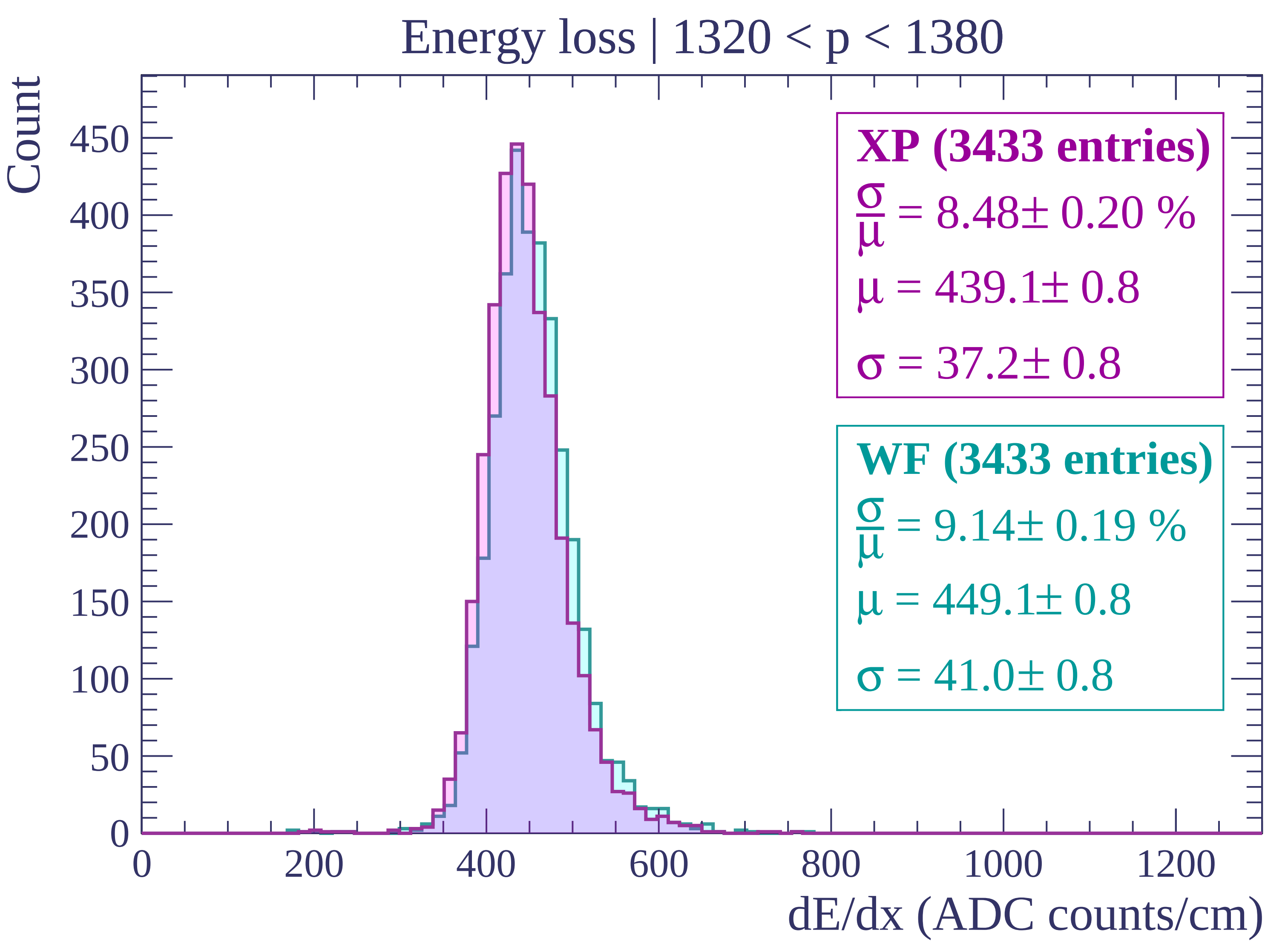
Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3925 entries)
σ/μ = 8.73± 0.19 %
μ = 426.4± 0.7
σ = 37.2± 0.7
WF (3925 entries)
σ/μ = 8.92± 0.19 %
μ = 434.3± 0.8
σ = 38.7± 0.7


Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3837 entries)
σ/μ = 8.87± 0.18 %
μ = 429.5± 0.7
σ = 38.1± 0.7
WF (3837 entries)
σ/μ = 8.98± 0.17 %
μ = 438.3± 0.8
σ = 39.4± 0.7

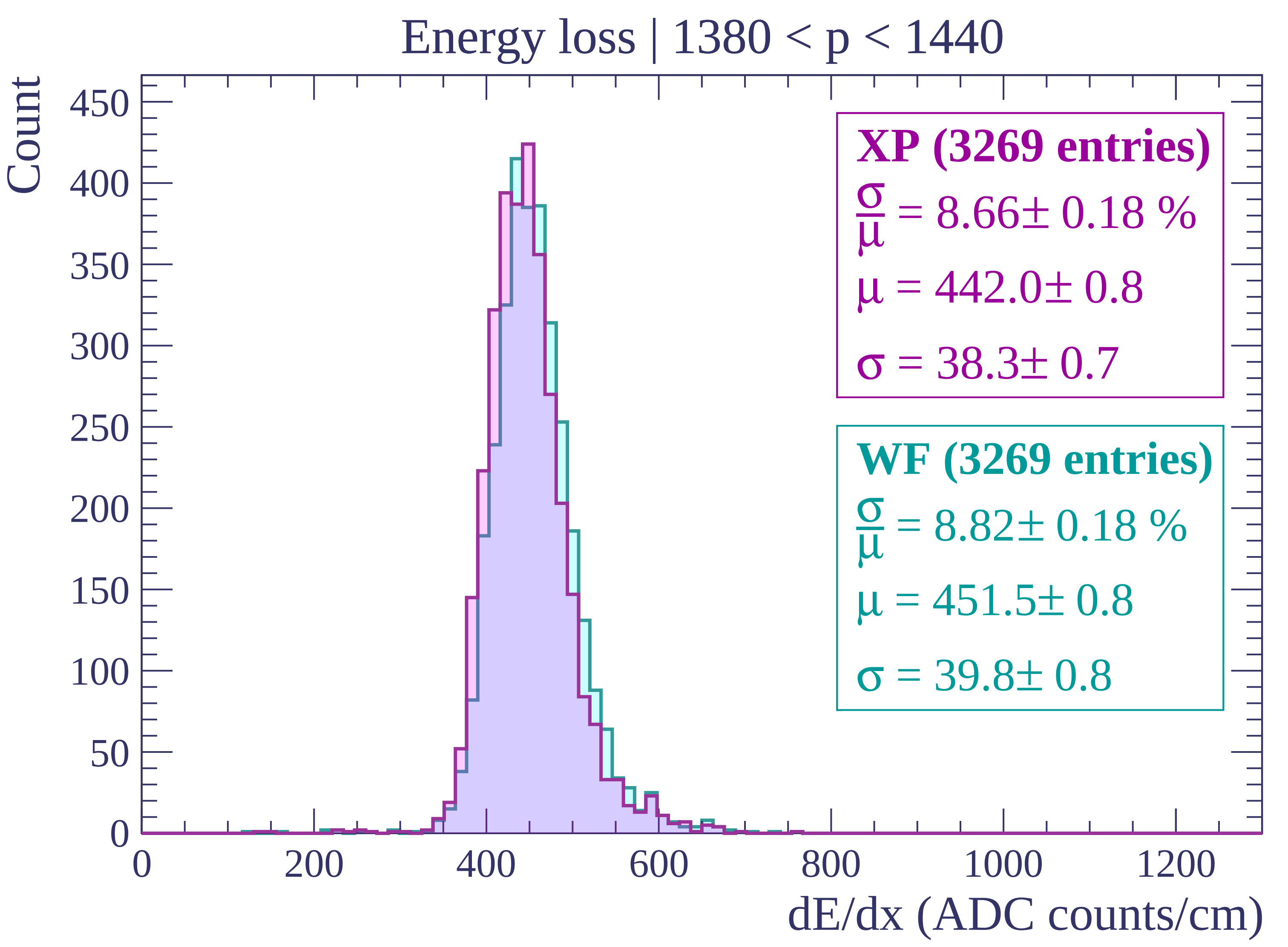

Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3722 entries)
$\frac{\sigma}{\mu}$ = 8.45± 0.17 %
μ = 432.5± 0.7
σ = 36.5± 0.7
WF (3722 entries)
$\frac{\sigma}{\mu}$ = 9.05± 0.19 %
μ = 441.2± 0.8
σ = 39.9± 0.8


Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
100
200
300
400
500
XP (3773 entries)
σ/μ = 8.62± 0.17 %
μ = 435.6± 0.7
σ = 37.5± 0.7
WF (3773 entries)
σ/μ = 9.00± 0.19 %
μ = 444.3± 0.8
σ = 40.0± 0.8

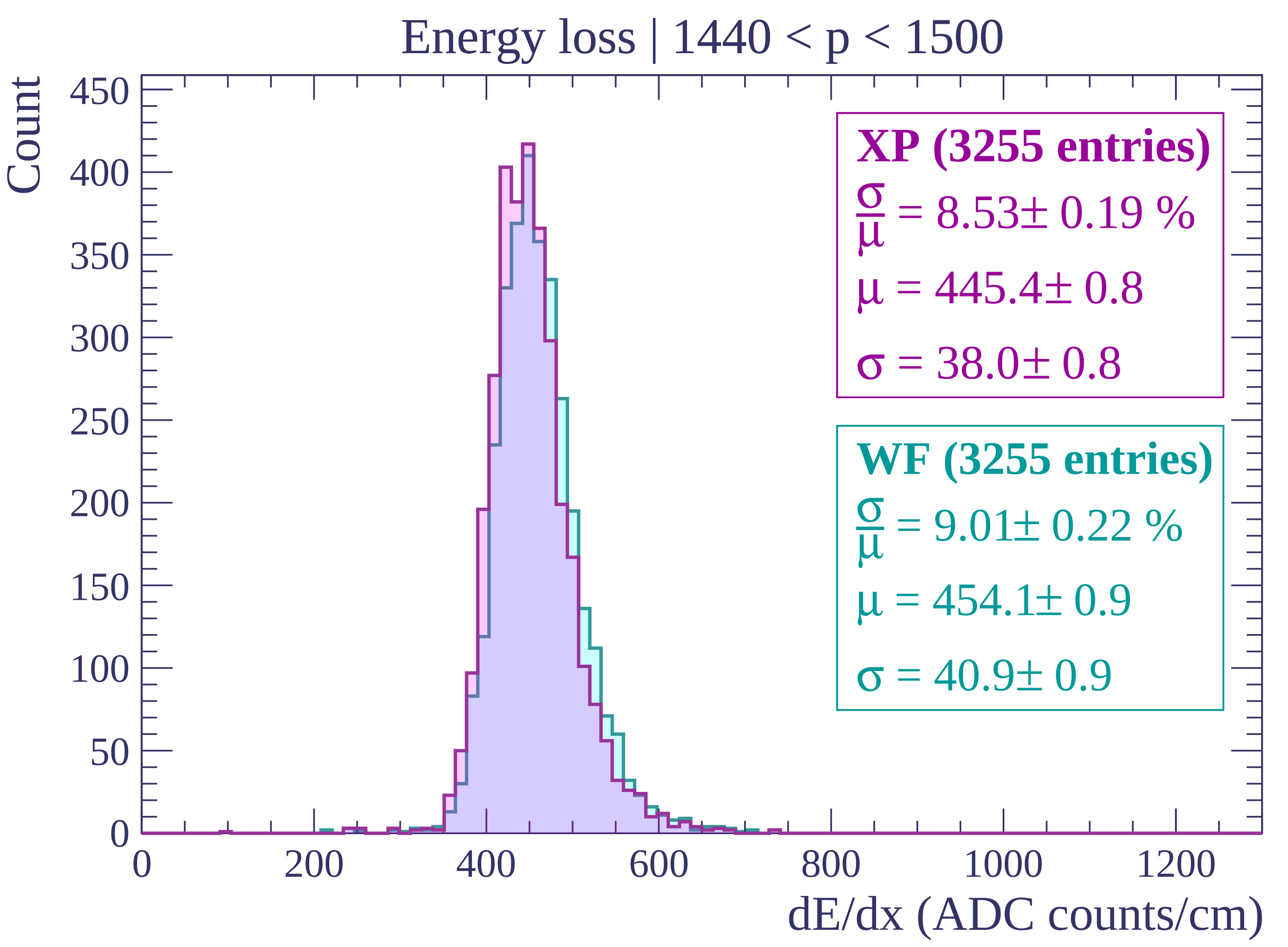

Energy loss | 1260 < p < 1320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3478 entries)
$\frac{\sigma}{\mu}$ = 8.78± 0.21 %
μ = 437.4± 0.8
σ = 38.4± 0.9
WF (3478 entries)
$\frac{\sigma}{\mu}$ = 9.00± 0.20 %
μ = 446.0± 0.8
σ = 40.1± 0.8



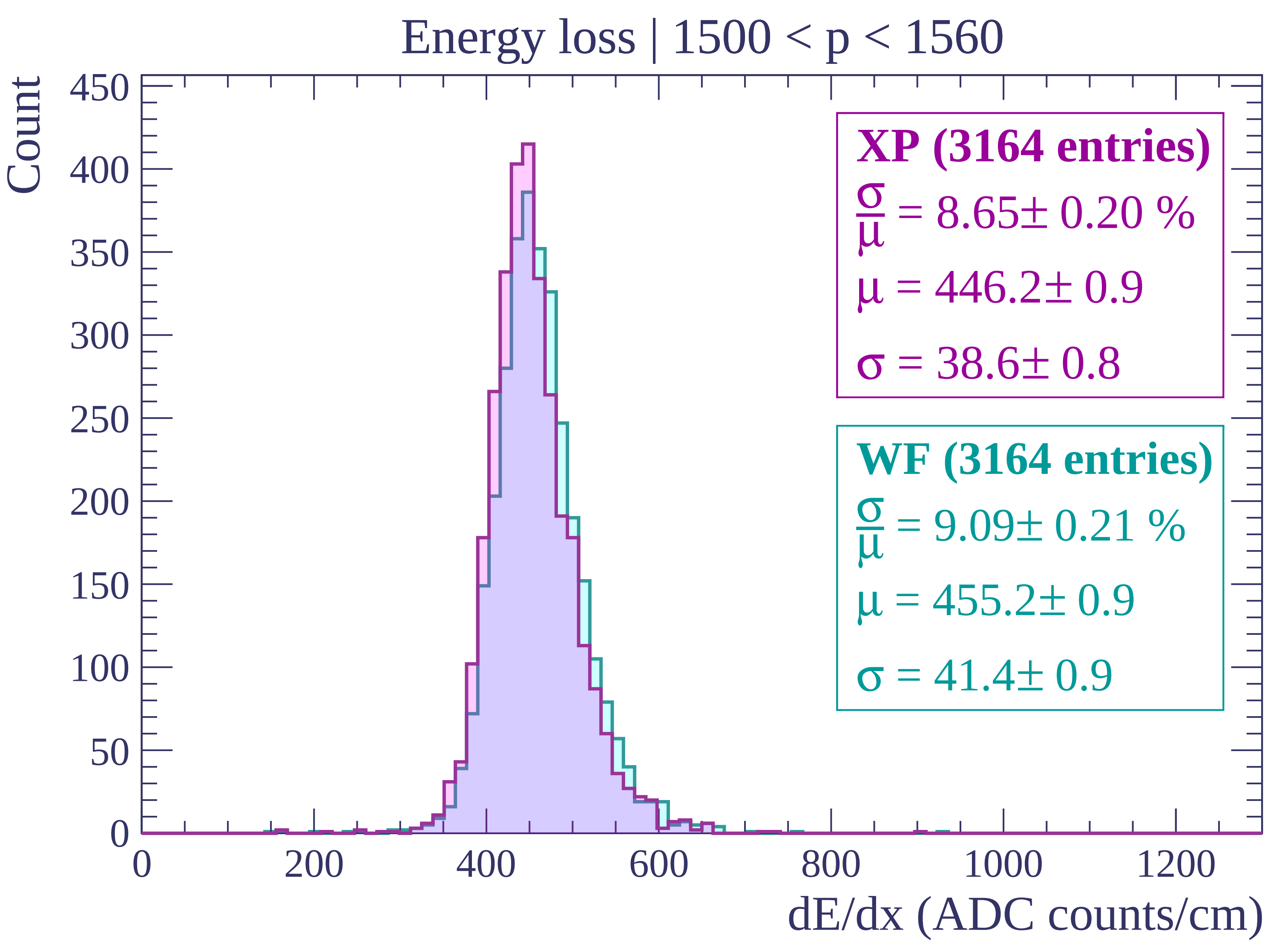
Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
XP (3433 entries)
$\frac{\sigma}{\mu}$ = 8.48± 0.20 %
μ = 439.1± 0.8
σ = 37.2± 0.8
WF (3433 entries)
$\frac{\sigma}{\mu}$ = 9.14± 0.19 %
μ = 449.1± 0.8
σ = 41.0± 0.8
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200



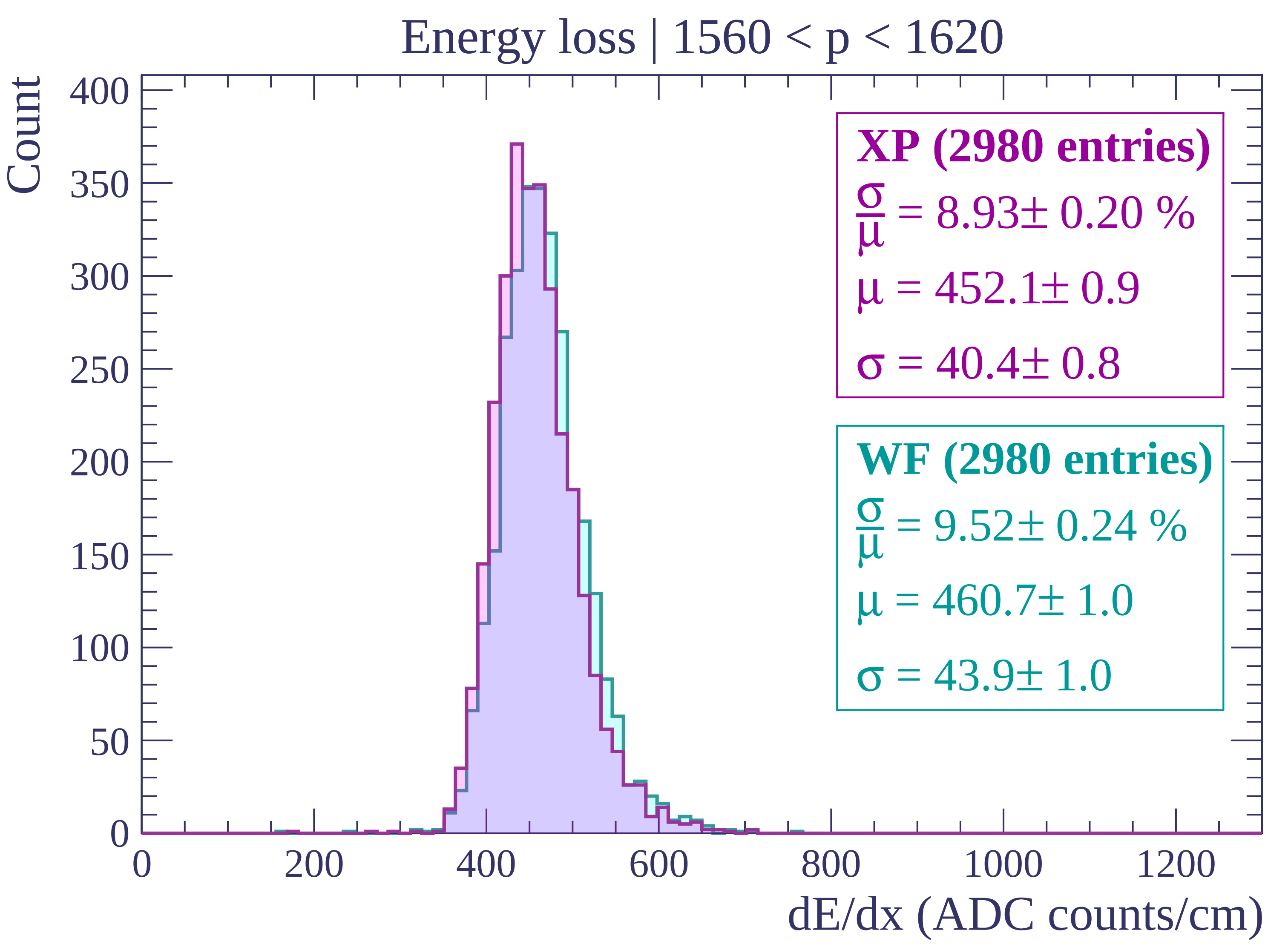
Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
450
XP (3269 entries)
$\frac{\sigma}{\mu}$ = 8.66± 0.18 %
μ = 442.0± 0.8
σ = 38.3± 0.7
WF (3269 entries)
$\frac{\sigma}{\mu}$ = 8.82± 0.18 %
μ = 451.5± 0.8
σ = 39.8± 0.8



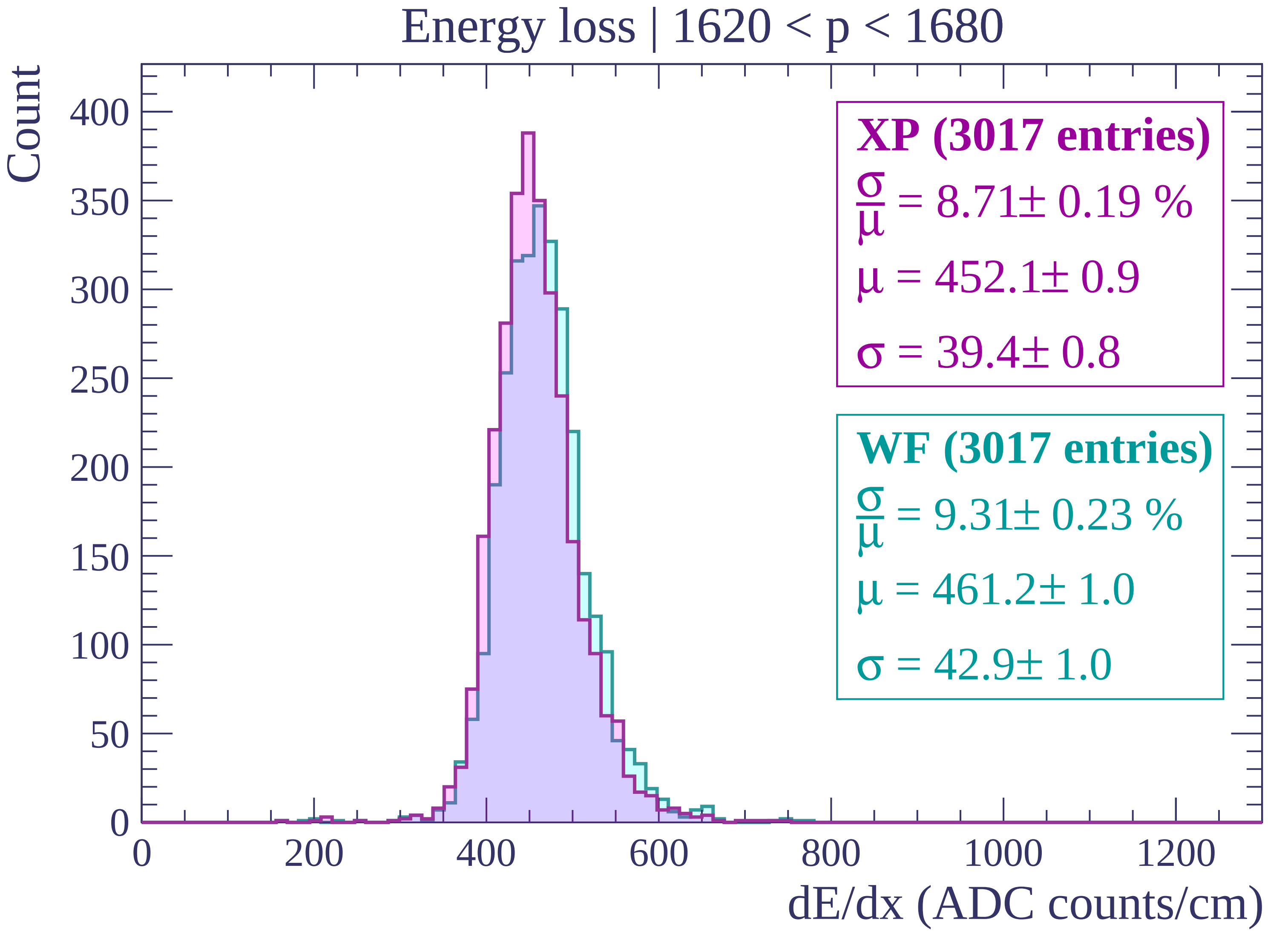
Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3255 entries)
σ/μ = 8.53± 0.19 %
μ = 445.4± 0.8
σ = 38.0± 0.8
WF (3255 entries)
σ/μ = 9.01± 0.22 %
μ = 454.1± 0.9
σ = 40.9± 0.9

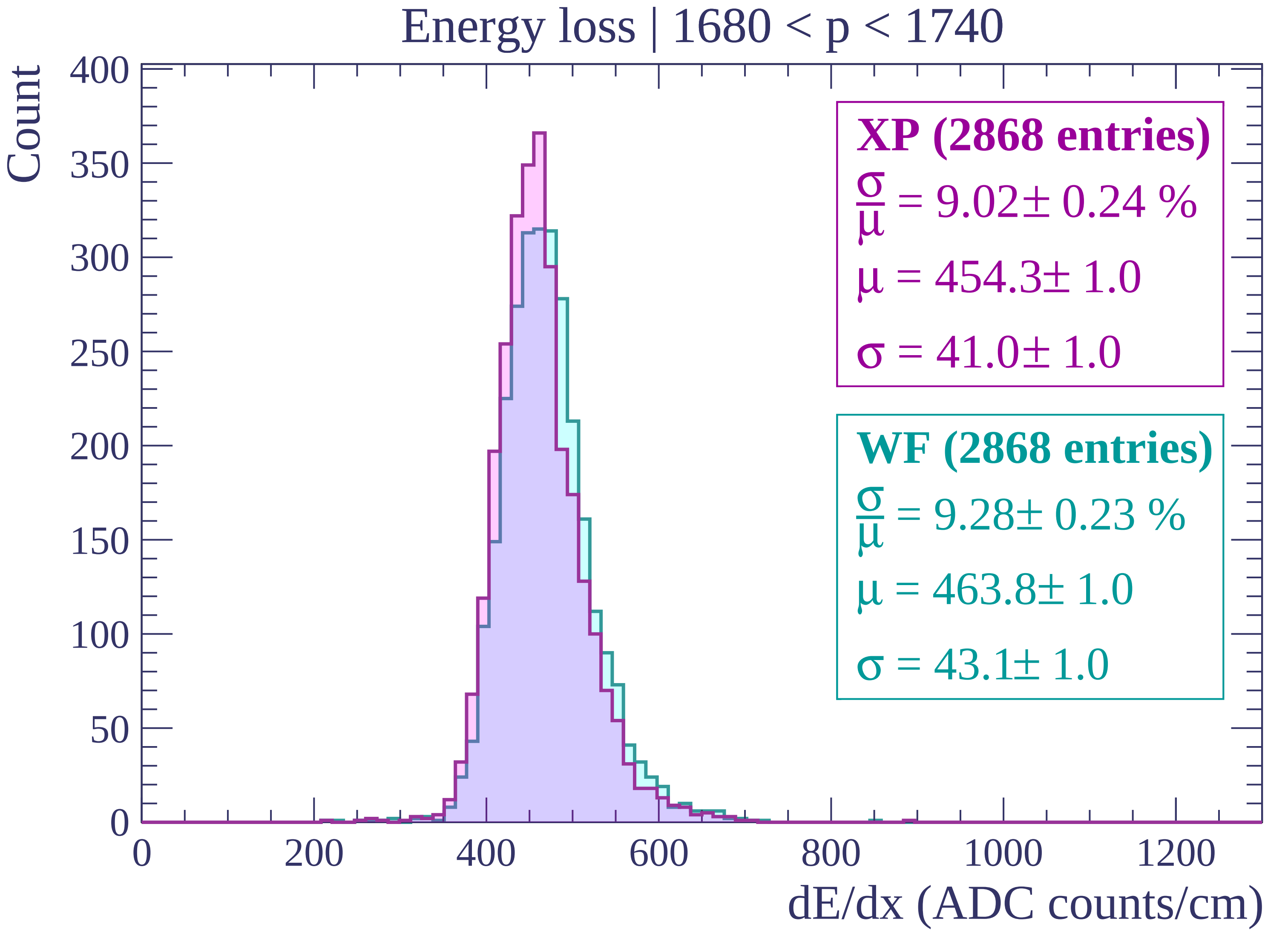


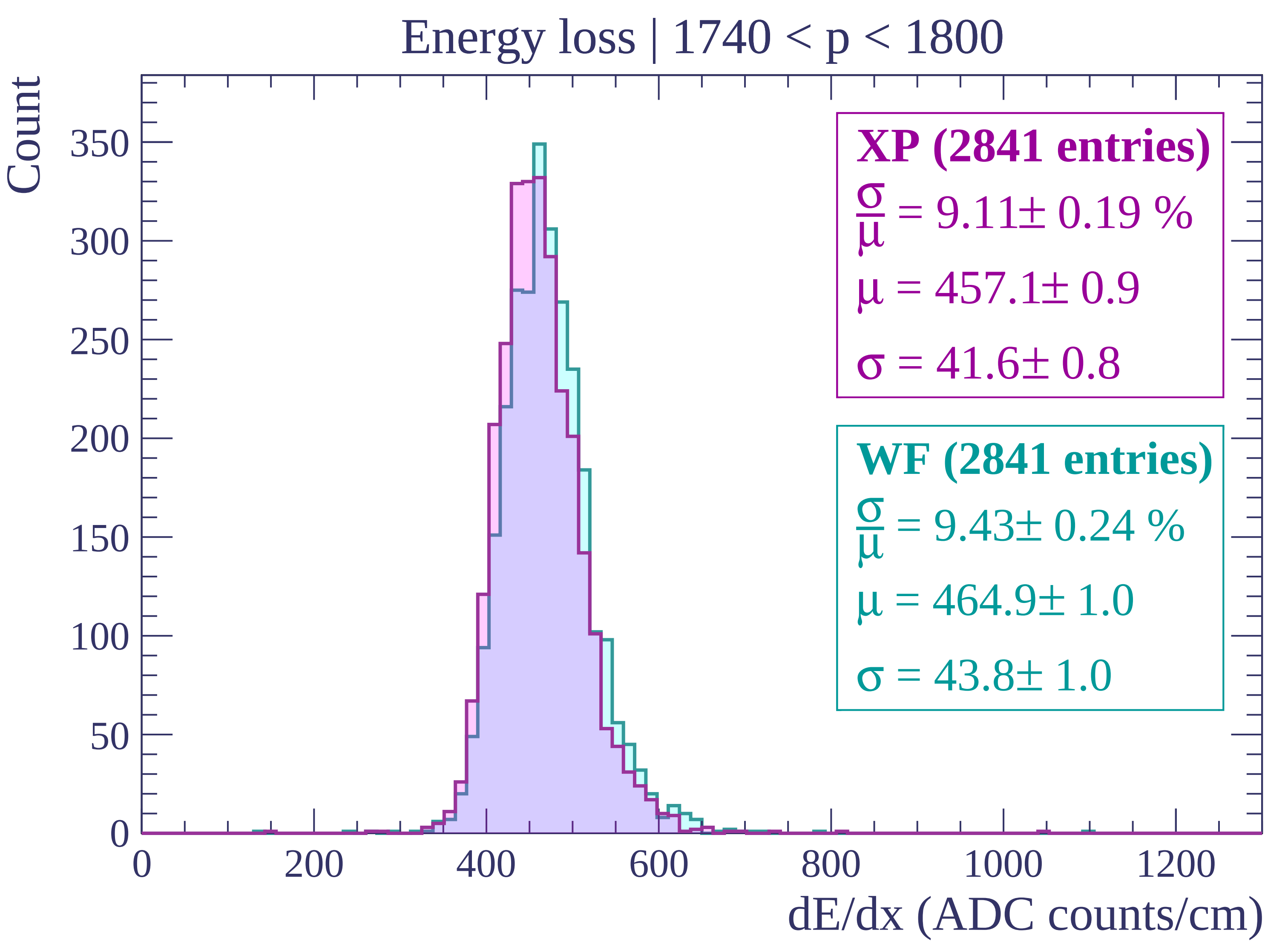
Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
XP (3164 entries)
σ/μ = 8.65± 0.20 %
μ = 446.2± 0.9
σ = 38.6± 0.8
WF (3164 entries)
σ/μ = 9.09± 0.21 %
μ = 455.2± 0.9
σ = 41.4± 0.9

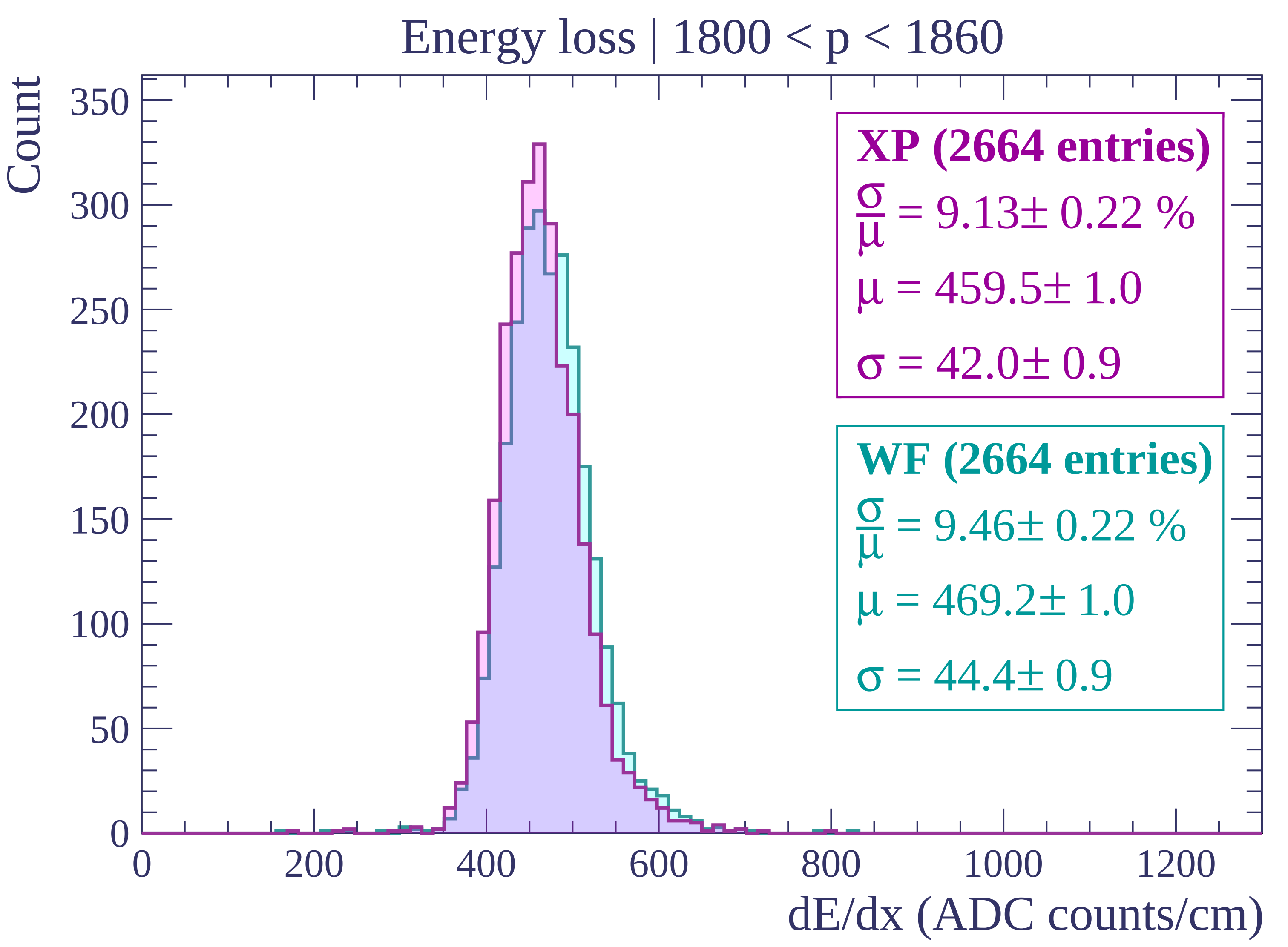


Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
XP (2980 entries)
σ/μ = 8.93± 0.20 %
μ = 452.1± 0.9
σ = 40.4± 0.8
WF (2980 entries)
σ/μ = 9.52± 0.24 %
μ = 460.7± 1.0
σ = 43.9± 1.0
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

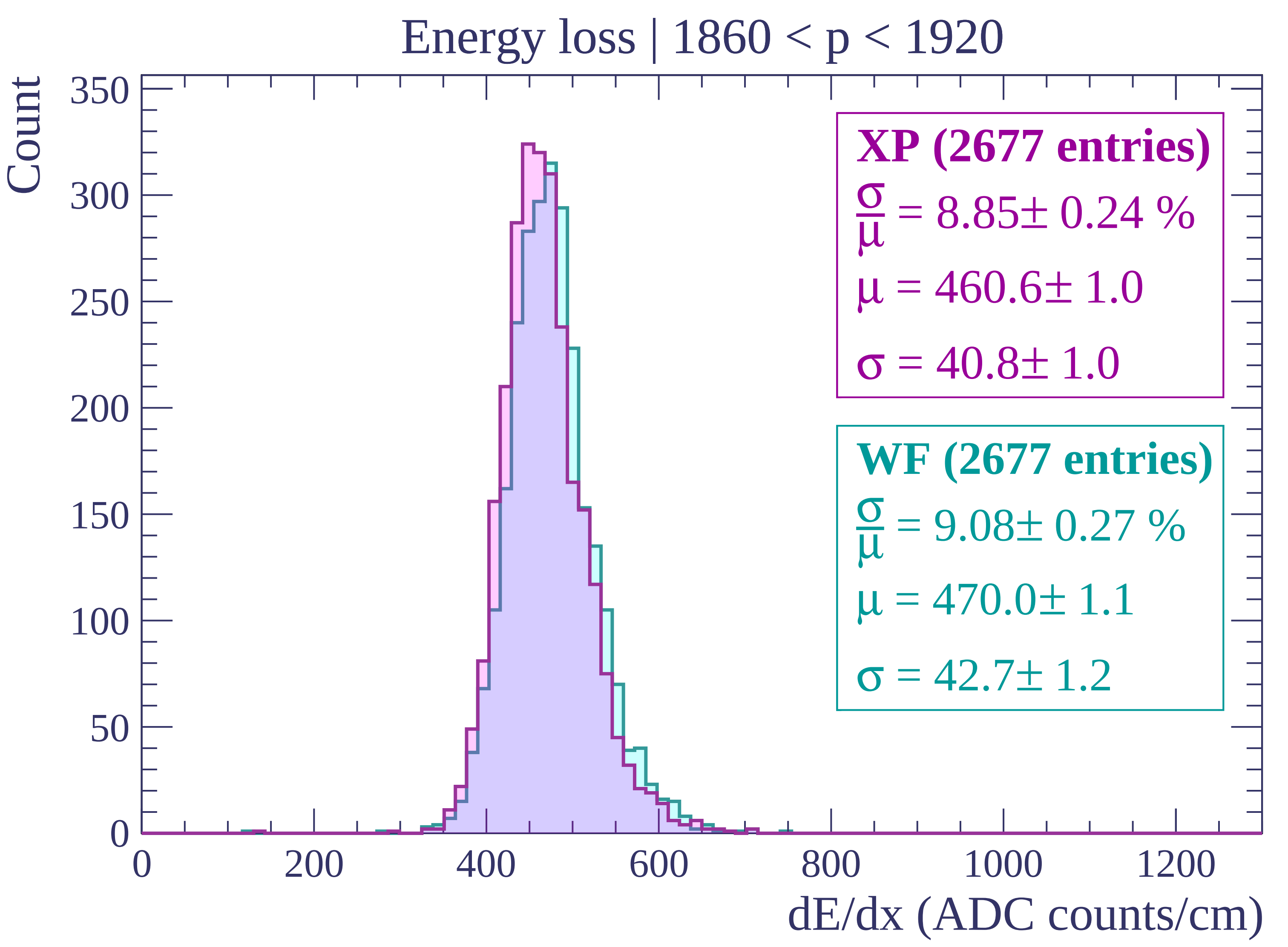

Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
XP (3017 entries)
σ/μ = 8.71± 0.19 %
μ = 452.1± 0.9
σ = 39.4± 0.8
WF (3017 entries)
σ/μ = 9.31± 0.23 %
μ = 461.2± 1.0
σ = 42.9± 1.0

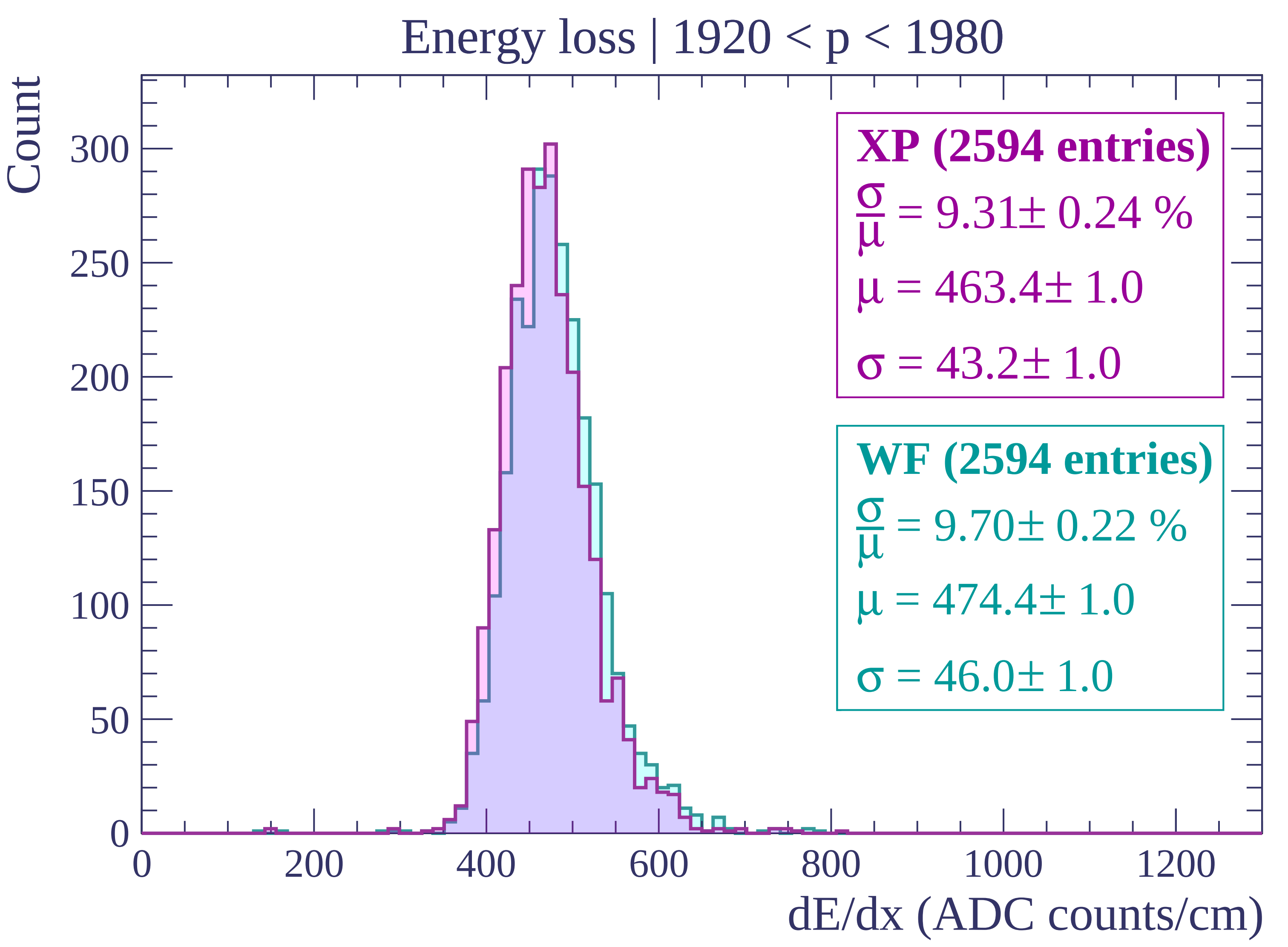

Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (2868 entries)
$\frac{\sigma}{\mu}$ = 9.02± 0.24 %
μ = 454.3± 1.0
σ = 41.0± 1.0
WF (2868 entries)
$\frac{\sigma}{\mu}$ = 9.28± 0.23 %
μ = 463.8± 1.0
σ = 43.1± 1.0

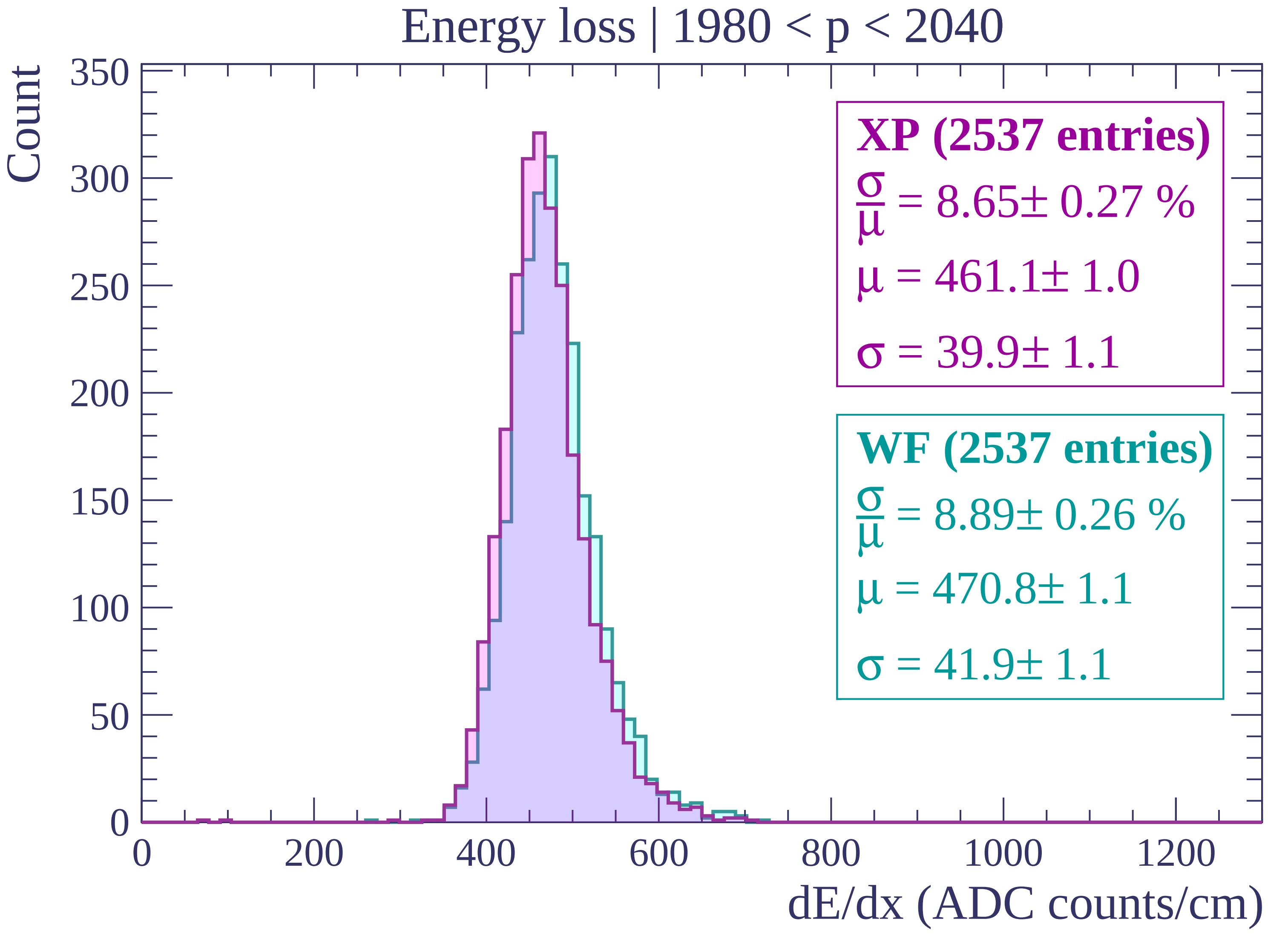

Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2841 entries)
$\frac{\sigma}{\mu}$ = 9.11± 0.19 %
μ = 457.1± 0.9
σ = 41.6± 0.8
WF (2841 entries)
$\frac{\sigma}{\mu}$ = 9.43± 0.24 %
μ = 464.9± 1.0
σ = 43.8± 1.0



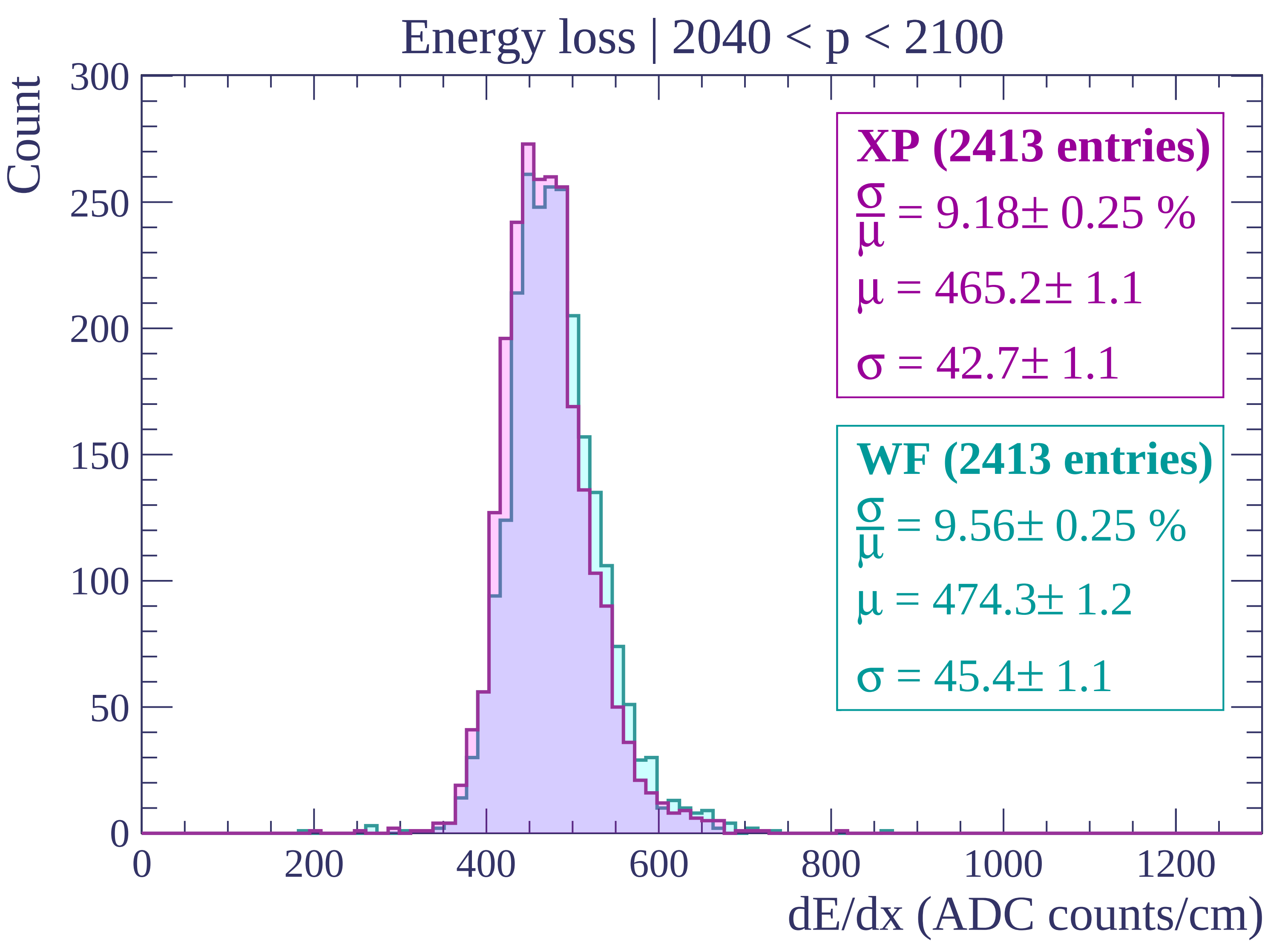
Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2664 entries)
σ/μ = 9.13± 0.22 %
μ = 459.5± 1.0
σ = 42.0± 0.9
WF (2664 entries)
σ/μ = 9.46± 0.22 %
μ = 469.2± 1.0
σ = 44.4± 0.9


Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2677 entries)
$\frac{\sigma}{\mu}$ = 8.85± 0.24 %
μ = 460.6± 1.0
σ = 40.8± 1.0
WF (2677 entries)
$\frac{\sigma}{\mu}$ = 9.08± 0.27 %
μ = 470.0± 1.1
σ = 42.7± 1.2

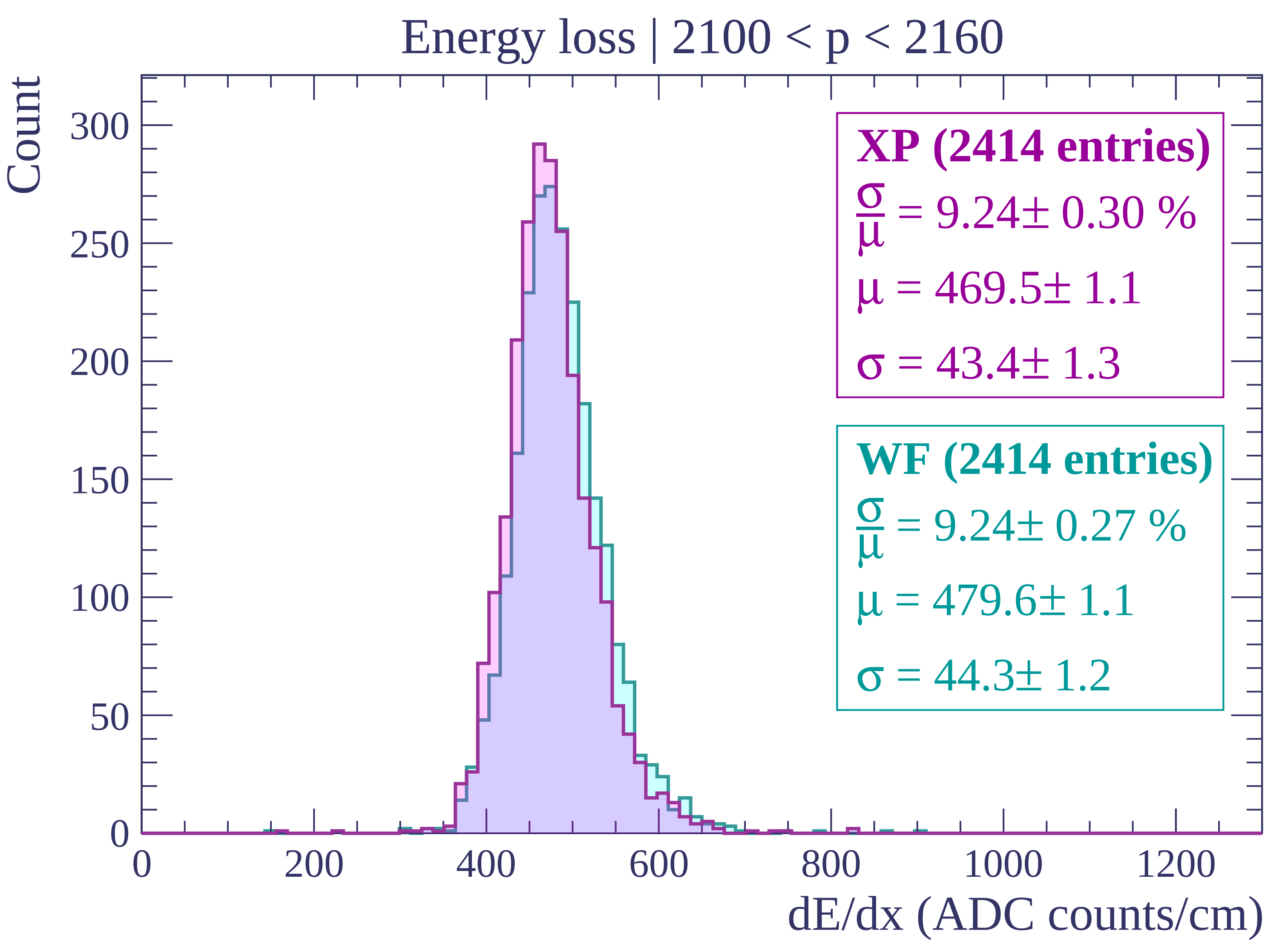

Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2594 entries)
$\frac{\sigma}{\mu}$ = 9.31± 0.24 %
μ = 463.4± 1.0
σ = 43.2± 1.0
WF (2594 entries)
$\frac{\sigma}{\mu}$ = 9.70± 0.22 %
μ = 474.4± 1.0
σ = 46.0± 1.0



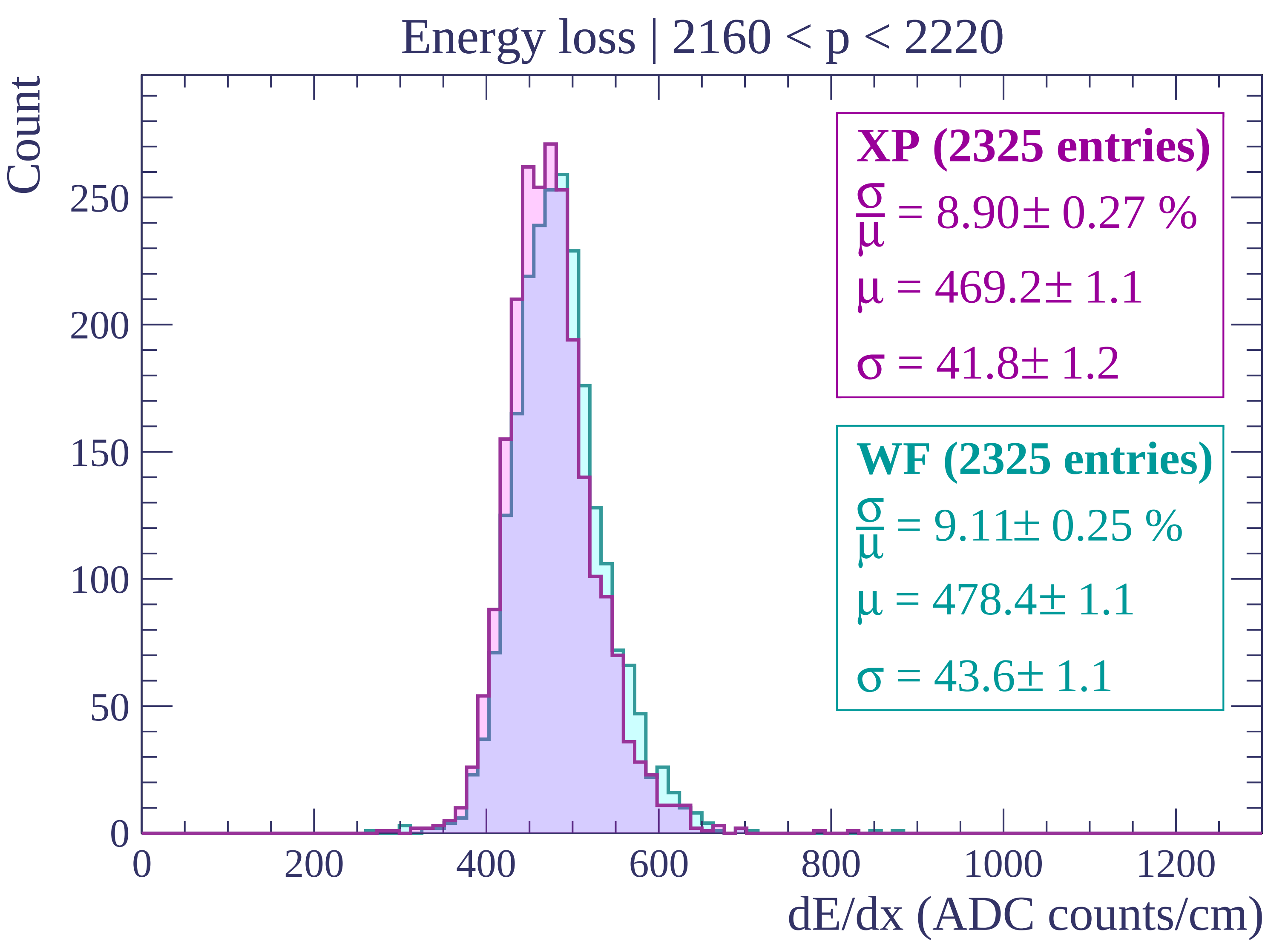
Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2537 entries)
σ/μ = 8.65± 0.27 %
μ = 461.1± 1.0
σ = 39.9± 1.1
WF (2537 entries)
σ/μ = 8.89± 0.26 %
μ = 470.8± 1.1
σ = 41.9± 1.1

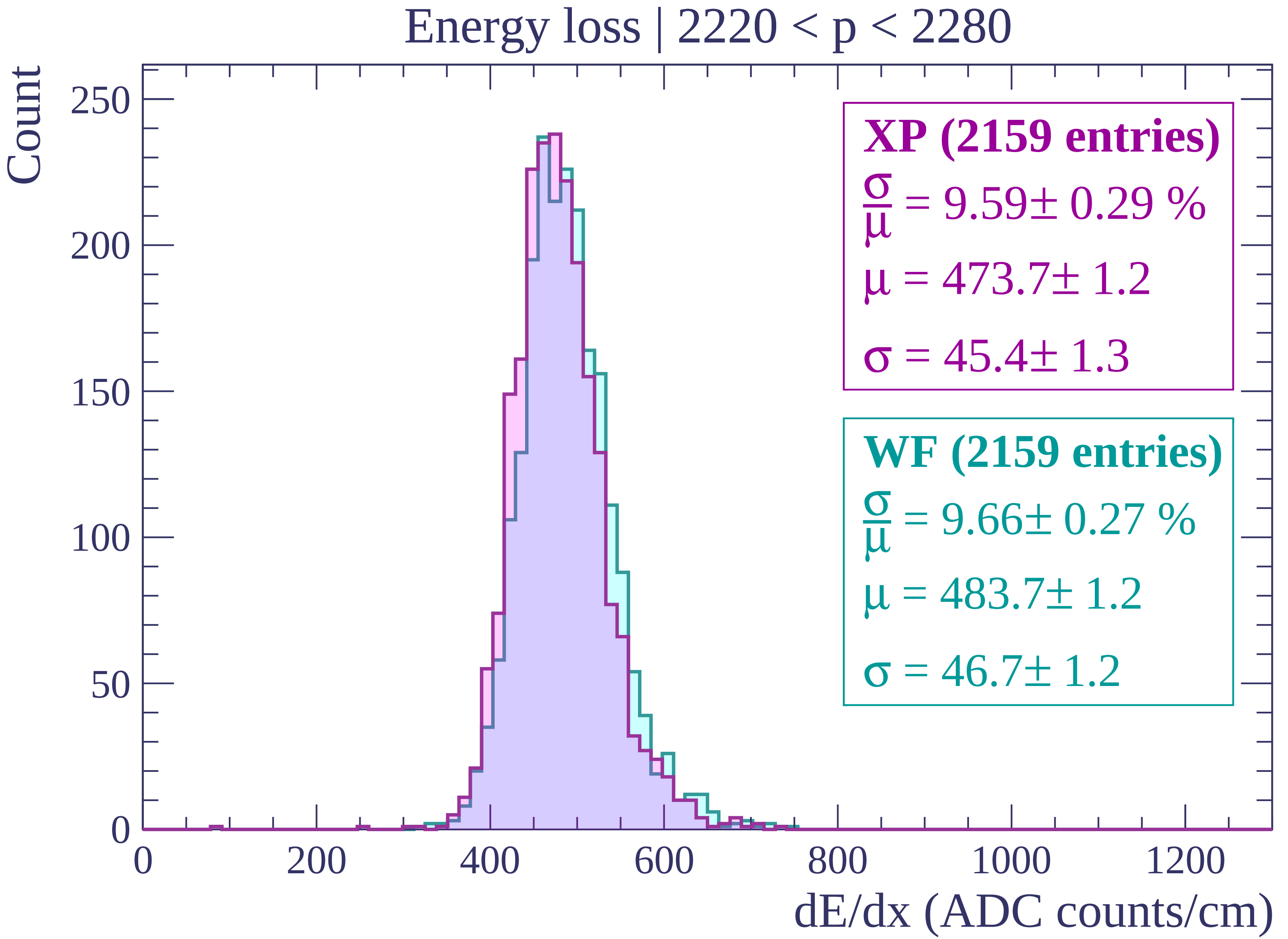


Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2413 entries)
$\frac{\sigma}{\mu}$ = 9.18± 0.25 %
μ = 465.2± 1.1
σ = 42.7± 1.1
WF (2413 entries)
$\frac{\sigma}{\mu}$ = 9.56± 0.25 %
μ = 474.3± 1.2
σ = 45.4± 1.1

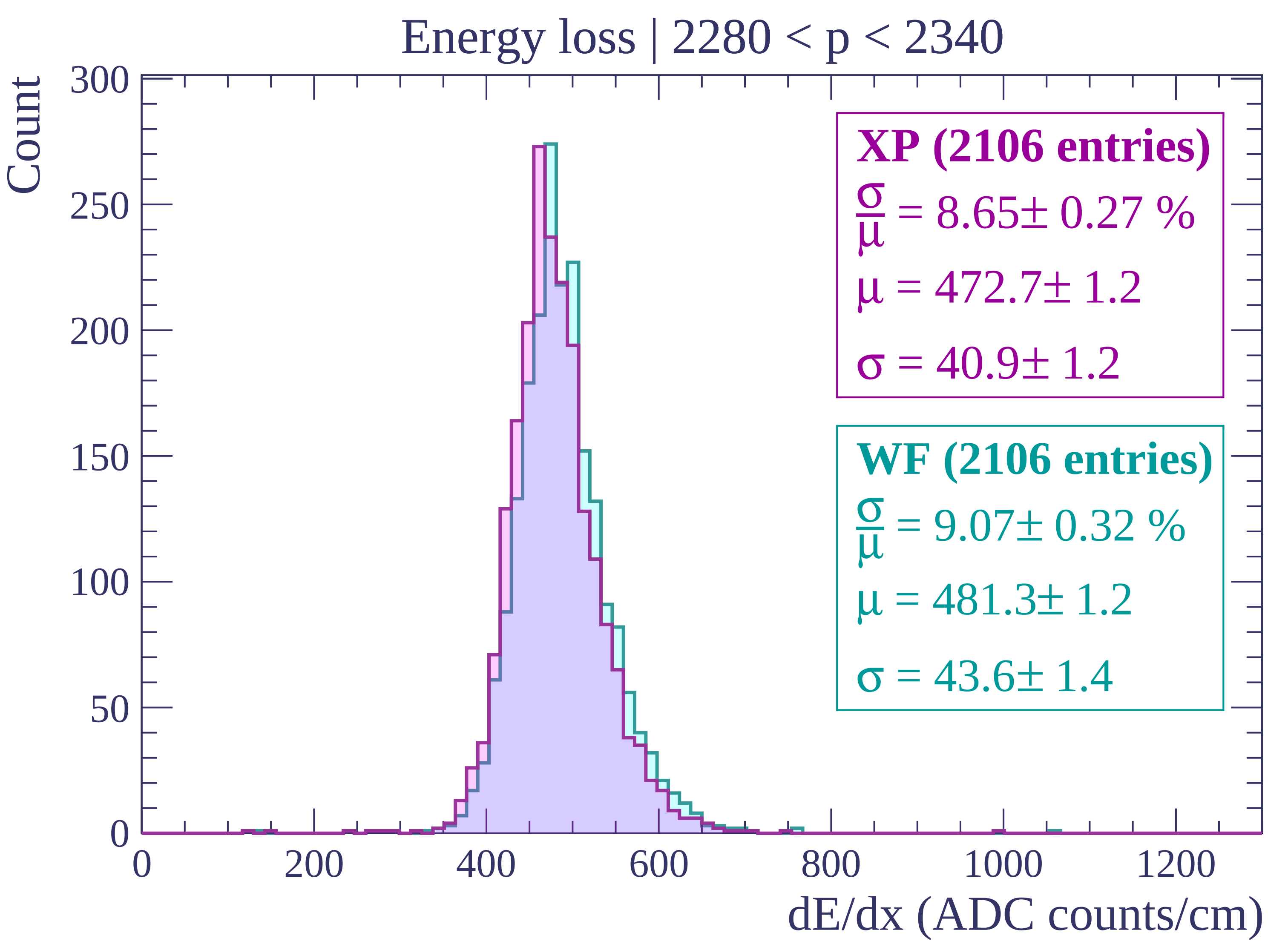


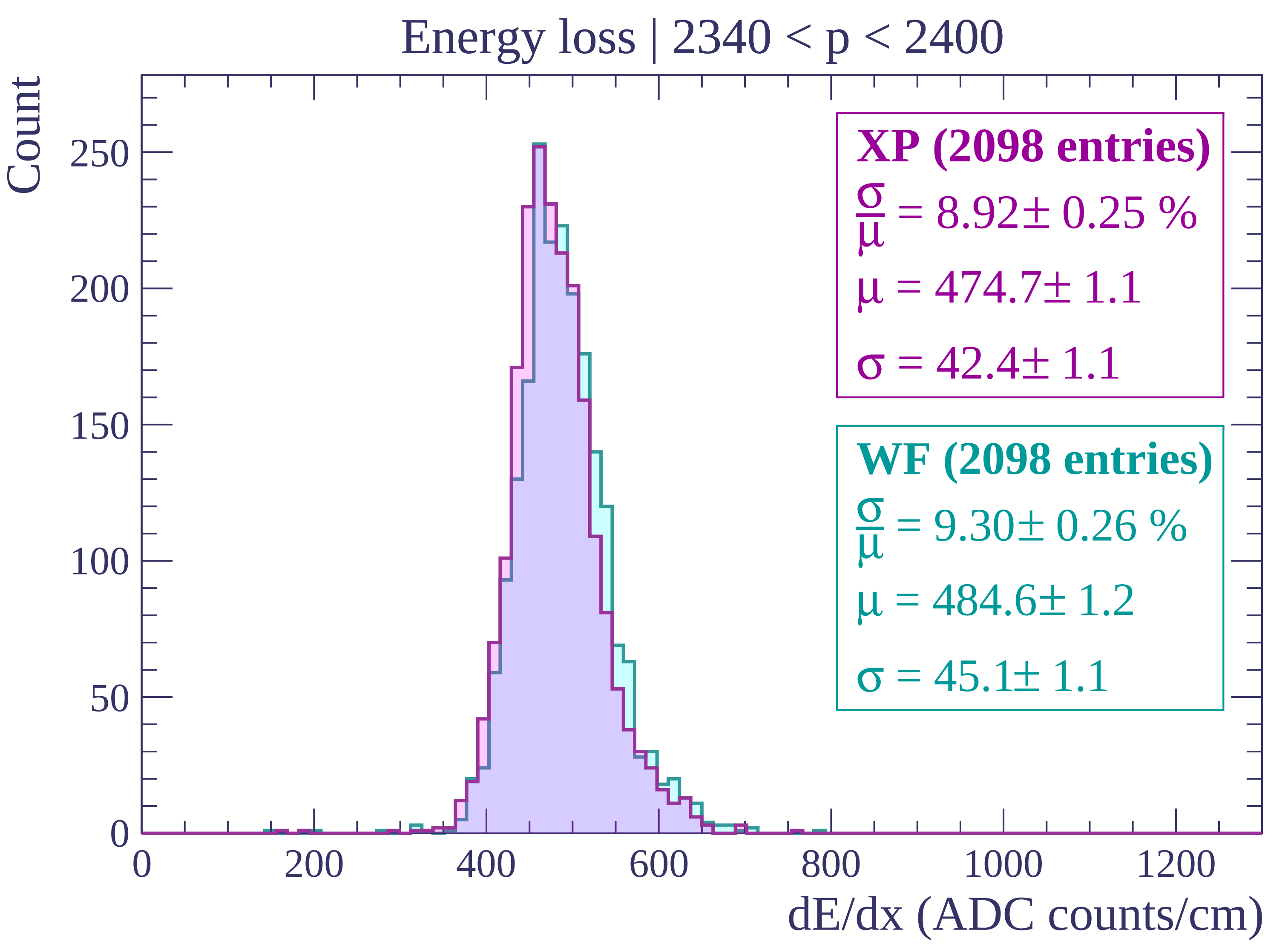
Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2414 entries)
σ/μ = 9.24± 0.30 %
μ = 469.5± 1.1
σ = 43.4± 1.3
WF (2414 entries)
σ/μ = 9.24± 0.27 %
μ = 479.6± 1.1
σ = 44.3± 1.2


Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2325 entries)
σ/μ = 8.90± 0.27 %
μ = 469.2± 1.1
σ = 41.8± 1.2
WF (2325 entries)
σ/μ = 9.11± 0.25 %
μ = 478.4± 1.1
σ = 43.6± 1.1

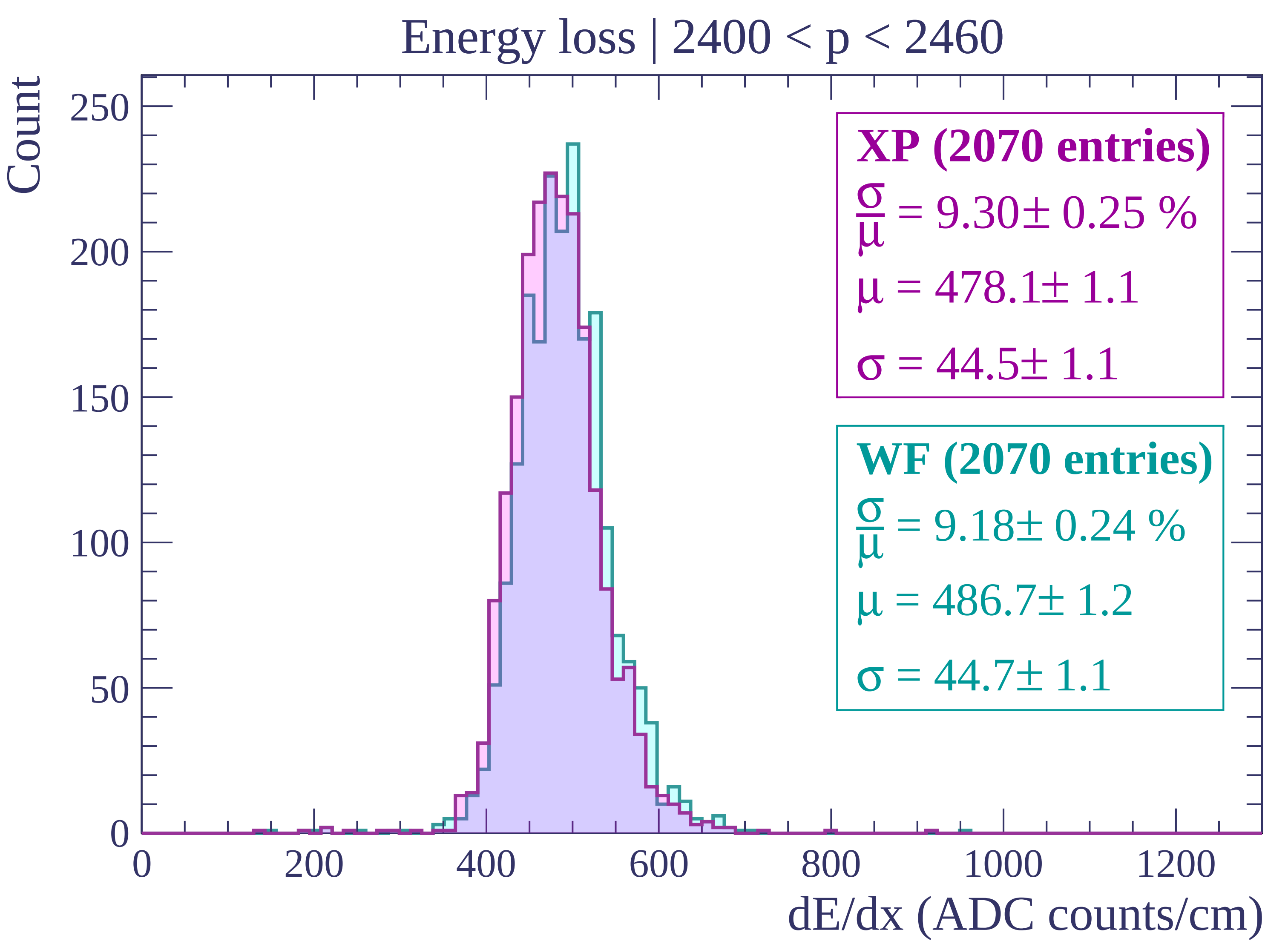


Energy loss | 2220 < p < 2280
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2159 entries)
σ/μ = 9.59± 0.29 %
μ = 473.7± 1.2
σ = 45.4± 1.3
WF (2159 entries)
σ/μ = 9.66± 0.27 %
μ = 483.7± 1.2
σ = 46.7± 1.2

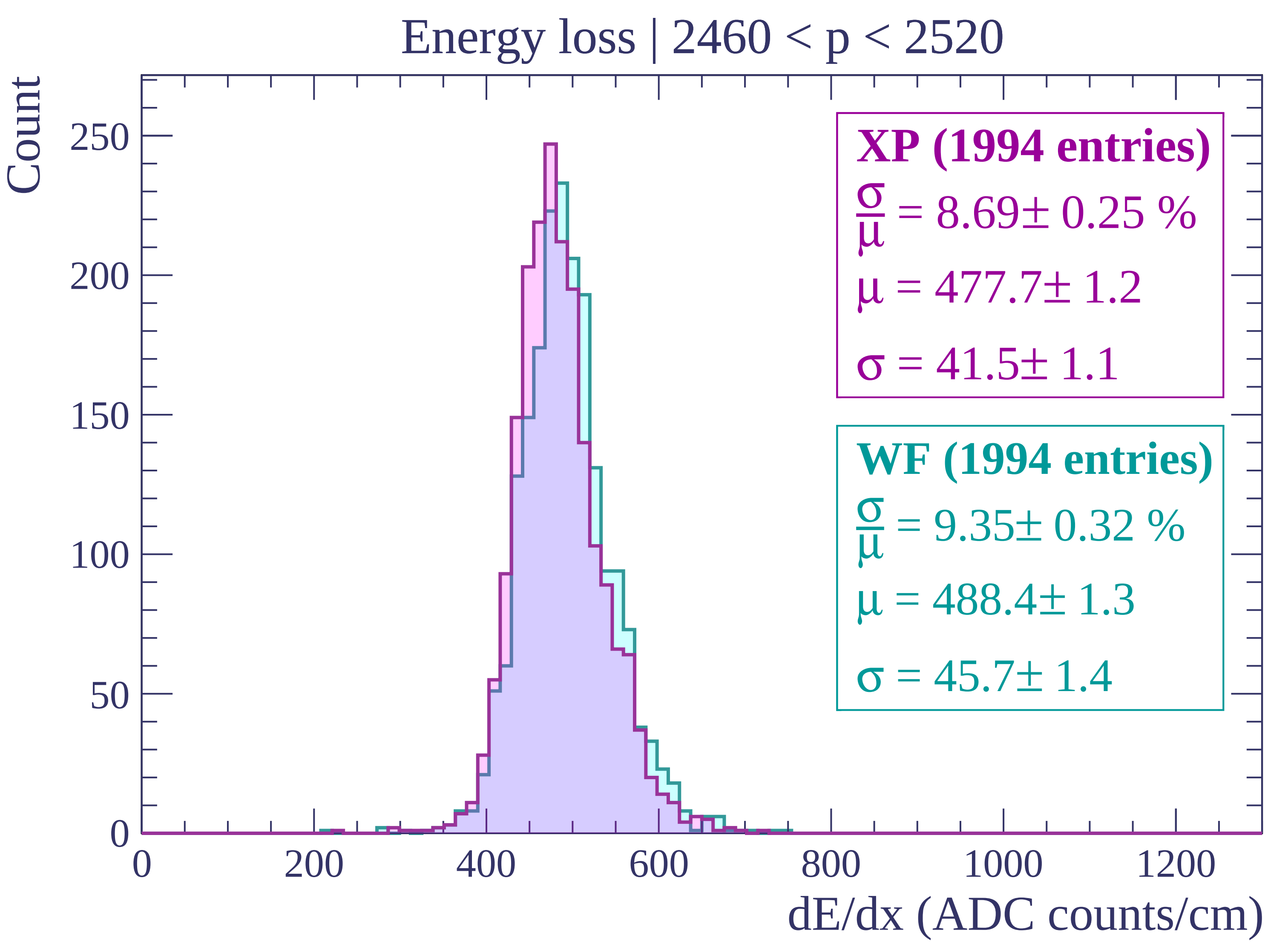

Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2106 entries)
σ/μ = 8.65± 0.27 %
μ = 472.7± 1.2
σ = 40.9± 1.2
WF (2106 entries)
σ/μ = 9.07± 0.32 %
μ = 481.3± 1.2
σ = 43.6± 1.4

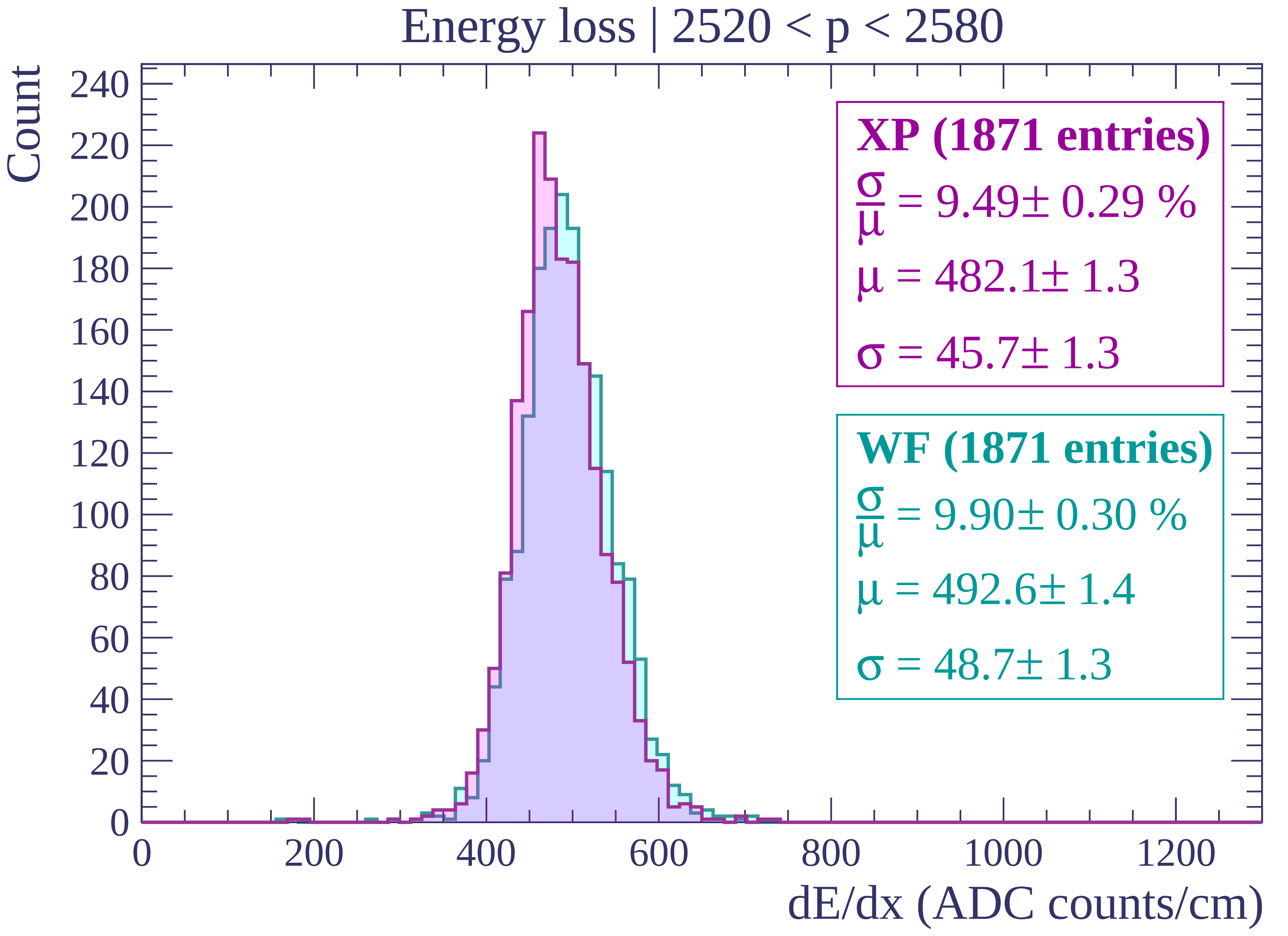


Energy loss | 2340 < p < 2400
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2098 entries)
$\frac{\sigma}{\mu}$ = 8.92± 0.25 %
μ = 474.7± 1.1
σ = 42.4± 1.1
WF (2098 entries)
$\frac{\sigma}{\mu}$ = 9.30± 0.26 %
μ = 484.6± 1.2
σ = 45.1± 1.1

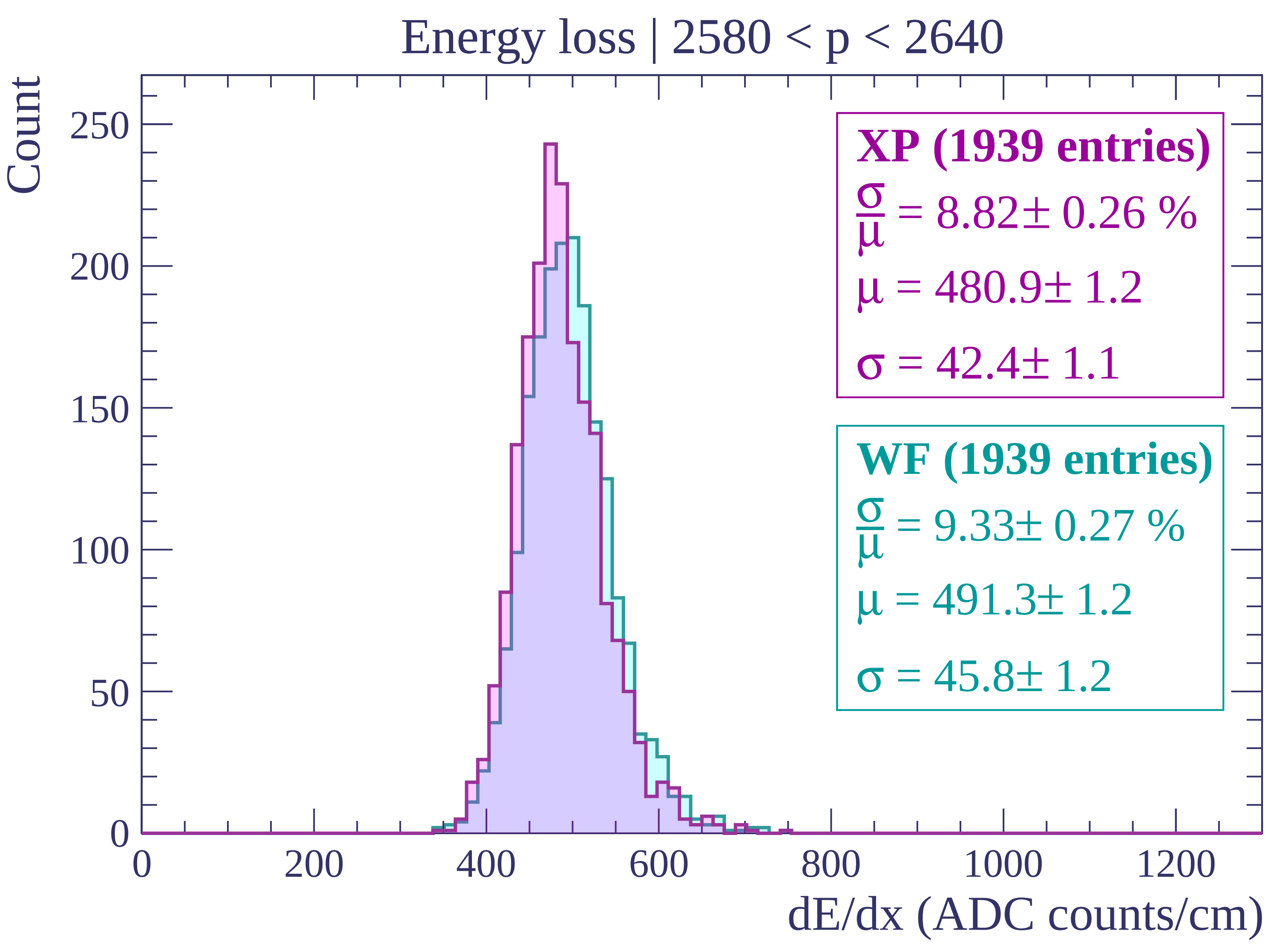


Energy loss | 2400 < p < 2460
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2070 entries)
$\frac{\sigma}{\mu}$ = 9.30± 0.25 %
μ = 478.1± 1.1
σ = 44.5± 1.1
WF (2070 entries)
$\frac{\sigma}{\mu}$ = 9.18± 0.24 %
μ = 486.7± 1.2
σ = 44.7± 1.1

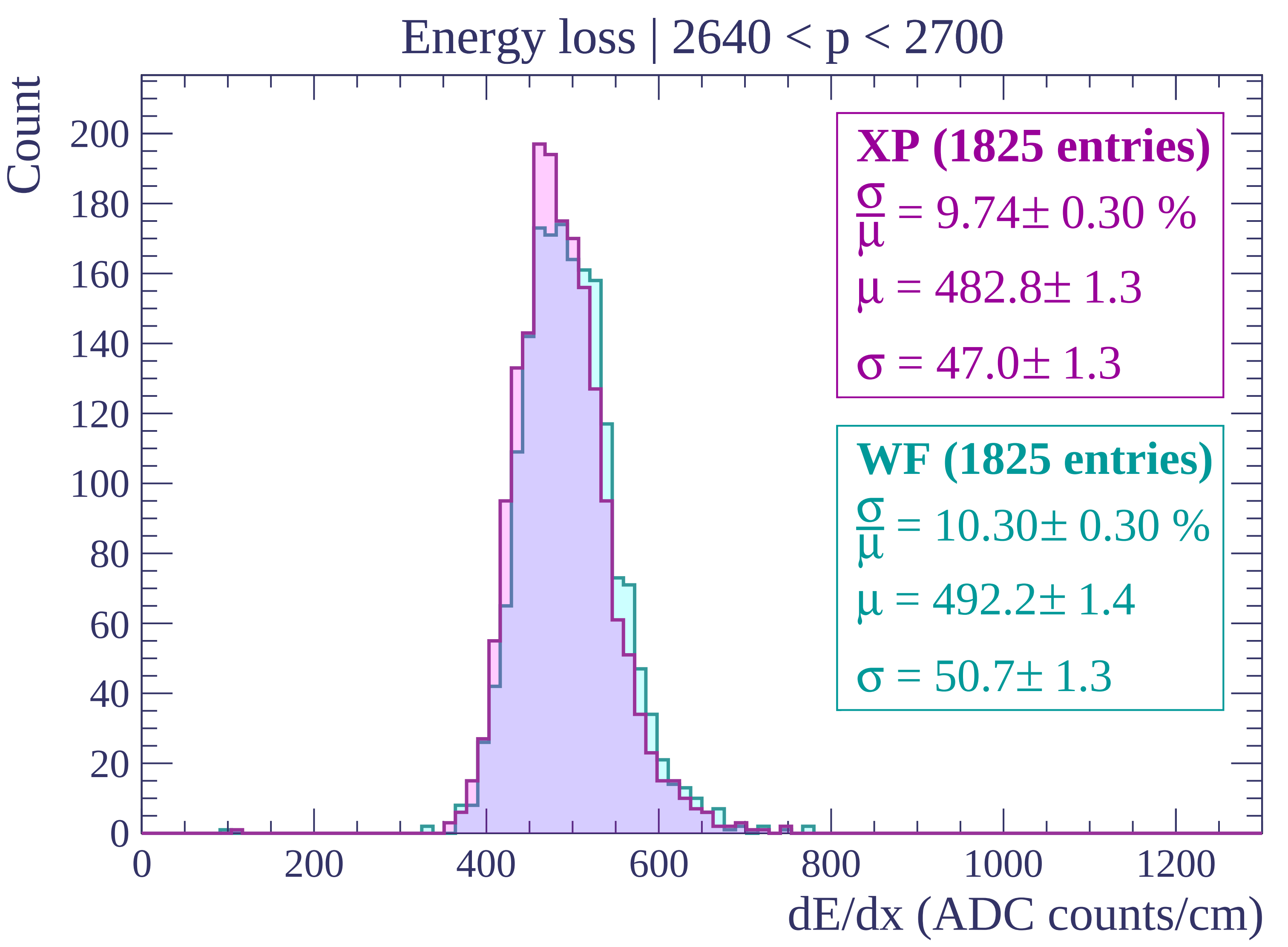

Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
XP (1994 entries)
$\frac{\sigma}{\mu}$ = 8.69± 0.25 %
μ = 477.7± 1.2
σ = 41.5± 1.1
WF (1994 entries)
$\frac{\sigma}{\mu}$ = 9.35± 0.32 %
μ = 488.4± 1.3
σ = 45.7± 1.4
0
50
100
150
200
250
0
200
400
600
800
1000
1200

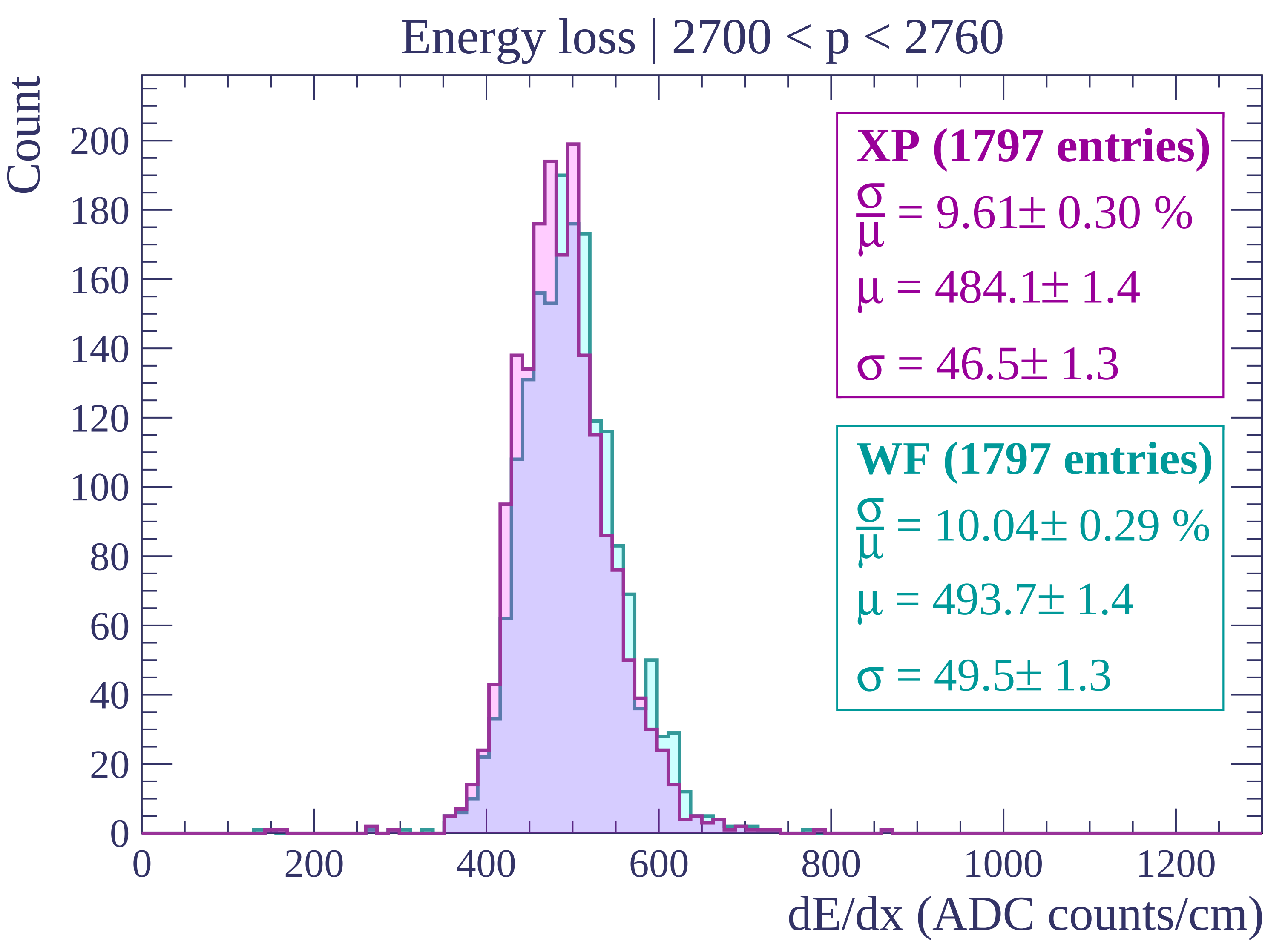


Energy loss | 2520 < p < 2580
Count
240
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1871 entries)
σ/μ = 9.49± 0.29 %
μ = 482.1± 1.3
σ = 45.7± 1.3
WF (1871 entries)
σ/μ = 9.90± 0.30 %
μ = 492.6± 1.4
σ = 48.7± 1.3

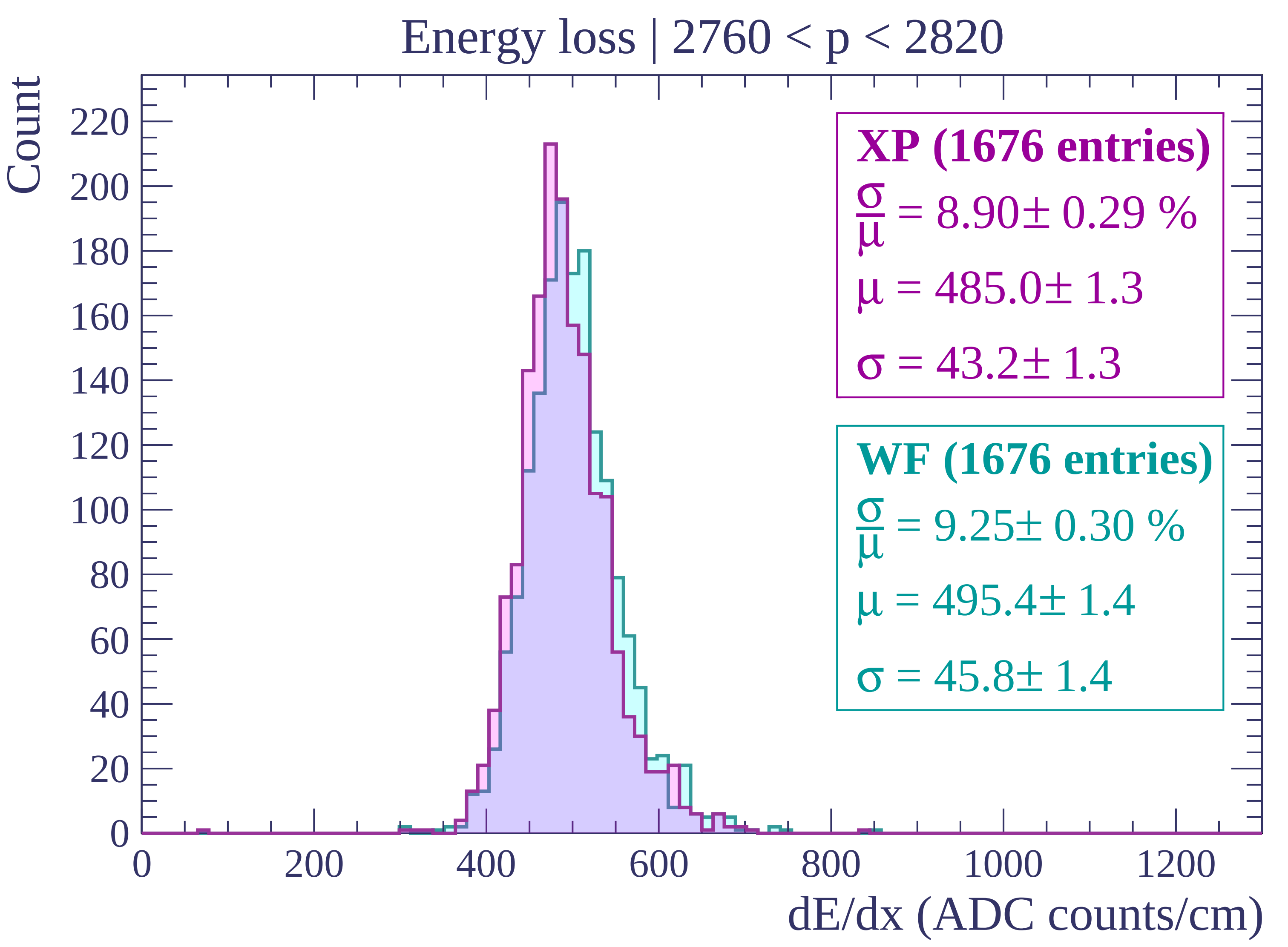


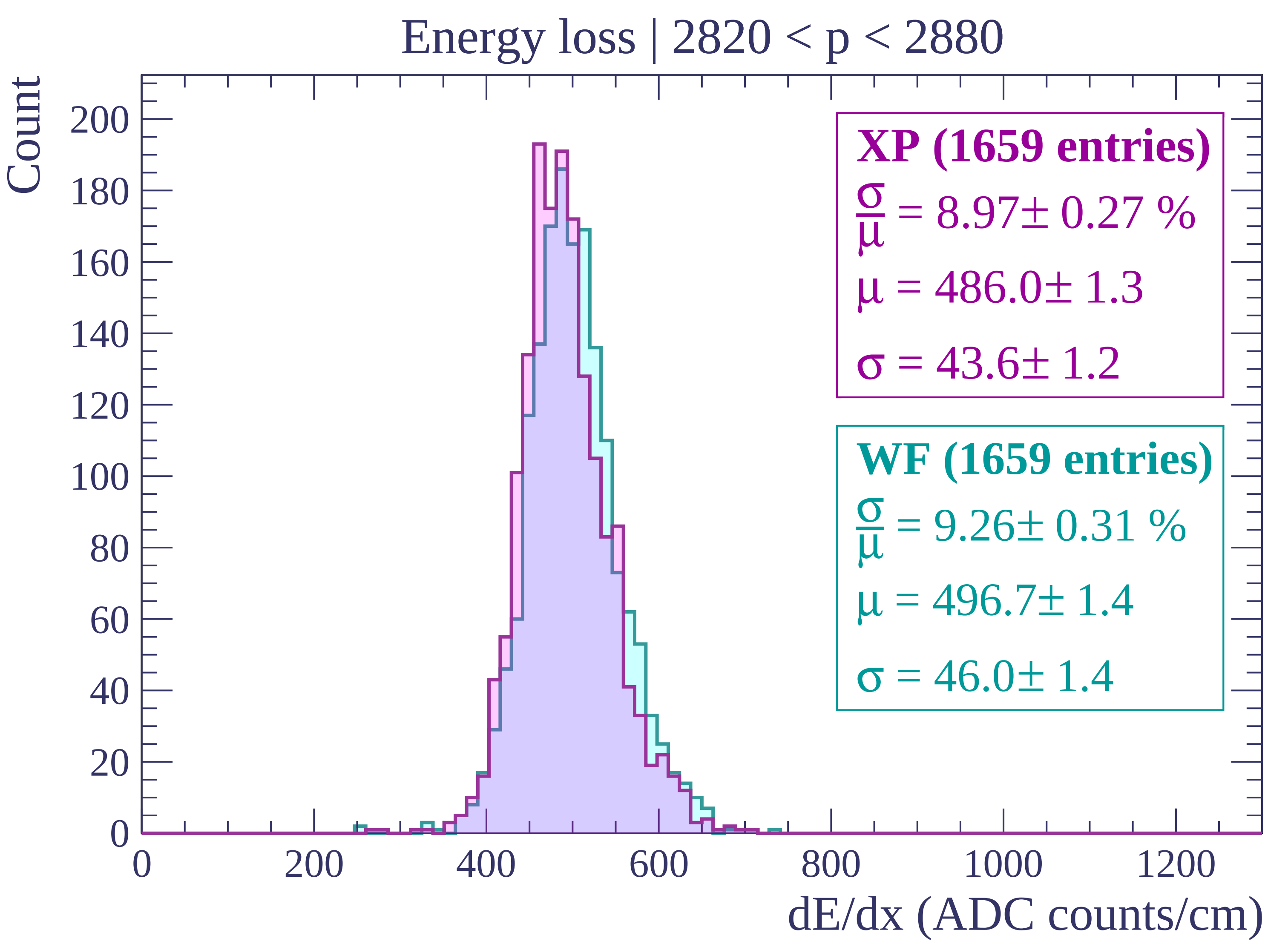
Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP (1939 entries)
σ/μ = 8.82± 0.26 %
μ = 480.9± 1.2
σ = 42.4± 1.1
WF (1939 entries)
σ/μ = 9.33± 0.27 %
μ = 491.3± 1.2
σ = 45.8± 1.2


Energy loss | 2640 < p < 2700
Count
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1825 entries)
σ/μ = 9.74± 0.30 %
μ = 482.8± 1.3
σ = 47.0± 1.3
WF (1825 entries)
σ/μ = 10.30± 0.30 %
μ = 492.2± 1.4
σ = 50.7± 1.3

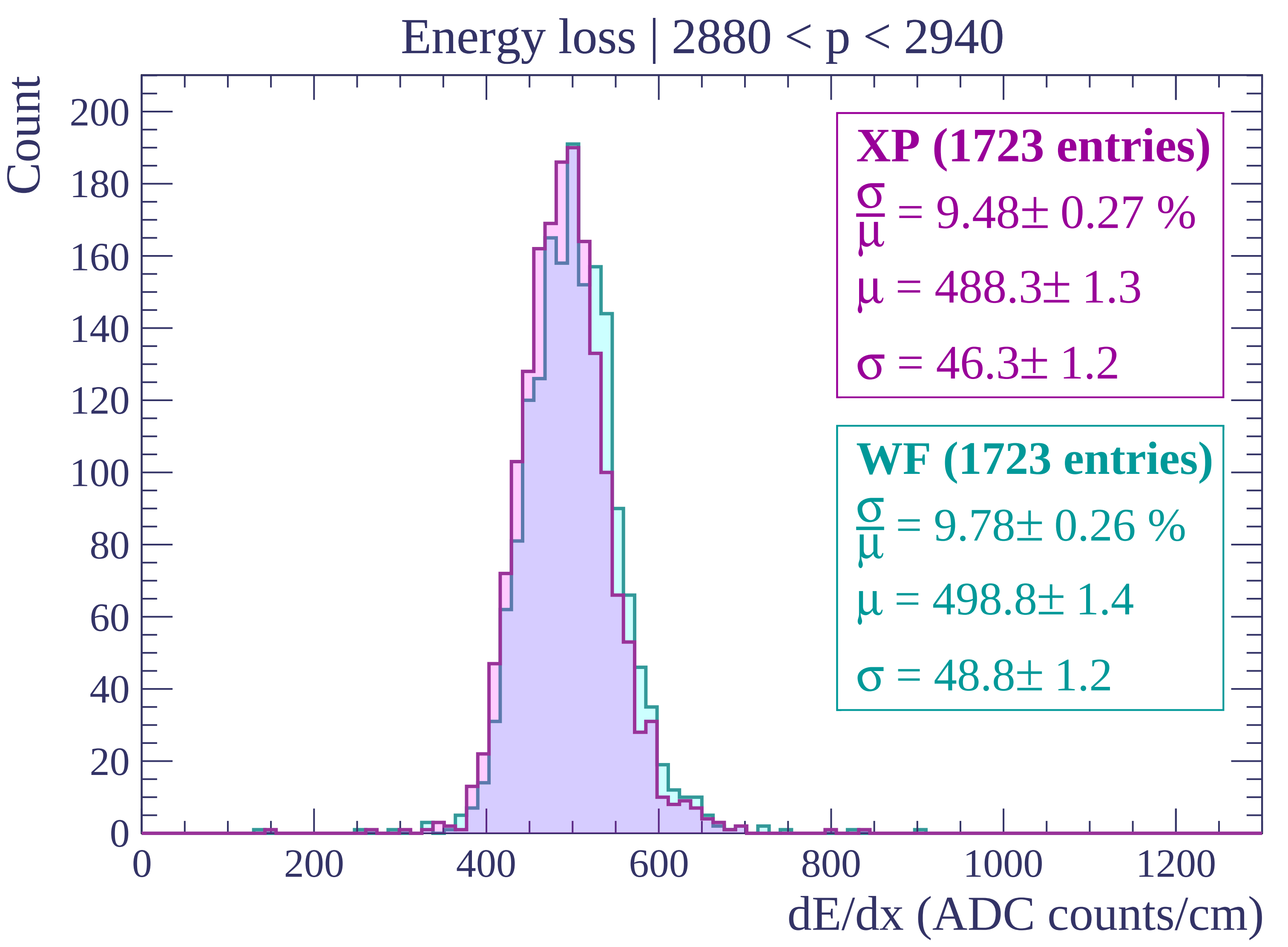


Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
XP (1797 entries)
σ/μ = 9.61± 0.30 %
μ = 484.1± 1.4
σ = 46.5± 1.3
WF (1797 entries)
σ/μ = 10.04± 0.29 %
μ = 493.7± 1.4
σ = 49.5± 1.3

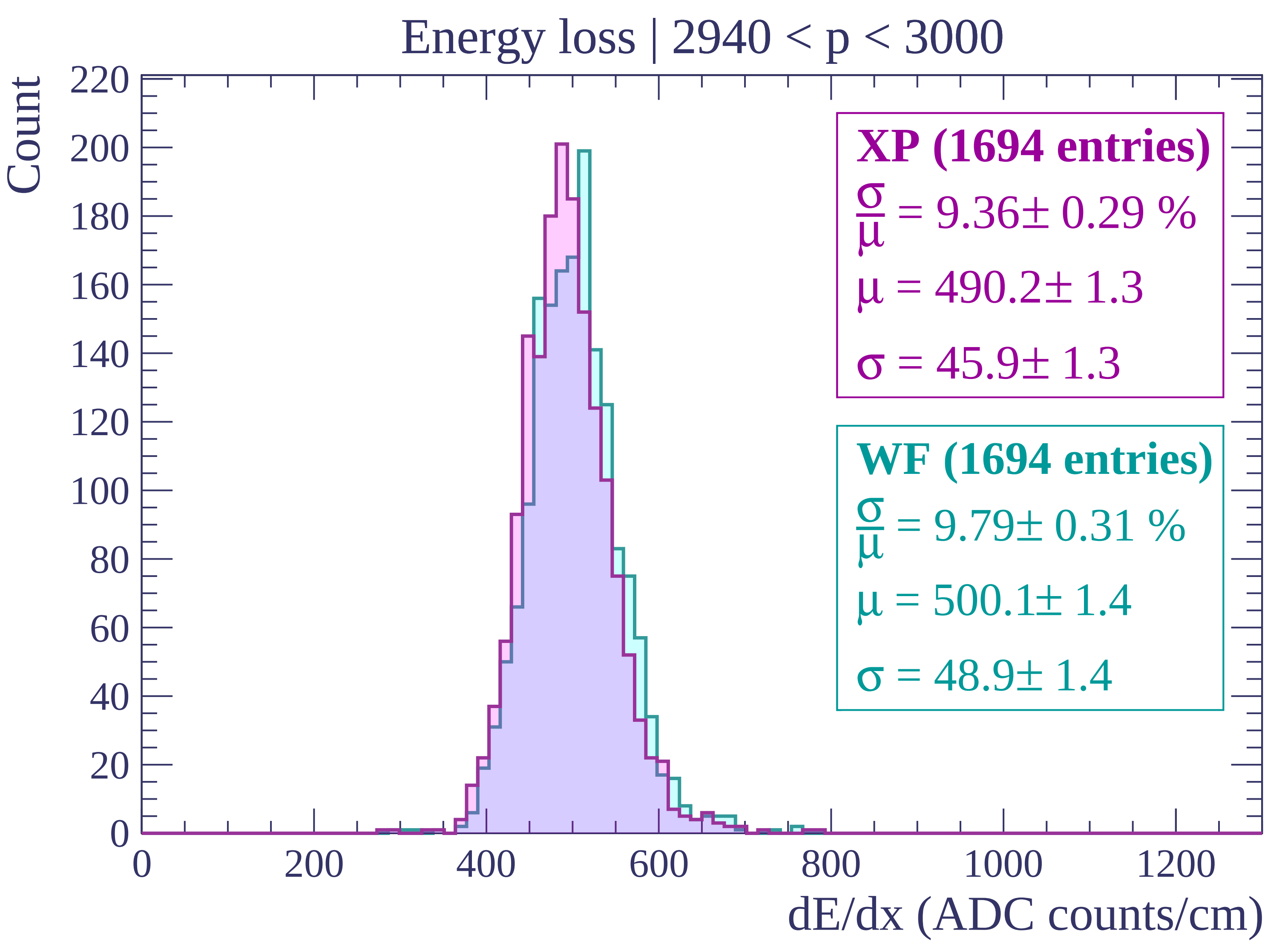

Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP (1676 entries)
σ/μ = 8.90± 0.29 %
μ = 485.0± 1.3
σ = 43.2± 1.3
WF (1676 entries)
σ/μ = 9.25± 0.30 %
μ = 495.4± 1.4
σ = 45.8± 1.4



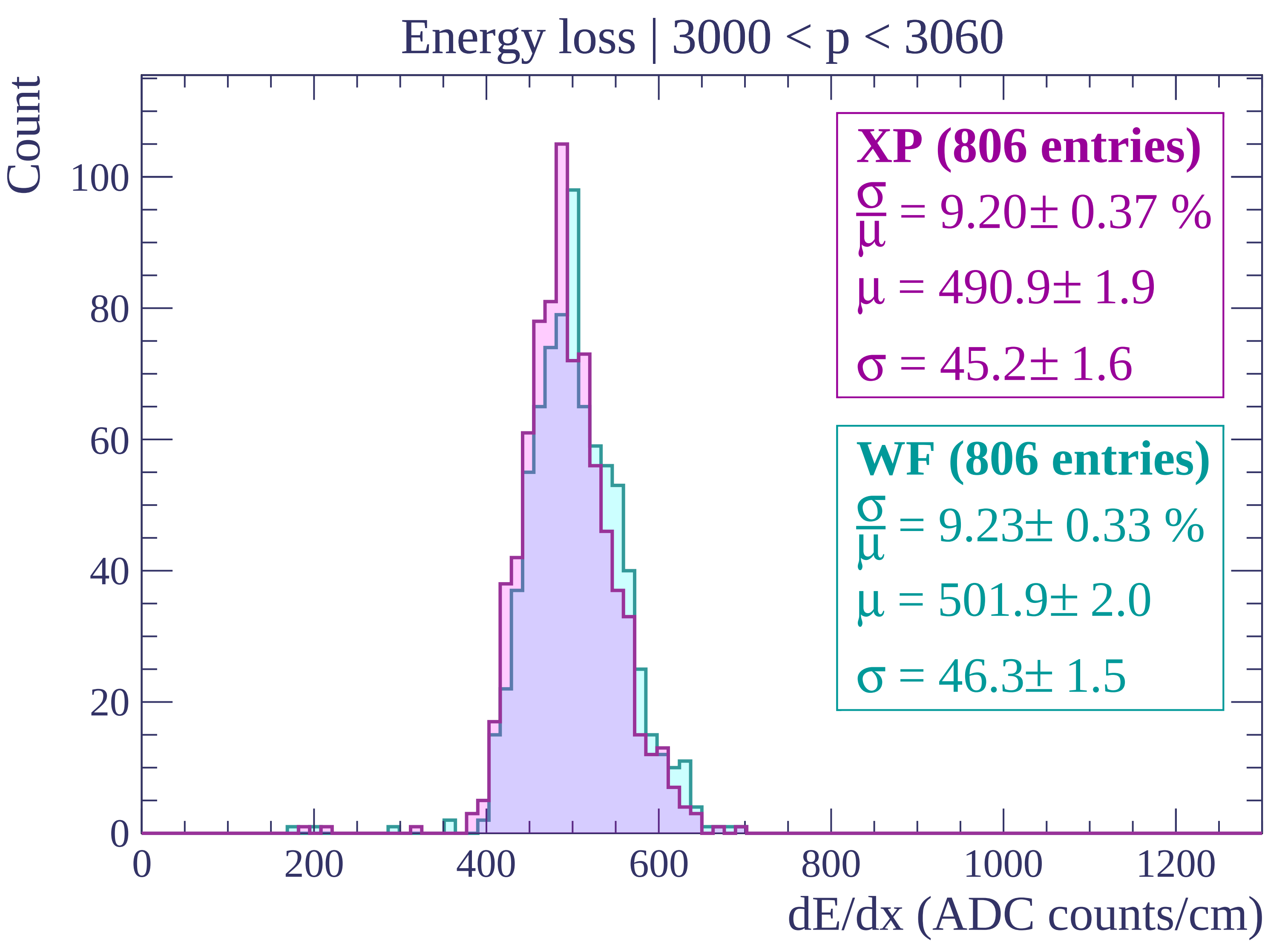
Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
XP (1659 entries)
σ/μ = 8.97± 0.27 %
μ = 486.0± 1.3
σ = 43.6± 1.2
WF (1659 entries)
σ/μ = 9.26± 0.31 %
μ = 496.7± 1.4
σ = 46.0± 1.4



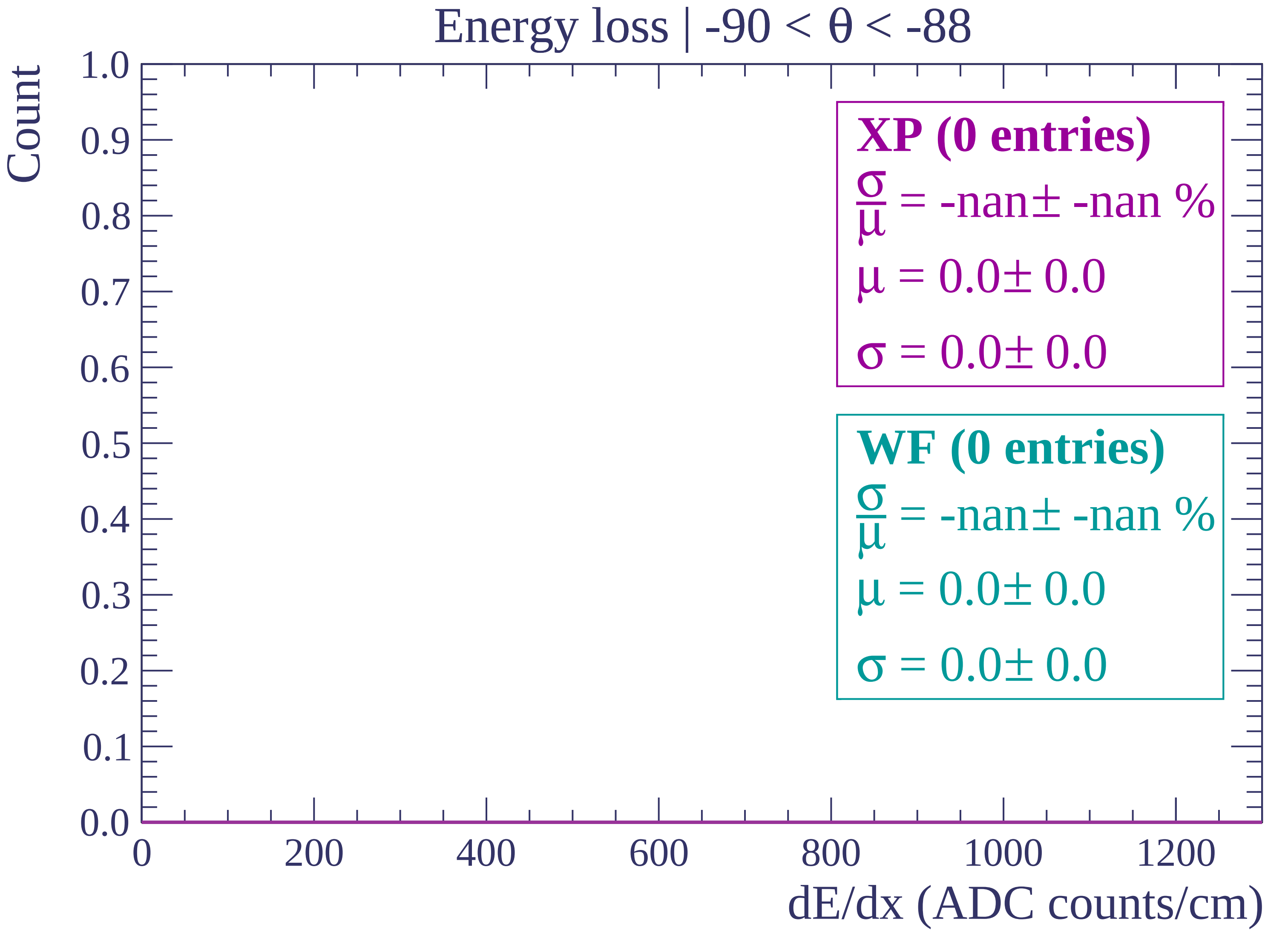
Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (1723 entries)
$\frac{\sigma}{\mu}$ = 9.48± 0.27 %
μ = 488.3± 1.3
σ = 46.3± 1.2
WF (1723 entries)
$\frac{\sigma}{\mu}$ = 9.78± 0.26 %
μ = 498.8± 1.4
σ = 48.8± 1.2



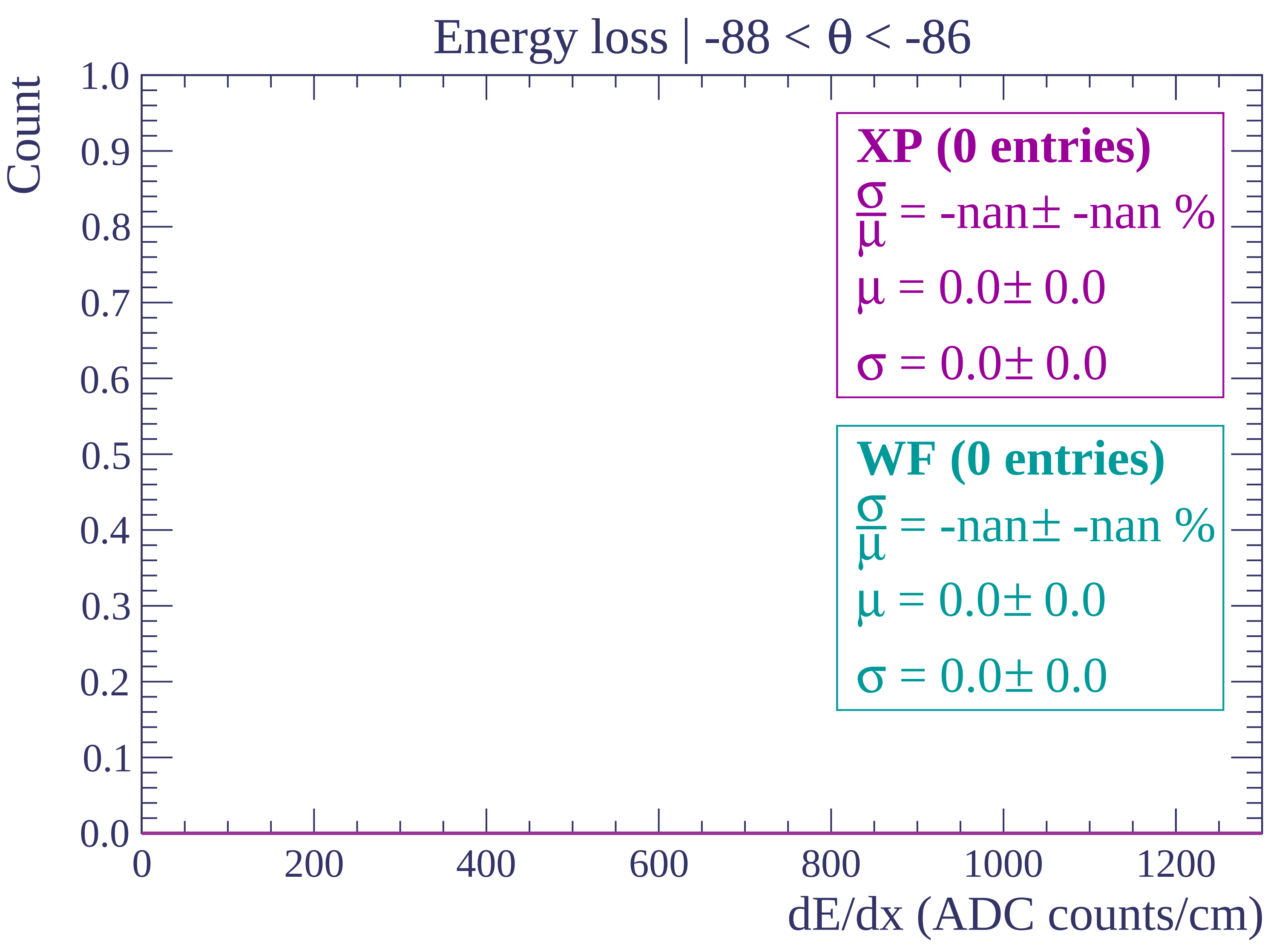
Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (1694 entries)
σ/μ = 9.36± 0.29 %
μ = 490.2± 1.3
σ = 45.9± 1.3
WF (1694 entries)
σ/μ = 9.79± 0.31 %
μ = 500.1± 1.4
σ = 48.9± 1.4



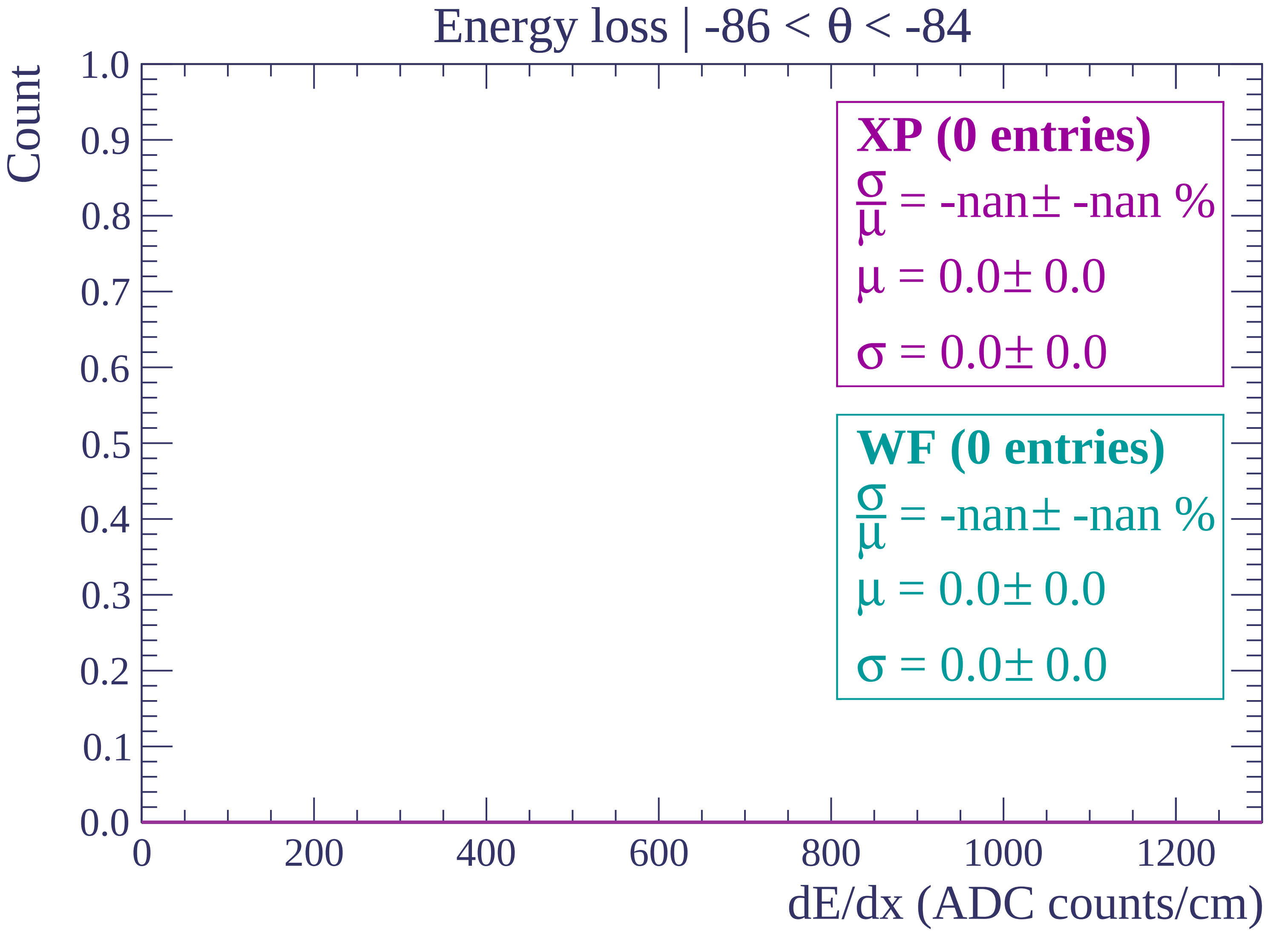
Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (806 entries)
$\frac{\sigma}{\mu}$ = 9.20± 0.37 %
μ = 490.9± 1.9
σ = 45.2± 1.6
WF (806 entries)
$\frac{\sigma}{\mu}$ = 9.23± 0.33 %
μ = 501.9± 2.0
σ = 46.3± 1.5



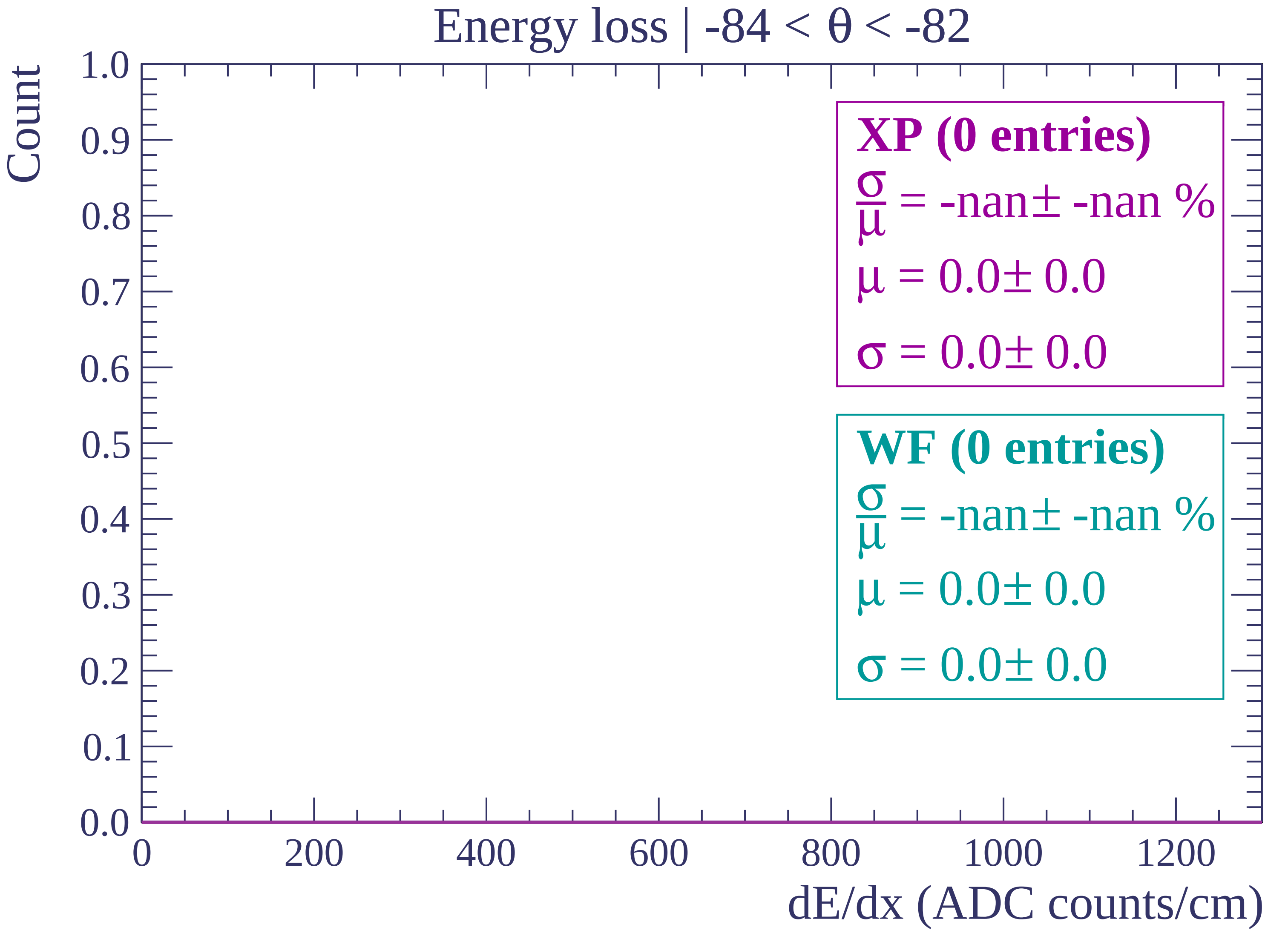
Energy loss | -90 < θ < -88
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

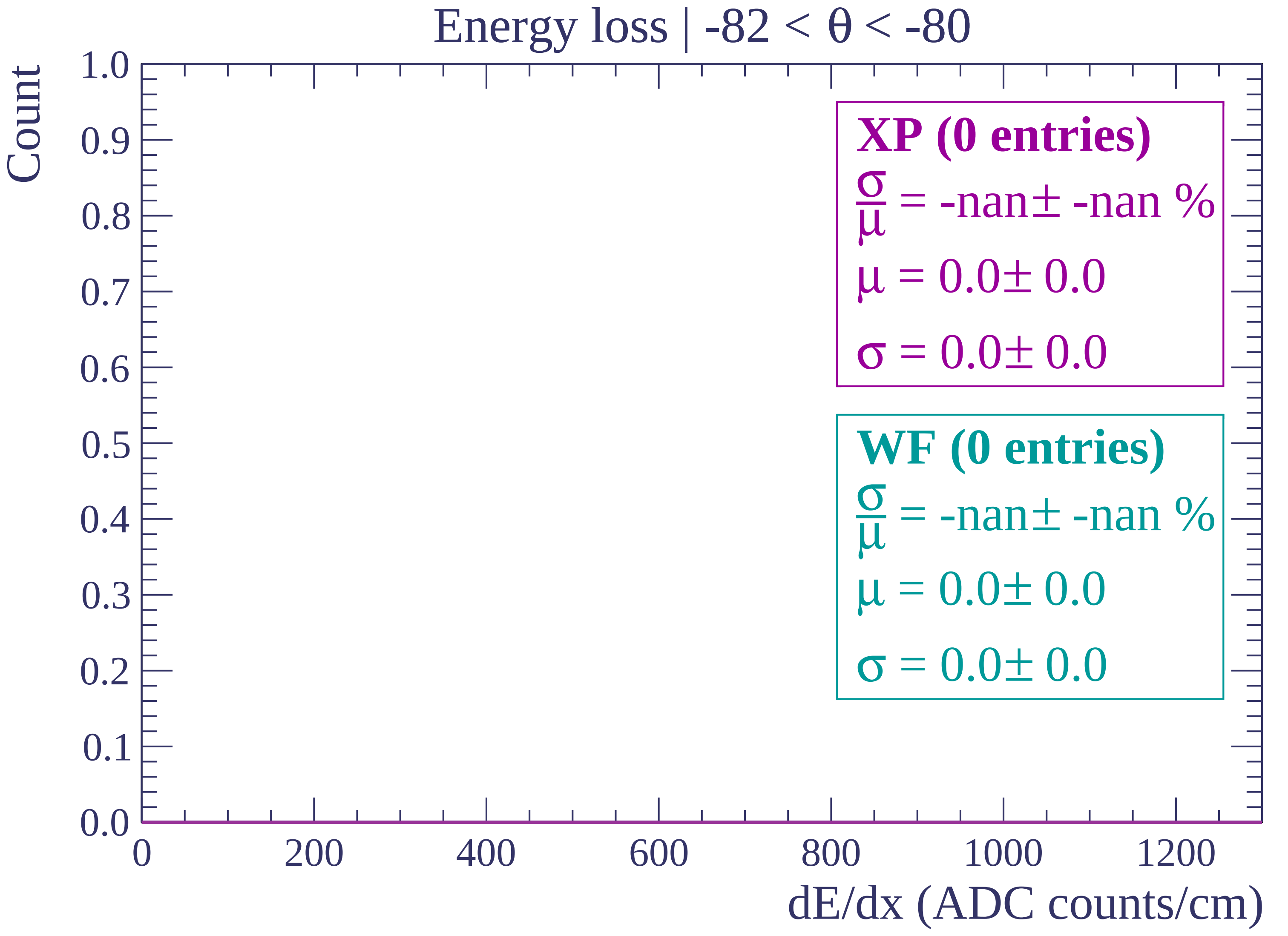


Energy loss | -88 < θ < -86
Count
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200

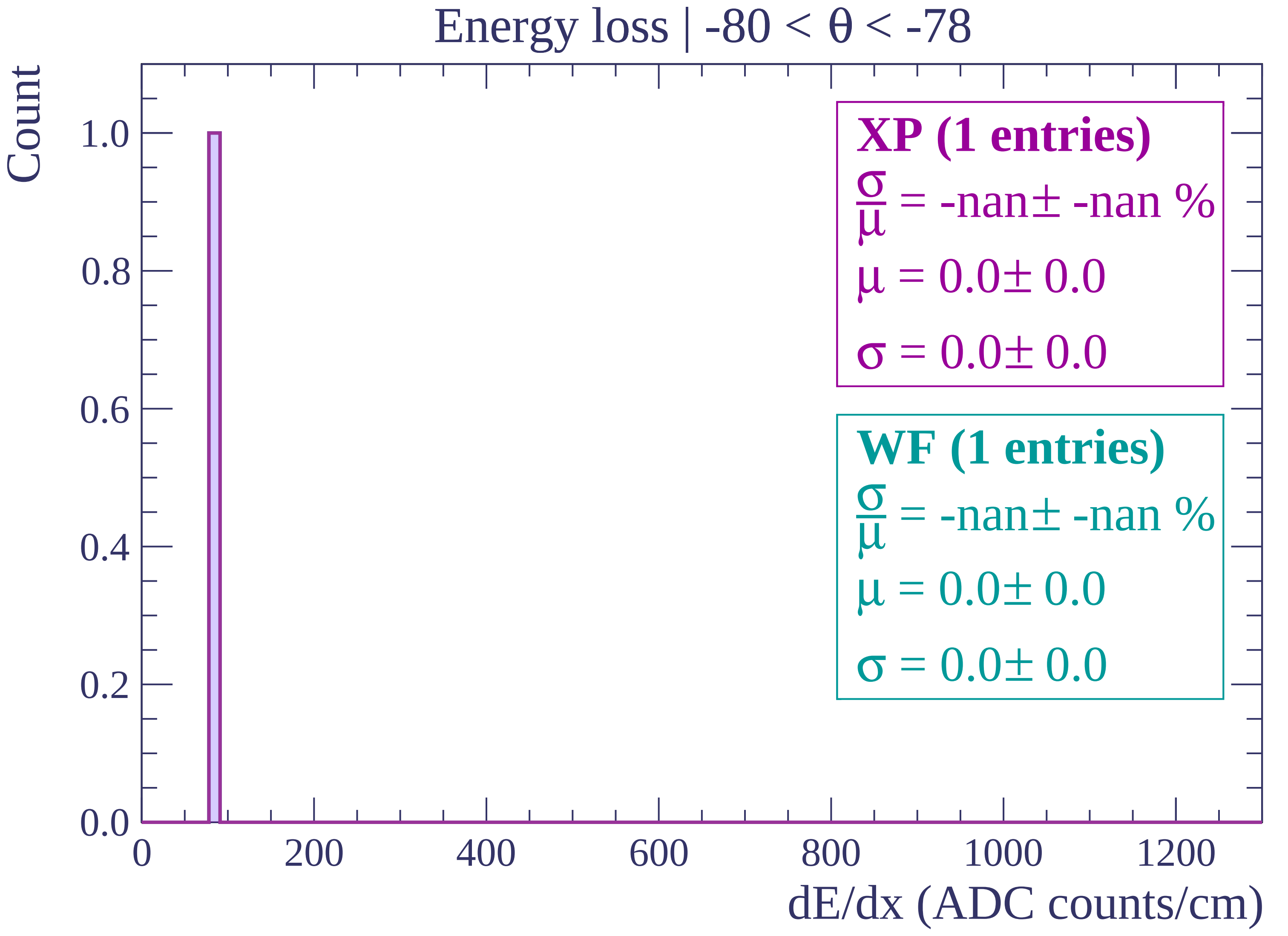


Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

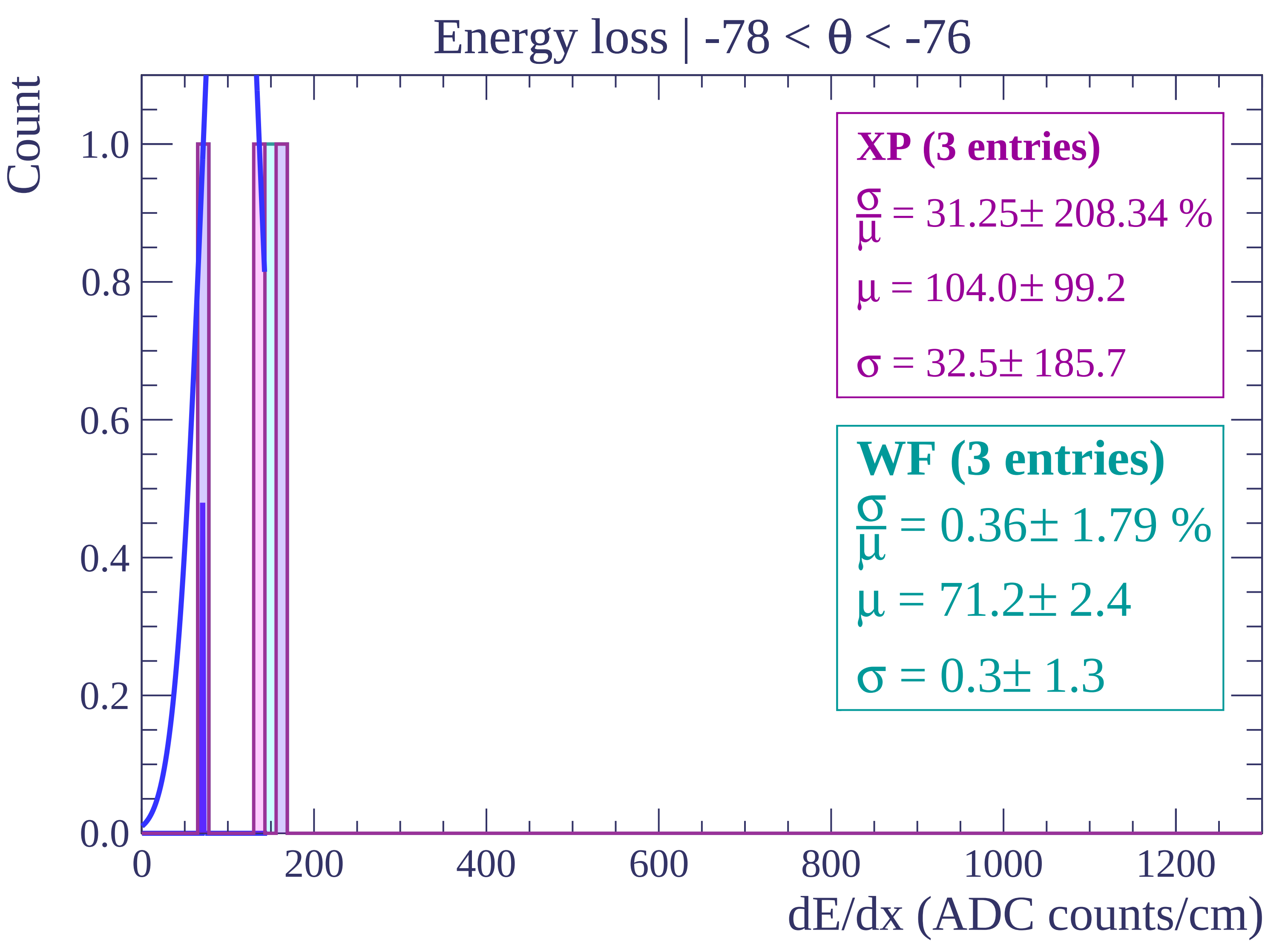

Energy loss | -84 < θ < -82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

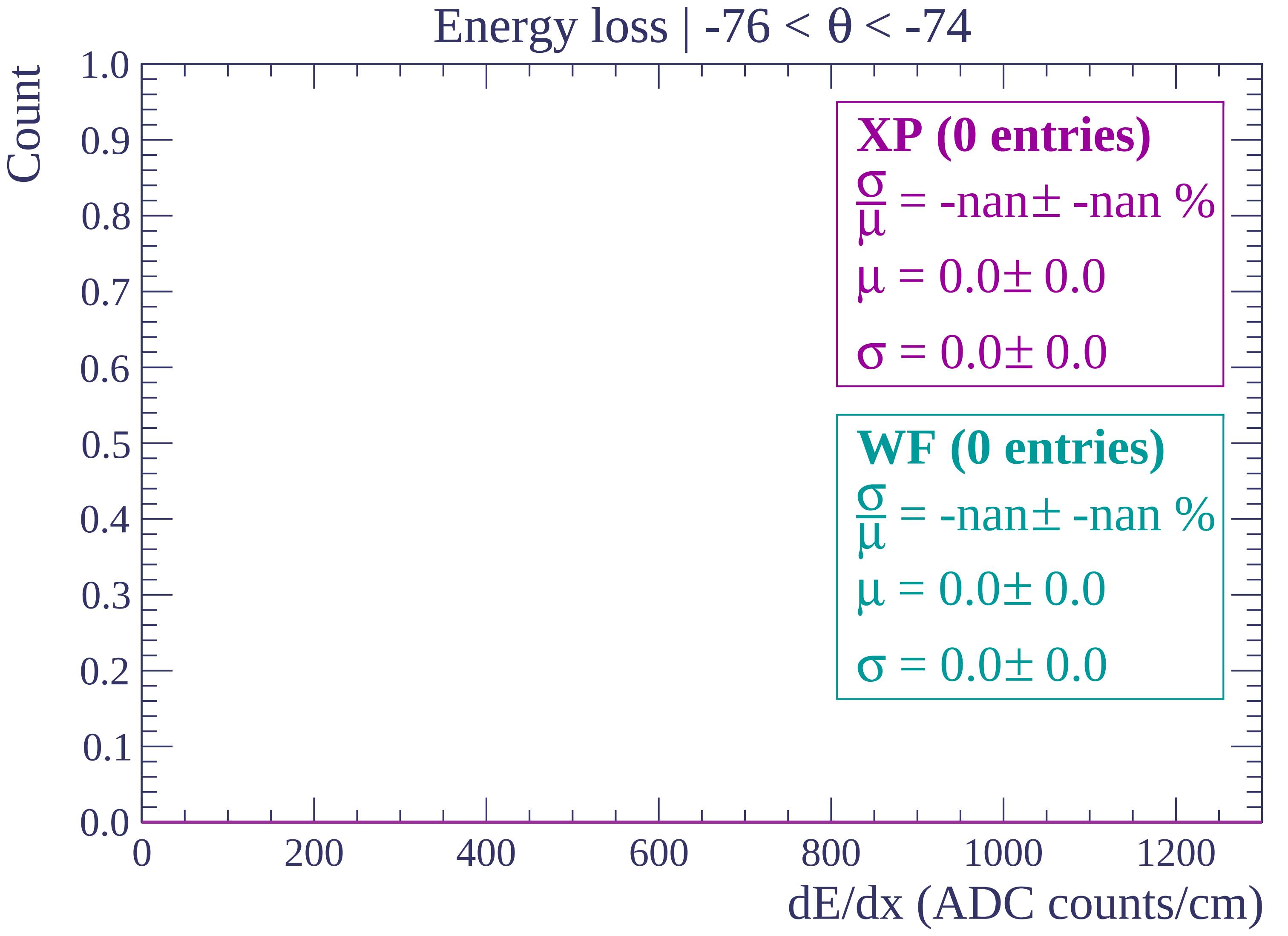


Energy loss | -82 < θ < -80
Count
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200

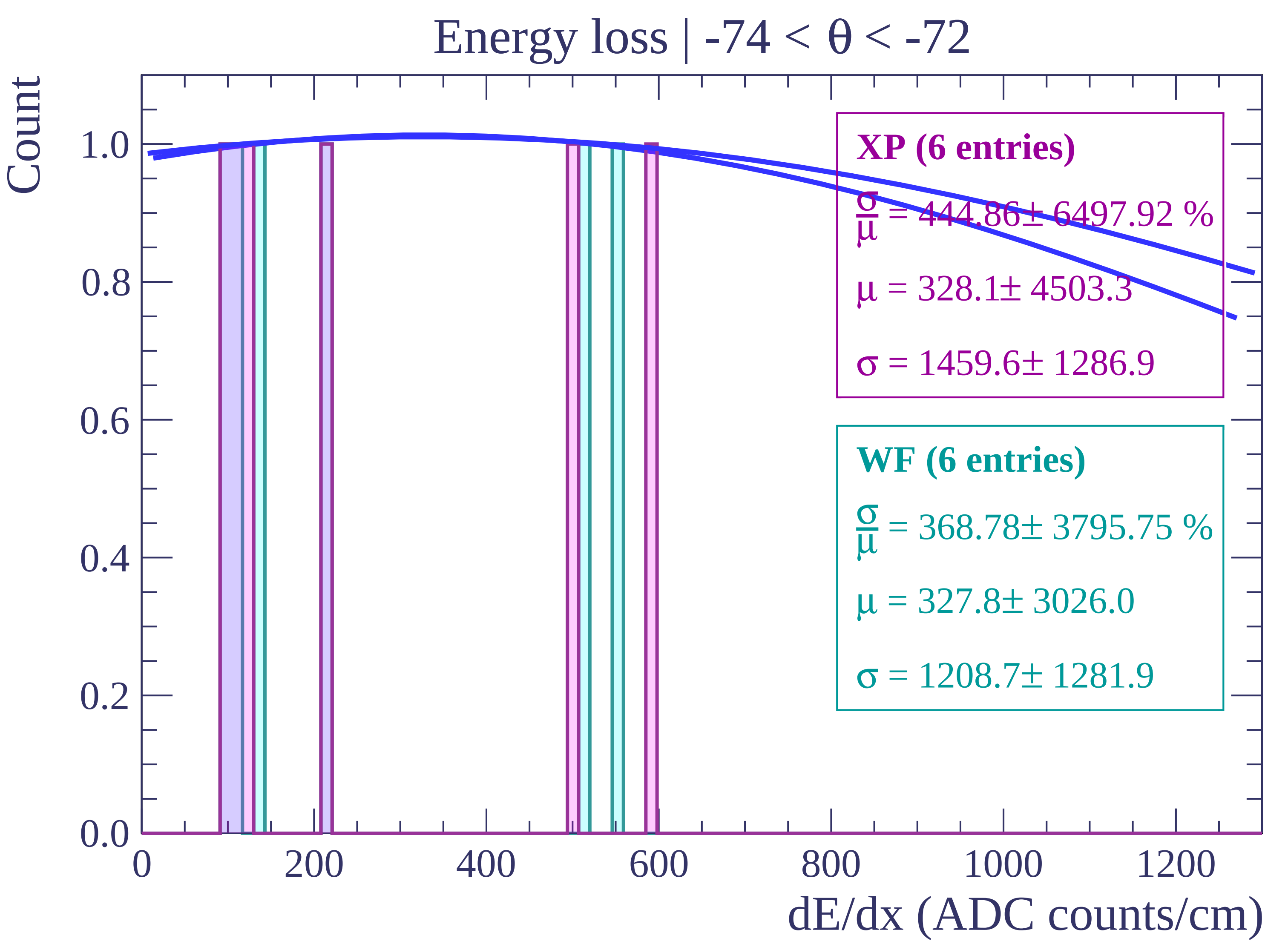


Energy loss | -80 < θ < -78
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

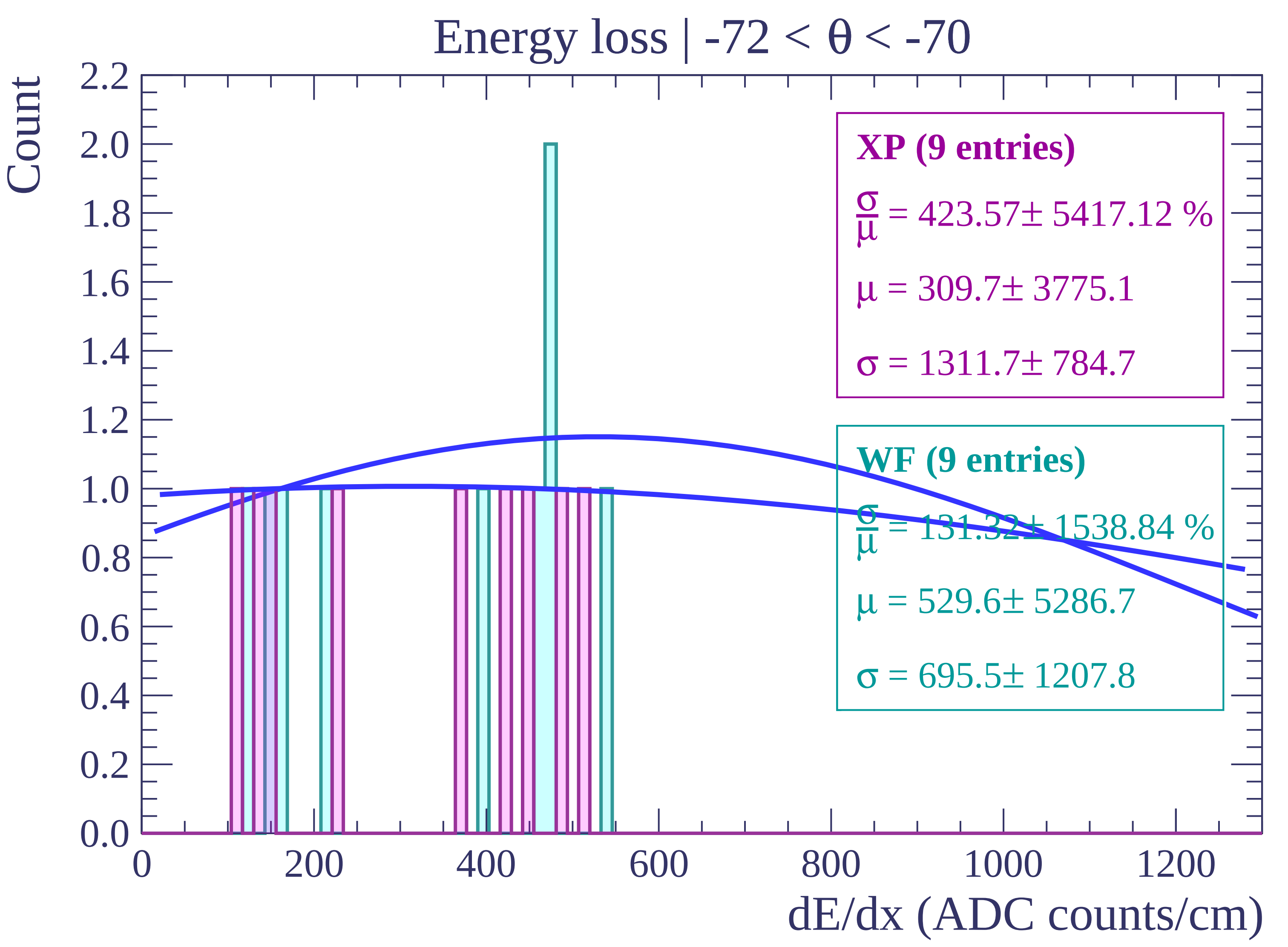

Energy loss | -78 < θ < -76
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3 entries)
σ/μ = 31.25± 208.34 %
μ = 104.0± 99.2
σ = 32.5± 185.7
WF (3 entries)
σ/μ = 0.36± 1.79 %
μ = 71.2± 2.4
σ = 0.3± 1.3



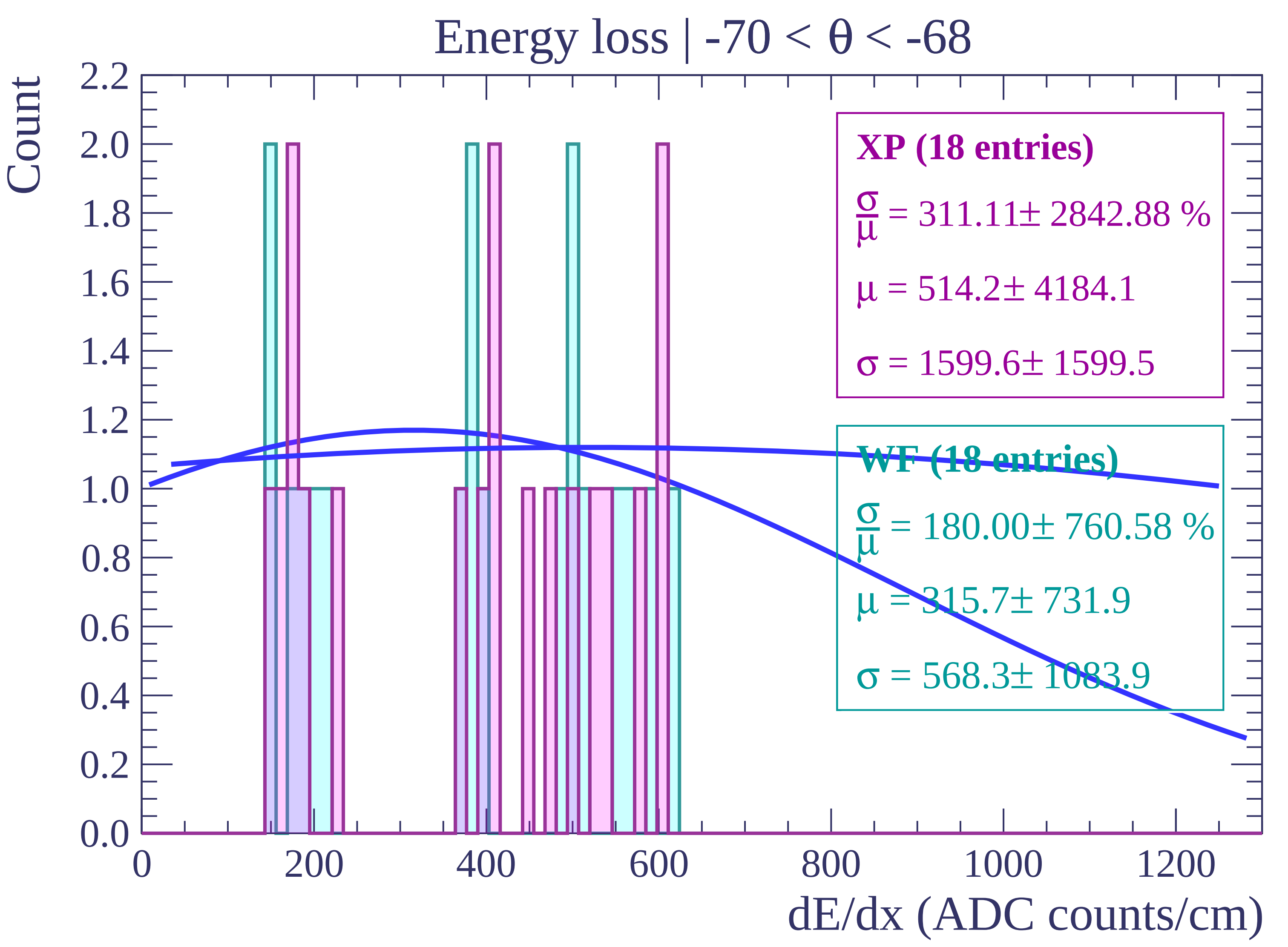
Energy loss | -76 < θ < -74
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



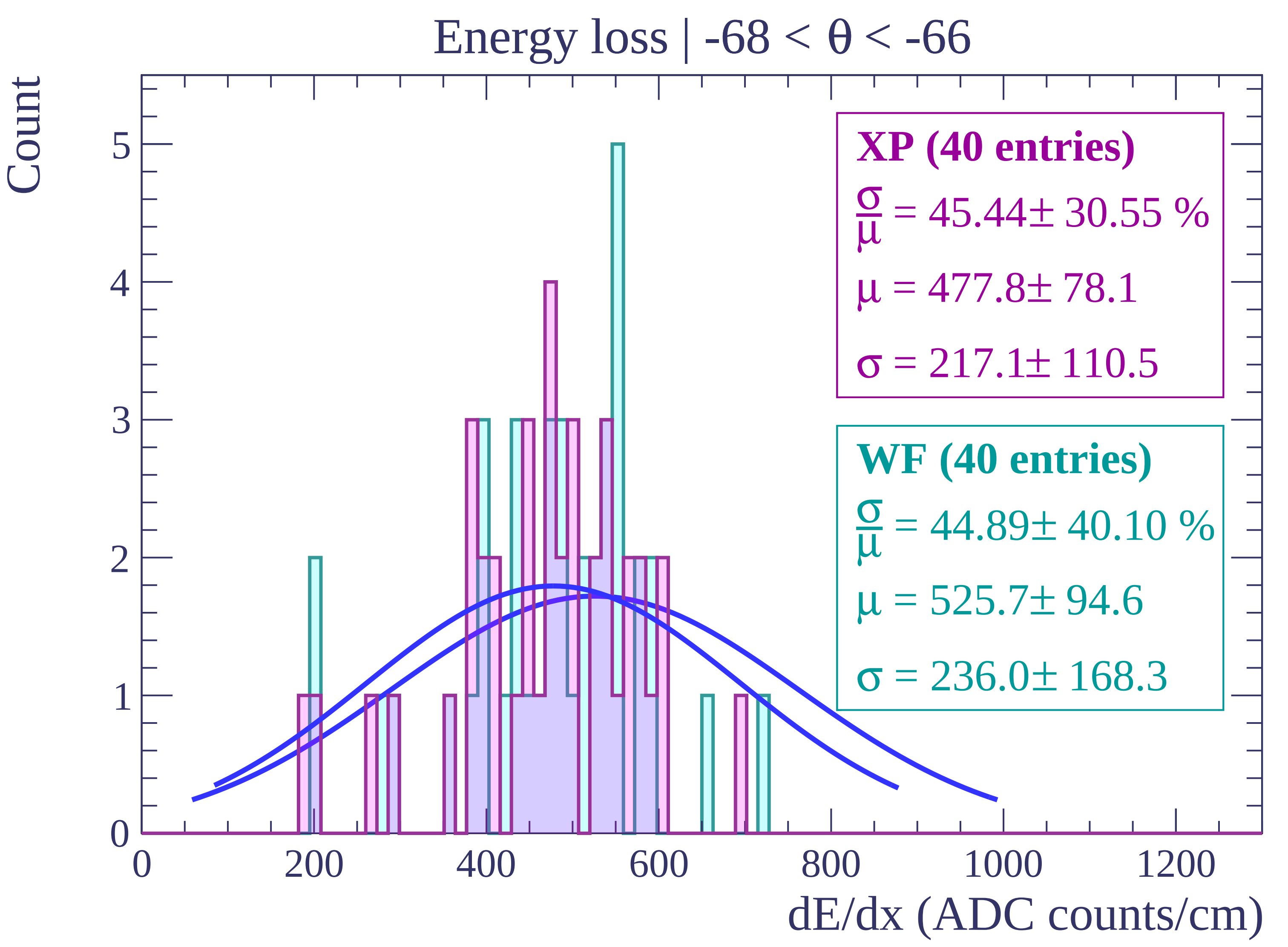
Energy loss | -74 < θ < -72
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6 entries)
σ/μ = 444.86± 6497.92 %
μ = 328.1± 4503.3
σ = 1459.6± 1286.9
WF (6 entries)
σ/μ = 368.78± 3795.75 %
μ = 327.8± 3026.0
σ = 1208.7± 1281.9

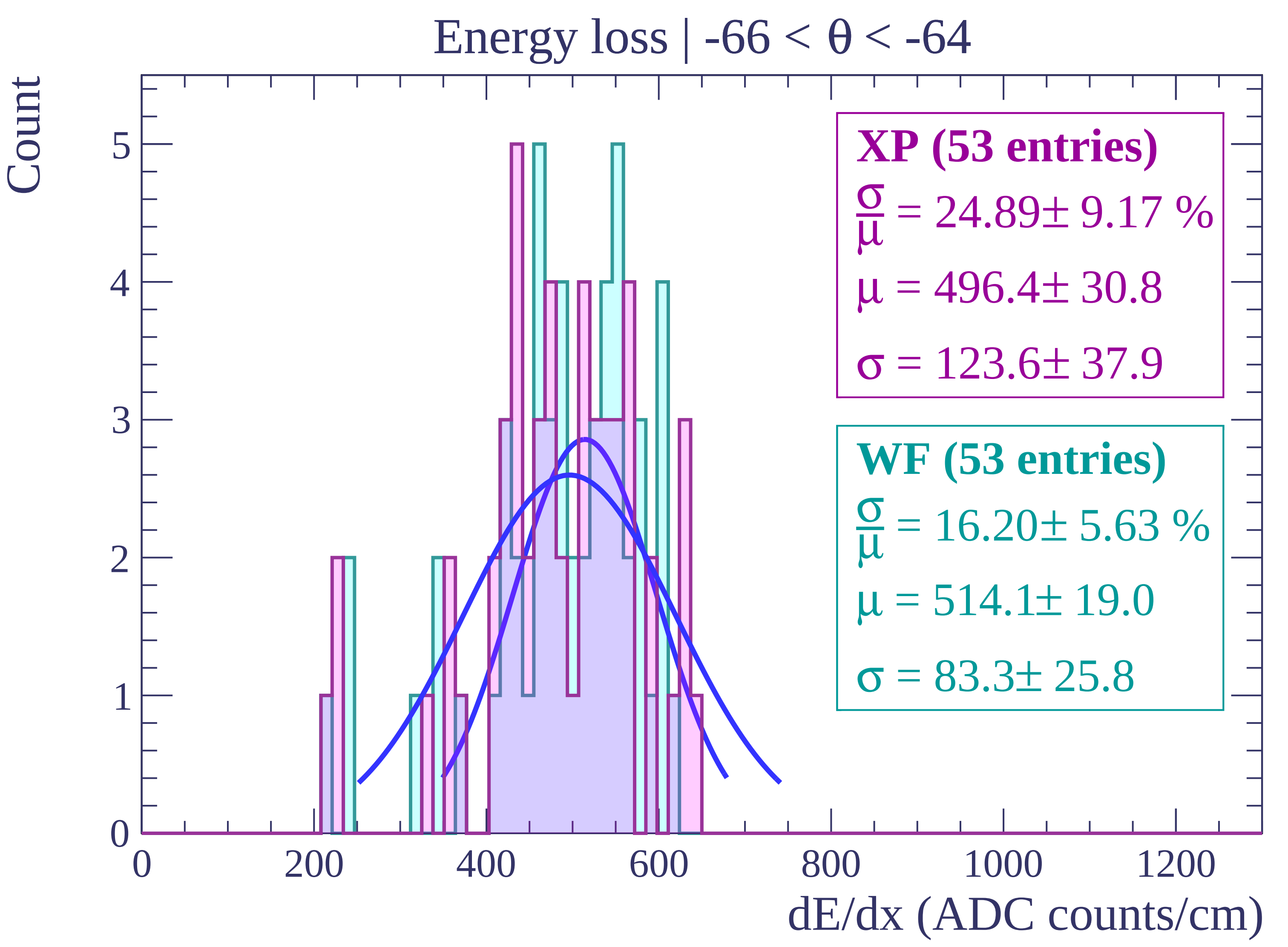


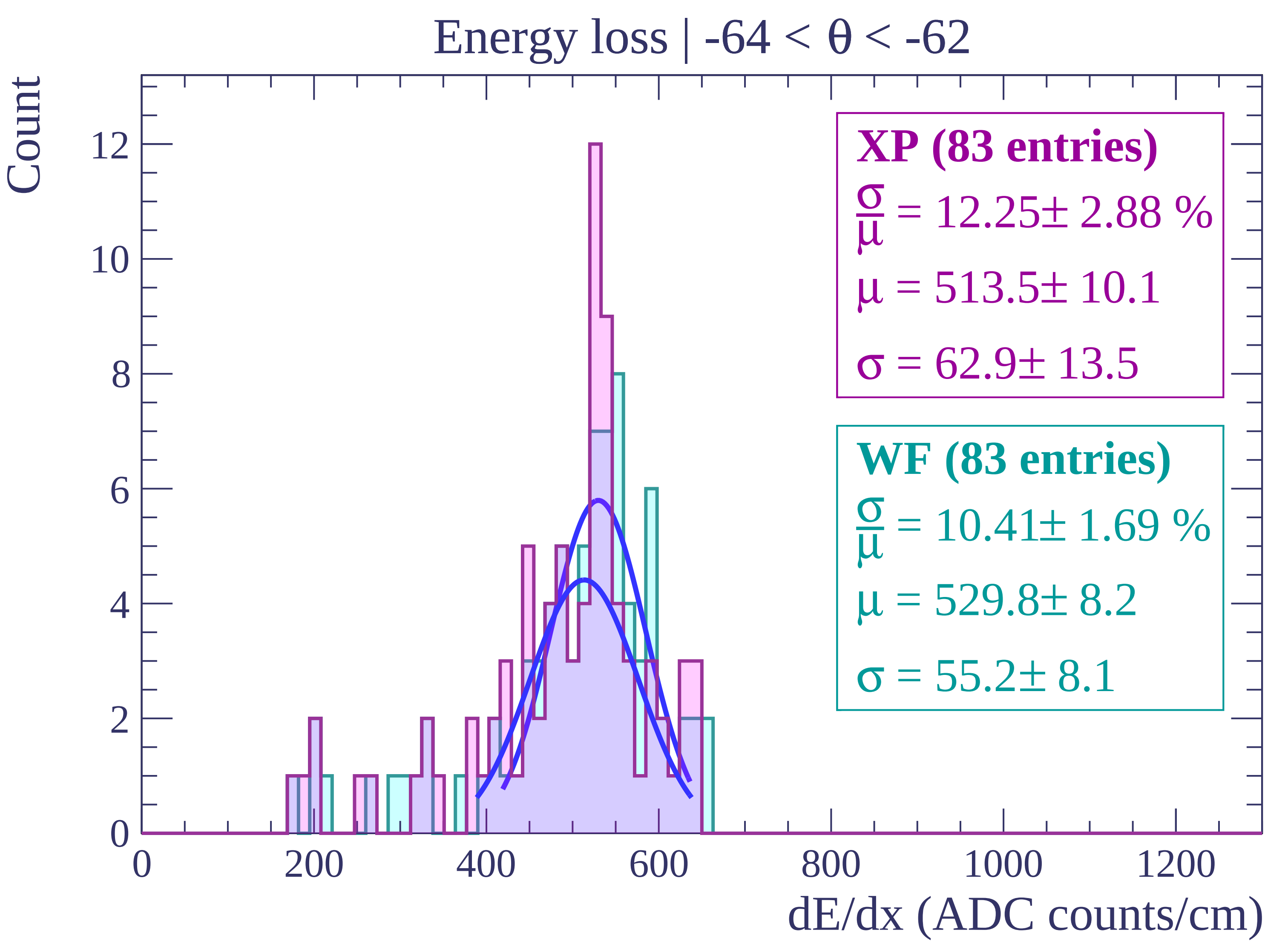
Energy loss | -72 < θ < -70
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (9 entries)
σ/μ = 423.57± 5417.12 %
μ = 309.7± 3775.1
σ = 1311.7± 784.7
WF (9 entries)
σ/μ = 131.32± 1538.84 %
μ = 529.6± 5286.7
σ = 695.5± 1207.8



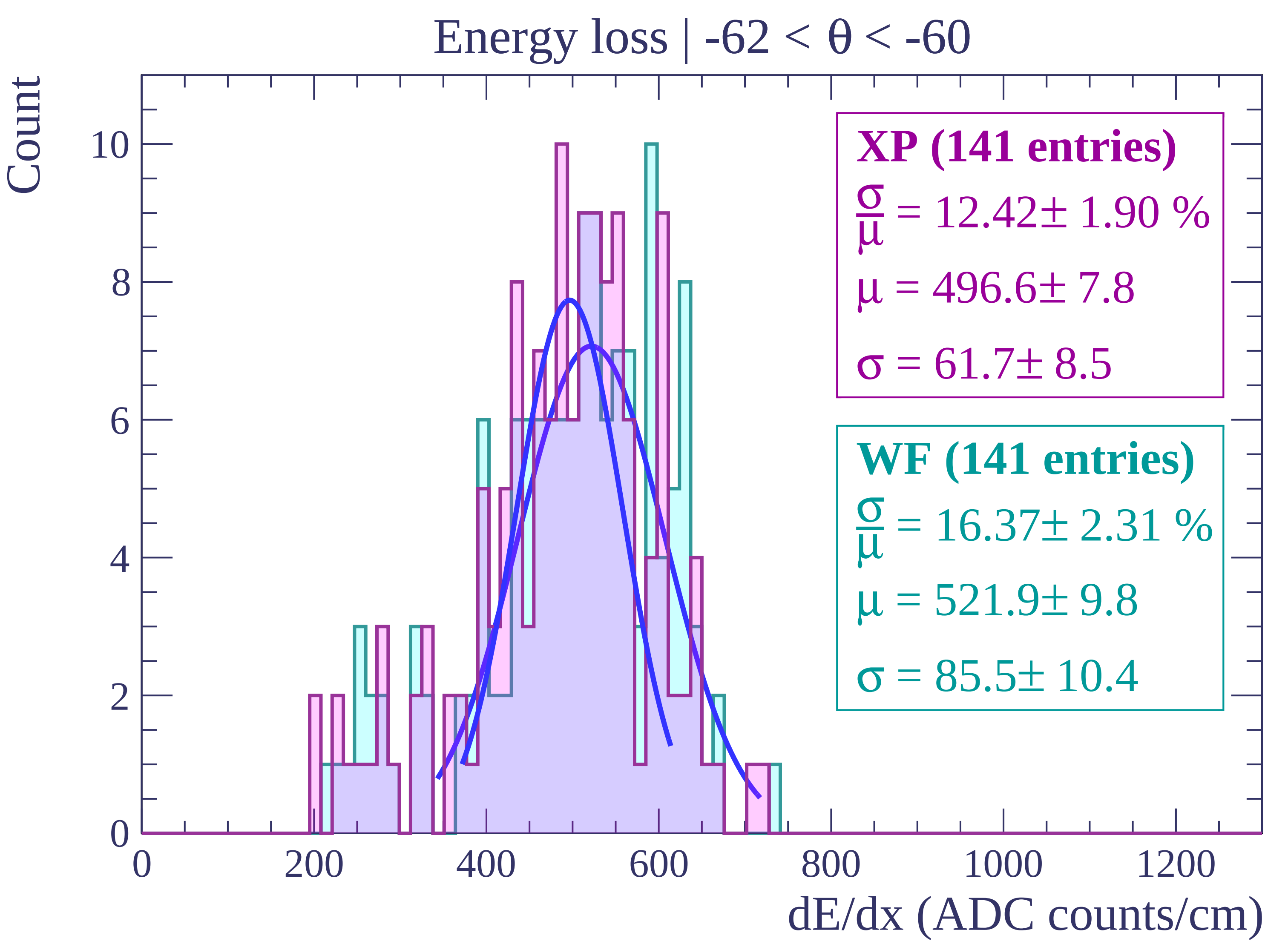
Energy loss | -70 < θ < -68
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (18 entries)
σ/μ = 311.11± 2842.88 %
μ = 514.2± 4184.1
σ = 1599.6± 1599.5
WF (18 entries)
σ/μ = 180.00± 760.58 %
μ = 315.7± 731.9
σ = 568.3± 1083.9


Energy loss | -68 < θ < -66
Count
0
1
2
3
4
5
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (40 entries)
$\frac{\sigma}{\mu}$ = 45.44± 30.55 %
μ = 477.8± 78.1
σ = 217.1± 110.5
WF (40 entries)
$\frac{\sigma}{\mu}$ = 44.89± 40.10 %
μ = 525.7± 94.6
σ = 236.0± 168.3

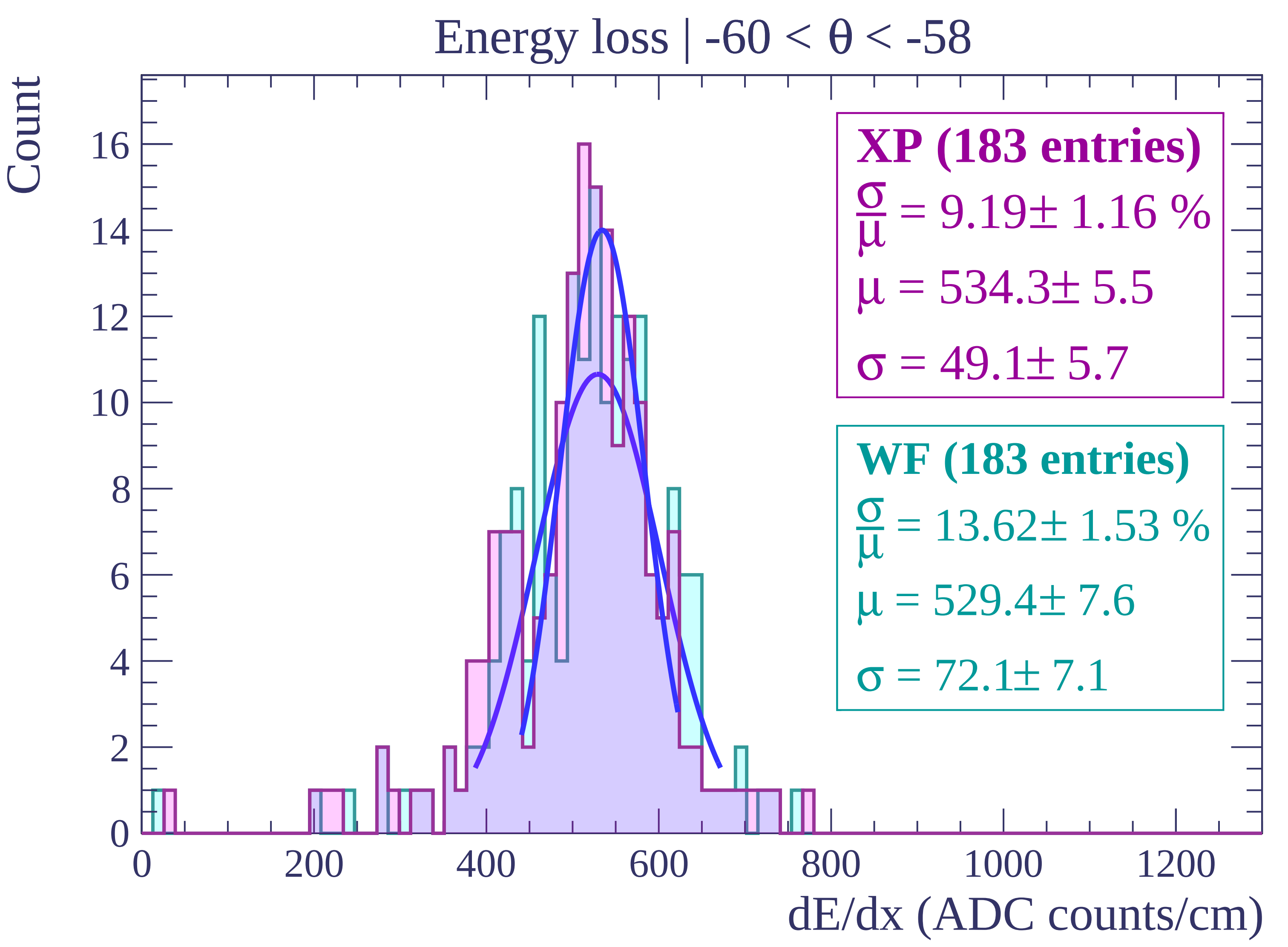

Energy loss | -66 < θ < -64
Count
XP (53 entries)
σ/μ = 24.89± 9.17 %
μ = 496.4± 30.8
σ = 123.6± 37.9
WF (53 entries)
σ/μ = 16.20± 5.63 %
μ = 514.1± 19.0
σ = 83.3± 25.8
0
1
2
3
4
5
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

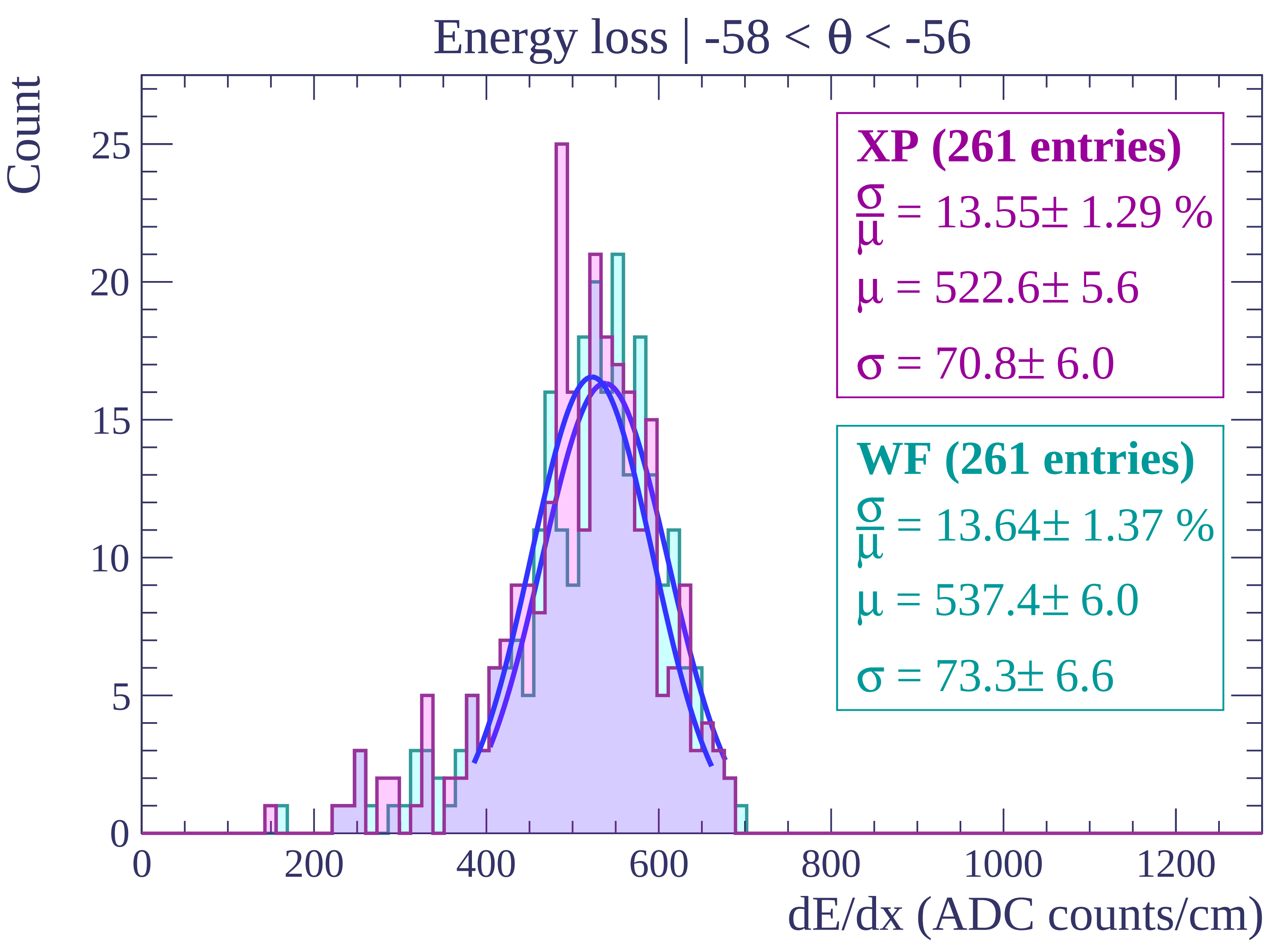


Energy loss | -64 < θ < -62
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP (83 entries)
σ/μ = 12.25± 2.88 %
μ = 513.5± 10.1
σ = 62.9± 13.5
WF (83 entries)
σ/μ = 10.41± 1.69 %
μ = 529.8± 8.2
σ = 55.2± 8.1

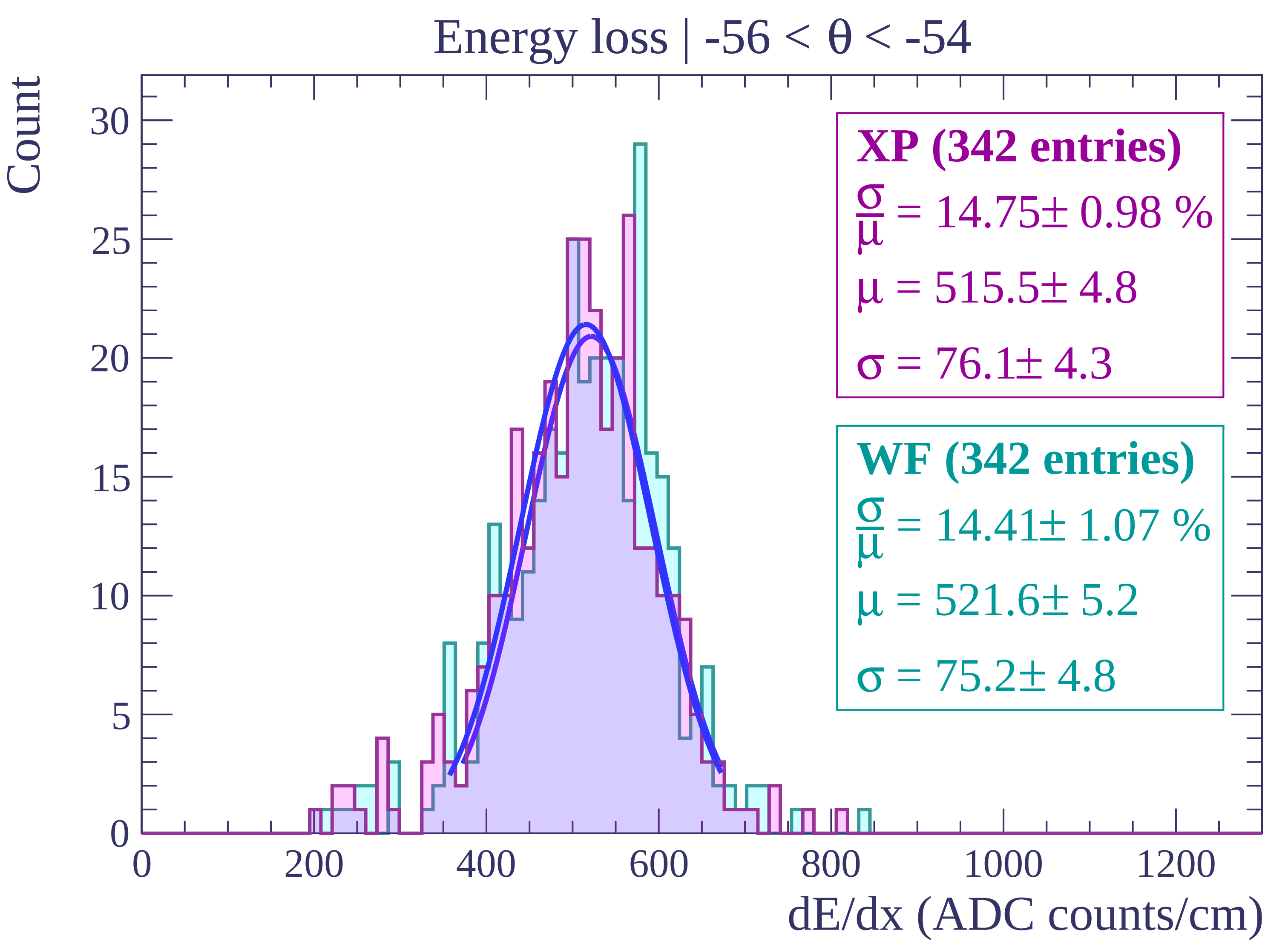


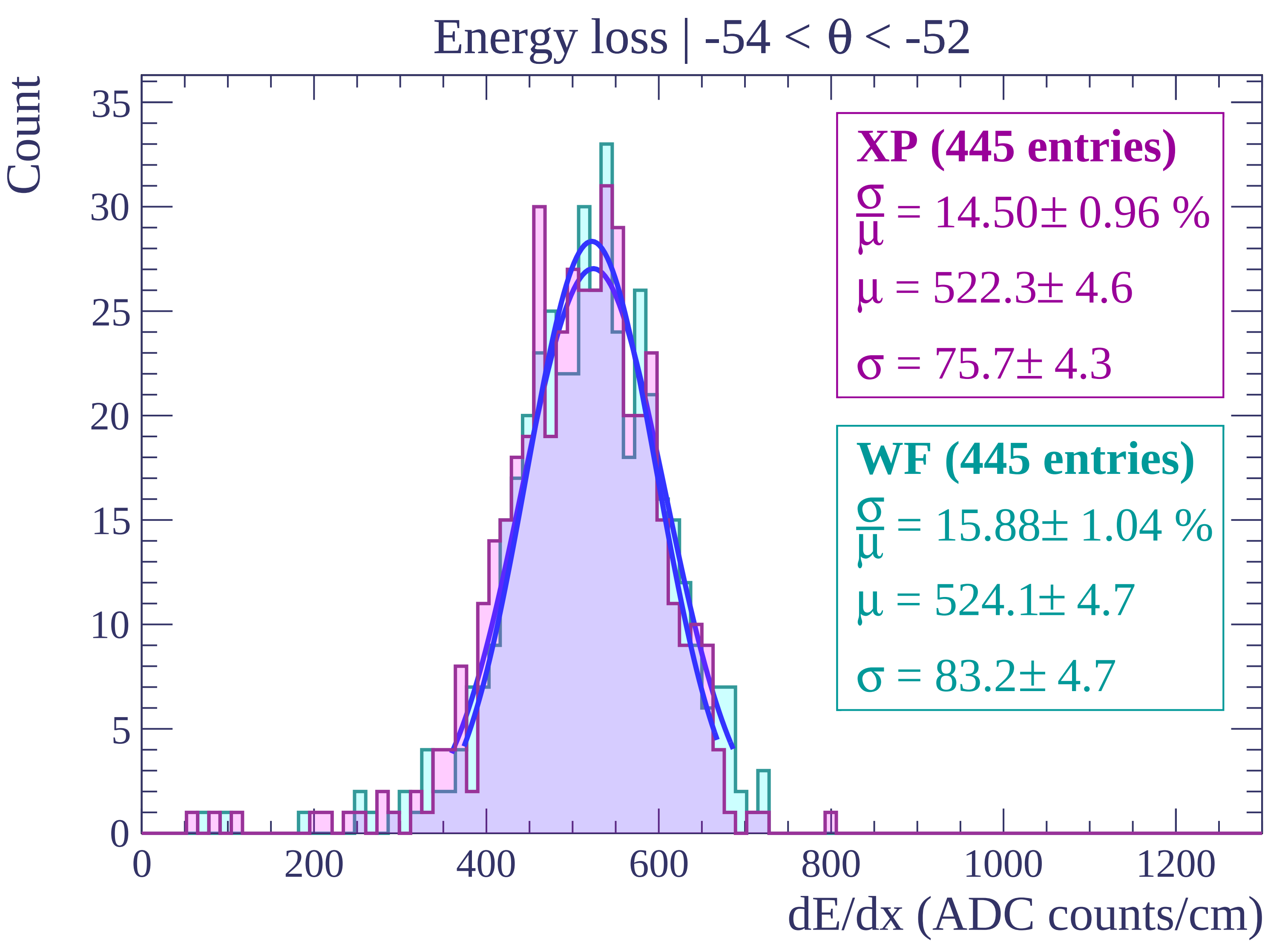
Energy loss | -62 < θ < -60
Count
0
2
4
6
8
10
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (141 entries)
$\frac{\sigma}{\mu}$ = 12.42± 1.90 %
μ = 496.6± 7.8
σ = 61.7± 8.5
WF (141 entries)
$\frac{\sigma}{\mu}$ = 16.37± 2.31 %
μ = 521.9± 9.8
σ = 85.5± 10.4


Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP (183 entries)
σ/μ = 9.19± 1.16 %
μ = 534.3± 5.5
σ = 49.1± 5.7
WF (183 entries)
σ/μ = 13.62± 1.53 %
μ = 529.4± 7.6
σ = 72.1± 7.1

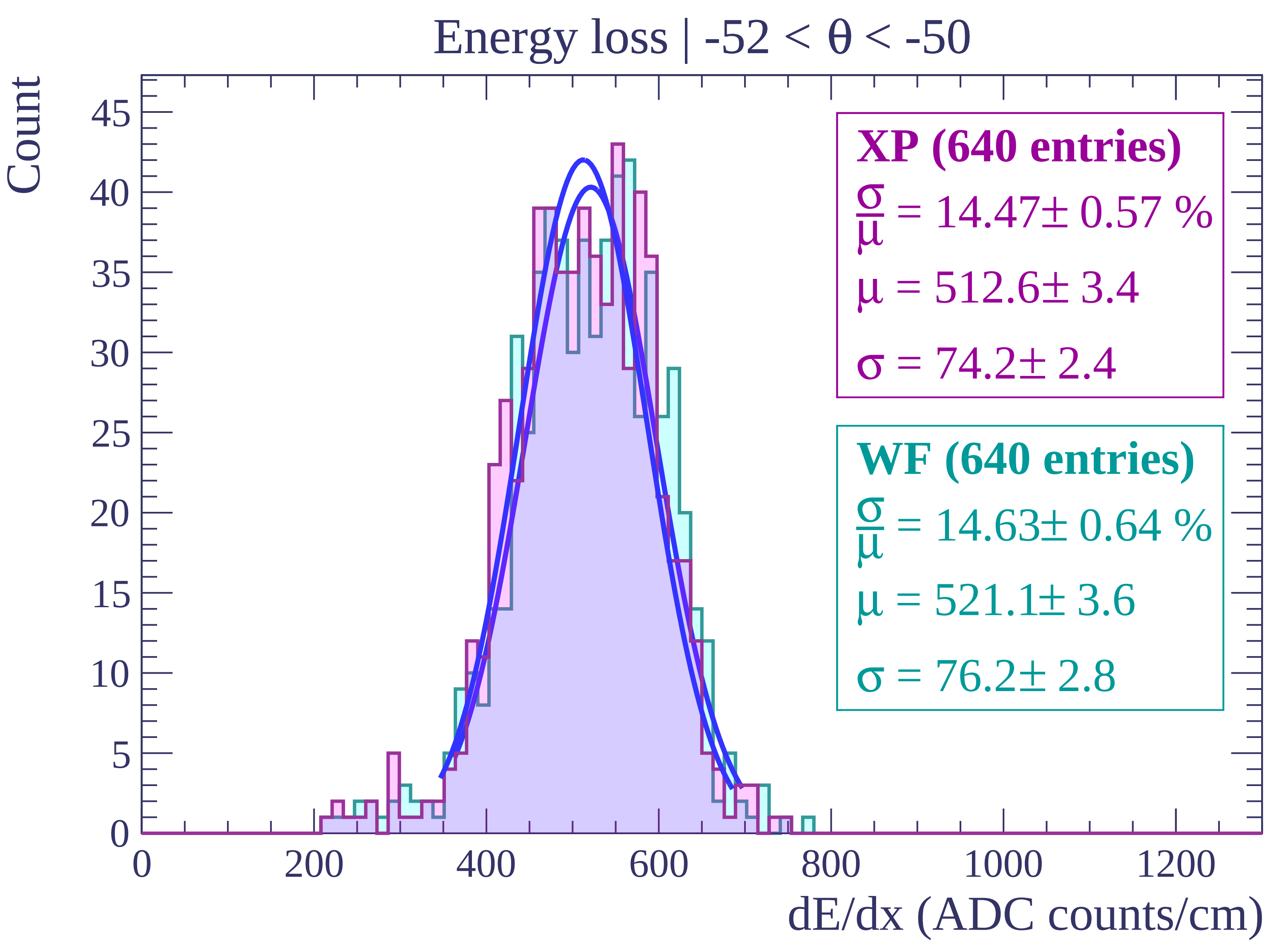


Energy loss | -58 < θ < -56
Count
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (261 entries)
σ/μ = 13.55± 1.29 %
μ = 522.6± 5.6
σ = 70.8± 6.0
WF (261 entries)
σ/μ = 13.64± 1.37 %
μ = 537.4± 6.0
σ = 73.3± 6.6

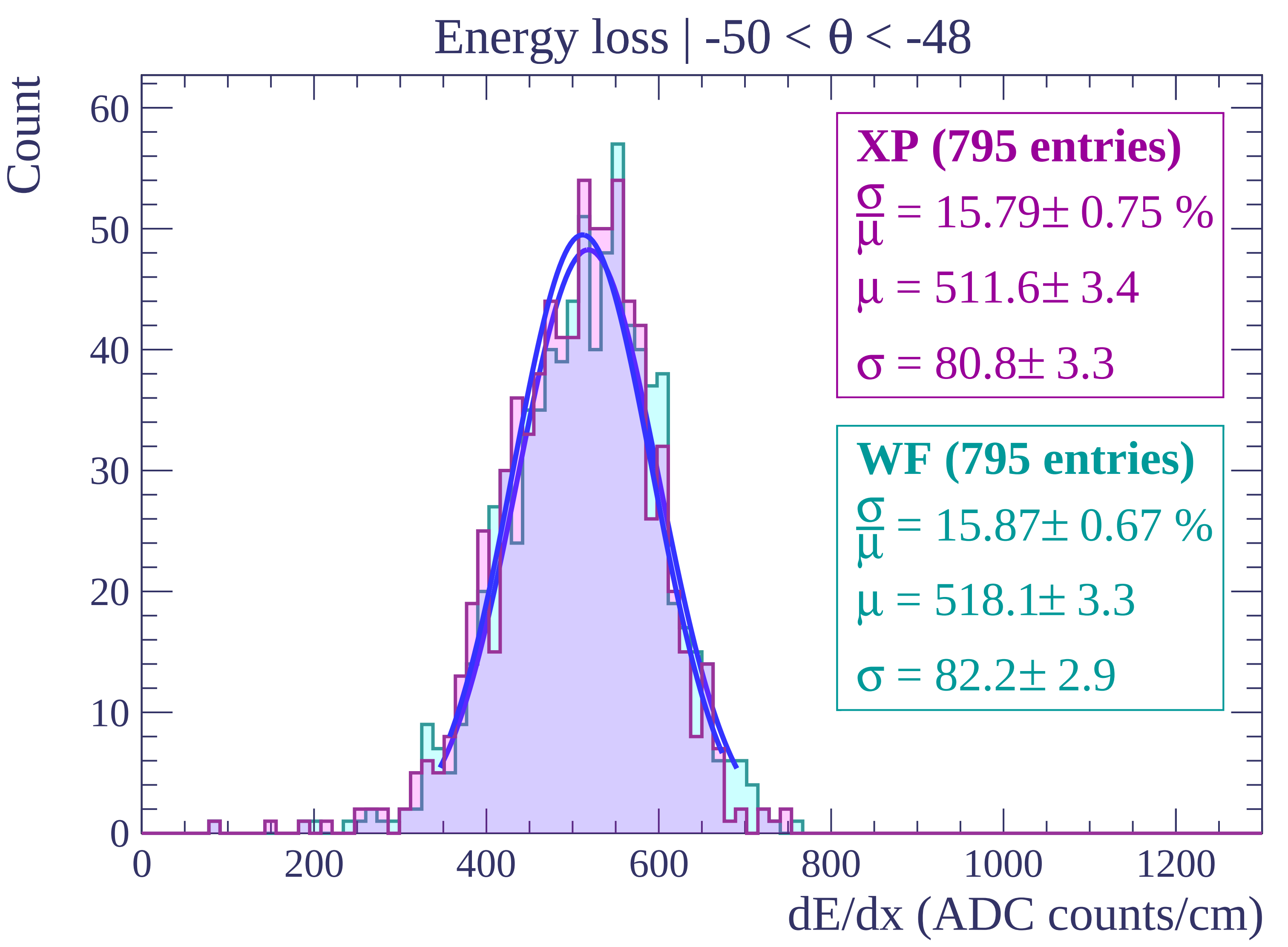

Energy loss | -56 < θ < -54
Count
XP (342 entries)
σ/μ = 14.75± 0.98 %
μ = 515.5± 4.8
σ = 76.1± 4.3
WF (342 entries)
σ/μ = 14.41± 1.07 %
μ = 521.6± 5.2
σ = 75.2± 4.8
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



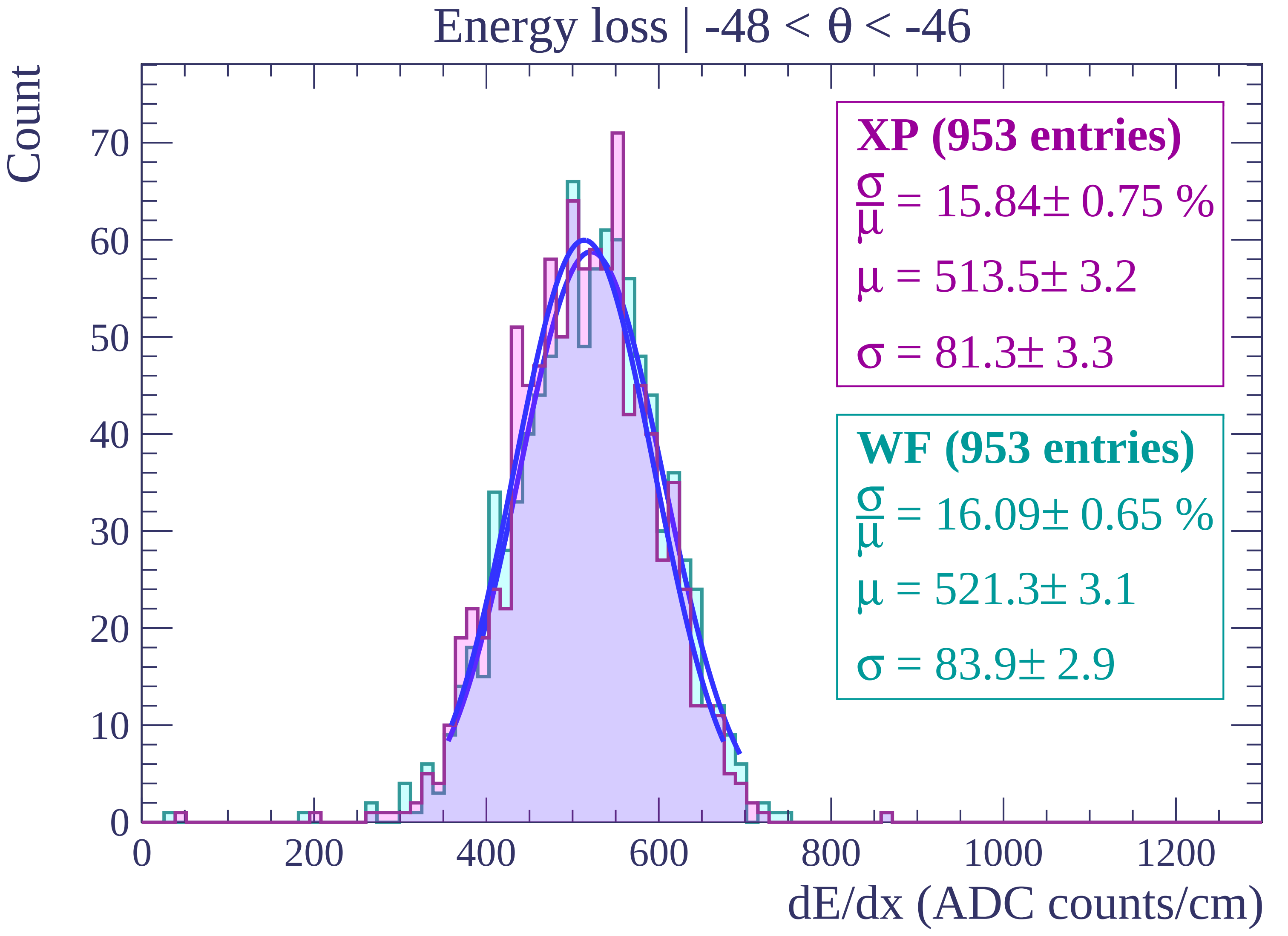
Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
XP (445 entries)
σ/μ = 14.50± 0.96 %
μ = 522.3± 4.6
σ = 75.7± 4.3
WF (445 entries)
σ/μ = 15.88± 1.04 %
μ = 524.1± 4.7
σ = 83.2± 4.7

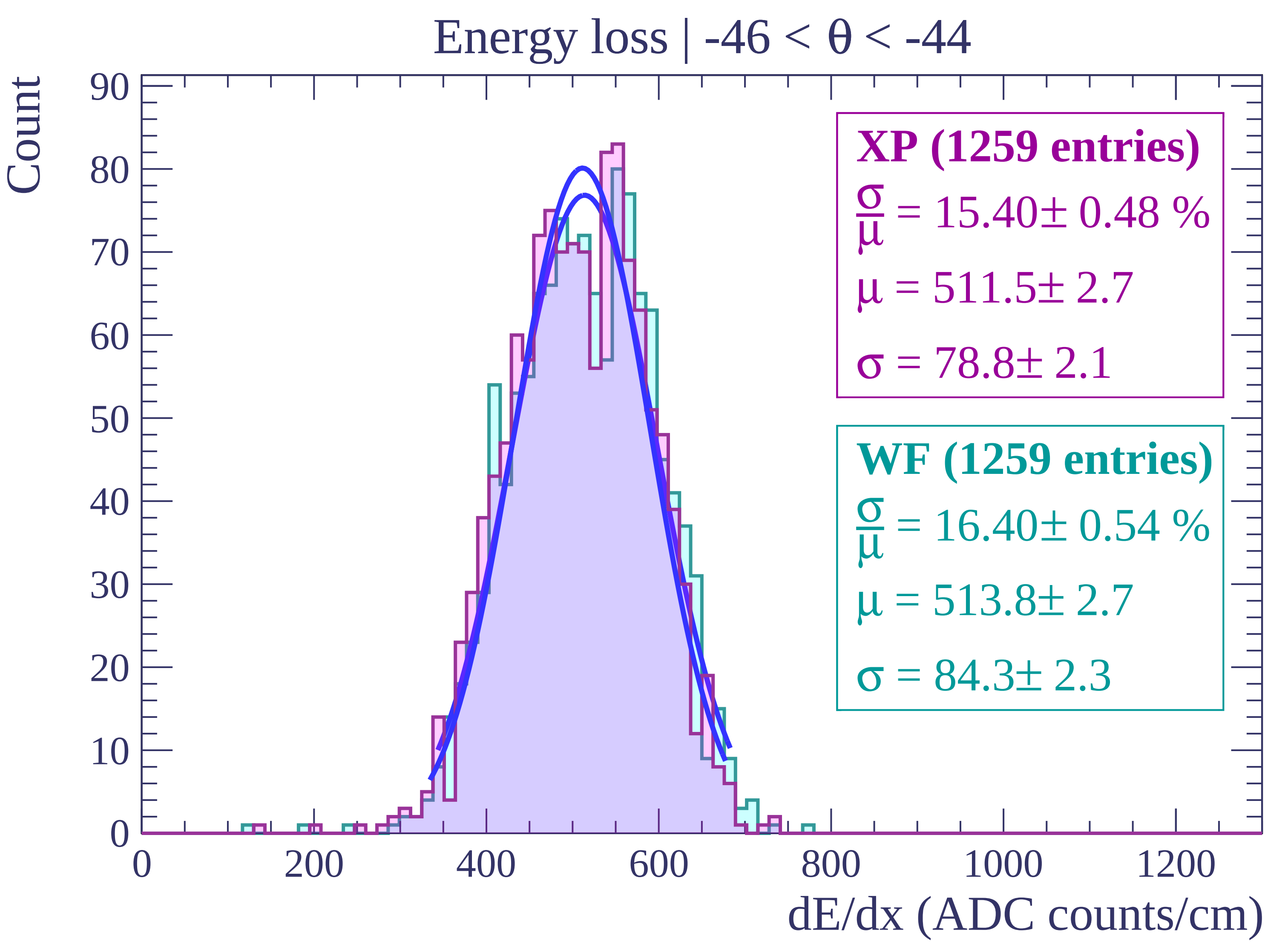


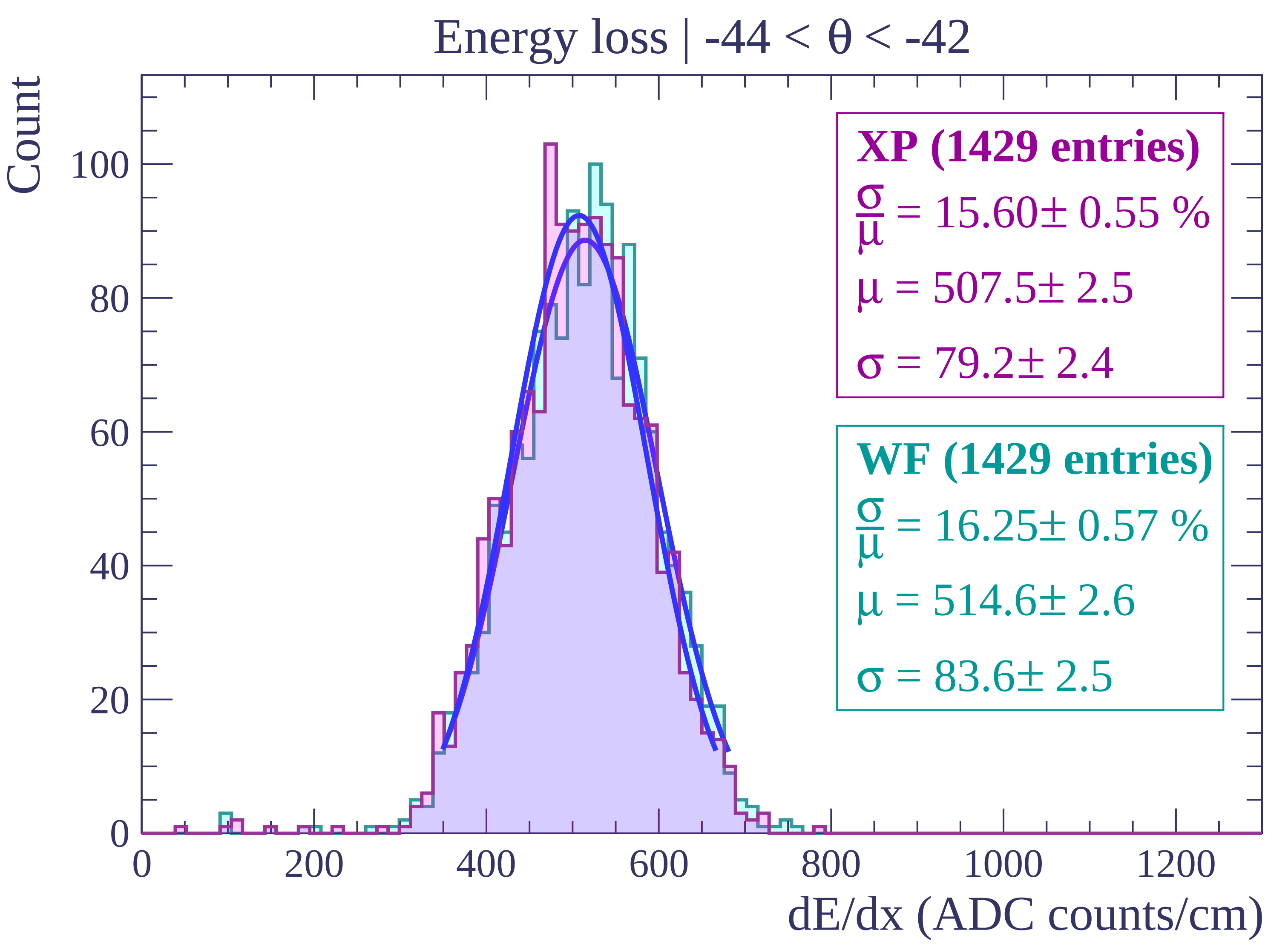
Energy loss | -52 < θ < -50
Count
dE/dx (ADC counts/cm)
XP (640 entries)
$\frac{\sigma}{\mu}$ = 14.47± 0.57 %
μ = 512.6± 3.4
σ = 74.2± 2.4
WF (640 entries)
$\frac{\sigma}{\mu}$ = 14.63± 0.64 %
μ = 521.1± 3.6
σ = 76.2± 2.8



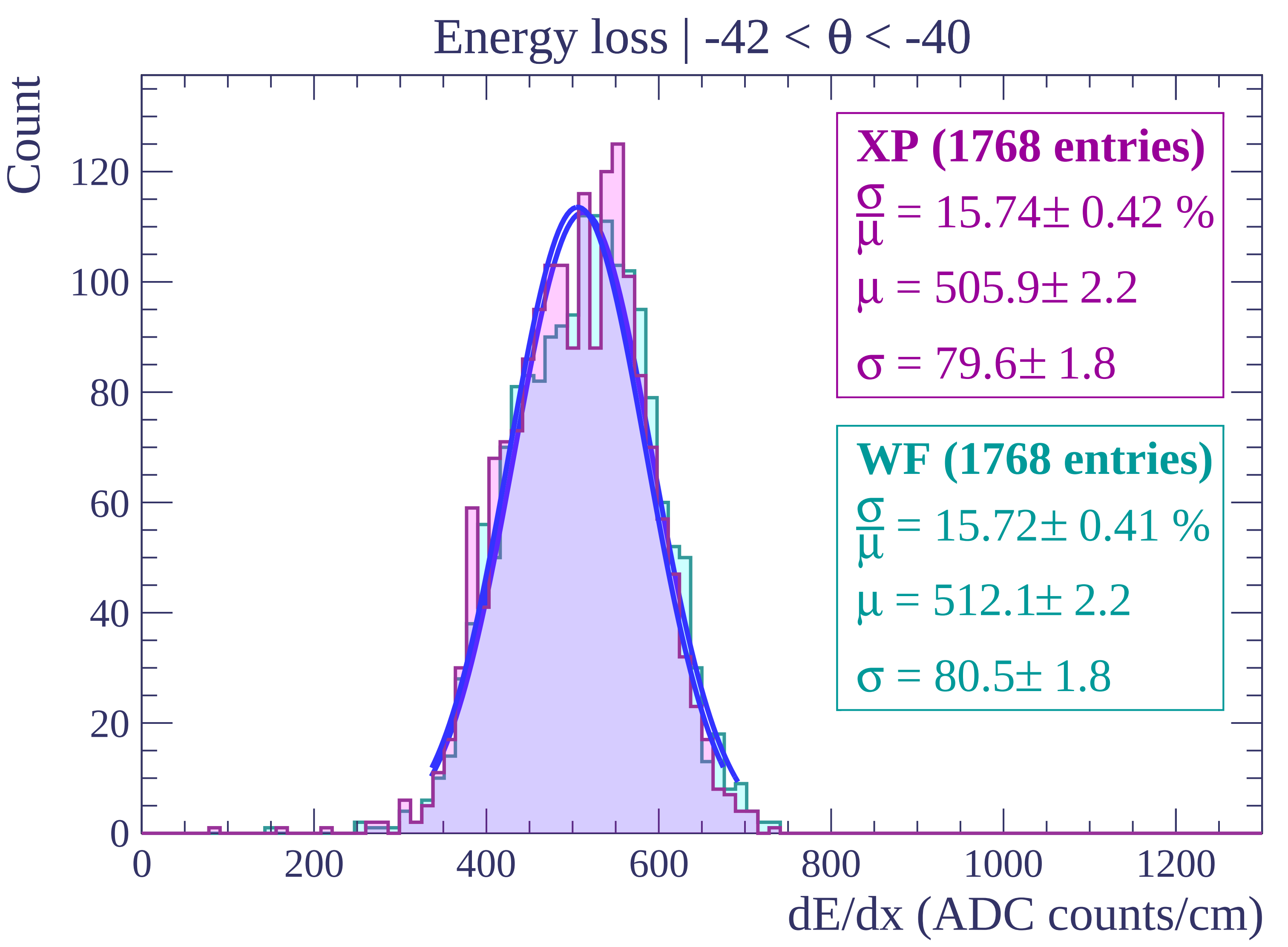
Energy loss | -50 < θ < -48
Count
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (795 entries)
σ/μ = 15.79± 0.75 %
μ = 511.6± 3.4
σ = 80.8± 3.3
WF (795 entries)
σ/μ = 15.87± 0.67 %
μ = 518.1± 3.3
σ = 82.2± 2.9


Energy loss | -48 < θ < -46
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (953 entries)
σ/μ = 15.84± 0.75 %
μ = 513.5± 3.2
σ = 81.3± 3.3
WF (953 entries)
σ/μ = 16.09± 0.65 %
μ = 521.3± 3.1
σ = 83.9± 2.9

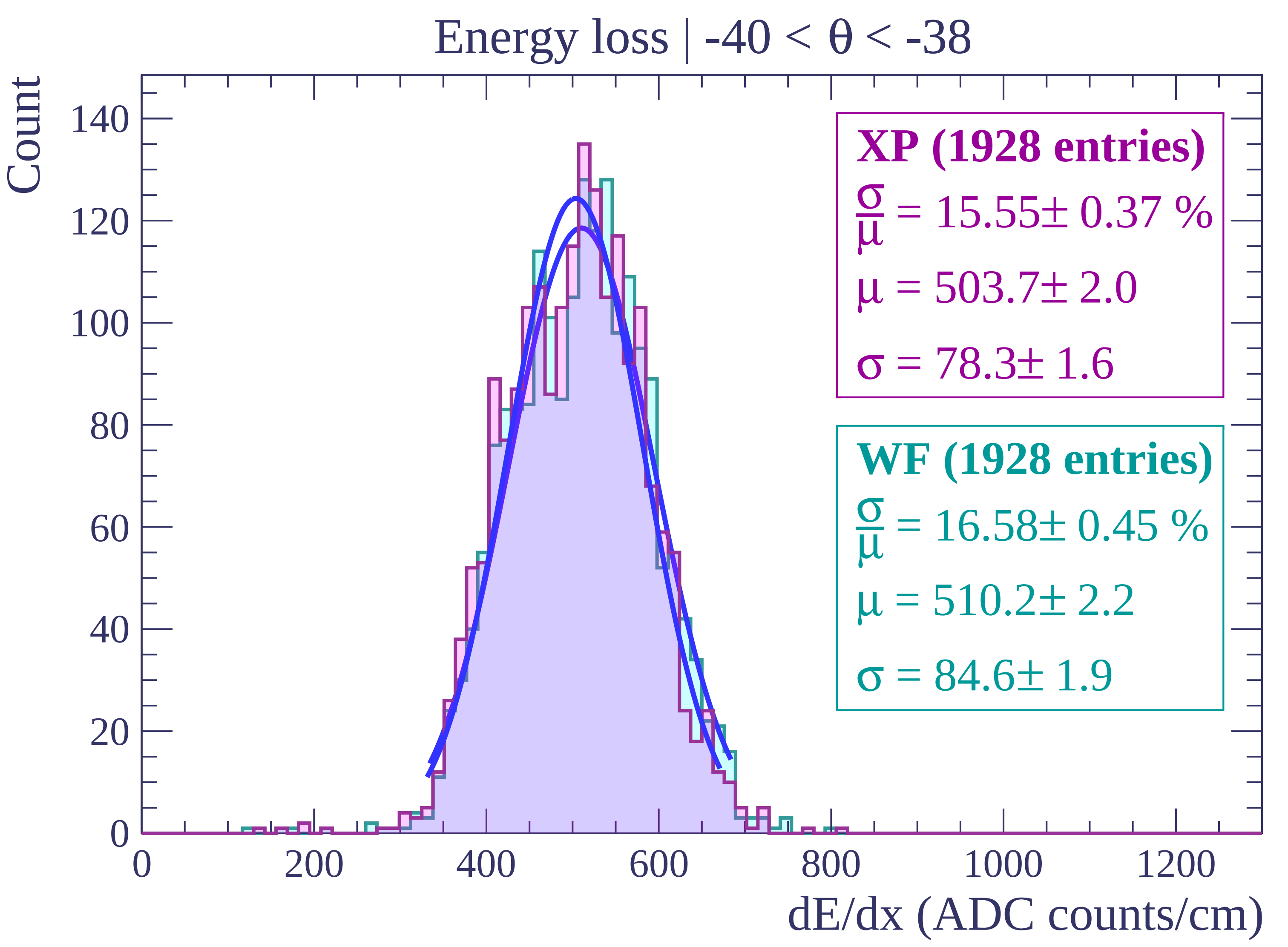

Energy loss | -46 < θ < -44
Count
dE/dx (ADC counts/cm)
XP (1259 entries)
$\frac{\sigma}{\mu}$ = 15.40± 0.48 %
μ = 511.5± 2.7
σ = 78.8± 2.1
WF (1259 entries)
$\frac{\sigma}{\mu}$ = 16.40± 0.54 %
μ = 513.8± 2.7
σ = 84.3± 2.3
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200



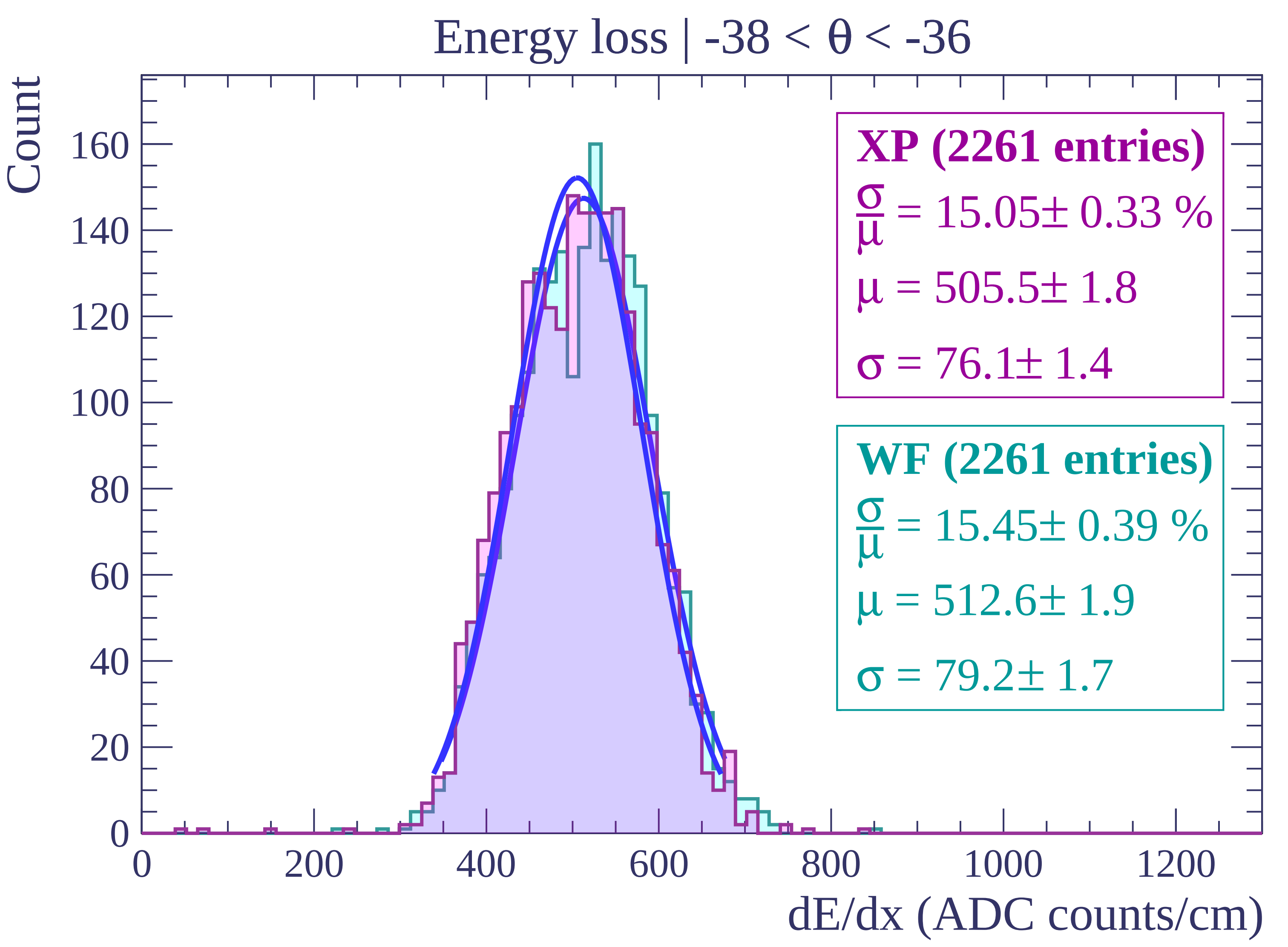
Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP (1429 entries)
$\frac{\sigma}{\mu}$ = 15.60± 0.55 %
μ = 507.5± 2.5
σ = 79.2± 2.4
WF (1429 entries)
$\frac{\sigma}{\mu}$ = 16.25± 0.57 %
μ = 514.6± 2.6
σ = 83.6± 2.5



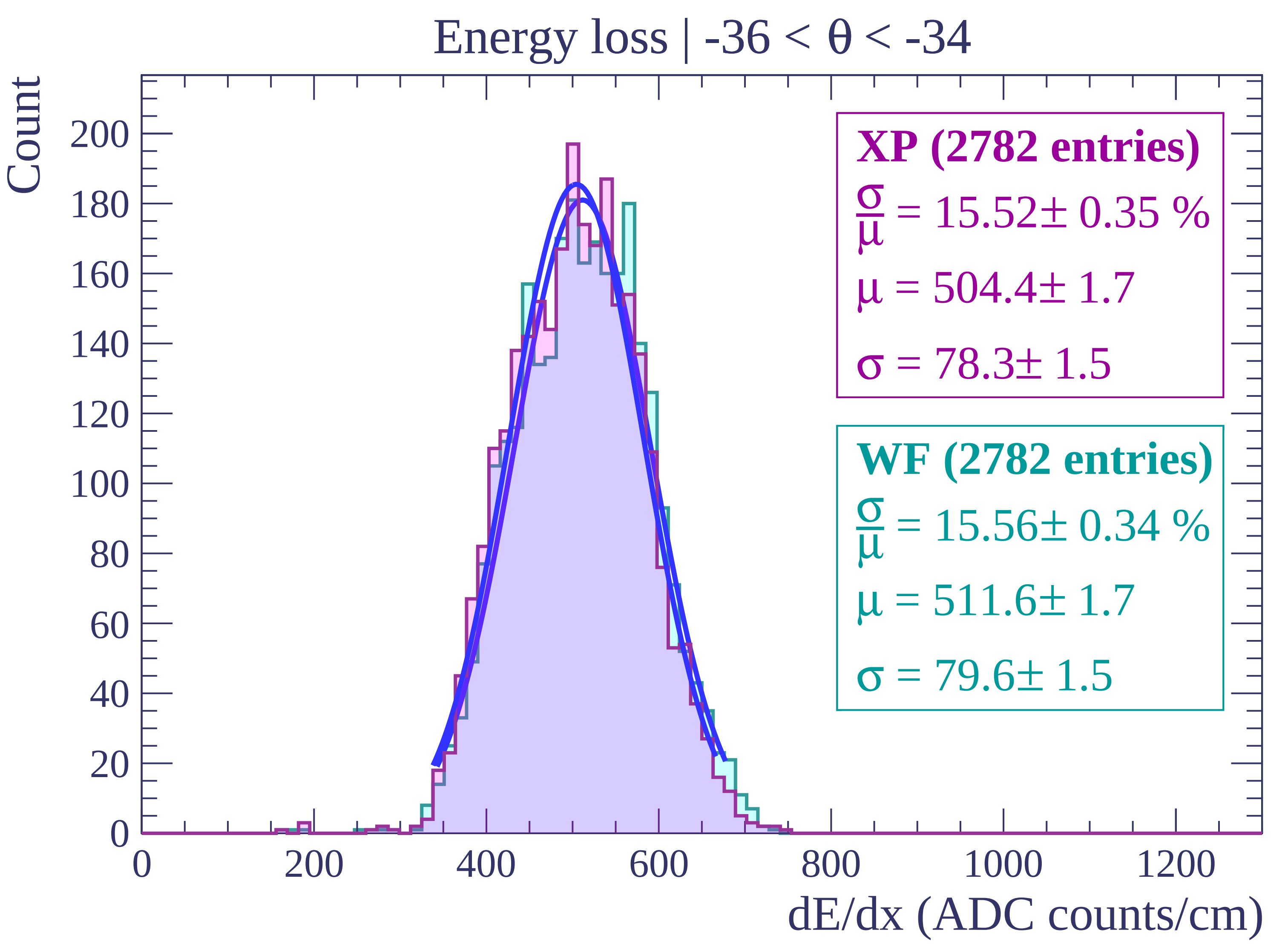
Energy loss | -42 < θ < -40
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1768 entries)
σ/μ = 15.74± 0.42 %
μ = 505.9± 2.2
σ = 79.6± 1.8
WF (1768 entries)
σ/μ = 15.72± 0.41 %
μ = 512.1± 2.2
σ = 80.5± 1.8



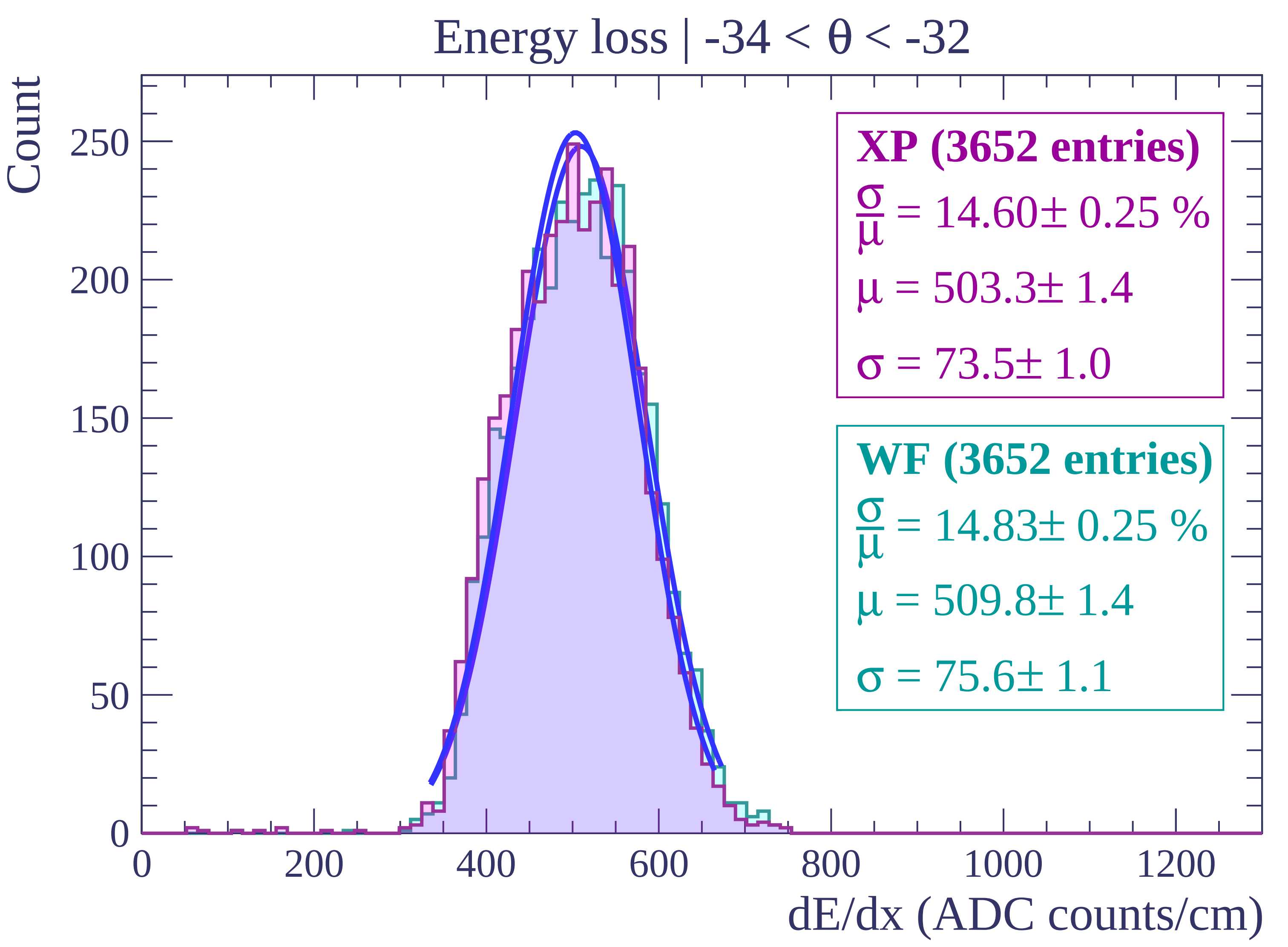
Energy loss | -40 < θ < -38
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1928 entries)
σ/μ = 15.55± 0.37 %
μ = 503.7± 2.0
σ = 78.3± 1.6
WF (1928 entries)
σ/μ = 16.58± 0.45 %
μ = 510.2± 2.2
σ = 84.6± 1.9

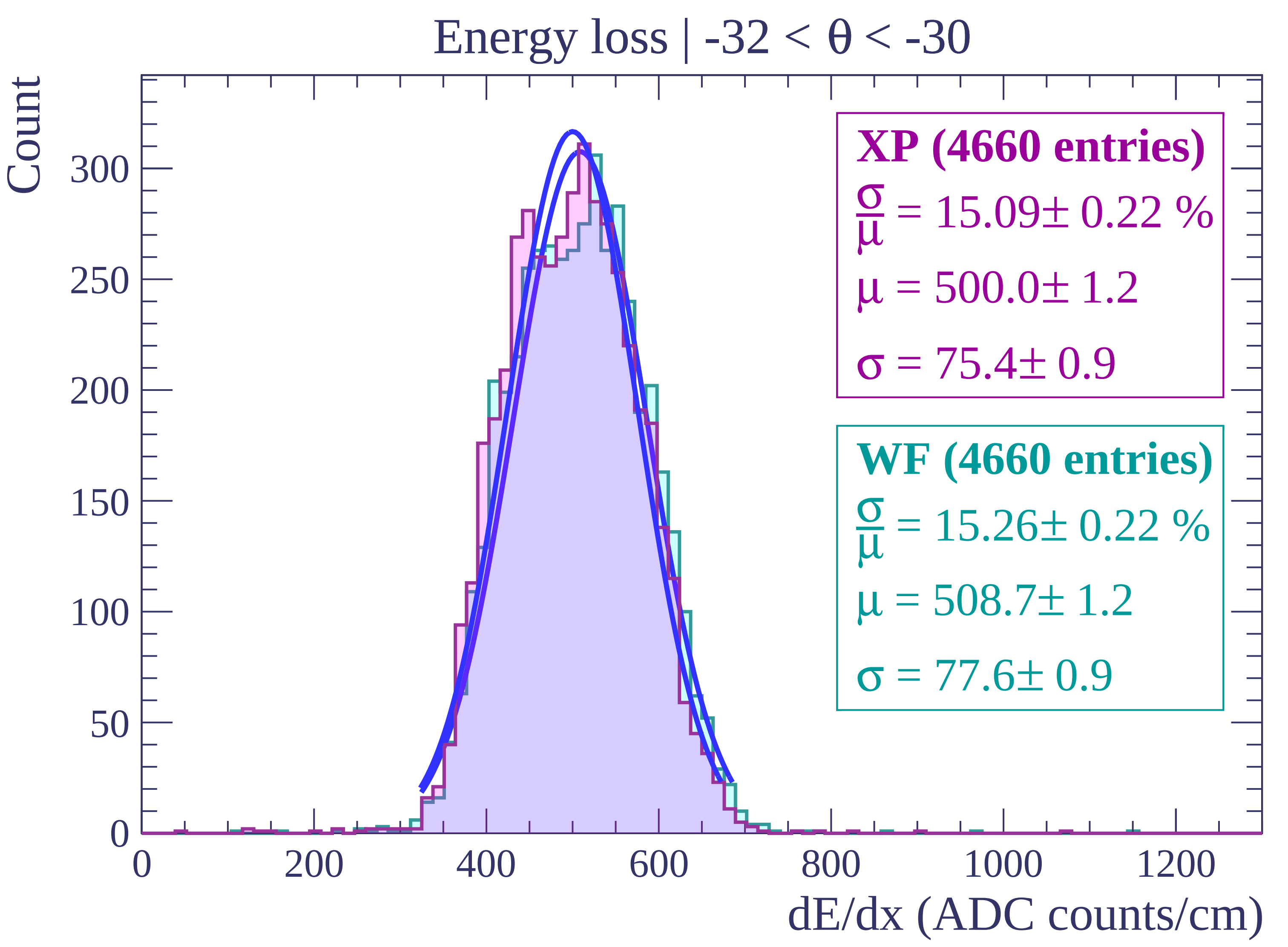


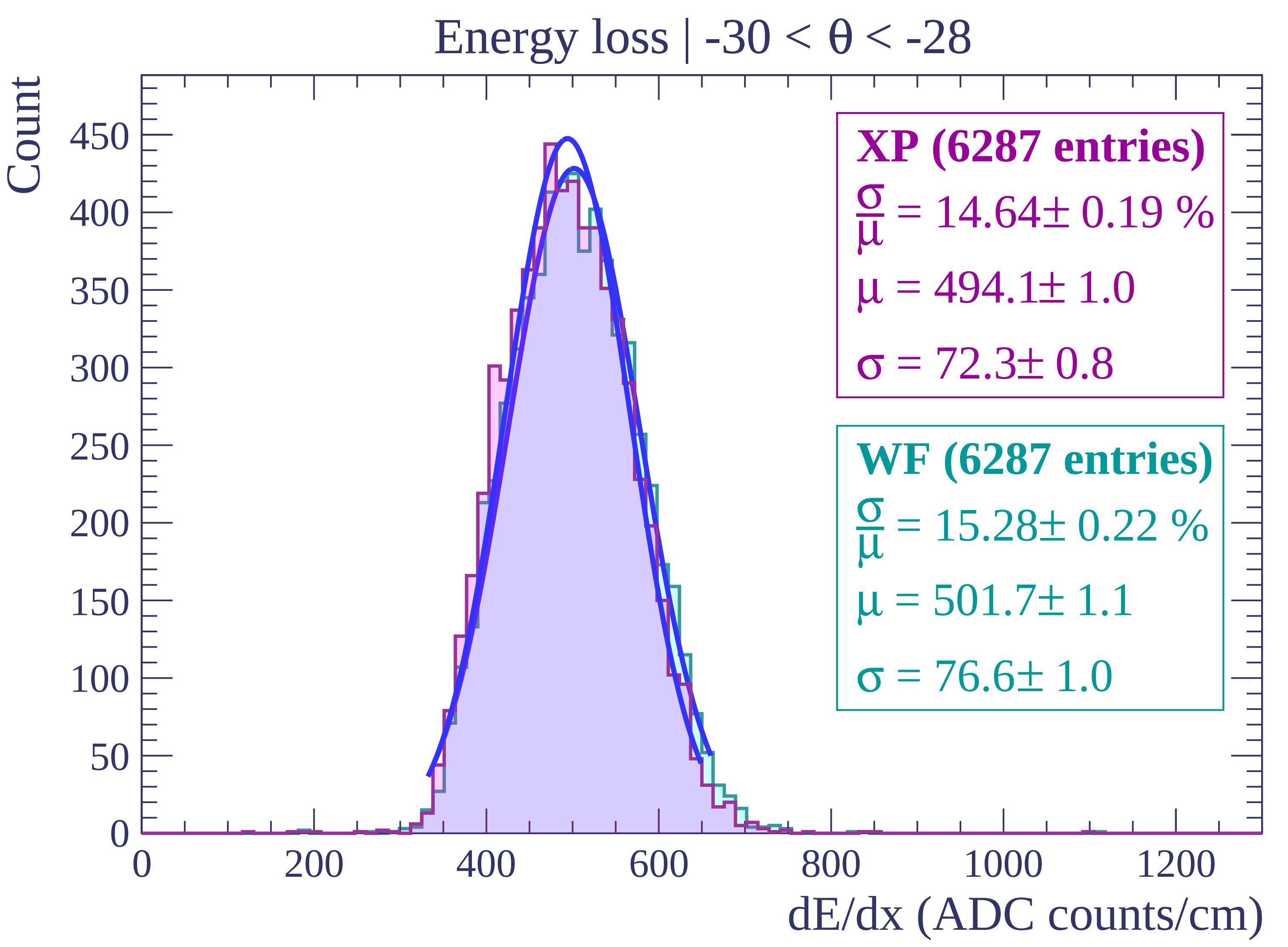
Energy loss | -38 < θ < -36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
160
XP (2261 entries)
$\frac{\sigma}{\mu}$ = 15.05± 0.33 %
μ = 505.5± 1.8
σ = 76.1± 1.4
WF (2261 entries)
$\frac{\sigma}{\mu}$ = 15.45± 0.39 %
μ = 512.6± 1.9
σ = 79.2± 1.7

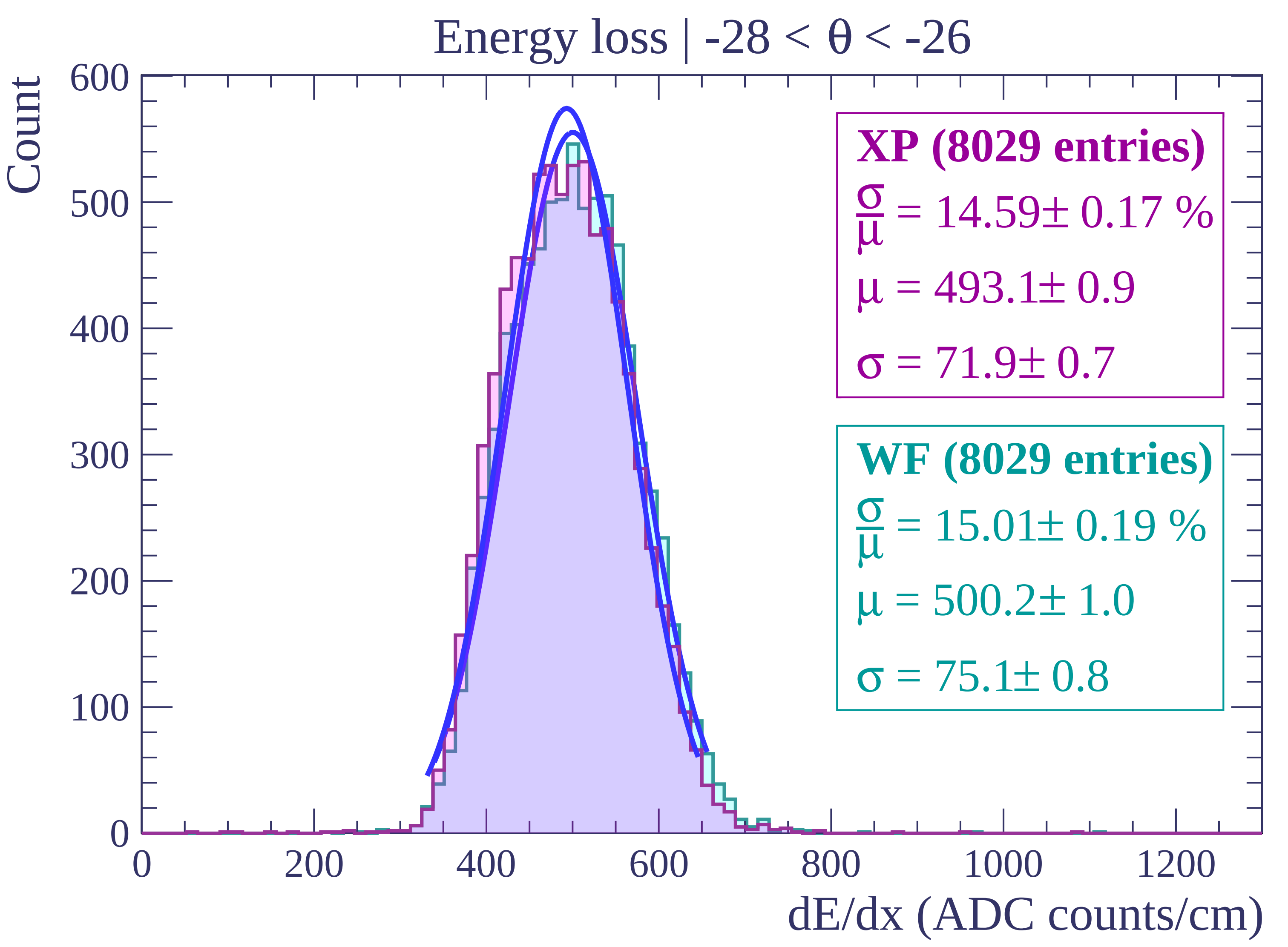


Energy loss | -36 < θ < -34
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (2782 entries)
σ/μ = 15.52± 0.35 %
μ = 504.4± 1.7
σ = 78.3± 1.5
WF (2782 entries)
σ/μ = 15.56± 0.34 %
μ = 511.6± 1.7
σ = 79.6± 1.5

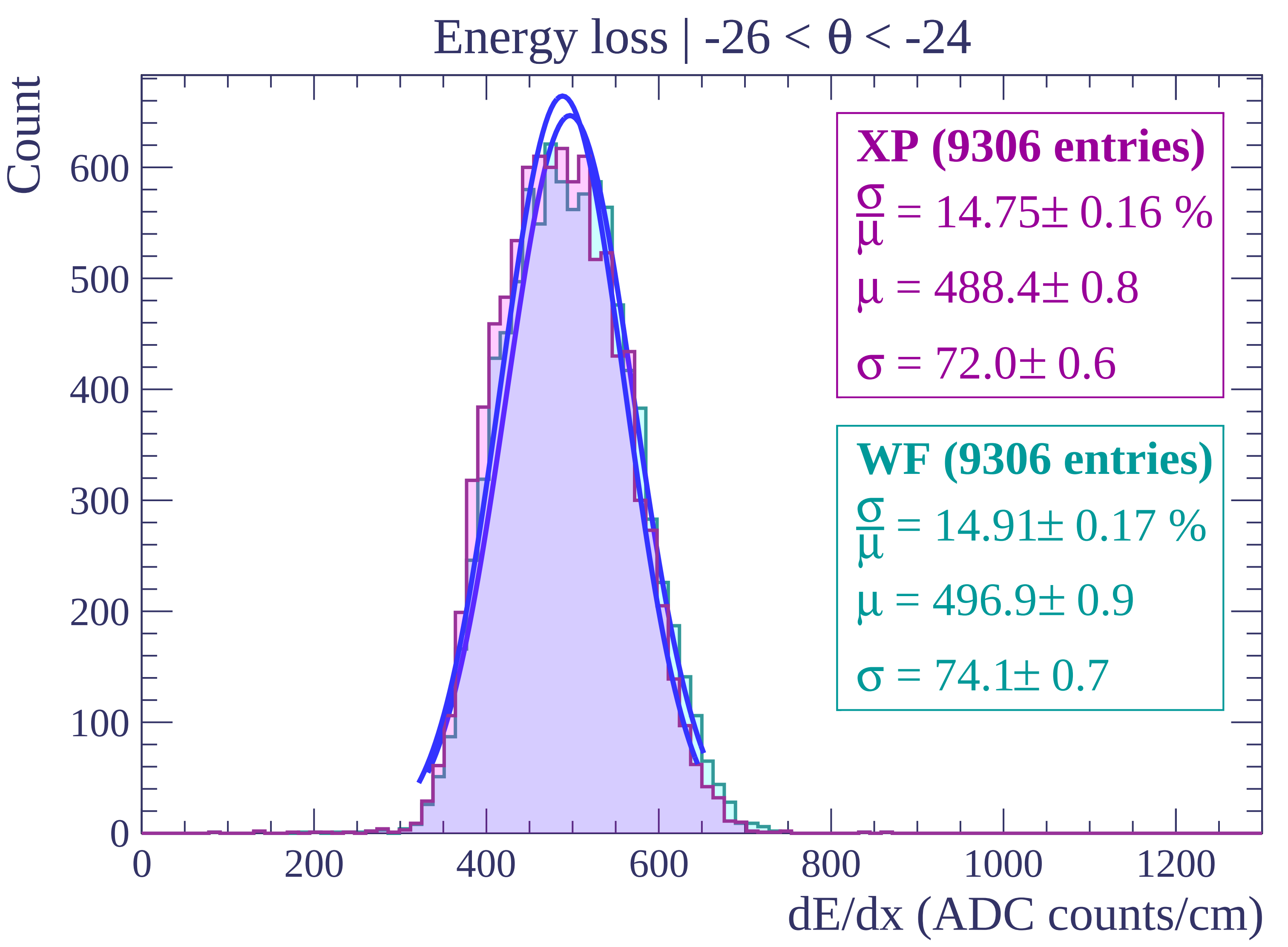

Energy loss | -34 < θ < -32
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3652 entries)
σ/μ = 14.60± 0.25 %
μ = 503.3± 1.4
σ = 73.5± 1.0
WF (3652 entries)
σ/μ = 14.83± 0.25 %
μ = 509.8± 1.4
σ = 75.6± 1.1


Energy loss | -32 < θ < -30
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (4660 entries)
σ/μ = 15.09± 0.22 %
μ = 500.0± 1.2
σ = 75.4± 0.9
WF (4660 entries)
σ/μ = 15.26± 0.22 %
μ = 508.7± 1.2
σ = 77.6± 0.9

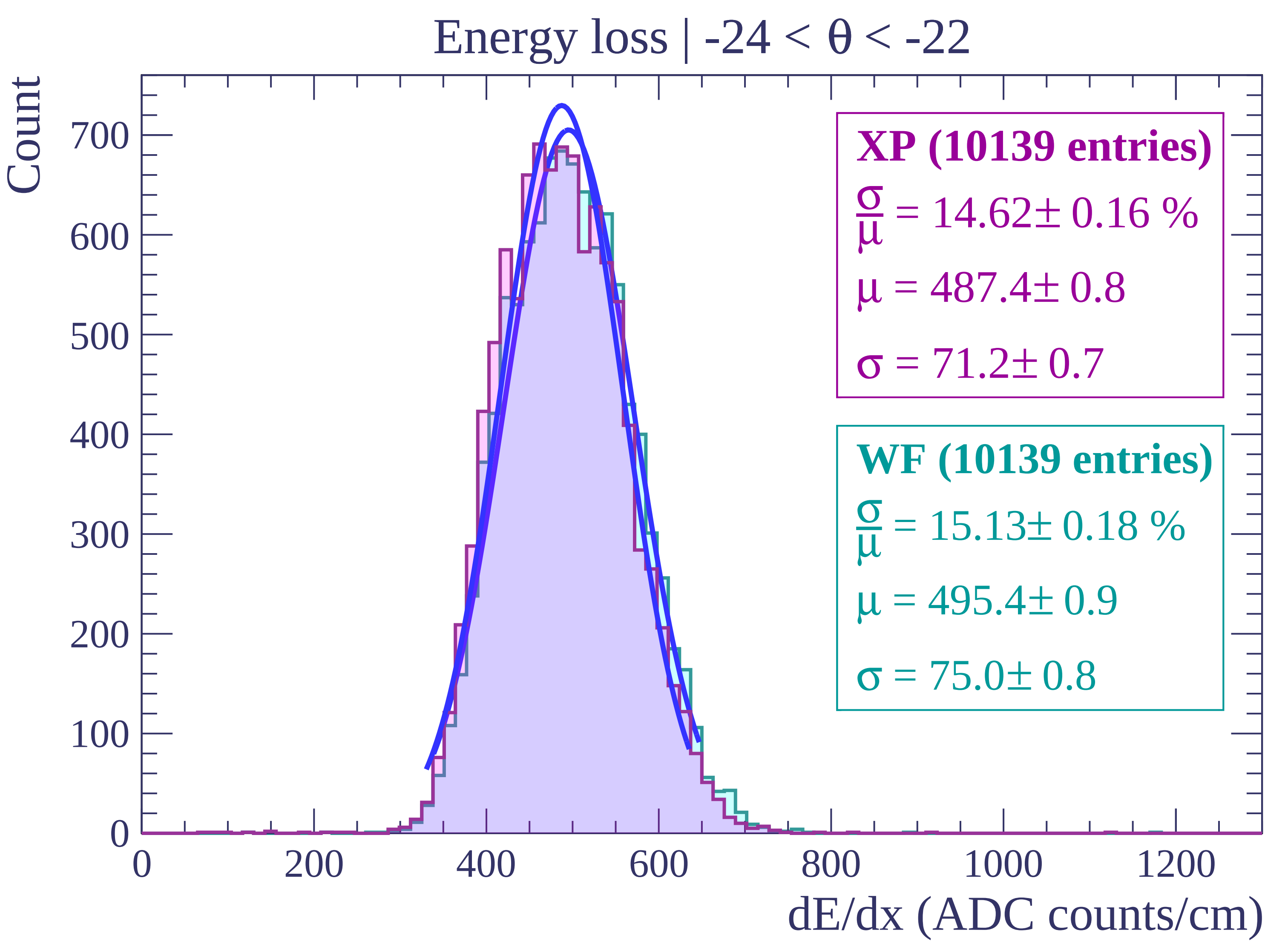

Energy loss | -30 < θ < -28
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6287 entries)
σ/μ = 14.64± 0.19 %
μ = 494.1± 1.0
σ = 72.3± 0.8
WF (6287 entries)
σ/μ = 15.28± 0.22 %
μ = 501.7± 1.1
σ = 76.6± 1.0

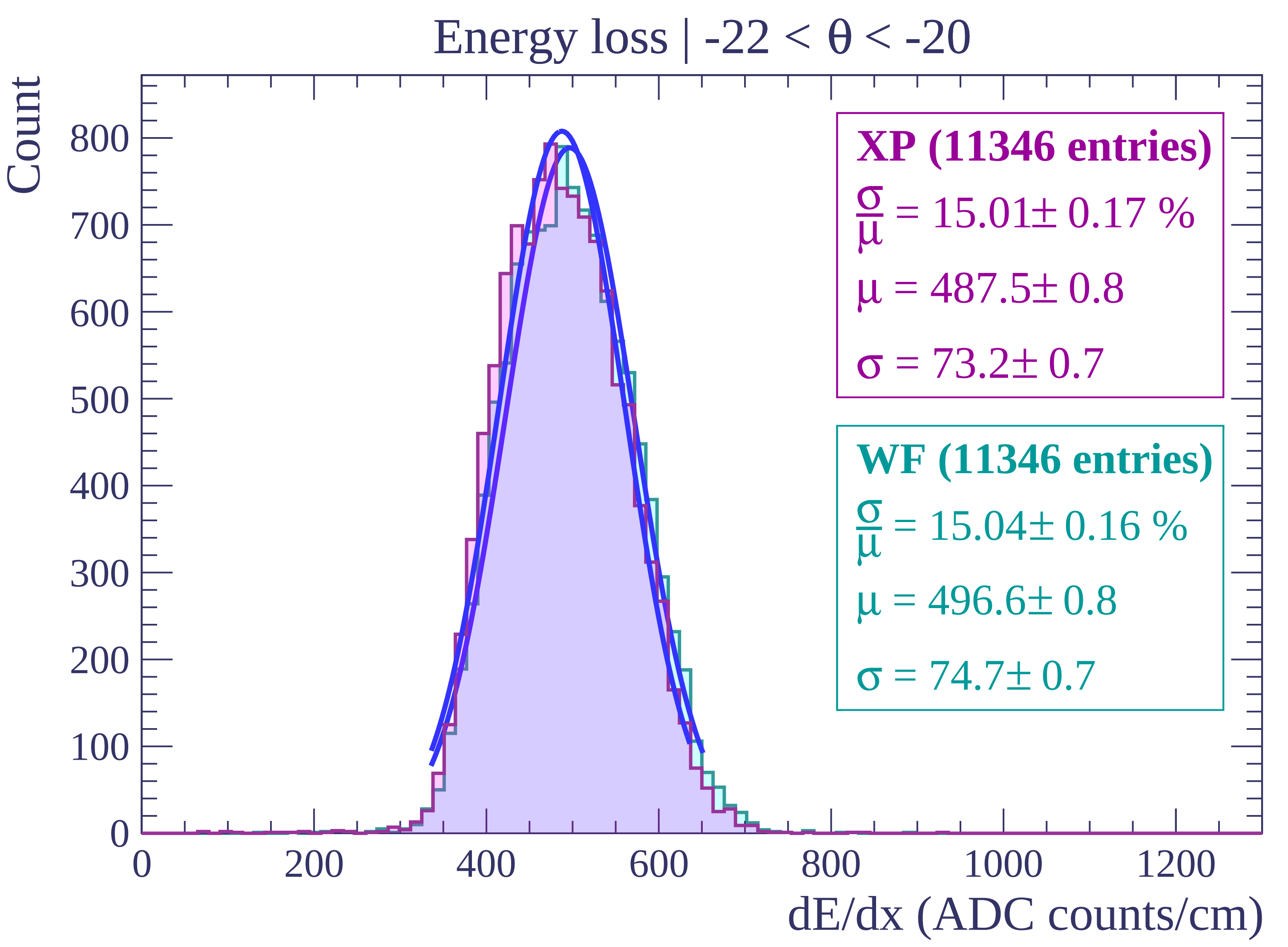


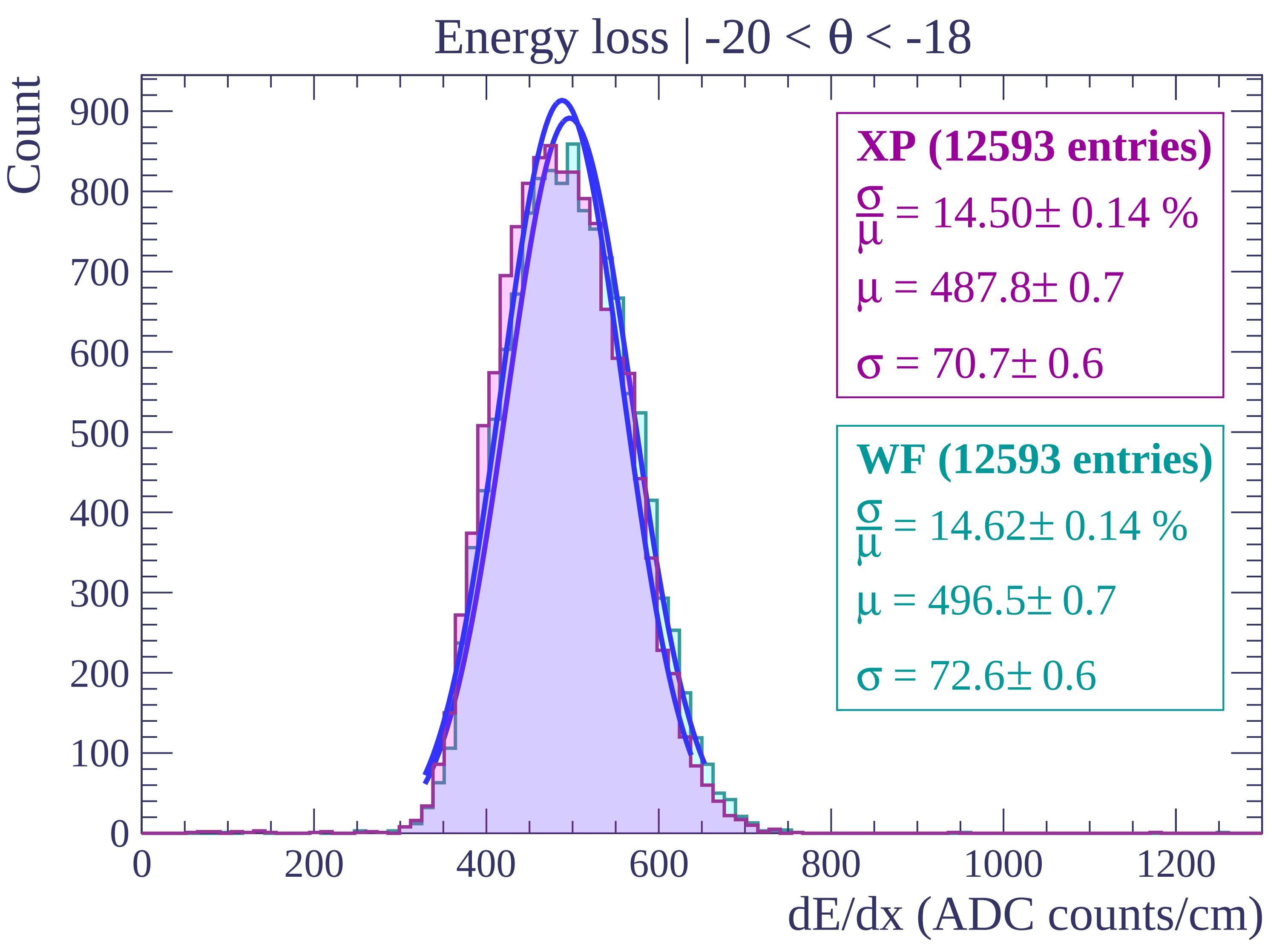
Energy loss | -28 < θ < -26
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8029 entries)
σ/μ = 14.59± 0.17 %
μ = 493.1± 0.9
σ = 71.9± 0.7
WF (8029 entries)
σ/μ = 15.01± 0.19 %
μ = 500.2± 1.0
σ = 75.1± 0.8

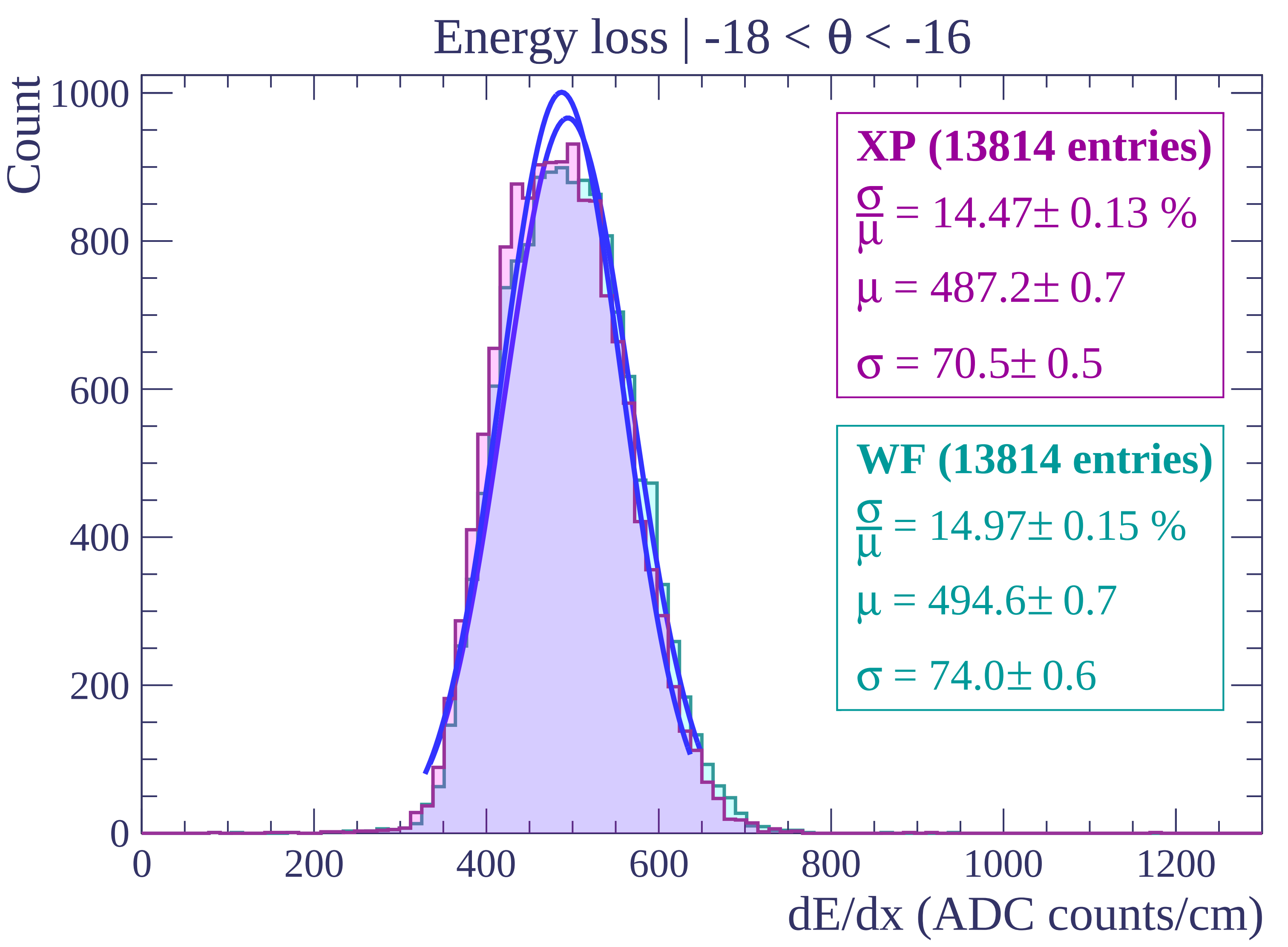


Energy loss | -26 < θ < -24
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (9306 entries)
σ/μ = 14.75± 0.16 %
μ = 488.4± 0.8
σ = 72.0± 0.6
WF (9306 entries)
σ/μ = 14.91± 0.17 %
μ = 496.9± 0.9
σ = 74.1± 0.7

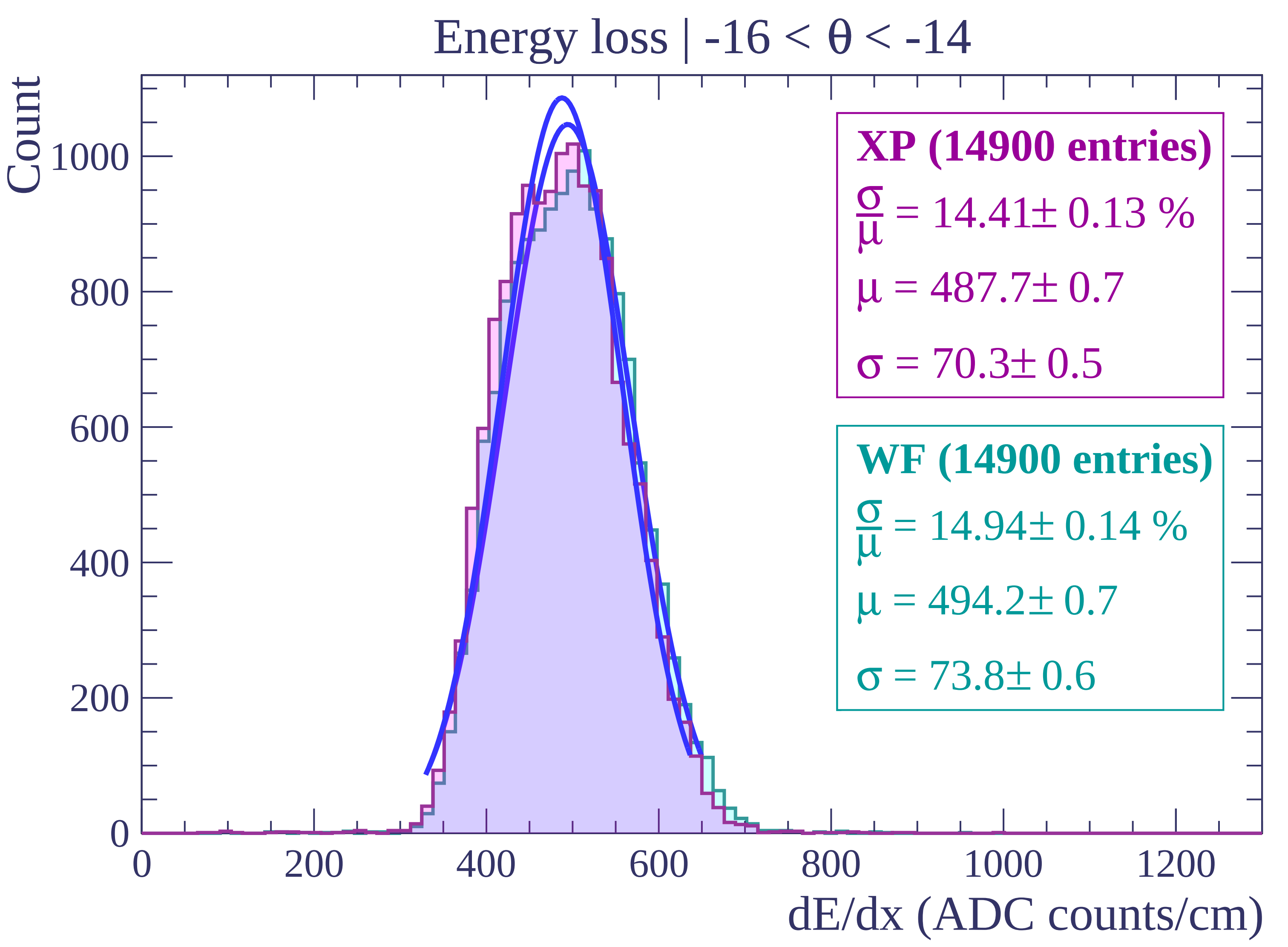

Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (10139 entries)
σ/μ = 14.62± 0.16 %
μ = 487.4± 0.8
σ = 71.2± 0.7
WF (10139 entries)
σ/μ = 15.13± 0.18 %
μ = 495.4± 0.9
σ = 75.0± 0.8


Energy loss | -22 < θ < -20
Count
dE/dx (ADC counts/cm)
XP (11346 entries)
σ/μ = 15.01± 0.17 %
μ = 487.5± 0.8
σ = 73.2± 0.7
WF (11346 entries)
σ/μ = 15.04± 0.16 %
μ = 496.6± 0.8
σ = 74.7± 0.7
0
200
400
600
800
1000
1200
100
300
500
700

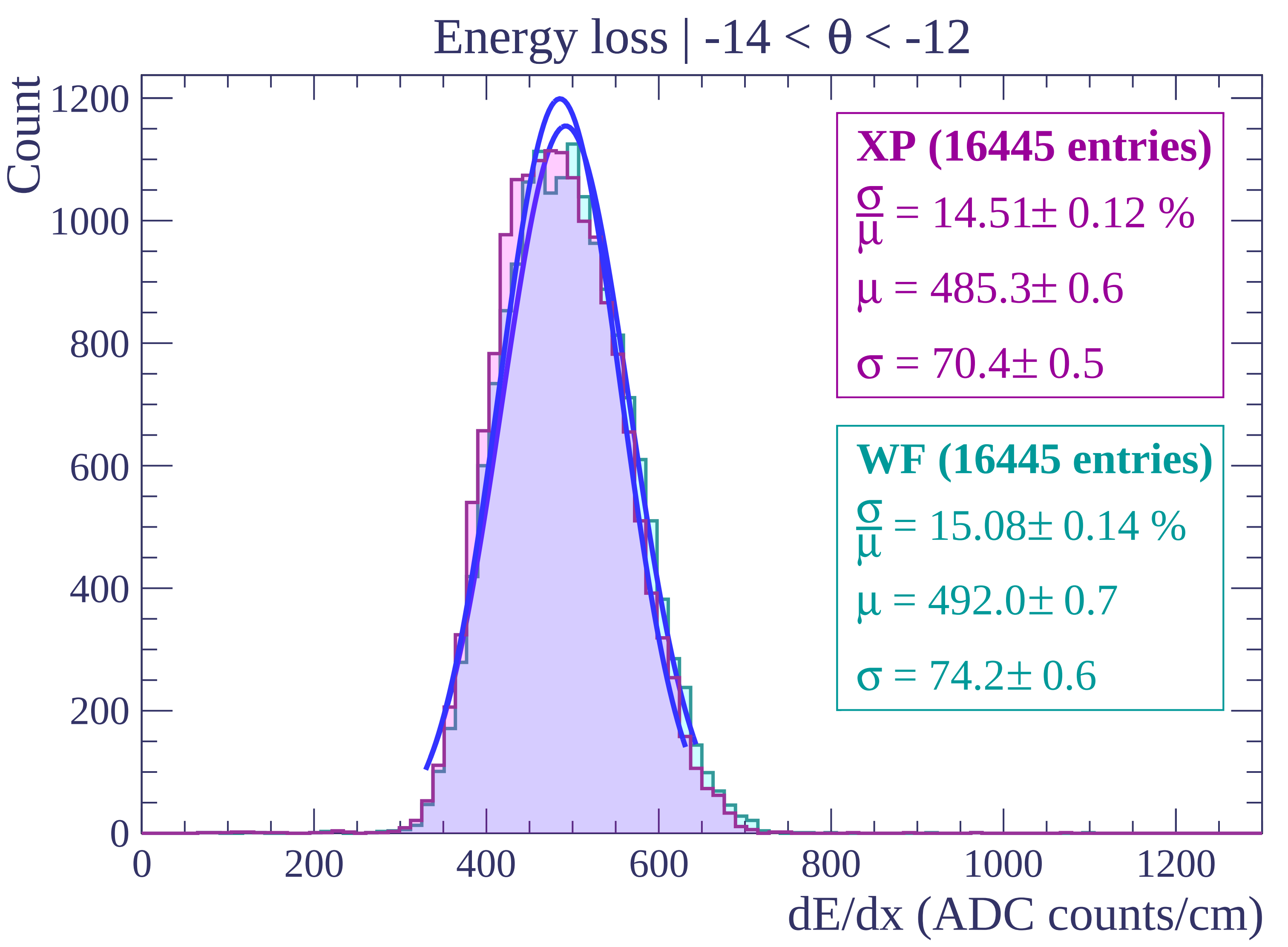

Energy loss | -20 < θ < -18
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (12593 entries)
σ/μ = 14.50± 0.14 %
μ = 487.8± 0.7
σ = 70.7± 0.6
WF (12593 entries)
σ/μ = 14.62± 0.14 %
μ = 496.5± 0.7
σ = 72.6± 0.6



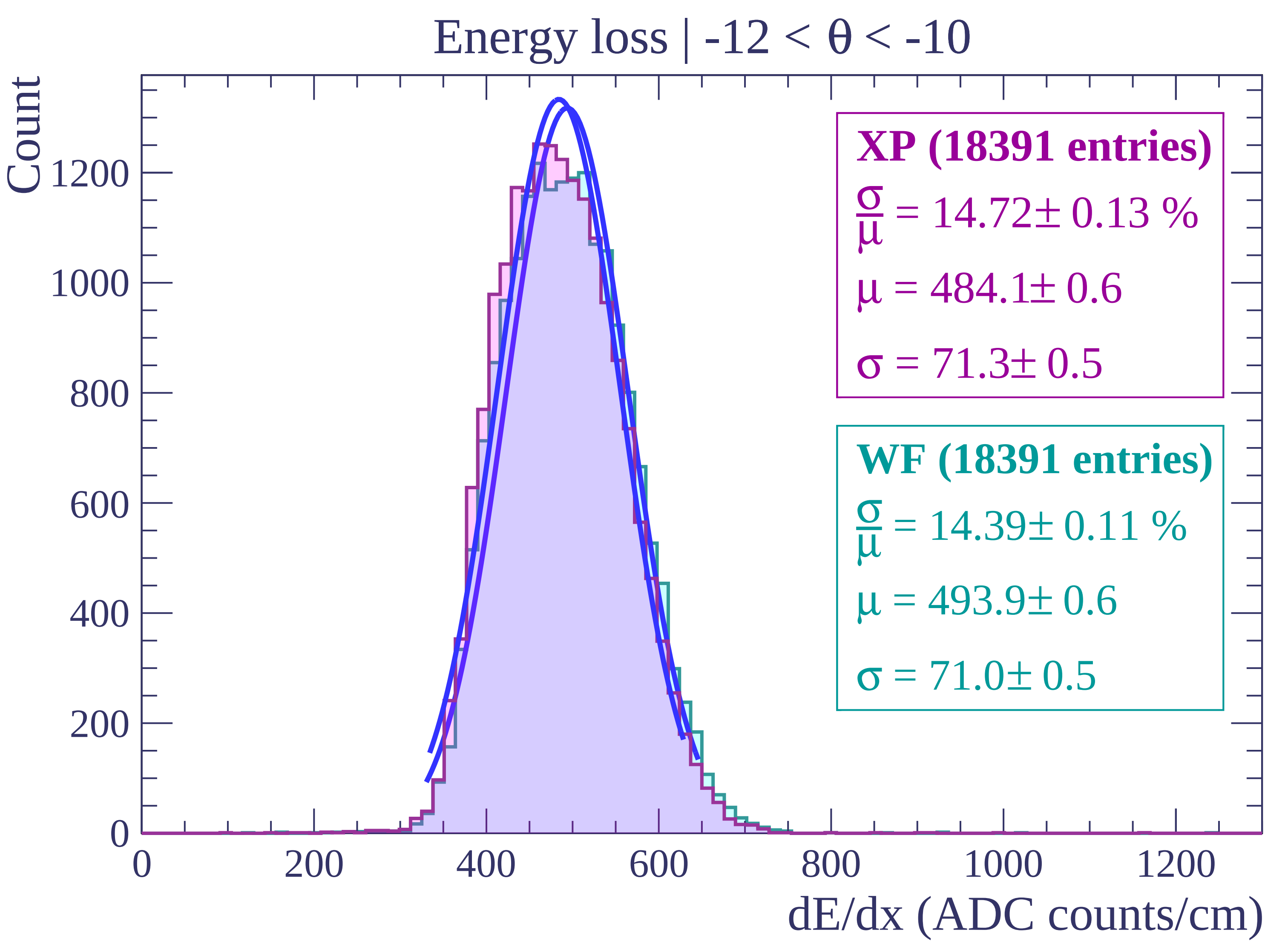
Energy loss | -18 < θ < -16
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (13814 entries)
σ/μ = 14.47± 0.13 %
μ = 487.2± 0.7
σ = 70.5± 0.5
WF (13814 entries)
σ/μ = 14.97± 0.15 %
μ = 494.6± 0.7
σ = 74.0± 0.6

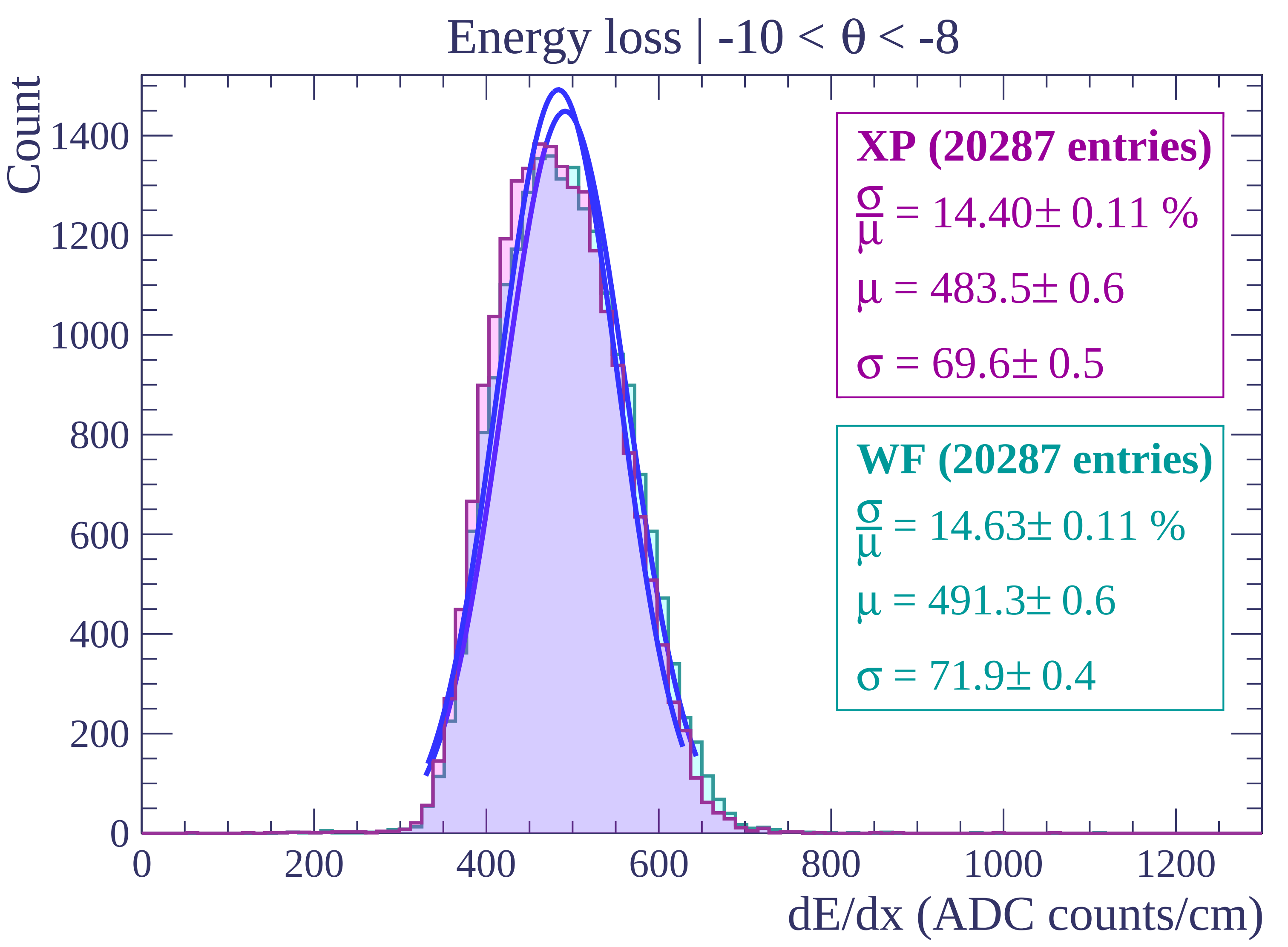


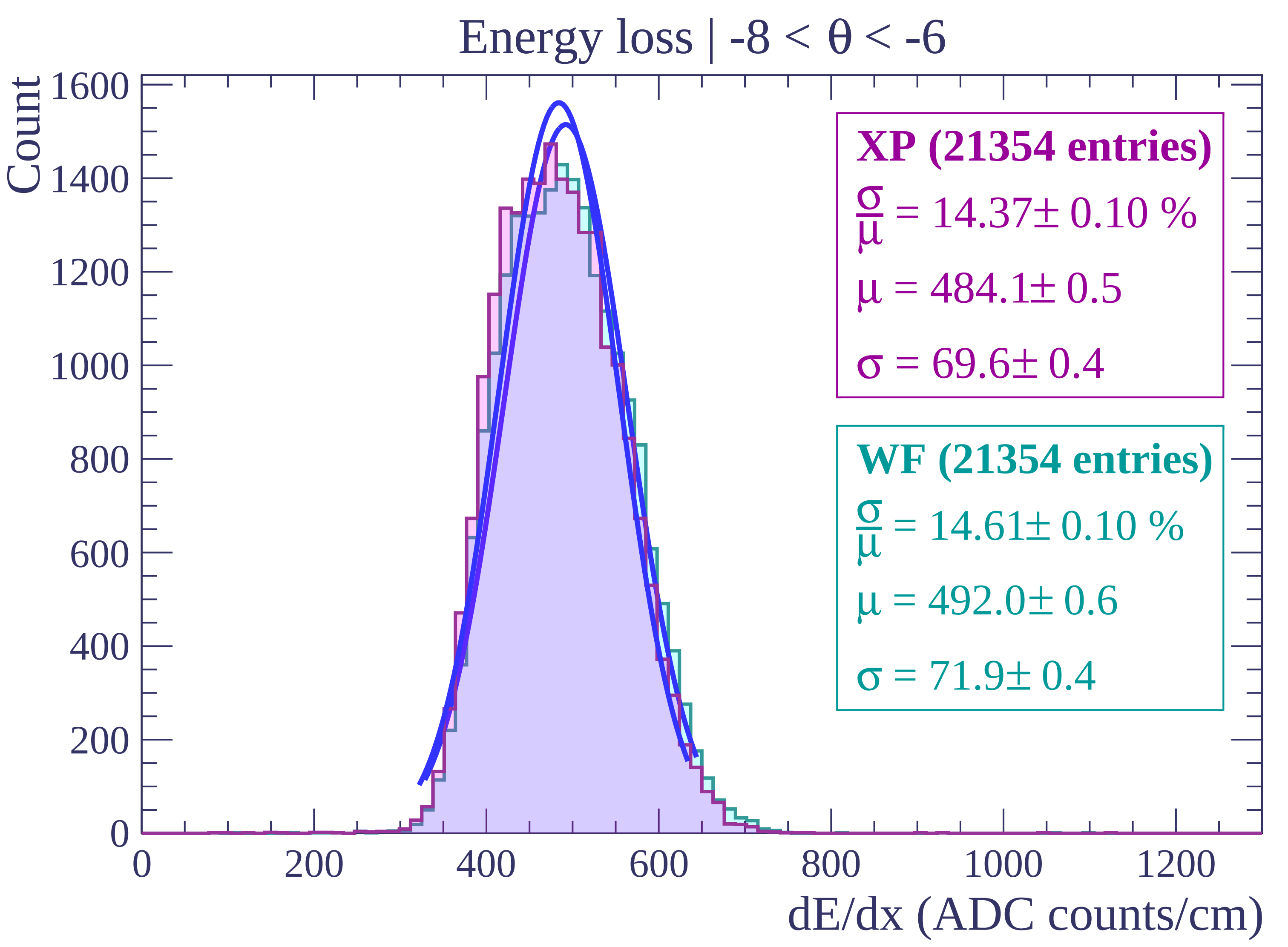
Energy loss | -16 < θ < -14
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (14900 entries)
σ/μ = 14.41± 0.13 %
μ = 487.7± 0.7
σ = 70.3± 0.5
WF (14900 entries)
σ/μ = 14.94± 0.14 %
μ = 494.2± 0.7
σ = 73.8± 0.6



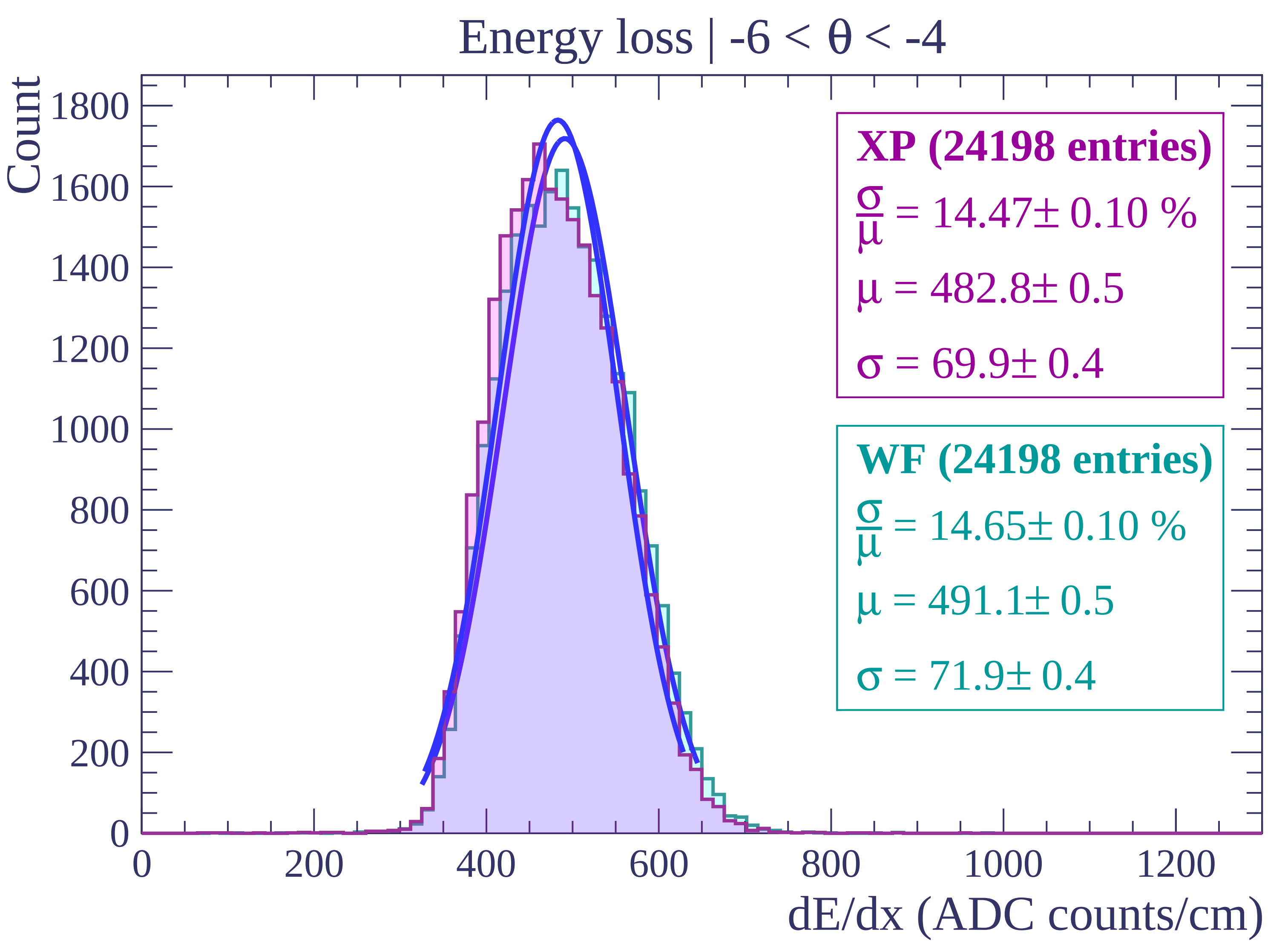
Energy loss | -14 < θ < -12
Count
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (16445 entries)
σ/μ = 14.51± 0.12 %
μ = 485.3± 0.6
σ = 70.4± 0.5
WF (16445 entries)
σ/μ = 15.08± 0.14 %
μ = 492.0± 0.7
σ = 74.2± 0.6

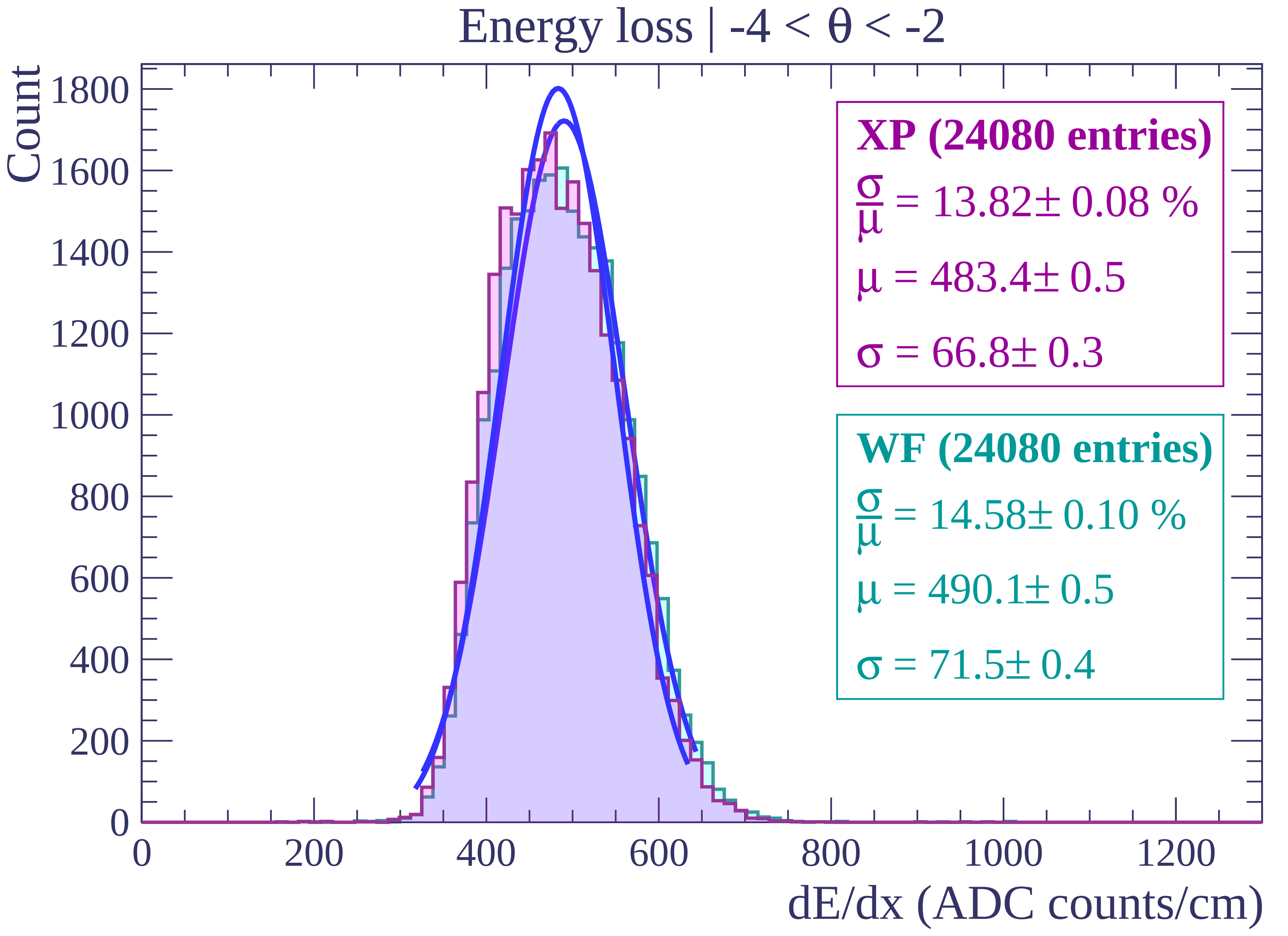

Energy loss | -12 < θ < -10
Count
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (18391 entries)
σ/μ = 14.72± 0.13 %
μ = 484.1± 0.6
σ = 71.3± 0.5
WF (18391 entries)
σ/μ = 14.39± 0.11 %
μ = 493.9± 0.6
σ = 71.0± 0.5

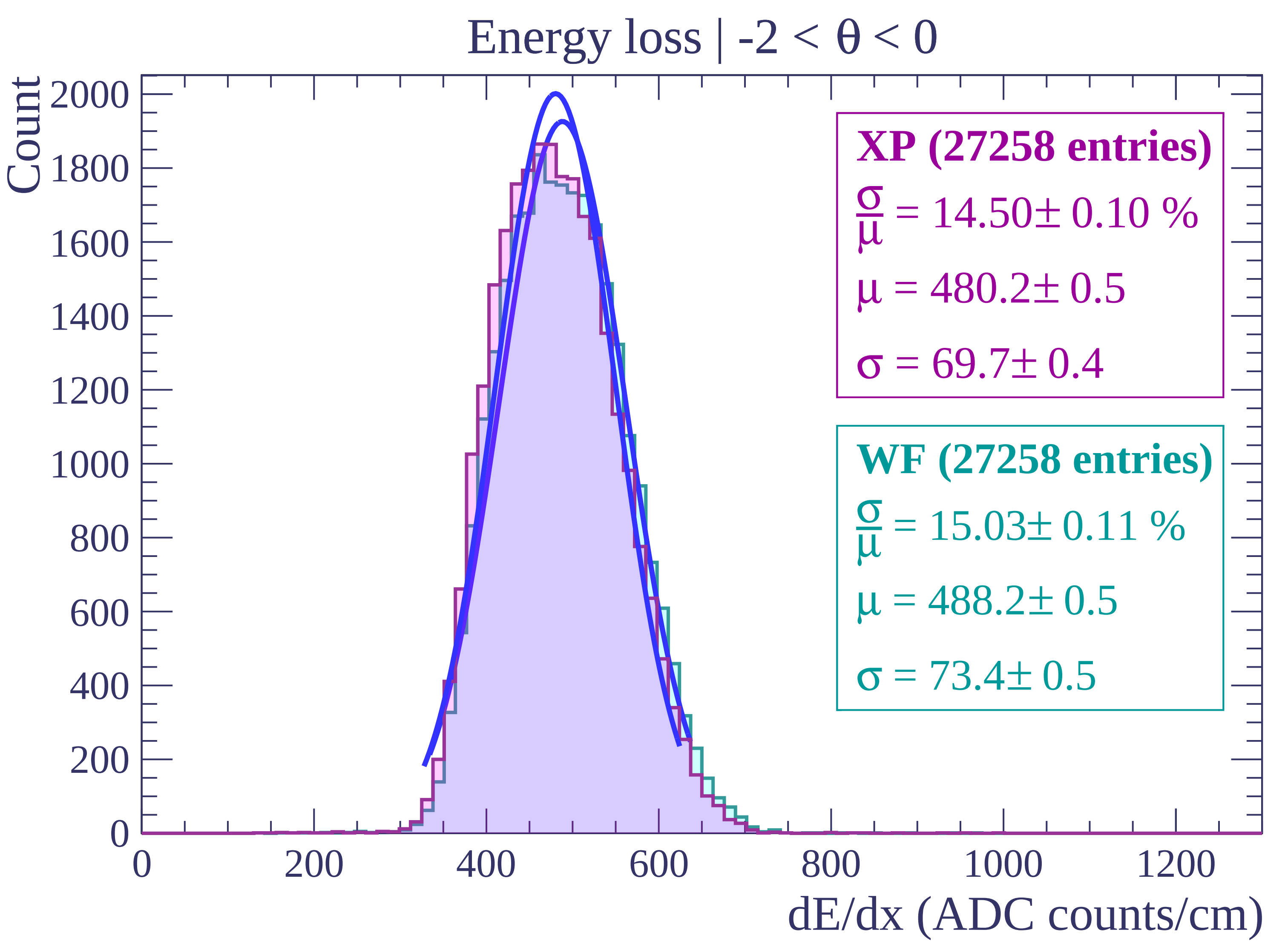

Energy loss | -10 < θ < -8
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP (20287 entries)
σ/μ = 14.40± 0.11 %
μ = 483.5± 0.6
σ = 69.6± 0.5
WF (20287 entries)
σ/μ = 14.63± 0.11 %
μ = 491.3± 0.6
σ = 71.9± 0.4



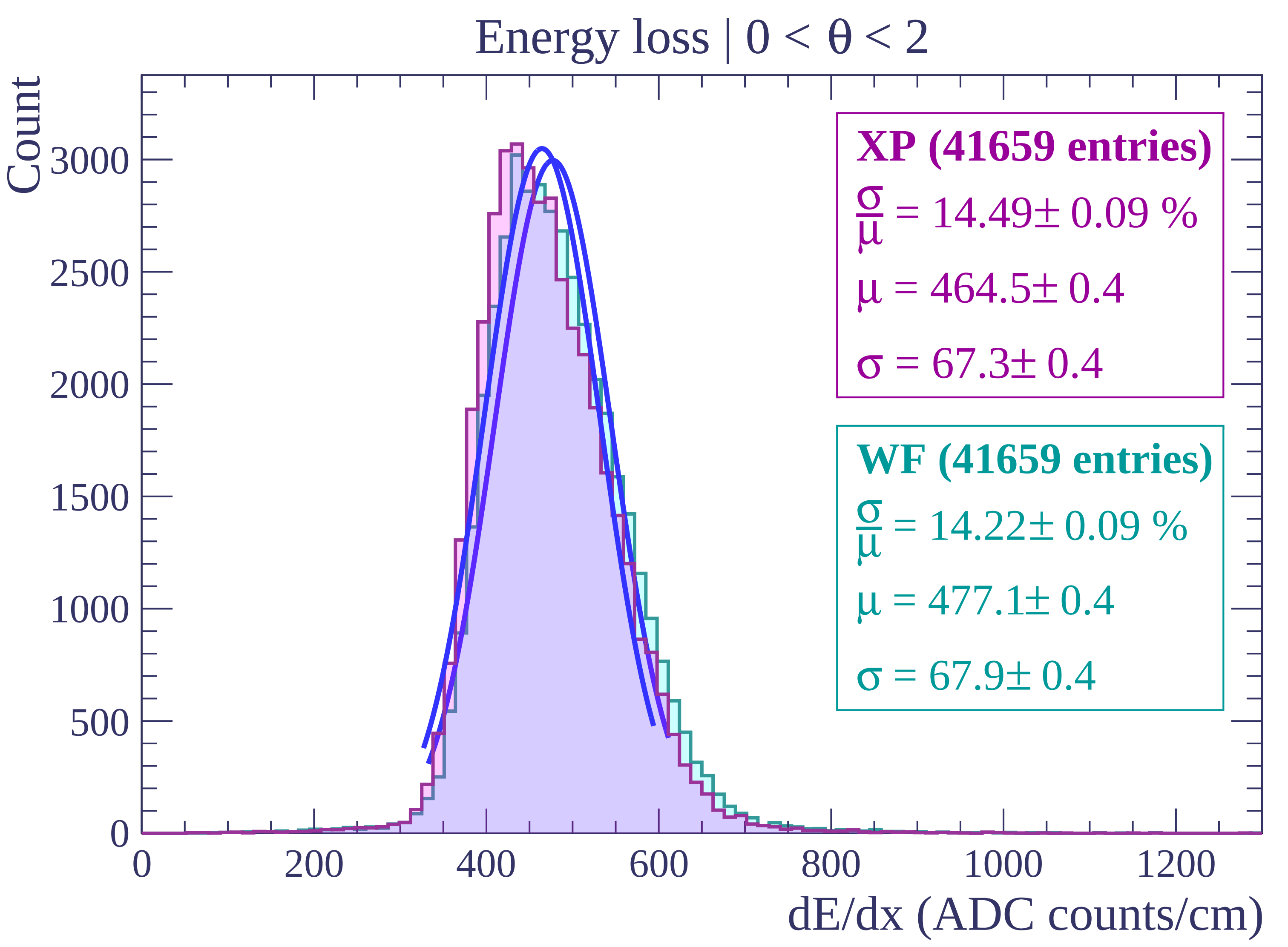
Energy loss | -8 < θ < -6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200
XP (21354 entries)
$\frac{\sigma}{\mu}$ = 14.37± 0.10 %
μ = 484.1± 0.5
σ = 69.6± 0.4
WF (21354 entries)
$\frac{\sigma}{\mu}$ = 14.61± 0.10 %
μ = 492.0± 0.6
σ = 71.9± 0.4


Energy loss | -6 < θ < -4
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200
XP (24198 entries)
σ/μ = 14.47± 0.10 %
μ = 482.8± 0.5
σ = 69.9± 0.4
WF (24198 entries)
σ/μ = 14.65± 0.10 %
μ = 491.1± 0.5
σ = 71.9± 0.4

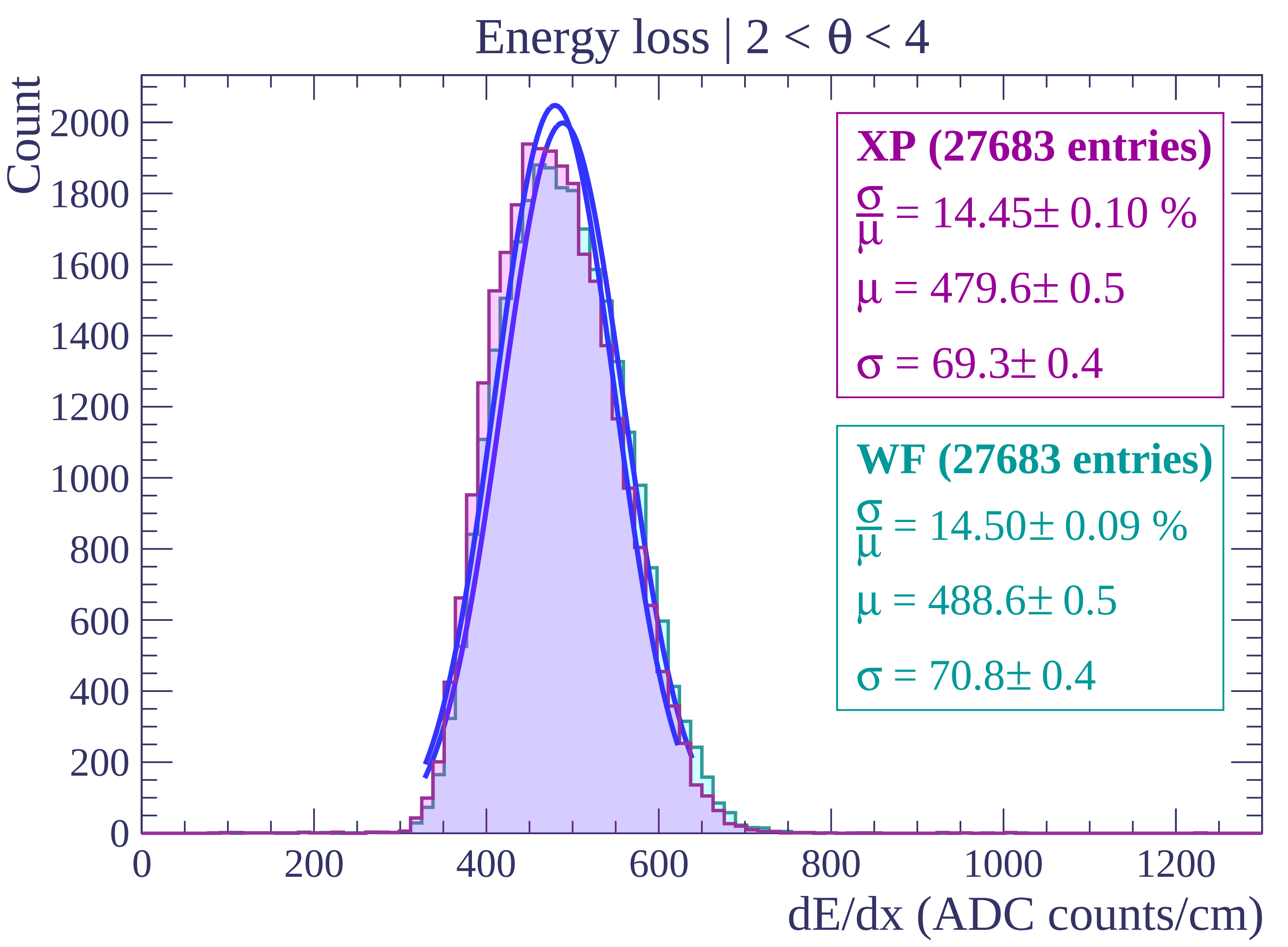

Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
XP (24080 entries)
σ/μ = 13.82± 0.08 %
μ = 483.4± 0.5
σ = 66.8± 0.3
WF (24080 entries)
σ/μ = 14.58± 0.10 %
μ = 490.1± 0.5
σ = 71.5± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800

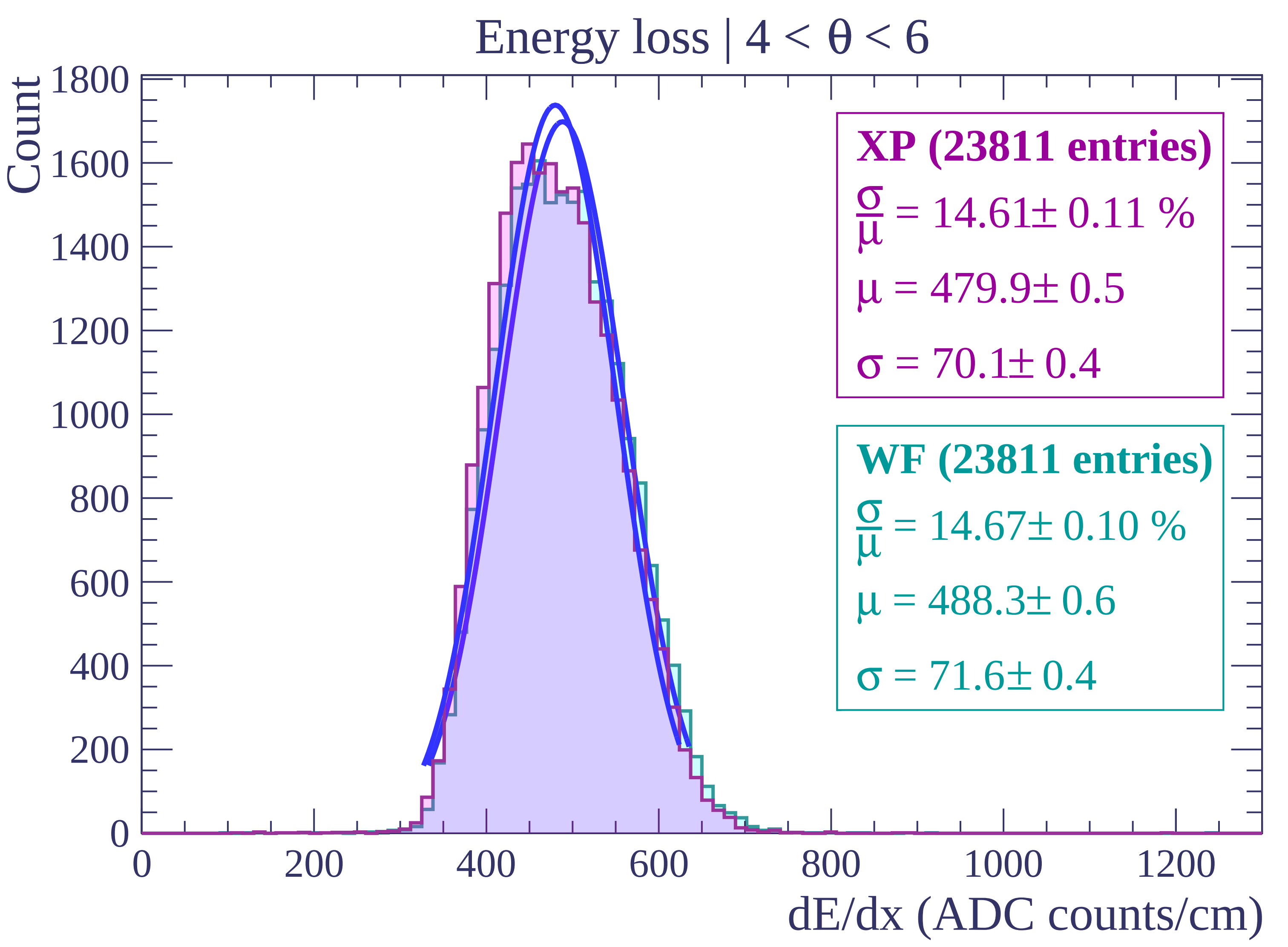


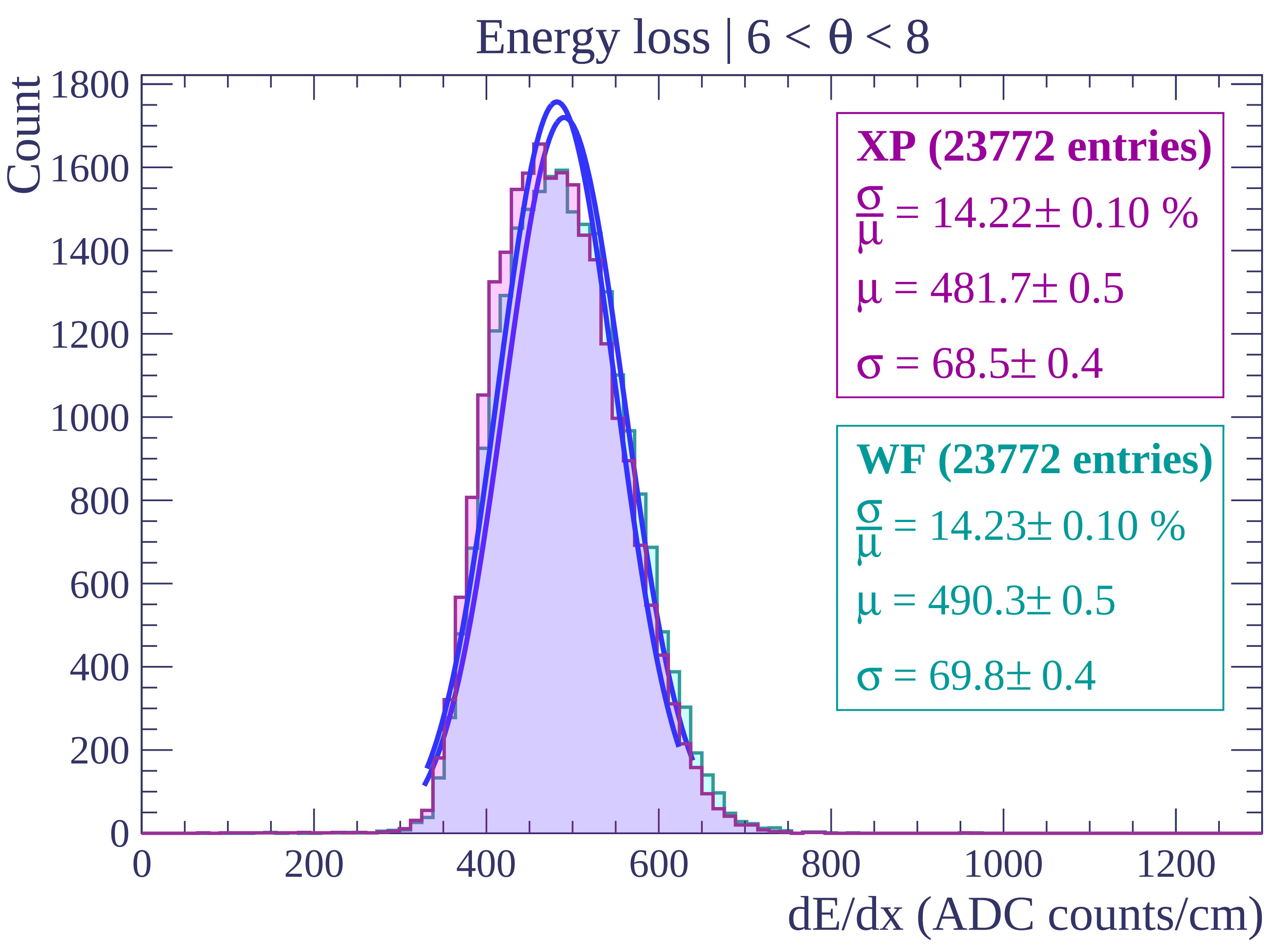
Energy loss | -2 < θ < 0
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200
XP (27258 entries)
$\frac{\sigma}{\mu}$ = 14.50± 0.10 %
μ = 480.2± 0.5
σ = 69.7± 0.4
WF (27258 entries)
$\frac{\sigma}{\mu}$ = 15.03± 0.11 %
μ = 488.2± 0.5
σ = 73.4± 0.5

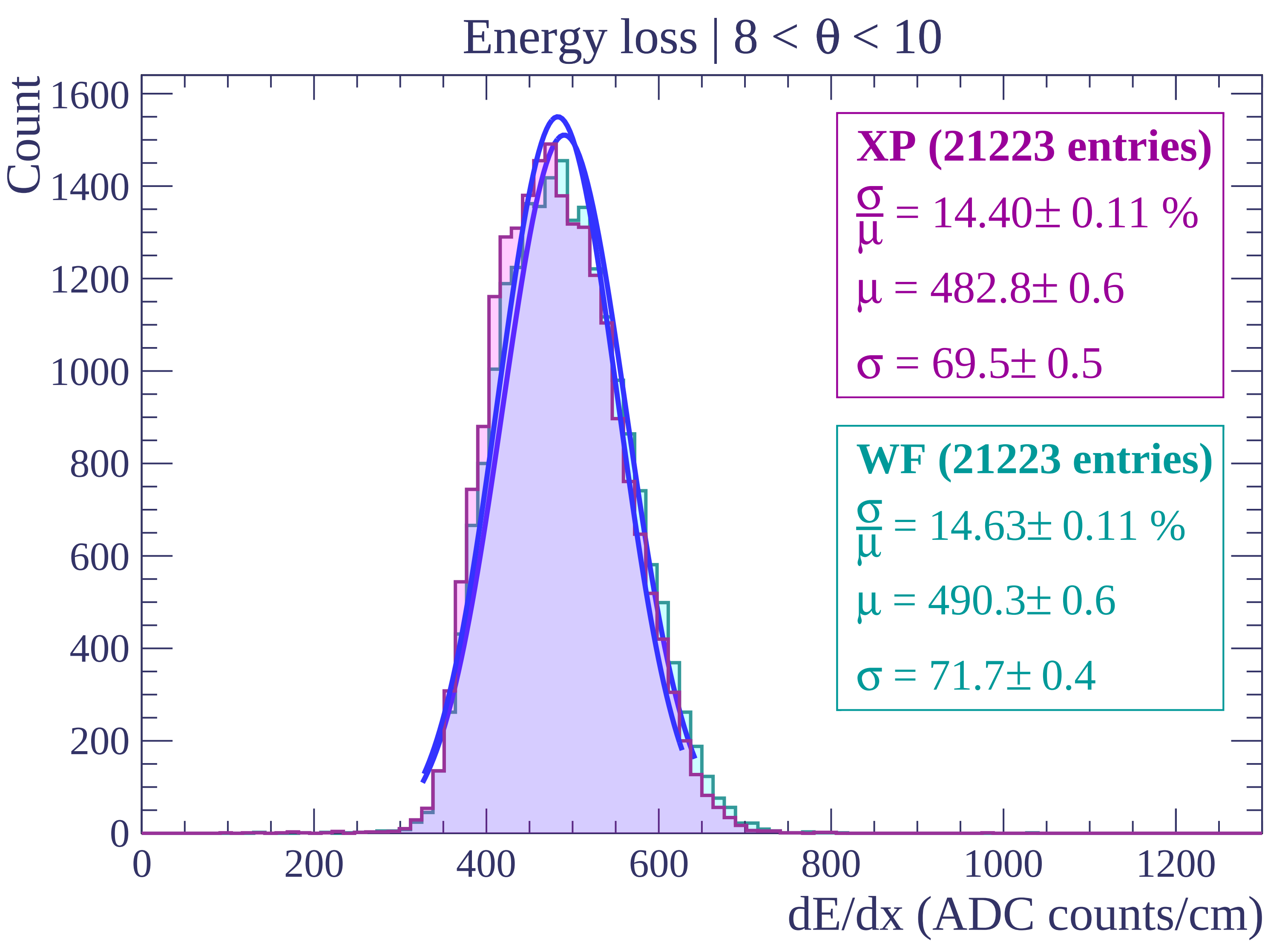


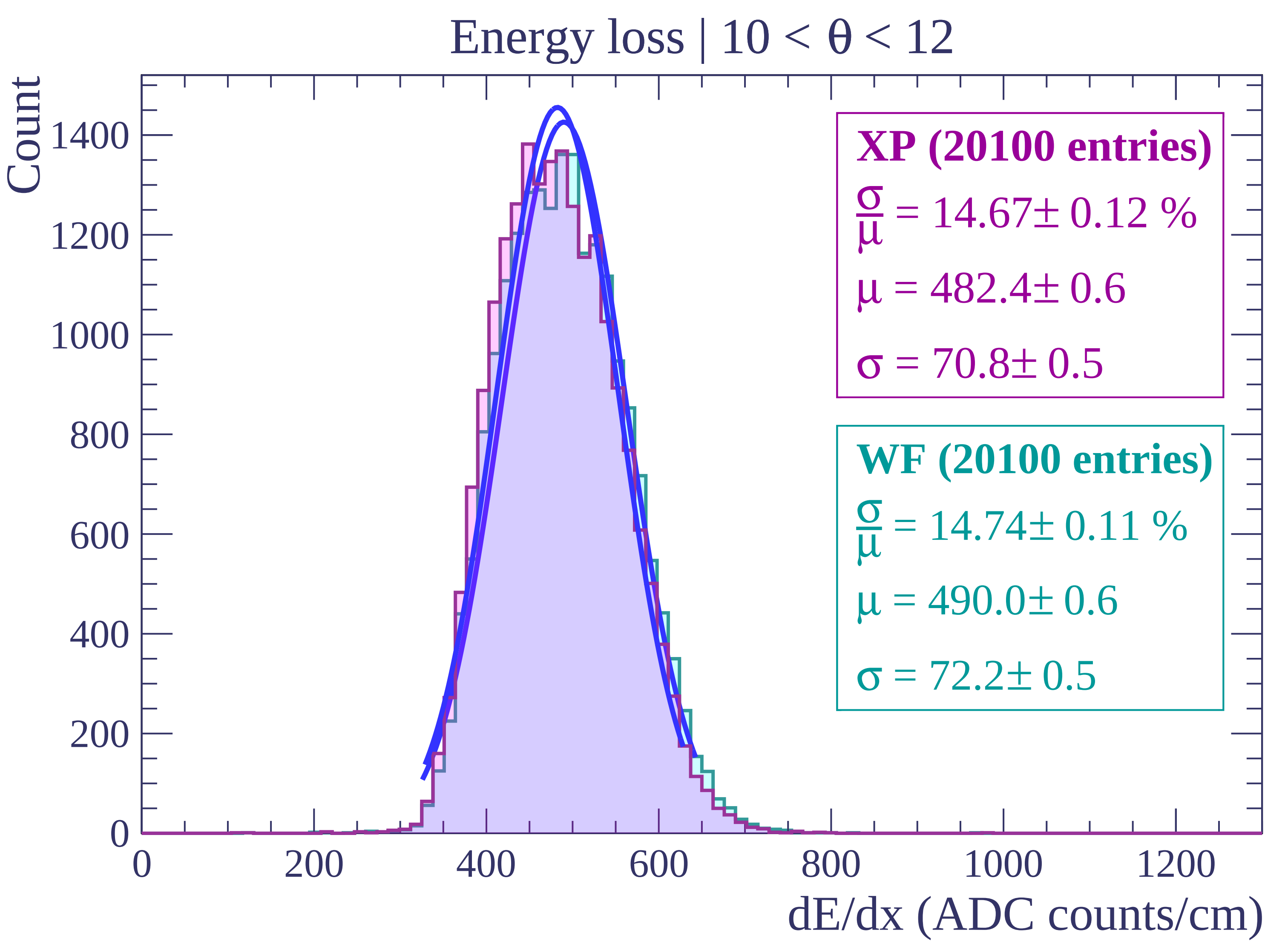
Energy loss | 0 < θ < 2
Count
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (41659 entries)
σ/μ = 14.49± 0.09 %
μ = 464.5± 0.4
σ = 67.3± 0.4
WF (41659 entries)
σ/μ = 14.22± 0.09 %
μ = 477.1± 0.4
σ = 67.9± 0.4


Energy loss | 2 < θ < 4
Count
dE/dx (ADC counts/cm)
XP (27683 entries)
σ/μ = 14.45± 0.10 %
μ = 479.6± 0.5
σ = 69.3± 0.4
WF (27683 entries)
σ/μ = 14.50± 0.09 %
μ = 488.6± 0.5
σ = 70.8± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200

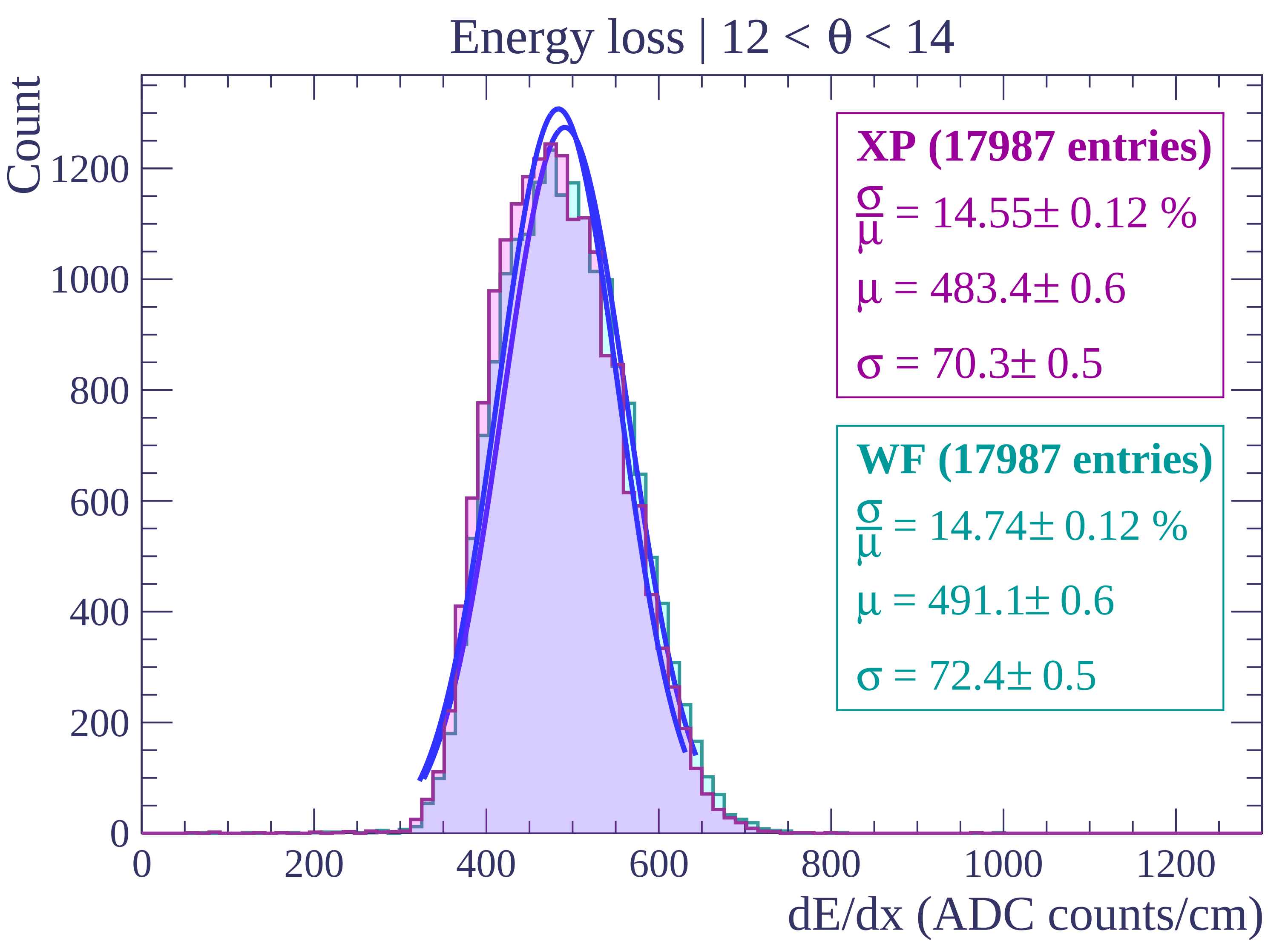

Energy loss | 4 < θ < 6
Count
dE/dx (ADC counts/cm)
XP (23811 entries)
σ/μ = 14.61± 0.11 %
μ = 479.9± 0.5
σ = 70.1± 0.4
WF (23811 entries)
σ/μ = 14.67± 0.10 %
μ = 488.3± 0.6
σ = 71.6± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800


Energy loss | 6 < θ < 8
Count
dE/dx (ADC counts/cm)
XP (23772 entries)
σ/μ = 14.22± 0.10 %
μ = 481.7± 0.5
σ = 68.5± 0.4
WF (23772 entries)
σ/μ = 14.23± 0.10 %
μ = 490.3± 0.5
σ = 69.8± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800

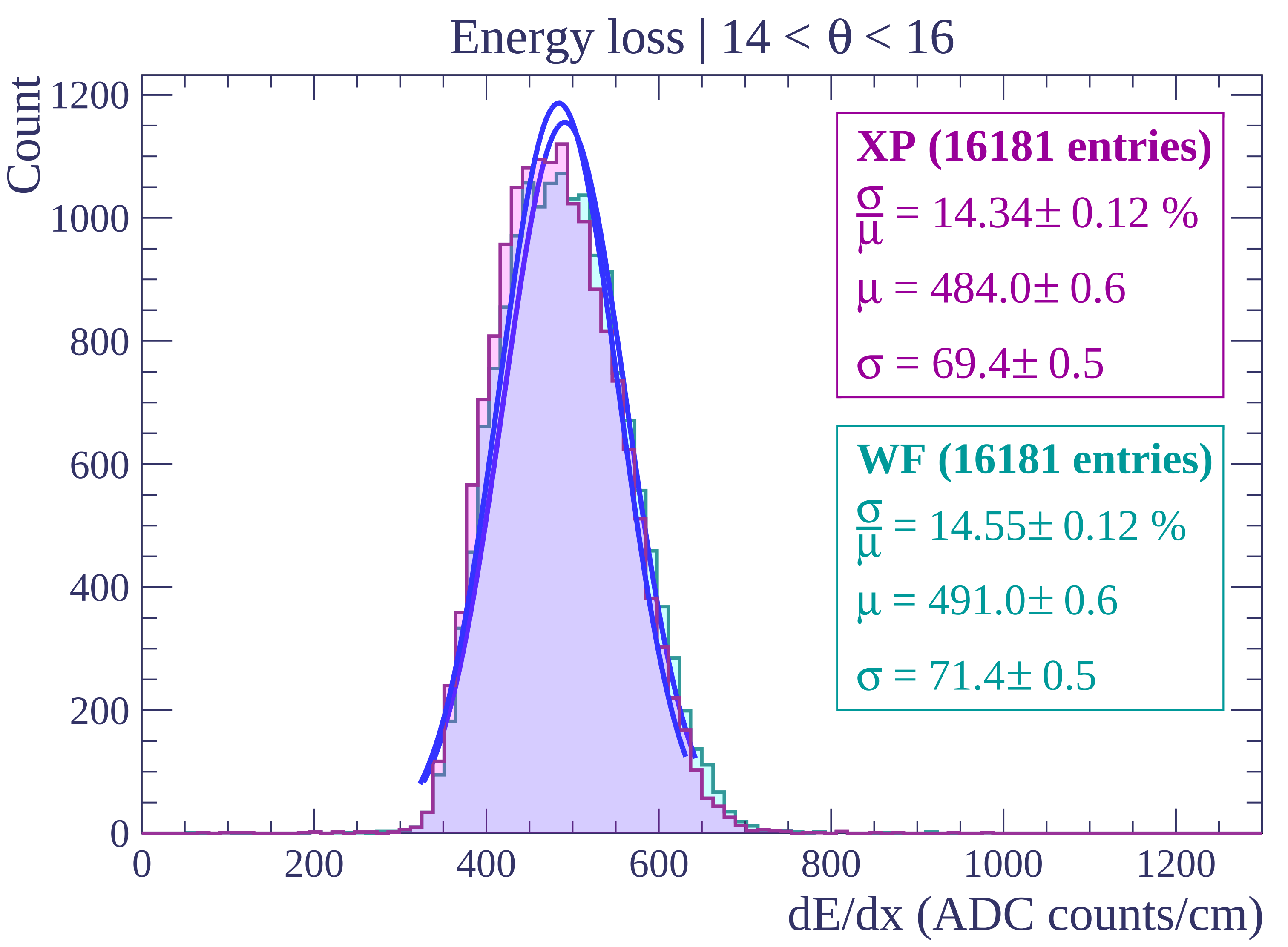

Energy loss | 8 < θ < 10
Count
dE/dx (ADC counts/cm)
XP (21223 entries)
σ/μ = 14.40± 0.11 %
μ = 482.8± 0.6
σ = 69.5± 0.5
WF (21223 entries)
σ/μ = 14.63± 0.11 %
μ = 490.3± 0.6
σ = 71.7± 0.4
0
200
400
600
800
1000
1200
1400
1600



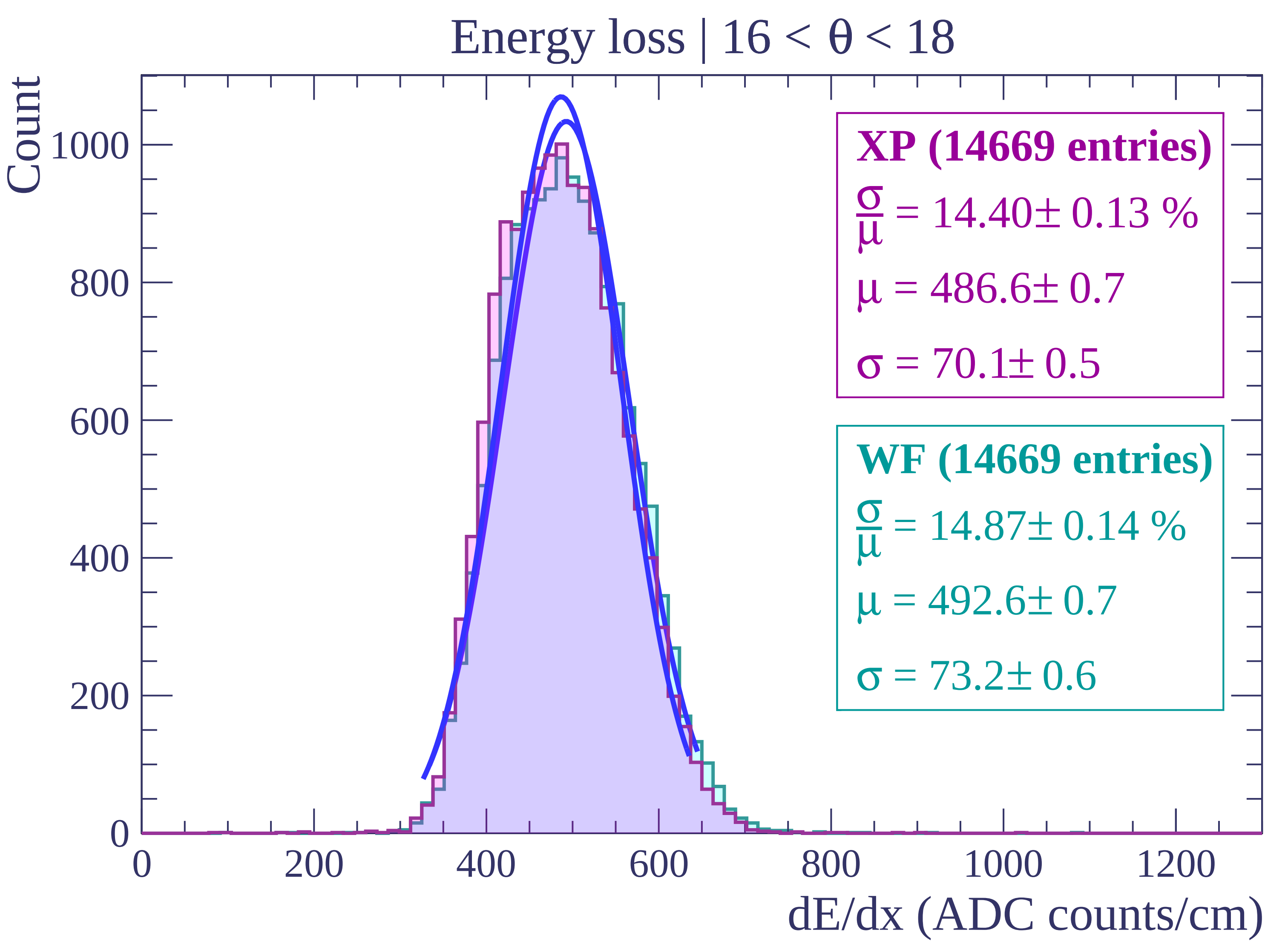
Energy loss | 10 < θ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP (20100 entries)
σ/μ = 14.67± 0.12 %
μ = 482.4± 0.6
σ = 70.8± 0.5
WF (20100 entries)
σ/μ = 14.74± 0.11 %
μ = 490.0± 0.6
σ = 72.2± 0.5

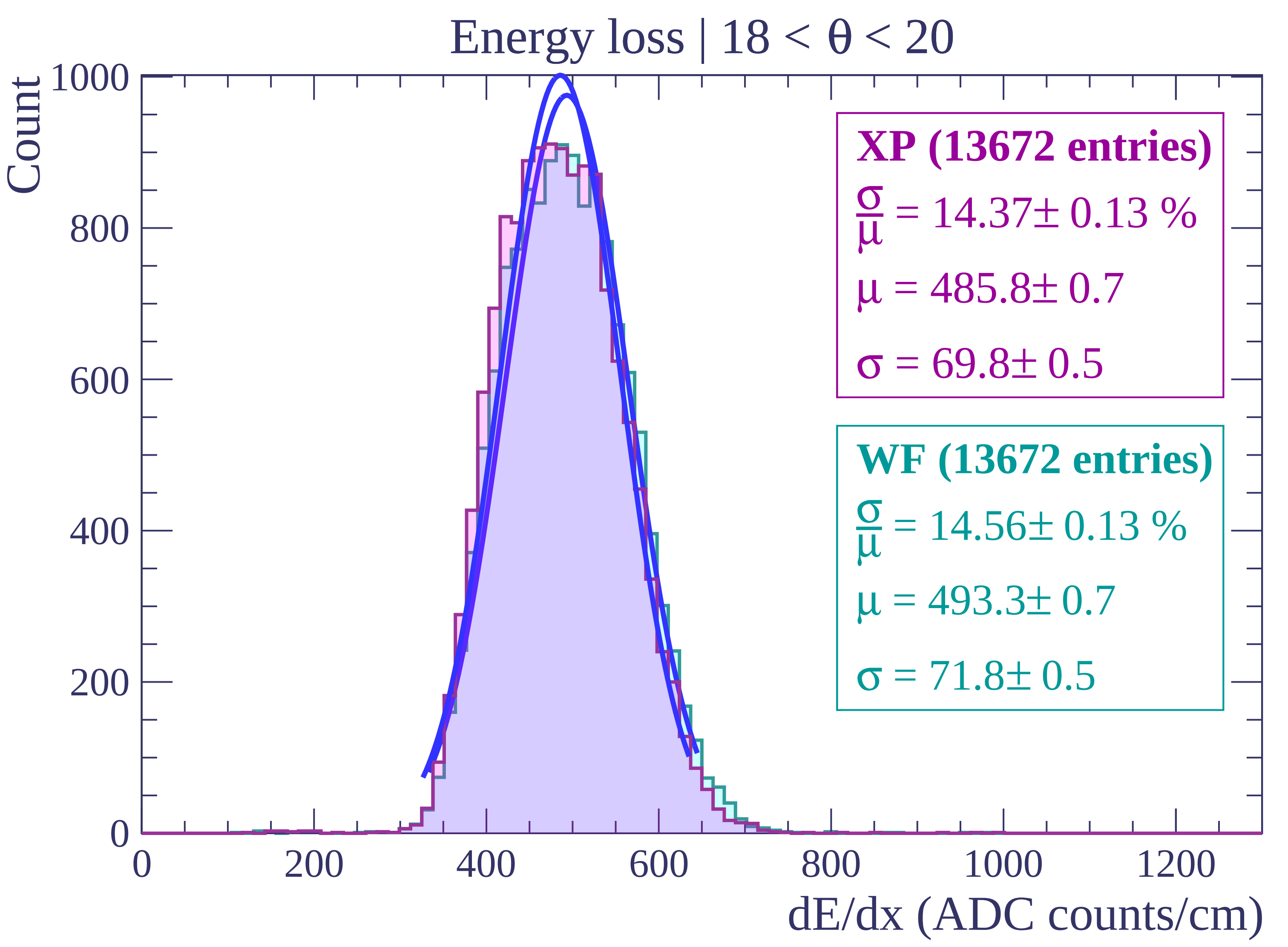


Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (17987 entries)
σ/μ = 14.55± 0.12 %
μ = 483.4± 0.6
σ = 70.3± 0.5
WF (17987 entries)
σ/μ = 14.74± 0.12 %
μ = 491.1± 0.6
σ = 72.4± 0.5

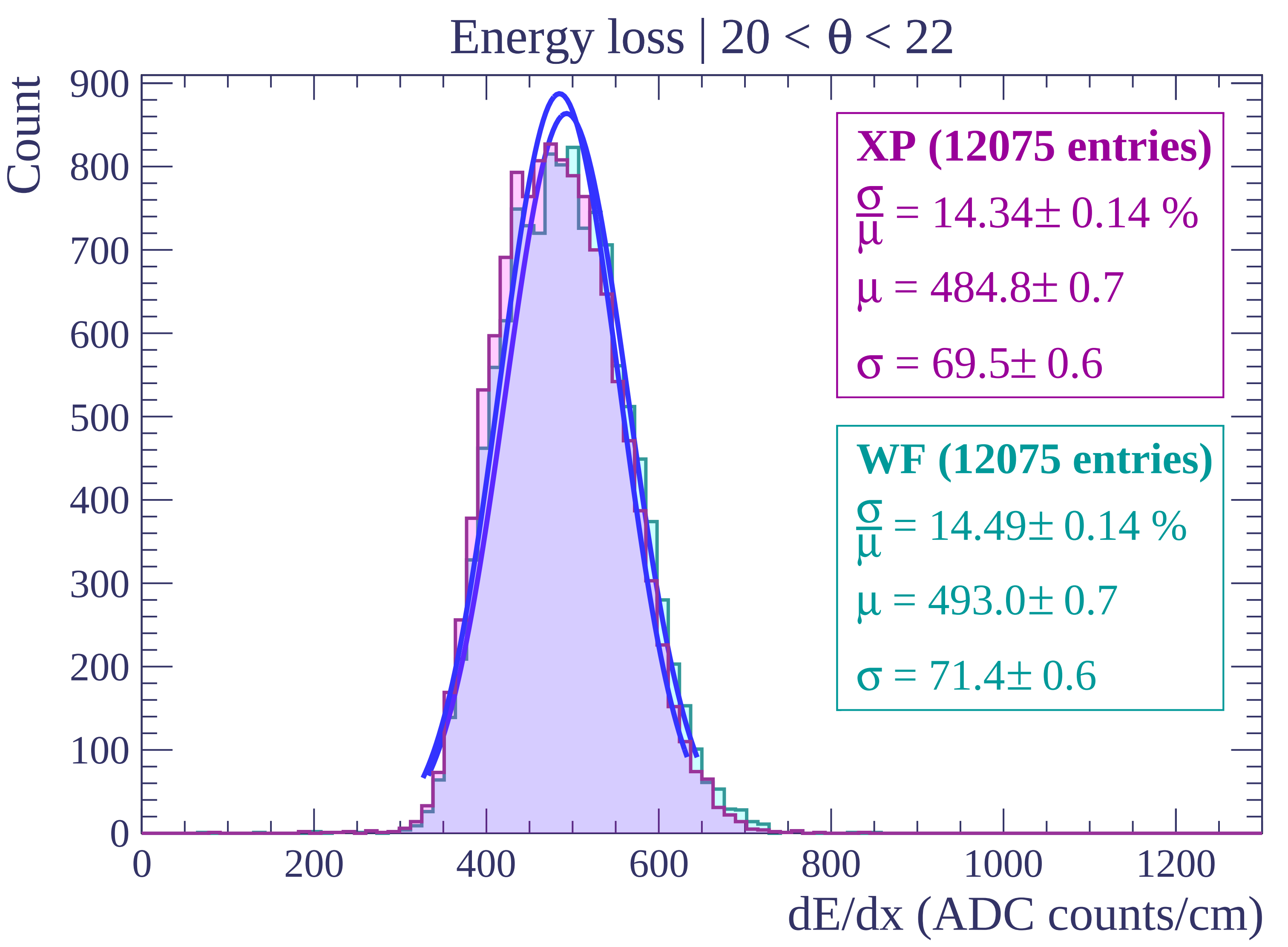


Energy loss | 14 < θ < 16
Count
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (16181 entries)
σ/μ = 14.34± 0.12 %
μ = 484.0± 0.6
σ = 69.4± 0.5
WF (16181 entries)
σ/μ = 14.55± 0.12 %
μ = 491.0± 0.6
σ = 71.4± 0.5

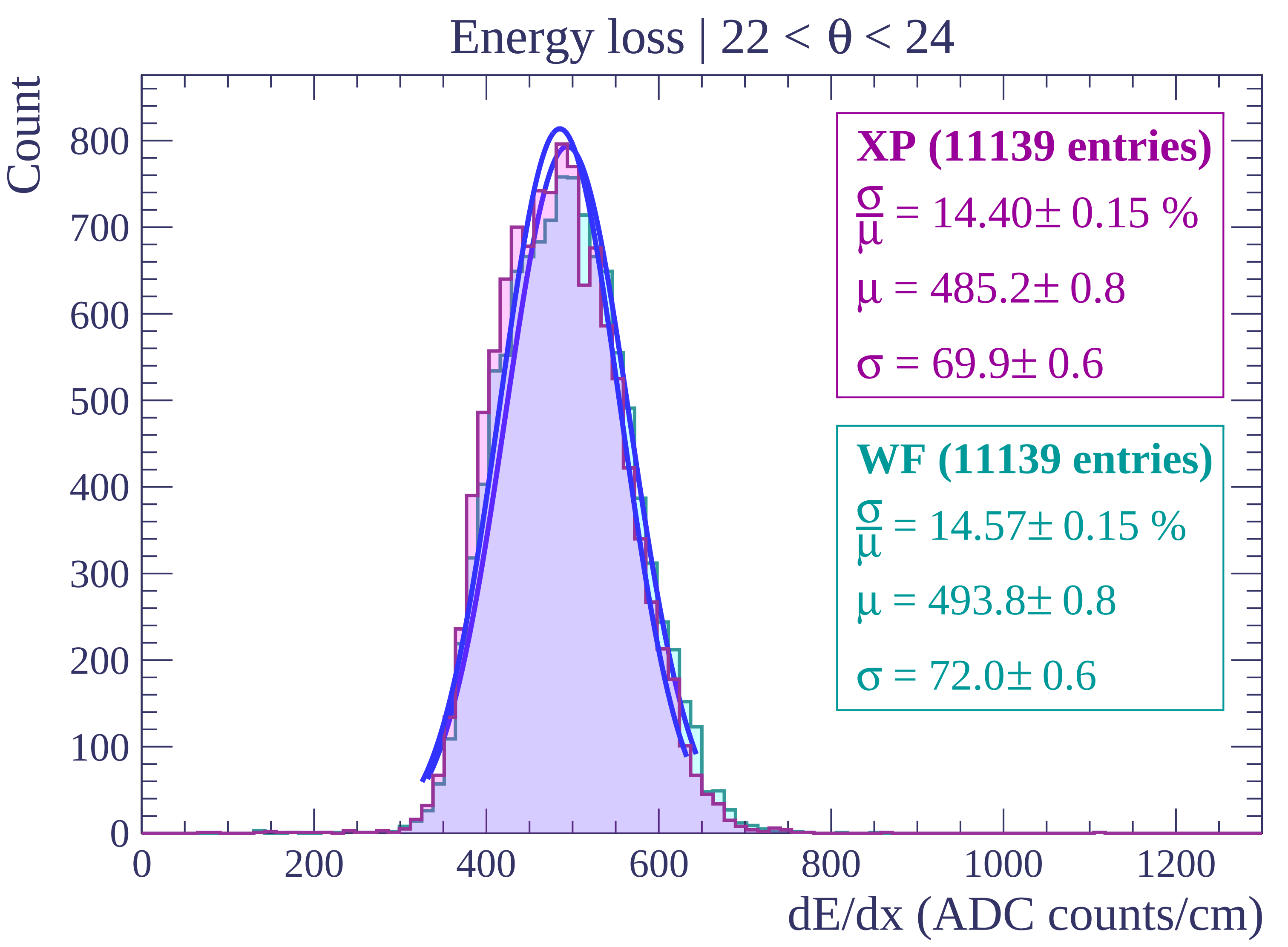

Energy loss | 16 < θ < 18
Count
dE/dx (ADC counts/cm)
XP (14669 entries)
σ/μ = 14.40± 0.13 %
μ = 486.6± 0.7
σ = 70.1± 0.5
WF (14669 entries)
σ/μ = 14.87± 0.14 %
μ = 492.6± 0.7
σ = 73.2± 0.6
0
200
400
600
800
1000
1200

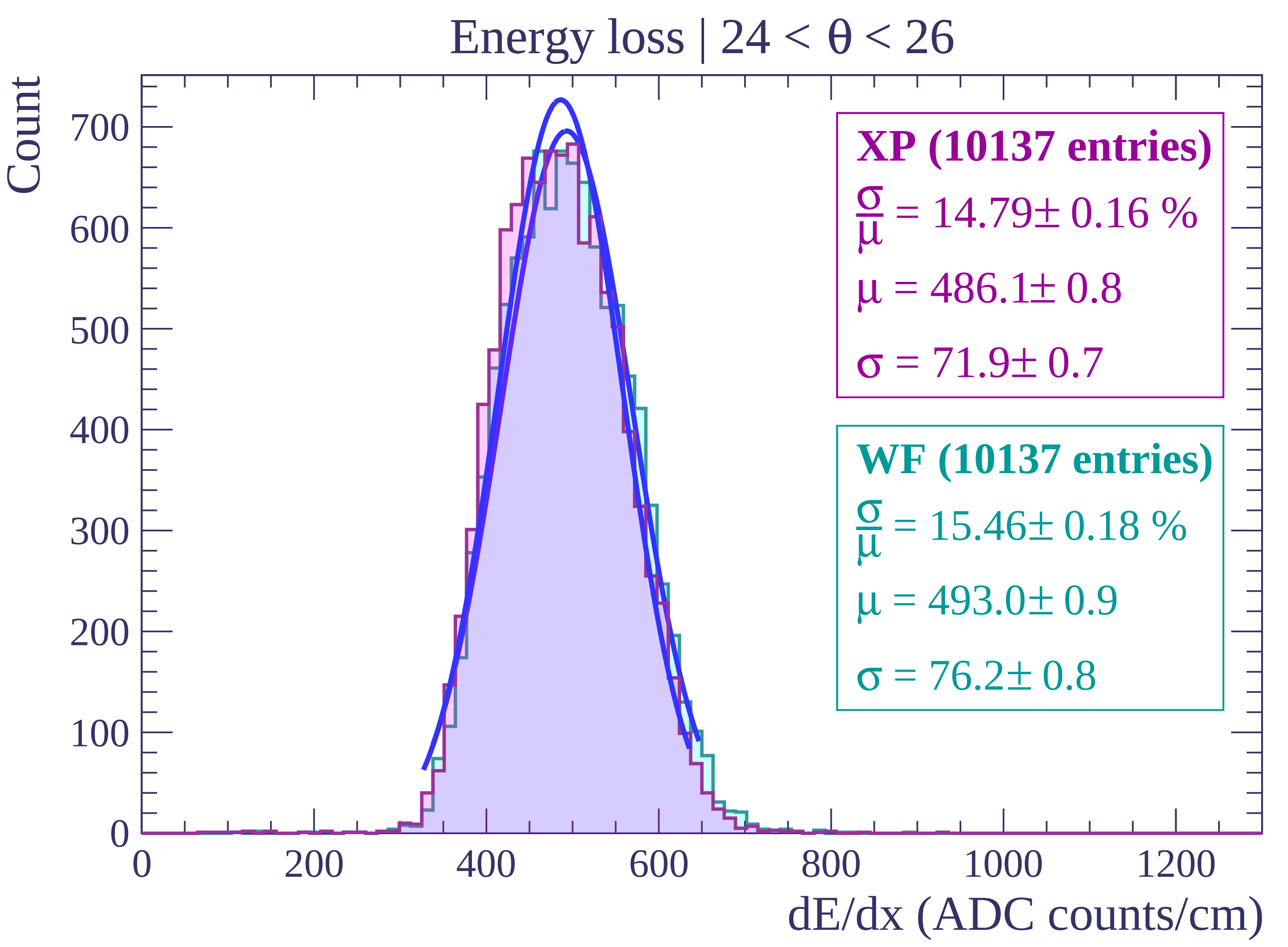


Energy loss | 18 < θ < 20
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (13672 entries)
σ/μ = 14.37± 0.13 %
μ = 485.8± 0.7
σ = 69.8± 0.5
WF (13672 entries)
σ/μ = 14.56± 0.13 %
μ = 493.3± 0.7
σ = 71.8± 0.5

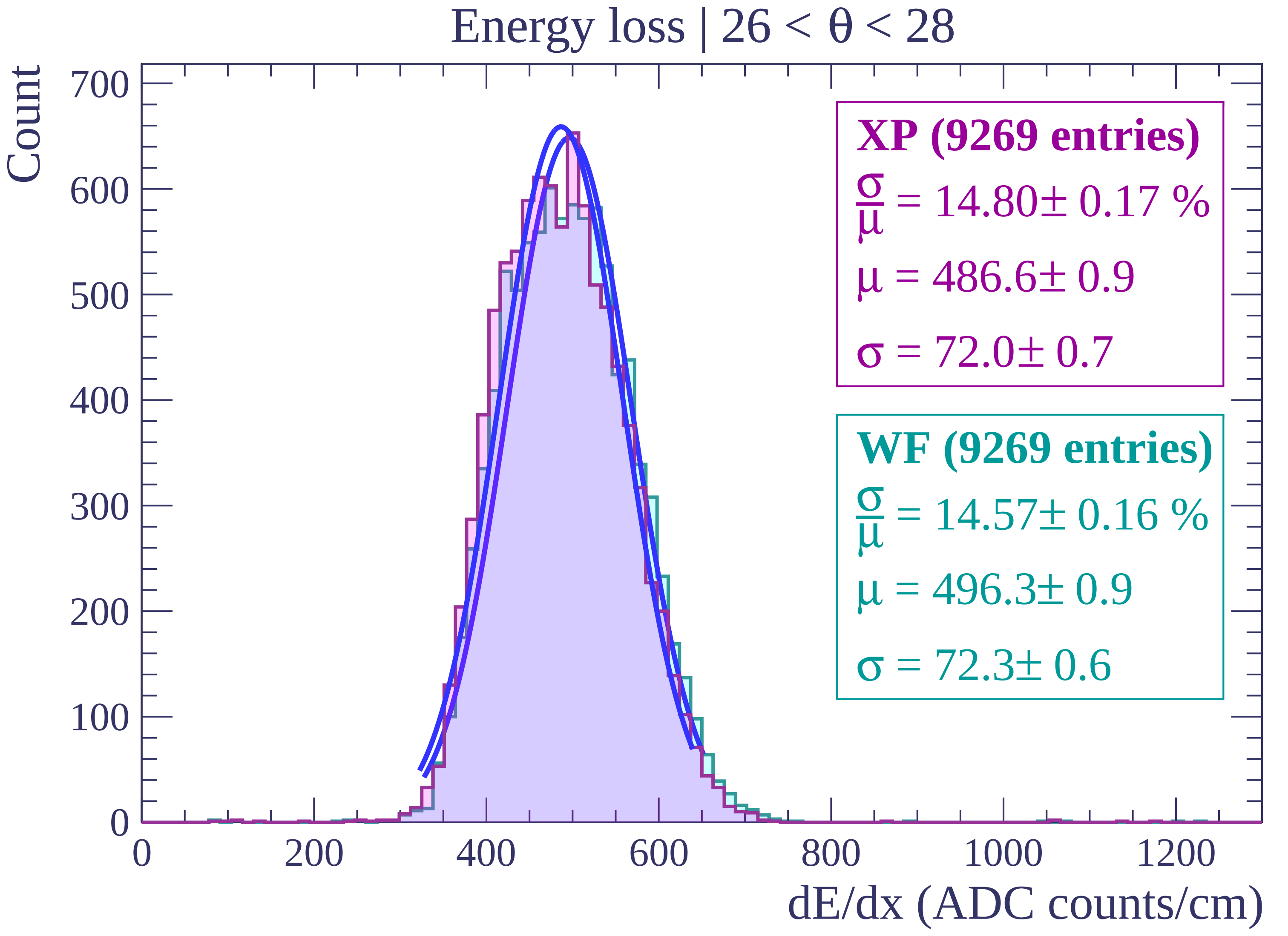


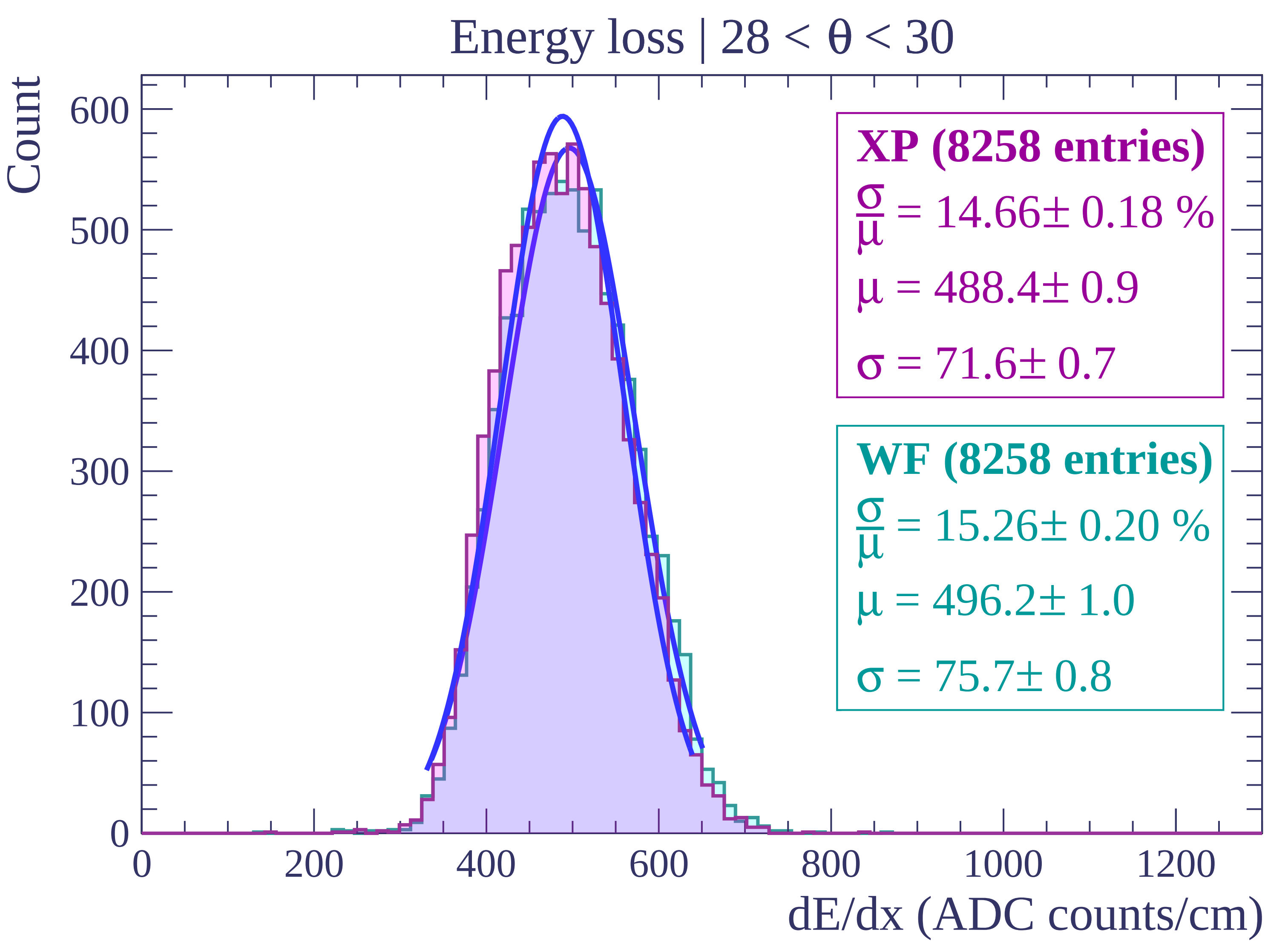
Energy loss | 20 < θ < 22
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (12075 entries)
σ/μ = 14.34± 0.14 %
μ = 484.8± 0.7
σ = 69.5± 0.6
WF (12075 entries)
σ/μ = 14.49± 0.14 %
μ = 493.0± 0.7
σ = 71.4± 0.6


Energy loss | 22 < θ < 24
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (11139 entries)
σ/μ = 14.40± 0.15 %
μ = 485.2± 0.8
σ = 69.9± 0.6
WF (11139 entries)
σ/μ = 14.57± 0.15 %
μ = 493.8± 0.8
σ = 72.0± 0.6

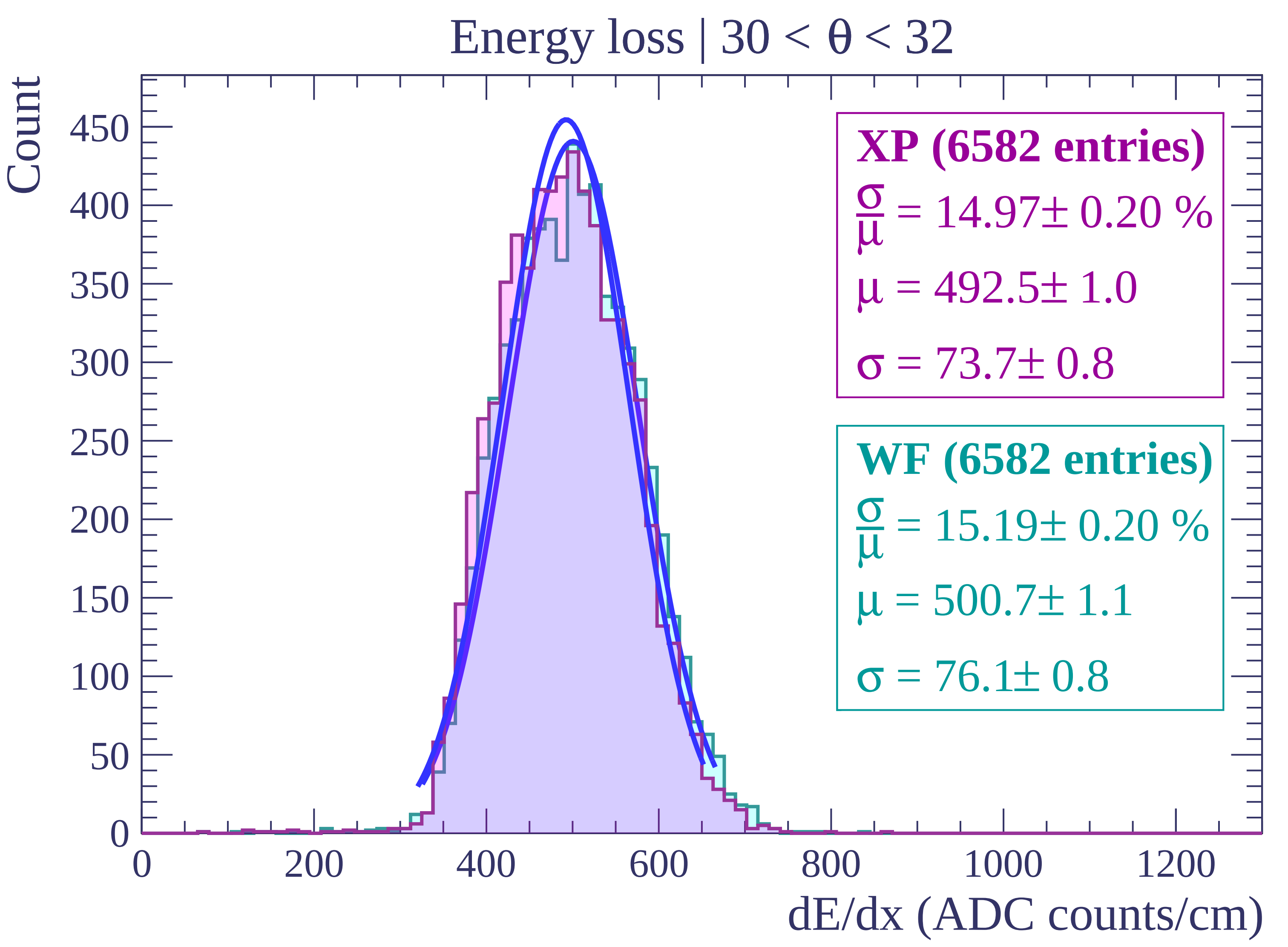


Energy loss | 24 < θ < 26
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (10137 entries)
σ/μ = 14.79± 0.16 %
μ = 486.1± 0.8
σ = 71.9± 0.7
WF (10137 entries)
σ/μ = 15.46± 0.18 %
μ = 493.0± 0.9
σ = 76.2± 0.8

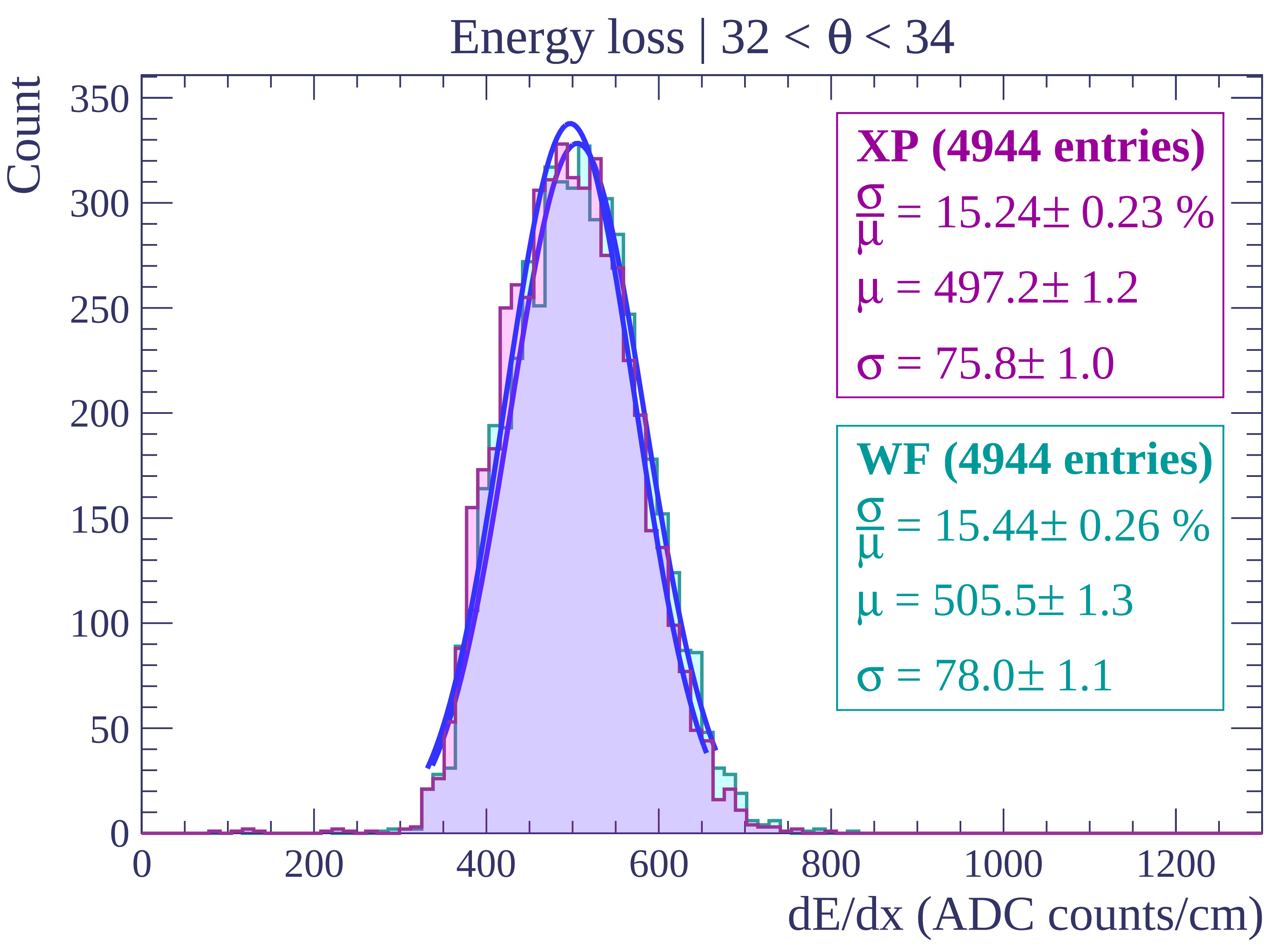

Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (9269 entries)
σ/μ = 14.80± 0.17 %
μ = 486.6± 0.9
σ = 72.0± 0.7
WF (9269 entries)
σ/μ = 14.57± 0.16 %
μ = 496.3± 0.9
σ = 72.3± 0.6



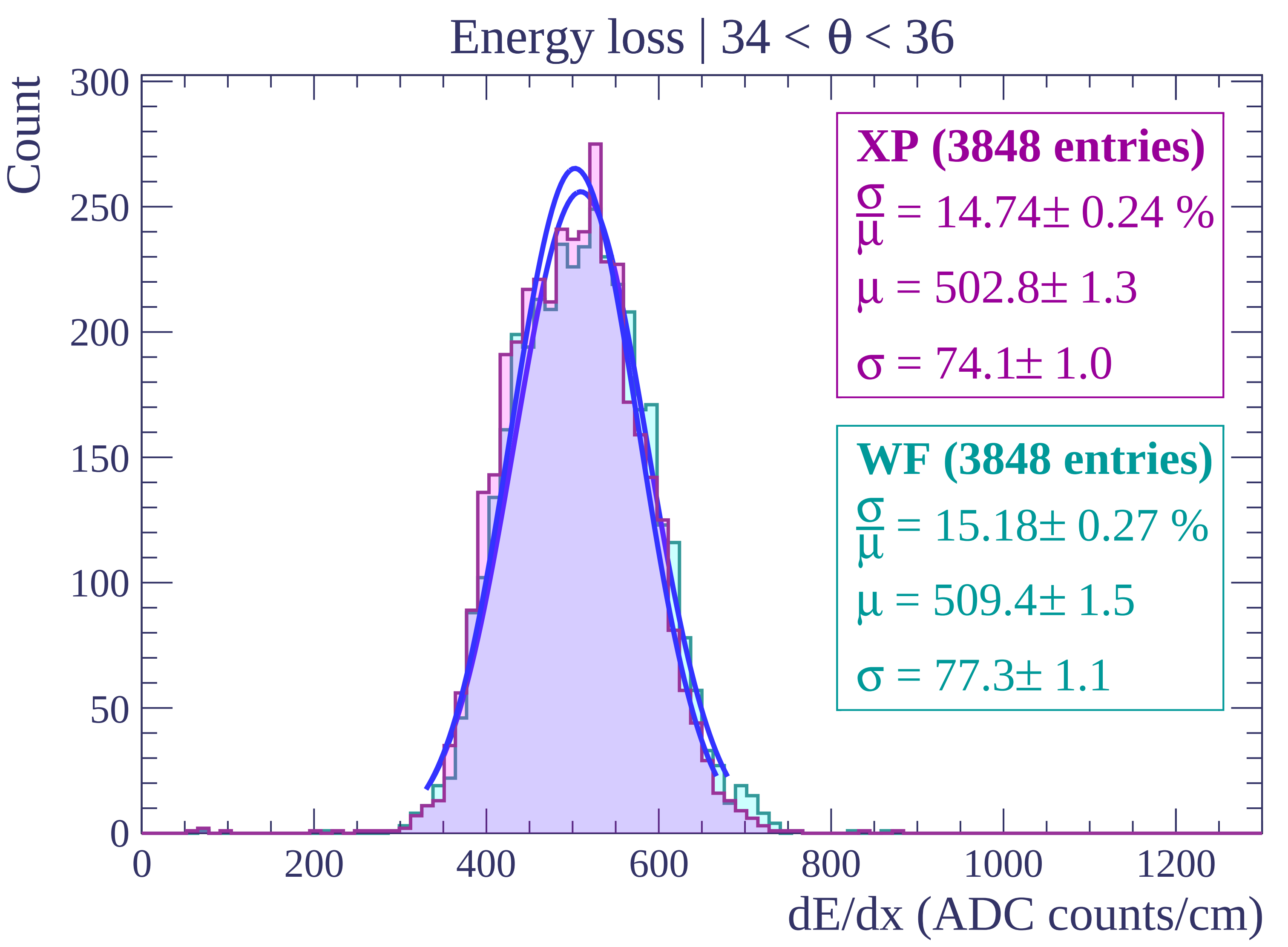
Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (8258 entries)
$\frac{\sigma}{\mu}$ = 14.66± 0.18 %
μ = 488.4± 0.9
σ = 71.6± 0.7
WF (8258 entries)
$\frac{\sigma}{\mu}$ = 15.26± 0.20 %
μ = 496.2± 1.0
σ = 75.7± 0.8



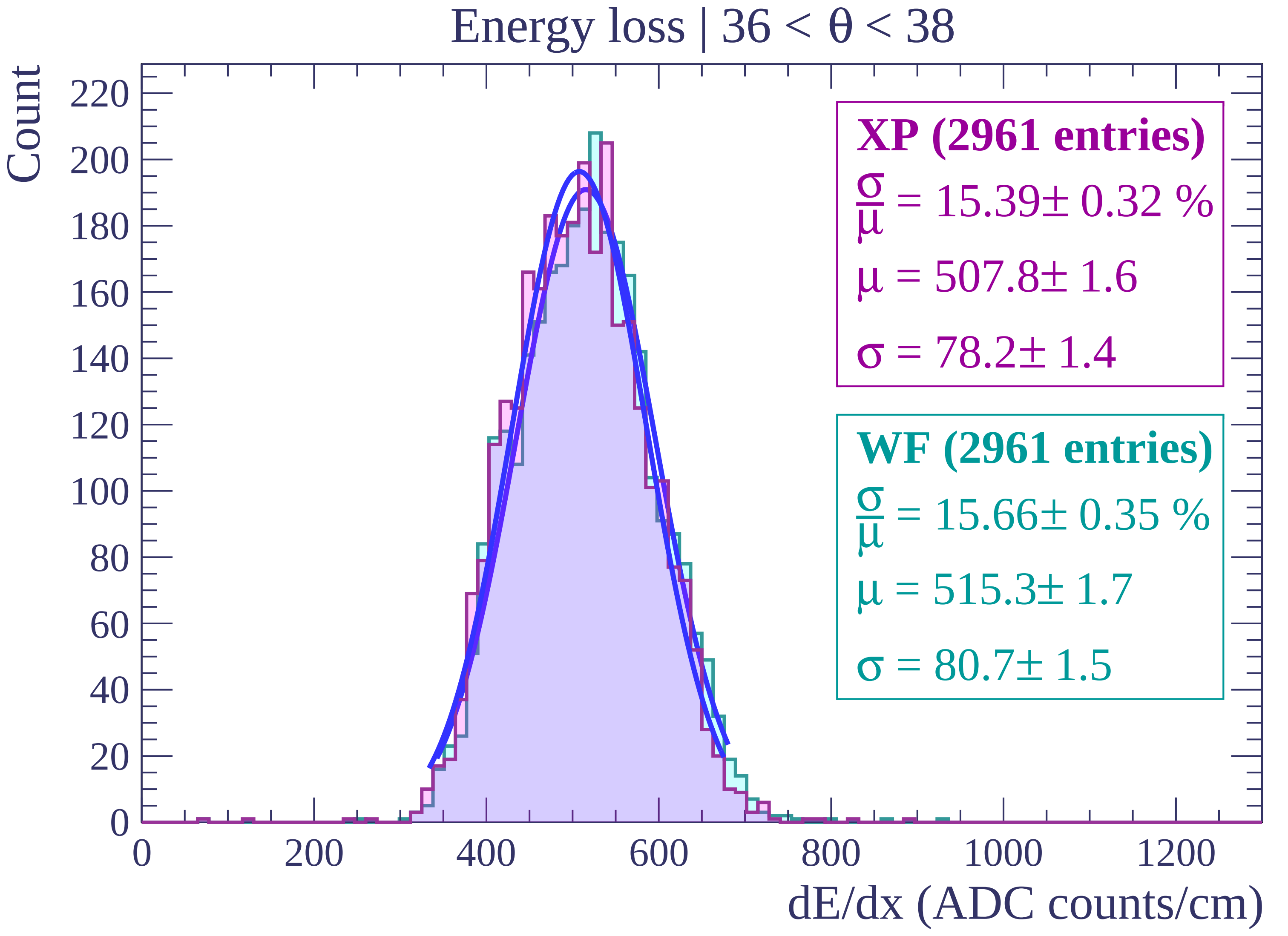
Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
XP (6582 entries)
σ/μ = 14.97± 0.20 %
μ = 492.5± 1.0
σ = 73.7± 0.8
WF (6582 entries)
σ/μ = 15.19± 0.20 %
μ = 500.7± 1.1
σ = 76.1± 0.8

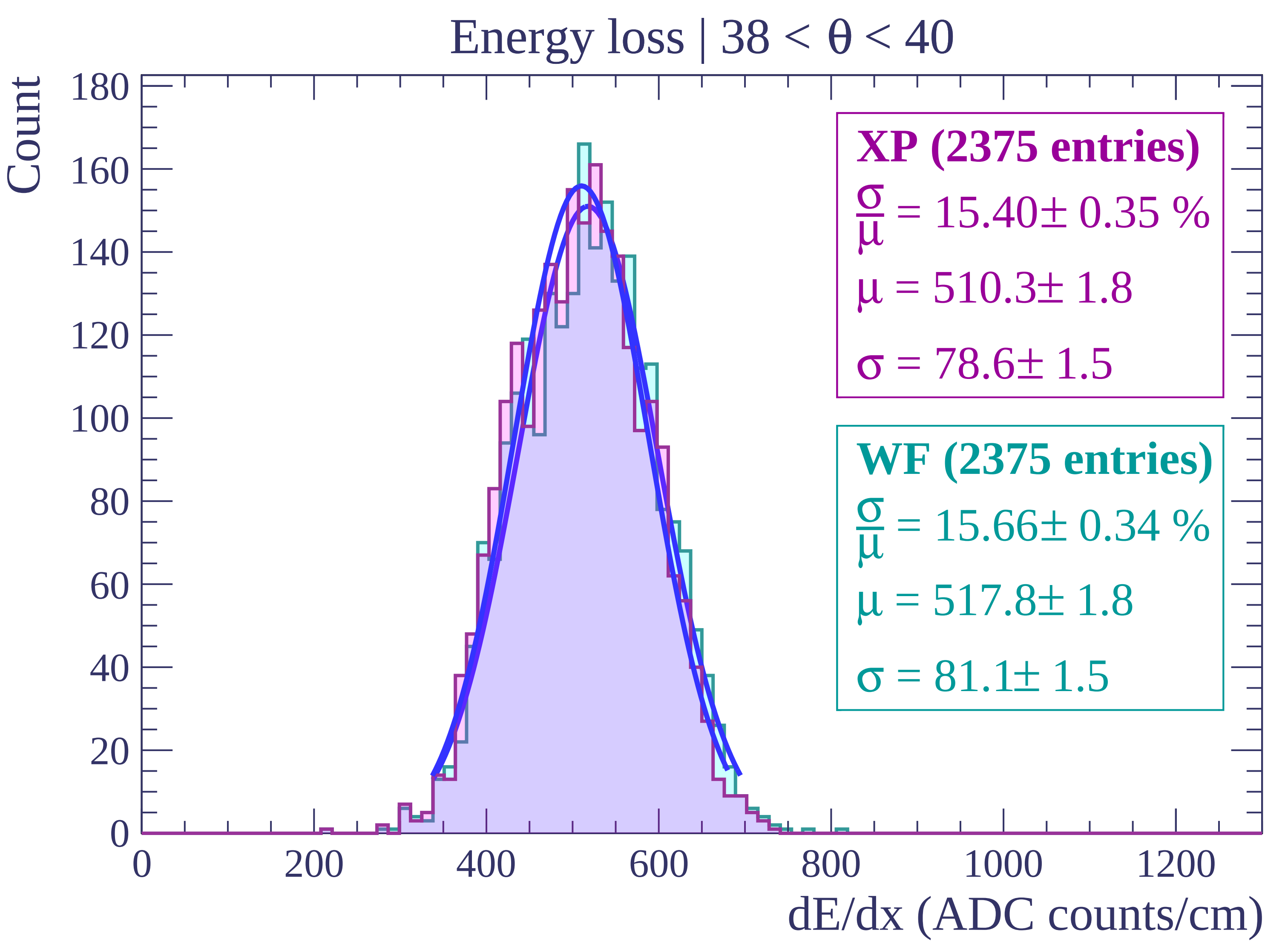


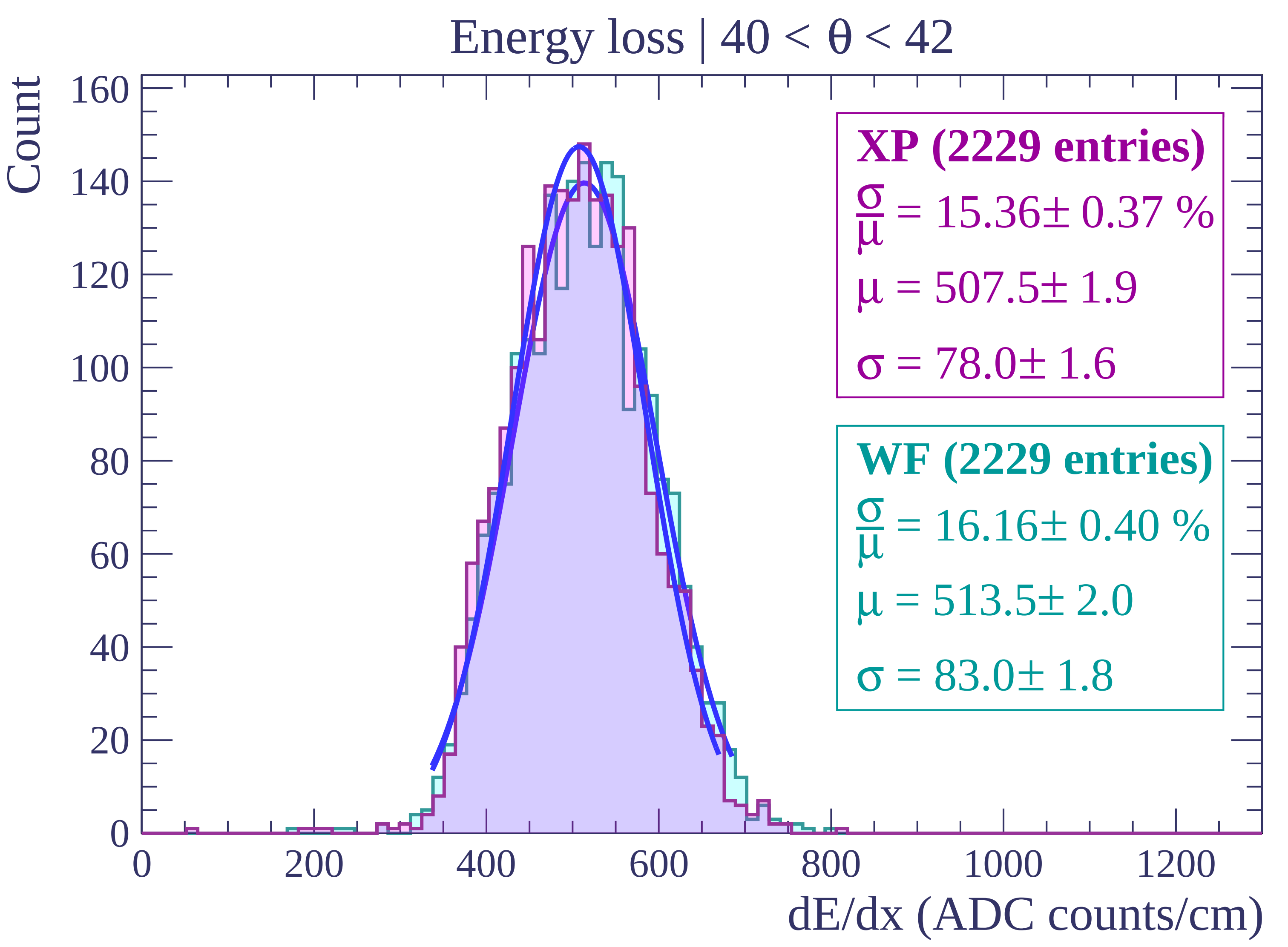
Energy loss | 32 < θ < 34
Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4944 entries)
$\frac{\sigma}{\mu}$ = 15.24± 0.23 %
μ = 497.2± 1.2
σ = 75.8± 1.0
WF (4944 entries)
$\frac{\sigma}{\mu}$ = 15.44± 0.26 %
μ = 505.5± 1.3
σ = 78.0± 1.1



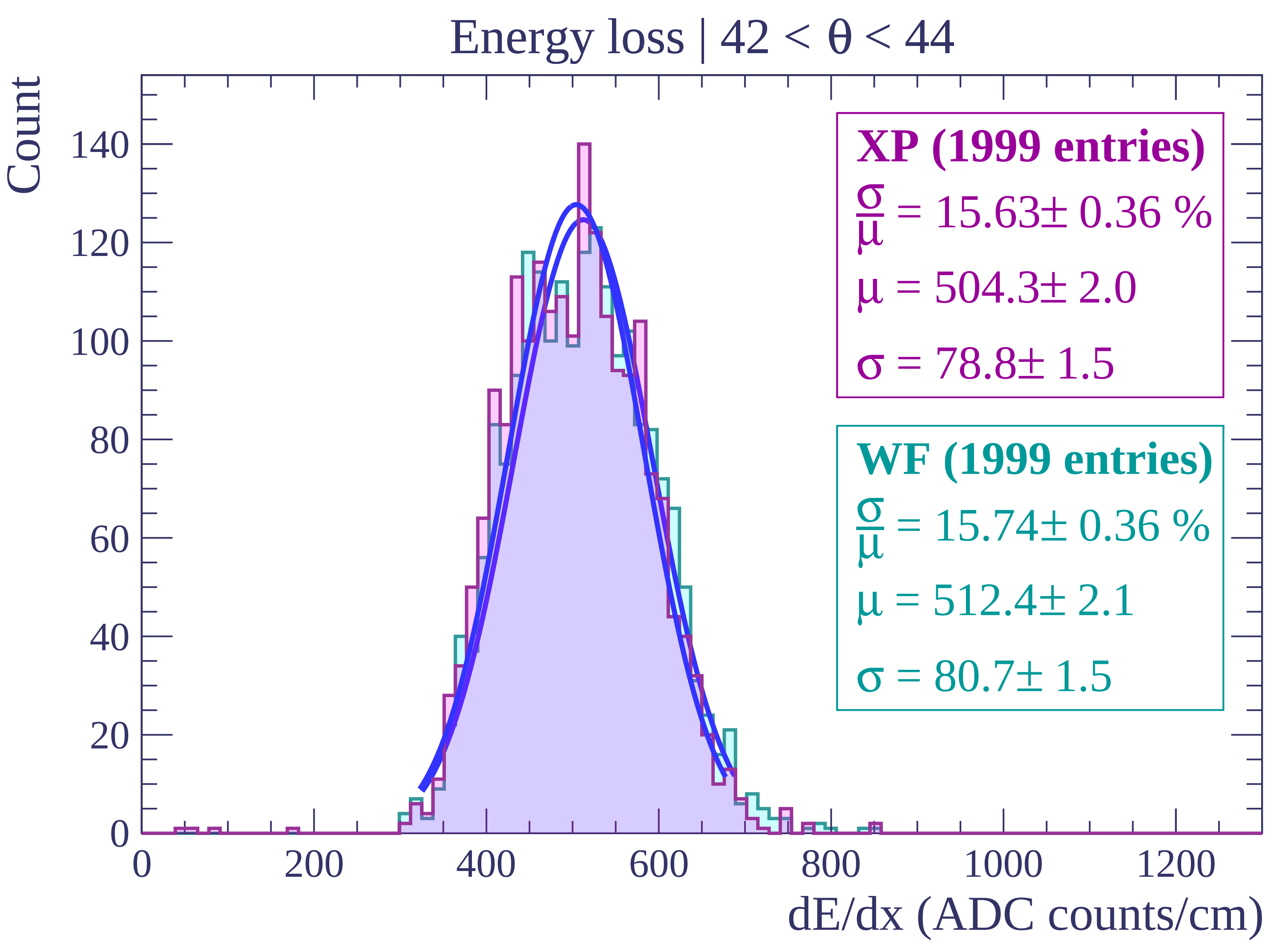
Energy loss | 34 < θ < 36
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3848 entries)
σ/μ = 14.74± 0.24 %
μ = 502.8± 1.3
σ = 74.1± 1.0
WF (3848 entries)
σ/μ = 15.18± 0.27 %
μ = 509.4± 1.5
σ = 77.3± 1.1


Energy loss | 36 < θ < 38
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2961 entries)
σ/μ = 15.39± 0.32 %
μ = 507.8± 1.6
σ = 78.2± 1.4
WF (2961 entries)
σ/μ = 15.66± 0.35 %
μ = 515.3± 1.7
σ = 80.7± 1.5

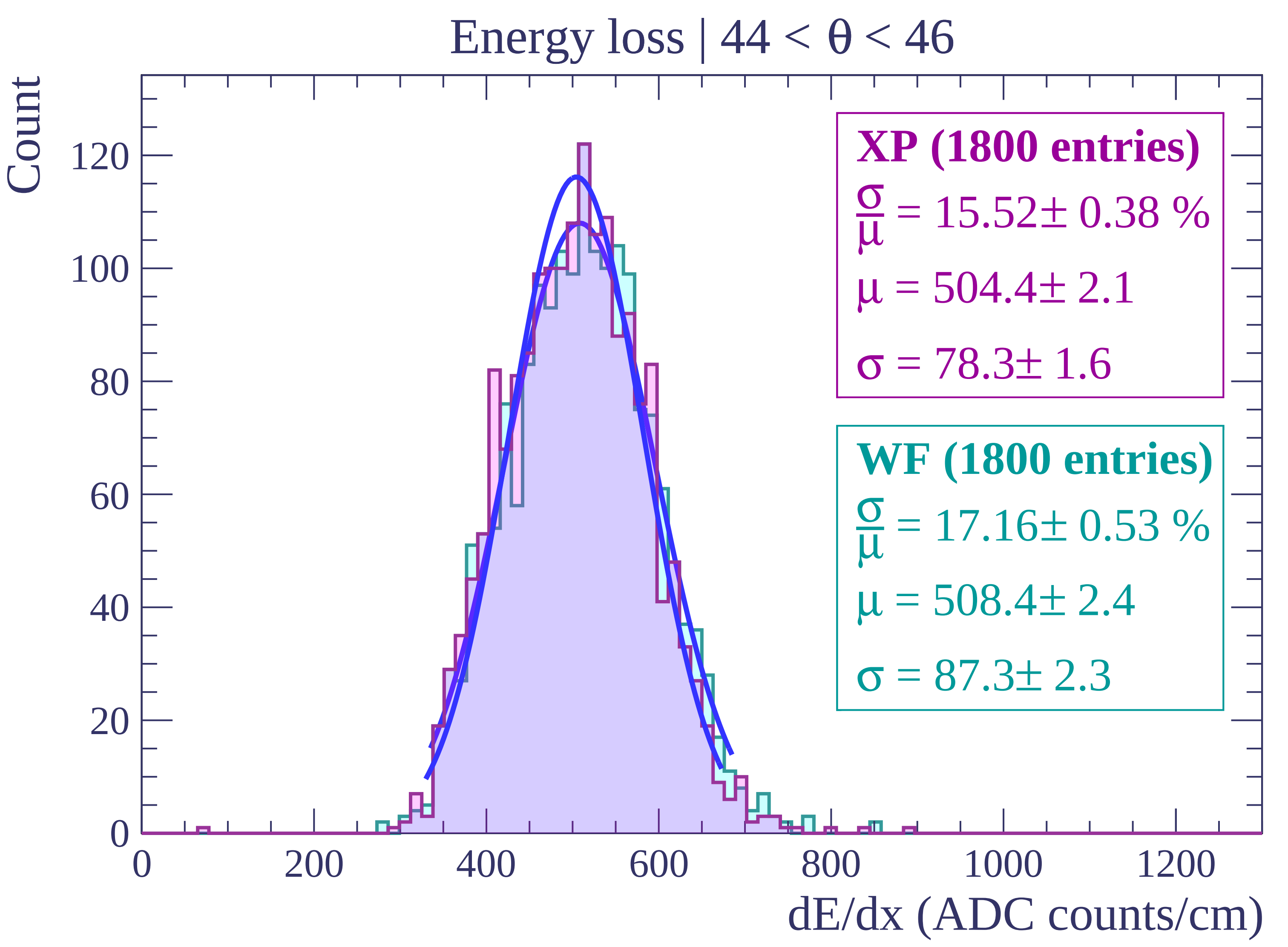

Energy loss | 38 < θ < 40
Count
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2375 entries)
σ/μ = 15.40± 0.35 %
μ = 510.3± 1.8
σ = 78.6± 1.5
WF (2375 entries)
σ/μ = 15.66± 0.34 %
μ = 517.8± 1.8
σ = 81.1± 1.5



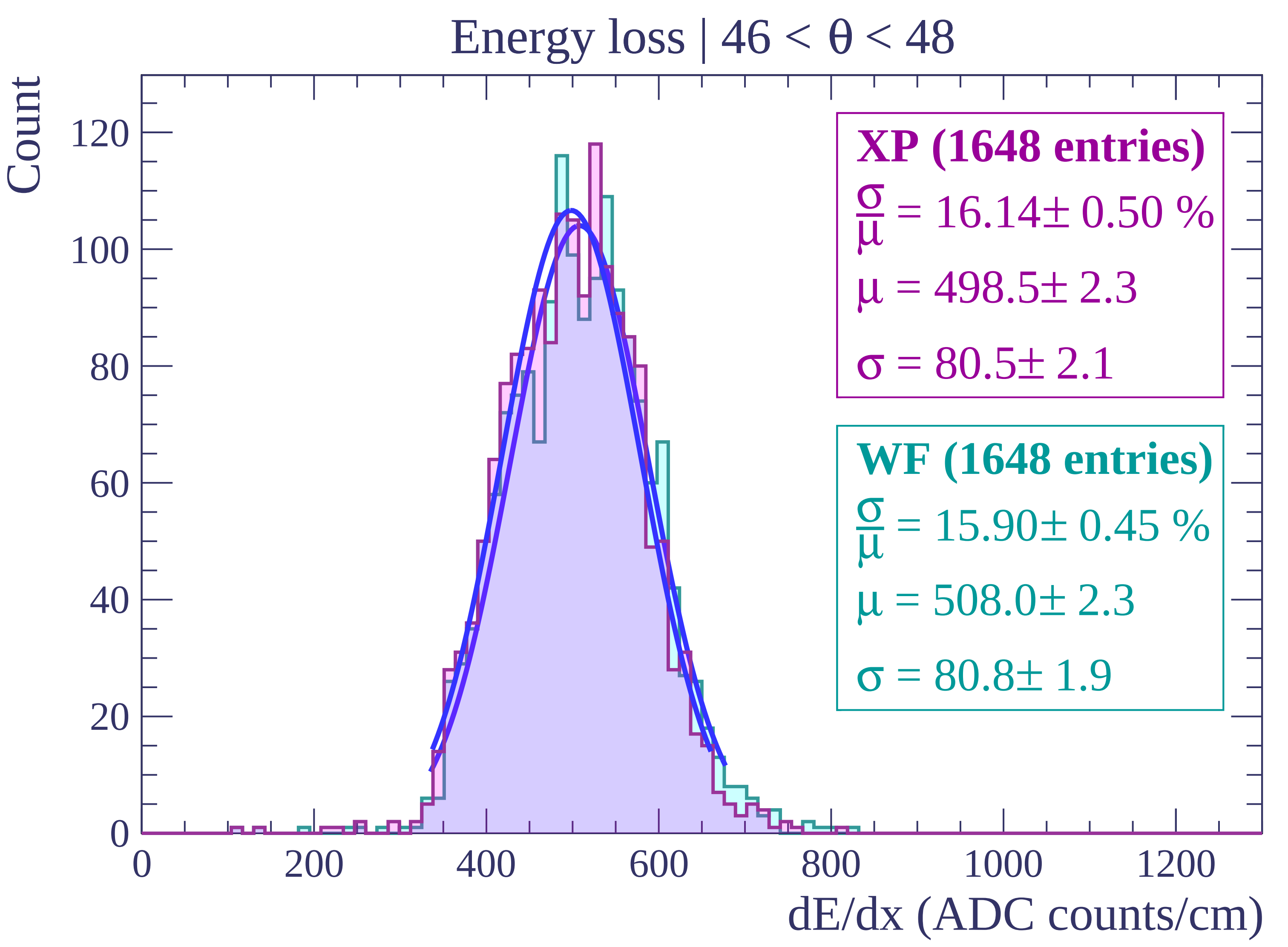
Energy loss | 40 < θ < 42
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2229 entries)
$\frac{\sigma}{\mu}$ = 15.36± 0.37 %
μ = 507.5± 1.9
σ = 78.0± 1.6
WF (2229 entries)
$\frac{\sigma}{\mu}$ = 16.16± 0.40 %
μ = 513.5± 2.0
σ = 83.0± 1.8

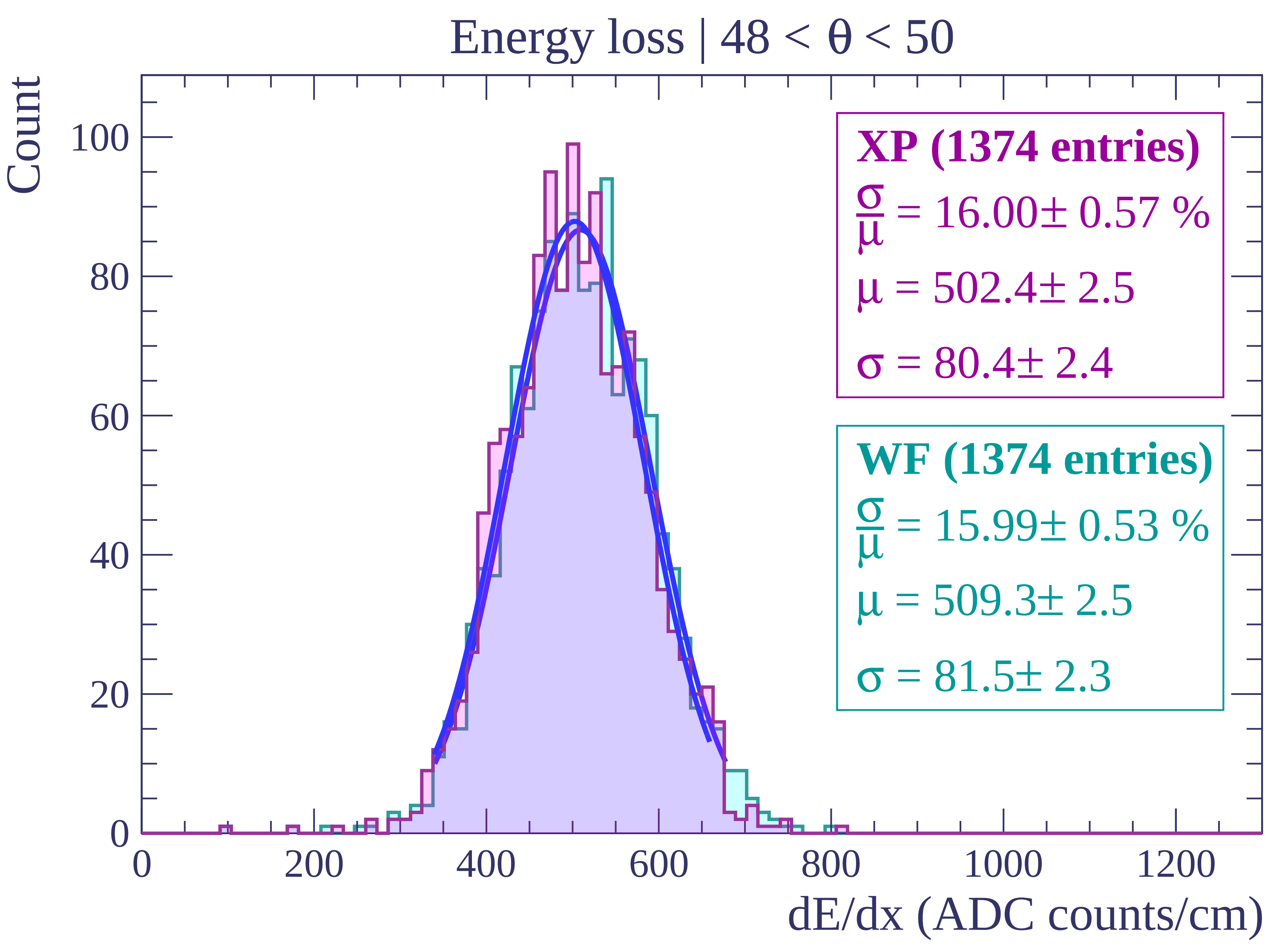


Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1999 entries)
σ/μ = 15.63± 0.36 %
μ = 504.3± 2.0
σ = 78.8± 1.5
WF (1999 entries)
σ/μ = 15.74± 0.36 %
μ = 512.4± 2.1
σ = 80.7± 1.5

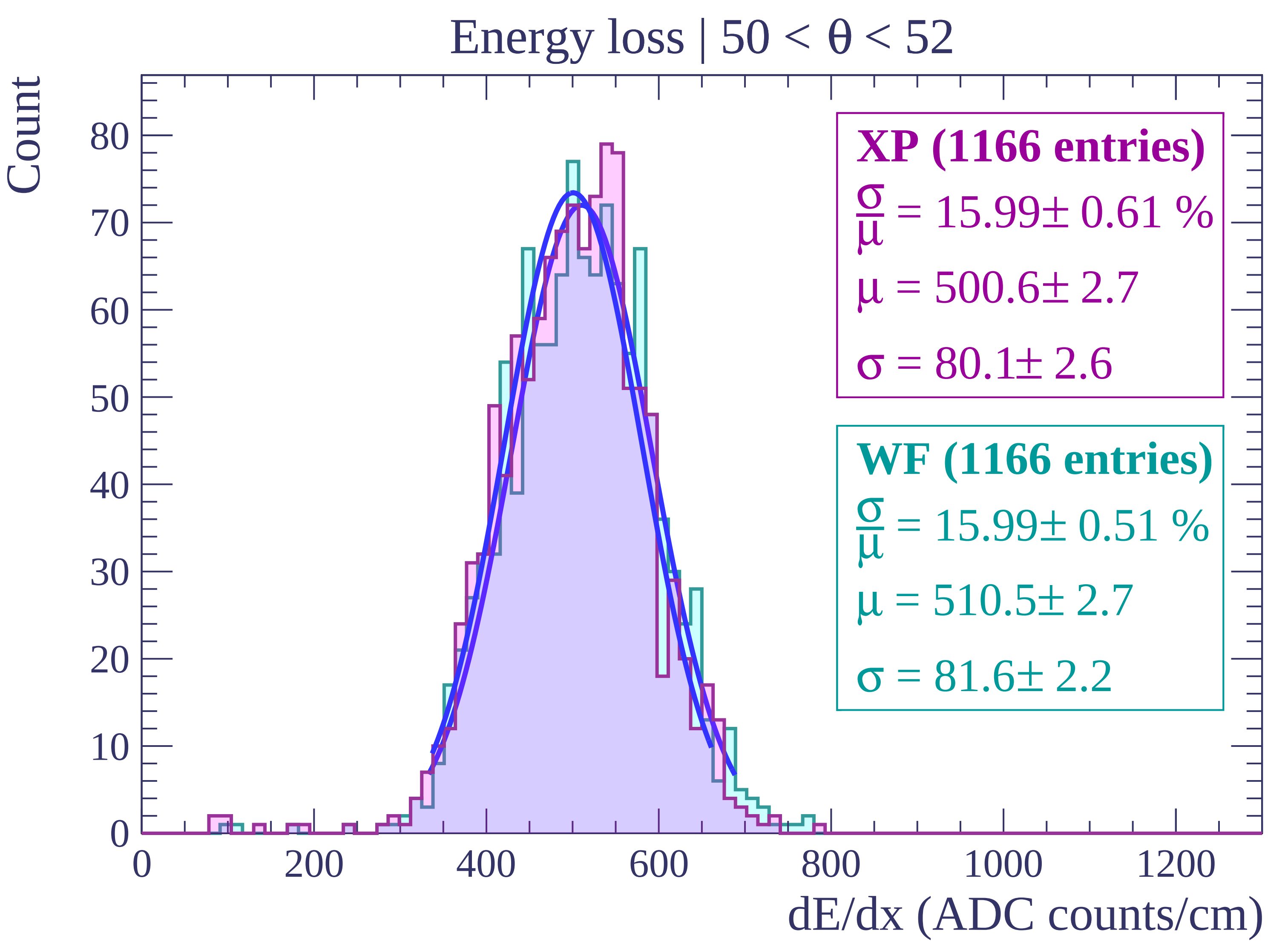


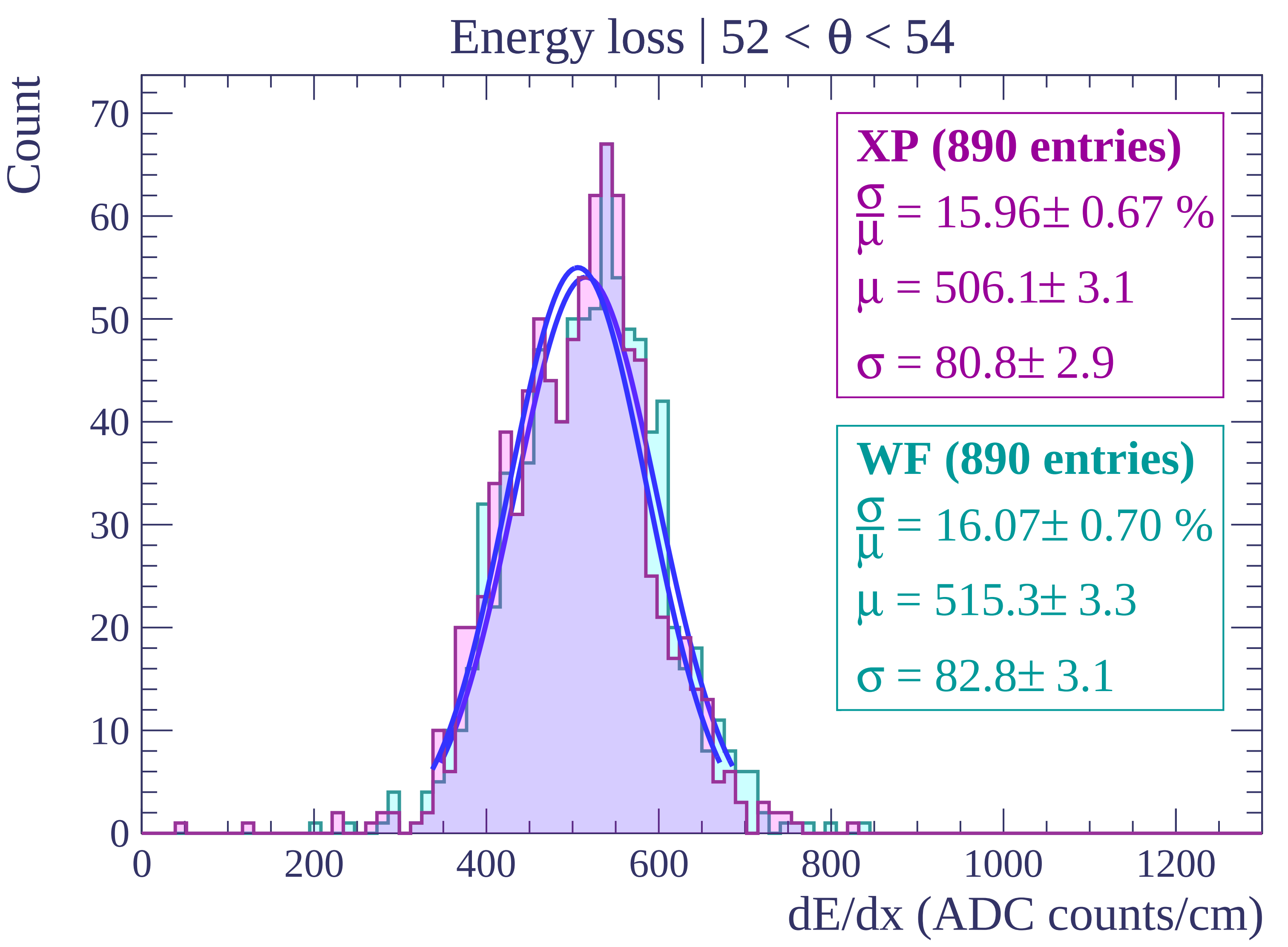
Energy loss | 44 < θ < 46
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1800 entries)
σ/μ = 15.52± 0.38 %
μ = 504.4± 2.1
σ = 78.3± 1.6
WF (1800 entries)
σ/μ = 17.16± 0.53 %
μ = 508.4± 2.4
σ = 87.3± 2.3

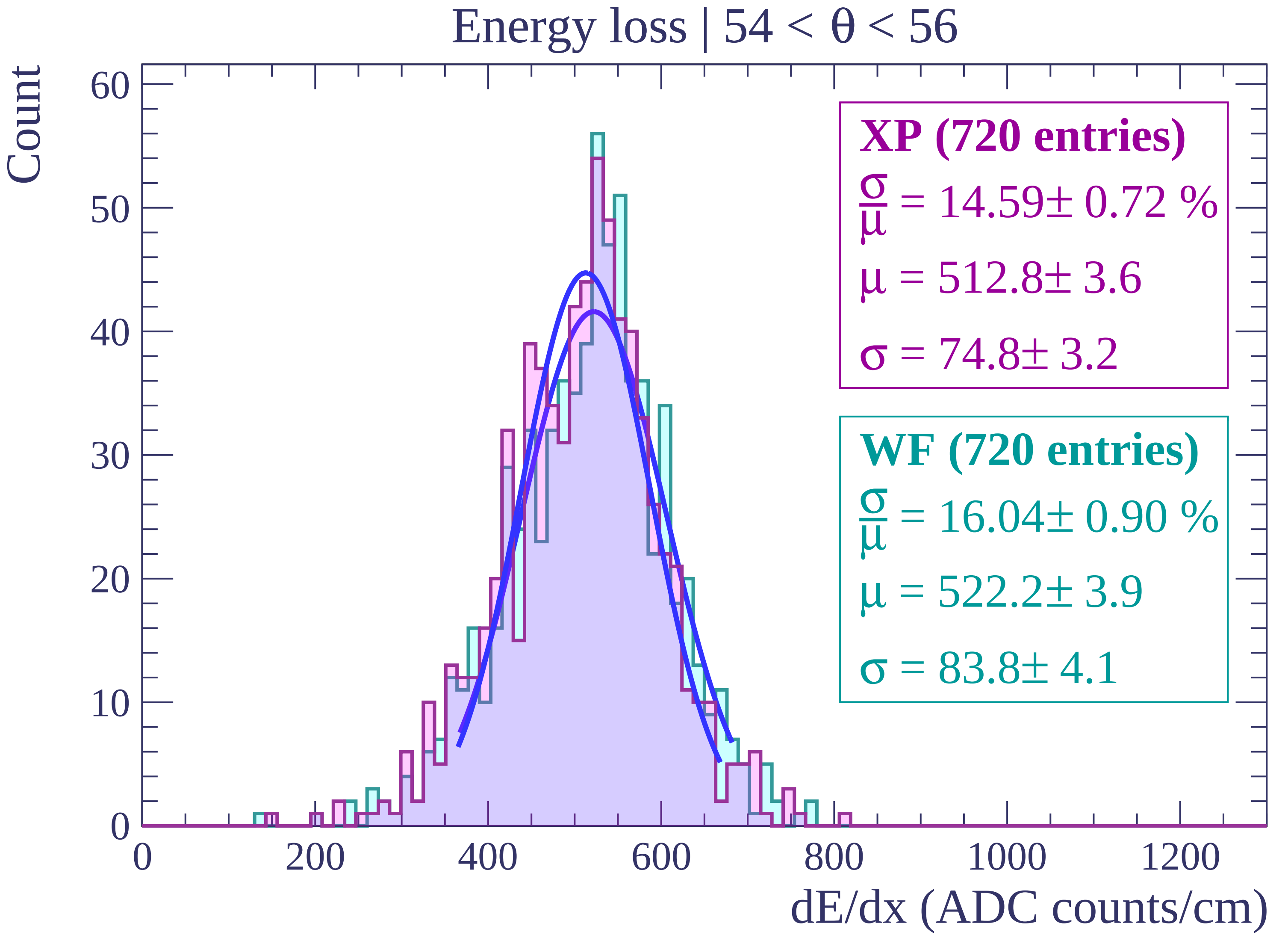

Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
XP (1648 entries)
σ/μ = 16.14± 0.50 %
μ = 498.5± 2.3
σ = 80.5± 2.1
WF (1648 entries)
σ/μ = 15.90± 0.45 %
μ = 508.0± 2.3
σ = 80.8± 1.9
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

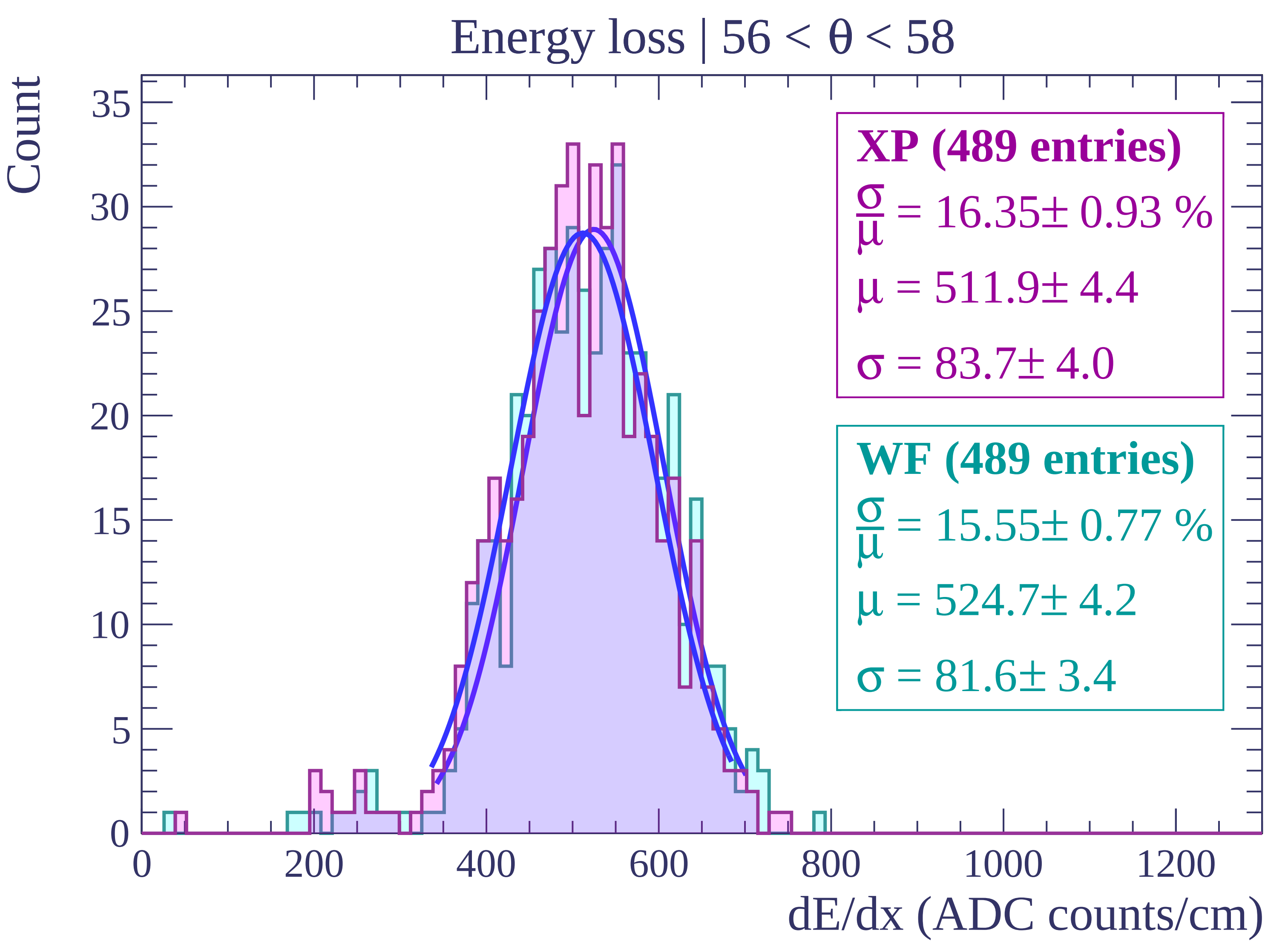


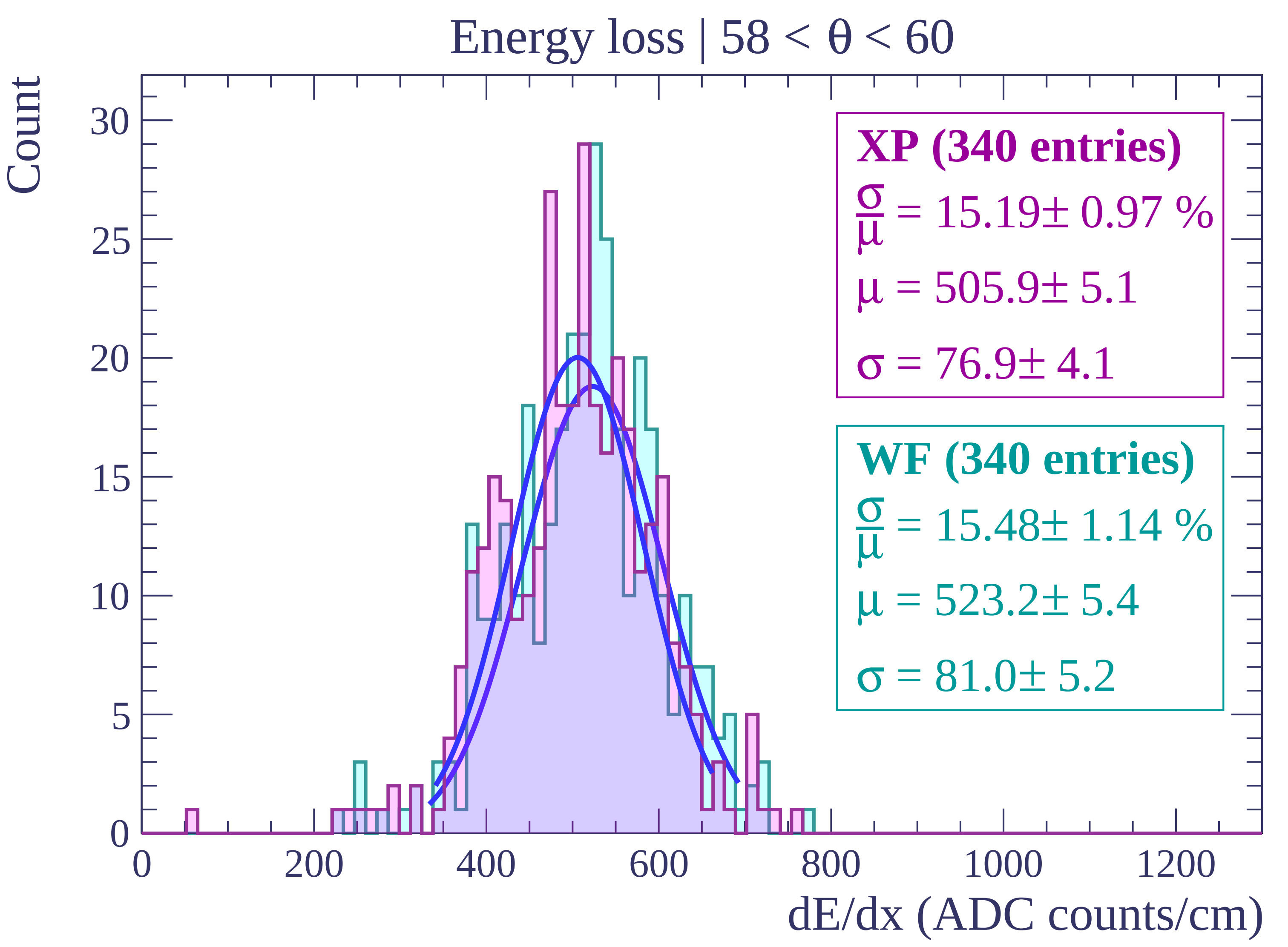
Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1374 entries)
σ/μ = 16.00± 0.57 %
μ = 502.4± 2.5
σ = 80.4± 2.4
WF (1374 entries)
σ/μ = 15.99± 0.53 %
μ = 509.3± 2.5
σ = 81.5± 2.3


Energy loss | 50 < θ < 52
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP (1166 entries)
σ/μ = 15.99± 0.61 %
μ = 500.6± 2.7
σ = 80.1± 2.6
WF (1166 entries)
σ/μ = 15.99± 0.51 %
μ = 510.5± 2.7
σ = 81.6± 2.2

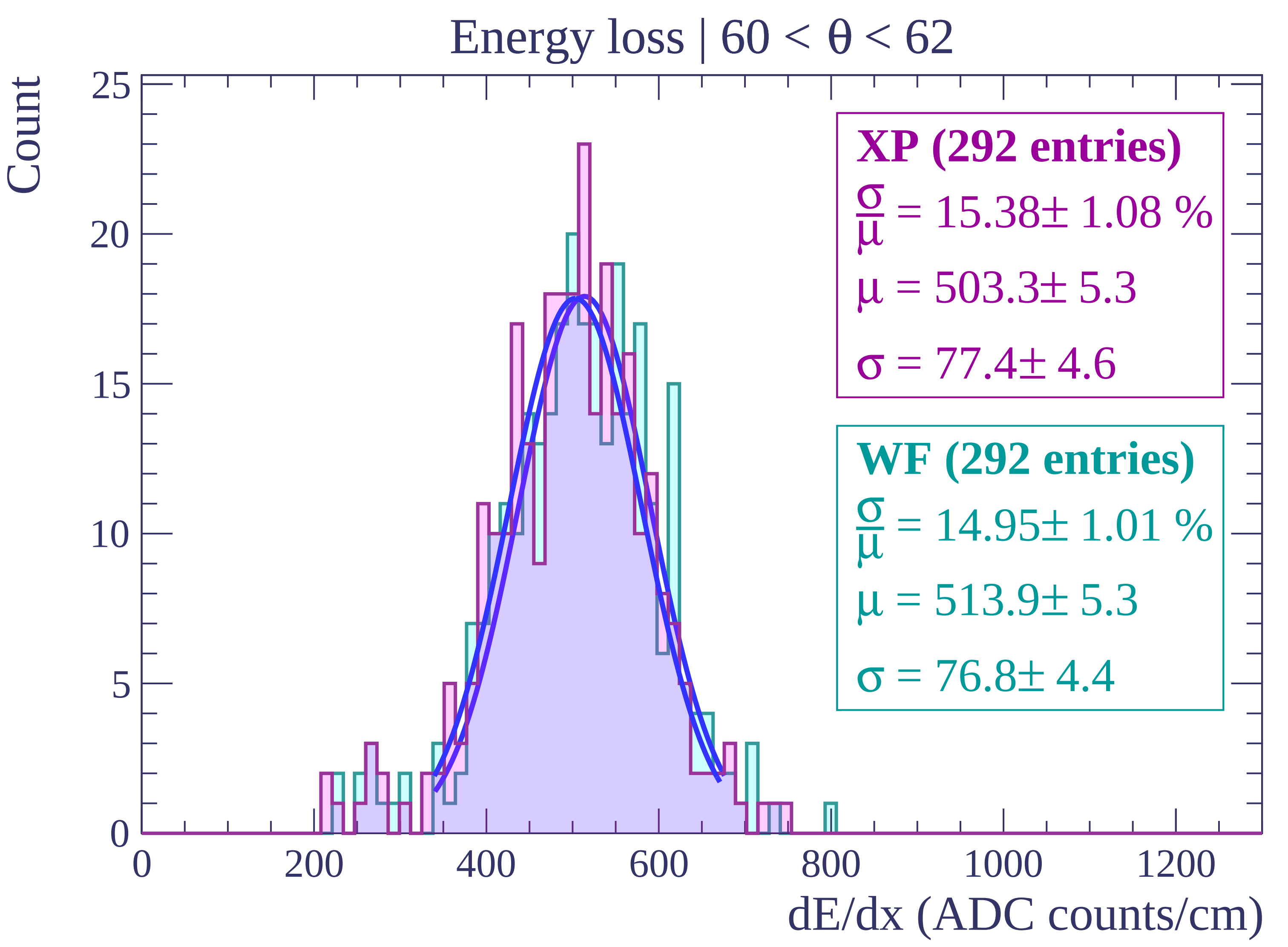

Energy loss | 52 < θ < 54
Count
dE/dx (ADC counts/cm)
XP (890 entries)
σ/μ = 15.96± 0.67 %
μ = 506.1± 3.1
σ = 80.8± 2.9
WF (890 entries)
σ/μ = 16.07± 0.70 %
μ = 515.3± 3.3
σ = 82.8± 3.1
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200


Energy loss | 54 < θ < 56
Count
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (720 entries)
σ/μ = 14.59± 0.72 %
μ = 512.8± 3.6
σ = 74.8± 3.2
WF (720 entries)
σ/μ = 16.04± 0.90 %
μ = 522.2± 3.9
σ = 83.8± 4.1

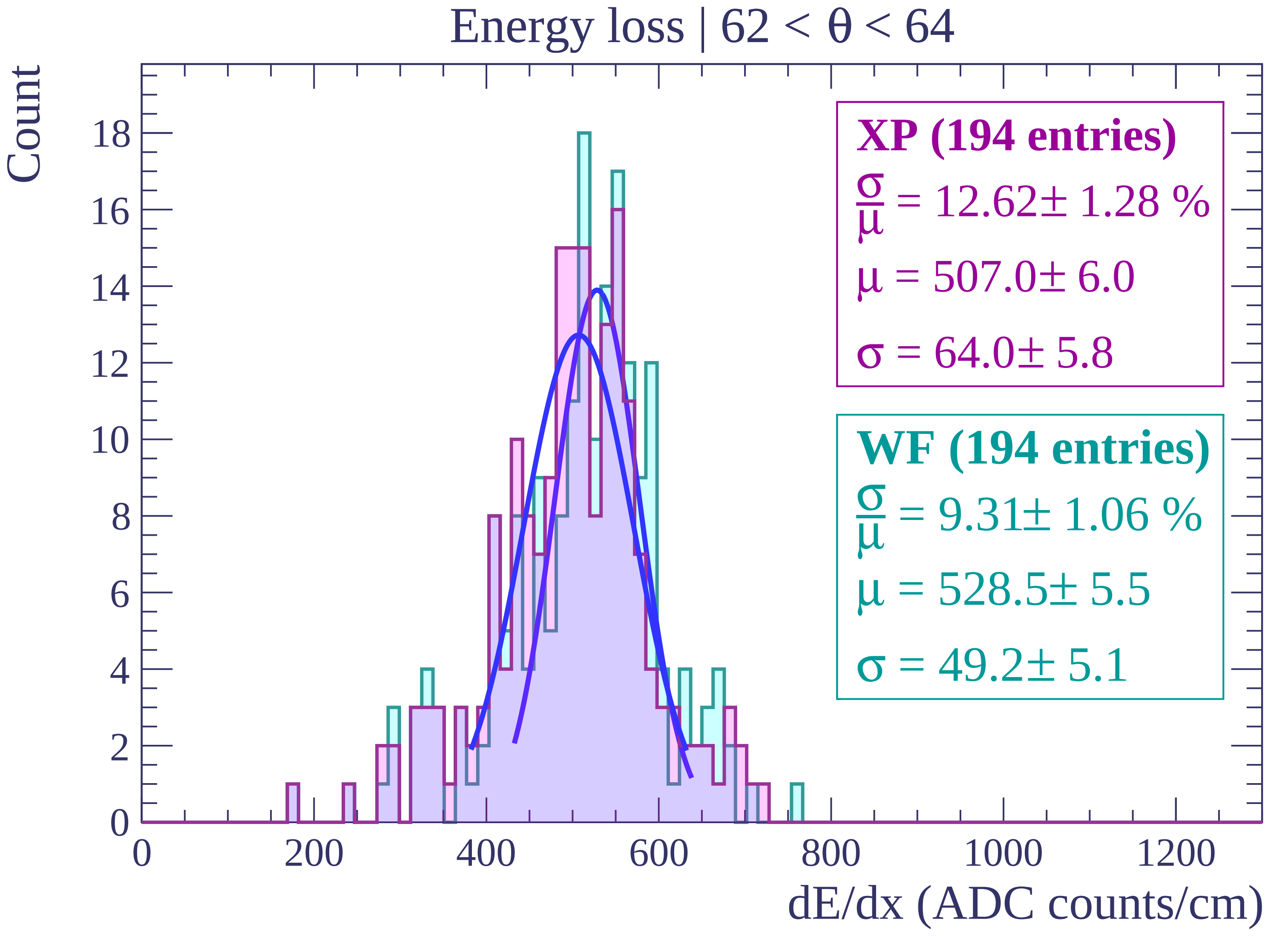

Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP (489 entries)
σ/μ = 16.35± 0.93 %
μ = 511.9± 4.4
σ = 83.7± 4.0
WF (489 entries)
σ/μ = 15.55± 0.77 %
μ = 524.7± 4.2
σ = 81.6± 3.4

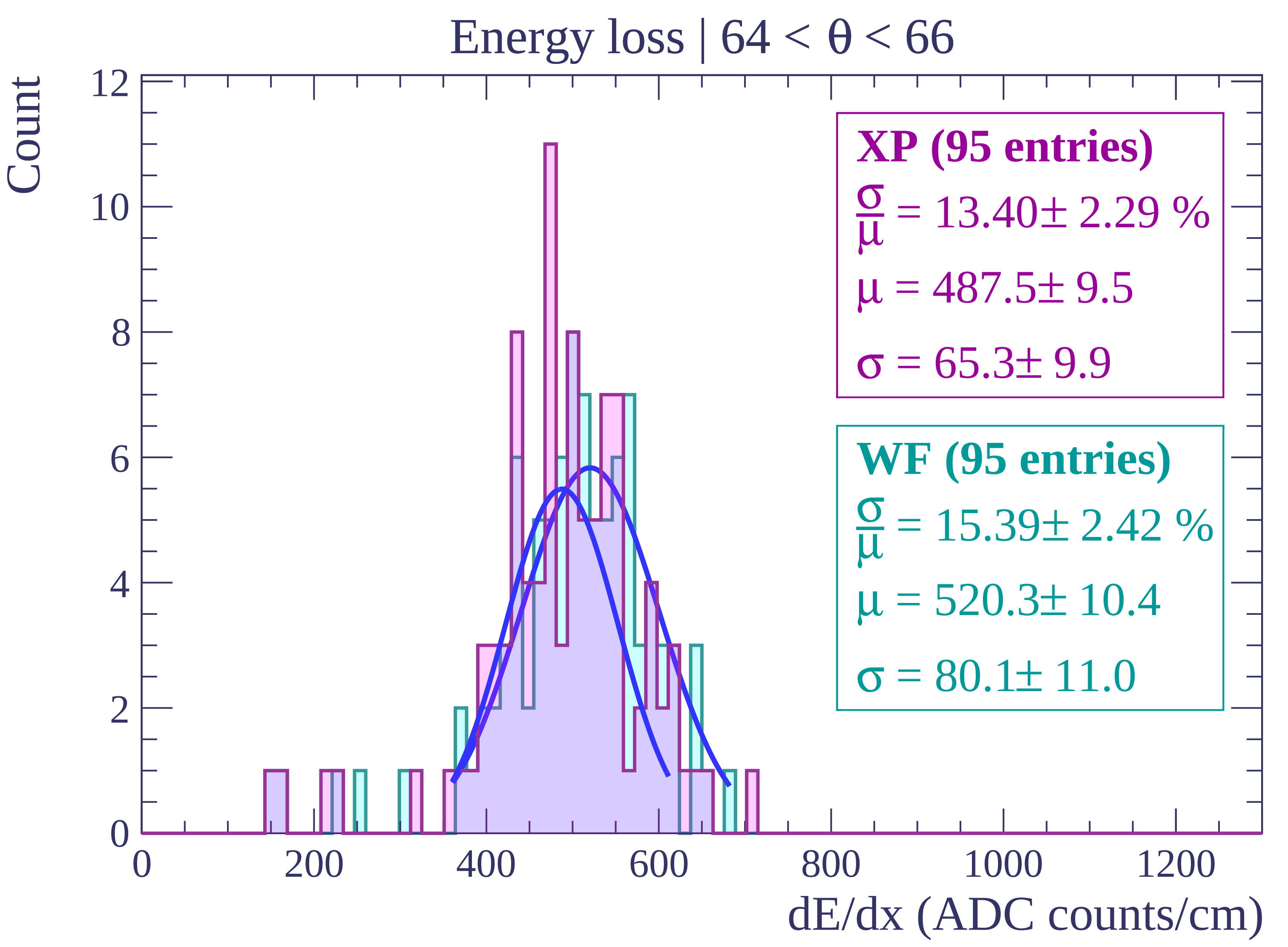


Energy loss | 58 < θ < 60
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP (340 entries)
σ/μ = 15.19± 0.97 %
μ = 505.9± 5.1
σ = 76.9± 4.1
WF (340 entries)
σ/μ = 15.48± 1.14 %
μ = 523.2± 5.4
σ = 81.0± 5.2

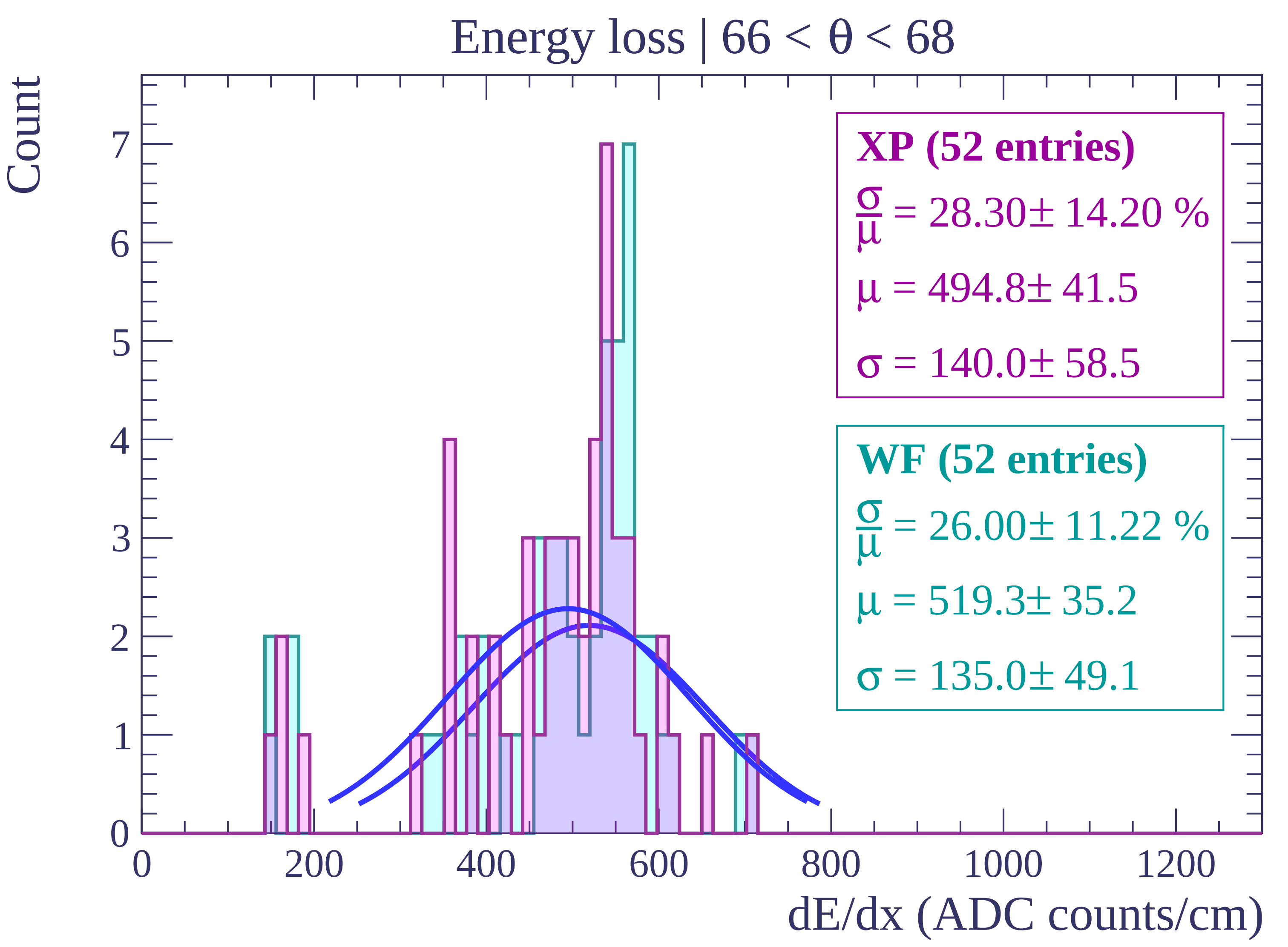


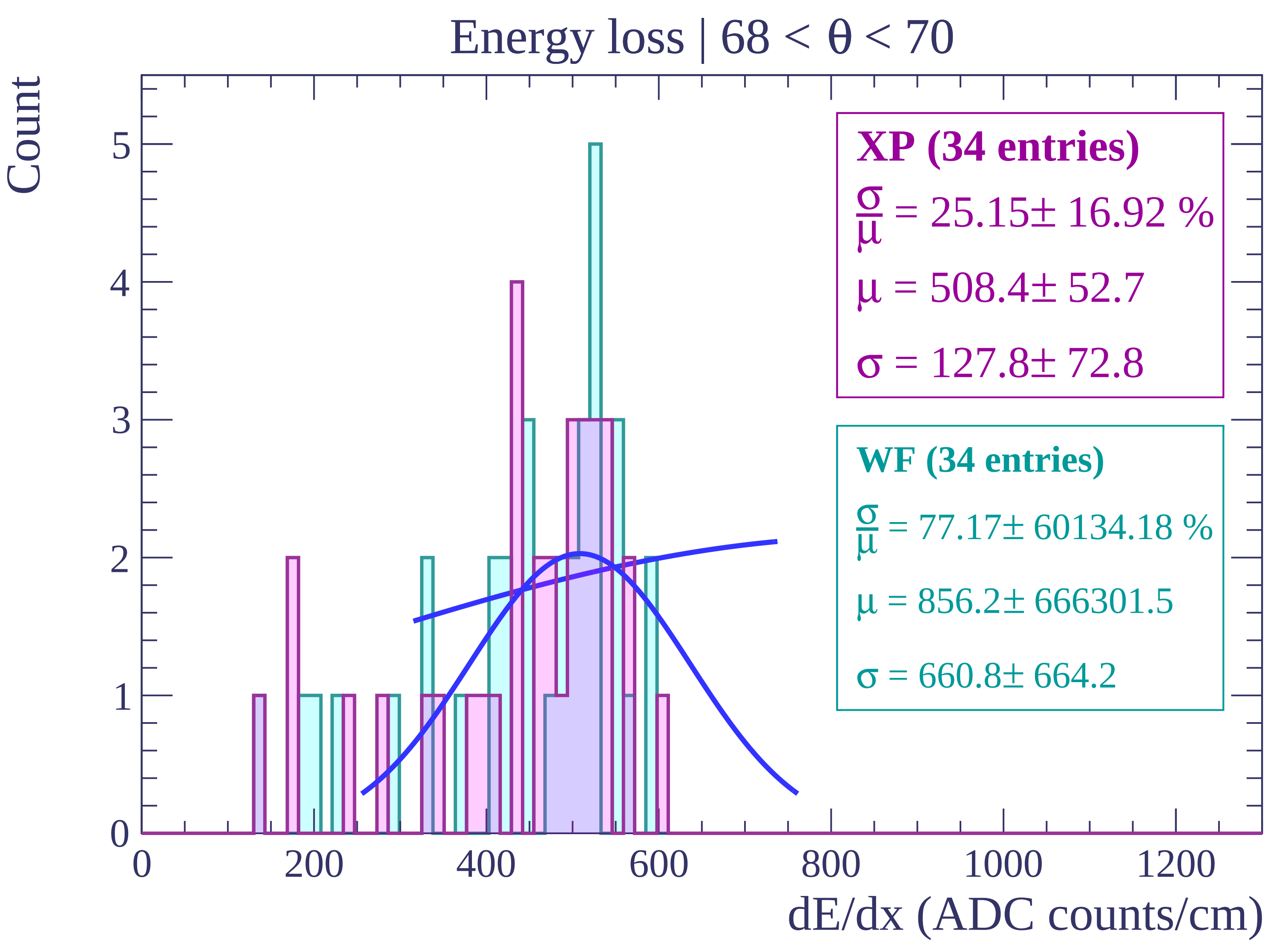
Energy loss | 60 < θ < 62
Count
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (292 entries)
σ/μ = 15.38± 1.08 %
μ = 503.3± 5.3
σ = 77.4± 4.6
WF (292 entries)
σ/μ = 14.95± 1.01 %
μ = 513.9± 5.3
σ = 76.8± 4.4


Energy loss | 62 < θ < 64
Count
dE/dx (ADC counts/cm)
XP (194 entries)
σ/μ = 12.62± 1.28 %
μ = 507.0± 6.0
σ = 64.0± 5.8
WF (194 entries)
σ/μ = 9.31± 1.06 %
μ = 528.5± 5.5
σ = 49.2± 5.1
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200

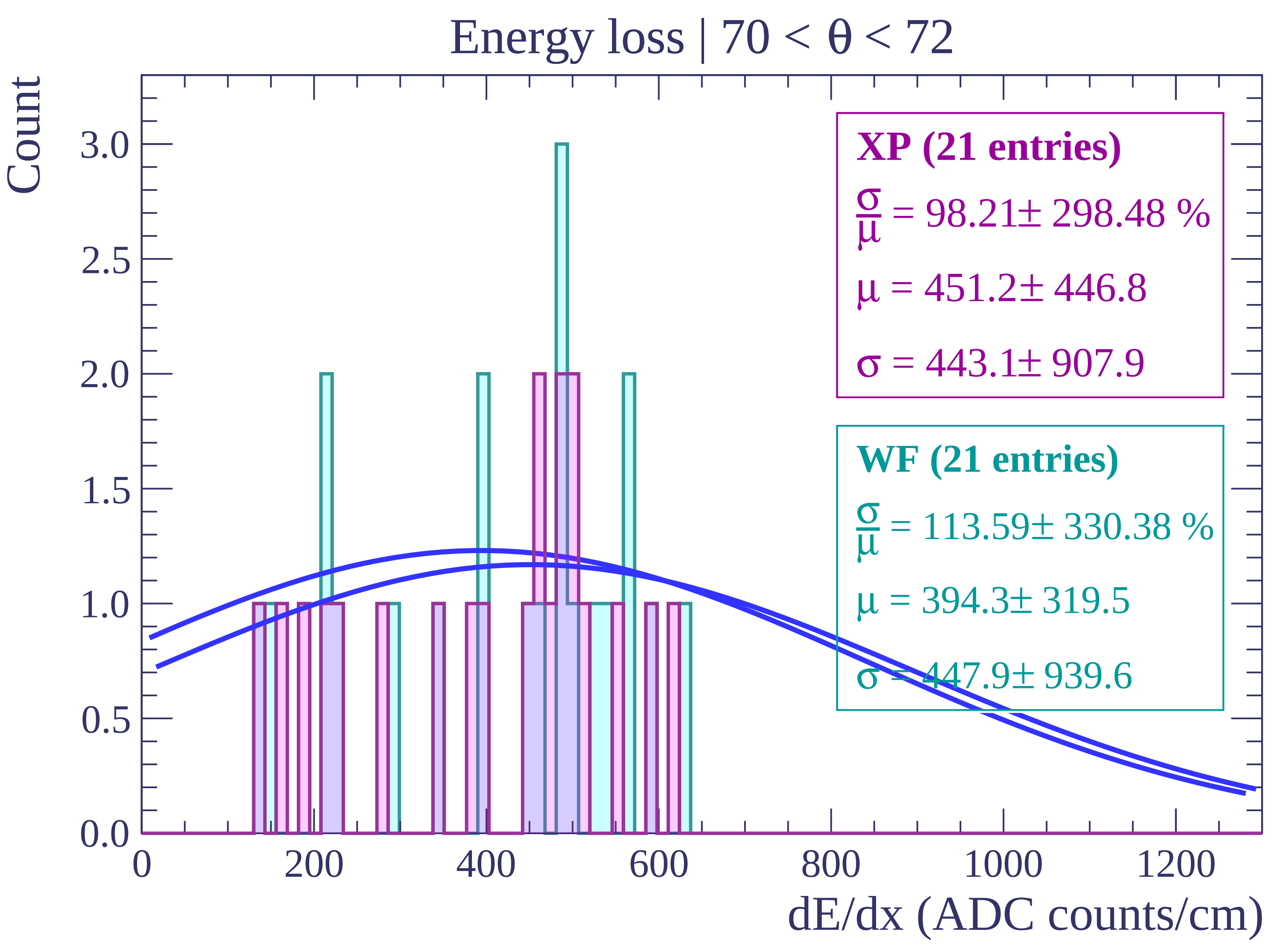


Energy loss | 64 < θ < 66
Count
dE/dx (ADC counts/cm)
XP (95 entries)
σ/μ = 13.40± 2.29 %
μ = 487.5± 9.5
σ = 65.3± 9.9
WF (95 entries)
σ/μ = 15.39± 2.42 %
μ = 520.3± 10.4
σ = 80.1± 11.0
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

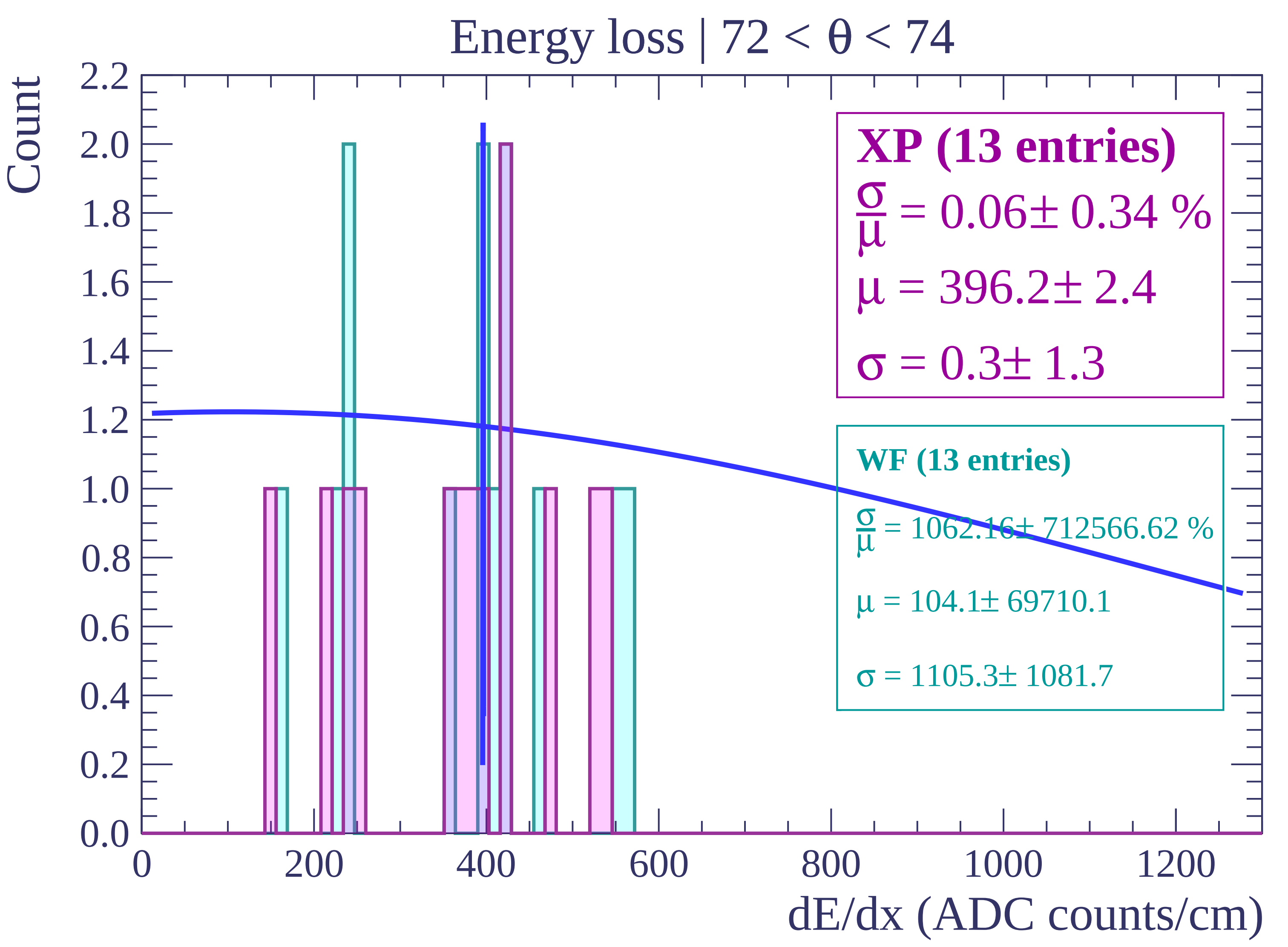

Energy loss | 66 < θ < 68
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
XP (52 entries)
σ/μ = 28.30± 14.20 %
μ = 494.8± 41.5
σ = 140.0± 58.5
WF (52 entries)
σ/μ = 26.00± 11.22 %
μ = 519.3± 35.2
σ = 135.0± 49.1

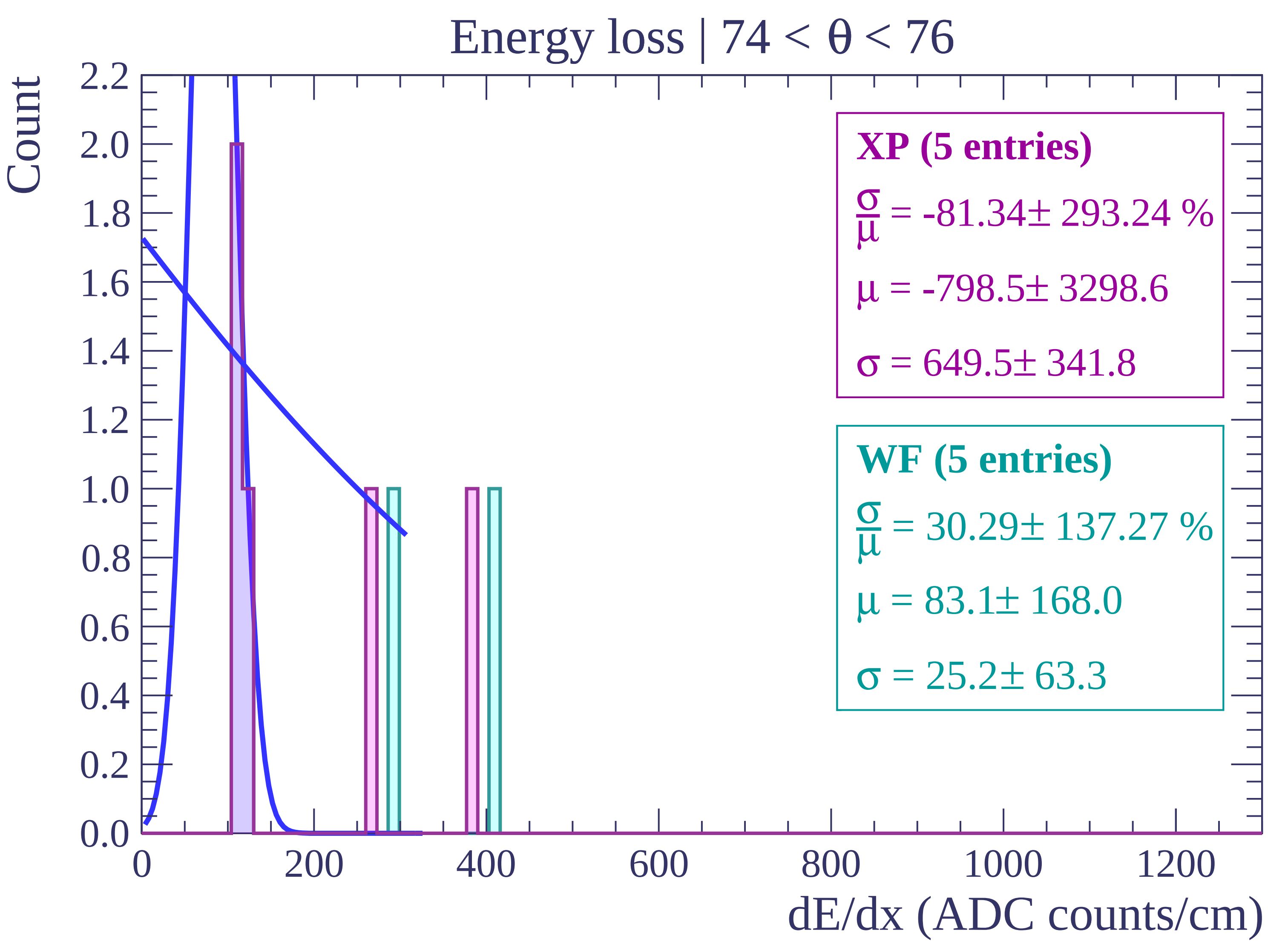


Energy loss | 68 < θ < 70
Count
dE/dx (ADC counts/cm)
XP (34 entries)
σ/μ = 25.15± 16.92 %
μ = 508.4± 52.7
σ = 127.8± 72.8
WF (34 entries)
σ/μ = 77.17± 60134.18 %
μ = 856.2± 666301.5
σ = 660.8± 664.2
0
1
2
3
4
5
0
200
400
600
800
1000
1200

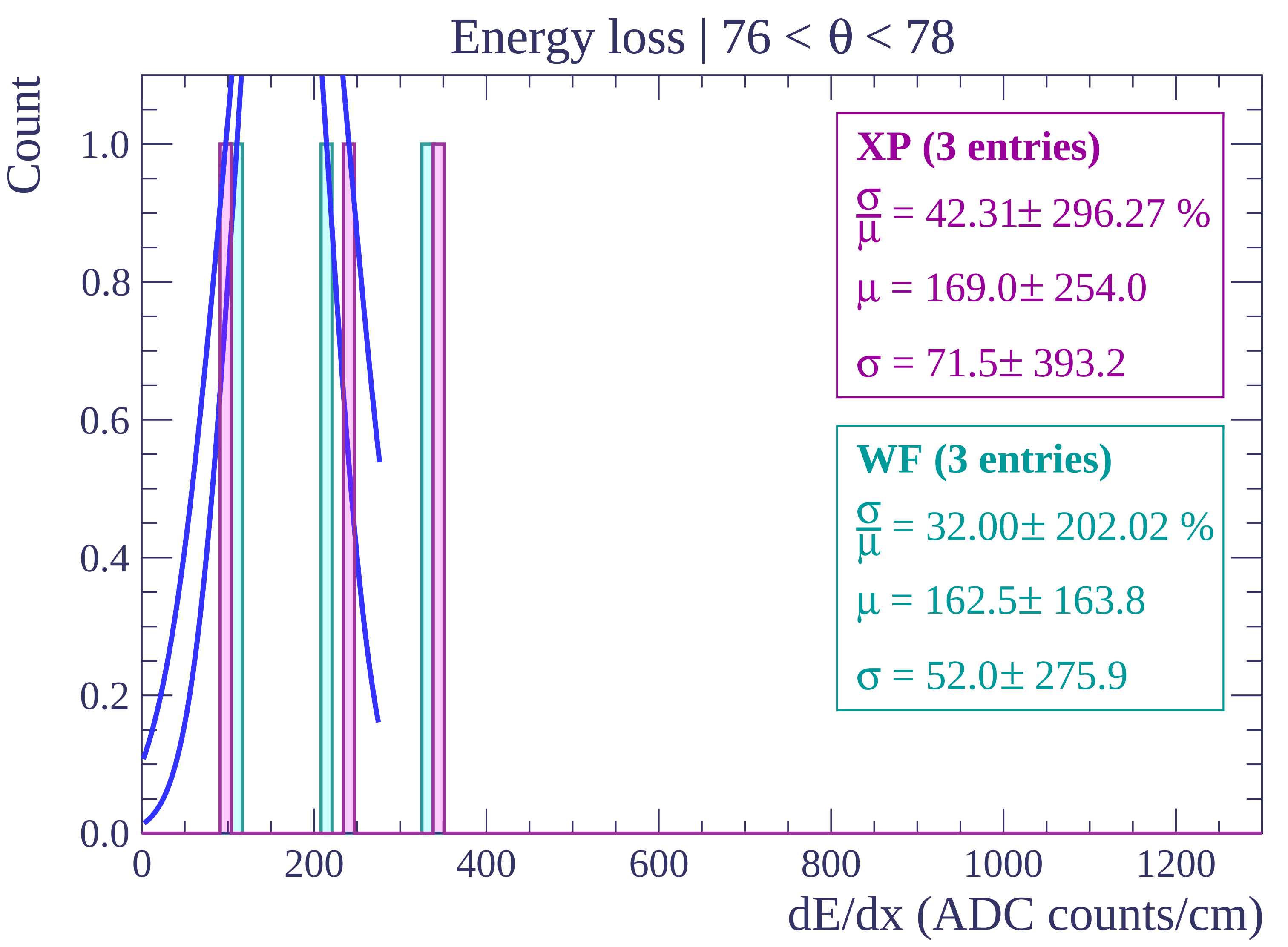


Energy loss | 70 < θ < 72
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (21 entries)
σ/μ = 98.21± 298.48 %
μ = 451.2± 446.8
σ = 443.1± 907.9
WF (21 entries)
σ/μ = 113.59± 330.38 %
μ = 394.3± 319.5
σ = 447.9± 939.6

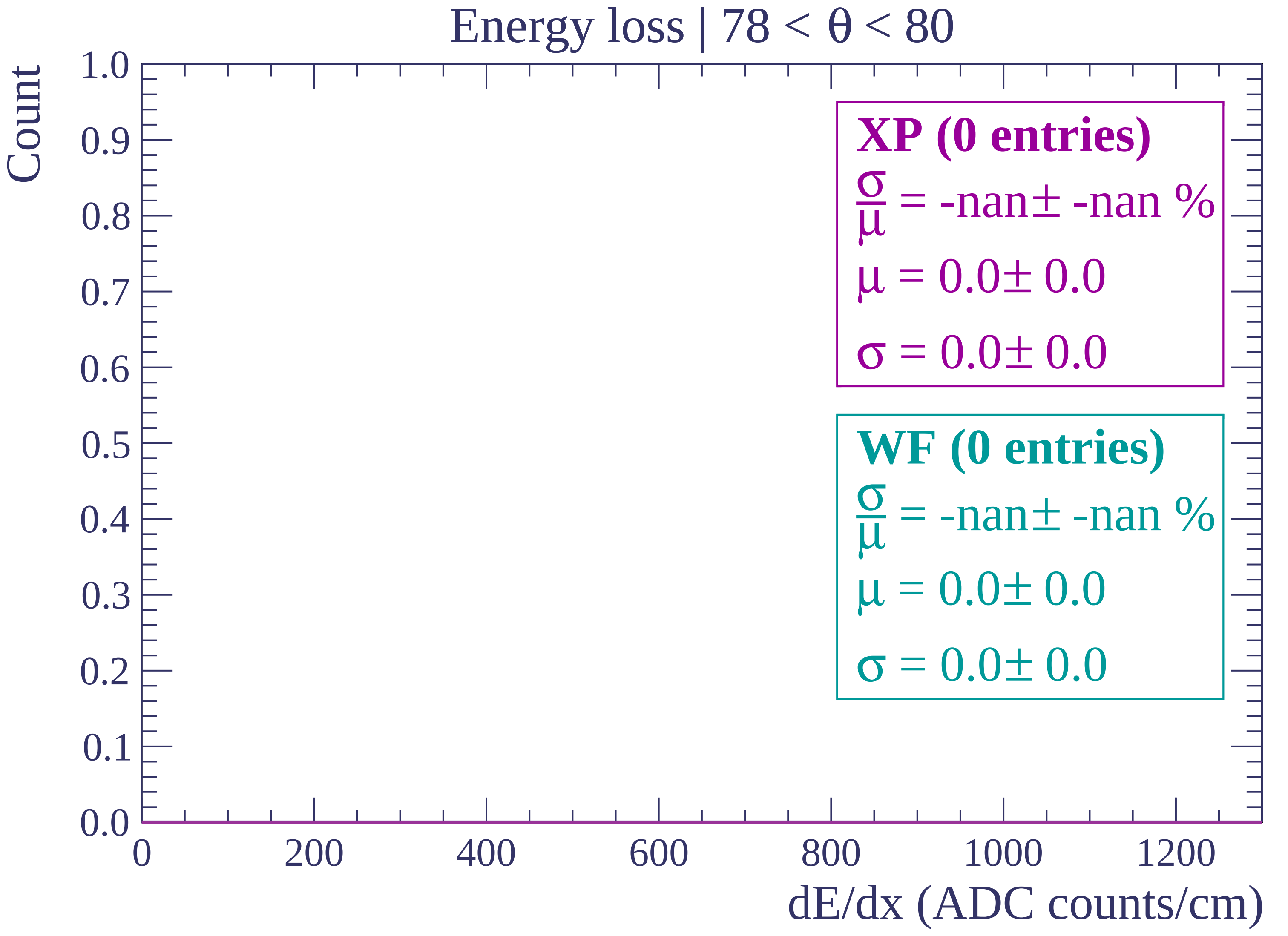

Energy loss | 72 < θ < 74
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP (13 entries)
σ/μ = 0.06± 0.34 %
μ = 396.2± 2.4
σ = 0.3± 1.3
WF (13 entries)
σ/μ = 1062.16± 712566.62 %
μ = 104.1± 69710.1
σ = 1105.3± 1081.7



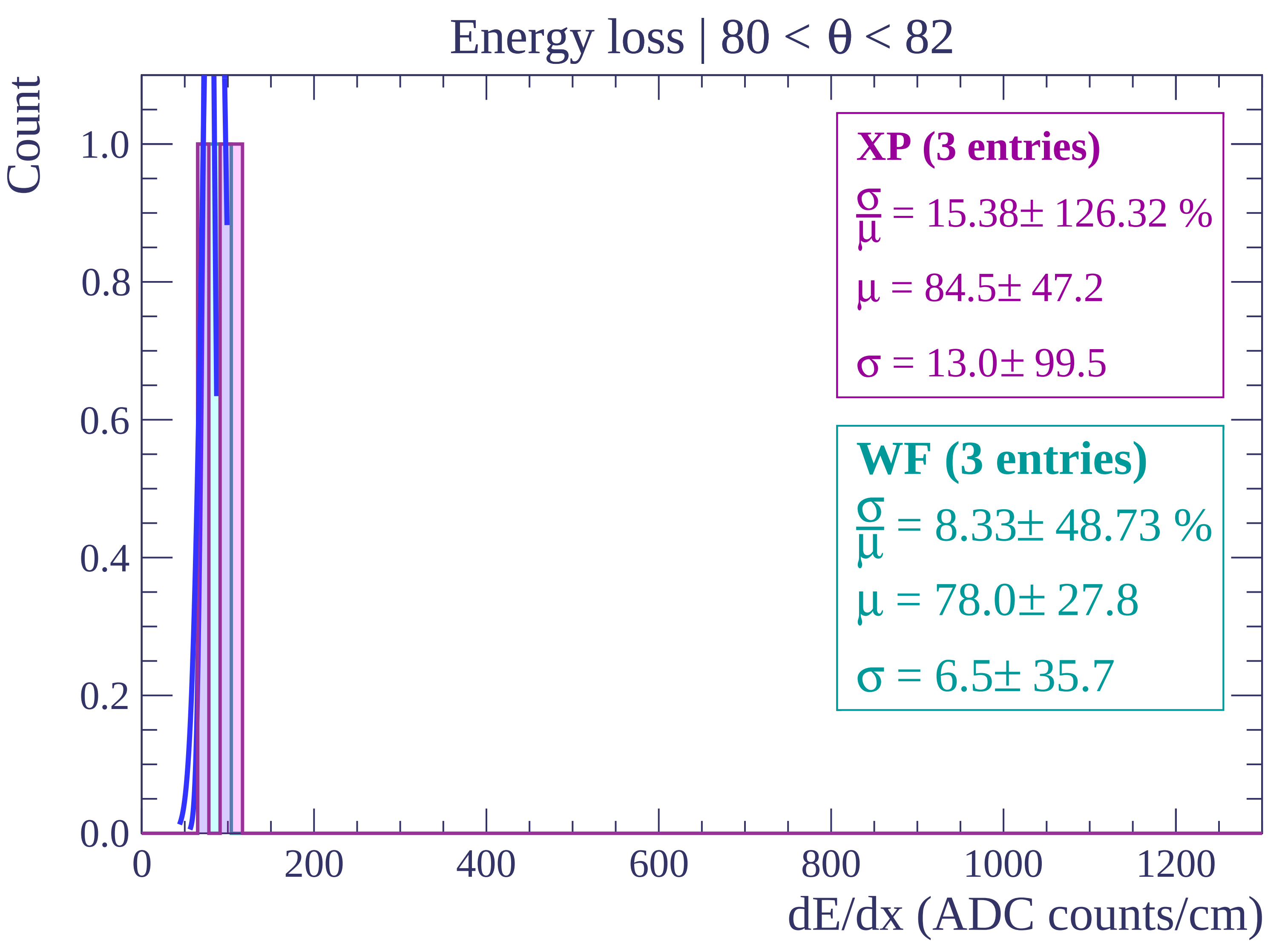
Energy loss | 74 < θ < 76
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5 entries)
σ/μ = -81.34± 293.24 %
μ = -798.5± 3298.6
σ = 649.5± 341.8
WF (5 entries)
σ/μ = 30.29± 137.27 %
μ = 83.1± 168.0
σ = 25.2± 63.3

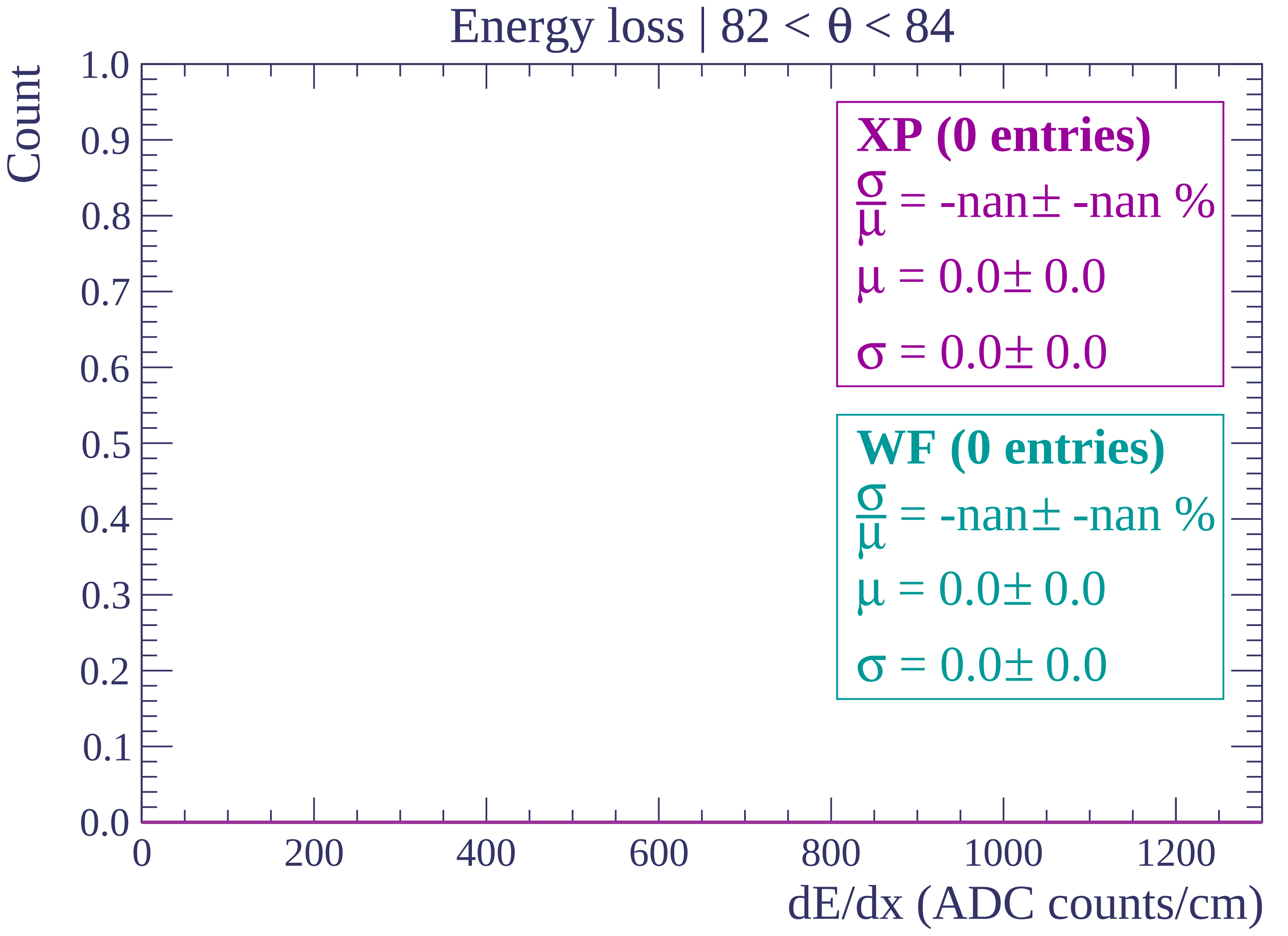


Energy loss | 76 < θ < 78
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP (3 entries)
σ/μ = 42.31± 296.27 %
μ = 169.0± 254.0
σ = 71.5± 393.2
WF (3 entries)
σ/μ = 32.00± 202.02 %
μ = 162.5± 163.8
σ = 52.0± 275.9

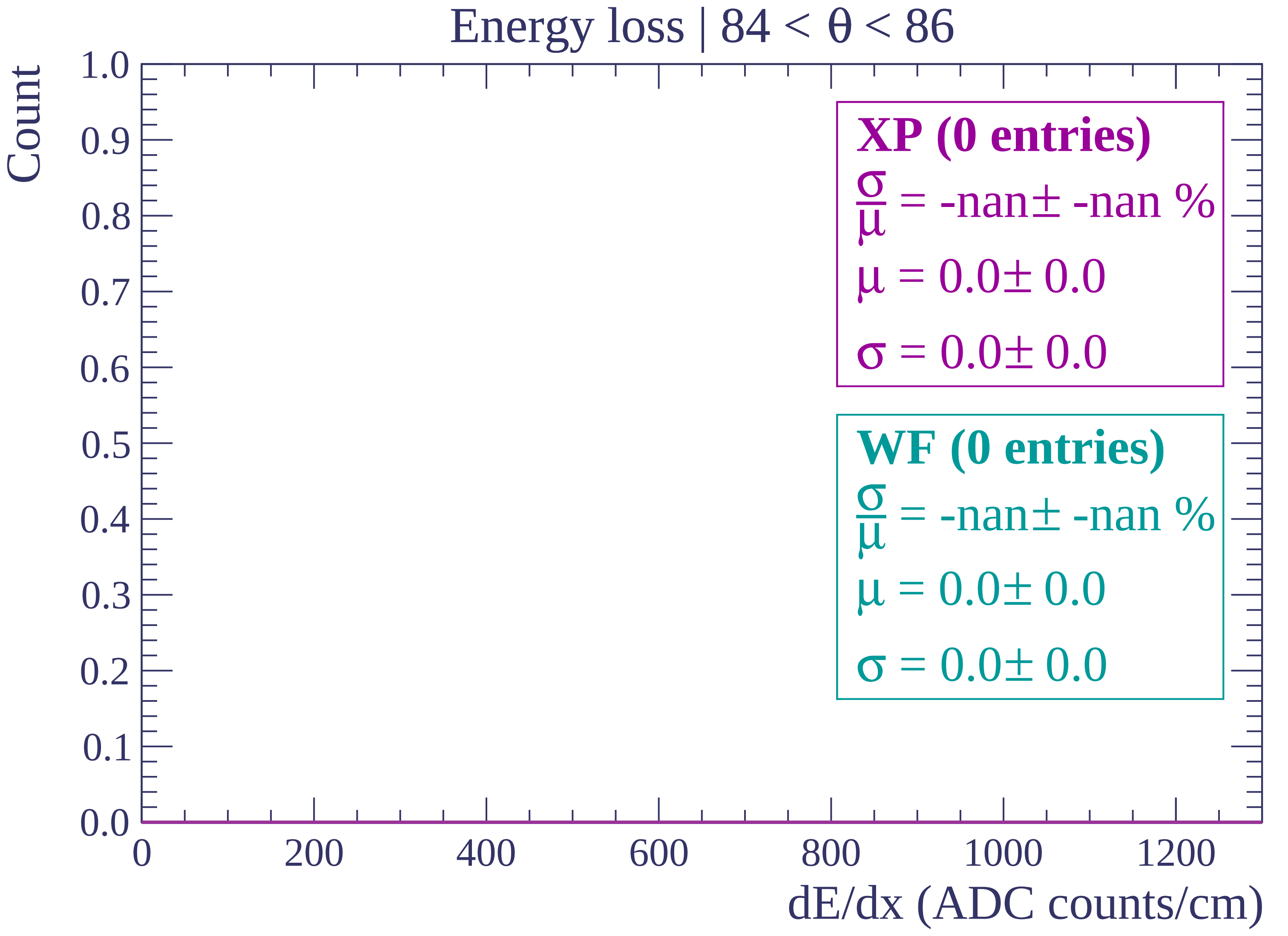


Energy loss | 78 < θ < 80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

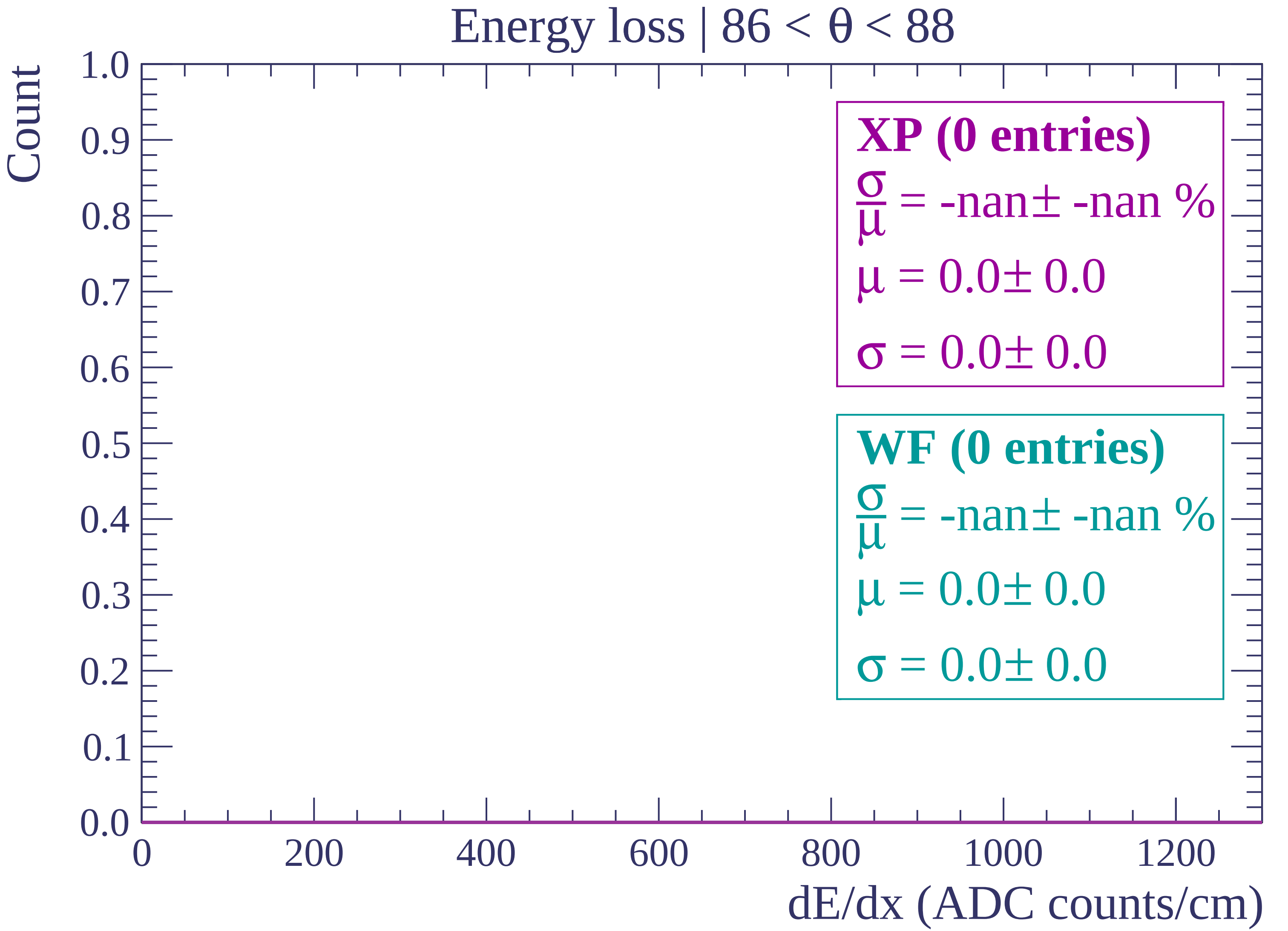

Energy loss | 80 < θ < 82
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3 entries)
σ/μ = 15.38± 126.32 %
μ = 84.5± 47.2
σ = 13.0± 99.5
WF (3 entries)
σ/μ = 8.33± 48.73 %
μ = 78.0± 27.8
σ = 6.5± 35.7



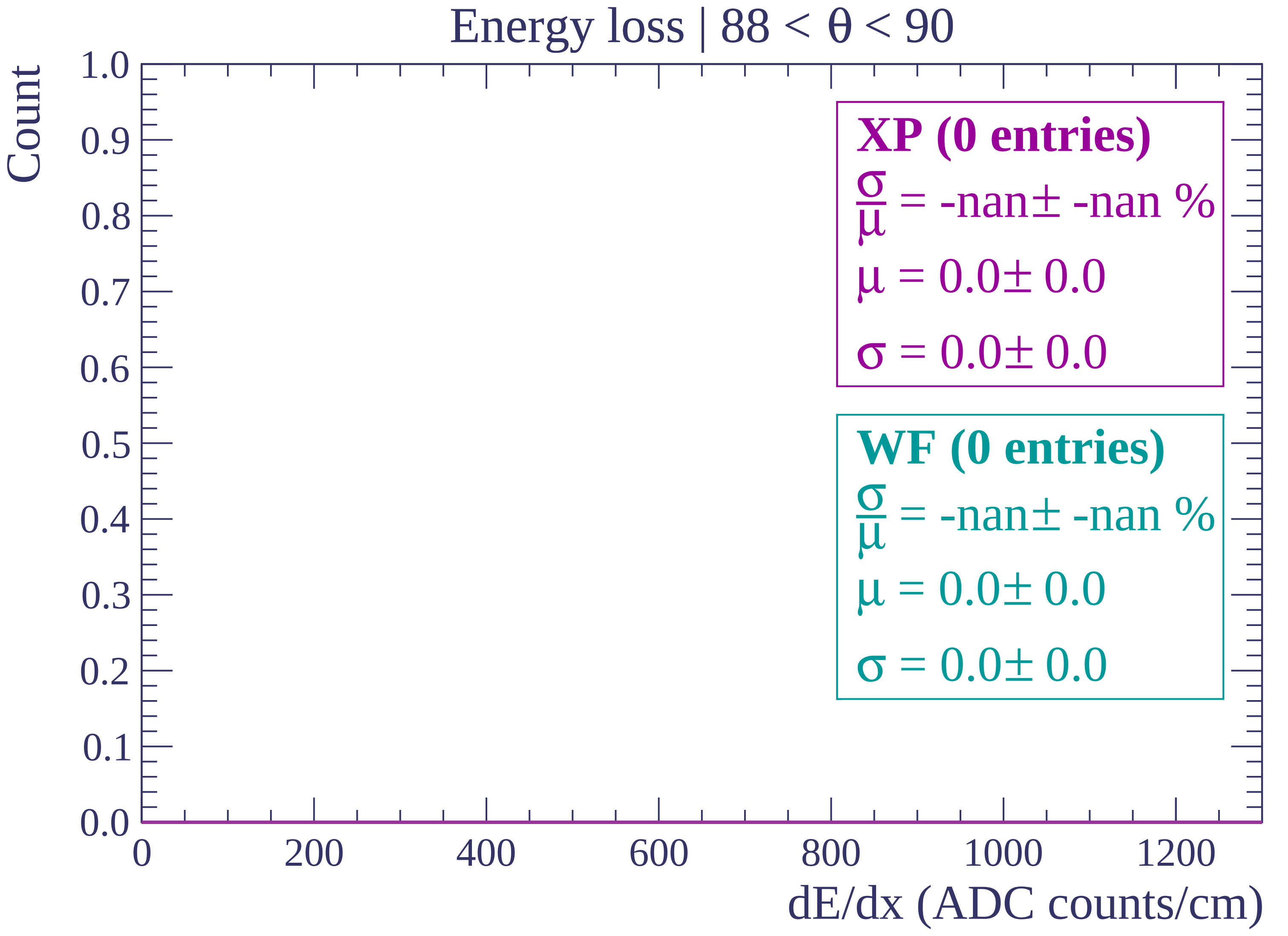
Energy loss | 82 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



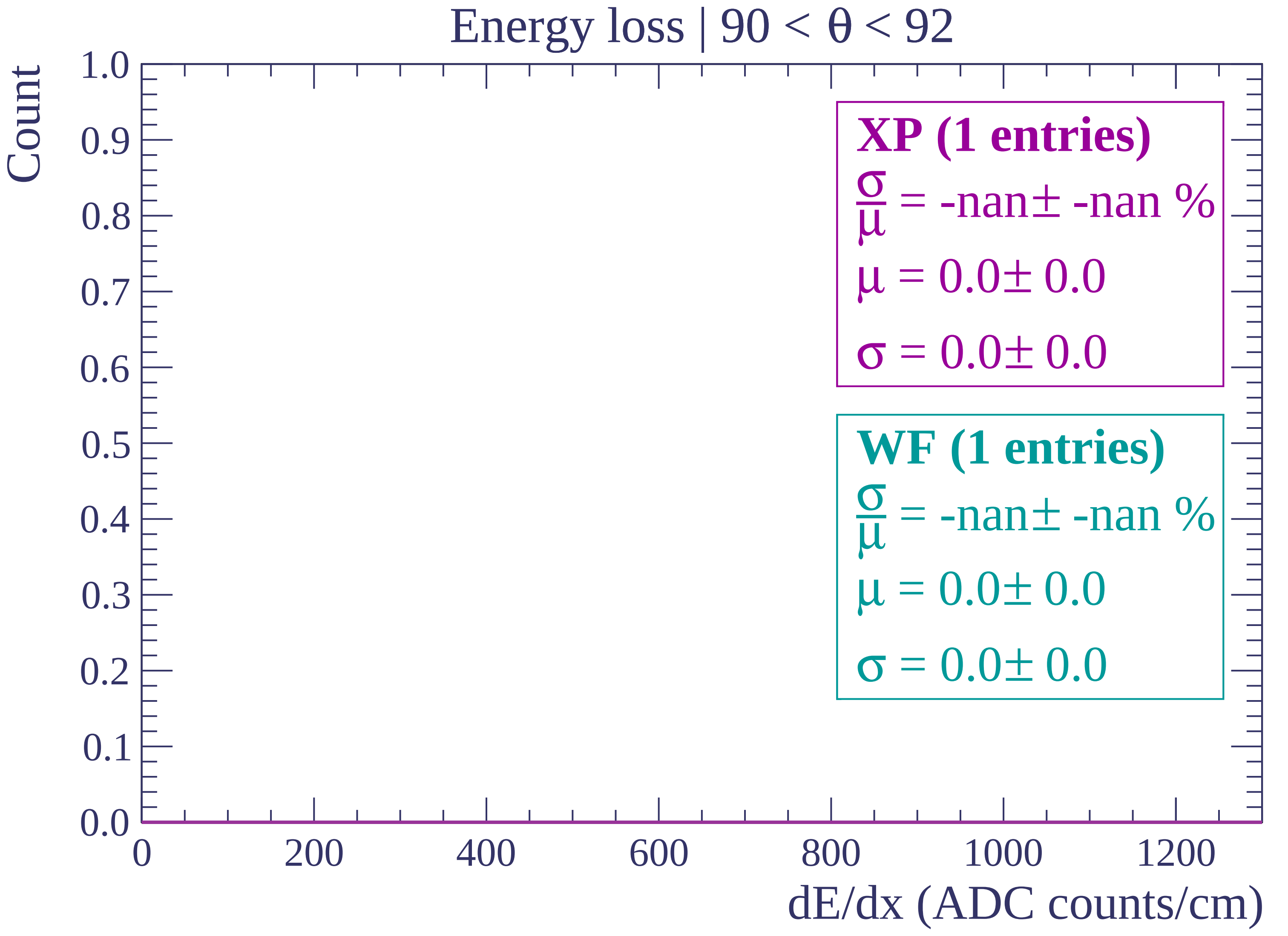
Energy loss | 84 < θ < 86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



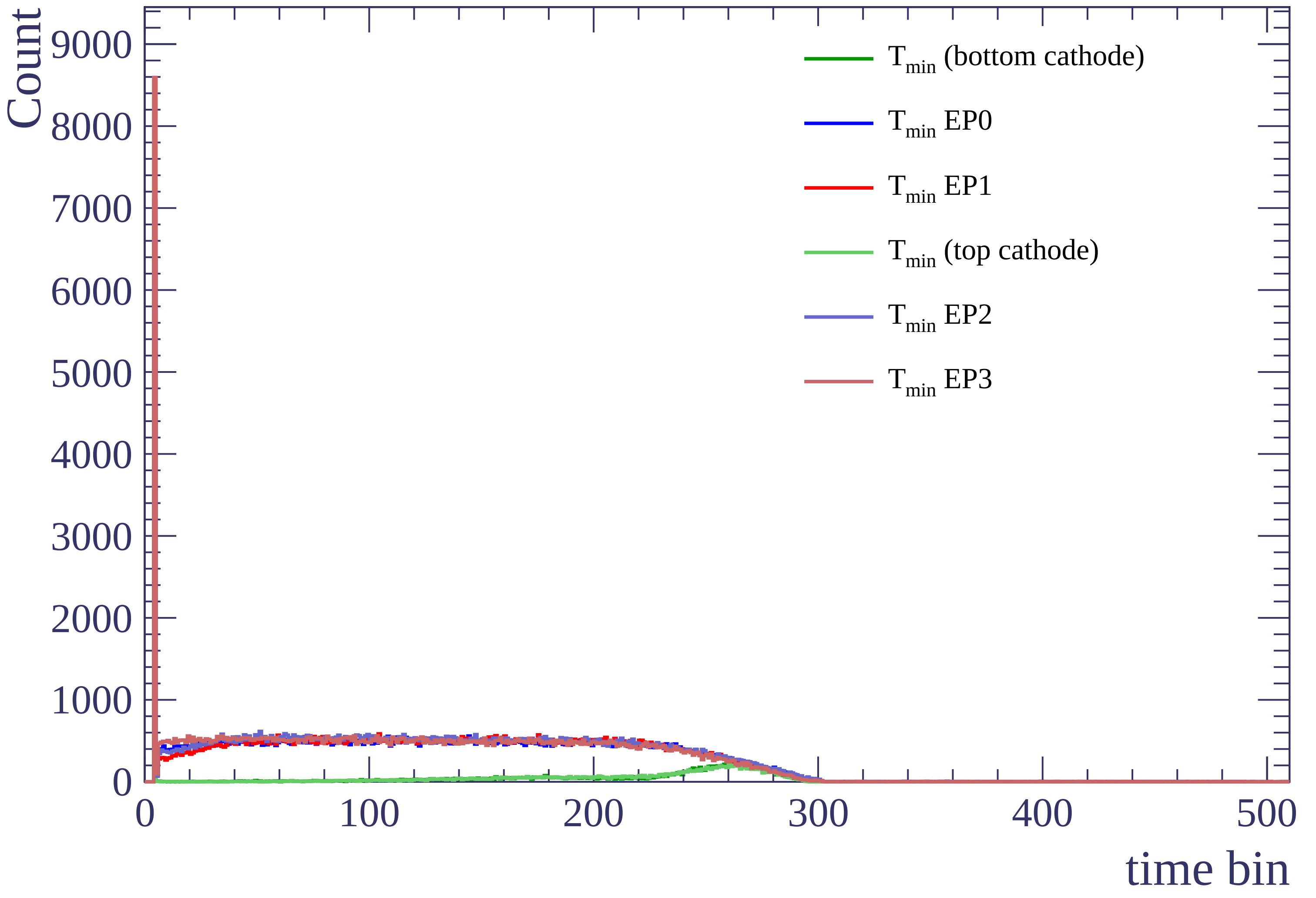
Energy loss | 86 < θ < 88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

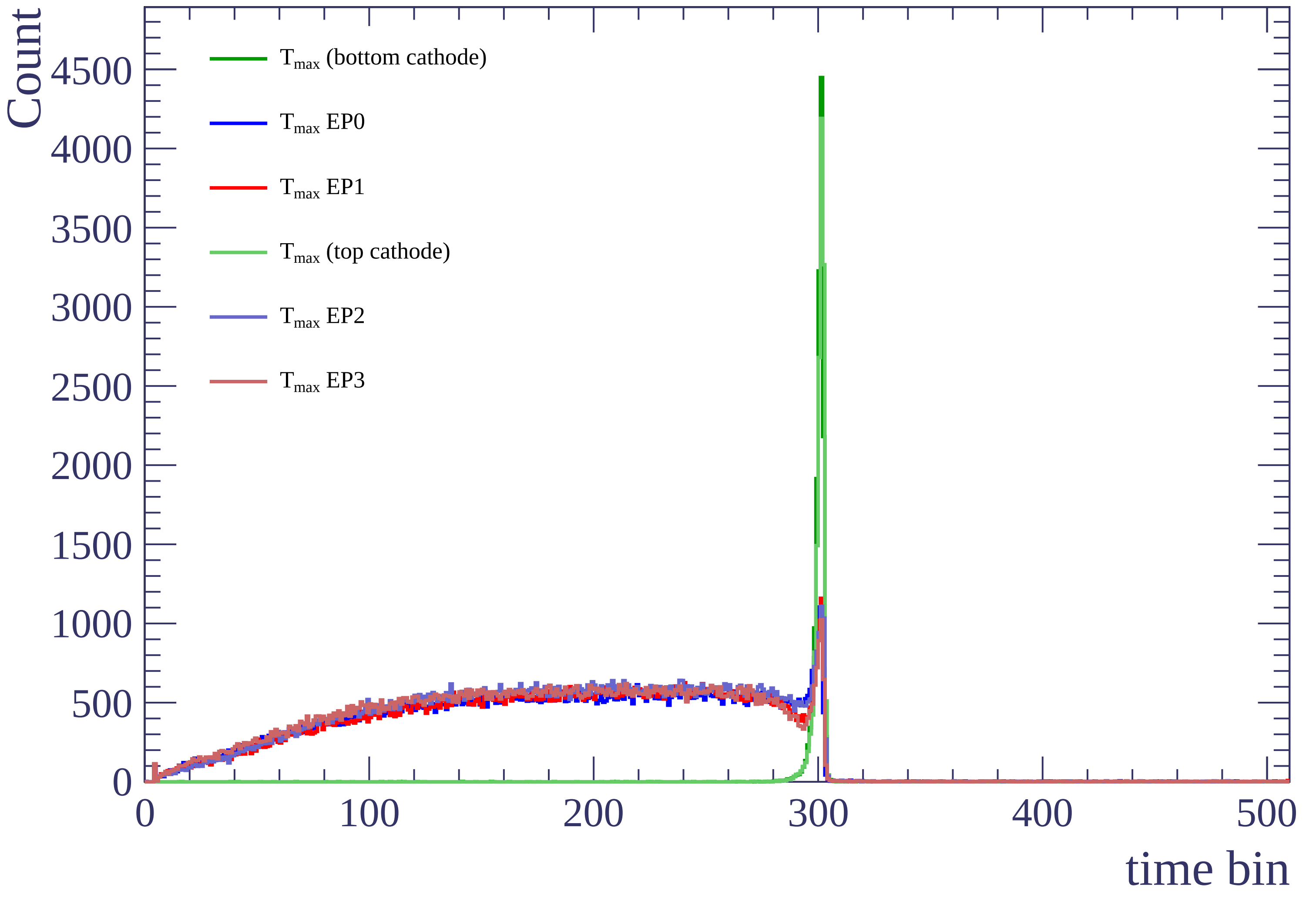


Energy loss | 88 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 90 < θ < 92
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (1 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
500
1000
1500
2000
2500
3000
3500
4000
4500
0
100
200
300
400
500